

论文证明材料汇编

老师您好！点击下面的目录可以跳转到对应文件

一、已发表论文		
01	Information Processing & Management 第一作者 中科院1区 • TOP • SSCI Q1 • IF 8.1 • FMS/CCF B	2
02	Humanities and Social Sciences Communications 第一作者 中科院2区 • SSCI Q1 • IF 4.8	26
03	British Journal of Educational Technology 共同第一作者 中科院1区 • TOP • SSCI Q1 • IF 13.0	42
04	Journal of Professional Capital and Community 共同第一作者 SSCI Q2 • IF 2.0	72
05	Business Process Management Journal 通讯作者 SSCI Q1 • IF 5.2	94
06	Aslib Journal of Information Management 通讯作者 SSCI Q1 • IF 3.5	116
07	建筑与文化 通讯作者 省级期刊	142
08	Journal of Innovation and Social Science Research 第一作者	151
09	Water Environment Research 第二作者 SCI Q1 • IF 2.8	152
10	Journal of Information Science 第三作者 FMS B • SSCI Q2 • IF 2.6	171
11	International Journal of Mentoring and Coaching in Education 第三作者 ESCI Q2 • IF 1.6	190

二、在投稿论文（同行评审证明）		
12	经济理论与经济管理 第三作者 • 同行评审中	210
13	Journal of Business Research 第一作者 • 同行评审中	211
14	Engineering, Construction and Architectural Management 第一作者 • 同行评审中	212
15	Information Processing & Management 通讯作者 • 同行评审中	213
16	The Internet and Higher Education 第二作者 • 同行评审中	214
17	Technology Analysis & Strategic Management 第一作者 • 返修中	215
18	Business Ethics, the Environment & Responsibility 第一作者 • 同行评审中	216
三、会议论文与学术交流		
19	AERA 2026 Conference Paper	217
20	ICIET 2022 录用证明	229



Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/infoproman

Official or Influencer? An AI-Enhanced Analytical Framework for Decoding Multimodal Persuasion in Government Marketing Videos

Yuying Luo^a, Xue Lei^a, Wenxuan Zhang^b, Hao Zheng^b, Xueguo Xu^a, Jun Xu^{b,*}

^a School of Management, Shanghai University, Shanghai, China

^b Department of Educational Information Technology, East China Normal University, Shanghai, China

ARTICLE INFO

Keywords:

Multimodal Content Analysis
Human-Computer Collaboration
Government Communication
Digital Persuasion
Official Influencers

ABSTRACT

Local officials increasingly use short-video platforms for direct-to-consumer marketing, creating tension between bureaucratic identity and platform logic. This multimodal persuasion context is difficult to analyze at scale without losing theoretical depth. To address this gap, we propose and validate an AI-enhanced analytical framework centered on a "Human-AI Dynamic Cognitive Dialogue" protocol, which turns the Multimodal Large Language Model (MM-LLM) from an automated "black box" into a theory-guided analytical mechanism and reduces the text-centric bias of traditional computational methods. Crucially, it uses theory-driven human oversight to guide AI-assisted induction, preserving methodological transparency while eliciting strategy configurations beyond researcher priors. Based on 779 Douyin videos from Chinese officials, the study identifies a "bifurcation of authority": authority cues are differentially associated with engagement depending on personified presentation (e.g., official attire) rather than institutionalized context (e.g., a government office). Engagement-cost differentiation is also evident: higher-cost actions such as sharing correspond to a synergy of personified authority, narrative resonance, and high cognitive value. Smaller accounts show a "legitimacy compensation" pattern, with strong authority signals associated with higher engagement among accounts lacking established social capital. The study establishes a replicable protocol for decoding unstructured video data and offers a bounded extension of the Elaboration Likelihood Model by showing how dynamic multimodal cues correspond to differentiated engagement patterns in algorithmic short-video environments.

1. Introduction

Amid the global tide of digitalization, the interaction between governments and the public is undergoing a profound shift toward "personalization," as officials increasingly act as charismatic communication agents (Bangero, 2025) who engage citizens directly via social media (e.g., Ecker-Ehrhardt, 2023; Ozdemir et al., 2025). In China's distinctive governance context, grassroots officials have extended this trend by using short videos for direct-to-consumer (D2C) marketing, thereby making it a more common practice (Cheng et al., 2024). They must continually negotiate between the "bureaucratic logic" that sustains institutional authority and the "influencer logic" that caters to platform algorithms and high engagement (Chen & Gong, 2025). This complex identity negotiation is rendered

* Corresponding author.

E-mail address: xujun1569@gmail.com (J. Xu).

<https://doi.org/10.1016/j.ipm.2026.105068>

Received 2 October 2025; Received in revised form 15 June 2026; Accepted 18 July 2026

Available online 24 July 2026

0306-4573/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

entirely through the multimodal signals of the short-video medium.

However, investigating this phenomenon presents a dual challenge that intertwines theoretical ambiguity with methodological limitations. Theoretically, while the Elaboration Likelihood Model (ELM) explains persuasion via central and peripheral routes, it was originally conceptualized for static media (Chandrasekaran et al., 2025). It remains unclear how conflicting cues, such as the "bureaucratic" need for authority and the "influencer" need for authenticity, interact dynamically within a short video to shape user engagement (Lu & Shen, 2023). Specifically, existing literature has not sufficiently explored how these multimodal strategies are differentially associated with behaviors with varying cognitive costs (e.g., the low cost of a "like" versus the high social cost of a "share") (Song et al., 2026). Without addressing this, current theoretical frameworks may struggle to fully account for the temporal nuances of persuasion specific to algorithm-driven streaming media environments.

Methodologically, bridging this theoretical gap is hindered by the intrinsic complexity of unstructured video data (Yin et al., 2025). Traditional content analysis struggles to capture the dynamic interplay of visual and auditory cues at scale (Kirilenko & Stepchenkova, 2025), while rule-based computational pipelines often operate in a "semantic vacuum" lacking theoretical guidance (Tong et al., 2022). Furthermore, although Multimodal Large Language Models (MM-LLM) offer promise, deploying them as unsupervised "black boxes" introduces systematic biases and epistemic uncertainties (e.g., Hu et al., 2023; Qi et al., 2023). Consequently, scholars face a methodological bottleneck. Without a replicable protocol to analyze video data at scale, it remains difficult to empirically examine how conflicting identity cues interact over time and relate to user behavior.

To address these challenges and move beyond treating AI as an automated "black box," this study develops an AI-enhanced multimodal content analysis framework that positions LLMs as analytic mechanisms within structured human-AI cognitive collaboration. Guided by human theoretical judgment, the framework counteracts modality bias, supports rigorous qualitative procedures (Gao et al., 2024), and uses an iterative "dynamic cognitive dialogue" protocol to induce and validate communication strategies from video data. Its value lies in combining scale with human-AI complementarity, allowing emergent strategy configurations beyond researchers' priors and supporting deeper theoretical insight than either agent could achieve alone (Kirilenko & Stepchenkova, 2025; Lu & Zhang, 2024; Xu et al., 2026).

To systematically bridge the theoretical gaps identified above using this methodological innovation, this study pursues three specific research objectives: (1) To construct a theoretically grounded, human-in-the-loop AI framework for decoding multimodal government communication; (2) To empirically identify the strategy configurations used by officials to navigate the tension between bureaucratic and platform logics; (3) To examine how these strategies are differentially associated with public engagement behaviors across varying levels of cognitive cost. Guided by these objectives, we propose the following research questions:

- RQ1: What key multimodal combinations of strategies do grassroots officials employ in D2C marketing videos to balance their dual identities as "government officials" and "influencers"?
- RQ2: How are combinations of strategies differentially associated with public engagement behaviors across varying cost levels (likes, comments, shares)?
- RQ3: To what extent can the AI-enhanced analytical framework reliably and validly parse complex video content?

To achieve these objectives, this study grounds its analysis in a dataset of 779 videos from Chinese officials on Douyin, utilizing a validated "Human-AI Dynamic Cognitive Dialogue" protocol to extract high-dimensional semantic features. Our empirical results document a "bifurcation of authority," where personified cues are positively associated with engagement while institutional contexts are negatively associated with it, and support the interpretation of distinct "cost ladders" for digital persuasion. By establishing this replicable methodological blueprint, the study contributes to Information Management by offering a valid protocol for unstructured video analysis, and extends the Elaboration Likelihood Model into the algorithmic era by elucidating the dynamic interplay of conflicting persuasion cues.

2. Theoretical Background

2.1. Identity Negotiation Dilemmas in Platformized Governance

When grassroots officials enter digital platforms as content creators, they are immediately situated between two fundamentally conflicting institutional logics: a traditional bureaucratic logic and an emergent platform logic. Bureaucratic logic, rooted in hierarchy and procedural compliance, prioritizes authoritative and orderly information transmission; platform logic, governed by algorithmic visibility mechanisms, rewards personalized and affectively charged content that elicits high engagement. This institutional complexity compels grassroots officials to engage in continuous identity work, seeking a dynamic equilibrium between maintaining the legitimacy of their bureaucratic role and meeting the platform's expectation of "authenticity" (J. Kim & Kim, 2024). Strategically, they cultivate a hybrid identity that preserves the authority of an official representative while exhibiting personal qualities that can evoke emotional resonance.

Existing scholarship has notable blind spots in explaining this phenomenon. Research on government social media has examined policy communication at the macro level (Qin et al., 2025) and governance objectives (Guo et al., 2025), yet it often treats individual officials as passive executors of institutional will, overlooking their agentic, micro-level strategic choices. Meanwhile, scholarship on digital platforms has analyzed how algorithms shape content production (Chen & Shi, 2025) but rarely considers the distinctive dynamics that arise when communicators wield public authority. More critically, the literature has not established a coherent analytical chain from the macro-sociological question of why officials must negotiate identity (Chen & Gong, 2025) to the

micro-communication question of how they accomplish this negotiation through concrete information strategies (Grantham et al., 2025). To bridge this gap, we first turn to established persuasion theory to map the mechanisms available to these officials.

2.2. The Elaboration Likelihood Model in Digital Persuasion

To systematically decode the specific strategies officials employ, we adopt the Elaboration Likelihood Model (ELM) as our foundational theoretical lens. As a dominant framework in computer-mediated communication, ELM posits that persuasion occurs through two distinct cognitive pathways based on the audience's motivation and ability to process information (Petty & Cacioppo, 1986).

The central route involves high cognitive elaboration, where audiences critically evaluate the quality of arguments and the substance of the message. In this study, a construct is treated as central-route when its persuasive relevance depends primarily on audiences' processing of argument quality, informational density, message substance, or cognitive utility. In the context of government D2C marketing, this route corresponds not only to the strength of rational policy appeals but also to the density of high-utility product information (Wang et al., 2025). Both dimensions represent substantive cognitive value, whether for institutional legitimacy or transactional utility, and require sustained viewer attention and cognitive effort to process. Conversely, the peripheral route operates when cognitive elaboration is low. A construct is treated as peripheral-route when its persuasive relevance is expected to operate mainly through source heuristics, visual authority, affective resonance, scene context, editing rhythm, or directive prompts that can be evaluated with limited elaboration. Here, audiences rely on heuristic cues, such as official uniforms, visual aesthetics, or emotional resonance, to make quick judgments without scrutinizing the underlying arguments (Xu et al., 2025). Ideally, this dual-process model allows officials to balance their strategies: using central cues to convey bureaucratic legitimacy and peripheral cues to meet platform demands for engagement. However, while ELM provides a structural framework for classifying these cues, it does not fully account for the relational dynamics inherent to social media. Consequently, we extend this theoretical lens by incorporating social identity, perceived social presence, and trust formation as explanatory mechanisms (Li et al., 2024). This integration clarifies why specific authoritative or affective cues succeed or fail within ELM's cognitive routes depending on the official's evolving relationship with the audience. These criteria provide the conceptual basis for the construct mapping used in the empirical analysis.

2.3. The Temporal Challenge: Reassessing Persuasion in Dynamic Media

However, applying this classic framework to the attention-scarce environment of short videos presents fundamental theoretical challenges. The Limited Capacity Model of Motivated Mediated Message Processing (LC4MP) suggests that before any ELM-based processing can occur, content must first survive an initial attentional filter. Specific low-level audiovisual features—such as rapid cutting, color contrast, or abrupt sound changes—trigger orienting responses that forcibly allocate cognitive resources (Lang, 2000). Operating largely outside conscious control, these perceptual cues modulate visual search efficiency and directly shape initial information encoding (Cardamone et al., 2025; Clayton & Myrick, 2025).

Consequently, a digital paradox emerges. The perceptual cues that capture immediate attention often map onto peripheral-route cues in ELM. This creates a potential conflict with officials' intention to communicate authoritative, reasoned messages via the central route. If officials over-rely on "influencer-style" peripheral cues to clear the attentional hurdle, they risk eroding the central-route authority required for bureaucratic legitimacy.

Yet, because the orienting response is transient and fleeting (Potter & Bolls, 2012), static variable coding fails to capture how attention is sustained or lost along the video timeline. This reveals a deeper challenge: classic persuasion theories were essentially developed for static information (e.g., print ads or text posts) and thus face limitations when applied to short video, a medium characterized by temporal dependency. The persuasive force of short video may not arise from a static bundle of multimodal cues alone, but from their temporal evolution and cumulative patterns (Sim et al., 2024). For instance, a cut from a government office to an agricultural field derives its persuasive impact from the dynamic shift itself, which cannot be reconstructed by simply summing separate static codings of the two scenes (Wu et al., 2025). This dynamic complexity exposes an internal deficit in existing theory: current frameworks lack the constructs needed to describe and explain how persuasive effects accumulate and transform over time.

Temporal dependency is therefore not merely a methodological obstacle but a profound theoretical challenge—current persuasion theories remain ill-equipped to fully account for information processing in dynamic media environments. To resolve these theoretical frictions, our AI-enhanced framework shifts the unit of interpretation from isolated static feature sets to temporally ordered multimodal cue configurations. This refinement does not reject ELM's dual-route logic; rather, it specifies how central and peripheral cues unfold, combine, and change salience across short-video sequences. This shift allows us to analyze more directly how multimodal strategies evolve and interact over time, rather than viewing them as static signals.

3. Data

To test the framework, our data strategy constructed an empirical corpus reflecting direct-to-consumer (D2C) marketing by "government official influencers" in a complex digital media environment. We selected Douyin, China's leading short-video platform, because its algorithmically driven distribution and large user base provide a natural setting for observing digital engagement behaviors (Chen & Shi, 2025). This setting is analytically appropriate but platform-specific: Douyin's recommendation logic, creator norms, and short-video culture may shape which official videos become visible and how users interact with them. Against the policy backdrop of rural revitalization, many grassroots officials use the platform personally to promote local economies, creating naturally occurring data on the negotiation between bureaucratic and influencer identities. Our multi-stage, theory-driven sampling process (Fig. 1)

supports the analytic validity and consistency of the final sample within this defined platform context.

Corpus construction began with purposive sampling to identify and isolate official accounts aligned with our research constructs. We adopted three stringent inclusion criteria: first, the account had to bear an official institutional verification badge or be verifiably affiliated with public office through open sources; second, its content had to focus on substantive promotion of local economic resources rather than purely policy publicity or personal life-logging; third, the account needed to remain active throughout the observation window (from June to August 2025). Using a combination of algorithmic recommendations and systematic keyword search, we initially identified 159 qualifying accounts. To extract content highly relevant to D2C marketing from the tens of thousands of videos posted by these accounts, we implemented a capped extraction strategy: for each account, we retrieved up to the top 20 videos ranked by public engagement (sum of likes, comments, and shares). For accounts with a total volume below this threshold, the full available corpus was retained to ensure comprehensive coverage, resulting in an initial pool of 3,149 videos (with an average of 19.8 videos per account due to the capped extraction strategy and varying account activity levels). This strategy prioritizes analytically relevant and visible D2C communication, but it may also overrepresent content that had already attracted engagement or algorithmic amplification and underrepresent low-exposure or less discoverable videos. We then addressed the challenge of detecting marketing intent by implementing a hierarchical, lexicon-based weighted scoring heuristic to filter textual metadata (descriptions and hashtags). This deterministic strategy assigned cumulative relevance scores based on three positive tiers, specific commodity entities (weight +3), explicit transactional markers (e.g., "buy," "link"; weight +2), and general rural revitalization contexts (weight +1), while penalizing unrelated governance topics (weight -2) and provisionally retaining entries with insufficient information to minimize false negatives, reducing the candidate set to 847. To validate algorithmic screening, two trained research assistants independently verified this subset, achieving excellent inter-coder reliability (Krippendorff's $\alpha = 0.947$), with disagreements adjudicated by a third coder. Finally, to bound computational scale and define analytic scope, we excluded videos longer than 300 seconds. The resulting analytic sample comprised 779 valid videos spanning 23 provincial-level regions across China, providing a robust empirical foundation for the subsequent analyses within the stated sampling scope.

For multimodal analysis, we performed systematic feature engineering for each video in the final sample. First, all audio tracks were extracted and standardized to facilitate automatic speech transcription and acoustic feature extraction. On the visual side, to balance the capture of key narrative shifts with the computational cost of AI analysis, we implemented a duration-adaptive keyframe sampling strategy. Rather than uniform or random sampling, this strategy systematically targets key visual information near the opening, middle, and closing segments, thereby preserving core temporal cues while optimizing efficiency—an especially important consideration for dynamic video data. Let T denote the total video duration (in seconds). The set of keyframe timestamps $S(T)$ is defined as:

$$\mathcal{S}(T) = \{2, 5, 10, 15\} \cup \{t \in \{30, 45, 60, 90, 120, 150, 180\} | t < T\}$$

$$\cup \{45k | 225 \leq 45k < T\} \cup \{T - 1\}$$

In parallel, we collected platform-visible engagement metrics for each video (counts of likes, comments, and shares), metadata

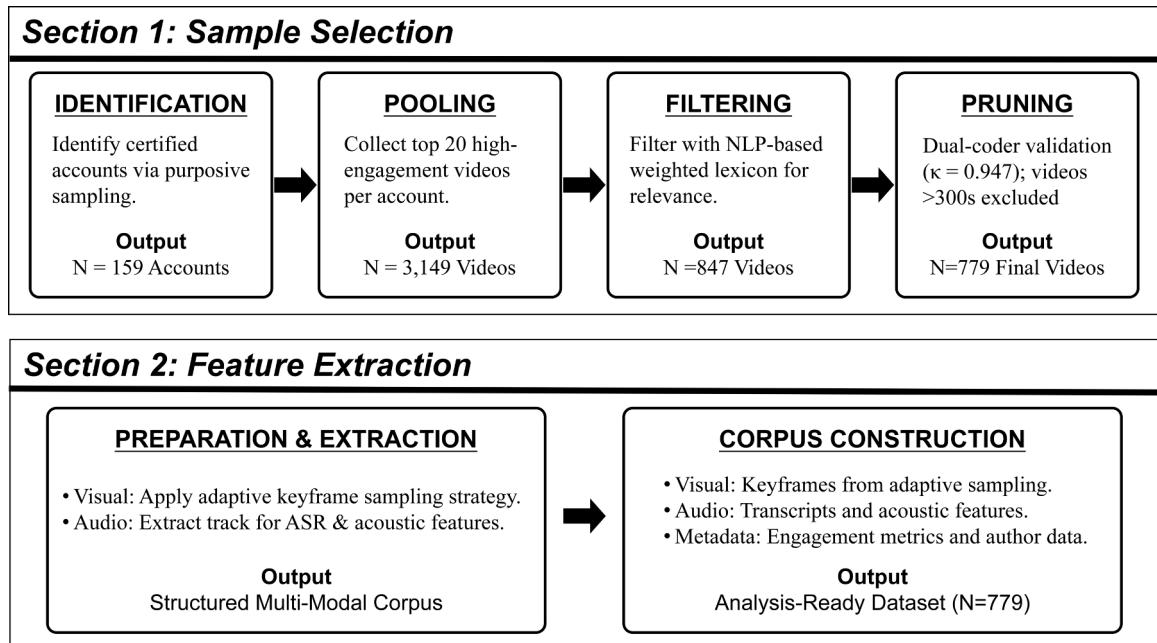


Fig. 1. Data Collection and Sampling Pipeline. Note: The filtration process reduced 159 qualifying accounts to 779 videos through purposive sampling, engagement-based pooling, and NLP-based filtering.

(titles and tags), and publisher information. These measures capture observable engagement behaviors on Douyin rather than direct indicators of complete public attitudes, persuasion outcomes, or offline behavioral change. We also cross-validated and compiled background information for each official— such as region and administrative rank; these characteristics were included as control variables in the empirical models. Taken together, these procedures yielded a high-dimensional, analysis-ready multimodal dataset that provides a reliable empirical basis for implementing our AI-enhanced framework and examining associations between multimodal strategies and observable platform engagement.

4. Methodology: An AI-Enhanced Multimodal Content Analysis Framework

To analyze grassroots officials' multimodal persuasion strategies and their associations with public engagement, we developed a five-stage AI-enhanced analytical framework. Fig. 2 shows how the framework integrates theory-driven deduction and data-driven induction through structured human-AI collaboration (see Appendix Table A1) to extract and validate strategies from 779 videos. For procedural transparency, the analytical chain is: raw videos → adaptive keyframes and audio transcripts → MM-LLM semantic features → BERTopic topics → HAC-supported strategy constructs → ELM theory anchoring → validated coding rules → full-corpus variables → empirical models and reported findings. Each stage specifies its input, AI task, human oversight, output, and validation function in the text or appendix, treating AI assistance as an auditable workflow rather than an automated decision process.

4.1. From Visual Signals to Semantic Features: Dynamic Cognitive Dialogue

To capture the temporal dependency overlooked by conventional static-frame analysis, the first stage implements a dynamic cognitive dialogue protocol designed to induce features directly from the data. From a theory-driven sample of 78 high-engagement videos, we extracted keyframe sequences using our adaptive strategy.

As illustrated in Fig. 3, we use Gemini 2.5 Pro as the multimodal inference engine within a sequential context-injection workflow to process sampled frames together with the audio track. We selected this model for its strong multimodal performance and long-context capability, which enables the same analytical episode to retain an audio-derived semantic baseline while moving through the keyframe sequence (Comanici et al., 2025). This design minimizes information loss that can arise when modalities are encoded separately. Operationally, the model first used the transcript/audio and metadata to establish the product category, communicative setting, and emotional tone; it then interpreted each keyframe against this baseline and the accumulated prior-frame context. Crucially, to capture temporal evolution, the model performs a dynamic contrastive analysis: for each frame at time *t*, it compares the new visual evidence against the immediately preceding frame at time *t*–1, whose analysis is provided as context (see Appendix Table A2). The comparison focused on newly appearing, disappearing, intensified, or weakened cues relevant to the coding rules, rather than treating each frame as an isolated image. This procedure captures meaning as it unfolds over time and yields 4,680 structured features. Each visual event—whether a stable attribute or a change—was annotated with its inferred communicative function, providing a fine-grained semantic foundation for subsequent analysis.

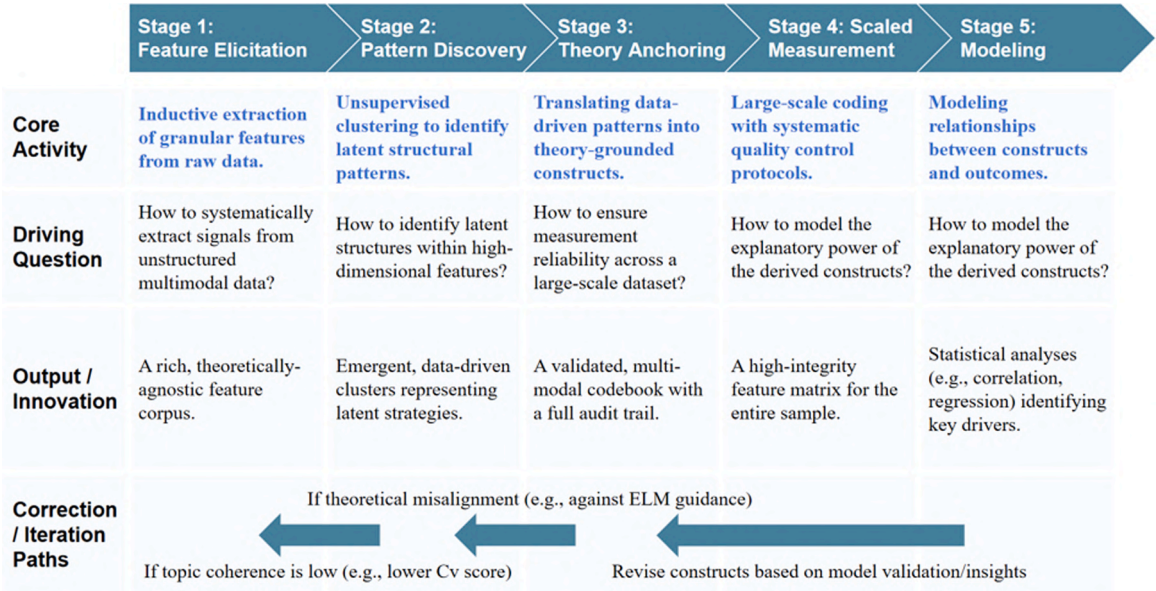


Fig. 2. The Five-Stage AI-Enhanced Analytical Framework for Multimodal Content. Note: The workflow integrates inductive feature elicitation with deductive measurement. Each stage specifies the analytical input, AI operation, human review function, output, and validation or correction path.

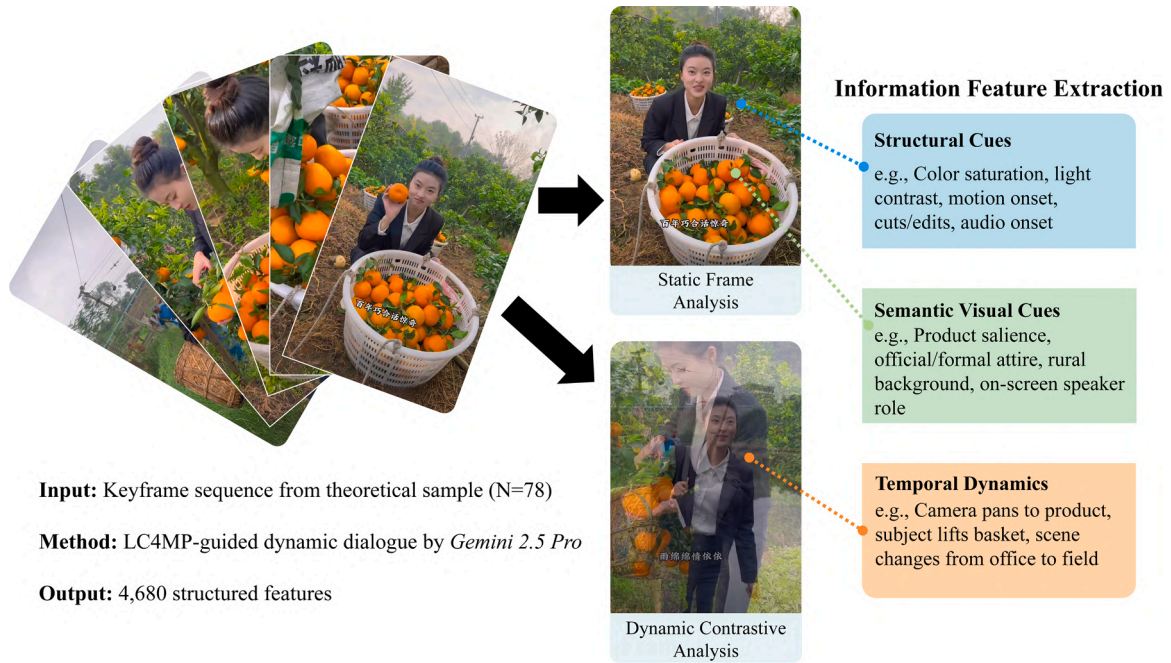


Fig. 3. The AI-Enhanced Feature Extraction Process via Dynamic Contrastive Analysis.

4.2. From Semantic Features to Strategy Combinations: Unsupervised Pattern Discovery

This stage aims to convert unstructured visual signals into interpretable strategy categories without imposing theoretical biases. The input to this stage was not the raw video itself but the set of MM-LLM-generated semantic annotations produced in Stage 1, which are short-text-like, semantically dense descriptions of visual, audio, and temporal cues. To minimize researcher priors and efficiently identify strategy combinations from large volumes of atomic features, the second stage adopts a data-driven inductive procedure. We selected BERTopic over traditional probabilistic frameworks, such as Latent Dirichlet Allocation (LDA), specifically to mitigate the data sparsity inherent in short-text analysis (Benz et al., 2025). This choice is a methodological fit argument rather than a claim of universal superiority: bag-of-words topic models are less suitable when concise annotations contain semantically related but lexically diverse descriptions, whereas BERTopic's contextual embeddings are better aligned with the study's intermediate data structure. Unlike the bag-of-words assumptions of LDA, BERTopic leverages S-BERT contextual embeddings to capture deep semantic dependencies, ensuring robust topic representation even within concise visual annotations.

Model quality was evaluated using C_v topic coherence. The algorithm initially identified 85 topics with an average coherence of 0.504, indicating meaningful latent structure. This score was used as a diagnostic indicator of latent semantic structure rather than as the sole criterion for accepting or rejecting topics. To rigorously refine these clusters, we implemented a dual-filter protocol (Son & Park, 2025): Quantitatively, we excluded outlier topics that failed to meet the minimum document frequency threshold; qualitatively, we manually pruned "noise topics" stemming from OCR errors or irrelevant background descriptors (e.g., generic "sky") that lacked communicative intent. Topic refinement therefore combined quantitative topic size, representative documents, semantic interpretability, and researcher review of communicative relevance, rather than relying mechanically on the coherence score alone.

The remaining 47 validated topics were then synthesized into nine preliminary strategic constructs via Hierarchical Agglomerative Clustering (HAC). For this step, the clustering unit was the validated topic rather than the individual video or raw semantic annotation. Each topic was represented as a TF-IDF topic-document vector derived from its representative topic documents. We employed Ward's linkage with Euclidean distance to maximize within-cluster homogeneity, effectively forming compact higher-order candidate groups from semantically proximate topics. The hierarchical tree was cut to yield nine candidate clusters, with the corresponding HAC dendrogram reported in Appendix Fig. A1. This solution was retained because it provided the clearest balance among semantic coherence, non-redundancy, interpretability, and fit with the ELM-oriented strategy distinctions used in the following theory-anchoring stage. HAC served as an organizing device for grouping semantically proximate topics; final construct boundaries were further audited through c-TF-IDF keyword inspection, representative examples, and preparatory semantic review before the formal ELM theory anchoring described later. Finally, to substantiate the semantic validity of this aggregation, we inspected the top class-based TF-IDF (c-TF-IDF) keywords associated with representative topics. For instance, the construct "Source Official Identity Salience" was anchored by high-weight terms such as "uniform," "secretary," and "inspection," whereas "Engaging Narrative Style" emerged from clusters defined by "plot twist," "funny," and "contrast plot," confirming that the derived constructs possess distinct and theoretically coherent boundaries.

4.3. From Strategy Combinations to Theoretical Constructs: Human-AI Collaborative Theory Anchoring

To systematically connect data-driven discoveries to established theory, the third stage uses a guided theory-building protocol that converts preliminary strategy constructs into measurement tools testable within the Elaboration Likelihood Model (ELM). The protocol guides the large language model through three hierarchical steps: first, empirical deconstruction of each preliminary construct to identify its core elements; second, distillation of the underlying strategy theme and hypothesis generation regarding its psychological mechanism; and third, mapping the strategy to ELM's central or peripheral route and producing operational definitions and measurement scales. The route assignment followed the criteria: central-route constructs were those whose persuasive relevance depended on argument quality, informational density, message substance, or cognitive utility, whereas peripheral-route constructs were those expected to operate primarily through source heuristics, visual authority, affective resonance, scene context, editing rhythm, or directive prompts. This mapping was completed during the theory-anchoring stage, before scaled coding of the full corpus and before statistical modeling; it was therefore not derived from the direction, magnitude, or significance of the regression coefficients. This procedure ensures that each transformation—from raw features to final constructs—has explicit decision rationales, establishing a complete audit trail. For example, the formation of Source Official Identity Salience progresses from the atomic feature "uniform present" to the theme of "authoritative visual cues," which is then mapped to the ELM peripheral route via source-credibility mechanisms and operationalized as a binary variable (see Table 1).

4.4. Measurement Validation and Scaled Application

To ensure measurement robustness and realize the benefits of human-AI complementarity, we implemented a systematic validation and application workflow.

4.4.1. Triangulated Validation and Optimization Based on Cognitive Complementarity

Guided by the principle of human-AI cognitive complementarity, we conducted triangulated reliability diagnostics on a 78-video subset (AI-AI, human-human, and AI-human reliability) to delineate the optimal division of labor. Fig. 4 visualizes the inter-rater reliability diagnostics (refer to the Figure Note for interpretation thresholds), revealing that AI exhibited superior internal consistency on structured visual constructs ($\alpha_{AI-AI} > 0.950$). Specifically, the model consistently outperformed human coders in detecting objective patterns such as Source Official Identity Salience and Scene Official Context Salience. This advantage reflects AI's strict adherence to visual patterns; for instance, in judging Product Visual Salience, AI classified 81.78% of cases as "not salient," compared with 75.56% for humans, indicating a more conservative yet highly consistent standard.

By contrast, when tasks required contextual understanding and affective inference, human cognition remained indispensable. Engaging Narrative Style most clearly manifested this difference: human-human reliability ($\alpha_{Human-Human} = 0.825$) far exceeded AI's internal reliability ($\alpha_{AI-AI} = 0.642$), the latter falling below acceptable thresholds. Qualitative diagnostics suggested distinct judgment mechanisms: humans could register tacit cues such as intonation and pausing, identifying narrative engagement in 72.37% of cases versus 67.11% for AI.

These diagnostics informed iterative optimization. For constructs on which AI underperformed, we decomposed abstract concepts into concrete behavioral indicators that AI could reliably detect (e.g., breaking Engaging Narrative Style into first-person narration and intonational variation), raising AI internal reliability above 0.85. In parallel, we established a dedicated audio analysis pipeline, using

Table 1
Preliminary Theoretical Constructs and ELM Mapping.

Construct Name	Object of Assessment	ELM Route & Interpretation
Source Official Identity Salience	Visual appearance of the speaker/uploader	Peripheral route. An official identity is a classic peripheral cue representing source authority and credibility. When audiences are unwilling or unable to engage in deep content processing, this cue can rapidly establish trust.
Scene Official Context Salience	Visual background of the video's focal scene	Peripheral route. As a contextual cue, the scene amplifies the source's official attributes and provides a legitimacy endorsement for the content without requiring further evidence.
Content Rational Appeal Strength	Textual properties of the complete content (audio and script)	Central route. When audiences have sufficient motivation and ability, stronger rational argumentation is perceived as more compelling evidence.
Engaging Narrative Style	Full audio transcript	Peripheral route. Through affective expression, it directly elicits audience liking and resonance, fostering shallow interactions such as likes and shares.
Editing Frequency	Visual editing techniques of the video	Peripheral route. A highly dynamic visual presentation functions as a sensory stimulus that effectively attracts and sustains audience attention.
Call to Action	Complete video content (audio and on-screen text)	Peripheral route. As a direct, low-cognition directive cue, a call to action is intended to bypass elaborate processing.
Product Visual Salience	Visual focal point within the frame	Peripheral route. By making the product the primary visual focus, the video simplifies its persuasive message and reduces the audience's cognitive processing costs.
Content Information Complexity	Audio content	Central route. Higher informational complexity implies greater cognitive resources are required for processing.
Marketing Information Intensity	Audio content	Central route. This variable directly gauges the argument strength of the core message (the product).

Note: ELM route labels were assigned before scaled coding and modeling. Central-route constructs involve argument quality, informational density, message substance, or cognitive utility; peripheral-route constructs rely on source, affective, visual-contextual, temporal, or directive heuristics.

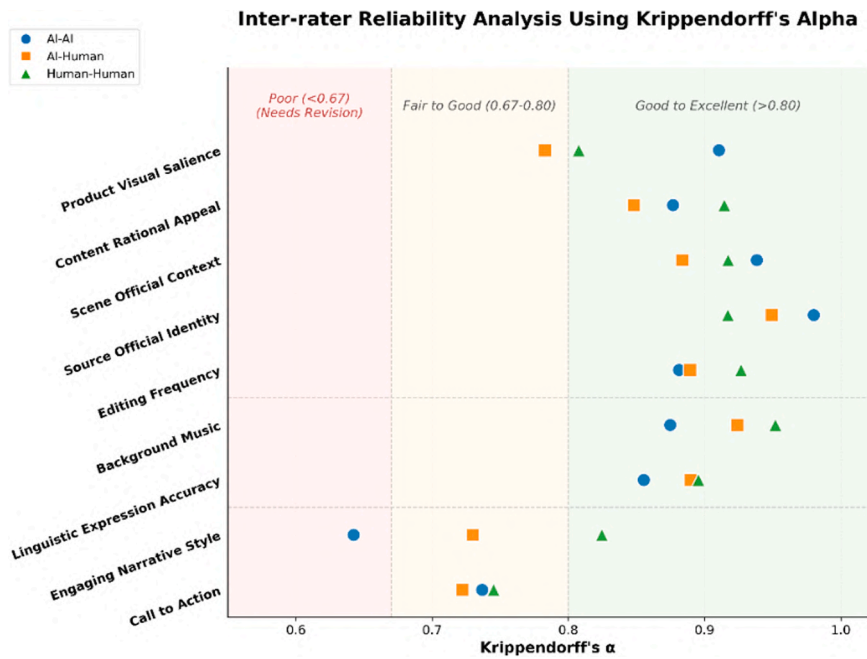


Fig. 4. Inter-rater Reliability Analysis Using Krippendorff's Alpha. Note: Marker distance indicates the AI-human cognitive-complementarity gap; wider gaps on subjective constructs indicate greater need for human adjudication.

Table 2
The Theoretical Construct System of Video Strategies.

Construct Name	Object of Assessment	Quantification Rule Summary	ELM Route & Interpretation
Source Official Identity	Visual appearance of the speaker/uploader	Binary: 1 (salient), the speaker wears a uniform, formal attire, or official insignia; 0 (non-salient), casual or ethnic attire.	Peripheral route. Official identity functions as an authority cue that rapidly fosters trust in low-involvement contexts.
Scene Official Salience	Visual background of the video's focal scene	Binary: 1 (formal), government/institutional spaces or displays of official symbols; 0 (informal), everyday life or production settings.	Peripheral route. Contextual cues strengthen the source's official attributes and provide a legitimacy endorsement.
Content Rational Appeal	Textual properties of the complete content (audio + subtitle)	Binary: 1 (rational); primarily conveys objective information; 0 (emotional) — primarily aims at affective arousal.	Central route. The strength of rational argumentation drives deep cognitive processing and trust formation.
Engaging Narrative Style	Full audio transcript	Binary (AI-coded): 1 (present): overtly passionate or emotional expression; 0 (absent) — calm, matter-of-fact narration.	Peripheral route. Affective expression evokes resonance and promotes shallow interactions.
Editing Frequency	Visual editing of the video	Binary: 1 (high dynamism): ≥ 2 scene switches; 0 (low dynamism) — < 2 switches.	Peripheral route. Visual dynamism sustains attention and enhances entertainment value.
Call to Action	Complete video content (audio and on-screen text)	Binary: 1 (present): contains cues that guide specific behaviors; 0 (absent) — no directive cues.	Peripheral route. Direct instructions bypass elaborate processing and prompt targeted behavior.
Product Visual Salience	Visual focal point within the frame	Binary: 1 (salient): the product is the sole visual subject and no human face appears; 0 (non-salient) — otherwise.	Peripheral route. Visual simplification reduces cognitive processing costs and accelerates attention.
Content Information Complexity	Audio content	Continuous: number of unique words $\div$ total words (lexical diversity).	Central route. Higher informational complexity requires more cognitive resources and deeper processing.
Marketing Information Intensity	Audio content	Continuous: count of product-introduction words $\div$ total words (share of product-intro text).	Central route. The density of core product information shapes deliberative decision-making.
Linguistic Expression Accuracy	Quality of speech-to-text transcript	Binary: 1 (problematic): semantically incoherent; 0 (normal) — semantically clear.	Control variable. Used to account for potential bias due to transcription/data quality.

Note: Binary coding is used for visually salient or directive cues whose meaning depends on presence or absence; continuous coding is used for linguistic and informational constructs whose meaning depends on degree, density, or diversity. ELM labels follow the criteria introduced above and summarized in Table 1.

natural language processing to derive richer linguistic features from transcripts. This optimization, grounded in a principled division of cognitive labor, preserves human insight while maximizing AI's strengths in scalable, consistent processing and thus constitutes the core mechanism safeguarding study reliability. Accordingly, the simplified indicators should be understood as theory-anchored operational proxies rather than arbitrary reductions of complex constructs. For visually anchored cues, binary coding captures whether a salient cue is perceptually available to viewers; for linguistically rich constructs, continuous measures preserve meaningful variation in lexical diversity or product-information density. The resulting construct system (Table 2) reflects this strategy: visually anchored constructs use efficient binary coding, whereas linguistically rich constructs employ finer-grained measurement scales.

4.4.2. AI Double-Blind Coding with Human Adjudication

To systematically decode the multimodal corpus, we engineered a "Sequential Context-Injection Protocol" within the AI coding workflow. The input for each coding case consisted of metadata, the complete transcript/audio context, and the sampled keyframe sequence; the AI task was to apply the finalized codebook to this evidence package, while the human role was to design the codebook, inspect reliability diagnostics, and adjudicate divergent or context-dependent cases. In this protocol, the MM-LLM was mandated to ingest and process the complete audio track in the initial conversational turn, implicitly performing automatic speech recognition and acoustic scene analysis to extract semantic keywords and emotional tones (Comanici et al., 2025). This process established a latent semantic schema (e.g., inferring the specific product category or official context) before the model received any visual frames. By leveraging this pre-loaded auditory context as a cognitive anchor, the protocol ensured cross-modal semantic consistency, aligning the interpretation of subsequent visual cues with the established narrative timeline and significantly improving stability for ambiguous features (e.g., reliability for Editing Frequency rose to 0.932). To ensure reproducibility, we enforced a deterministic generation protocol with strict hyperparameter constraints to maximize output stability (Dewan et al., 2026) (see Appendix Table A2 for detailed prompt logic and settings).

Applying this rigorous protocol to the full set of 779 videos, two independent AI instances coded the corpus separately under identical prompts and coding rules and achieved high inter-rater reliability (average = 0.893). Despite this high agreement, a subset of cases (approximately 10.7%), primarily involving context-dependent constructs, exhibited divergence and required human adjudication using optimized coding rules. Adjudication was reserved for these divergent or context-dependent cases: researchers returned to the transcript/audio context, sampled keyframes, and coding rules, and final codes were assigned based on explicit textual, visual, or audiovisual evidence rather than by averaging AI outputs. Crucially, to address the limitation of framework dependency on a specific model (instrumental validity), we conducted a robustness check on a stratified subsample (N=78) using the efficiency-optimized Gemini 2.5 Flash. Applying identical prompts and logic, the results indicated strong convergence with the final dataset (mean agreement = 94.3%; mean κ = 0.861; see Appendix Fig. A2), supporting procedural stability across two Gemini model variants. This consistency indicates that the framework's reliability derives from the structured "Context-Injection" protocol and rigorous construct definitions rather than the specific idiosyncrasies of a single foundation model. Accordingly, the validation should be read as a diagnostic cross-model check, not as evidence that the procedure is independent of all possible foundation models. Through these quality-control mechanisms, we completed feature extraction for all 779 valid samples.

Table 3
Descriptive Statistics of Raw Engagement Outcomes and Explanatory Variables.

Variable	Mean	SD	Min	Max
Likes	5696.619	24590.528	27	378151
Comments	766.225	4313.593	2	77769
Shares	1641.516	16550.657	1	357835
Content Information Complexity	0.578	0.148	0	1
Marketing Information Intensity	0.226	0.159	0	1
Content Rational Appeal Strength	0.553	0.497	0	1
Source Official Identity Salience	0.281	0.448	0	1
Scene Official Context Salience	0.155	0.362	0	1
Engaging Narrative Style	0.754	0.43	0	1
Editing Frequency	0.777	0.417	0	1
Call to Action	0.44	0.497	0	1
Product Visual Salience	0.228	0.412	0	1
Background Music	0.926	0.263	0	1
Speech Rate	3.476	1.225	0.182	9.567
Linguistic Expression Accuracy	0.084	0.277	0	1
Video Duration (In seconds)	3.424	0.675	1.386	5.704
Follower Count (In followers)	10.507	1.24	7.457	15.569
Publisher Age	38.217	6.864	24	56
Publisher Gender	0.172	0.376	0	1
Publisher Province	2.099	0.823	1	3

Note: Likes, Comments, and Shares are platform-visible raw count outcomes used in the negative binomial models. Video Duration and Follower Count are reported on the log scale; binary variables are coded 0/1.

5. Empirical Analysis

The empirical analysis uses the validated variables produced by the preceding workflow: semantic features extracted from key-frames and transcripts were organized into topics, strategy constructs, coding rules, and full-corpus measures, which are summarized in Table 3 and then entered into the regression models. Our analysis of 779 marketing videos produced by grassroots officials describes how the hybrid "official-influencer" identity materializes as concrete communication strategies. Descriptive statistics (Table 3) first quantify these observable patterns. The evidence indicates a marked de-emphasis on visual symbols associated with the traditional bureaucratic identity: only 28.1% of videos display official attire, and 15.5% use officially coded scene backgrounds. In sharp contrast, creators broadly adopt platform-native elements: 75.4% of videos employ an Engaging Narrative Style, and 77.7% use high Editing Frequency. This strategic pivot from authority-laden symbols toward affective, entertainment-oriented audiovisual elements reflects creators' identity negotiation to meet algorithmic incentives and user expectations. Yet the information-strategy layer exhibits a subtler balance: more than half of the videos (55.3%) retain Content Rational Appeal Strength, and Content Information Complexity remains moderately high ($M=0.578$). These patterns suggest that, while embracing platform culture, grassroots officials do not fully relinquish their informational role as public servants. For consistency with the count-model specification, Table 3 reports likes, comments, and shares as raw platform-visible counts. Fig. 5 separately presents log-transformed engagement counts only for visual readability under extreme right skew; these transformed values are not used as the negative binomial regression outcomes.

The correlation matrix visualized in Fig. 6 reports pairwise correlations among the 17 modeled independent variables summarized in Table 3. These variables include the nine ELM-anchored strategy constructs listed in Table 1, together with two additional audio/presentation features (Background Music and Speech Rate), one expression-quality indicator, and the video-, account-, publisher-, and region-level controls used in the regression models. The matrix therefore supports the subsequent modeling strategy by showing generally weak-to-moderate correlations among the modeled variables, thereby alleviating multicollinearity concerns. It also clarifies theoretically meaningful relationships among strategy constructs, modeled features, and covariates; for example, Content Rational Appeal Strength correlates positively with Video Duration ($r=0.410$) and Speech Rate ($r=0.400$), suggesting that more rational presentations tend to appear in longer and more verbally dense videos. A notable pattern is the "decoupling" of Source Official Identity Salience from most other modeled variables, with correlations close to zero overall, suggesting that personified authority functions as a relatively independent visual cue.

5.1. Main Effects Analysis

This section examines which strategy combinations are most strongly associated with user engagement across different levels of cognitive effort. To examine these strategy–engagement relationships, we estimate a set of nested regression models grounded in ELM's dual-route framework. The dependent variables (likes, comments, shares) are non-negative count outcomes with pronounced right skew and over-dispersion, making a count-data specification more appropriate than linear regression. The NB models therefore

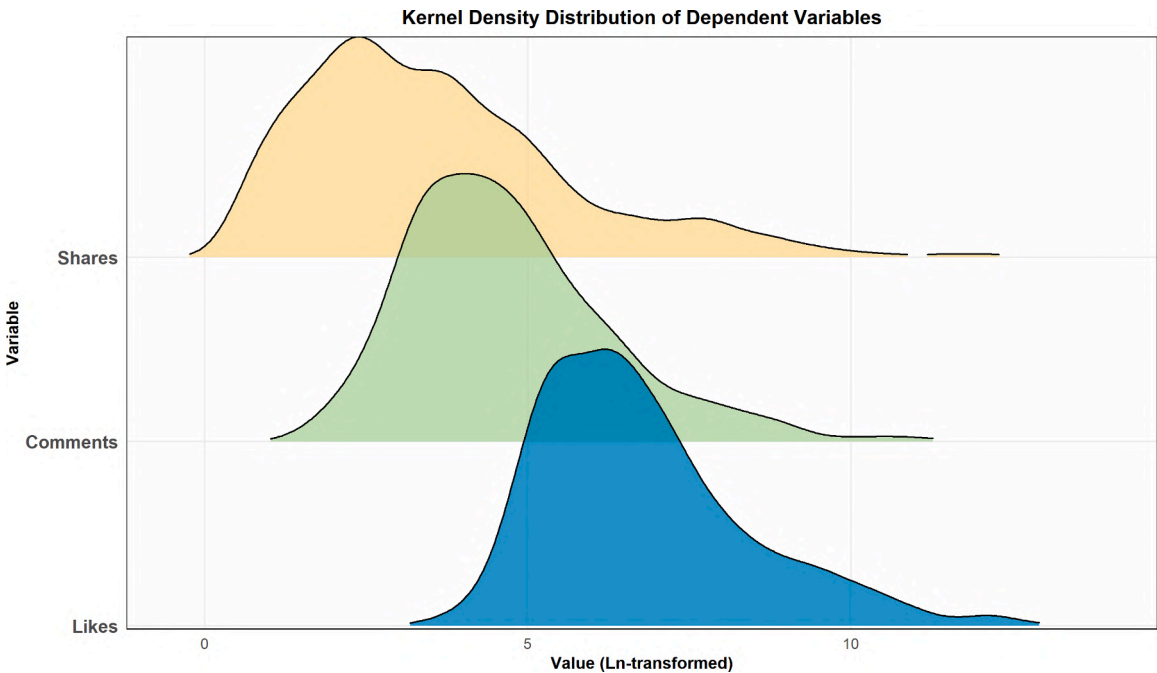


Fig. 5. Kernel Density Distribution of Log-Transformed Engagement Counts. Note: The horizontal axis shows log-transformed engagement counts for visual readability under right skew; regression models use raw non-negative counts as dependent variables.

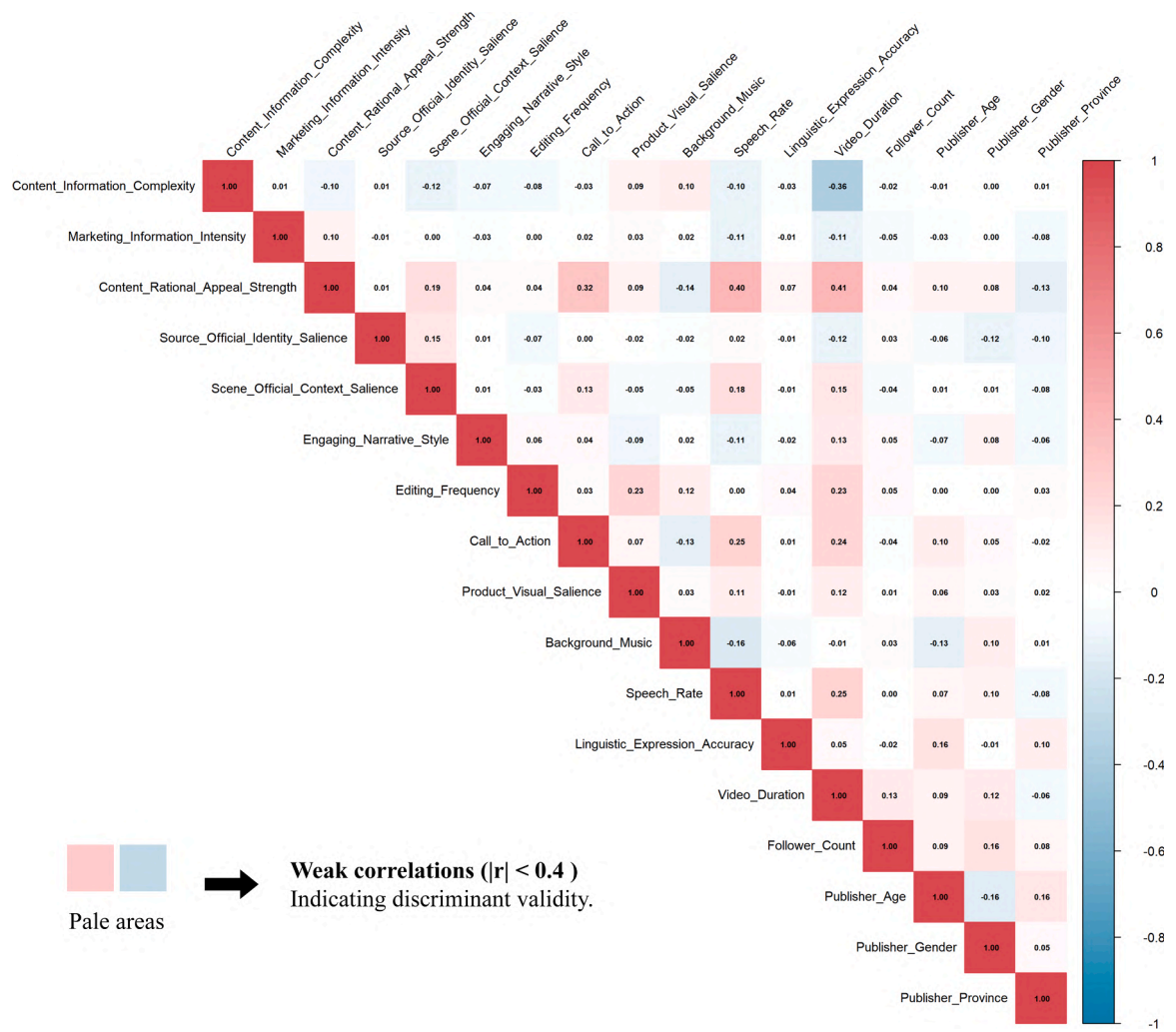


Fig. 6. Correlation Matrix Heatmap of Modeled Independent Variables. **Note:** The matrix includes the 17 independent variables used in the regression models, covering the nine ELM-anchored strategy constructs and the additional features and controls listed in Table 3.

use the raw non-negative counts of likes, comments, and shares, not the log-transformed values shown descriptively in Fig. 5. Specifically, each model estimates expected engagement counts under a log link, with over-dispersion handled through the NB variance structure. Consequently, conventional OLS assumptions are violated, rendering standard linear estimates potentially biased (Ver Hoef & Boveng, 2007). We therefore adopt Negative Binomial (NB) regression as our primary specification. We also attempted Zero-Inflated Negative Binomial (ZINB) models, but they did not converge across multiple reasonable specifications, suggesting that a distinct zero-inflation process is not supported by the data; accordingly, we retain the standard NB model to address over-dispersion. In the main models, we report standard errors clustered at the publisher level. As robustness checks, Appendix Table A4 additionally reports NB estimates with HC3-robust (sandwich) standard errors and PPML results, which remain consistent with the main findings (MacKinnon & White, 1985).

To mitigate algorithmic confounds, we implement a multi-dimensional control strategy. First and foremost, we explicitly control for Follower Count. In the context of algorithmic distribution, this variable serves as a structural proxy for the account's baseline "cold start" traffic pool (Chen & Shi, 2025). By holding this variable constant, we estimate strategy-engagement associations net of pre-existing social capital that supports initial visibility.

Furthermore, we introduce Video Duration as a critical covariate. Given that recommendation systems prioritize metrics such as completion rate—which are intrinsically correlated with length—this control helps distinguish multimodal-cue associations from the mechanical biases associated with video format. Finally, by including Publisher Province and demographic characteristics, we account for observed sources of heterogeneity arising from regional digital disparities. This specification reduces the risk that observed engagement differences reflect structural, temporal, or environmental advantages.

The regression results (Table 4) clarify observed strategy-engagement patterns in grassroots officials' persuasion practices on digital platforms, yielding four principal findings. In interpreting these results, we retain the route classifications established before

Table 4
Negative Binomial Regression Results for Communication Strategies and Public Engagement (N=779).

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Central Route (ELM)	Content Information	0.160** (0.065)	0.156** (0.063)	0.133* (0.068)	0.151** (0.066)	0.078* (0.042)	0.067** (0.032)
	Complexity						
	Marketing Information	0.019 (0.053)	-0.084* (0.051)	0.165** (0.065)	-0.004 (0.051)	0.158*** (0.026)	0.138*** (0.039)
	Intensity						
	Content Rational Appeal Strength	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
Peripheral Route (ELM)	Source Official Identity		0.152*** (0.050)		0.150** (0.059)		0.237*** (0.056)
	Saliency						
	Scene Official Context Saliency		-0.190*** (0.051)		-0.222*** (0.052)		-0.343*** (0.057)
	Engaging Narrative Style		0.105* (0.054)		0.107** (0.054)		0.258*** (0.057)
	Editing Frequency		0.104** (0.053)		0.136** (0.053)		0.277*** (0.059)
	Call to Action		-0.140** (0.058)		-0.117** (0.059)		-0.286*** (0.061)
	Product Visual Saliency		-0.170*** (0.051)		-0.245*** (0.063)		-0.318*** (0.055)
	Background Music		0.256*** (0.051)		0.253*** (0.052)		0.258*** (0.053)
	Speech Rate		-0.445*** (0.062)		-0.572*** (0.057)		-0.542*** (0.053)
	Linguistic Expression	-0.052 (0.053)	-0.050 (0.050)	0.021 (0.055)	0.114* (0.061)	-0.036 (0.032)	-0.037 (0.043)
	Accuracy						
	Video Duration	-0.224*** (0.062)	-0.072 (0.062)	-0.225*** (0.065)	-0.055 (0.063)	0.134*** (0.039)	0.070** (0.035)
	Follower Count	1.399*** (0.053)	0.986*** (0.050)	0.862*** (0.055)	0.570*** (0.051)	0.452*** (0.043)	0.363*** (0.054)
	Publisher Age	-0.174*** (0.054)	-0.008 (0.055)	-0.239*** (0.057)	-0.050 (0.053)	-0.249*** (0.041)	-0.112** (0.051)
	Publisher Gender	-0.220*** (0.054)	-0.101* (0.052)	-0.345*** (0.056)	-0.125** (0.052)	-0.478*** (0.044)	-0.353*** (0.054)
	Publisher Province	-0.035 (0.053)	0.004 (0.051)	-0.118** (0.056)	-0.107** (0.054)	-0.212*** (0.041)	-0.188*** (0.046)
Model Fit	Intercept	8.083*** (0.051)	7.914*** (0.048)	6.097*** (0.054)	5.836*** (0.049)	5.971*** (0.043)	4.842*** (0.054)
	Pseudo-R ²	0.301	0.391	0.275	0.406	0.235	0.409

Note: Dependent variables are raw counts of likes, comments, and shares. Standard errors are in parentheses. *p < 0.1; **p < 0.05; ***p < 0.01.

modeling in Table 2; coefficient signs are used to assess whether a theoretically assigned cue is positively or negatively associated with engagement, not to determine its ELM route.

First, the central-route results indicate a striking paradox in which substantive information outperforms traditional didacticism. Content Information Complexity is consistently and positively associated with all forms of engagement, suggesting that audiences differentiate substantive cognitive value from bureaucratic didacticism and are willing to invest cognitive resources to endorse high-quality information. Although this variable is positive across all engagement metrics, its estimated coefficient remains numerically lower for shares ($\beta=0.067$, $p<0.05$) than for likes ($\beta=0.156$, $p<0.05$) and comments ($\beta=0.151$, $p<0.05$).

Notably, Marketing Information Intensity exhibits a robust positive association specifically with sharing ($\beta=0.138$, $p<0.01$) while showing no significant relationship with likes or comments, contrasting sharply with the consistent negative associations of Content Rational Appeal Strength. This nuance is critical: audiences do not appear to reject detailed product information (Intensity) or high-value knowledge (Complexity); rather, the pattern suggests lower engagement for the didactic, prescriptive tone associated with "Rational Appeal." Thus, successful central-route persuasion appears to benefit from decoupling high-value information from formulaic bureaucratic framing.

Second, authority, a canonical peripheral cue, exhibits a pronounced bifurcation in observed engagement patterns, which is contingent on personalization rather than institutionalization. This finding, summarized as "authority resides in persons, not places," is evidenced by divergent behavioral outcomes. Authority personified through an official's attire (Source Official Identity Saliency) acts as a powerful trust proxy that is positively associated with engagement. In contrast, its institutionalized expression through bureaucratic backdrops (Scene Official Context Saliency) is negatively associated with interaction and is consistent with psychological-distance interpretations. This supports the interpretation that effective identity negotiation retains the personalized carrier of authority while stripping away its colder institutional context.

Third, the utility of peripheral cues follows a differentiated spectrum centered on experience enhancement versus intention revelation. Not all peripheral tactics are beneficial. Audiences distinguish experience-enhancing cues (e.g., Engaging Narrative Style, high Editing Frequency) from intention-revealing cues (e.g., explicit Call to Action, overly strong Product Visual Saliency). The former

carry hedonic value that is associated with higher engagement by optimizing sensory and affective experience (Hughes et al., 2019; Pan et al., 2025). By contrast, intention-revealing cues—such as Call to Action ($\beta=-0.286$, $p<0.01$) and Product Visual Salience ($\beta=-0.318$, $p<0.01$)—make commercial or persuasive intent overly explicit, a pattern consistent with defensive responses and lower engagement (Agarwal et al., 2025).

Finally, a stepwise logic underlies the observed engagement patterns at different depths. From the baseline to the extended models, adding audiovisual peripheral strategies substantially increases explanatory power for likes, comments, and shares (Pseudo- R^2 increases by 29.9%, 47.6%, and 74.0% respectively), indicating that peripheral cues account for a large share of variance in user engagement. More importantly, the observed associations diverge by engagement depth. Low-cost interactions such as likes are more strongly associated with a set of experience-oriented peripheral cues, whereas sharing, the highest-cost form of deep endorsement, is associated with a more demanding combination: strong source credibility (high Source Official Identity Salience), deep affective resonance (high Engaging Narrative Style), and exceptional informational value (high Content Information Complexity) acting in concert.

5.2. Robustness Checks and Heterogeneity Analysis

To examine the boundary conditions of our findings, we assessed how video duration, account size, and regional context condition the association between persuasion strategies and sharing behavior. We further estimated full-sample negative binomial interaction models by interacting each subgroup indicator with the strategy variables; the interaction-based coefficient-difference tests are reported in Appendix Table A7 and provide statistical evidence for several key heterogeneity patterns emphasized below. To assess the statistical robustness of our findings, we implemented a multidimensional validation strategy (see Appendix Tables A3–A5 and A7).

First, regarding model specification, we tested Zero-Inflated Negative Binomial (ZINB) models (Hilbe, 2011); however, their persistent non-convergence did not provide stable evidence for a distinct zero-inflation process in this dataset, so we retained the standard Negative Binomial specification. Second, to address potential heteroskedasticity and estimator dependence, we re-estimated the models using the Poisson Pseudo-Maximum Likelihood (PPML) estimator (Silva & Tenreiro, 2006). As shown in Table A4, the results remained broadly consistent with the main findings in sign and substantive interpretation.

Third, to mitigate observable selection bias and assess robustness to observed-covariate imbalance, we used Propensity Score Matching (PSM) as a robustness check to estimate the ATT with logit-based propensity scores and nearest-neighbor matching (Rosenbaum & Rubin, 1983). For each strategy treatment, propensity scores used the same observable video-, publisher-, account-, regional-, and expression-quality controls as the main models: video duration, publisher gender, publisher age, follower count, publisher province, and linguistic expression accuracy. These covariates capture observable differences in content form, publisher characteristics, account scale, regional context, and expression quality that could jointly shape strategy adoption and engagement. Table A5 supports the robustness of the estimated associations after adjustment for observed-covariate imbalance, and Appendix Fig. A3 reports treatment-specific SMD diagnostics showing that post-matching imbalance is substantially reduced and falls within or close to the conventional $\pm 10\%$ threshold. Table A6 summarizes the analytical transparency and alternative validation logic.

Table 5
Heterogeneity Analysis of Shares by Video Duration, Account Size, and Regional Governance Context (Negative Binomial Regression).

Variable Classification	Variables	Video Long	Video Short	Fans Large	Fans Small	Gov Good	Gov Poor
Central Route	Content Information Complexity	0.227*** (0.064)	-0.145* (0.082)	-0.176* (0.090)	0.458*** (0.102)	-0.217*** (0.061)	0.772*** (0.135)
	Marketing Information Intensity	-0.260*** (0.083)	0.140* (0.075)	0.035 (0.070)	-0.134** (0.065)	0.145* (0.076)	-0.246*** (0.074)
	Content Rational Appeal Strength	-0.389*** (0.044)	-0.212*** (0.052)	-0.333*** (0.082)	-0.214*** (0.043)	-0.205*** (0.073)	-0.865*** (0.087)
Peripheral Route	Source Official Identity Salience	0.847*** (0.155)	0.128*** (0.048)	-0.082* (0.045)	0.746*** (0.183)	-0.082* (0.046)	0.768*** (0.135)
	Scene Official Context Salience	-0.209*** (0.045)	-0.270*** (0.074)	-0.172** (0.075)	-0.425*** (0.044)	-0.215*** (0.075)	0.128*** (0.042)
	Engaging Narrative Style	-0.127*** (0.034)	0.066 (0.052)	0.168* (0.090)	0.286*** (0.039)	0.195*** (0.070)	0.107*** (0.029)
	Editing Frequency	0.507*** (0.036)	0.200*** (0.061)	0.196** (0.083)	0.423*** (0.046)	0.148** (0.075)	0.525*** (0.077)
	Call to Action	-0.446*** (0.064)	-0.148* (0.078)	-0.107 (0.077)	-0.209*** (0.033)	-0.195*** (0.070)	-0.175*** (0.058)
	Product Visual Salience	-0.560*** (0.034)	-0.353*** (0.072)	-0.210*** (0.075)	-0.402*** (0.043)	-0.210*** (0.057)	-0.863*** (0.099)
	Background Music	0.357*** (0.031)	0.200*** (0.070)	0.350*** (0.066)	0.198*** (0.042)	0.240*** (0.060)	0.208*** (0.034)
	Speech Rate	-0.284*** (0.063)	-0.145*** (0.042)	-0.578*** (0.077)	-0.186*** (0.061)	-0.413*** (0.083)	-0.283*** (0.071)
Controls	Included	Yes	Yes	Yes	Yes	Yes	Yes
Model Fit	Pseudo- R^2	0.558	0.323	0.426	0.532	0.393	0.401

Note: Standard errors are in parentheses. Subgroups use median splits for Video Duration/Follower Count and categorical Gov Good/Poor; Table A7 reports interaction tests. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To identify theoretical boundary conditions, we stratified the heterogeneity analysis by video duration, account follower size, and regional digital governance level, using shares as the dependent variable because sharing represents the highest threshold of user endorsement and social reputation commitment (Table 5).

First, Video Duration shows a theoretical divergence consistent with temporal constraints on central-route processing. The two central-route strategies exhibit inverse patterns: Marketing Information Intensity is positively associated with sharing in short videos ($\beta=0.140$, $p<0.1$) but negatively associated with sharing in longer formats ($\beta=-0.260$, $p<0.01$), suggesting that prolonged exposure to dense sales cues may trigger persuasion resistance. Conversely, Content Information Complexity is positively associated with sharing in longer videos ($\beta=0.227$, $p<0.01$) but marginally negative in short videos ($\beta=-0.145$, $p<0.1$). This pattern is consistent with the idea that central-route persuasion is more plausible when sufficient cognitive opportunity exists: in videos under 30 seconds, efficient product cues are easier to process, whereas higher complexity may become a cognitive burden.

Second, the analysis of Follower Count is consistent with a "legitimacy compensation" pattern. For smaller accounts ("Fans Small"), strong authority cues—specifically Source Official Identity Salience ($\beta=0.746$, $p<0.01$) and Content Information Complexity ($\beta=0.458$, $p<0.01$)—are strongly associated with sharing. Conversely, for established influencers ("Fans Large"), these cues are largely ineffective or even counterproductive (Identity: $\beta=-0.082$, $p<0.1$; Complexity: $\beta=-0.176$, $p<0.1$). This suggests that grassroots officials with lower social capital appear to rely more on "hard" institutional symbols to compensate for a lack of initial credibility, whereas established accounts appear more vulnerable to an "authenticity penalty" where such formal cues distance them from their audience.

Third, the Regional Governance Context is consistent with an informational scarcity pattern. In regions with lagging digital governance development, authoritative and complex information is more strongly associated with sharing (Complexity: $\beta=0.772$, $p<0.01$; Identity: $\beta=0.768$, $p<0.01$). Crucially, even institutionalized settings—which are typically negatively associated with engagement—show a positive association in this context (Scene Official Context: $\beta=0.128$, $p<0.01$). This pattern suggests that in information-scarce environments, audiences prioritize the credibility signaled by official backdrops over platform-native aesthetics, treating official videos as scarce sources of reliable public data. In contrast, in regions with advanced digital governance, where users are likely saturated with official messaging, these traditional signals are negatively associated with sharing (Complexity: $\beta=-0.217$, $p<0.01$; Identity: $\beta=-0.082$, $p<0.1$), indicating a higher demand for novel, platform-native presentation styles over redundant official ones (Xu et al., 2025).

6. Discussion

This study suggests that, on algorithmically driven short-video platforms, observable engagement with government communication does not hinge on optimizing a single persuasive route. Rather, the findings are consistent with a pattern in which multidimensional strategy cues drawn from both the central and peripheral routes align with the cost ladder of public engagement. The following sections detail the empirical findings, their theoretical contributions to information management and social science, and actionable strategies for practitioners.

6.1. Summary of Key Findings

First, the central-route pattern shifts from traditional didactic rationality to a coupling of cognitive value and narrative experience. High informational density, especially Marketing Information Intensity, is positively associated with engagement mainly in efficient short-form contexts, whereas stereotypically didactic "Rational Appeal" is consistently negatively associated with engagement, especially for sharing. In an attention-scarce environment, audiences do not reject valuable information; they appear to reject monotonous, high-load presentation. High-value information therefore appears more likely to remain engaging when embedded in narrative and sensory experiences (Li et al., 2026). Experience-enhancing peripheral cues, especially Engaging Narrative Style and Background Music, may provide a cognitive escort by increasing affective immersion, reducing immediate counter-arguing (Li et al., 2024), and increasing perceived social presence and parasocial familiarity with the official communicator (H.-J. Kim et al., 2025).

Second, the analysis supports a "bifurcation of authority," where the engagement value of officialdom appears to hinge on its mode of presentation rather than its mere presence. While our results indicate that personified cues (e.g., uniforms) act as effective trust proxies, institutionalized contexts (e.g., bureaucratic offices) are negatively associated with engagement. This divergence can be explained through the lens of social identity theory: personified authority cues facilitate the recategorization of the official as a proximate, approachable community member, whereas static institutional backdrops accentuate bureaucratic category boundaries and amplify "outgroup" perception (Peña et al., 2026). Effective identity negotiation, therefore, depends on retaining the individualized carrier of authority while stripping away cues that foreground institutional distance.

Third, the utility of peripheral cues follows a differentiated spectrum centered on experience enhancement versus intention revelation. Not all peripheral tactics are beneficial. Audiences distinguish experience-enhancing cues (e.g., Engaging Narrative Style, high Editing Frequency) from intention-revealing cues (e.g., explicit Call to Action, overly strong Product Visual Salience). The former carry hedonic value that is associated with higher engagement by optimizing sensory and affective experience (Hughes et al., 2019; Pan et al., 2025); the latter prematurely and excessively expose persuasive or commercial intent, readily activating persuasion knowledge, vigilance and defense against perceived attempts to influence, and thus backfiring (Agarwal et al., 2025).

Fourth, these patterns cohere in a unified dynamic-matching framework in which strategy–engagement associations depend on their alignment with the costs of engagement behaviors (the "Cost Ladder"). Likes, which represent low-effort feedback (Yang et al., 2020), are more strongly associated with experience-enhancing peripheral cues. Comments require greater emotional involvement and cognitive effort: users seek to discuss content with informational depth but resist one-way didactic messaging (Garz et al., 2020).

Shares, the highest-cost form of deep endorsement, consume cognitive resources and function as social backing that puts personal reputation at stake. Crossing this stricter persuasive threshold is associated with a concerted bundle: personalized authority (high Source Official Identity Salience), strong affective resonance (high Engaging Narrative Style), and exceptional cognitive value (high Content Information Complexity) acting in synergy consistent with transitions from passive reception to active diffusion (Deng et al., 2025; Li et al., 2024).

Consistent with the interaction-based coefficient-difference checks reported in Appendix Table A7, our heterogeneity analysis (Table 5) provides evidence for several association-based boundary patterns that characterize this synergistic matching. First, temporal constraints shape the feasibility of central-route processing; higher Information Complexity is positively associated with sharing in longer videos (>30s) where users have sufficient opportunity for elaboration, but corresponds to lower sharing in shorter clips due to cognitive overload (Nguyen et al., 2025). Second, the follower-count results are consistent with a legitimacy compensation pattern. Smaller accounts rely more heavily on formal symbols such as uniforms to establish initial, institution-based trust, whereas larger and more established accounts incur an authenticity penalty in which the same symbols reduce engagement (Alibakhshi & Srivastava, 2025). This pattern aligns with stage-contingent trust formation (Y. A. Kim & Ahmad, 2013): early-stage accounts lacking interaction history appear to rely more on "hard" institutional anchors to secure initial credibility, whereas mature accounts thrive on experience-based relational trust, making rigid formality incompatible with the accumulated expectations of authenticity. Third, information scarcity modulates value perception; in regions with lower digital governance development, authoritative and complex content is more strongly associated with sharing (Jiang et al., 2025), a pattern that appears weaker or reversed in saturated, high-governance environments.

6.2. Theoretical Implications

This study bridges the epistemological gap between computational efficiency and theoretical depth, offering distinct contributions to research perspectives, methodological protocols in Information Management, and digital persuasion theory in Social Science.

6.2.1. Shifting Research Perspectives: From Static Text to Dynamic Multimodality

Traditional government information research has predominantly relied on text mining of policy documents or social media posts (Qin et al., 2025). This study supports a shift in research perspective toward dynamic multimodal analysis. The persuasive force of short video may not arise from a static bundle of multimodal cues alone, but from their temporal evolution and cumulative patterns (Sim et al., 2024). By operationalizing constructs such as "dynamic contrastive" visual changes (e.g., the shift from office to field), we show that observed persuasion-related patterns can be interpreted as temporally patterned, defined not just by static visual elements, but by the temporal sequence of sensory stimuli. This challenges the "bag-of-features" assumption prevalent in previous computer-mediated communication research and supports the importance of temporal dynamics for interpreting platformized digital governance communication.

6.2.2. Methodological Contributions: A Human-AI Symbiosis Protocol

For the field of Information Management, the AI-enhanced analytical framework proposed and validated here establishes a rigorous and operable paradigm for computational social science in the era of generative AI. As visualized in Fig. 2, this protocol addresses the "black box" problem of MM-LLM through core methodological principles:

First, it operationalizes "cognitive complementarity" to redefine reliability standards. In the AI era, reliability must extend beyond human-human agreement to encompass a plural verification system. As evidenced by the reliability diagnostics in Fig. 4, AI demonstrates superior consistency on patterned visual features, whereas humans excel when contextual understanding and affective inference are required. Consequently, our workflow allocates structured pattern recognition to AI and conceptual adjudication to humans, maximizing both scalability and validity.

Second, it enforces structural "auditability" for unstructured data. Given the inherent opacity of foundation models, traditional code-and-data replication is insufficient to guarantee credibility. Our framework necessitates a clear, traceable audit trail from raw data to final claims, mandating systematic documentation of decision rationales at every step of human-AI interaction. This creates a standard for procedural transparency essential for the rigorous replication of future video-based studies (Naeem et al., 2025).

6.2.3. Advancing Digital Persuasion Theory

In Social Science, this study offers a bounded extension of the Elaboration Likelihood Model (ELM) for algorithmic governance. Rather than proposing a separate persuasion model, it specifies how ELM's central and peripheral routes can be interpreted when cues are multimodal, temporally sequenced, and embedded in platformized relationships. The central insight is that authority as a peripheral cue is bifurcated: its association with engagement depends less on "authority" per se than on whether authority is personified or institutionalized. The "Cost Ladder" refinement further suggests that central and peripheral cues are complementary rather than competitive: experience-enhancing peripheral cues can be interpreted as the "entry ticket" for low-cost engagement, while high-value central cues serve as the "currency" for high-cost endorsement. This refinement shifts ELM application from static cue classification to temporally informed cue interpretation, from isolated cues to cue configurations, and from purely cognitive processing to processing conditioned by social identity, perceived social presence, and trust formation. Thus, ELM remains a useful organizing framework in algorithmic governance, but its explanatory value improves when paired with identity-based boundary processes and stage-contingent trust formation.

6.3. Practical Implications

For local-government practitioners and content creators who operate in comparable platformized governance-marketing contexts and navigate the "official-influencer" dilemma, this study offers evidence-bounded guidance on identity construction, strategy matching, and lifecycle management. First, creators can address the central tension of their hybrid role by strategically "decoupling authority from bureaucracy." The empirical evidence suggests that while authority remains a potent trust proxy, its observed association with engagement varies by mode of presentation. Officials may emphasize personified authority cues, such as wearing official attire or insignia, which can establish immediate credibility without alienating the audience. Simultaneously, they may avoid over-reliance on institutionalized contexts, such as filming purely within static government offices, which are negatively associated with engagement in our models and may create psychological distance. By positioning themselves as visible representatives who are nonetheless accessible individuals, practitioners may better align professional legitimacy with the authenticity and intimacy norms of comparable algorithmically mediated short-video environments.

Furthermore, in comparable algorithmically mediated short-video environments, content strategies can be calibrated to the "cost ladder" of public engagement to support communicative effectiveness. Officials may clarify their engagement goals and deploy distinct cue combinations accordingly. To encourage broad, low-cost visibility such as likes, creators can emphasize experience-enhancing cues, leveraging rhythmic editing, background music, and narrative twists to support sensory immersion. Conversely, to elicit deep social diffusion in the form of shares, content appears to benefit from high Information Complexity and substantive cognitive value rather than empty rhetoric. Crucially, this information delivery is more likely to remain engaging when decoupled from the didactic tone often associated with "rational preaching," which is negatively associated with engagement in our models (see Table 4, M4). Instead, high-value information can be embedded within conversational narratives to reduce defensive processing, so that the audience perceives the content as valuable knowledge rather than administrative instruction.

Finally, the deployment of authority cues can be adapted according to the account's lifecycle stage. Newer or smaller official accounts often face a "legitimacy deficit" and may place greater emphasis on "hard" symbols, such as uniforms and badges, to draw on the observed "legitimacy compensation" pattern for initial trust building (cold start). However, as the account matures and accumulates social capital, these formal symbols can be gradually relaxed. Continued over-reliance on rigid authority markers in established accounts may be associated with an "authenticity penalty," where excessive formality distances the creator from an audience that has already granted them trust based on personal connection.

Therefore, while established accounts should transition toward a personal digital brand, for resource-limited practitioners who may find the full research workflow challenging, we recommend a **Simplified Application Workflow**. Rather than strictly replicating the inductive theory-building phases, government media teams can implement a three-step protocol: (1) *Adaptation*: modify the validated codebook (Table 2) to local policy contexts via prompt engineering; (2) *Calibration*: refine prompts on a small pilot sample to verify output consistency; and (3) *Deployment*: execute a "Direct-Inference Mode" using cost-effective models (e.g., Gemini Flash). This approach renders high-precision monitoring accessible to grassroots departments with minimal computational overhead.

7. Conclusions, Limitations, and Future Directions

7.1. Conclusions and Contributions

This study makes a dual contribution to Information Management and Government Communication by examining Chinese grassroots officials' D2C marketing videos on Douyin. Theoretically, it systematically examines persuasion patterns of "government official influencers" within this platformized short-video setting by identifying the "bifurcation of authority" and the "legitimacy compensation" pattern. Crucially, this study refines the application of the Elaboration Likelihood Model to short-video environments by specifying how central and peripheral cues can be interpreted under temporal and multimodal conditions. We find evidence that, within this algorithmically mediated setting, persuasion can be interpreted as a dynamic matching process rather than simply a fixed accumulation of cues. Specifically, the observed associations of authority and affect are not intrinsic to the cue itself but vary with the temporal "cost ladder" of engagement, with deeper platform-visible engagement associated with higher-order cue combinations.

Methodologically, we develop and implement an AI-enhanced analytical framework that addresses the trade-off between theoretical depth and scalability in large-scale multimodal video analysis. In contrast to studies that use models primarily as automated classifiers for feature counting, our protocol specifies a structured, theory-informed human-in-the-loop procedure for extracting and validating multimodal cues. This design supports the interpretation of complex semantic relationships rather than simple signal detection, enabling rigorous analysis under the constraints of rapidly evolving platform media.

7.2. Limitations and Future Directions

Notwithstanding these contributions, the conclusions should be read within specific boundary conditions that, in turn, delineate promising avenues for further inquiry. First, while the framework relies on generative AI, our cross-model validation (Appendix Fig. A2) indicates that the identified strategies are robust to variations in model parameter size. This consistency suggests that the framework possesses "instrumental resilience," where reliability derives from the structured "Context-Injection" protocol and rigorous construct definitions rather than the idiosyncratic weights of a specific foundation model. While this diagnostic check supports the protocol's robustness, future work could further extend this validation across heterogeneous model families (e.g., Qwen or Llama) to isolate potential algorithmic biases inherent to specific architectures. Furthermore, we anticipate that emerging models

trained on broader multimodal corpora will likely yield enhanced consistency and a more refined capacity to decode the subtle, human-like attentional cues that underpin persuasion.

Second, the empirical setting, China’s grassroots governance on Douyin, limits generalizability, especially regarding cultural contingencies and platform-induced biases. Although the tension between bureaucratic and platform logics may recur across contexts, cultural dimensions such as power distance can shape how audiences interpret institutional symbols and authority cues (Daniels & Greguras, 2014). Douyin should therefore be treated not as a neutral distribution channel but as a platform environment in which recommendation mechanisms, trending dynamics, creator norms, and short-video culture shape which official videos become visible, interacted with, and observable. Because the capped extraction strategy prioritized top-engagement videos within each account, the corpus is well suited to analyzing visible D2C communication but may underrepresent low-exposure, less algorithmically amplified, or less platform-native content. Cross-cultural replication across platforms and political systems is needed to assess when institutional contexts function as legitimacy signals and when they increase psychological distance. Although our models account for observed covariates and overdispersed engagement counts, unobserved real-time algorithmic boosting may still introduce residual confounding. Likewise, likes, comments, and shares should be interpreted as platform-visible behavioral traces rather than direct measures of public attitudes, persuasion outcomes, or offline action; future work could address this limitation through controlled field experiments.

Third, our theoretical framing is organized around ELM, and the quantitative models do not directly operationalize identity-based or relational psychological states. Accordingly, social identity, perceived social presence, and parasocial perspectives are used as interpretive lenses to clarify the observed patterns rather than as empirically measured mechanisms. Future research should directly measure these constructs to test the proposed psychological pathways and to evaluate the strength of these mechanisms across contexts.

Building on these conclusions and boundaries, future research can proceed along three interrelated avenues. On the theoretical front, the observed bifurcation of authority calls for moving beyond traditional source-credibility models toward a dynamic digital source-credibility theory that integrates constructs such as personified authority and social presence. In application, the framework should be deployed for cross-cultural comparative validation. Future research could apply this protocol to datasets from distinct political contexts (e.g., Western officials on TikTok) to empirically verify whether the “Cost Ladder” and “Authority Bifurcation” persist across divergent civic engagement cultures. Methodologically, the most forward-looking direction is to evolve AI from an efficient “measurement tool” into a genuine “partner in theory discovery.” This entails systematically designing experiments and analytical workflows to explore and verify the potential of large multimodal models for abductive reasoning and original theory building, an agenda that would both deepen understanding of complex social phenomena and help reshape the research paradigm of computational social science.

Ethical approval

Our study did not require further ethics committee approval as it did not involve animal or human clinical trials and was not unethical.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit authorship contribution statement

Yuying Luo: Writing – original draft, Visualization, Validation, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Xue Lei:** Validation, Supervision, Resources, Project administration, Investigation. **Wenxuan Zhang:** Validation, Software, Methodology, Conceptualization. **Hao Zheng:** Validation, Supervision, Investigation, Data curation. **Xueguo Xu:** Validation, Supervision, Funding acquisition, Conceptualization. **Jun Xu:** Validation, Software, Methodology, Data curation, Conceptualization.

Declaration of conflicting interest

The authors declare no competing financial interest.

Appendix

Table A1
Structured Prompting Protocol for the AI-Enhanced Analytical Framework.

Phase	Core Instruction / Prompt Logic	Key Input Variables	Mechanism & Function
-------	---------------------------------	---------------------	----------------------

(continued on next page)

Table A1 (continued)

Phase	Core Instruction / Prompt Logic	Key Input Variables	Mechanism & Function
1. Exploratory Feature Emergence	Role: Cognitive Visual Analysis Expert (guided by LC4MP). Task: Perform frame-by-frame analysis covering: (1) Pre-Attentive Processing, (2) Attention Allocation, (3) Information Encoding. Constraint: Focus on dynamic changes relative to $t - 1$ rather than static description.	author_profile video_description key_frame_image	Simulates open-coding to generate atomic, descriptive features for inductive analysis.
2. Inductive Strategy Interpretation	Task: Evaluate topic cluster validity (Coherence/Relevance on 1-5 scale). Logic: Classify based on ELM and "Identity Orientation" (Authority vs. Engagement). Output: Recommendation (Accept/Review/Reject) with academic justification.	topic_keywords representative_docs coherence_score	Bridges unsupervised clustering (BERTopic) with theoretical filtering to remove noise.
3. Theoretical Construct Synthesis	Protocol: Three-Step Deduction. Step 1: Empirical Deconstruction (Actors, Context, Tone). Step 2: Thematic Abstraction (Hypothesize persuasion mechanism). Step 3: Operationalization (Map to ELM route; define binary/continuous code).	topic_summary validation_scores	Abductive reasoning to transform raw patterns into measurable, theoretically grounded constructs.
4. Large-Scale Quantitative Coding	Role: Quantitative Content Analyst. Protocol: Sequential Context-Injection. 1. Ingest Audio/Meta -> 2. Code Visual Frames -> 3. Holistic Synthesis. Constraint: Strict adherence to codebook; "Consistency Mandate" applies.	video_metadata audio_transcript key_frame_sequence	Automated application of the finalized codebook with maximum reproducibility (Temp=0.0).

Table A2
Technical Specifications and Prompt Engineering Mechanics.

Component	Configuration / Core Template Structure
1. Model Hyperparameters	Model: Gemini 2.5 Pro Temperature: 0.0 (Measurement Phase); 0.4 (Discovery Phase) Top-p: 0.95 Safety Settings: BLOCK_ONLY_HIGH
2. System Persona	Instruction: "You are a quantitative content analysis assistant. Apply the codebook exactly and base all decisions only on the provided evidence."
3. Audio-First Context Injection	Logic: Establish semantic baseline before visual decoding. Template: "Input: [Audio_Transcript] + [Metadata]. Task: Analyze this context to infer the core product category and emotional tone. Await response before processing visual frames."
4. Dynamic Contrastive Protocol	Logic: Differential Analysis (t vs. $t - 1$). Template: "Frame Timestamp: [T]. Context: [Previous_Frame_Analysis]. Instruction: Identify strictly what is NEW or DIFFERENT in visual elements/narrative compared to the previous frame."
5. Chain-of-Thought (CoT)	Logic: Internal Reasoning Trace required for all variables. Template: "For variable [X], perform silent reasoning: (1) Check definition, (2) Scan evidence, (3) Determine score. Output: JSON format only."

Table A3
Robustness Check using OLS (N=779).

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Central Route (ELM)	Content Information	0.757***	0.751***	0.728**	0.649*	0.945**	0.853**
	Complexity	(0.283)	(0.272)	(0.350)	(0.377)	(0.433)	(0.391)
	Marketing Information	-0.105	-0.367	0.548	0.206	1.350***	0.975***
	Intensity	(0.339)	(0.328)	(0.348)	(0.332)	(0.484)	(0.371)
	Content Rational Appeal	-0.803***	-0.364***	-0.892***	-0.385***	-0.971***	-0.424**
Peripheral Route (ELM)	Strength	(0.119)	(0.124)	(0.122)	(0.126)	(0.170)	(0.179)
	Source Official Identity		0.213*		0.200*		0.348**
	Salience		(0.116)		(0.118)		(0.167)
	Scene Official Context Salience		-0.320**		-0.372**		-0.401**
			(0.145)		(0.147)		(0.194)
	Engaging Narrative Style		0.203*		0.201*		0.375**
			(0.121)		(0.122)		(0.173)
	Editing Frequency		0.167		0.104		0.354*
			(0.128)		(0.130)		(0.185)
	Call to Action		-0.476***		-0.455***		-0.485***
			(0.110)		(0.111)		(0.158)
	Product Visual Salience		-0.311**		-0.296**		-0.396**
			(0.129)		(0.130)		(0.185)
	Background Music		0.404**		0.538***		0.524*

(continued on next page)

Table A3 (continued)

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Control Variables	Speech Rate		(0.200) -0.255*** (0.047)		(0.203) -0.317*** (0.048)		(0.288) -0.366*** (0.068)
	Linguistic Expression	-0.096 (0.194)	-0.168 (0.185)	0.012 (0.200)	-0.06 (0.188)	0.023 (0.277)	-0.065 (0.266)
	Accuracy	-0.052 (0.094)	0.028 (0.095)	0.026 (0.097)	0.113 (0.096)	0.071 (0.135)	0.134 (0.137)
	Video Duration	0.482*** (0.040)	0.475*** (0.038)	0.355*** (0.042)	0.342*** (0.040)	0.205*** (0.035)	0.198*** (0.032)
	Follower Count	-0.017** (0.008)	-0.007 (0.008)	-0.016** (0.008)	-0.006 (0.008)	-0.042*** (0.012)	-0.029*** (0.011)
	Publisher Age	0.025 (0.146)	0.118 (0.142)	-0.122 (0.150)	-0.026 (0.144)	-0.535** (0.208)	-0.208 (0.209)
	Publisher Gender	-0.080 (0.066)	-0.081 (0.064)	-0.116* (0.068)	-0.122* (0.064)	-0.311*** (0.095)	-0.309*** (0.091)
	Publisher Province	7.674*** (0.543)	7.563*** (0.581)	5.334*** (0.557)	5.331*** (0.589)	5.078*** (0.774)	4.597*** (0.851)
	Intercept	0.170	0.26	0.133	0.246	0.124	0.205
	R ²						

Table A4
Other model validation & robustness.

Variable Classification	Variables	NB estimates with HC3-robust (sandwich) SEs			PPML Poisson Pseudo-Maximum Likelihood		
		Likes (M1)	Comments (M2)	Shares (M3)	Likes (M4)	Comments (M5)	Shares (M6)
Central Route (ELM)	Content Information Complexity	0.156** (0.071)	0.151* (0.082)	0.067* (0.038)	0.034*** (0.011)	0.053*** (0.015)	0.080*** (0.025)
	Marketing Information Intensity	-0.084 (0.061)	-0.004 (0.063)	0.138*** (0.045)	-0.004 (0.008)	0.014 (0.012)	0.047** (0.019)
Peripheral Route (ELM)	Content Rational Appeal Strength	-0.307*** (0.080)	-0.327*** (0.075)	-0.515*** (0.053)	-0.030*** (0.010)	-0.049*** (0.014)	-0.075*** (0.025)
	Source Official Identity Salience	0.152** (0.064)	0.150* (0.079)	0.237*** (0.072)	0.038* (0.019)	0.048* (0.027)	0.099** (0.045)
	Scene Official Context Salience	-0.190*** (0.057)	-0.222*** (0.056)	-0.343*** (0.060)	-0.049** (0.021)	-0.091*** (0.030)	-0.067 (0.055)
	Engaging Narrative Style	0.105 (0.078)	0.107 (0.070)	0.258*** (0.078)	0.045* (0.023)	0.059** (0.029)	0.089** (0.039)
	Editing Frequency	0.104 (0.067)	0.136** (0.066)	0.277*** (0.076)	0.025 (0.019)	0.019 (0.028)	0.099** (0.049)
	Call to Action	-0.140* (0.075)	-0.117 (0.073)	-0.286*** (0.074)	-0.076*** (0.017)	-0.101*** (0.025)	-0.131*** (0.043)
	Product Visual Salience	-0.170*** (0.065)	-0.245*** (0.061)	-0.318*** (0.064)	-0.055** (0.021)	-0.076** (0.030)	-0.127** (0.053)
	Background Music	0.256*** (0.051)	0.253*** (0.052)	0.258*** (0.057)	0.092*** (0.029)	0.146*** (0.041)	0.169** (0.083)
	Speech Rate	-0.445*** (0.075)	-0.572*** (0.073)	-0.542*** (0.072)	-0.048*** (0.011)	-0.092*** (0.016)	-0.134*** (0.028)
	Linguistic Expression Accuracy	-0.050 (0.058)	0.114* (0.069)	-0.037 (0.053)	-0.025 (0.027)	-0.020 (0.040)	-0.045 (0.075)
Control Variables	Video Duration	-0.072 (0.072)	-0.055 (0.079)	0.070* (0.041)	0.014 (0.010)	0.034** (0.014)	0.041 (0.025)
	Follower Count	0.986*** (0.048)	0.570*** (0.049)	0.363*** (0.052)	0.052*** (0.010)	0.055*** (0.014)	0.093*** (0.024)
	Publisher Age	-0.008 (0.066)	-0.050 (0.069)	-0.112* (0.061)	-0.001 (0.010)	-0.003 (0.013)	-0.048* (0.024)
	Publisher Gender	-0.101 (0.061)	-0.125** (0.056)	-0.353*** (0.063)	0.012 (0.020)	-0.015 (0.029)	-0.125** (0.049)
	Publisher Province	0.004 (0.068)	-0.107 (0.076)	-0.188*** (0.064)	-0.011 (0.010)	-0.023 (0.015)	-0.075*** (0.025)
Model Fit	Intercept	7.914*** (0.066)	5.836*** (0.066)	4.842*** (0.070)	1.830*** (0.042)	1.451*** (0.059)	1.207*** (0.113)
	Pseudo-R ²	0.391	0.406	0.409	0.248	0.243	0.217

Table A5
Propensity Score Matching (PSM)–Based Estimates of the Average Treatment Effect on the Treated (ATT).

Strategy Variable	Likes (ATT)	Comments (ATT)	Shares (ATT)
-------------------	-------------	----------------	--------------

(continued on next page)

Table A5 (continued)

Strategy Variable	Likes (ATT)	Comments (ATT)	Shares (ATT)
Content_Information_Complexity	0.272**	0.257**	0.416***
Marketing_Information_Intensity	-0.265*	0.049	0.243***
Content_Rational_Appeal_Strength	-1.119***	-1.266***	-1.174***
Source_Official_Identity_Salience	0.118***	0.116**	0.365***
Scene_Official_Context_Salience	-1.062***	-1.108***	-0.727***
Engaging_Narrative_Style	0.652*	0.670**	0.688***
Editing_Frequency	0.566***	0.643***	0.914***
Call_to_Action	-0.908***	-0.785*	-1.037***
Product_Visual_Salience	-0.771***	-1.506***	-0.414***
Background_Music	0.754*	1.553***	0.752*

Table A6
Analytical Transparency and Alternative Validation Logic.

Analytical step	Primary choice in this study	Alternative or baseline considered	Validation logic and boundary
Multimodal feature extraction	Sequential context-injection with dynamic contrastive analysis comparing frame t with frame t−1.	Static or independent frame coding.	Chosen to preserve temporal cue changes in short videos. This design improves analytical fit for dynamic media but should not be interpreted as a causal identification strategy.
Topic discovery	BERTopic applied to MM-LLM-generated semantic annotations.	LDA-style bag-of-words topic modeling.	Chosen because the annotations are short-text-like, semantically dense, and lexically diverse. This is a methodological fit argument; no formal numerical claim is made that BERTopic outperforms all topic models.
Topic-to-construct synthesis	C _v topic coherence diagnostics, topic-size filtering, qualitative pruning, HAC with Ward’s linkage, and researcher review.	Accepting raw topic labels or relying only on coherence scores.	The 85 topics → 47 validated topics → nine constructs chain combines quantitative diagnostics with semantic interpretability, representative examples, c-TF-IDF inspection, and preparatory review before formal theory anchoring.
Scaled coding reliability	Two independent AI coding instances plus human adjudication for divergent or context-dependent cases.	Single AI run, unadjudicated AI output, or simple averaging of AI outputs.	AI-AI, human-human, and AI-human diagnostics identify where AI is reliable and where human judgment is necessary; final adjudication is evidence-based and rule-guided.
Model-specific stability	Gemini 2.5 Pro as the main inference model, with Gemini 2.5 Flash validation on a stratified subsample.	Reliance on a single foundation model.	Cross-model validation on N = 78 shows strong convergence with the final dataset, supporting procedural stability across two Gemini variants while not claiming universal model independence.

Table A7
Full-Sample Interaction-Based Coefficient-Difference Tests for Heterogeneity Analysis.

Variables	Video Long - Video Short	Fans Large - Fans Small	Gov Good - Gov Poor
Content Information Complexity	0.412*** (0.145)	-0.494* (0.280)	-0.857*** (0.291)
Marketing Information Intensity	-0.508*** (0.161)	0.549 (0.352)	0.391** (0.165)
Content Rational Appeal Strength	-0.190 (0.441)	-0.055 (0.636)	0.739 (0.503)
Source Official Identity Salience	0.421** (0.207)	-0.828*** (0.223)	-0.752*** (0.230)
Scene Official Context Salience	0.600* (0.344)	0.227 (0.448)	-0.343** (0.145)
Engaging Narrative Style	-0.607* (0.347)	-0.103 (0.474)	0.174 (0.476)
Editing Frequency	0.630* (0.362)	-0.603** (0.291)	-0.375 (0.572)
Call to Action	-0.551* (0.314)	0.033 (0.548)	-0.361 (0.497)
Product Visual Salience	-0.214 (0.462)	0.242 (0.472)	0.300 (0.409)
Background Music	0.128 (0.233)	0.388 (0.377)	0.385 (0.377)
Speech Rate	-0.215 (0.399)	-0.586 (0.479)	-0.611 (0.495)

Note: Cells report interaction coefficients from full-sample negative binomial models predicting shares. Column labels indicate contrast direction. Standard errors are in parentheses; controls are included. *p < 0.1; **p < 0.05; ***p < 0.01.

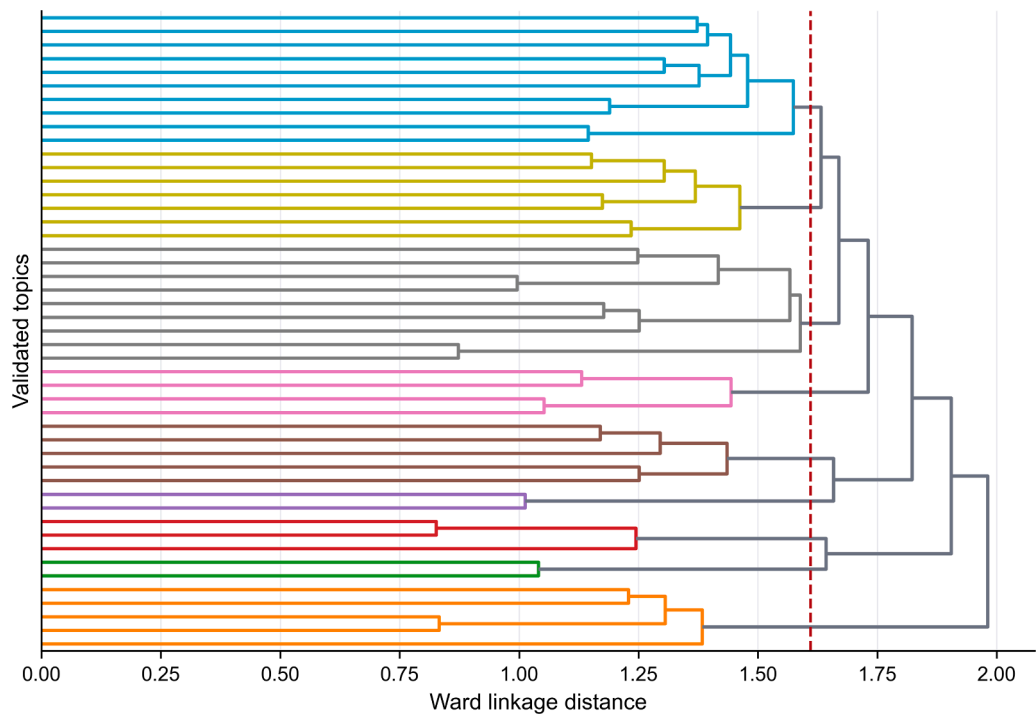


Fig. A1. HAC dendrogram for the 47 validated topics. Note. The dendrogram was generated using Ward’s linkage with Euclidean distance on TF-IDF topic-document vectors. The dashed vertical line indicates the tree cut used to obtain nine candidate clusters. Colors denote HAC candidate clusters prior to ELM-based theory anchoring and construct-label auditing.

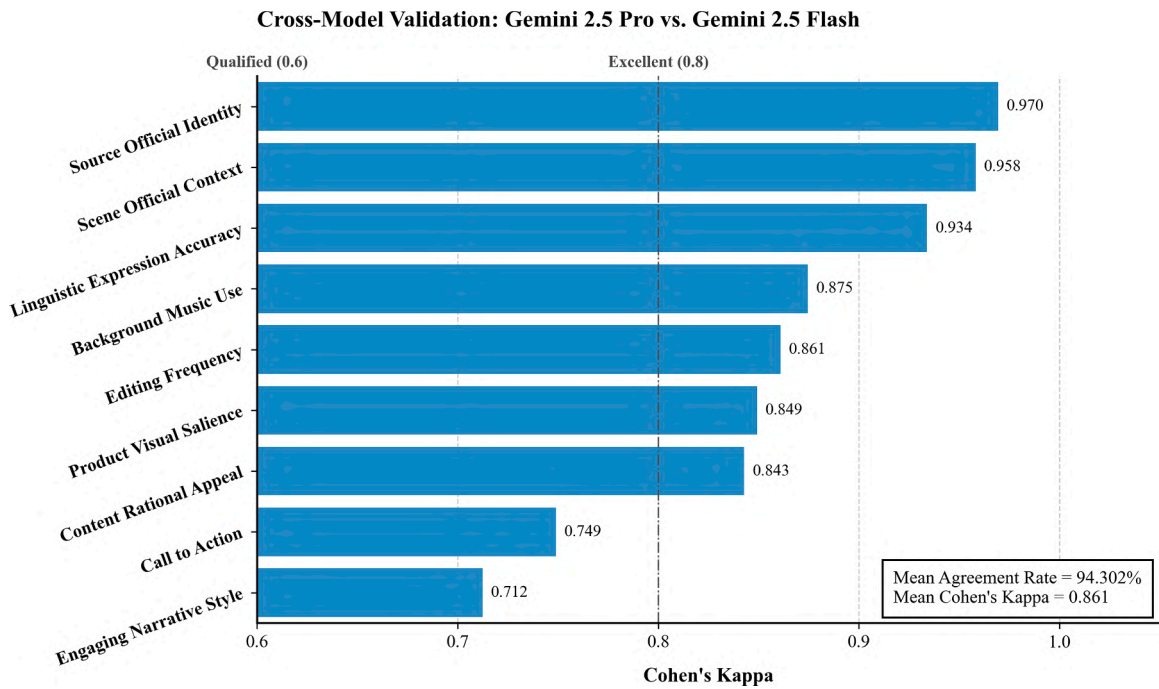


Fig. A2. Cross-Model Validation Metrics.

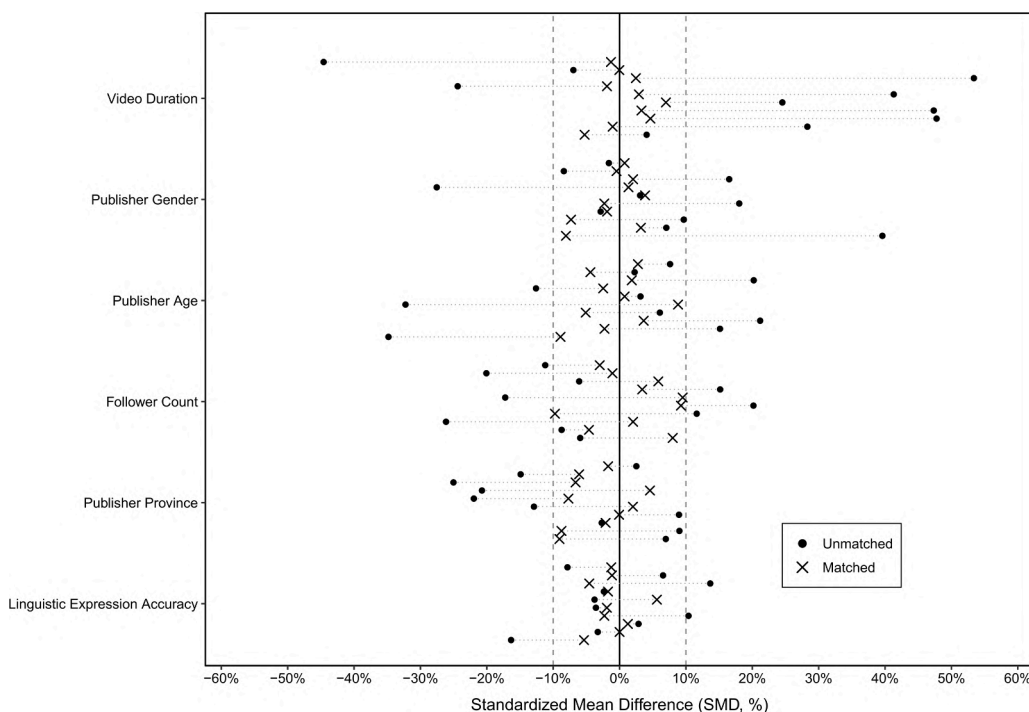


Fig. A3. Treatment-Specific Covariate Balance Diagnostics After Propensity Score Matching. Note: Solid circles indicate unmatched SMDs and crosses indicate matched SMDs for treatment-specific covariate balance. Dashed vertical lines mark the conventional $\pm 10\%$ threshold.

¹Cost Analysis: The total cost for the double-blind analysis of 779 videos (1,558 requests) was approximately \$43.86. This includes 10.13 million input tokens (video/audio/text context at \$1.25/1M) and 3.12 million output tokens (structured coding data and Chain-of-Thought at \$10.00/1M).

Data availability

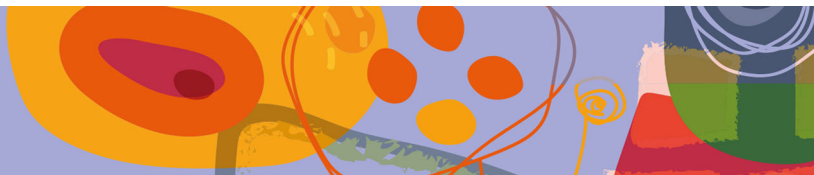
The data that support the findings of this study are available from the author upon reasonable request.

References

- Agarwal, A., Lee, S.-Y., & Whinston, A. B. (2025). The Effect of Popularity Cues and Peer Endorsements on Assertive Social Media Ads. *Information Systems Research*, 36(2), 1167–1182. <https://doi.org/10.1287/isre.2021.0606>
- Alibakhshi, R., & Srivastava, S. C. (2025). How Sentiment Divergence in Influencers' Multimodal Social Media Posts Shapes Follower Engagement? *Journal of the Association for Information Systems*, 26(4), 1081–1137.
- Bangero, H. B. (2025). Best practices in e-government communication: Lessons from the local Governments' use of official facebook pages. *Government Information Quarterly*, 42(1), Article 102010. <https://doi.org/10.1016/j.giq.2025.102010>
- Benz, P., Pradier, C., Kozłowski, D., Shokida, N. S., & Larivière, V. (2025). Mapping the unseen in practice: Comparing latent Dirichlet allocation and BERTopic for navigating topic spaces. *Scientometrics*, 130(7), 3839–3870. <https://doi.org/10.1007/s11192-025-05339-6>
- Cardamone, E., Miceli, G. "Nino", & Raimondo, M. A. (2025). Color Inside and Outside the Lines: Evidence From Eye-Tracking Studies on Conformity to and Differentiation From Category Color Codes. *Psychology & Marketing*, 42(5), 1330–1345. <https://doi.org/10.1002/mar.22180>
- Chandrasekaran, S., Annamalai, B., Yoshida, M., RV, S., & Pathak, A. A. (2025). Engaging head and heart: Effect of marketer-generated content on social media engagement. *Behaviour & Information Technology*, 44(19), 4713–4733. <https://doi.org/10.1080/0144929X.2025.2486586>
- Chen, J., & Shi, C. (2025). Proactive governance by official administrators on Chinese social media platforms: Boundary discourse and governance legitimacy. *Media, Culture & Society*, Article 01634437251336153. <https://doi.org/10.1177/01634437251336153>
- Chen, Y., & Gong, Q. (2025). Bureaucrats on Douyin: The Official Appropriation of Egao in Promoting Tourism on Chinese Social Media. *Asian Studies Review*, 49(1), 134–155. <https://doi.org/10.1080/10357823.2024.2366294>
- Cheng, G., Li, W., He, M., & Liao, L. (2024). Exploring consumer responses to official endorsement: Roles of credibility and attractiveness attributes in live streaming. *Frontiers in Psychology*, 15. <https://doi.org/10.3389/fpsyg.2024.1371343>
- Clayton, R. B., & Myrick, J. G. (2025). What Can the Heart Tell Us About Thinking? A Three-Decade Review of Heart Rate Measurement and Its Future Applications to Advertising Research. *Journal of Advertising*, 1–16. <https://doi.org/10.1080/00913367.2025.2515905>
- Comanici, G., Bieber, E., Schaekermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., Marris, L., Petulla, S., Gaffney, C., Aharoni, A., Lintz, N., Pais, T. C., Jacobsson, H., Szepektor, I., Jiang, N.-J., ... Helmholtz, W. (2025). Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities (arXiv:2507.06261; Version 5). arXiv. <https://doi.org/10.48550/arXiv.2507.06261>
- Daniels, M. A., & Greguras, G. J. (2014). Exploring the Nature of Power Distance: Implications for Micro- and Macro-Level Theories, Processes, and Outcomes. *Journal of Management*, 40(5), 1202–1229. <https://doi.org/10.1177/0149206314527131>

- Deng, Z., Tang, Y., Wu, M., & Zhang, X. (2025). Investigating the effects of clickbait on user engagement in health communication: A mixed-method study. *Information & Management*, 62(8), Article 104231. <https://doi.org/10.1016/j.im.2025.104231>
- Dewan, M., Liu, J., & Shah, C. (2026). TRUE: A Reproducible Framework for LLM-Driven Relevance Judgment in Information Retrieval. *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, 1130–1134. <https://doi.org/10.1145/3773966.3779397>.
- Ecker-Ehrhardt, M. (2023). Public Legitimation by “Going Personal”? The Ambiguous Role of International Organization Officials on Social Media. *Politics and Governance*, 11(3), 213–225.
- Gao, J., Guo, Y., Lim, G., Zhang, T., Zhang, Z., Li, T. J.-J., & Perrault, S. T. (2024). CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–29. <https://doi.org/10.1145/3613904.3642002>
- Garz, M., Sörensen, J., & Stone, D. F. (2020). Partisan selective engagement: Evidence from Facebook. *Journal of Economic Behavior & Organization*, 177, 91–108. <https://doi.org/10.1016/j.jebo.2020.06.016>
- Grantham, S., Cervi, L., & Iachizzi, M. (2025). Double tap democracy: Political authenticity in the TikTok era. *Media International Australia*. London, England: Sage UK. <https://doi.org/10.1177/1329878x251327232>
- Guo, D., Qu, R., Wang, Y., & Xu, J. (2025). Social media and citizen mobilization: How Chinese government agencies use Weibo to advance the 2030 agenda goals on sustainability. *Public Relations Review*, 51(3), Article 102579. <https://doi.org/10.1016/j.pubrev.2025.102579>
- Hilbe, J. M. (2011). *Negative Binomial Regression* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/CBO9780511973420>
- Hu, W., Xu, Y., Li, Y., Li, W., Chen, Z., & Tu, Z. (2023). BLIVA: A Simple Multimodal LLM for Better Handling of Text-Rich Visual Questions (arXiv:2308.09936). arXiv. <https://doi.org/10.48550/arXiv.2308.09936>.
- Hughes, C., Swaminathan, V., & Brooks, G. (2019). Driving Brand Engagement Through Online Social Influencers: An Empirical Investigation of Sponsored Blogging Campaigns. *Journal of Marketing*, 83(5), 78–96. <https://doi.org/10.1177/0022242919854374>
- Jiang, T., Xu, Y., Li, Y., & Xia, Y. (2025). Integration of public libraries and cultural tourism in China: An analysis of library attractiveness components based on tourist review mining. *Information Processing & Management*, 62(2), Article 104000.
- Kim, H.-J., Ahn, S., & Ye, S. (2025). Exploring AI assistant in luxury brands: How social presence and emotional appeal drive technology adoption. *Journal of Retailing and Consumer Services*, 87, Article 104409. <https://doi.org/10.1016/j.jretconser.2025.104409>
- Kim, J., & Kim, H. (2024). Unlocking Creator-AI Synergy: Challenges, Requirements, and Design Opportunities in AI-Powered Short-Form Video Production. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI '24, 1–23. <https://doi.org/10.1145/3613904.3642476>
- Kim, Y. A., & Ahmad, M. A. (2013). Trust, distrust and lack of confidence of users in online social media-sharing communities. *Knowledge-Based Systems*, 37, 438–450. <https://doi.org/10.1016/j.knsys.2012.09.002>
- Kirilenko, A. P., & Stepchenkova, S. (2025). Facilitating topic modeling in tourism research: Comprehensive comparison of new AI technologies. *Tourism Management*, 106, Article 105007. <https://doi.org/10.1016/j.tourman.2024.105007>
- Lang, A. (2000). The limited capacity model of mediated message processing. *Journal of Communication*, 50(1), 46–70.
- Li, H., Luo, Y., Wang, J., Ye, Y., & Xu, J. (2026). From access to adherence: Fostering professional capital and continued engagement in blended teacher workshops. *Journal of Professional Capital and Community*, 1–22. <https://doi.org/10.1108/JPPC-12-2025-0145>
- Li, H., Xu, J., & Luo, Y. (2024). The impact of mobile social media on knowledge sharing among vocational school teachers: A social cognitive career perspective. *Journal of Information Science*. <https://doi.org/10.1177/01655515241293754>, 01655515241293754.
- Li, M., Li, G., & Yang, S. (2024). Correction by distraction: How high-tempo music enhances medical experts' debunking TikTok videos. *Journal of Computer-Mediated Communication*, 29(5), Article zmae007. <https://doi.org/10.1093/jcmc/zmae007>
- Lu, T., & Zhang, Y. (2024). 1 + 1 > 2? Information, Humans, and Machines. *Information Systems Research*. <https://doi.org/10.1287/isre.2023.0305> (world).
- Lu, Y., & (Cindy) Shen, C. (2023). Unpacking Multimodal Fact-Checking: Features and Engagement of Fact-Checking Videos on Chinese TikTok (Douyin). *Social Media & Society*, 9(1), Article 20563051221150406. <https://doi.org/10.1177/20563051221150406>
- MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7)
- Naem, M., Smith, T., & Thomas, L. (2025). Thematic Analysis and Artificial Intelligence: A Step-by-Step Process for Using ChatGPT in Thematic Analysis. *International Journal of Qualitative Methods*, 24, Article 16094069251333886. <https://doi.org/10.1177/16094069251333886>
- Nguyen, L., Walters, J., Paul, S., Monreal Ijurco, S., Rainey, G. E., Parekh, N., Blair, G., & Darrah, M. (2025). Feeds, feelings, and focus: A systematic review and meta-analysis examining the cognitive and mental health correlates of short-form video use. *Psychological Bulletin*, 151(9), 1125.
- Özdemir, S. F., Graneng, K., & Wilde, P. de (2025). EU executives on social media: Assessing the potential and trajectories of legitimation via public communication on Twitter. *European Union Politics*. London, England: Sage UK. <https://doi.org/10.1177/14651165251320881>
- Pan, M., Blut, M., Ghiassaleh, A., & Lee, Z. W. Y. (2025). Influencer marketing effectiveness: A meta-analytic review. *Journal of the Academy of Marketing Science*, 53(1), 52–78. <https://doi.org/10.1007/s11747-024-01052-7>
- Peña, J., Allen, C., Tseng, J. T., Chen, V. H. H., Ibasco, G. C., & Koek, W. J. D. (2026). Intergroup contact in virtual reality: The influence of avatar identity, social identity, and contact valence on prejudice. *New Media & Society*. , Article 14614448251411262. <https://doi.org/10.1177/14614448251411262>
- Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In *Advances in experimental social psychology*, 19 pp. 123–205). Elsevier. <https://www.sciencedirect.com/science/article/pii/S0065260108602142>.
- Potter, R. F., & Bolls, P. (2012). *Psychophysiological measurement and meaning: Cognitive and emotional processing of media*. Routledge. <https://www.taylorfrancis.com/books/mono/10.4324/9780203181027/psychophysiological-measurement-meaning-paul-bolls-robert-potter>.
- Qi, S., Cao, Z., Rao, J., Wang, L., Xiao, J., & Wang, X. (2023). What is the limitation of multimodal LLMs? A deeper look into multimodal LLMs through prompt probing. *Information Processing & Management*, 60(6), Article 103510. <https://doi.org/10.1016/j.ipm.2023.103510>
- Qin, Q., Wen, Z., Zhou, Z., & Fan, B. (2025). Leveraging social media for new energy vehicle policy diffusion in China: A central-local government interaction analysis. *Transportation Research Part A: Policy and Practice*, 195, Article 104459. <https://doi.org/10.1016/j.tra.2025.104459>
- Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55. <https://doi.org/10.1093/biomet/70.1.41>
- Silva, J. M. C. S., & Tenreiro, S. (2006). The Log of Gravity. *The Review of Economics and Statistics*, 88(4), 641–658. <https://doi.org/10.1162/rest.88.4.641>
- Sim, J., Choi, K., Han, S. P., & Cho, D. (2024). In-Consumption Information Cues and Online Video Consumption. *Management Information Systems Quarterly*, 48(2), 645–678.
- Son, H., & Park, Y. E. (2025). Agenda-setting effects for covid-19 vaccination: Insights from 10 million textual data from social media and news articles using BERTopic. *International Journal of Information Management*, 83, Article 102907.
- Song, Y. (Amy), Wu, J., Mookherjee, S., & Bose, I. (2026). Advertising green products on brands' official social media accounts or mega-influencer accounts? *Information & Management*, 63(1), Article 104259. <https://doi.org/10.1016/j.im.2025.104259>
- Tong, X., Chen, Y., Zhou, S., & Yang, S. (2022). How background visual complexity influences purchase intention in live streaming: The mediating role of emotion and the moderating role of gender. *Journal of Retailing and Consumer Services*, 67, Article 103031. <https://doi.org/10.1016/j.jretconser.2022.103031>
- Ver Hoef, J. M., & Boveng, P. L. (2007). QUASI-POISSON VS. NEGATIVE BINOMIAL REGRESSION: HOW SHOULD WE MODEL OVERDISPERSED COUNT DATA? *Ecology*, 88(11), 2766–2772. <https://doi.org/10.1890/07-0043.1>
- Wang, Q., Yao, X., Li, X., Yan, X., & Li, R. (2025). When peripheral route meets central route: An elaboration likelihood model of sales performance in live commerce. *Journal of Retailing and Consumer Services*, 84, Article 104218. <https://doi.org/10.1016/j.jretconser.2024.104218>
- Wu, P., Zou, S., Chen, C., & Song, Y. (2025). Hotbed of stigmatization or source of support: A multimodal analysis of mental health-related videos on Douyin. *Computers in Human Behavior*, 172, Article 108716. <https://doi.org/10.1016/j.chb.2025.108716>
- Xu, J., Liu, F., Luo, Y., & Xu, X. (2025). Stage-Specific Impacts of Digital Technologies on Supply-Chain Resilience: Meta-Analytic Evidence for Continuous Process Improvement. *Business Process Management Journal*. <https://doi.org/10.1108/BPMJ-04-2025-0550>

- Xu, J., Luo, Y., Wang, C., Wang, M., & Wu, Y. (2026). AI support in self-regulated learning: A decade of technological evolution and meta-analysis. *British Journal of Educational Technology*. <https://doi.org/10.1111/bjet.70058>. bjet.70058.
- Xu, X., Wang, S., Liu, J., Wang, X. L., & Morgan, N. (2025). How does the cuteness of persuasive communication about sustainable tourism shape tourists' environmentally responsible behavioral intentions? *Journal of Sustainable Tourism*, 33(11), 2367–2395. <https://doi.org/10.1080/09669582.2025.2478187>
- Yang, M., Ren, Y., & Adomavicius, G. (2020). Engagement by Design: An Empirical Study of the "Reactions" Feature on Facebook Business Pages. *ACM Trans. Comput.-Hum. Interact.*, 27(6), 43. <https://doi.org/10.1145/3412844>, 1-43:35.
- Yin, M., Chen, W., Zhu, D., & Jiang, J. (2025). Enhancing video rumor detection through multimodal deep feature fusion with time-sync comments. *Information Processing & Management*, 62(1), Article 103935. <https://doi.org/10.1016/j.ipm.2024.103935>



ARTICLE



<https://doi.org/10.1057/s41599-026-06673-7>

OPEN

Smart city strategy, China's urban innovation and policy effectiveness

Yuying Luo¹, Juanfeng Zhang²✉, Rui Han²✉ & Jun Xv³

Smart city strategies are the key policy in China's modernization. Yet how such strategies influence urban innovation remains insufficiently understood. This study employs a multi-period difference-in-differences approach within the Technology-Organization-Environment (TOE) framework to analyze panel data from 283 prefecture-level cities in China from 2006 to 2021. It seeks to evaluate the impact of smart city pilot policies on urban innovation levels and their mechanisms of action. The findings suggest that these policies are associated with higher urban innovation, with particularly pronounced effects in central and western regions and in small and medium-sized cities. The mediation effect model shows that improvements in information and communication technology, transportation infrastructure, digital government and social capital are potential channels linking smart city development to higher innovation, and that these mechanisms differ across regions. Spatial analysis further indicates the spillover effects of smart city policies, whereby innovation activities in pilot cities positively affect neighboring cities through regional networks. Overall, the research provides causal evidence and an analytical framework for evaluating place-based innovation policies and their effects on regional development disparities.

¹ Shanghai University, Shanghai, China. ² Zhejiang University of Technology, Hangzhou, China. ³ East China Normal University, Shanghai, China.
✉email: zhangjf@zjut.edu.cn; hanr@zjut.edu.cn

Introduction

In recent years, with the rapid advancement of information technology, the concept of smart cities has emerged, becoming a significant direction for global urban development (Javed et al., 2022). By integrating advanced information and communication technologies (ICT), smart cities can enhance urban management and service efficiency, thereby promoting sustainable urban development (Sorri et al., 2024; K.-L. Wang et al., 2025). Globally, numerous countries and regions have initiated smart city projects to improve the urban quality of life, optimize resource allocation, and stimulate economic growth through technological innovation.

As a novel urban development model, a smart city approach not only aims to improve citizens' quality of life but can also enhance a city's innovation capacity and competitiveness. Therefore, many countries regard fostering innovation and entrepreneurship as a central goal of smart city construction (Giffinger et al., 2007). European initiatives, exemplified by Finland and Italy, make use of open urban data platforms to encourage collaboration among citizens, firms and research institutions in addressing urban challenges, thereby generating a wide range of data-driven innovation projects. This bottom-up, open and collaborative model of digital innovation not only enhances the efficiency of urban service delivery, but also supports the growth of start-ups developing applications such as transport navigation and environmental monitoring (Jaakola et al., 2015; Marchesani & Ceci, 2025). By contrast, Singapore's Smart Nation initiative is state-led and has established a range of grant and tax incentive schemes to support frontier technologies such as artificial intelligence and big data, while the National Research Foundation (NRF) provides ongoing investment to strengthen research and its translation into industry (D. Das & Kwek, 2024). At the same time, smart-city projects in the Middle East, such as Masdar City, have emerged as capital-intensive test beds for clean technologies and as flagship projects for projecting a global image of urban innovation (Cugurullo, 2016; Randeree & Ahmed, 2019). Against this diverse global backdrop, China's strategy is distinct: it is neither purely bottom-up nor confined to a few flagship projects. Instead, China employs a massive, centrally coordinated "pilot" system—a quasi-natural experiment designed to test and diffuse digital governance across hundreds of existing cities simultaneously. Since 2012, the central government has elevated this to a national strategic priority, launching three successive batches of pilot projects covering nearly 300 cities (C. Guo et al., 2024; Yan et al., 2023). Unlike the organic evolution of smart cities in some Western contexts, China's approach represents a form of institutional change, where central authorities leverage pilot status to induce local innovation and industrial upgrading (W. Wang et al., 2025; F. Wu, 2018). By mandating the integration of digital technologies into urban management, these pilots aim to create an innovation-friendly environment and support the shift from factor-driven to innovation-driven growth (Zhu et al., 2024).

Although case studies generally acknowledge the positive link between smart city development and urban innovation, empirical findings remain mixed. Caragliu and Del Bo (2019) report that European cities with smart-city policies generate more patents than their counterparts without such policies. However, other scholars argue that certain smart-city projects have failed to deliver meaningful innovation or inclusive change, and their outcomes vary substantially across cities and contexts (Echebarria et al., 2021; Hollands, 2015). Consequently, key research gaps persist. First, do smart city strategies genuinely catalyze urban innovation, or do they merely create sophisticated digital infrastructures without meaningful innovative capacity? Second, the concrete mechanisms through which smart-city pilots translate

into tangible innovation remain under-explored. Third, the heterogeneity of policy effects across diverse regions and city types is poorly understood; China's vast territory encompasses pronounced disparities in economic development, scientific capacity, and institutional quality that may condition the effectiveness of national smart-city initiatives (Xu et al., 2022; Q. Yang & Elbakri, 2024). Fourth, potential spatial spillover effects of smart-city pilots on the innovation performance of neighboring non-pilot cities warrant systematic investigation.

The rest of this paper is organized as follows: Section "Literature review" provides a systematic review of the foundational literature and pertinent theories concerning smart cities and urban innovation. Section "Hypotheses, Models and Variables" discusses the development of the research hypotheses, the construction of the analytical model, and the selection of the variables. Section "Empirical Results" first presents the empirical findings on the influence of smart cities on urban innovation and then delves into the underlying mechanisms and spatial implications. The discussion section summarizes the main findings and offers policy insights based on the study's results. The final section concludes.

Literature review

Smart-city governance and China's pilot model. Smart cities have gradually evolved from a predominantly technology-centred imaginary into a widely diffused governance paradigm for addressing complex urban challenges, but the institutional configurations through which this paradigm is implemented differ markedly across places (Mora et al., 2023a). Recent comparative studies emphasise the importance of institutional embeddedness, arguing that the success of smart cities depends less on the mere deployment of digital technologies than on how these technologies are articulated with local governance structures, human capital and social institutions (Kummitha & Crutzen, 2019; Marchesani et al., 2023). Kummitha (Kummitha, 2018a) analyses entrepreneurial, corporate-led smart-city visions and warns that treating digital technology as a "technological panacea" raises fundamental concerns about technology appropriation, social polarisation, switching costs and data management. Similarly, Marchesani and Ceci (Marchesani & Ceci, 2025) show that in many European cases, the value of digitalisation is unlocked only when combined with collaborative governance in a "quadruple helix" involving government, industry, academia and citizens. Taken together, these international perspectives suggest that smart-city policies are not merely technical upgrades but complex social innovations that reconfigure urban governance arrangements.

Departing from the more commonly decentralised, market- or community-driven governance models highlighted in this literature, China's smart-city trajectory has been structured around a centrally steered logic of top-level design plus pilot experimentation. The policy sequence began with the Ministry of Housing and Urban–Rural Development's (MOHURD) National Smart City Pilot Work Notice in 2012, which authorised the first batch of 90 pilot cities and framed smart-city construction as a means of integrating digital technologies into urban planning, infrastructure and public-service delivery (K.-L. Wang et al., 2025). Subsequent documents—most notably the National New-type Urbanization Plan (2014–2020) and a series of guidelines on smart-city standard systems—embedded these pilots within a broader agenda of coordinated, people-centred urbanisation and codified technical architectures, data-sharing protocols and evaluation procedures, thereby institutionalising smart-city development as part of China's national digital-governance and

territorial management regime (Yin & Anantsuksomsri, 2022; Yuan et al., 2020).

The designation of smart-city pilots was the outcome of a strategic selection process rather than random assignment. Official documents and empirical analyses indicate that MOHURD prioritised cities with strong governance and high demonstration potential when approving pilot applications; early cohorts were therefore disproportionately composed of provincial capitals, sub-provincial cities and economically advanced prefectures, with later batches gradually extending to smaller and less-developed jurisdictions (C. Guo et al., 2024; Yan et al., 2023). This stratified selection of pilots distinguishes pilot from non-pilot jurisdictions and grants the former selective policy preferences, additional resources and greater latitude for regulatory experimentation, thereby endowing pilot cities with specific political status and institutional advantages (Yi & Shi, 2025). Within this state-led experimental framework, pilot designation forms part of an “experimentation under hierarchy” cycle, in which the centre sets strategic objectives and steers local governments to conduct policy trials that can later be scaled up nationally, aligning local development agendas with national priorities rather than fully autonomous, bottom-up agenda-setting (Heilmann, 2008). In this context, the innovation outcomes of China’s smart-city pilots are shaped not only by technological adoption, but also by the mobilisation of organisational, fiscal and regulatory resources in line with central mandates, as highlighted in broader analyses of state entrepreneurialism and performance-based incentives in China (F. Wu, 2018).

Theoretical framework. Evaluating the innovation effects of smart-city policies requires not just static benchmarking indicators but an explanatory framework that links policy interventions to the contextual conditions under which they operate. Early industry and policy frameworks—developed, for example, by IBM’s “Smarter Cities” programme, the European Union’s smart-city initiatives, and ISO standards such as ISO 37120—primarily focus on descriptive performance indicators, including broadband coverage, sensor deployment, energy consumption or service accessibility (ISO & IEC, 2019). While these metrics are useful for cross-city comparison, they remain limited in at least three important respects. They remain heavily technology- and output-oriented, privileging infrastructure and service efficiency over organisational capacities, governance processes and local institutional contexts (Angelidou, 2015; Neirotti et al., 2014). They are also largely static and cross-sectional: they record a city’s level of “smartness” at a given moment but say little about the mechanisms through which smart-city strategies generate, or fail to generate, innovation over time (Dolmans et al., 2023; Hollands, 2015). In addition, they tend to treat the city as a homogeneous bundle of assets rather than as a dynamic socio-technical ecosystem in which digital technologies interact with diverse organisations, multi-level governance arrangements and uneven market and regulatory environments (Nastjuk et al., 2022; Rodríguez Bolívar et al., 2023). As a result, indicator-based approaches are ill-equipped to explain why ostensibly similar smart-city projects can yield divergent innovation trajectories across different urban and institutional settings, especially in contexts—such as China—where smart-city programmes operate as state-designed policy pilots rather than purely market-driven initiatives.

To address these analytical limitations, this study adopts the Technology–Organization–Environment (TOE) framework as its primary theoretical lens. The TOE framework conceptualizes innovation adoption and impact as the result of interactions

among three interdependent contexts: the technological base, organizational capabilities, and surrounding environmental conditions (Drazin, 1991). Contemporary empirical scholarship has demonstrated that the TOE framework is particularly efficacious for interrogating complex digital transformations, as it eschews technological determinism in favor of a configurational perspective that explicitly links technology deployment to broader organizational and institutional conditions (Benchis et al., 2025). Within the smart city domain, this perspective is instrumental in framing governance not merely as a technical upgrade, but as a mechanism of institutional change wherein digital infrastructures reconfigure administrative routines and stakeholder relationships to enhance urban livability (Chen et al., 2025). This analytical approach is particularly salient for evaluating China’s national smart city pilot programme, where pilots function as centrally designed policy bundles that reshape local incentive structures; applying the TOE framework enables a granular examination of how unified national policies interact with city-specific technological endowments and environmental pressures to engender differentiated construction efficiency and innovation outcomes (Liao & Chen, 2025).

Smart-city innovation effects and research gaps. Several studies report that smart-city policies can stimulate urban innovation by upgrading ICT infrastructure, enabling digital governance, and enhancing overall city efficiency (Barba-Sánchez et al., (2019a), Barba-Sánchez et al., (2019b); Caragliu & Del Bo, 2019a). Robust digital infrastructure and open-data platforms associated with smart-city projects facilitate rapid information exchange and reduce barriers to innovation, fostering knowledge sharing, entrepreneurial activity, and collaborative creativity (Barba-Sánchez et al., (2019a), Barba-Sánchez et al., (2019b); Du & Wang, 2024). However, other research suggests that performance-assessment regimes can lock smart-city programmes into KPI-driven, short-term outputs, while their policy focus often prioritises immediate service improvements over more fundamental technological breakthroughs (Dolmans et al., 2023). Thus, although enhanced ICT infrastructure and governance processes may raise the sheer quantity of innovative activity, the depth and long-term impact of that activity are less clear, and smart-city projects may achieve only limited progress in terms of technological-innovation outcomes (Caragliu & Del Bo, 2019a).

Regional heterogeneity in policy outcomes further merits attention. Prior work shows that cities differ markedly in innovative capacity owing to variations in economic development, infrastructure maturity, human capital, and governance quality (Rodríguez-Pose et al., 2021; Y. Yang et al., 2023). Highly developed urban areas—benefiting from greater absorptive capacity and infrastructural readiness—may realise more immediate gains from smart-city investments, whereas less-developed regions could experience delayed or divergent effects; some studies even suggest that lagging areas may reap relatively larger benefits given their lower initial baselines (Sun et al., (2023a), Sun et al., (2023b)). Systematic empirical analyses that explicitly examine such regional disparities under smart-city policies, however, remain scarce. Moreover, because smart cities often function as regional hubs, it is still unclear whether pilot-oriented smart-city policies generate positive knowledge and innovation spillovers to neighbouring cities or instead produce a “siphon effect” that concentrates resources and capabilities in pilot areas (M. Wang & Zhou, 2025; T. Wu et al., 2025). Clarifying these spatial dynamics is essential for evaluating the broader regional impact and overall effectiveness of smart-city investments.

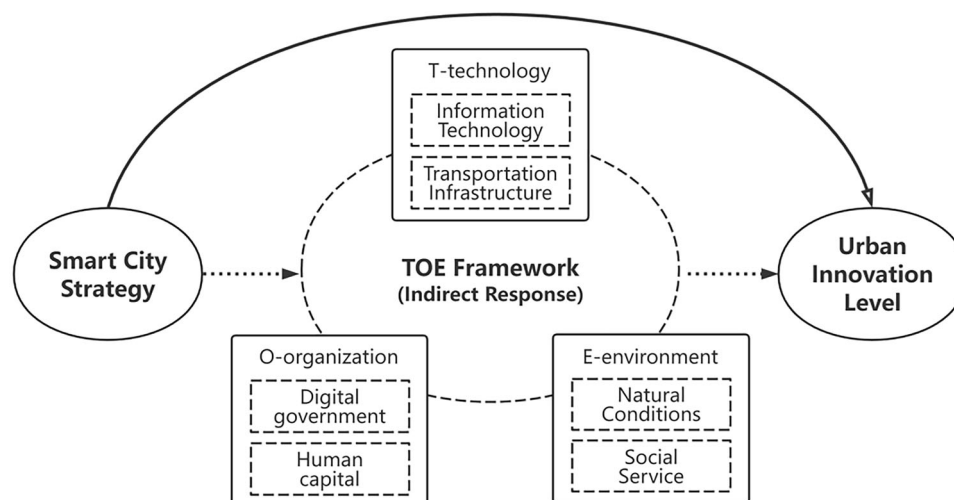


Fig. 1 Theoretical mechanism of smart city strategies impact on urban innovation.

This study seeks to bridge these gaps by systematically evaluating the relationship between China's smart-city pilot policy and urban-innovation-quality. By empirically exploring heterogeneous responses across China's diverse regions and city sizes, as well as potential spatial spillover effects, we offer a comprehensive account of the conditions under which smart-city strategies can effectively foster substantive urban innovation. These insights advance the theoretical understanding of smart-city policy effectiveness and provide valuable guidance for optimising future urban-innovation strategies.

Hypotheses, models and variables

Theoretical framework and hypotheses. As illustrated in Fig. 1, the theoretical model examines both the direct impact of smart city construction on urban innovation levels and the mechanisms through which it operates. The model, which is based on the TOE framework, details how smart city strategies influence urban innovation through the technological, organizational, and environmental dimensions. The technological dimension includes information technology and transportation infrastructure, the organizational dimension encompasses digital government and transportation infrastructure, and the environmental dimension covers the natural and social environments. These elements, when enriched through a smart city strategy, can enhance urban innovation capabilities and determine the policy's effectiveness. This study also considers the spatial spillover effects of smart city construction, addressing the inter-regional innovation impacts arising from policy implementation.

Based on this theoretical model, four research hypotheses are proposed.

Smart cities foster urban innovation and sustainable development by leveraging ICT to provide technical support and facilitate enhanced intelligent city management and services (Agboola et al., 2023; Neirotti et al., 2014). This development accelerates knowledge-sharing and the application of innovative technologies through open data platforms and innovation incubators (Komninos, 2019). The abundant resources, technical support, and conducive environment created by smart city initiatives attract innovative enterprises and high-tech talent, thereby establishing a robust innovation ecosystem (Caragliu et al., 2011; Zeng et al., 2023). Furthermore, smart city development not only upgrades infrastructure and public services but also optimizes resource allocation, stimulates innovation, and promotes entrepreneurship, creating favorable conditions for

innovative and entrepreneurial activities (Maulan et al., 2023; Mitra et al., 2022). Based on these observations, this study proposes the following hypothesis:

Hypothesis 1: Smart city construction positively influences urban innovation levels.

China's vast geography results in significant regional variations in economic development, technological infrastructure, population density, and socio-cultural aspects. This regional heterogeneity leads to differential impacts of smart city construction on urban innovation. For instance, eastern coastal cities, which are characterized by higher economic development and greater openness, are more likely to attract talent and technology, potentially giving them an advantage in promoting urban innovation compared to less developed regions. Additionally, larger cities with greater talent pools and resources, as well as higher population densities, face higher demands and therefore have stronger incentives to innovate, further driving urban innovation (Balland et al., 2020; Shukai et al., 2021). Based on this analysis, the following hypothesis is proposed:

Hypothesis 2: The impact of smart city construction on urban innovation varies across different regions.

The multidimensional characteristics of smart city construction imply a complex and multifaceted impact on urban innovation. From a technological perspective, smart city construction accelerates the adoption of next-generation information technologies such as big data and 5 G, enhancing urban informatization, providing enterprises with abundant innovation resources and technological platforms, and fostering data-driven decision-making and innovation (Bibri & Krogstie, 2020; Gohar & Nencioni, 2021). In terms of organizational collaboration, smart city initiatives optimize human capital structures, attract and concentrate highly skilled talent, and enhance collaboration and communication between firms and government, thus providing robust organizational support for innovation projects (Lim et al., 2023; Pereira et al., 2017). Additionally, smart city initiatives improve urban ecological environments, increase livability, and create favorable conditions for civic citizen well-being and innovation activities (Addas, 2023). Based on these insights, the following hypothesis is proposed:

Hypothesis 3: Smart city initiatives influence urban innovation through technological, organizational, and environmental mediators.

The successful implementation of smart city pilots creates regional centers of knowledge and technology. These pilots often spearhead infrastructure development and high-tech industry growth in the region, endowing neighboring cities with enhanced economic development conditions and innovative environments (Zeng et al., 2023). Moreover, a smart city's construction and policy implementation efforts foster talent concentration and increase innovation activities, which subsequently diffuse to neighboring regions through various networks and partnerships. For instance, cross-city project collaborations enable technological resources and policies to spill over administrative boundaries, promoting co-development among multiple cities in the region through strategic alliances or innovation networks, thereby enhancing the region's overall innovation capacity (Rădulescu et al., 2020). Based on this analysis, the following hypothesis is proposed:

Hypothesis 4: Smart city pilot policies have spatial spillover effects on urban innovation.

Models

Baseline model. The multi-period difference-in-differences (DID) method is a robust econometric tool for assessing the impact of policy changes or events on specific outcomes. We adopt a staggered multi-period DID as the baseline identification strategy, because smart city pilots were introduced in several batches and the design allows us to trace within-city changes in innovation and compare them with those in non-pilot cities over time. Compared with dynamic panel approaches, this framework does not require strong assumptions about the exact lag structure of innovation responses and focuses instead on differences in pre- and post-treatment trends between treated and untreated cities. To mitigate concerns about heterogeneous treatment timing, we use an event-study specification to examine pre-trends and dynamic effects. Spatial dependence is then examined in a subsequent step using a spatial Durbin model. This modelling strategy is consistent with recent quasi-natural experiment studies that evaluate China's digital economy and smart city policies using staggered DID designs (Q. Guo & Zhong, 2022; Sun et al., 2023a), Sun et al., (2023b)). Following the methodology of Callaway and Sant'Anna (2021), this study employs interaction terms to identify policy effects, constructing the multi-period DID model as follows:

$$\ln inn_{it} = \alpha_0 + \alpha_1 DID_{it} + \sum \gamma_j CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (1)$$

Where $\ln inn_{it}$ is the logarithmic form of the explanatory variable of urban innovation level. The core explanatory variable DID_{it} is a dummy variable indicating whether city i implemented a smart city pilot policy in year t . CV_{jit} represents control variables affecting the city's innovation level in year t . Crucially, to address endogeneity concerns stemming from omitted variable bias, the model incorporates high-dimensional fixed effects: μ_{city} absorbs time-invariant unobserved heterogeneity (e.g., geographical location and cultural endowment), v_t captures common macro-economic shocks, and λ_{it} is the random error term.

Mediation effects model. This study constructs a mediation model to investigate the mechanisms through which smart city pilots influence urban innovation levels. Initially, a series of mediating variables are treated as dependent variables, with smart city pilot as the primary explanatory variable in a regression analysis to test its impact on these mediating factors. If the estimated coefficient of smart city pilot is significant, it indicates that the smart city pilot variable affects the mediating variables. Next, both the mediating variables and the smart city pilot variable are included

in the model as explanatory variables, with the urban innovation index as the dependent variable. If the coefficients of both the pilot variable and the mediating variables are significant, this suggests that the mediating variables play a partial mediating role in the effect of a smart city pilot program on urban innovation. If the coefficient of the pilot variable is not significant but the coefficient of the mediating variable is, this indicates that the mediating variable has a full mediating effect. The mediation effect model is specified as follows:

$$M_{it} = \beta_0 + \beta_1 DID_{it} + \sum \delta CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (2)$$

$$\ln inn_{it} = \beta_2 + \beta_3 DID_{it} + \beta_4 M_{it} + \sum \gamma_j CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (3)$$

M represents the mediating variable, and the other variables are defined as previously mentioned.

Spatial econometric model. Cities do not exist in isolation, and they interact with and depend on each other economically, socially, and technologically. Ignoring spatial interactions between regions when examining the effects of smart city pilot policies can thus lead to inaccurate and unreliable conclusions. Therefore, this paper tests for spatial spillover effects to determine whether the impacts of smart city pilots on urban innovation extend to neighboring regions through various channels. First, the Lagrange multiplier test, Hausman test, and likelihood ratio test are conducted. Based on the results and following the study by Wang L. and Shao (2024), a difference-in-differences spatial Durbin model is introduced as follows:

$$W_{ij} = \begin{cases} \frac{1}{d_{ij}^2}, & i \neq j \\ 0, & i = j \end{cases} \quad (4)$$

$$Y_{it} = \theta_0 + \theta_1 DID_{it} + \theta_2 W * INN_{it} + \theta_3 W_{ij} DID_{it} + \sum_{j=1}^n \gamma_j W_{ij} CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (5)$$

Where W is the spatial weight matrix. Due to data collection challenges and the exclusion of non-compliant cities, the selected sample cities are not completely contiguous. Therefore, an inverse geographic distance spatial matrix is constructed using the inverse square of the geographic distance between cities, as shown in equation (4). $W * INN_{it}$, $W_{ij} DID_{it}$ and $W_{ij} CV_{it}$ denote the spatial lag terms of the dependent, independent, and control variables, respectively. Y_{it} represents the dependent variable; θ_0 is the constant term; θ_1 represents the regression coefficients of the independent variables; θ_2 , θ_3 and γ_j denote the spatial lagged regression coefficients of the dependent, independent, and control variables, respectively. The remaining symbols retain their meanings as defined in equation (1).

Variables

Dependent variable. Considering that previous studies have typically relied on indicators such as financial investment in science and technology or the number of patents granted, which may not fully capture the breadth of urban innovation, this study adopts a more comprehensive measure, the Urban Innovation Index, to assess a city's innovation capacity. Traditional metrics such as patent counts are informative, but they generally ignore heterogeneity in patent quality, and financial indicators capture innovation inputs rather than realized outputs. Accordingly, we use the Urban Innovation Index developed by the Fudan Institute of Industrial Development, which is constructed from invention patent data provided by the China National Intellectual Property Administration and a structural patent-renewal model that

Table 1 Indicators of the intermediary variables.

TOE Framework	Primary Indicator	Secondary Indicator	Original Indicator Data
Technical Conditions	Information and Communication Technology	Internet Penetration	Number of International Internet Users per 100 People
		Communication Penetration	Mobile Phone Penetration Rate
		Information Infrastructure	Long-Distance Optical Cable Density
	Transportation Infrastructure	Internet Infrastructure	Per Capita Broadband Internet Access Ports
		Road Construction	Per Capita Road Area
		Road Coverage and Connectivity	Highway Mileage
Organizational Conditions	Digital Government	Rail Transit	Number of High-Speed Rail Stations Opened
		Public Transport Benefit	Number of Public Transport Vehicles per 10,000 People
		Digital Intelligence Emphasis	Frequency of Government Work Report
		Government Transparency	Government Fiscal Transparency
	Human Capital	Scientific Support	Government Science Expenditure
		Government Performance	Government Website Performance Evaluation Index
		High-level Talent Efficacy	Per capita gross output value of employees in high-tech zones
		Citizen Income Level	Per Capita Disposable Income
		Education Level	Number of Students in Higher Education per 1,000 People
		Scientific and Technological Talent Resources	Proportion of Employees in Information Transmission, Computer, and Software Industries
Environmental Conditions	Urban Natural Environment	Urban Air Quality	Annual Average PM2.5 Concentration
		Urban Environmental Attractiveness	Per Capita Park Green Area
		Urban Water Source Status	Centralized Treatment Rate of Sewage Treatment Plants
	Social Service Environment	Environmental Governance	Environmental Regulation Strength
		Basic Education	Student-teacher ratio in regular secondary schools
		Medical and Health Care	Number of hospital beds per 100 people
		Cultural Services	Number of public library books per 10,000 people
		Social Security	Proportion of participants in endowment insurance to the year-end total population

assigns age-specific value weights to granted patents (Kou & Liu, 2017). In each year, the estimated values of all valid invention patents are aggregated at the city level to obtain the Urban Innovation Index, which provides a more comprehensive and quality-adjusted measure of urban innovation.

Independent variables. The primary explanatory variable in this study is smart city pilot policy, defined as a dummy variable. For cities designated as pilots, the variable is set to 1 for the year the policy is implemented and subsequent years; for non-pilot cities or years before policy implementation, it is set to 0.

Control variables. To mitigate omitted variable bias, we adopt the Regional Innovative Capacity framework (Furman et al., 2002), which attributes regional innovation performance to three dimensions: common innovation infrastructure, the cluster-specific environment, and the quality of linkages. First, regarding the common innovation infrastructure, we control for Economic Development Level and City Classification. Consistent with the RIC framework, a strong economic foundation (measured by log per capita GDP) provides the resources necessary to buffer R&D risks (Furman et al., 2002). In China's hierarchical administrative system, high-ranking cities typically enjoy expanded powers relative to ordinary prefecture-level cities; we therefore include a dummy variable to capture these institutional advantages (Lu & Tsai, 2021). Second, to capture the cluster-specific environment, we control for Urban Agglomeration, measured by log population density. Consistent with agglomeration theory, higher spatial density reduces

knowledge transmission costs and fosters the cross-fertilization of ideas by facilitating face-to-face interactions and labor matching (Matsiuk et al., 2024). Third, reflecting the quality of linkages to the global frontier, we control for the Level of Openness, measured by the ratio of FDI to GDP. FDI acts as a conduit for international technology transfer, promoting domestic innovation through demonstration effects and labor mobility (Y. Wang et al., 2025).

Mediator variables. This study utilizes the TOE theoretical framework, as depicted in Fig. 1, encompassing technology, organization, and environment components. This framework is designed to explore the interactions and causal relationships among technological innovations within a specific context. Technological advancements can foster new forms of urban innovation, organizational capabilities determine the efficiency and direction of innovation activities, and environmental factors shape the opportunities and constraints for urban innovation. Hence, this theoretical framework is well-suited to the study's theme. Drawing on established literature, we align the TOE components with the six-dimension smart-city construct (Table 1): technological conditions (transport and ICT infrastructure), organizational conditions (human capital and digital government), and environmental conditions (social and natural resources) (Caragliu et al., 2011; Marchesani & Ceci, 2025; Shao & Min, 2025).

Data and descriptive statistics. Considering the timeline of policy implementation and data availability, this study selects 283

Table 2 Descriptive statistics for key variables.							
Variable	Variable name	Variable symbol	N	mean	sd	min	max
Explained Variables	Urban Innovation Index	CII	4528	0.650	2.046	−5.273	7.975
Explanatory Variables	Smart City Pilot	DID	4528	0.207	0.405	0	1
Control Variables	City Clustering	PD	4521	5.727	0.941	0.446	7.882
	City Ranking	CC	4528	0.106	0.308	0	1
	Economic Development	ED	4516	10.49	0.724	4.595	13.06
	Openness to the Outside World	FO	4490	1.706	1.498	0.161	19.783

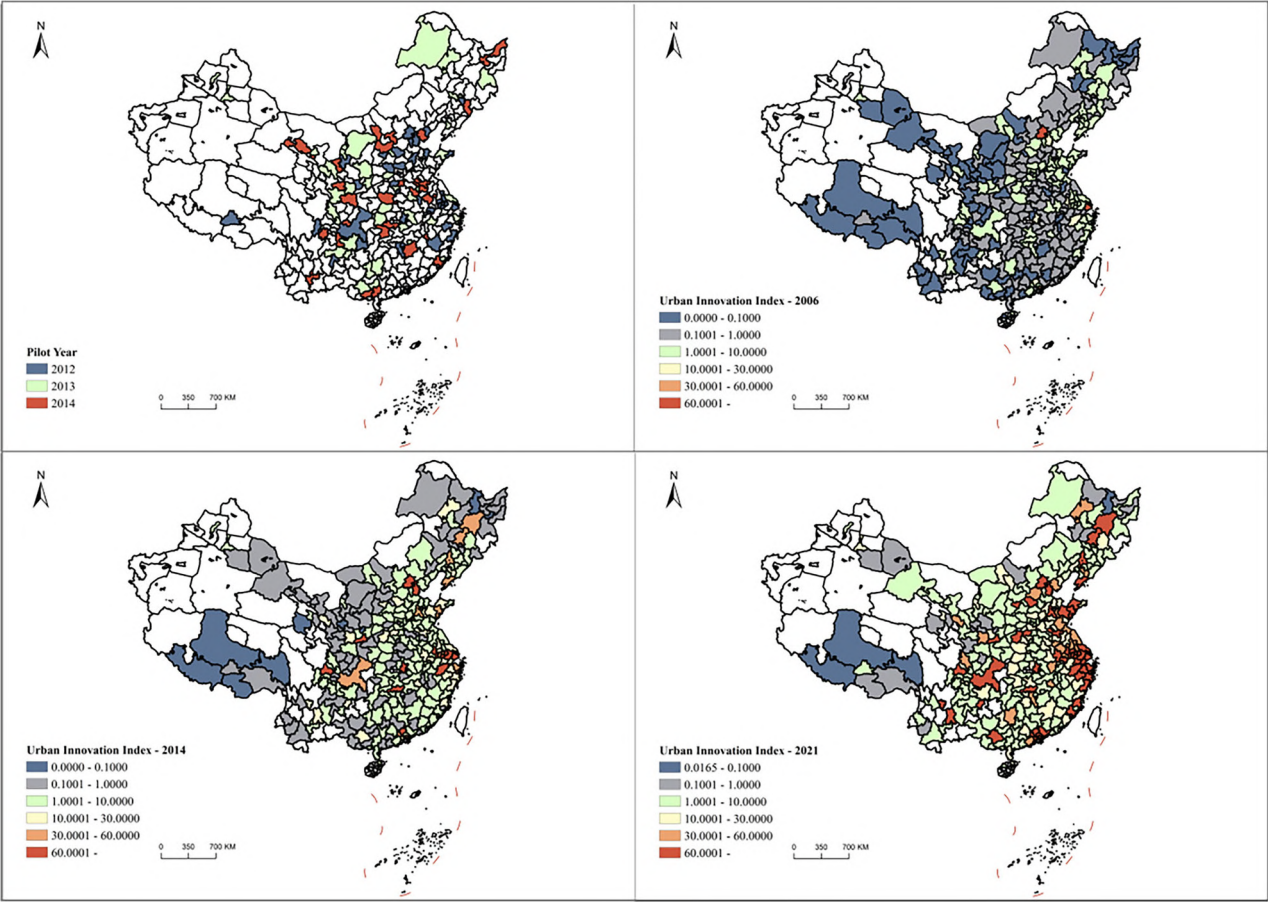


Fig. 2 China’s smart city pilots and urban innovation index in 2006, 2014, 2021.

prefecture-level cities in China from 2006 to 2021 as the research sample. The variables used in the model and their descriptive statistics are presented in Table 2. Data sources include the “Report on the Innovative Power of Chinese Cities and Industries” by Fudan University, the three batches of the “Smart City Pilot List” from the Ministry of Housing and Urban-Rural Development, the China Land Market Network, the “Statistical Yearbook of Land and Resources,” and Peking University’s Digital Financial Inclusion Index. Data for the control and mediating variables are sourced from various authoritative statistical yearbooks and databases.

As shown in Fig. 2, smart city pilots are primarily concentrated in the central and eastern regions, especially in eastern China. From 2006 to 2021, the urban innovation index has shown an overall upward trend, with the eastern region exhibiting higher innovation capacity compared to the central, western, and northeastern regions. This “high in the east, lower in the west” distribution pattern of innovation levels is similar to the distribution of smart city pilots, indicating that smart city

projects may be having a positive effect on enhancing urban innovation capacity.

Empirical results

Baseline regression. Prior to the baseline estimation, we verified the parallel trend assumption to ensure the validity of the multi-period DID strategy (Figure A1). The event study results indicate that the treatment and control groups exhibited similar pre-treatment trends. Subsequently, the impact of smart city pilot construction on urban innovation levels is analyzed using equation (1), with the benchmark regression results shown in Table 3. The first column presents the impact of smart city pilot policies on urban innovation without the control variables, while column (2) shows the results with the control variables. Both models account for city and time fixed effects. The results indicate that the estimated coefficient for the explanatory variable is 0.156 without the control variables and 0.121 with the control variables, both significant at the 1% level. These results suggest that smart

ARTICLE

city pilot construction is positively associated with urban innovation capacity, supporting Hypothesis 1. The models' R^2 values of 0.972 and 0.973 indicate a high degree of goodness of fit. This may be attributed to the fact that intelligent manufacturing and smart governance—pivotal components of smart cities—provide policy support for intelligent transformation and industrial upgrading, reduce information asymmetry, and optimize resource allocation, thereby fostering collaborative innovation in cities (Branco et al., 2023; Jin & Chen, 2024).

In addition, placebo tests, alternative sample periods and innovation measures, outlier trimming, and controls for other major national urban policies (reported in Appendix Table A2-4) yield similar results, supporting the robustness of these baseline estimates.

The analysis of control variables shows that higher population density, capital city status and stronger economic development

are associated with higher urban innovation. However, foreign investment – a proxy for openness – negatively impacts urban innovation. This might be because foreign direct investment is often concentrated in highly profitable sectors rather than innovative ones, leading to uneven regional innovation stimulation. Investments prioritizing efficiency and short-term gains over long-term innovation development exacerbate this issue. Moreover, foreign investment can lead firms to rely excessively on external support, with high-tech companies potentially prioritizing the acquisition of advanced foreign technologies over independent R&D (Li, 2011). Additionally, when foreign capital repatriates more profits to its home country and reduces investment in the host country, it can diminish the host country's R&D and innovation capacity (Wang Y. et al., 2007).

Heterogeneity analysis. China is a vast country with prominent issues of unbalanced and insufficient regional development, resulting in significant disparities between cities (Hu & Zheng, 2021; Zhuang & Ye, 2020). Therefore, this paper conducts a heterogeneity test along two dimensions: location and size, as detailed in Table 4. First, there are notable differences between eastern China and central and western China regarding economic development levels, infrastructure development, policy support strength, and openness (Yu & Shang, 2018). Based on the geographical division standard published by the National Bureau of Statistics, the sample cities are divided into eastern and central-western regions. As shown result, smart city construction is statistically significantly associated with higher urban innovation in the central and western regions, but no statistically significant effect was observed in the eastern region. This suggests that in the eastern region, where innovation ecosystems are already mature, the marginal impact of the pilot policy may be limited (Hu & Zheng, 2021; Zhuang & Ye, 2020). In contrast, the less developed central and western regions appear to benefit more from smart city projects, as new technologies and ideas introduced during construction boost urban innovation.

Second, considering the heterogeneity of city sizes (Zhang et al., 2022), the sample cities are categorised into large cities and small to medium-sized cities according to the “Notice on

Table 3 Benchmark regression results.

VARIABLES	(1) CII	(2) CII
<i>DID</i>	0.156*** (0.022)	0.121*** (0.022)
<i>PD</i>		0.110*** (0.031)
<i>CC</i>		0.064** (0.026)
<i>ED</i>		0.254*** (0.030)
<i>FO</i>		−0.018*** (0.005)
<i>Constant</i>	0.625*** (0.007)	−2.635*** (0.353)
<i>Observations</i>	4528	4512
<i>R-squared</i>	0.972	0.973
<i>Time fixed effects</i>	YES	YES
<i>Individual fixed effect</i>	YES	YES

Note: **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 4 Heterogeneity test results.

VARIABLES	City Location		City Size		
	Midwest Region CII	Eastern Region CII	Small and Medium-sized Cities CII	Large Cities CII	Add to Interaction CII
<i>DID</i>	0.160*** (0.036)	0.063 (0.042)	0.189*** (0.039)	0.096* (0.051)	0.076** (0.036)
<i>MCDID</i>					0.111*** (0.036)
<i>PD</i>	0.078** (0.033)	−0.197 (0.161)	0.042 (0.051)	0.140*** (0.036)	0.108*** (0.031)
<i>CC</i>	0.094* (0.049)	0.066* (0.035)	0.089** (0.038)	0.057* (0.031)	0.062** (0.026)
<i>ED</i>	0.328*** (0.035)	0.221*** (0.055)	0.308*** (0.039)	0.225*** (0.049)	0.257*** (0.030)
<i>FO</i>	0.026*** (0.006)	−0.044*** (0.007)	0.010 (0.008)	−0.034*** (0.006)	−0.017*** (0.005)
<i>Constant</i>	−3.709*** (0.390)	0.650 (1.248)	−3.612*** (0.456)	−1.102* (0.577)	−2.658*** (0.353)
<i>Observations</i>	3117	1375	2896	1596	4512
<i>R-squared</i>	0.965	0.983	0.951	0.975	0.973
<i>Time fixed effects</i>	YES	YES	YES	YES	YES
<i>Individual fixed effect</i>	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Adjusting the Criteria for the Classification of City Scale.” The regression analysis suggests that smart city construction is more strongly associated with innovation in small and medium-sized cities than in large cities. Specifically, the coefficient of the interaction term between population size and the pilot variable is significantly positive. This phenomenon may be due to the Chinese government’s recent emphasis on coordinated urban economic development, promoting the growth of small to medium-sized cities through policy and financial support, allowing these cities to benefit more from smart city construction (Yu & Shang, 2018; Liu et al., 2022). Additionally, smart city construction in small to medium-sized cities can effectively attract talent, promote knowledge exchange and technology transfer, and accelerate the development and improvement of the urban innovation system. Conversely, in large cities, the high population density results in a smaller marginal effect of smart city construction on talent attraction.

In summary, the heterogeneity test reveals differentiated associations between smart city construction and innovation across regions and city sizes, with relatively stronger effects in central and western regions and in small and medium-sized cities. Beyond these two dimensions, we further examine heterogeneity by administrative hierarchy, fiscal capacity and industrial structure; the corresponding estimates are reported in Appendix Table A1.

Mediation test. Based on the theoretical framework and the validated constructs above, this section analyzes the internal mechanisms by which smart city pilot construction influences urban innovation capacity. Following the methodology of Caragliu and Del Bo (2019), the four secondary indicators in each category in Table 1 are subjected to principal component analysis to extract composite indicators. Each composite indicator is then treated as a mediating variable for the mediation effect test. The results of the heterogeneity analysis reveal that the impact of smart city pilots on urban innovation varies by geographic location, but the differences due to city size are less pronounced. Therefore, this mechanism analysis focuses on the geographic location differences. The mediation mechanism analysis is based on the TOE framework outlined in Section “Theoretical framework and hypotheses”.

Crucially, to address potential concerns that mediating variables within the TOE framework (e.g., ICT infrastructure and digital government) might represent redundant underlying processes, we conducted a collinearity diagnostic prior to the regression analysis. We generated a correlation matrix heatmap (Appendix Fig. A2) to visually inspect the relationships between mediators. Furthermore, the Variance Inflation Factor (VIF) results (see Appendix Table A5) show a mean VIF of 2.18 and a maximum of 4.06, well below the critical threshold of 10. These statistics, combined with the theoretical distinction between technical connectivity (ICT, transportation) and organizational capabilities (digital government, human capital), confirm that the mediators capture distinct mechanisms rather than alternative measures of the same construct.

The results of the heterogeneity analysis reveal that the impact of smart city pilots on urban innovation varies by geographic location, but the differences due to city size are less pronounced. Therefore, this mechanism analysis focuses on the geographic location differences. The mediation mechanism analysis is based on the TOE framework outlined in Section “Theoretical framework and hypotheses”.

Technological effect. The technological effect is divided into two aspects: ICT and transportation infrastructure. The results in

Table 5 Results of the mediation test for technological conditions.

VARIABLES	Information Technology			Eastern Information Technology			Central and Western Information Technology			Transportation Infrastructure			Eastern Transportation Infrastructure			Central and Western Transportation Infrastructure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)						
	IT	CII	IT	CII	IT	CII	TI	CII	TI	CII	TI	CII						
DID	0.292*** (0.111)	0.120*** (0.022)	0.104 (0.258)	0.064* (0.034)	0.308** (0.128)	0.161*** (0.026)	0.937*** (0.213)	0.108*** (0.022)	0.224 (0.363)	0.063* (0.033)	1.150*** (0.259)	0.144*** (0.026)						
Mediating variables		0.053*** (0.023)		0.006* (0.003)		0.045** (0.021)		0.010*** (0.002)		0.010*** (0.003)		0.011*** (0.002)						
Control variable	control	control	control	control	control	control	control	control	control	control	control	control						
Constant	15.838*** (1.930)	-2.681*** (0.356)	82.939*** (9.646)	0.110 (1.282)	13.575*** (1.926)	-3.685*** (0.393)	7.973** (3.501)	-2.640*** (0.354)	6.218*** (2.426)	-0.040 (1.247)	8.442** (3.911)	-3.738*** (0.391)						
Observations	4506	4492	1388	1375	3118	3117	4475	4474	1373	1373	3102	3101						
R-squared	0.884	0.973	0.784	0.983	0.897	0.965	0.892	0.973	0.825	0.983	0.906	0.966						
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES						
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES						

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are standard errors.

Table 6 Results of the mediation test for organizational conditions.												
VARIABLES	Digital Government		Eastern Digital Government		Central and Western Digital Government		Human Capital		Eastern Human Capital		Central and Western Human Capital	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	DG	CII	DG	CII	DG	CII	HC	CII	HC	CII	HC	CII
DID	0.413*** (0.078)	0.117*** (0.022)	0.038 (0.139)	0.064* (0.034)	0.562*** (0.095)	0.158*** (0.026)	0.069* (0.040)	0.114** (0.052)	0.254*** (0.082)	0.045** (0.021)	0.026 (0.031)	0.162** (0.078)
Mediating variables		0.009** (0.004)		0.003 (0.007)		0.004** (0.002)		0.105** (0.042)		0.078** (0.036)		0.005* (0.003)
Control variable	control	control	control	control	control	control	control	control	control	control	control	control
Constant	1.421 (1.285)	−2.648*** (0.353)	1.055 (5.208)	0.645 (1.248)	0.116 (1.426)	−3.710*** (0.390)	4.573*** (0.563)	−3.210*** (1.086)	3.962** (1.850)	0.335* (0.195)	3.836*** (0.515)	−3.818*** (1.226)
Observations	4506	4492	1388	1375	3118	3117	4506	4492	1388	1375	3118	3117
R-squared	0.873	0.973	0.720	0.983	0.876	0.965	0.924	0.971	0.926	0.982	0.923	0.962
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.												

Table 5 indicate that, overall, smart city construction significantly promotes the development of ICT and transportation infrastructure at the 1% level. The enhancement of these technological aspects, in turn, is associated with higher urban innovation levels. Sub-regional regression results show that in the eastern region, technological conditions do not mediate the effect of smart cities on urban innovation. However, in the central and western regions, smart city construction can enhance urban innovation by advancing technological development.

The differences between the aforementioned regions may be attributed to the more mature technological development in the eastern region, leading to smaller marginal benefits from a smart city pilot policy. Conversely, the central and western regions, with relatively lagging technological infrastructure, can benefit significantly from the high-tech advancements and management models introduced by a smart city pilot policy. Furthermore, smart city construction significantly improves transportation accessibility in the central and western regions through the implementation and promotion of smart transportation. A robust transportation system not only enhances urban connectivity but also facilitates the efficient flow of information, talent, and resources.

Organizational effect. The mediation results for organisational conditions reported in Table 6 indicate that digital government and human capital together form an important channel linking the smart city strategy to urban innovation. In the full sample, improvements in digital government significantly transmit policy effects to higher innovation, while the human capital mediator is also positive but becomes weaker and sometimes loses significance once additional controls are introduced. In the eastern region, the estimated effect of human capital is larger and more robust than that of digital government, whereas the organisational channel is clearly weaker in central and western cities. This pattern suggests that eastern cities, which host dense clusters of high-tech firms and well-developed labour markets, rely more strongly on market-based mechanisms to translate smart city investments into better matching between skilled workers and innovative enterprises. Smart city projects that enhance data availability, transparency and online services reinforce these market processes and increase the attractiveness of leading cities

for mobile talent and innovative companies(Woods et al., 2024), while at the same time raising questions about the social consequences of entrepreneurial, technology-led urban development (Khurana & Dutta, 2024).

By contrast, in central and western cities the mediating effect of digital government is statistically significant, while the human capital channel is small and often not significant. This suggests a threshold effect: in regions where high-tech industries and skilled labour pools exist but are less concentrated and innovation networks are thinner, smart city initiatives first tend to strengthen basic administrative capacity, information transparency and online public services. The productivity gains associated with more efficient allocation of human resources are likely to materialise more gradually as industrial clusters deepen and local innovation systems mature. In these contexts, the main organisational contribution of the smart city strategy is to improve government coordination and reduce transaction costs, which is consistent with evidence that more capable smart city governance and open, multi-actor innovation ecosystems create more favourable conditions for inclusive and innovation-oriented urban development (Kummitha, 2018b; Mora et al., 2025).

Environmental effect. Regarding the environmental effects, this study focuses on natural conditions and social services, with the test results shown in Table 7. The mediating role of environmental conditions is comparatively modest in the baseline results, and the overall coefficient is not robustly significant. This outcome can be understood in light of the fiscal and temporal characteristics of investments in environmental quality and public service provision: smart city projects often prioritise digital platforms and visible information infrastructure, whereas improvements in environmental indicators and basic services require long-term budget commitments and cross-sector coordination. Recent work on smart cities and inequality finds that such initiatives can alleviate disparities, but that their distributional effects are heterogeneous and depend on whether policies explicitly address social equity alongside technological upgrading (Caragliu & Del Bo, 2022; Z. Yang et al., 2024).

At the same time, the regional estimates indicate that environmental conditions play a more pronounced mediating role in central and western cities, where the coefficient for the

Table 7 Results of the mediation test for environmental conditions.

VARIABLES	Natural Conditions		Eastern Natural Conditions		Central and Western Natural Conditions		Social Services		Eastern Social Services		Central and Western Social Services	
	(1) NC	(2) CII	(3) NC	(4) CII	(5) NC	(6) CII	(7) SS	(8) CII	(9) SS	(10) CII	(11) SS	(12) CII
DID	0.690** (0.340)	0.119*** (0.022)	2.971*** (0.779)	0.074** (0.034)	0.017 (0.356)	0.160*** (0.026)	0.047 (0.040)	0.106*** (0.035)	−0.037 (0.079)	0.077** (0.036)	0.079* (0.046)	0.138*** (0.046)
Mediating variables		0.003*** (0.001)		0.003** (0.001)		0.003*** (0.001)		0.015** (0.007)		0.041* (0.023)		−0.034* (0.020)
Control variable	control	control	control	control	control	control	control	control	control	control	control	control
Constant	14.265** (5.583)	−2.677*** (0.353)	70.108** (29.103)	0.876 (1.248)	−2.218 (5.356)	−3.701*** (0.389)	6.840*** (0.848)	−2.854** (1.176)	18.079*** (4.218)	0.512* (0.299)	6.137*** (0.894)	−3.091** (1.306)
Observations	4506	4492	1388	1375	3118	3117	4485	4484	1371	1371	3114	3113
R-squared	0.880	0.973	0.933	0.983	0.642	0.965	0.903	0.979	0.946	0.986	0.824	0.968
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 8 Moran's Index.

Research Item	Moran's I	z-value	p-value	Research Item	Moran's I	z-value	p-value
Y2006	0.045	4.8	0.000	Y2014	0.076	6.4	0.000
Y2007	0.047	5.2	0.000	Y2015	0.08	6.2	0.000
Y2008	0.051	5.6	0.000	Y2016	0.086	6.6	0.000
Y2009	0.061	6.0	0.000	Y2017	0.086	6.8	0.000
Y2010	0.063	6.4	0.000	Y2018	0.09	7.0	0.000
Y2011	0.064	6.2	0.000	Y2019	0.095	7.2	0.000
Y2012	0.068	6.6	0.000	Y2020	0.094	7.0	0.000
Y2013	0.069	6.8	0.000	Y2021	0.099	6.6	0.000

social service environment is positive and statistically significant. This pattern is consistent with a compensatory effect: in regions with a historically weaker provision of public services, the smart city pilot is more likely to be accompanied by efforts to expand basic services and improve the local living environment, which helps to retain and gradually attract skilled workers. Research on urban amenities shows that cities investing in the quality of everyday services are better able to attract and keep high-skilled residents, thereby supporting local innovation capacity (Glaeser et al., 2001). At the same time, the negative coefficient observed in some specifications suggests that in fiscally constrained cities, short-term increases in spending on public services may crowd out resources that could otherwise be devoted to research and development, underlining the importance of balancing investments in digital infrastructure, environmental quality and social services within broader urban innovation strategies.

Taken together, the three mediating mechanisms suggest that China's smart city strategy currently operates in a predominantly technology- and efficiency-oriented mode. Technological conditions appear to act as the most stable and sizeable transmission channel from policy pilots to urban innovation across all regions. Organisational conditions, especially human capital and digital government, contribute more strongly in eastern cities, where market institutions and high-tech ecosystems are already relatively mature, while their mediating role is weaker in central and western regions. Environmental and social service conditions remain the least developed mechanism, with limited and heterogeneous effects that are stronger in less-developed regions and in some cases associated with short-

term resource crowding-out. These results are consistent with comparative evidence that smart city policies generate stronger innovation gains when they are embedded in supportive institutional settings, coupled with long-term investments in human and social capital and framed by robust governance arrangements (Caragliu & Del Bo, 2019b; Kummitha & Crutzen, 2017; Mora et al., 2023b).

Spatial spillover effect test

Spatial auto-correlation. Using spatial econometric models for empirical analysis requires an initial assessment of the spatial correlation of the dependent variable. Moran's I statistic is employed to test the spatial auto-correlation of the innovation indices for the 283 Chinese cities from 2006 to 2021. The results in Table 8 reveal significant positive spatial correlation between cities' innovation levels, with a clear geographic clustering effect, indicating substantial spatial spillover effects. The global Moran's I increases annually from 2006 to 2021, suggesting increased spatial dependence. This may be due to cities becoming more interconnected through improved transportation networks and digital communication infrastructure, facilitating the spillover of innovation activities to neighboring cities. Additionally, regional development strategies have strengthened interregional linkages by creating channels for knowledge and resource sharing (Lopes & Franco, 2019; Qin, 2022).

This study follows J. P. Elhorst's approach for model selection and validation (Elhorst, 2014). The spatial Lagrange multiplier (LM) test results show that the p-values are significant at the 1% level, indicating the presence of spatial error and spatial lag

Table 9 Spatial Panel Durbin Model Spillover Effects and Decomposition.

VARIABLES	(1) Main	(2) Wx	(3) Direct	(4) Indirect	(5) Total
DID	0.353*** (0.025)	0.464* (0.272)	0.438*** (0.038)	0.256*** (0.095)	0.694*** (0.102)
PD	0.290*** (0.027)	0.215 (0.448)	0.322*** (0.036)	0.280*** (0.105)	0.602*** (0.111)
CC	0.021* (0.012)	-0.040 (0.029)	0.025* (0.014)	0.032 (0.025)	0.057** (0.029)
ED	0.305*** (0.019)	0.249* (0.155)	0.354*** (0.023)	0.410*** (0.120)	0.764*** (0.122)
FO	-0.095*** (0.029)	-0.243*** (0.050)	-0.107*** (0.030)	-0.285** (0.117)	-0.392*** (0.121)
rho	0.387*** (0.052)				
sigma2_e	0.185*** (0.004)				
Observations	4528	4528	4528	4528	4528
R-squared	0.572	0.572	0.572	0.572	0.572
Number of CITY	283	283	283	283	283

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are standard errors.

effects, leading to the initial selection of the spatial Durbin model, which incorporates both effects. Subsequent post hoc tests, including the Hausman test, confirm that the spatial regression model applies to fixed effects. Finally, the likelihood ratio (LR) test and Wald test results reject the null hypothesis of spatial Durbin model degradation at the 1% significance level, further validating the model's applicability.

Spatial durbin model estimation results. The spatial spillover effects are tested using the difference-in-differences spatial Durbin model, with the results presented in Table 9. Column 1 indicates that the spatial autoregressive coefficient for urban innovation, ρ , is significantly positive at the 1% level, demonstrating the positive spatial dependence of urban innovation. The results show that the smart city pilot policy is statistically significantly associated with higher innovation capacity in neighbouring cities, exhibiting notable spatial spillover effects and verifying Hypothesis 4. The decomposition of the total effect reveals that both direct and indirect effects are significantly positive, indicating that the implementation of the smart city pilot policy not only promotes innovation within the city but also exerts an influence on surrounding cities. Neighboring cities can learn from the pilot regions' experiences in high-tech applications, urban governance models, and talent development strategies to bolster their own innovation capacity and development potential. This cross-regional dissemination of knowledge and experience fosters overall technological advancement and economic growth, enhancing regional sustainable development and the construction of innovation ecosystems (Bevilacqua et al., 2021; Stöber et al., 2023).

Although smart city strategies have generated positive spillovers, the spatiotemporal evolution depicted in Fig. 2 reveals a persistent innovation divide. Inland regions adjacent to high-innovation coastal clusters and Northeast regions near the capital have not manifested the anticipated trickle-down effect, remaining at low innovation levels. In contrast, certain non-adjacent cities with robust economic foundations exhibit synchronized advancement. This visual evidence aligns with the robustness check based on the economic-geographic nested matrix (see Appendix Table A6), and suggests that spillover effects are conditioned by economic similarity. This implies that while geographical proximity serves as a transmission conduit,

absorptive capacity is the critical determinant of reception efficacy (Miguélez & Moreno, 2015a, Miguélez & Moreno, 2015b; Stettler et al., 2025). Given that similar stages of economic development typically correspond to compatible institutional environments and governance capacities (Rodríguez-Pose, 2013), the low-innovation traps observed in Fig. 2 can be attributed to institutional mismatches; specifically, less developed neighbors struggle to effectively interface with the smart systems of pilot cities due to a lack of compatible digital governance infrastructure or administrative efficiency. Where significant governance gradients exist, the siphon effect may overshadow the spillover effect, highlighting that optimizing regional institutional arrangements to reduce barriers to knowledge flow is as critical as investment in physical infrastructure (Mora et al., 2023a).

Conclusions and limitations

Conclusions and discussion. Drawing on a quasi-natural experiment of smart city pilots, this paper combines theoretical analysis with empirical research, utilizing panel data from 283 prefecture-level cities across China from 2006 to 2021. Empirical studies in Europe and other regions have often evaluated smart city policies through cross-sectional comparisons between cities with and without such initiatives, which provides static evidence on innovation outcomes but offers limited insight into dynamic policy effects (Barba-Sánchez et al., (2019a), Barba-Sánchez et al., (2019b)). By employing a multi-period difference-in-differences design, this study instead traces how urban innovation evolves before and after pilot designation and contrasts these trajectories with those of non-pilot cities. It further combines the Technology–Organization–Environment (TOE) framework with mediation analysis to examine the channels through which smart city pilots influence innovation and embeds this setting in a spatial Durbin model to identify spillover effects on neighbouring cities.

First, smart city pilot policies are positively associated with urban innovation, with notable regional heterogeneity. The baseline regressions show a robust and statistically significant positive impact of pilot designation on innovation, which remains stable under multiple robustness checks. However, the effects of these policies vary by region and city size. In the central and western regions, the estimated coefficients are larger and

statistically significant, echoing broader findings that digital and innovation-oriented policies can especially enhance performance in structurally weaker or less diversified cities (Sun et al., (2023a), Sun et al., (2023b)). In contrast, the estimated effects for eastern cities are smaller and statistically insignificant. This implies that where the local innovation base is strong and governance modes rely more heavily on competitive markets and private initiative, the additional marginal value of the national pilot programme may be limited. Regarding city size, smart city policies improve urban innovation across all types of cities, with particularly pronounced effects in small and medium-sized cities. Additional heterogeneity tests also indicate that cities with more diversified industrial structures respond more strongly to the pilot, which underlines the importance of both governance modes and the existing innovation base in translating smart city strategies into measurable innovation gains.

Second, the results show that smart city strategies operate through technological, organisational and environmental conditions, with their relative importance varying across regions. At the aggregate level, the mediation analysis indicates that improvements in technological conditions, organisational capacities and natural environmental indicators are all associated with higher innovation, whereas the mediating role of social service is weak. For eastern cities, the most important mediating channels are human capital and natural environmental quality, while the technological mediator is relatively modest, which differs from many empirical studies that emphasise the role of digital infrastructure and ICT adoption in driving innovation (Du et al., 2023). A plausible interpretation is that, in technologically advanced eastern cities where digital infrastructure is already dense, incremental infrastructure upgrades contribute less to innovation than policies that reshape skills, environmental quality and governance modes—an evolution that aligns with arguments that mature smart cities increasingly adopt human-centred and governance-oriented approaches to development (T. Ji et al., 2021).

In central and western regions, by contrast, technological upgrading and digital government constitute the main mediating channels, while the indirect effect of human capital is not significant and the coefficient on the social-service environment is negative. This pattern suggests that, at the current stage, smart city investments in these areas are still concentrated on basic digital infrastructure, administrative modernisation and core public services, and in some cases may temporarily divert fiscal and managerial resources away from longer-term innovation-oriented activities (Ji et al., 2024). The asymmetric mediating structure between eastern and non-eastern regions therefore, adds nuance to international debates on smart city heterogeneity, illustrating that similar policy labels can correspond to different combinations of governance modes and innovation-supporting investments.

Finally, the spatial econometric analysis indicates that smart city pilots are associated with positive spillover effects for neighbouring cities. When the spatial weights are redefined using an economic–geographic nested matrix, the spillover effect remains positive. Taking the geographic distribution of innovation into account, it is apparent that this effect is concentrated among cities that are both geographically close and economically similar, while some adjacent but economically weaker neighbours do not experience significant gains. These findings are consistent with research on knowledge flows and absorptive capacity, which emphasises that geographic proximity must be combined with a compatible economic environment and governance capacity for spillovers to materialise (Miguélez & Moreno, 2015a, Miguélez & Moreno, 2015b).

The empirical results also suggest several implications for the design of smart city policies. First, given the stronger average

effects in central and western regions and in small and medium-sized cities, extending smart city-related support to cities with a relatively weak innovation base but adequate governance capacity may provide a useful way to reduce regional gaps, provided that local conditions and fiscal constraints are carefully assessed. Second, the pronounced differences in mediating channels between regions highlight the value of flexible policy packages: in less-developed cities, strengthening basic digital infrastructure and digital government may remain a priority, whereas in more advanced cities, policies that emphasise human capital, collaborative innovation and environmental quality are likely to be more effective. Third, the coexistence of positive spillover effects and persistent innovation divides suggests that regional coordination mechanisms, such as cross-city data platforms or joint innovation projects, can help transform pilot cities into effective demonstration hubs for neighbouring jurisdictions that share a similar economic and governance profile.

Limitations and prospects. Despite the robust findings, this study is subject to several limitations that should be acknowledged.

First, regarding measurement, while the study utilizes robust datasets, reliance on patent-based indicators may not fully capture non-technological innovations—such as improvements in public services and governance—that are central to smart city objectives. This limitation in data coverage may affect the comprehensive assessment of “smart” outcomes and the generalization of the findings.

Second, the binary difference-in-differences design treats pilot status as a uniform intervention, whereas policy implementation differs across regions. Consequently, this empirical design cannot isolate the marginal impact of varying investment intensities or the specific composition of policy bundles, which broad groupings may conceal. Future studies could employ granular data to disentangle the dose-response relationship of policy intensity.

Third, the city-level analysis summarizes average local and spillover effects but cannot uncover the micro-foundations of innovation. The current design identifies macro-channels (TOE) but cannot reveal the detailed mechanisms through which individual firms—the primary agents of innovation—alter their behavioral strategies in response to policy signals.

Finally, these findings are conditioned by China’s state-led institutional context, and mechanisms may operate differently in market-driven models. Further research is warranted to comprehensively understand these micro-dynamics and cross-contextual differences. This understanding will aid in the formulation of integrated smart city strategies and coordinated urban policies, which are essential for fostering broader regional innovation and development.

Data availability

The data used in this study are available from the corresponding author upon reasonable request.

Received: 19 June 2025; Accepted: 30 January 2026;

Published online: 11 February 2026

References

- Addas A (2023) The concept of smart cities: A sustainability aspect for future urban development based on different cities. *Front Environ Sci*. <https://doi.org/10.3389/fenvs.2023.1241593>
- Agboola OP, Bashir FM, Dodo YA, Mohamed MAS, Alsadun ISR (2023) The influence of information and communication technology (ICT) on stakeholders’ involvement and smart urban sustainability. *Environ Adv* 13: 100431. <https://doi.org/10.1016/j.envadv.2023.100431>

- Angelidou M (2015) Smart cities: a conjuncture of four forces. *Cities* 47:95–106. <https://doi.org/10.1016/j.cities.2015.05.004>
- Balland P-A, Jara-Figueroa C, Petralia SG, Steijn MPA, Rigby DL, & Hidalgo CA (2020). Complex economic activities concentrate in large cities. *Nat Human Behav*. <https://doi.org/10.1038/s41562-019-0803-3>
- Barba-Sánchez V, Arias-Antúnez E, Orozco-Barbosa L (2019a) Smart cities as a source for entrepreneurial opportunities: evidence for Spain. *Technol Forecast Soc Change* 148: 119713. <https://doi.org/10.1016/j.techfore.2019.119713>
- Barba-Sánchez V, Arias-Antúnez E, Orozco-Barbosa L (2019b) Smart cities as a source for entrepreneurial opportunities: evidence for Spain. *Technol Forecast Soc Change* 148: 119713. <https://doi.org/10.1016/j.techfore.2019.119713>
- Benchis MP, Shahzad K, Dan S (2025) Comparative analysis of blockchain adoption in the public and private sectors. A technology-organization-environment (TOE) framework approach. *J Innov Knowl* 10(4):100746. <https://doi.org/10.1016/j.jik.2025.100746>
- Bevilacqua C, Calabrò F, & Della Spina L (Eds.). (2021). *New Metropolitan Perspectives: Knowledge Dynamics and Innovation-driven Policies Towards Urban and Regional Transition Volume 2* (Vol. 178). Springer International Publishing. <https://doi.org/10.1007/978-3-030-48279-4>
- Bibri SE, Krogstie J (2020) The emerging data-driven Smart City and its innovative applied solutions for sustainability: The cases of London and Barcelona. *Energy Inform* 3(1):5. <https://doi.org/10.1186/s42162-020-00108-6>
- Branco T, Fonseca IC, & Ugeda L (2023). Smart Governance Models and Solutions. 2023 18th Iberian Conference on Information Systems and Technologies (CISTI), 1–6. <https://doi.org/10.23919/CISTI58278.2023.10211734>
- Callaway B, Sant'Anna PHC (2021) Difference-in-differences with multiple time periods. *J Econ* 225(2):200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>
- Caragliu A, Del Bo CF (2019a) Smart innovative cities: the impact of Smart City policies on urban innovation. *Technol Forecast Soc Change* 142:373–383. <https://doi.org/10.1016/j.techfore.2018.07.022>
- Caragliu A, Del Bo CF (2019b) Smart innovative cities: The impact of Smart City policies on urban innovation. *Technol Forecast Soc Change* 142:373–383. <https://doi.org/10.1016/j.techfore.2018.07.022>
- Caragliu A, Del Bo CF (2022) Smart cities and urban inequality. *Regional Stud* 56(7):1097–1112. <https://doi.org/10.1080/00343404.2021.1984421>
- Caragliu A, Del Bo C, Nijkamp P (2011) Smart cities in Europe. *J Urban Technol* 18(2):65–82. <https://doi.org/10.1080/10630732.2011.601117>
- Chen Y, Jiang C, Peng L, Zhao S, Chen C (2025) Digital infrastructure construction and urban industrial chain resilience: Evidence from the “Broadband China” strategy. *Sustain Cities Soc* 121: 106228. <https://doi.org/10.1016/j.scs.2025.106228>
- Cugurullo F (2016) Urban eco-modernisation and the policy context of new city projects: Where Masdar City fails and why. *Urban Stud* 53(11):2417–2433. <https://doi.org/10.1177/0042098015588727>
- Das D, Kwek B (2024) AI and data-driven urbanism: the Singapore experience. *Digit Geogr Soc* 7: 100104. <https://doi.org/10.1016/j.diggeo.2024.100104>
- Dolmans SAM, van Galen WPL, Walrave B, den Ouden E, Valkenburg R, Romme AGL (2023) A dynamic perspective on collaborative innovation for smart city development: the role of uncertainty, governance, and institutional logics. *Organ Stud* 44(10):1577–1601. <https://doi.org/10.1177/01708406231169422>
- Drazin R (1991) The processes of technological innovation. *J Technol Transf* 16(1):45–46. <https://doi.org/10.1007/BF02371446>
- Du Y, Wang Q, Zhou J (2023) How does digital inclusive finance affect economic resilience: Evidence from 285 cities in China. *Int Rev Financ Anal* 88: 102709. <https://doi.org/10.1016/j.irfa.2023.102709>
- Du Z-Y, Wang Q (2024) Digital infrastructure and innovation: Digital divide or digital dividend?. *J Innov Knowl* 9(3):100542. <https://doi.org/10.1016/j.jik.2024.100542>
- Echebarria C, Barrutia, Jose M, Aguado-Moralejo I (2021) The Smart City journey: A systematic review and future research agenda. *Innov: Eur J Soc Sci Res* 34(2):159–201. <https://doi.org/10.1080/13511610.2020.1785277>
- Elhorst JP (2014) *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*. Springer. <https://doi.org/10.1007/978-3-642-40340-8>
- Furman JL, Porter ME, Stern S (2002) The determinants of national innovative capacity. *Res Policy* 31(6):899–933. [https://doi.org/10.1016/S0048-7333\(01\)00152-4](https://doi.org/10.1016/S0048-7333(01)00152-4)
- Giffinger R, Fertner C, Kramar H, & Meijers E (2007) City-ranking of European Medium-Sized Cities. <https://www.semanticscholar.org/paper/City-ranking-of-European-Medium-Sized-Cities-Giffinger-Fertner/fcd2292c028ef31f24d750aef302f6fa1716fc6>
- Glaeser EL, Kolko J, Saiz A (2001) Consumer city. *J Econ Geogr* 1(1):27–50. <https://doi.org/10.1093/jeg/1.1.27>
- Gohar A, & Nencioni G (2021). The Role of 5G Technologies in a Smart City: The Case for Intelligent Transportation System. *Sustainability*, 13(9), Article 9. <https://doi.org/10.3390/su13095188>
- Guo C, Wang Y, Hu Y, Wu Y, Lai X (2024) Does smart city policy improve corporate green technology innovation? Evidence from Chinese listed companies. *J Environ Plan Manag* 67(6):1182–1211. <https://doi.org/10.1080/09640568.2022.2157708>
- Guo Q, Zhong J (2022) The effect of urban innovation performance of smart city construction policies: Evaluate by using a multiple period difference-in-differences model. *Technol Forecast Soc Change* 184: 122003. <https://doi.org/10.1016/j.techfore.2022.122003>
- Heilmann S (2008) Policy experimentation in China's economic rise. *Stud Comp Int Dev* 43(1):1–26. <https://doi.org/10.1007/s12116-007-9014-4>
- Hollands RG (2015) Critical interventions into the corporate smart city. *Camb J Reg, Econ Soc* 8(1):61–77
- Hu Q, Zheng Y (2021) Smart city initiatives: a comparative study of American and Chinese cities. *J Urban Aff* 43(4):504–525. <https://doi.org/10.1080/07352166.2019.1694413>
- ISO, & IEC. (2019) Information technology—Smart city ICT indicators. <https://www.iso.org/obp/ui/en/#iso:std:70302:en>
- Jaakola A, Kekkonen H, Lahti T, & Manninen A (2015). Open data, open cities: Experiences from the Helsinki Metropolitan Area. Case Helsinki Region Infoshare www.hri.fi. Statistical Journal of the IAOS, 31(1), 117–122. <https://doi.org/10.3233/SJI-150873>
- Javed AR, Shahzad F, Rehman Sur, Zikria YB, Razzak I, Jalil Z, Xu G (2022) Future smart cities: Requirements, emerging technologies, applications, challenges, and future aspects. *Cities* 129: 103794. <https://doi.org/10.1016/j.cities.2022.103794>
- Ji M, Jin M, Chen L, Liu Y, Tian Y (2024) Promoting urban innovation through smart cities: evidence from a Quasi-natural experiment in China. *Land* 13(3):319. <https://doi.org/10.3390/land13030319>
- Ji T, Chen J-H, Wei H-H, Su Y-C (2021) Towards people-centric smart city development: Investigating the citizens' preferences and perceptions about smart-city services in Taiwan. *Sustain Cities Soc* 67: 102691. <https://doi.org/10.1016/j.scs.2020.102691>
- Jin M, Chen Y (2024) Has green innovation been improved by intelligent manufacturing?—Evidence from listed Chinese manufacturing enterprises. *Technol Forecast Soc Change* 205: 123487. <https://doi.org/10.1016/j.techfore.2024.123487>
- Khurana I, Dutta DK (2024) From place to space: The emergence and evolution of sustainable entrepreneurial ecosystems in smart cities. *Small Bus Econ* 62(2):541–569. <https://doi.org/10.1007/s11187-023-00765-3>
- Komninos N (2019) *Smart Cities and Connected Intelligence: Platforms, Ecosystems and Network Effects*. Routledge. <https://doi.org/10.4324/9780367823399>
- Kou Z, & Liu X (2017) FIND Report on City and Industrial Innovation in China (2017). Fudan Institute of Industrial Development, School of Economics, Fudan University
- Kummitha RKR (2018a) Entrepreneurial urbanism and technological panacea: Why Smart City planning needs to go beyond corporate visioning?. *Technol Forecast Soc Change* 137:330–339. <https://doi.org/10.1016/j.techfore.2018.07.010>
- Kummitha RKR (2018b) Entrepreneurial urbanism and technological panacea: Why Smart City planning needs to go beyond corporate visioning?. *Technol Forecast Soc Change* 137:330–339. <https://doi.org/10.1016/j.techfore.2018.07.010>
- Kummitha RKR, Crutzen N (2017) How do we understand smart cities? An evolutionary perspective. *Cities* 67:43–52. <https://doi.org/10.1016/j.cities.2017.04.010>
- Kummitha RKR, Crutzen N (2019) Smart cities and the citizen-driven internet of things: A qualitative inquiry into an emerging smart city. *Technol Forecast Soc Change* 140:44–53. <https://doi.org/10.1016/j.techfore.2018.12.001>
- Li X (2011) Sources of external technology, absorptive capacity, and innovation capability in Chinese state-owned high-tech enterprises. *World Dev* 39:1240–1248. <https://doi.org/10.1016/j.WORLDDEV.2010.05.011>
- Liao Z, Chen K (2025) What drives market-oriented trading of resource and environmental elements: an analysis based on the technology–organization–environment framework. *Sustain Dev* 33(4):5998–6012. <https://doi.org/10.1002/sd.3449>
- Lim Y, Edelenbos J, Gianoli A (2023) Dynamics in the governance of smart cities: Insights from South Korean smart cities. *Int J Urban Sci* 27(sup1):183–205. <https://doi.org/10.1080/12265934.2022.2063158>
- Liu S, Jiang G, Chang L, & Wang L (2022) Can the smart city pilot policy promote high-quality economic development? A quasi-natural experiment based on 239 cities in China. *Sustainability*. <https://doi.org/10.3390/su142316005>
- Lopes J, Franco M (2019) Review about regional development networks: an ecosystem model proposal. *J Knowl Econ* 10(1):275–297. <https://doi.org/10.1007/s13132-017-0464-2>
- Lu WW, Tsai KS (2021) Picking places and people: centralizing provincial governance in China. *China Q* 248(1):957–986. <https://doi.org/10.1017/S0305741021000606>
- Marchesani F, Ceci F (2025) A quadruple helix view on smart city: exploring the effect of internal and external open innovation on public services digitalization. *Technovation* 139: 103141. <https://doi.org/10.1016/j.technovation.2024.103141>
- Marchesani F, Masciarelli F, Bikfalvi A (2023) Smart city as a hub for talent and innovative companies: Exploring the (dis) advantages of digital technology

- implementation in cities. *Technol Forecast Soc Change* 193: 122636. <https://doi.org/10.1016/j.techfore.2023.122636>
- Matsiuk N, Doloreux D, Shearmur R (2024) Beyond “buzz”: Knowledge interactions, innovation, and neighborhood characteristics. *J Econ Geogr* 24(6):821–842. <https://doi.org/10.1093/jeg/lbae026>
- Maulan RY, Yanti O, & Sukmawati N (2023). The emergence of smart governance in local administrations: A new approach to Jambi city policy innovations. *JPII (Jurnal Penelitian Pendidikan Indonesia)*. <https://doi.org/10.29210/020232922>
- Miguélez E, Moreno R (2015a) Knowledge flows and the absorptive capacity of regions. *Res Policy* 44(4):833–848. <https://doi.org/10.1016/j.respol.2015.01.016>
- Miguélez E, Moreno R (2015b) Knowledge flows and the absorptive capacity of regions. *Res Policy* 44(4):833–848. <https://doi.org/10.1016/j.respol.2015.01.016>
- Mitra S, Kumar H, Gupta M, & Bhattacharya J (2022). Entrepreneurship in smart cities: Elements of Start-up Ecosystem. *J Sci Technol Policy Manag*. <https://doi.org/10.1108/jstpm-06-2021-0078>
- Mora L, Gerli P, Ardito L, Messeni Petruzzelli A (2023a) Smart city governance from an innovation management perspective: theoretical framing, review of current practices, and future research agenda. *Technovation* 123:102717. <https://doi.org/10.1016/j.technovation.2023.102717>
- Mora L, Gerli P, Ardito L, Messeni Petruzzelli A (2023b) Smart city governance from an innovation management perspective: Theoretical framing, review of current practices, and future research agenda. *Technovation* 123: 102717. <https://doi.org/10.1016/j.technovation.2023.102717>
- Mora L, Gerli P, Batty M, Binet Royall E, Carfi N, Coenegrachts K-F, De Jong M, Fachchina M, Janssen M, Meijer A, Pasi G, Perrino M, Raven R, Sagar A, Sancino A, Santi P, Sharp D, Trencher G, Van Zoonen L, Westerberg P, Woods O, Zhang X, Ziemer G (2025) Confronting the smart city governance challenge. *Nat Cities* 2(2):110–113. <https://doi.org/10.1038/s44284-024-00168-9>
- Nastjuk I, Trang S, Papageorgiou EI (2022) Smart cities and smart governance models for future cities. *Electron Mark* 32(4):1917–1924. <https://doi.org/10.1007/s12525-022-00609-0>
- Neirotti P, De Marco A, Cagliano AC, Mangano G, Scorrano F (2014) Current trends in Smart City initiatives: some stylised facts. *Cities* 38:25–36. <https://doi.org/10.1016/j.cities.2013.12.010>
- Pereira G, Cunha MA, Lampoltshammer TJ, Parycek P, Testa MG (2017) Increasing collaboration and participation in smart city governance: a cross-case analysis of smart city initiatives. *Inf Technol Dev* 23:526–553. <https://doi.org/10.1080/02681102.2017.1353946>
- Qin X (2022) The Impact of Interregional Collaboration on Multistage R&D Productivity and Their Interregional Gaps in Chinese Provinces. *Mathematics*, 10(8), Article 8. <https://doi.org/10.3390/math10081310>
- Rădulescu CM, Slava S, Rădulescu AT, Toader R, Toader D-C, & Boca GD (2020). A Pattern of Collaborative Networking for Enhancing Sustainability of Smart Cities. *Sustainability*, 12(3), Article 3. <https://doi.org/10.3390/su12031042>
- Randeree K, Ahmed N (2019) The social imperative in sustainable urban development: The case of Masdar City in the United Arab Emirates. *Smart Sustain Built Environ* 8(2):138–149. <https://doi.org/10.1108/SASBE-11-2017-0064>
- Rodríguez Bolívar MP, Alcaide Muñoz L, Alcaide Muñoz C (2023) Identifying patterns in smart initiatives’ planning in smart cities. An empirical analysis in Spanish smart cities. *Technol Forecast Soc Change* 196:122781. <https://doi.org/10.1016/j.techfore.2023.122781>
- Rodríguez-Pose A (2013) Do Institutions Matter for Regional Development?. *Regional Stud* 47(7):1034–1047. <https://doi.org/10.1080/00343404.2012.748978>
- Rodríguez-Pose A, Wilkie C, Zhang M (2021) Innovating in “lagging” cities: a comparative exploration of the dynamics of innovation in Chinese cities. *Appl Geogr* 132: 102475. <https://doi.org/10.1016/j.apgeog.2021.102475>
- Shao J, Min B (2025) Sustainable development strategies for Smart Cities: Review and development framework. *Cities* 158: 105663. <https://doi.org/10.1016/j.cities.2024.105663>
- Shukai C, Haochen W, Xiaohong Z (2021) Do City Size and Population Density Influence Regional Innovation Output Evidence from China? *Wirel Commun Mob Comput* 2021:e3582053. <https://doi.org/10.1155/2021/3582053>
- Sorri K, Rjónkoski K, Seppänen M (2024) Smart cities, smarter values: Unpacking the ecosystem of urban innovation. *Technol Soc* 77: 102499. <https://doi.org/10.1016/j.techsoc.2024.102499>
- Stettler TR, Moosauer EJ, Schweiger SA, Baldauf A, Audretsch D (2025) Absorptive capacity in a more (or less) absorptive environment: A meta-analysis of contextual effects on firm innovation. *J Prod Innov Manag* 42(1):18–47. <https://doi.org/10.1111/jipm.12758>
- Stöber LF, Boesino M, Pyka A, & Schuenemann F (2023). Bioeconomy Innovation Networks in Urban Regions: The Case of Stuttgart. *Land*, 12(4), Article 4. <https://doi.org/10.3390/land12040935>
- Sun J, Zhai C, Dong X, Li C, Wang Z, Li D, Sun Y (2023a) How does digital infrastructure break the resource curse of cities? Evidence from a quasi-natural experiment in China. *Resour Policy* 86:104302. <https://doi.org/10.1016/j.resourpol.2023.104302>
- Sun J, Zhai C, Dong X, Li C, Wang Z, Li D, Sun Y (2023b) How does digital infrastructure break the resource curse of cities? Evidence from a quasi-natural experiment in China. *Resour Policy* 86: 104302. <https://doi.org/10.1016/j.resourpol.2023.104302>
- Wang K-L, Qiao Y-J, Xu R-Y, Cheng Y-H, Miao Z (2025) Smart city construction and urban livability: Evidence from a quasi-natural experiment in China. *Humanit Soc Sci Commun* 12(1):1–16. <https://doi.org/10.1057/s41599-025-05096-0>
- Wang L, & Shao J (2024) Digital economy and urban green development: A quasi-natural experiment based on national big data comprehensive pilot zone. *Energy Environ*, 0958305X241238348. <https://doi.org/10.1177/0958305X241238348>
- Wang M, Zhou T (2025) Rethinking the effects of smart city implementation on industrial structure change. *Soc Sci Comput Rev* 43(4):897–923. <https://doi.org/10.1177/08944393241269414>
- Wang W, Wu F, Zhang F, Yang T (2025) State-led smart city with Chinese characteristics: The case of Xiong’an New Area, China. *Int J Urban Sci* 29(sup1):115–135. <https://doi.org/10.1080/12265934.2024.2407785>
- Wang Y, Li Y, Ding P, Guo B (2025) Technology transfer and innovation efficiency in a large emerging economy: an integrative perspective of absorptive capacity and the technology ladder. *J Technol Transf* 50(5):2340–2376. <https://doi.org/10.1007/s10961-024-10184-5>
- Wang Y, Liu X, Zhang Y (2007) Cross-Border Mergers and Acquisitions: Innovative Capacity and National Economic Security. *J Econ Policy Reform* 10(4):263–281. <https://doi.org/10.1080/17487870701551998>
- Woods O, Bunnell T, Kong L (2024) Insourcing the smart city: assembling an ideotechnical ecosystem of talent, skills, and civic-mindedness in Singapore. *Urban Geogr* 45(5):735–754. <https://doi.org/10.1080/02723638.2023.2233353>
- Wu F (2018) Planning centrality, market instruments: Governing Chinese urban transformation under state entrepreneurialism. *Urban Stud* 55(7):1383–1399. <https://doi.org/10.1177/0042098017721828>
- Wu T, Xu W, Kung C-C (2025) The impact of data elements on urban sustainable development: Evidence from the big data policy in China. *Technol Soc* 81: 102800. <https://doi.org/10.1016/j.techsoc.2024.102800>
- Xu A, Qiu K, Jin C, Cheng C, Zhu Y (2022) Regional innovation ability and its inequality: Measurements and dynamic decomposition. *Technol Forecast Soc Change* 180: 121713. <https://doi.org/10.1016/j.techfore.2022.121713>
- Yan Z, Sun Z, Shi R, Zhao M (2023) Smart city and green development: empirical evidence from the perspective of green technological innovation. *Technol Forecast Soc Change* 191:122507. <https://doi.org/10.1016/j.techfore.2023.122507>
- Yang Q, Elbakri I (2024) Low-carbon innovation efficiency of high-tech industries and its influencing factors. *Int J Low-Carbon Technol* 19:517–525. <https://doi.org/10.1093/ijlct/ctad140>
- Yang Y, Chen W, Gu R (2023) How does digital infrastructure affect industrial eco-efficiency? Considering the threshold effect of regional collaborative innovation. *J Clean Prod* 427:139248. <https://doi.org/10.1016/j.jclepro.2023.139248>
- Yang Z, Gao W, Han Q, Qi L (2024) Aggravating or alleviating? Smart city construction and urban inequality in China. *Technol Soc* 77:102562. <https://doi.org/10.1016/j.techsoc.2024.102562>
- Yi D, Shi Y (2025) Strategy-embedded diffusion and policy reproduction: How China’s special access policies for drugs and medical devices evolve through three-tiered interactions. *Asian Rev Political Econ* 4(1):16. <https://doi.org/10.1007/s44216-025-00058-4>
- Yin M, & Anantsuksomsri S (2022) Development of Smart City Policies in First-tier Cities in China. *ASR: Chiang Mai University Journal of Social Sciences and Humanities*, 9(2). <https://doi.org/10.12982/CMUJASR.2022.015>
- Yu L, Shang S (2018) The impact of policy intervention on regional economic growth in China. *Mod Econ* 09:2224–2232. <https://doi.org/10.4236/me.2018.912137>
- Yuan J, Xie H, Yang D, Xiahou X, Skibniewski MJ, Huang W (2020) Strategy formulation for the sustainable development of smart cities: a case study Of Nanjing, China. *Int J Strateg Prop Manag* 24(6):379–399. <https://doi.org/10.3846/ijspm.2020.13345>
- Zeng Y, Zhang Z, Ye Z, Li L (2023) Regional innovation effect of smart city construction in China. *PLOS ONE* 18(2):e0281862. <https://doi.org/10.1371/journal.pone.0281862>
- Zhang X, Fu Y, & Zhuang H (2022). City size, technological innovation, and urban income inequality: Empirical evidence from China family panel studies. *Front Environ Sci*, 10. <https://doi.org/10.3389/fenvs.2022.1065941>
- Zhu K, Zhang F, Wu F, Feng Y (2024) Governing innovation-driven development under state entrepreneurialism in China. *Cities* 152: 105194. <https://doi.org/10.1016/j.cities.2024.105194>
- Zhuang L, Ye C (2020) Changing imbalance: Spatial production of national high-tech industrial development zones in China (1988–2018). *Land Use Policy* 94: 104512. <https://doi.org/10.1016/j.landusepol.2020.104512>

ARTICLE

HUMANITIES AND SOCIAL SCIENCES COMMUNICATIONS | <https://doi.org/10.1057/s41599-026-06673-7>

Acknowledgements

This research was supported by National Social Science Fund of China (23FJYB009), Soft Science Research Program of Zhejiang Province (2026C15002).

Author contributions

YL and RH wrote the main manuscript text. JZ and RH provided idea, funding, and modifications. JX did the data analysis. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1057/s41599-026-06673-7>.

Correspondence and requests for materials should be addressed to Juanfeng Zhang or Rui Han.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026

Received: 18 February 2025 | Accepted: 15 February 2026

DOI: 10.1111/bjet.70058

REVIEW ARTICLE

British Journal of
Educational Technology 

AI support in self-regulated learning: A decade of technological evolution and meta-analysis

Jun Xu¹  | Yuying Luo²  | Chengliang Wang¹  |
Mengji Wang¹ | Yonghe Wu¹

¹Faculty of Education, Department of Education Information Technology, East China Normal University, Shanghai, China

²Department of Management Science and Engineering, School of Management, Shanghai University, Shanghai, China

Correspondence

Yonghe Wu, Faculty of Education, Department of Education Information Technology, East China Normal University, Shanghai, China.
Email: yhwu@deit.ecnu.edu.cn

Funding information

the National Social Science Foundation of China, Grant/Award Number: 21 and ZD328

Abstract

This meta-analysis systematically examines 35 empirical studies (2013–2025) investigating artificial intelligence applications within Zimmerman's cyclical model of self-regulated learning (SRL). Three principal discoveries emerge: (1) Technological progression has evolved through three co-existing paradigms: rule-based architectures, data-driven adaptive systems and generative AI ecosystems, that demonstrate increasingly sophisticated capabilities for human-AI collaboration. (2) While AI-supported SRL interventions yield a moderate overall effect size ($g=0.507$), their impact is uneven; AI is significantly more effective during the task performance phase ($g=0.574$) than in the preparatory forethought phase ($g=0.401$). Notably, generative AI shows markedly superior efficacy across all phases (e.g. $g=0.709$ for forethought, $g=0.938$ for performance), though high heterogeneity suggests these effects are heavily contingent on specific instructional designs. (3) Moderator analysis identifies optimal contexts in secondary education, natural science disciplines, fully online settings and interventions of medium duration (2–10 weeks), while also revealing that effects are substantially larger when measured by behavioural traces compared to self-reports. Critically, these findings highlight a persistent performance-competence divide, suggesting that AI's capacity to scaffold immediate task performance may outpace its current ability to cultivate durable, transferable self-regulatory competence. The study discusses the implications of

Jun Xu and Yuying Luo contributed equally to this work and share first authorship.

© 2026 British Educational Research Association.

this divide and proposes a research agenda focused on designing AI systems that foster genuine learner autonomy.

KEYWORDS

artificial intelligence, educational technology, large language models, meta-analysis, self-regulated learning

Practitioner notes

What is already known about this topic

- AI technologies have shown promise in supporting self-regulated learning (SRL) through personalized feedback and metacognitive scaffolding.
- Previous studies report inconsistent outcomes of AI interventions across SRL phases (forethought, performance, self-reflection), with limited exploration of phase-specific impacts
- Existing meta-analyses often treat SRL as a unified construct rather than examining how different AI types support distinct self-regulatory processes.

What this paper adds

- Demonstrates moderate overall effectiveness of AI-supported SRL ($g=0.507$) with differential impacts across phases: strongest during task performance ($g=0.574$), weaker in forethought ($g=0.401$) and self-reflection ($g=0.464$)
- Maps three coexisting AI paradigms—rule-based, data-driven and generative AI—revealing that generative AI achieves superior outcomes across all SRL phases, though effectiveness depends on alignment between AI affordances and specific self-regulatory processes.
- Identifies optimal implementation contexts through moderator analysis: secondary education settings, natural science disciplines, fully online environments and medium-duration interventions (2–10 weeks) yield stronger effects.
- Articulates a critical performance-competence divide, substantiated by quantifying how AI's impact on observable behaviours ($g=0.751$) is more than double its effect on learners' self-reported perceptions ($g=0.369$).

Implications for practice and/or policy

- Integrate AI paradigms strategically rather than viewing them as replacements: combine rule-based systems' stable scaffolding with generative AI's adaptive dialogue to support the full SRL cycle.
- Design interventions of medium duration (2–10 weeks) to optimize skill acquisition while maintaining engagement, avoiding both novelty effects and scaffold dependency.
- Adapt AI implementation to disciplinary demands: structured procedural support works well in natural sciences, while social sciences require enhanced support for open-ended inquiry and critical analysis.
- Develop assessment frameworks that measure delayed, unsupported performance alongside immediate gains to ensure AI fosters genuine self-regulatory competence rather than temporary performance enhancement.

INTRODUCTION

The rapid advancement of artificial intelligence (AI) and its growing impact on the workforce has intensified the need for lifelong learning, positioning self-regulated learning (SRL) as an essential competency for navigating societal and technological change (de Mooij et al., 2025). Supporting learners to become effective self-regulators remains a persistent pedagogical challenge, largely because SRL processes are complex, cyclical and often covert (Winne & Hadwin, 1998). AI technologies offer a potential solution by providing novel data-driven methods to make these internal learning processes more visible and supportable (Sposato, 2025). However, in traditional classroom settings, successful SRL often relies on learners' intrinsic motivation and teachers' individualized guidance (de Ruig et al., 2023; Li et al., 2024). As the demand for independent student learning increases and resources become relatively limited—particularly in online environments—maintaining effective SRL becomes increasingly difficult without timely interaction and emotional support (Jin et al., 2023).

Against this backdrop, AI technologies have demonstrated considerable promise for supporting SRL by addressing the scalability limitations of human scaffolding. These capabilities address many of the challenges identified in traditional learning environments, offering learners immediate assistance with goal setting, metacognitive monitoring and reflective practices (Chang & Sun, 2024; Rawas, 2024). However, research examining the effectiveness of AI interventions across the forethought, performance and self-reflection phases has yielded inconsistent findings. Current empirical studies typically focus on specific AI tools or individual SRL phases, lacking systematic quantitative evaluation across the complete regulatory cycle (Wu, 2024). This fragmented research landscape has resulted in a lack of integrated evidence, constraining the development of a comprehensive understanding of how AI's evolving affordances differentially support SRL processes.

To address these gaps, this study undertakes a systematic review of intervention-based research on AI-supported SRL over the past decade, employing the meta-analytic framework proposed by Pigott and Polanin (2020). By synthesizing empirical evidence, our goal was to trace the developmental trajectory of AI in SRL, evaluate its overall effectiveness and investigate variations in effect sizes across different paradigms of technological evolution and SRL processes. While recognizing that SRL unfolds recursively in practice, this phase-based analysis serves as a necessary heuristic to identify where AI support is most effective and where gaps remain. Specifically, the following research questions guide our inquiry:

RQ1. How has the trajectory of AI development in supporting SRL evolved over the past decade?

RQ2. What are the effect sizes of AI interventions across the forethought, performance and self-reflection phases of SRL?

RQ3. From the perspective of technological evolution, do different types of AI exhibit varying effect sizes in supporting SRL?

RQ4. What potential moderating variables significantly influence the intervention effectiveness of AI in SRL overall?

THEORETICAL FRAMEWORK AND PRIOR RESEARCH

Development of self-regulated learning theory and the focus of this study

The concept of SRL emphasizes learners' proactive management of cognition, motivation and behaviour to achieve specific learning goals (Zimmerman, 1989). Since the 1980s, SRL theory has progressed from an individualized, linear perspective to more dynamic, socially contextualized approaches. Bandura's (1991) social cognitive theory underscores the interplay between environment and self-efficacy, Pintrich's (2000) four-phase model highlights the essential role of motivation, Winne and Hadwin (1998) explore SRL from an information-processing and metacognitive standpoint, and Hadwin et al.'s (2017) socially shared regulation of learning (SSRL) framework focuses on self-regulation in collaborative group settings. Despite differing emphases, all the aforementioned theoretical frameworks converge on the notion that phased self-regulation can enhance academic performance.

In technology-enhanced learning, Zimmerman's three-phase model (forethought, performance and self-reflection) remains a widely accepted operational framework (Wong et al., 2019). As AI evolves into a learning partner, it introduces more complex, co-regulatory dynamics; however, the efficacy of such human–AI partnerships is predicated on the individual's core self-regulatory competencies, as effective shared regulation is built upon the skills of its individual members (Sharma et al., 2024). Zimmerman's (2002) cyclical model provides an essential, process-oriented lens to systematically map varied interventions onto the core components of individual learning. This framework's parsimonious structure and widespread adoption in AI-SRL research (Lee et al., 2025) enable the methodological consistency required for meta-analytic synthesis across diverse technological paradigms.

Compared to traditional contexts where SRL depends largely on learners' self-awareness or teacher guidance, AI technologies offer innovative means to assist SRL processes. Drawing on data from the Web of Science database retrieved in June 2025, Figure 1 shows the annual publication volume and citation trends related to SRL. This trajectory highlights significant growth periods, including an initial rise around 2007, a broadening of research scope around 2014 and a pronounced surge peaking in 2023. In the early 21st century, the

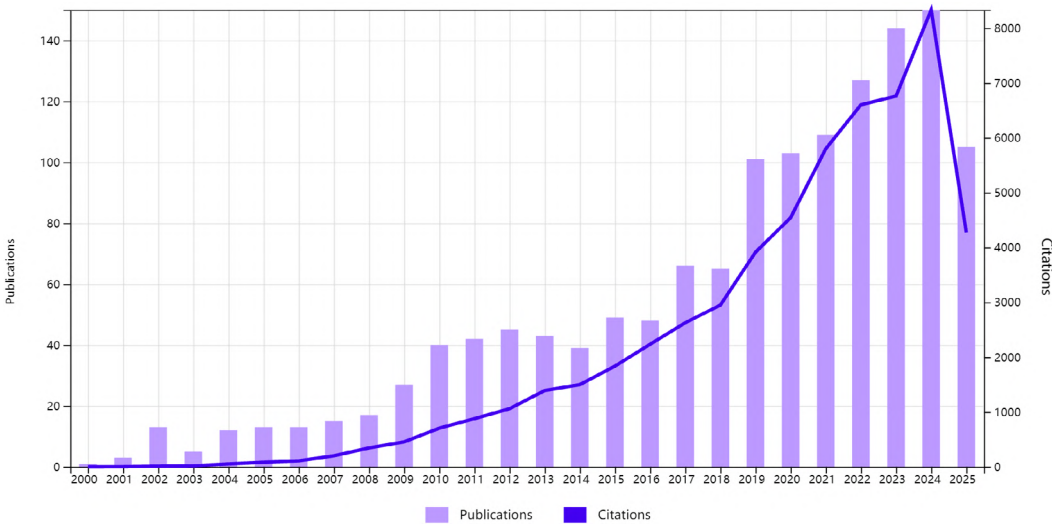


FIGURE 1 Annual publications and citation trends in self-regulated learning since the 21st century.

proliferation of internet and mobile devices enhanced access to digital learning resources (Dabbagh & Kitsantas, 2013). Subsequently, learning analytics and social media data broadened the reach of SRL research, while the recent emergence of generative AI has enabled highly personalized, real-time support, spurring a growing interest in using emerging technologies to strengthen SRL (Gao et al., 2024). As a result, SRL research has shifted from predominantly conceptual exploration to more data-driven empirical validation (Molenaar et al., 2023).

Meta-analytic research on technology-supported SRL: Progress and limitations

In recent years, numerous studies have examined the impact of various technologies on SRL and summarized their findings through meta-analyses. Overall, this body of work affirms SRL's crucial role in improving academic performance and supporting learner autonomy (Dent & Koenka, 2016; Dignath & Büttner, 2018). Researchers have further noted AI's potential to simulate teacher guidance, personalize strategy application and perform advanced learning analytics (Akamatsu et al., 2019; Dahri et al., 2024).

However, few studies have systematically focused on SRL as a skill to be cultivated, particularly with regard to emerging AI paradigms (Luckin et al., 2022). Existing meta-analyses often concentrate on specific technologies (Chen et al., 2024), leaving AI's distinct role in different SRL phases underexamined (Lu & Lin, 2024). Additionally, most research treats SRL as a unified construct rather than investigating how AI can support its forethought, performance and self-reflection phases individually. Drawing on Zimmerman's three-phase SRL model, this study quantitatively explores AI's impact on each phase, taking into account the varying developmental paradigms of AI technologies. By doing so, it addresses current gaps in meta-analytic SRL research and provides insights into how AI's evolving capabilities can enhance SRL across diverse educational contexts.

METHODS

This study adopts Zimmerman's SRL model as its theoretical basis and conducts a meta-analysis of empirical research published between 2013 and 2025 that applied AI technologies to SRL interventions. In line with the meta-analytic procedures outlined by Pigott and Polanin (2020), the research process involved (1) systematically searching and screening the literature, (2) coding and evaluating the quality of the selected studies, (3) calculating effect sizes using appropriate statistical methods and (4) interpreting the results comprehensively.

Given that AI-supported SRL spans a wide range of educational contexts and technological paradigms, maintaining high methodological standards is essential to minimize potential biases. To ensure research quality while maximizing inclusiveness, this study included peer-reviewed journal articles indexed in SSCI, SCIE, Scopus or EI, as well as conference proceedings from Scopus and EI databases that met rigorous quality criteria. All included studies underwent systematic quality assessment using the Medical Education Research Study Quality Instrument (MERSQI), which has been established as a reliable tool for evaluating methodological rigour in technology-enhanced learning reviews (Tian & Zheng, 2024). Following evidence that abstracts with MERSQI scores below 10 have significantly lower publication rates (Sawatsky et al., 2015), we adopted this score as our minimum threshold to ensure a methodologically robust evidence base.

Literature search and inclusion criteria

Adhering to the PRISMA guidelines, a systematic search was conducted in four major databases (ERIC, Ei Compendex, Scopus and Web of Science) to ensure thoroughness and quality (Xu et al., 2025). The search focused on SRL-related studies, with the time frame set from 2013 to 2025 to capture recent developments. For instance, in Web of Science, the search query was:

TS=('self-regulat*' AND learn*) AND PY=2013-2025. This query included various forms of 'self-regulation' and 'self-regulated', without strictly limiting the term 'AI', thereby encompassing diverse AI applications. Manual screening followed to identify studies that explicitly used AI technologies—such as intelligent tutoring systems, adaptive learning platforms or conversational AI tools—to support SRL.

Based on the above search strategy and the study's aims, the following inclusion criteria were established: (1) The study examined AI-supported SRL interventions, addressing at least one of the three SRL phases (forethought, performance or self-reflection). (2) It used experimental or quasi-experimental designs, comparing outcomes between an experimental group and a control group. (3) The intervention's effect was assessed using measures of SRL processes. Outcomes such as academic achievement were included only when theoretically justified as a direct consequence of the supported regulatory processes. To clarify the distinction between subject-matter and process-focused support, interventions were included only if their feedback was designed to function as an SRL scaffold—that is, to prompt regulatory processes such as self-evaluation and strategy adjustment, rather than merely to deliver content corrections. (4) It offered sufficient data (e.g. means, standard deviations, sample sizes) to calculate effect sizes. (5) It achieved a MERSQI score of 10 or higher, indicating acceptable methodological quality.

Following the initial screening, quality assessment was conducted on all potential studies using MERSQI, which evaluates six domains: study design, sampling, type of data, validity of evaluation instrument, data analysis and outcomes. Two studies with MERSQI scores below 10 were excluded due to methodological limitations. The final sample comprised 35 articles (mean MERSQI=13.07, SD=1.27), including 31 journal articles and 4 conference proceedings, all meeting the established criteria for quality and comparability. The complete list of included studies is provided in [Appendix A](#). As shown in [Figure 2](#), the final screening yielded 35 articles meeting the criteria for quality and comparability.

Study characteristics and intervention coding

To enable precise data extraction and facilitate comparison across studies, a systematic coding scheme was employed for study characteristics, AI interventions and SRL outcome measures ([Table 1](#)). First, participant details (e.g. geographical region, educational level, subject area and learning environment) were recorded to capture the context of each study. Additionally, recognizing the critical distinction between learners' perceptions of their self-regulation and actual regulatory behaviours (Winne, 2022), we coded each outcome's measurement type as either self-report or behavioural/trace. Next, AI interventions were classified according to their developmental paradigms and specific types (e.g. intelligent tutoring systems, adaptive learning platforms, conversational AI, intelligent feedback tools), and each intervention's targeted SRL phase was identified.

To investigate AI's impact on distinct SRL processes, relevant variables—such as goal setting, strategy planning, self-efficacy, metacognitive monitoring, time management and self-evaluation—were aligned with the appropriate SRL phases ([Table 2](#)),

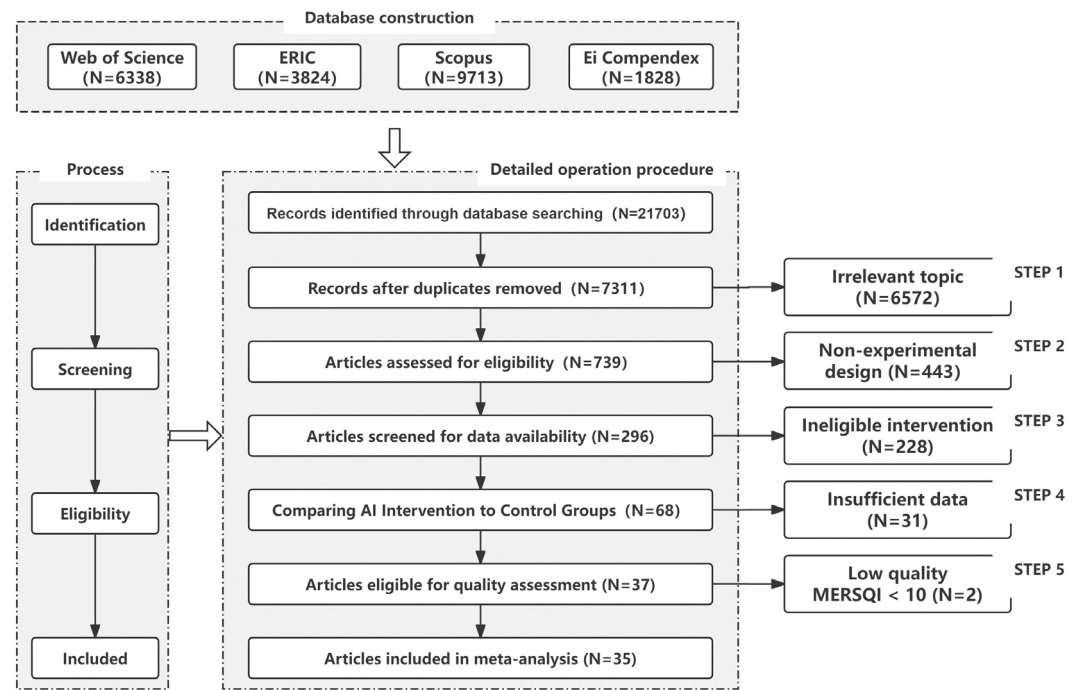


FIGURE 2 Search flow diagram.

with each distinct outcome treated as an independent effect size, ultimately yielding 133 effect sizes from the 35 included studies. While acknowledging that SRL processes are recursive and overlapping in authentic practice, this phased analysis is employed as an essential analytical heuristic. For a meta-synthesis of diverse literature, such a framework is methodologically necessary to systematically categorize disparate outcome measures and distil meaningful patterns from heterogeneous evidence (e.g. Hemmler & Ifenthaler, 2024). The classification of each variable was determined by its conceptual definition and its measurement point within a study's design (e.g. pre-task for planning, post-task for reflection), ensuring alignment with Zimmerman's (2002) cyclical model. To enhance analytical precision and directly address the critical distinction between perceived and actual regulation, we further coded each outcome measure by assessment type, distinguishing between self-report instruments (68% of effect sizes) that capture learners' perceptions and beliefs about their SRL, and behavioural/trace measures (32%) that document observable regulatory actions (Winne, 2022). This coding enabled a more transparent interpretation of whether the observed effects reflect changes in metacognitive awareness or actual behavioural manifestations of self-regulation.

In addition, to track how AI-supported SRL has evolved over the past decade, a longitudinal cluster analysis was conducted based on technological features, publication timelines and research methods (see Section 'A preliminary examination of AI development in the field of self-regulated learning (for RQ1)'). This analysis grouped AI tools into different stages of development. To ensure coding accuracy, two researchers independently coded 20 of the selected articles. Following Landis and Koch's (1977) guidelines, Cohen's kappa coefficients of 0.88 and 0.85 indicate strong interrater reliability; discrepancies were resolved through discussion, yielding a final coding scheme that is both accurate and reliable.

TABLE 1 Lists of codes for the analysis of selected articles.

Dimension	Item	Description
Participant characteristics	Region	Text records, for example, the United States, China, Japan, etc.
	Sample size	(1) 1–50; (2) 51–100; (3) 101–300.
	Educational level	(1) Primary school; (2) Junior and senior high school; (3) Institutions of higher education.
	Discipline	(1) Natural Science (including science, mathematics, physics, biology, geography); (2) Social Science (including politics, education, psychology, linguistics).
Content characteristics	Learning environment	(1) Online learning settings; (2) Blended learning settings.
	AI intervention method	A system that uses computational models (e.g. rule-based logic, machine learning, generative modeling) to provide adaptive or personalized feedback and scaffolding for SRL.
	SRL variables	Variables related to SRL recorded in the literature, such as self-efficacy, metacognition, attention focus and time management.

RESULTS

A preliminary examination of AI development in the field of self-regulated learning (for RQ1)

Analysis of the 35 studies included in this research reveals a technological trajectory defined by three co-existing paradigms. This evolution does not represent a linear displacement of earlier technologies but rather an expansion of the pedagogical landscape, progressing from structured, rule-based guidance to data-driven personalization and, most recently, to dialogic co-regulation with generative AI. Figure 3 illustrates the distribution of the 133 effect sizes from these studies across these AI paradigms and chronological periods, visually confirming that established rule-based and data-driven systems persist alongside emerging generative models rather than being displaced.

Paradigm 1: Rule-based intelligent tutoring systems

Foundational to the field and prominent since 2015, rule-based systems, such as MetaTutor (Duffy & Azevedo, 2015) and early intelligent tutoring systems (Mohammadzadeh & Sarkhosh, 2018) mainly depended on predefined rules and expert knowledge bases to provide structured but relatively rigid support. These systems helped learners with goal setting and strategy planning by offering metacognitive scaffolding and basic feedback (Jones et al., 2018). Although their capacity for contextual adaptation and deeper self-reflection was limited, they established a critical foundation for the broader, more versatile applications of subsequent AI technologies. Notably, such rule-based approaches have persisted in development and application, as evidenced by their continued presence in studies from 2021 to 2023 (Figure 3) and their ongoing utility in structured environments or for specific tasks where clarity and stability are paramount (Chu et al., 2023; Liang et al., 2024).

TABLE 2 Mapping of SRL indicators to phases.

Phase	Theoretical definition	Indicator	Definition
Forethought (pre-task)	The preparatory phase involving task analysis and mobilizing motivational beliefs	Goal setting and strategy planning	Establishing clear learning goals and implementation paths
		Self-efficacy	Confidence in one's ability to complete tasks
		Task value and interest	Awareness of the significance and enjoyment of learning tasks
		Outcome expectation	Predictions of the effort and potential benefits of learning
Performance (during task)	The execution phase involving self-control (e.g. strategy use) and self-observation	Task strategies	Applying cognitive and metacognitive skills (e.g. summarizing, monitoring)
		Attention focus and time management	Focusing on tasks and allocating study time effectively
		Self-observation and monitoring	Continuously tracking one's performance and strategy usage
		Seeking help	Identifying difficulties and actively seeking external assistance
Self-reflection (post-task)	The evaluative phase involving self-judgement of performance and adaptive self-reactions	Self-evaluation	Comparing outcomes with standards or initial expectations
		Causal attribution	Interpreting successes or failures as a basis for strategy modification or effort adjustment
		Adaptive inference and response	Adjusting future study behaviour based on reflective results
			Test results or indicators of academic performance
Comprehensive phase	An assessment of the overall effectiveness of an SRL-driven task, using measures not attributable to a single phase	Academic achievement	Application of skills and knowledge in new contexts
		Skill acquisition and transfer	Emotional responses and investment in the learning process
		Learner satisfaction and engagement	

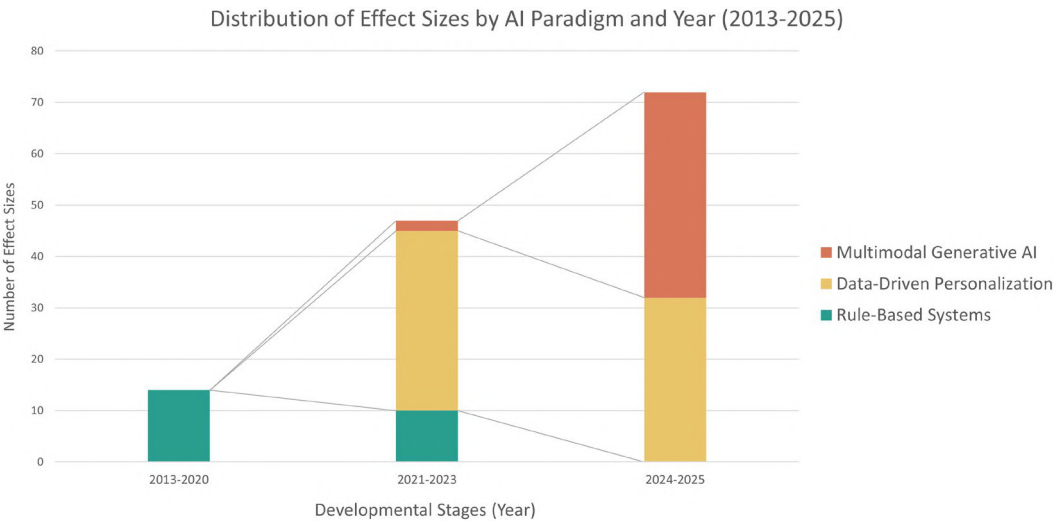


FIGURE 3 Distribution of effect sizes by AI paradigm and year ($k=133$ effect sizes).

Paradigm 2: The advancement of data-driven personalization

With the introduction of educational data mining, natural language processing (NLP) and complex system frameworks, AI tools expanded into a data-driven paradigm, emerging prominently from around 2021 and showing sustained presence (see Figure 3). These systems are distinguished by their capacity to learn from learner data to tailor support. More advanced learning analytics systems, such as ALEKS and Studybuddy, utilized real-time data to analyse learners' knowledge states and dynamically adjust learning paths and resources, guiding effective learning behaviours (Harati et al., 2021; Mejeh et al., 2024). Meanwhile, advancements in NLP and speech recognition enabled these tools to offer more flexible and personalized language instruction and instant feedback, significantly enhancing SRL's adaptability and contextual awareness (Chang et al., 2022; Wei, 2023). Additionally, the integration of complex systems theory provided researchers with new perspectives to understand learners' regulatory strategies and implement targeted interventions from a macro viewpoint (Hilpert et al., 2023). However, despite progress in personalization and adaptability, these tools still relied on pre-defined data and strategies, limiting the depth and creativity of interactions.

Paradigm 3: The integration of multimodal generative AI in intelligent educational ecosystems

Representing the most recent expansion of the SRL ecosystem, generative AI gained significant traction from 2023 onwards and is prominent in the 2024–2025 period within our sample (see Figure 3), AI tools incorporating large language models (LLMs), such as ChatGPT-Assisted Learning Aid and SRLbot, began to act as proactive cognitive partners, supporting student self-learning. By combining contextual prompts with personalized feedback, these tools facilitated autonomous problem-solving and higher order thinking skills (Ng et al., 2024). At the same time, virtual reality systems and multimodal feedback technologies that use image recognition and voice-based interactions contributed to multi-sensory learning experiences. These environments supported emotional regulation, encouraged creativity and promoted cyclical co-regulation processes (Hsu et al., 2023; Wang et al., 2024). Compared with earlier stages, these generative AI platforms focused on individual learners during instructional sessions, integrating

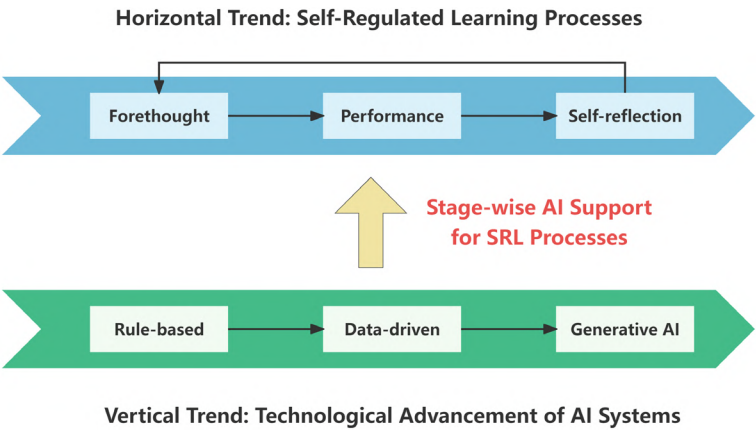


FIGURE 4 Conceptual framework: The intersection of AI evolution and SRL processes.

a range of inputs while placing the learner at the centre of the teaching and learning process. This enabled highly personalized, dynamically adaptive instruction (Liang et al., 2024).

Overall, the preliminary review suggests that AI support for SRL can be organized into three developmental paradigms. Figure 4 synthesizes this trajectory by mapping the technological evolution (vertical axis) against the cyclical nature of SRL processes (horizontal axis). Crucially, this vertical dimension represents a cumulative typology rather than a strictly chronological sequence, acknowledging that earlier rule-based approaches continue to function alongside emerging generative models. This framework facilitates a systematic examination of how the distinct affordances characterizing each paradigm intersect with specific phases of self-regulation.

Heterogeneity testing and effect size calculation (for RQ2)

Before calculating effect sizes, heterogeneity tests were performed to determine the most suitable meta-analysis model (Higgins & Thompson, 2002). Drawing on Zimmerman's SRL phases (forethought, performance, self-reflection and comprehensive evaluation), Cochran's Q -test and the I^2 statistic were used to examine potential heterogeneity. When the Q -test is significant ($p < 0.05$) and I^2 exceeds 75%, the data are considered to show high heterogeneity (Higgins, 2011). As shown in Table 3, all subgroups yielded Q -test significance, with I^2 values above 75%. This considerable heterogeneity indicates that AI's impact on SRL is not monolithic, confirming the need for a random-effects model and subsequent moderator analysis. Consequently, a random-effects model was adopted to address the substantial variability introduced by different research settings, participant characteristics, measurement approaches and the diverse nature of control group conditions (e.g. lab-based tasks, authentic classroom settings), where the common element was the absence of the specific AI intervention.

Next, Hedges' g was employed to standardize effect sizes for AI interventions in each SRL phase (Hedges & Tipton, 2010). This statistic is widely used for synthesizing results across studies (Hedges, 2008). Calculations were carried out in Stata 18, using precise estimation methods and small-sample corrections (Hamza et al., 2008) to reduce biases stemming from small sample sizes and high heterogeneity. These findings indicate that interventions exert a clear influence during the performance phase but have a comparatively weaker impact in the preparatory (forethought) and evaluative (self-reflection) stages.

Based on Cohen's (2013) thresholds, where g values of 0.2, 0.5 and 0.8 denote small, medium and large effects, the forethought ($g = 0.401$) and self-reflection ($g = 0.464$) phases both showed small effect sizes. In contrast, the performance ($g = 0.574$) and comprehensive

TABLE 3 Random-effects model for AI interventions in SRL phases.

Variables	Effect size and 95% CI				Heterogeneity test				Tau-squared				
	k	Hedges' g	SE		Lower	Upper	Q	df(Q)	p	I ²	Tau-squared	SE	Tau
Forethought phase	33	0.401	0.123		0.160	0.643	271.43	32	0.0000	88.95	0.415	0.133	0.644
Performance phase	51	0.574	0.110		0.358	0.789	533.81	50	0.0000	92.05	0.474	0.119	0.688
Self-Reflection phase	18	0.464	0.196		0.079	0.848	194.45	17	0.0000	91.81	0.594	0.256	0.771
Comprehensive phase	31	0.529	0.103		0.328	0.730	204.14	30	0.0000	81.93	0.337	0.117	0.581
Overall	133	0.507	0.063		0.384	0.630	1208.24	132	0.0000	89.81	0.432	0.069	0.657

($g=0.529$) phases reached medium levels. Collectively, the AI-supported interventions yielded a medium overall effect on SRL ($g=0.507$, 95% CI [0.384, 0.630]). The results confirm a pattern of differential effectiveness across the regulatory cycle: interventions exert a clear influence during the performance phase but have a comparatively weaker impact in the preparatory (forethought) and evaluative (self-reflection) stages. These latter phases may demand more complex, internally driven cognitive and metacognitive processes (e.g. proactive goal setting, nuanced self-critique and causal attribution) that current AI designs are still developing capabilities to effectively foster, pointing to a need for improved support across the full regulatory cycle.

Sensitivity analysis and publication bias

To confirm the stability of these findings, ‘leave-one-out’ sensitivity tests were conducted. After sequentially excluding each study, the overall effect size remained consistent, ranging from 0.491 to 0.519. Subgroup analyses for individual SRL phases also displayed a similar degree of stability, suggesting that no single study unduly influenced the overall results.

Publication bias was examined through multiple approaches, including funnel plots, Egger’s regression test, fail-safe N , and the trim-and-fill method.

While funnel plots for both the overall sample and individual phases appeared broadly symmetrical (Figures 5 and 6), Egger’s regression tests revealed significant asymmetry for the overall sample ($z=3.93$, $p=0.0001$) and the performance phase ($z=3.66$, $p=0.0003$). This asymmetry is particularly concerning for the performance phase, where we observed the highest effect sizes ($g=0.574$), suggesting that publication bias may be inflating our estimates precisely where AI appears most effective. The fail-safe N analysis indicated that 5908 null studies would be needed to invalidate the overall effect (exceeding Rosenthal’s (1979) $5k+10$ benchmark), with Orwin’s (1983) calculation requiring 4451 studies to reduce the effect below 0.01.

The concentration of publication bias in the performance phase reflects a broader pattern in educational technology research, where interventions demonstrating immediate,

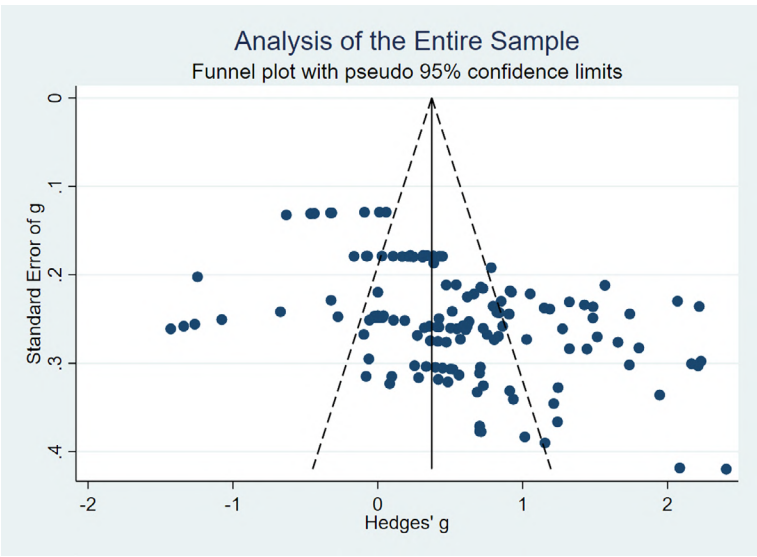


FIGURE 5 Funnel plot of overall sample.

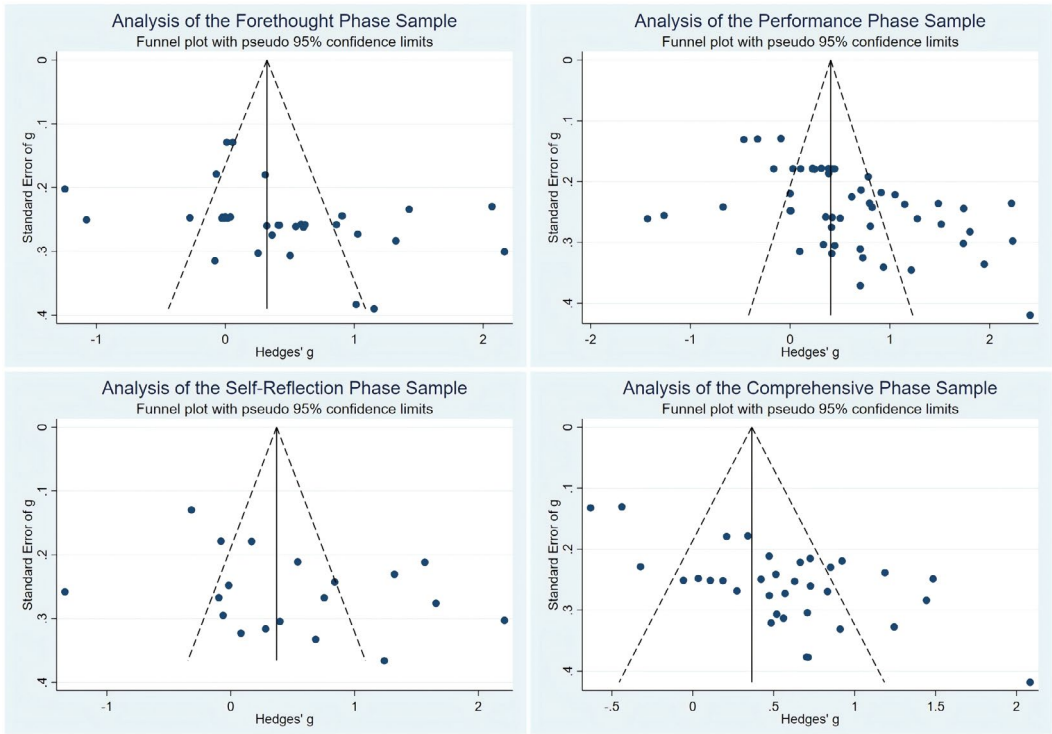


FIGURE 6 Funnel plots of the four SRL phases.

measurable gains are more likely to be published than those showing null or negative results (Cheung & Slavin, 2016). This selective reporting creates an overly optimistic picture of AI's capacity to enhance real-time learning performance, potentially misleading practitioners about the reliability of these interventions across diverse contexts.

The trim-and-fill method imputed three studies and adjusted the overall effect upwards from 0.507 to 0.569, suggesting minimal impact from missing studies. However, given the significant Egger's test results and the known limitations of trim-and-fill in distinguishing bias from heterogeneity (Peters et al., 2007), we interpret these effect sizes, particularly for the performance phase, as likely upper bounds rather than precise point estimates of AI's true impact. This interpretation aligns with current meta-analytic best practices (e.g. Wu & Yu, 2024) and underscores the need for systematic changes in publication practices, including pre-registration and commitment to publishing null results (Di Pietro & Castaño Muñoz, 2025), to build a more accurate evidence base for AI's role in supporting self-regulated learning (Table 4).

Subgroup analysis of AI technology evolution across SRL phases (for RQ3)

This study categorized the AI tools identified in the reviewed literature into three main developmental paradigms: rule-based systems, data-driven systems and GAI. To explore the significant heterogeneity observed in the initial analysis ($I^2=89.81\%$ overall), this section examines how intervention effectiveness varies as a function of the AI paradigm itself. Effect sizes were computed for both the overall dataset and for each phase of Zimmerman's SRL

TABLE 4 Measures of publication bias.

Self-regulated learning process	k	Egger's				Fail-safe N	
		Score	SD	z	p	Classic	Orwin
Forethought phase	33	5.15	2.270	2.27	0.0232	594	930
Performance phase	51	5.94	1.622	3.66	0.0003	3167	1945
Self-reflection phase	18	4.08	3.333	1.22	0.2213	218	555
Comprehensive phase	31	5.86	3.384	1.73	0.0834	843	930
Overall	133	4.66	1.186	3.93	0.0001	5908	4451

Note: k, number of studies; Classic Fail-Safe N: Represents the number of null-effect studies needed to raise the p-value above the significance threshold ($\alpha=0.05$); Orwin's Fail-Safe N: Represents the number of missing studies required to reduce the effect size to a predefined trivial level ($ES=0.01$).

TABLE 5 Combined effect sizes for AI developmental stages and SRL phases (Hedges' g).

SRL process	AI phase	k	Hedges' g	95% CI	I ²	p	Z
Forethought	Rule-based	2	—	—	—	—	—
	Data-driven	15	0.083	[−0.301, 0.466]	92.53%	0.6708	0.43
	Generative	16	0.709	[0.437, 0.981]	76.84%	0.0000	5.11
Performance	Rule-based	6	—	—	—	—	—
	Data-driven	32	0.352	[0.080, 0.624]	93.45%	0.0112	2.54
	Generative	13	0.938	[0.549, 1.328]	85.64%	0.0000	4.71
Self-reflection	Rule-based	2	—	—	—	—	—
	Data-driven	8	0.196	[−0.387, 0.779]	94.24%	0.5105	0.66
	Generative	8	0.826	[0.247, 1.405]	88.08%	0.0051	2.80
Comprehensive	Rule-based	13	0.429	[0.197, 0.662]	59.66%	0.0003	3.61
	Data-driven	13	0.580	[0.190, 0.971]	90.94%	0.0036	2.91
	Generative	8	0.682	[0.416, 0.938]	60.22%	0.0000	5.02
Overall	Rule-based	23	0.534	[0.339, 0.730]	66.77%	0.0000	5.34
	Data-driven	68	0.318	[0.134, 0.501]	93.07%	0.0006	3.41
	Generative	42	0.799	[0.604, 0.994]	82.34%	0.0000	8.03

Note: Caution is advised for subgroups with $k < 10$, while those with $k < 7$ may have limited reliability.

model (forethought, performance and self-reflection). Subgroups containing fewer than five studies were excluded due to limited interpretive value. Table 5 summarizes the combined effect sizes (Hedges' g) for each developmental stage across the SRL phases. The results show that overall effect sizes differ significantly across the three AI paradigms ($Q_b=15.73$, $p<0.001$), progressing from rule-based systems ($g=0.534$) and data-driven AI ($g=0.318$) to GAI ($g=0.799$).

Moderating effect analysis (for RQ4)

To identify potential sources of heterogeneity, this section reports the results of subgroup analyses on five study-level characteristics: educational level, discipline, learning space, duration and measurement type (Table 6). These analyses aim to reveal systematic patterns in intervention effectiveness and generate targeted hypotheses for future experimental validation.

TABLE 6 Subgroup analysis results of meta-analysis (Hedges' *g*).

Subgroup	<i>k</i>	<i>g</i>	95% CI	<i>Z</i> (<i>p</i>)	Between <i>p</i> *
<i>Educational level</i>					0.221 (3.02)
Primary school	14	0.304	[0.060, 0.548]	2.44 (0.015)	
Junior and senior high school	11	0.636	[0.284, 0.989]	3.54 (0.001)	
Higher education	108	0.517	[0.374, 0.660]	7.09 (<0.001)	
<i>Subject</i>					<0.001 (21.09)
Natural Science	73	0.753	[0.576, 0.929]	8.36 (<0.001)	
Social Science	56	0.211	[0.061, 0.360]	2.77 (0.006)	
Combined subject areas	129	0.516	[0.390, 0.643]	8.00 (<0.001)	
<i>Learning space</i>					0.087 (2.93)
Online learning	37	0.732	[0.376, 1.087]	4.04 (<0.001)	
Blended learning	96	0.409	[0.309, 0.509]	8.02 (<0.001)	
<i>Experimental duration</i>					0.002 (12.44)
Less than 2 weeks	45	0.776	[0.551, 1.002]	6.88 (<0.001)	
2–10 weeks	43	0.361	[0.222, 0.500]	7.64 (<0.001)	
>10 weeks	45	0.374	[0.139, 0.609]	0.32 (<0.001)	
Overall	133	0.507	[0.384, 0.630]	8.08 (<0.001)	—

DISCUSSION

This meta-analysis reveals that while AI interventions yield a moderate overall benefit to self-regulated learning, their effectiveness is highly variable and contingent on the type of AI, the specific SRL phase, the learning context and the method of measurement. The following discussion interprets these findings through a critical theoretical lens, focusing on the distinction between supporting in-task performance and fostering internalized competence. This performance-competence divide emerges as a central theme for understanding the current state and future trajectory of AI in SRL research.

The overall effectiveness of AI in supporting SRL in context

The overall medium effect size ($g=0.507$) confirms that AI-supported interventions are a valuable tool for enhancing SRL, with an impact comparable to other established educational interventions. Consistent with the medium effect size ($g=0.50$) reported by Guo (2022) for metacognitive prompt interventions in SRL, the findings here suggest that AI-supported interventions, through the delivery of specific instructional features such as personalized feedback or scaffolded pathways during task execution (the performance phase), can serve functions analogous to metacognitive prompts in guiding certain aspects of learners' self-regulatory skills. It is important to recognize that the effectiveness likely stems from these carefully designed instructional strategies facilitated by AI, rather than AI acting as a monolithic causal agent. However, this study's overall effect size is lower than the value ($g=0.897$) documented by D. Lee and Kwon (2024) in Korean K–12 classrooms—a discrepancy that may stem from SRL's long-term developmental trajectory or from variations in measurement approaches. This value is slightly higher than that reported by Dai et al. (2024) ($g=0.40$, focusing on AI-driven virtual agents for learners' general competencies), highlighting the

potential of appropriately designed AI-facilitated adaptive interventions to effectively support SRL, particularly when they are geared to foster personalized and self-directed learning processes.

The performance-competence divide: Interpreting effects across AI paradigms and SRL phases

The progression in overall effect sizes from rule-based systems ($g=0.534$) and data-driven AI ($g=0.318$) to GAI ($g=0.799$) is not merely a story of technological improvement. Instead, it must be interpreted through a critical theoretical lens that distinguishes between supporting in-task performance and fostering internalized regulatory competence. This distinction is particularly salient given that the synthesized effect sizes aggregate two distinct outcome families: learners' judgements about their SRL (via self-report) and their observable regulatory actions (via behavioural traces).

Because these measures capture different constructs and yield significantly different results, the apparent progression in overall effectiveness warrants a cautious interpretation regarding the enhancement of durable SRL competence. Rather than signalling definitive mastery, the data reflect a complex interplay between scaffolded performance and learners' perceptions of their capabilities, a tension we explore below.

Overall trends across AI paradigms

Rule-based AI: A foundational approach under continuous evolution

Although rule-based systems established a critical foundation ($g=0.534$) by providing stable scaffolding in structured environments (Duffy & Azevedo, 2015), their reliance on predefined prompts often limited their effectiveness in open-ended contexts (Chu et al., 2023). While early studies reported large effect sizes (Jones et al., 2018), later work revealed that fixed rule sets struggle to nurture transferable regulatory routines. Log-level evidence clarifies this mixed picture: although scripted prompts reliably triggered within-task planning and monitoring, subsequent segments of the same logfile showed a rapid decay of those behaviours once scaffolds disappeared, and the frequency of unprompted SRL actions rarely predicted posttest achievement (Bouchet et al., 2016; Long & Aleven, 2013). These findings suggest that while rule-based logic ensures clarity, its rigidity functions more as a procedural guide than a catalyst for adaptive self-regulation.

Data-driven AI: Advancing adaptive technology through dynamic analytics

Data-driven AI improved upon rule-based limitations by utilizing learning analytics to tailor interventions, yet its overall impact remains moderate and highly heterogeneous ($g\approx 0.32$, $I^2\approx 93\%$). For example, Duolingo showed notable success in supporting goal setting and performance monitoring (Qiao & Zhao, 2023; Wei, 2023). Such platforms typically achieve these outcomes by dynamically adapting curriculum content and task difficulty based on performance data (e.g. Weber et al., 2025), thereby influencing learning processes. At the same time, the effectiveness of data-driven systems can be closely tied to learners' self-regulation skills and comfort with technology. Some studies found that increased complexity lowered self-efficacy (Harati et al., 2021), especially during multidisciplinary or extended projects (Pereira & Díaz, 2021). Overall, data-driven AI excelled at capturing learning trajectories and offering immediate feedback (Dever et al., 2023), such as automated corrective feedback in language learning (Liu et al., 2025).

The divergence in outcomes likely stems from the gap between information availability and pedagogical mediation. While real-time dashboards heighten situational awareness, they may also overwhelm novices when analytics are surfaced without interpretive support (Yang et al., 2024). This divergence appears rooted in socio-emotional and cognitive responses, as novice learners facing high cognitive loads may hesitate to seek help due to feelings of embarrassment, a barrier that well-designed, on-demand AI scaffolding can effectively mitigate (Huang et al., 2025). Although survey data often register feelings of personalization, trace studies indicate that these perceptions do not reliably translate into durable regulatory routines (Cabı & Türkoğlu, 2025). This suggests sophisticated monitoring can amplify SRL only when paired with scaffolds that help learners interpret—and act on—algorithmic signals.

Generative AI: Personalized companionship and multimodal integration

Generative AI achieved superior outcomes ($g=0.799$) by functioning as a responsive learning companion rather than a mere delivery system, though high heterogeneity ($I^2=82.34\%$) indicates that efficacy depends heavily on instructional design. Unlike earlier paradigms, GAI's capacity for highly adaptive, dialogic interaction serves to motivate learners and scaffold complex problem-solving (Pan et al., 2025; Wang et al., 2024). However, this very sophistication introduces the risk of fostering a polished dependency. Recent behavioural evidence cautions that intrusive or overly helpful support may foster 'simulated SRL' rather than durable competence, as learners' prompted elaborations can vanish once the tutor is withdrawn (Huang et al., 2025). This aligns with findings that overreliance on LLM-based feedback may even hinder performance when the support is unavailable, suggesting that short-term gains can mask a failure to internalize regulatory skills (Sun et al., 2025).

This discrepancy highlights a critical performance-competence divide. Current research reliance on immediate, single-wave measures leaves the durability of reported gains largely untested (Guan et al., 2025). This challenge is particularly acute for far transfer, defined as the application of metacognitive skills to structurally dissimilar tasks requiring different cognitive strategies (Schuster et al., 2020), which is the true hallmark of internalized competence. The available evidence, which rarely assesses far transfer, tends to focus on near transfer, leaving AI's capacity to foster robust, generalizable regulation an open and critical question. Therefore, while GAI demonstrates significant potential, the high variance suggests its effectiveness is not intrinsic to the technology but is contingent on designs that mitigate dependency. Without robust evidence of skill transfer, the technical richness that enables personalized companionship risks eroding learner autonomy. To harness GAI's strengths, future research must move beyond measuring immediate performance and adopt longitudinal designs that triangulate trace data, delayed achievement, and qualitative evidence. Furthermore, AI systems should be engineered with adaptive mechanisms, such as the guidance-based approaches that provide hints over direct answers (Lee et al., 2024), or scaffolds that strategically fade over time.

Phase-specific effects across AI paradigms

The substantial within-phase heterogeneity confirms that effectiveness depends on how well specific AI affordances address the distinct cognitive bottlenecks of each SRL phase.

In the forethought phase, GAI's marked advantage ($g=0.709$) over data-driven AI ($g=0.083$) reflects its unique capacity to operationalize abstract intentions. GAI systems excel in goal-setting because they force learners to externalize and refine their objectives through dialogue (Pan et al., 2025), whereas data-driven systems typically

support only the review of past performance without actively scaffolding future planning (Yang et al., 2024). The limited rule-based AI evidence ($k=2$) suggests these systems can prompt basic planning behaviours but lack the flexibility to address the nuances of motivational beliefs.

During the performance phase, differential effects reveal how technology aligns with the demand for real-time adaptation. GAI's superior effect ($g=0.938$) indicates that its co-regulatory capabilities effectively reduce cognitive load during complex problem-solving (Ng et al., 2024). Data-driven AI's moderate but variable effect ($g=0.352$) reflects its strength in self-monitoring via dashboards but reveals a limitation in offering the immediate, corrective guidance required when learners encounter obstacles (Lim et al., 2023). Rule-based systems show promise here for structured strategy training through explicit metacognitive prompts (Dever et al., 2023), though their rigidity limits their utility in dynamic tasks.

In the self-reflection phase, the pattern illuminates a critical distinction between information availability and meaning-making. GAI's effectiveness ($g=0.826$) derives from scaffolding the interpretive process of self-evaluation and causal attribution (Lee et al., 2024). Conversely, data-driven AI's minimal impact ($g=0.196$) reveals that presenting performance metrics alone is insufficient; learners require support to translate raw data into adaptive inferences for future learning (Mejeh et al., 2024). This disparity underscores that effective support for self-reflection requires dialogic guidance that helps learners internalize the 'why' behind their performance. Ultimately, these analyses reveal that AI effectiveness depends not on technological sophistication per se, but on the alignment between specific AI affordances and particular self-regulatory processes.

Contextual and methodological moderators of AI effectiveness

Our moderator analysis reveals how study-level characteristics shape the effectiveness of AI interventions. As established in the results section, this analysis must be interpreted with considerable caution because these demographic-like variables represent distal factors and broad categories conceal significant internal heterogeneity. Thus, the purpose of this discussion is not to establish definitive causal pathways but to explore systematic patterns that signal underlying mechanisms and generate targeted hypotheses for future experimental validation.

Educational stage

The analysis revealed varying effectiveness across educational stages, with AI interventions yielding a small effect for primary school students ($g=0.304$), a larger effect for secondary students ($g=0.636$), and a moderate effect for higher education learners ($g=0.517$). This progression reflects developmental differences in metacognitive readiness and capacity for processing AI-generated feedback. Secondary students, at a critical juncture in developing self-regulatory skills, appear particularly responsive to the structured scaffolding AI provides (Ng et al., 2024). The pronounced effect at this level may also indicate that secondary students face greater challenges in SRL tasks without AI support, making the differential impact of AI interventions more apparent. Primary students' limited response stems from their developing cognitive maturity, which constrains their ability to independently utilize AI feedback without substantial teacher mediation (Liu, Hwang, et al., 2024). Higher education students, while possessing advanced SRL skills, still derive moderate benefits from AI support in managing complex cognitive tasks (Zhang, 2024).

Disciplinary domain

Natural science interventions demonstrated substantially larger effects ($g=0.753$) compared to social science interventions ($g=0.211$), with the latter approaching the threshold of small effect size. This marked disparity reflects the alignment between current AI capabilities and disciplinary task characteristics. Natural science problems typically feature structured procedures and clear solution paths, enabling AI to provide targeted diagnostic feedback and procedural scaffolding (Dever et al., 2023). In contrast, social science tasks involve open-ended inquiry, critical analysis of multiple perspectives and synthesis of ambiguous information—cognitive processes that challenge AI's capacity to offer nuanced support for dialectical reasoning and higher order thinking (Zhang & Hew, 2024). This suggests that effective AI scaffolding must be calibrated not only to content but also to the epistemological demands and ways of knowing inherent to different disciplines.

Learning space

Fully online environments yielded larger effects ($g=0.732$) than blended settings ($g=0.409$). This advantage stems from continuous AI tool availability in online contexts, enabling consistent engagement with regulatory support while minimizing external interferences (Zhang, 2024). However, this finding must be interpreted against the backdrop that online learning environments typically present heightened SRL challenges, potentially amplifying the relative benefit of AI support. Blended environments face the additional complexity of integrating AI-mediated and teacher-led instruction, where misalignment between online and offline components can diminish overall effectiveness (Zhang et al., 2023). This challenge is amplified when AI-driven feedback conflicts with teacher guidance, or when its autonomous nature clashes with the pedagogical norms of examination-oriented cultures, which may predispose students to specific, teacher-validated learning strategies (Pan et al., 2025). Optimal implementation requires careful orchestration of both modalities to create coherent learning experiences that leverage the strengths of each approach (Hsu et al., 2023).

Intervention duration

The counter-intuitive finding that brief interventions (<2 weeks) produced the largest effects ($g=0.776$) compared to medium (2–10 weeks, $g=0.361$) and extended durations (>10 weeks, $g=0.374$) aligns with patterns observed in some technology-enhanced learning research. This phenomenon likely reflects what Hooshyar et al. (2024) identify as novelty effects, where initial exposure to AI support generates heightened engagement and immediate performance gains that may not translate into durable SRL skill development. Medium-duration interventions provide a more realistic window for meaningful skill acquisition, offering adequate practice opportunities while maintaining engagement (Liu, Xu, et al., 2024). Extended interventions beyond 10 weeks may encounter diminishing returns due to technology fatigue, over-reliance on external scaffolding or what Buçinca et al. (2021) term 'cognitive forcing functions'—where prolonged AI support inadvertently undermines the development of autonomous regulation. This temporal pattern underscores that effective AI support for SRL requires not merely duration but carefully designed scaffolding that progressively fades to promote internalization of regulatory strategies.

Measurement type

A critical finding emerged from the analysis of measurement type: effects differed significantly by outcome family. Studies using behavioural/event traces showed a large effect ($g=0.751$), more than double the small-to-moderate effect from self-reports ($g=0.369$). Prior meta-analyses noted similar numerical trends but lacked sufficient behavioural studies to reach significance (Zhao et al., 2025). Our sample provides the statistical power to establish this difference. We attribute the larger behavioural effect to AI's capacity to orchestrate observable in-task regulation, which is captured in rich trace data (Winne, 2022). By contrast, the attenuated self-report effect likely reflects two complementary mechanisms. First, learners' conservative self-reports may reflect an accurate awareness that while performance improved, durable (Mejeh et al., 2024; Zhang, 2024), autonomous competence has not yet consolidated, a conclusion supported by evidence that AI scaffolds can yield 'simulated SRL' that decays once removed (Huang et al., 2025) or foster overreliance (Sun et al., 2025). Second, from a methodological standpoint, survey instruments themselves are reactive; the process of completing them can prompt metacognitive reflection across all conditions, thereby diluting the measured treatment effect. Collectively, this discrepancy, when considered alongside the field's general lack of long-term follow-up data, suggests that current tools may be more effective at boosting immediate, tool-dependent performance than at cultivating durable, transferable competence.

Conclusion and implications

This meta-analysis synthesized a decade of research on AI-supported self-regulated learning and revealed a fundamental tension: artificial intelligence demonstrates clear potential to enhance self-regulated learning. However, this potential is not a guaranteed outcome of technological sophistication. Instead, it is unevenly realized across different phases of the regulatory cycle and remains heavily contingent on specific technological, pedagogical and contextual factors. Beyond mere technological advancement, the observed evolutionary trajectory—from rule-based systems through data-driven approaches to generative AI—signifies a paradigm shift in conceptualizing technology's role in supporting learner autonomy. This progression suggests that effective AI support for self-regulated learning requires not merely sophisticated algorithms but careful alignment between technological affordances and the cognitive-motivational processes underlying self-regulation. The high heterogeneity observed even within the most advanced AI paradigms underscores that efficacy is less about the tool itself and more about the pedagogical precision of its implementation. Building on these conclusions, the following implications for future research and practice are proposed to address the field's most pressing challenges.

The future of traditional AI: From silos to smart integration

Many educational settings still employ rule-based and data-driven AI systems (Sarker et al., 2024). While these systems are less effective than GAI in supporting motivational and reflective processes (Pelánek et al., 2024), future research should explore intelligent integration strategies. A hybrid approach where rule-based systems provide stable structural scaffolding while GAI adds personalized, emotionally responsive elements could leverage the predictability of traditional AI as a foundation upon which more sophisticated interactions are built.

The paradox of generative AI: Balancing empowerment and dependence

Although GAI shows the highest pooled outcomes, the significant variance suggests that its human-like responsiveness can be a double-edged sword: it may maximally support performance or inadvertently deprive learners of necessary cognitive struggles that foster genuine self-regulation (Lin & Yu, 2025). The scaffolding that makes GAI most supportive may also create insidious forms of intellectual dependence. Future work must develop 'productive failure' mechanisms within GAI systems—strategic moments where support is deliberately withdrawn to promote autonomous problem-solving—and investigate designs that feature strategically fading scaffolds over time.

Context is key: Adapting AI to diverse educational ecologies

While our moderator analysis reveals significant variations across educational stages and disciplines, the effectiveness of AI-supported SRL also depends on unexplored sociocultural dynamics. Future research must move beyond treating context as demographic variables to examining the relational and cultural dimensions—such as the interplay between teacher guidance and AI feedback or cross-cultural responses to AI's emphasis on autonomy—that fundamentally determine whether AI scaffolding translates into meaningful learning support.

From supporting performance to fostering competence: Bridging the core divide

This synthesis culminates in a fundamental challenge for the field: bridging the gap between enhancing immediate, scaffolded task performance and developing transferable, internalized self-regulatory competence. This distinction, highlighted by the differential effects across SRL phases and measurement types, emerges as our most pressing research imperative. Future assessment frameworks must incorporate delayed measurements in unsupported environments and longitudinal tracking, shifting focus from 'Does AI improve learning outcomes?' to 'Does AI cultivate learners who can regulate their learning independently?'

LIMITATIONS AND SUGGESTIONS

Although this study made noteworthy progress by systematically reviewing and quantitatively synthesizing research on AI technologies that support SRL, its conclusions must be interpreted in light of several limitations. First, the findings are shaped by the methodological characteristics of the primary research synthesized. Many included studies were conducted in controlled laboratory settings over brief periods. This context, combined with our analytical aggregation of diverse outcome measures at the phase level, means that the reported effect sizes represent a high-level synthesis whose generalizability to real-world implementation requires careful consideration. This challenge is compounded by the considerable variability in how the primary studies operationalized transfer, which often remained undifferentiated between near- and far-transfer tasks—a critical distinction for evaluating authentic competence.

Furthermore, our analysis reveals a fundamental measurement heterogeneity that constrains interpretation. The predominance of self-report measures introduces potential reactivity, as survey instruments may function as a metacognitive intervention for all participants, thus potentially underestimating the AI's true effect. Conversely, while the significantly larger effects found in behavioural data confirm AI's capacity to scaffold in-task performance, this pattern simultaneously

suggests that the observed gains may reflect a scaffold-dependent performance that dissipates upon tool withdrawal. This tension between different measurement modalities underscores the central interpretive challenge in bridging the performance-competence divide.

Second, beyond the constraints inherited from the primary studies, a further limitation lies in the nature of the moderators themselves analysed in this synthesis. Broad categories such as 'disciplinary domain' mask substantial internal heterogeneity, which constrains the precision of our findings and precludes strong causal claims. Future inquiry should therefore shift focus from whether AI is effective in broad contexts to how specific AI affordances mediate the relationship between scaffolded performance and internalized competence. This requires investigating more proximal, process-oriented variables. For instance, future meta-analyses could code for the interplay between specific AI-driven scaffolds (e.g. the fading of prompts, the nature of dialogic feedback) and fine-grained learner actions captured in trace data (e.g. strategy revision, metacognitive monitoring). Such an approach would yield more targeted, actionable insights into designing AI systems that effectively bridge the performance-competence divide.

FUNDING INFORMATION

This study is supported by a 2021 major project from the National Social Science Foundation of China (grant number 21 and ZD328).

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest with respect to the research, authorship and publication of this article.

DATA AVAILABILITY STATEMENT

All data presented in this table were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS APPROVAL STATEMENT

This study does not involve human or animal subjects.

ORCID

Jun Xu  <https://orcid.org/0000-0002-0557-2046>

Yuying Luo  <https://orcid.org/0009-0004-1871-7381>

Chengliang Wang  <https://orcid.org/0000-0003-2208-3508>

REFERENCES

- Akamatsu, D., Nakaya, M., & Koizumi, R. (2019). Effects of metacognitive strategies on the self-regulated learning process: The mediating effects of self-efficacy. *Behavioral Science*, 9(12), 128. <https://doi.org/10.3390/bs9120128>
- Bandura, A. (1991). Social cognitive theory of self-regulation. *Organizational Behavior and Human Decision Processes*, 50(2), 248–287. [https://doi.org/10.1016/0749-5978\(91\)90022-L](https://doi.org/10.1016/0749-5978(91)90022-L)
- Bouchet, F., Harley, J. M., & Azevedo, R. (2016). Can adaptive pedagogical agents' prompting strategies improve students' learning and self-regulation? In A. Micarelli, J. Stamper, & K. Panourgia (Eds.), *Intelligent tutoring systems* (pp. 368–374). Springer International Publishing. https://doi.org/10.1007/978-3-319-39583-8_43
- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce over-reliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 188:1–188:21. <https://doi.org/10.1145/3449287>
- Cabi, E., & Türkoğlu, H. (2025). The impact of a learning analytics based feedback system on students' academic achievement and self-regulated learning in a flipped classroom. *The International Review of Research in Open and Distributed Learning*, 26(1), 175–196. <https://doi.org/10.19173/irrodl.v26i1.7924>
- Chang, C., Hwang, G., & Gau, M. (2022). Promoting students' learning achievement and self-efficacy: A mobile chatbot approach for nursing training. *British Journal of Educational Technology*, 53(1), 171–188. <https://doi.org/10.1111/bjet.13158>

- Chang, W.-L., & Sun, J. C.-Y. (2024). Evaluating AI's impact on self-regulated language learning: A systematic review. *System*, 126, 103484. <https://doi.org/10.1016/j.system.2024.103484>
- Chen, S., Cheng, H., & Huang, Y. (2024). Emotion recognition in self-regulated learning: Advancing metacognition through AI-assisted reflections. In D. Kourkoulou, A.-O. (O.) Tzirides, B. Cope, & M. Kalantzis (Eds.), *Trust and inclusion in AI-mediated education: WHERE human learning meets learning machines* (pp. 185–212). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-64487-0_9
- Cheung, A. C. K., & Slavin, R. E. (2016). How methodological features affect effect sizes in education. *Educational Researcher*, 45(5), 283–292. <https://doi.org/10.3102/0013189X16656615>
- Chu, S.-T., Hwang, G.-J., Chien, S.-Y., & Chang, S.-C. (2023). Incorporating teacher intelligence into digital games: An expert system-guided self-regulated learning approach to promoting EFL students' performance in digital gaming contexts. *British Journal of Educational Technology*, 54(2), 534–553. <https://doi.org/10.1111/bjet.13260>
- Cohen, J. (2013). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge. <https://doi.org/10.4324/9780203771587>
- Dabbagh, N., & Kitsantas, A. (2013). The role of social media in self-regulated learning. *International Journal of Web Based Communities*, 9(2), 256–273. <https://doi.org/10.1504/IJWBC.2013.053248>
- Dahri, N. A., Yahaya, N., Al-Rahmi, W. M., Aldraiweesh, A., Alturki, U., Almutairy, S., Shutaleva, A., & Soomro, R. B. (2024). Extended TAM based acceptance of AI-powered ChatGPT for supporting metacognitive self-regulated learning in education: A mixed-methods study. *Heliyon*, 10(8), e29317. <https://doi.org/10.1016/j.heliyon.2024.e29317>
- Dai, C.-P., Ke, F., Pan, Y., Moon, J., & Liu, Z. (2024). Effects of artificial intelligence-powered virtual agents on learning outcomes in computer-based simulations: A meta-analysis. *Educational Psychology Review*, 36(1), 31. <https://doi.org/10.1007/s10648-024-09855-4>
- de Mooij, S., Lämsä, J., Lim, L., Aksela, O., Athavale, S., Bistolfi, I., Jin, F., Li, T., Azevedo, R., Bannert, M., Gašević, D., Järvelä, S., & Molenaar, I. (2025). A systematic review of self-regulated learning through integration of multimodal data and artificial intelligence. *Educational Psychology Review*, 37(2), 54. <https://doi.org/10.1007/s10648-025-10028-0>
- de Ruig, N. J., de Jong, P. F., & Zee, M. (2023). Stimulating elementary school students' self-regulated learning through high-quality interactions and relationships: A narrative review. *Educational Psychology Review*, 35(3), 71. <https://doi.org/10.1007/s10648-023-09795-5>
- Dent, A. L., & Koenka, A. C. (2016). The relation between self-regulated learning and academic achievement across childhood and adolescence: A meta-analysis. *Educational Psychology Review*, 28(3), 425–474. <https://doi.org/10.1007/s10648-015-9320-8>
- Dever, D. A., Sonnenfeld, N. A., Wiedbusch, M. D., Schmorow, S. G., Amon, M. J., & Azevedo, R. (2023). A complex systems approach to analyzing pedagogical agents' scaffolding of self-regulated learning within an intelligent tutoring system. *Metacognition and Learning*, 18(3), 659–691. <https://doi.org/10.1007/s11409-023-09346-x>
- Di Pietro, G., & Castaño Muñoz, J. (2025). A meta-analysis on the effect of technology on the achievement of less advantaged students. *Computers & Education*, 226, 105197. <https://doi.org/10.1016/j.compedu.2024.105197>
- Dignath, C., & Büttner, G. (2018). Teachers' direct and indirect promotion of self-regulated learning in primary and secondary school mathematics classes – Insights from video-based classroom observations and teacher interviews. *Metacognition and Learning*, 13(2), 127–157. <https://doi.org/10.1007/s11409-018-9181-x>
- Duffy, M. C., & Azevedo, R. (2015). Motivation matters: Interactions between achievement goals and agent scaffolding for self-regulated learning within an intelligent tutoring system. *Computers in Human Behavior*, 52, 338–348. <https://doi.org/10.1016/j.chb.2015.05.041>
- Gao, L., Lu, J., Shao, Z., Lin, Z., Yue, S., leong, C., Sun, Y., Zauner, R. J., Wei, Z., & Chen, S. (2024). Fine-tuned large language model for visualization system: A study on self-regulated learning in education. In *IEEE Transactions on Visualization and Computer Graphics* (pp. 1–11). Institute of Electrical and Electronics Engineers. <https://doi.org/10.1109/TVCG.2024.3456145>
- Guan, R., Raković, M., Chen, G., & Gašević, D. (2025). How educational chatbots support self-regulated learning? A systematic review of the literature. *Education and Information Technologies*, 30(4), 4493–4518. <https://doi.org/10.1007/s10639-024-12881-y>
- Guo, L. (2022). Using metacognitive prompts to enhance self-regulated learning and learning outcomes: A meta-analysis of experimental studies in computer-based learning environments. *Journal of Computer Assisted Learning*, 38(3), 811–832. <https://doi.org/10.1111/jcal.12650>
- Hadwin, A., Järvelä, S., & Miller, M. (2017). Self-regulation, co-regulation, and shared regulation in collaborative learning environments. In *Handbook of self-regulation of learning and performance* (2nd ed.). Routledge.
- Hamza, T. H., van Houwelingen, H. C., & Stijnen, T. (2008). The binomial distribution of meta-analysis was preferred to model within-study variability. *Journal of Clinical Epidemiology*, 61(1), 41–51. <https://doi.org/10.1016/j.jclinepi.2007.03.016>

- Harati, H., Sujo-Montes, L., Tu, C.-H., Armfield, S. J. W., & Yen, C.-J. (2021). Assessment and learning in knowledge spaces (ALEKS) adaptive system impact on students' perception and self-regulated learning skills. *Education Sciences*, 11(10), 10. <https://doi.org/10.3390/educsci11100603>
- Hedges, L. V. (2008). What are effect sizes and why do we need them? *Child Development Perspectives*, 2(3), 167–171. <https://doi.org/10.1111/j.1750-8606.2008.00060.x>
- Hedges, L. V., & Tipton, E. (2010). Meta-analysis. In A. Steptoe (Ed.), *Handbook of behavioral medicine* (pp. 909–921). Springer New York. https://doi.org/10.1007/978-0-387-09488-5_58
- Hemmler, Y., & Ifenthaler, D. (2024). Self-regulated learning strategies in continuing education: A systematic review and meta-analysis. *Educational Research Review*, 45, 100629. <https://doi.org/10.1016/j.edurev.2024.100629>
- Higgins, J. P. (2011). *The Cochrane Collaboration's tool for assessing risk of bias in randomised trials*. The Cochrane Collaboration. <https://bibliotheek.ehb.be:2370/content/bmj/343/bmj.d5928.full.pdf>
- Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11), 1539–1558. <https://doi.org/10.1002/sim.1186>
- Hilpert, J. C., Greene, J. A., & Bernacki, M. (2023). Leveraging complexity frameworks to refine theories of engagement: Advancing self-regulated learning in the age of artificial intelligence. *British Journal of Educational Technology*, 54(5), 1204–1221. <https://doi.org/10.1111/bjet.13340>
- Hooshyar, D., Weng, X., Sillat, P. J., Tammets, K., Wang, M., & Hämäläinen, R. (2024). The effectiveness of personalized technology-enhanced learning in higher education: A meta-analysis with association rule mining. *Computers & Education*, 223, 1–22. <https://doi.org/10.1016/j.compedu.2024.105169>
- Hsu, T.-C., Chang, C., & Jen, T.-H. (2023). Artificial intelligence image recognition using self-regulation learning strategies: Effects on vocabulary acquisition, learning anxiety, and learning behaviours of English language learners. *Interactive Learning Environments*, 32(6), 1–19. <https://doi.org/10.1080/10494820.2023.2165508>
- Huang, A. Y. Q., Lin, C.-Y., Su, S.-Y., & Yang, S. J. H. (2025). The impact of GenAI-enabled coding hints on students' programming performance and cognitive load in an SRL-based python course. *British Journal of Educational Technology*, 56(5), 1942–1972. <https://doi.org/10.1111/bjet.13589>
- Jin, S.-H., Im, K., Yoo, M., Roll, I., & Seo, K. (2023). Supporting students' self-regulated learning in online learning using artificial intelligence applications. *International Journal of Educational Technology in Higher Education*, 20(1), 37. <https://doi.org/10.1186/s41239-023-00406-5>
- Jones, A., Bull, S., & Castellano, G. (2018). "I know that now, I'm going to learn this next" promoting self-regulated learning with a robotic tutor. *International Journal of Social Robotics*, 10(4), 439–454. <https://doi.org/10.1007/s12369-017-0430-y>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33, 159–174.
- Lee, D., & Kwon, H. (2024). Meta-analysis on effects of artificial intelligence education in K-12 south Korean classrooms. *Education and Information Technologies*, 29(17), 22859–22894. <https://doi.org/10.1007/s10639-024-12738-4>
- Lee, H.-Y., Chen, P.-H., Wang, W.-S., Huang, Y.-M., & Wu, T.-T. (2024). Empowering ChatGPT with guidance mechanism in blended learning: Effect of self-regulated learning, higher-order thinking skills, and knowledge construction. *International Journal of Educational Technology in Higher Education*, 21(1), 16. <https://doi.org/10.1186/s41239-024-00447-4>
- Lee, Y.-F., Hwang, G.-J., & Chen, P.-Y. (2025). Technology-based interactive guidance to promote learning performance and self-regulation: A chatbot-assisted self-regulated learning approach. *Educational Technology Research and Development*, 73, 2279–2304. <https://doi.org/10.1007/s11423-025-10478-x>
- Li, H., Xu, J., & Luo, Y. (2024). The impact of mobile social media on knowledge sharing among vocational school teachers: A social cognitive career perspective. *Journal of Information Science*, 01655515241293754. online ahead of print. <https://doi.org/10.1177/01655515241293754>
- Liang, H., Hwang, G., Hsu, T., & Yeh, J. (2024). Effect of an AI-based chatbot on students' learning performance in alternate reality game-based museum learning. *British Journal of Educational Technology*, 55(5), 2315–2338. <https://doi.org/10.1111/bjet.13448>
- Lim, L., Bannert, M., van der Graaf, J., Singh, S., Fan, Y., Surendrannair, S., Rakovic, M., Molenaar, I., Moore, J., & Gaevi, D. (2023). Effects of real-time analytics-based personalized scaffolds on students' self-regulated learning. *Computers in Human Behavior*, 139, 107547. <https://doi.org/10.1016/j.chb.2022.107547>
- Lin, Y., & Yu, Z. (2025). Learner perceptions of artificial intelligence-generated pedagogical agents in language learning videos: Embodiment effects on technology acceptance. *International Journal of Human-Computer Interaction*, 41(2), 1606–1627. <https://doi.org/10.1080/10447318.2024.2359222>
- Liu, C.-C., Hwang, G.-J., Yu, P., Tu, Y.-F., & Wang, Y. (2025). Effects of an automated corrective feedback-based peer assessment approach on students' learning achievement, motivation, and self-regulated learning conceptions in foreign language pronunciation. *Educational Technology Research and Development*, 73, 2403–2424. <https://doi.org/10.1007/s11423-025-10484-z>

- Liu, Y., Xu, G., Yuan, S., Zhou, C., & Wang, C. (2024). Assessment as learning: Evidence based on meta-analysis and quantitative ethnography research. *Studies in Educational Evaluation*, 83, 101423. <https://doi.org/10.1016/j.stueduc.2024.101423>
- Liu, Z.-M., Hwang, G.-J., Chen, C.-Q., Chen, X.-D., & Ye, X.-D. (2024). Integrating large language models into EFL writing instruction: Effects on performance, self-regulated learning strategies, and motivation. *Computer Assisted Language Learning*, 1–25. Online ahead of print. <https://doi.org/10.1080/09588221.2024.2389923>
- Long, Y., & Alevan, V. (2013). Supporting students' self-regulated learning with an open learner model in a linear equation tutor. In H. C. Lane, K. Yacef, J. Mostow, & P. Pavlik (Eds.), *Artificial intelligence in education* (pp. 219–228). Springer. https://doi.org/10.1007/978-3-642-39112-5_23
- Lu, W., & Lin, C. (2024). Meta-analysis of influencing factors on the use of artificial intelligence in education. *The Asia-Pacific Education Researcher*, 34, 611–627. <https://doi.org/10.1007/s40299-024-00883-w>
- Luckin, R., Cukurova, M., Kent, C., & du Boulay, B. (2022). Empowering educators to be AI-ready. *Computers and Education: Artificial Intelligence*, 3, 100076. <https://doi.org/10.1016/j.caeai.2022.100076>
- Mejeh, M., Sarbach, L., & Hascher, T. (2024). Effects of adaptive feedback through a digital tool – A mixed-methods study on the course of self-regulated learning. *Education and Information Technologies*, 29, 1–43. <https://doi.org/10.1007/s10639-024-12510-8>
- Mohammadzadeh, A., & Sarkhosh, M. (2018). The effects of self-regulatory learning through computer-assisted intelligent tutoring system on the improvement of EFL learners' speaking ability. *International Journal of Instruction*, 11(2), 167–184.
- Molenaar, I., de Mooij, S., Azevedo, R., Bannert, M., Jarvela, S., & Gaevi, D. (2023). Measuring self-regulated learning and the role of AI: Five years of research using multimodal multichannel data. *Computers in Human Behavior*, 139, 107540. <https://doi.org/10.1016/j.chb.2022.107540>
- Ng, D. T. K., Tan, C. W., & Leung, J. K. L. (2024). Empowering student self-regulated learning and science education through ChatGPT: A pioneering pilot study. *British Journal of Educational Technology*, 55(4), 1328–1353. <https://doi.org/10.1111/bjet.13454>
- Orwin, R. G. (1983). A fail-SafeN for effect size in meta-analysis. *Journal of Educational Statistics*, 8(2), 157–159. <https://doi.org/10.3102/10769986008002157>
- Pan, M., Lai, C., & Guo, K. (2025). Effects of GenAI-empowered interactive support on university EFL students' self-regulated strategy use and engagement in reading. *The Internet and Higher Education*, 65, 100991.
- Pelánek, R., Effenberger, T., & Jarušek, P. (2024). Personalized recommendations for learning activities in on-line environments: A modular rule-based approach. *User Modeling and User-Adapted Interaction*, 34(4), 1399–1430. <https://doi.org/10.1007/s11257-024-09396-z>
- Pereira, J., & Díaz, O. (2021). Struggling to keep tabs on capstone projects: A chatbot to tackle student procrastination. *ACM Transactions on Computing Education*, 22, 1–22. <https://doi.org/10.1145/3469127>
- Peters, J. L., Sutton, A. J., Jones, D. R., Abrams, K. R., & Rushton, L. (2007). Performance of the trim and fill method in the presence of publication bias and between-study heterogeneity. *Statistics in Medicine*, 26(25), 4544–4562. <https://doi.org/10.1002/sim.2889>
- Pigott, T. D., & Polanin, J. R. (2020). Methodological guidance paper: High-quality meta-analysis in a systematic review. *Review of Educational Research*, 90(1), 24–46. <https://doi.org/10.3102/0034654319877153>
- Pintrich, P. (2000). The role of goal orientation in self-regulated learning. In *Handbook of self-regulation* (pp. 451–502). Academic Press. <https://doi.org/10.1016/B978-012109890-2/50043-3>
- Qiao, H., & Zhao, A. (2023). Artificial intelligence-based language learning: Illuminating the impact on speaking skills and self-regulation in Chinese EFL context. *Frontiers in Psychology*, 14, 1255594. <https://doi.org/10.3389/fpsyg.2023.1255594>
- Rawas, S. (2024). ChatGPT: Empowering lifelong learning in the digital age of higher education. *Education and Information Technologies*, 29(6), 6895–6908. <https://doi.org/10.1007/s10639-023-12114-8>
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3), 638–641. <https://doi.org/10.1037/0033-2909.86.3.638>
- Sarker, I. H., Janicke, H., Ferrag, M. A., & Abuadba, A. (2024). Multi-aspect rule-based AI: Methods, taxonomy, challenges and directions toward automation, intelligence and transparent cybersecurity modeling for critical infrastructures. *Internet of Things*, 25, 101110.
- Sawatsky, A. P., Beckman, T. J., Edakkanambeth Varayil, J., Mandrekar, J. N., Reed, D. A., & Wang, A. T. (2015). Association between study quality and publication rates of medical education abstracts presented at the Society of General Internal Medicine Annual Meeting. *Journal of General Internal Medicine*, 30(8), 1172–1177. <https://doi.org/10.1007/s11606-015-3269-7>
- Schuster, C., Stebner, F., Leutner, D., & Wirth, J. (2020). Transfer of metacognitive skills in self-regulated learning: An experimental training study. *Metacognition and Learning*, 15(3), 455–477. <https://doi.org/10.1007/s11409-020-09237-5>
- Sharma, K., Nguyen, A., & Hong, Y. (2024). Self-regulation and shared regulation in collaborative learning in adaptive digital learning environments: A systematic review of empirical studies. *British Journal of Educational Technology*, 55(4), 1398–1436. <https://doi.org/10.1111/bjet.13459>

- Sposato, M. (2025). Artificial intelligence in educational leadership: A comprehensive taxonomy and future directions. *International Journal of Educational Technology in Higher Education*, 22(1), 20. <https://doi.org/10.1186/s41239-025-00517-1>
- Sun, D., Xu, P., Zhang, J., Liu, R., & Zhang, J. (2025). How self-regulated learning is affected by feedback based on large language models: Data-driven sustainable development in computer programming learning. *Electronics*, 14(1), 1. <https://doi.org/10.3390/electronics14010194>
- Tian, Q., & Zheng, X. (2024). Effectiveness of online collaborative problem-solving method on students' learning performance: A meta-analysis. *Journal of Computer Assisted Learning*, 40(1), 326–341. <https://doi.org/10.1111/jcal.12884>
- Wang, W.-S., Pedaste, M., Lin, C.-J., Lee, H.-Y., Huang, Y.-M., & Wu, T.-T. (2024). Signaling feedback mechanisms to promoting self-regulated learning and motivation in virtual reality transferred to real-world hands-on tasks. *Interactive Learning Environments*, 32, 1–16. <https://doi.org/10.1080/10494820.2024.2331151>
- Weber, F., Wambsganss, T., & Söllner, M. (2025). Enhancing legal writing skills: The impact of formative feedback in a hybrid intelligence learning environment. *British Journal of Educational Technology*, 56(2), 650–677. <https://doi.org/10.1111/bjet.13529>
- Wei, L. (2023). Artificial intelligence in language instruction: Impact on English learning achievement, L2 motivation, and self-regulated learning. *Frontiers in Psychology*, 14, 1261955. <https://doi.org/10.3389/fpsyg.2023.1261955>
- Winne, P., & Hadwin, A. (1998). Studying as self-regulated learning. In *Metacognition in educational theory and practice* (Vol. 93, pp. 277–304). Lawrence Erlbaum Associates.
- Winne, P. H. (2022). Learning analytics for self-regulated learning. In C. Lang, G. Siemens, & a. F. Wise (Eds.), *The handbook of learning analytics* (2nd ed., pp. 78–85). SOLAR. <https://doi.org/10.18608/hla22.008>
- Wong, J., Baars, M., Davis, D., Zee, T. V. D., Houben, G.-J., & Paas, F. (2019). Supporting self-regulated learning in online learning environments and MOOCs: A systematic review. *International Journal of Human-Computer Interaction*, 35, 356–373. <https://doi.org/10.1080/10447318.2018.1543084>
- Wu, R., & Yu, Z. (2024). Do AI chatbots improve students learning outcomes? Evidence from a meta-analysis. *British Journal of Educational Technology*, 55(1), 10–33. <https://doi.org/10.1111/bjet.13334>
- Wu, X.-Y. (2024). Exploring the effects of digital technology on deep learning: A meta-analysis. *Education and Information Technologies*, 29(1), 425–458. <https://doi.org/10.1007/s10639-023-12307-1>
- Xu, J., Liu, F., Luo, Y., & Xu, X. (2025). Stage-specific impacts of digital technologies on supply chain resilience: Meta-analytic evidence for continuous process improvement. *Business Process Management Journal*, 1–22. Online ahead of print, <https://doi.org/10.1108/BPMJ-04-2025-0550>
- Yang, C. C. Y., Wu, J.-Y., & Ogata, H. (2024). Learning analytics dashboard-based self-regulated learning approach for enhancing students' e-book-based blended learning. *Education and Information Technologies*, 30, 35–56. <https://doi.org/10.1007/s10639-024-12913-7>
- Zhang, L., & Hew, K. F. (2024). Leveraging unlabeled data: Fostering self-regulated learning in online education with semi-supervised recommender systems. *Education and Information Technologies*, 30, 7117–7142. <https://doi.org/10.1007/s10639-024-13111-1>
- Zhang, M. (2024). Enhancing self-regulation and learner engagement in L2 speaking: Exploring the potential of intelligent personal assistants within a learning-oriented feedback framework. *BMC Psychology*, 12(1), 421. <https://doi.org/10.1186/s40359-024-01917-0>
- Zhang, Z., Xu, Q., Koehler, A. A., & Newby, T. (2023). Comparing blended and online learners' self-efficacy, self-regulation, and actual learning in the context of educational technology. *Online Learning*, 27(4), 295–314.
- Zhao, Y., Li, Y., Ma, S., Xu, Z., & Zhang, B. (2025). A meta-analysis of the correlation between self-regulated learning strategies and academic performance in online and blended learning environments. *Computers & Education*, 230, 105279. <https://doi.org/10.1016/j.compedu.2025.105279>
- Zimmerman, B. J. (1989). A social cognitive view of self-regulated academic learning. *Journal of Educational Psychology*, 81(3), 329–339. <https://doi.org/10.1037/0022-0663.81.3.329>
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory Into Practice*, 41(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2

How to cite this article: Xu, J., Luo, Y., Wang, C., Wang, M., & Wu, Y. (2026). AI support in self-regulated learning: A decade of technological evolution and meta-analysis. *British Journal of Educational Technology*, 00, 1–30. <https://doi.org/10.1111/bjet.70058>

APPENDIX A

CHARACTERISTICS OF INCLUDED STUDIES

ID	Study (author and year)	Publication type	Title	N	AI paradigm	MERSQI
1	Bouchet, F., et al. (2013)	Conference	Impact of Different Pedagogical Agents' Adaptive Self-regulated Prompting Strategies on Learning with MetaTutor	40	Rule-based	12
2	Bouchet, F., et al. (2016)	Conference	Can adaptive pedagogical agents' prompting strategies improve students' learning and self-regulation?	58	Rule-based	14
3	Cabı, E., & Türkoğlu, H. (2025)	Journal	The Impact of a Learning Analytics Based Feedback System on Students' Academic Achievement and Self-Regulated Learning in a Flipped Classroom	127	Data-driven	13.5
4	Chang, C.-Y., et al. (2024)	Journal	Promoting students' learning achievement and self-efficacy: A mobile chatbot approach for nursing training	36	Data-driven	12.5
5	Chu, S.-T., et al. (2023)	Journal	Incorporating teacher intelligence into digital games: An expert system-guided self-regulated learning approach...	44	Rule-based	12.5
6	Dever, D. A., et al. (2023)	Journal	A complex systems approach to analysing pedagogical agents' scaffolding of self-regulated learning within an intelligent tutoring system	117	Data-driven	13
7	Duffy, M. C., & Azevedo, R. (2015)	Journal	Motivation matters: Interactions between achievement goals and agent scaffolding for self-regulated learning within an intelligent tutoring system	83	Rule-based	12.5
8	Fleur, D. S., et al. (2023)	Journal	IguideME: Supporting Self-Regulated Learning and Academic Achievement with Personalized Peer-Comparison Feedback in Higher Education	41	Data-driven	14.5
9	Harati, H., et al. (2021)	Journal	Assessment and Learning in Knowledge Spaces (ALEKS) Adaptive System Impact on Students' Perception and Self-Regulated Learning Skills	240	Data-driven	10
10	Hilpert, J. C., et al. (2023)	Journal	Leveraging complexity frameworks to refine theories of engagement: Advancing self-regulated learning in the age of artificial intelligence	143	Data-driven	14
11	Hsu, T.-C. (2023)	Journal	Artificial Intelligence image recognition using self-regulation learning strategies: effects on vocabulary acquisition...	47	Generative AI	13

ID	Study (author and year)	Publication type	Title	N	AI paradigm	MERSQI
12	Huang, A. Y. Q., et al. (2025)	Journal	The impact of GenAI-enabled coding hints on students' programming performance and cognitive load in an SRL-based Python course	31	Generative AI	14
13	Jones, A., et al. (2018)	Conference	'I Know That Now, I'm Going to Learn This Next' Promoting Self-regulated Learning with a Robotic Tutor	45	Rule-based	13
14	Lee, H.-Y., et al. (2024)	Journal	Empowering ChatGPT with guidance mechanism in blended learning: effect of self-regulated learning, higher-order thinking skills...	61	Generative AI	14
15	Liang, H.-Y., et al. (2024)	Journal	Effect of an AI-based chatbot on students' learning performance in alternate reality game-based museum learning	39	Generative AI	13.5
16	Liao, X., et al. (2024)	Journal	Design and implementation of an AI-enabled visual report tool as formative assessment to promote learning achievement...	125	Data-driven	12.5
17	Lim, L., et al. (2023)	Journal	Effects of real-time analytics-based personalized scaffolds on students' self-regulated learning	64	Rule-based	13.5
18	Liu, C.-C., et al. (2025)	Journal	Effects of an automated corrective feedback-based peer assessment approach on students' learning achievement, motivation, and self-regulated learning...	66	Data-driven	13.5
19	Liu, Z.-M., et al. (2024)	Journal	Integrating large language models into EFL writing instruction: effects on performance, self-regulated learning strategies, and motivation	65	Generative AI	14
20	Long, Y., & Alevén, V. (2013)	Conference	Supporting Students' Self-Regulated Learning with an Open Learner Model in a Linear Equation Tutor	56	Rule-based	12.5
21	Mejeh, M., et al. (2023)	Journal	Effects of adaptive feedback through a digital tool – a mixed-methods study on the course of self-regulated learning	66	Data-driven	12.5
22	Ng, D. T. K., et al. (2024)	Journal	Empowering student self-regulated learning and science education through ChatGPT: A pioneering pilot study	74	Generative AI	12.5
23	Pan, M., et al. (2025)	Journal	Effects of GenAI-empowered interactive support on university EFL students' self-regulated strategy use and engagement in reading	61	Generative AI	14
24	Pereira, J., & Díaz, Ó. (2021)	Journal	Struggling to Keep Tabs on Capstone Projects: A Chatbot to Tackle Student Procrastination	53	Data-driven	11

ID	Study (author and year)	Publication type	Title	N	AI paradigm	MERSQI
25	Qiao, H., & Zhao, A. (2023)	Journal	Artificial intelligence-based language learning: illuminating the impact on speaking skills and self-regulation in Chinese EFL context	93	Data-driven	15
26	Sarkhosh, M. (2018)	Journal	The Effects of Self-Regulatory Learning through Computer-Assisted Intelligent Tutoring System on the Improvement of EFL Learner' Speaking Ability	30	Rule-based	10.5
27	Su, J. M. (2014)	Conference	A Self-Regulated Learning System to Support Adaptive Scaffolding in Hypermedia-based Learning Environments	86	Rule-based	13
28	Sun, D., et al. (2025)	Journal	How Self-Regulated Learning Is Affected by Feedback Based on Large Language Models: Data-Driven Sustainable Development...	63	Generative AI	12.5
29	Wang, W.-S., et al. (2024)	Journal	Signaling feedback mechanisms to promoting self-regulated learning and motivation in virtual reality transferred to real-world...	71	Generative AI	15
30	Weber, F., et al. (2025)	Journal	Enhancing legal writing skills: The impact of formative feedback in a hybrid intelligence learning environment	43	Data-driven	14.5
31	Wei, L. (2024)	Journal	Artificial intelligence in language instruction: impact on English learning achievement, L2 motivation, and self-regulated learning	60	Data-driven	13.5
32	Wirzberger, M., et al. (2023)	Journal	Optimal feedback improves behavioral focus during self-regulated computer-based work	127	Data-driven	13.5
33	Yang, C. C. Y., et al. (2024)	Journal	Learning analytics dashboard-based self-regulated learning approach for enhancing students' e-book-based blended learning	83	Data-driven	13
34	Zhang, L., & Hew, K. F. (2024)	Journal	Leveraging unlabeled data: Fostering self-regulated learning in online education with semi-supervised recommender systems	84	Data-driven	10
35	Zhang, M. (2024)	Journal	Enhancing self-regulation and learner engagement in L2 speaking: exploring the potential of intelligent personal assistants...	54	Data-driven	13.5

From access to adherence: fostering professional capital and continued engagement in blended teacher workshops

Journal of
Professional
Capital and
Community

Haojun Li
School of Education, Zhejiang University of Technology, Hangzhou, China
Yuying Luo
School of Management, Shanghai University, Shanghai, China
Junyi Wang and Yiyang Ye
School of Education, Zhejiang University of Technology, Hangzhou, China, and
Jun Xu
Faculty of Education, East China Normal University, Shanghai, China

Received 20 December 2025
Revised 14 April 2026
20 May 2026
Accepted 24 May 2026

Abstract

Purpose – As teacher professional development migrates to digital formats, ensuring continuous engagement transcends mere technological adoption. Without sustained participation, online initiatives fail to impact teaching practice or build professional capital. This study investigates the factors driving teachers to continue participating in extensive online professional learning communities, specifically expert teacher workshops in China.

Design/methodology/approach – Viewing these workshops as distinct social ecosystems, this research integrated the technology acceptance model and expectation confirmation theory within a social support framework to model teacher motivation comprehensively. Data from 527 practicing teachers were analyzed using partial least squares structural equation modeling.

Findings – A dual pathway mechanism drives sustained engagement. A rational pathway is anchored in perceived usefulness, while an affective pathway is shaped by overall satisfaction. Multidimensional social support significantly bolsters both routes. Instrumental support lowers technical barriers, whereas informational and emotional support confirm platform value and foster collegiality.

Originality/value – Moving beyond single-theory explanations, this study identifies a socially embedded continuance mechanism that helps explain how digitally mediated participation may contribute to professional capital over time. The model does not directly measure professional capital; rather, it shows how sustained participation, supported by instrumental, informational and emotional support, creates conditions for the development of human, social and decisional capital within expert-led professional communities.

Keywords Professional learning communities, Social support, TAM-ECT, Continuance intention, Expert teacher workshops

Paper type Research article

1. Introduction

The rapid evolution of Fourth Industrial Revolution technologies, encompassing artificial intelligence, big data, and virtual reality, continues to fundamentally reshape the educational ecosystem. These shifts place increasingly rigorous demands on teachers' digital competencies and pedagogical perspectives (Teo *et al.*, 2021). Within the sphere of in-service teacher education, the sudden proliferation of generative AI tools compels educators not merely to master technical innovations rapidly but to immerse themselves in a professional development ecosystem that supports ongoing learning and adaptive resilience (Brandão *et al.*,



Journal of Professional Capital and
Community
© Emerald Publishing Limited
e-ISSN: 2056-9556
p-ISSN: 2056-9548
DOI 10.1108/JPC-12-2025-0145

Funding: This work was supported by the National Social Science Fund of China (Grant No. 16BTQ084).
Declaration of conflicting interest: The authors declare no competing financial interest.

2024). However, the variability in how teachers accept and integrate these new technologies (Chen *et al.*, 2025) creates unprecedented challenges for maintaining continuous professional growth.

Teacher professional development (TPD) is widely recognized as a primary driver for enhancing educational quality and student learning outcomes (Darling-Hammond *et al.*, 2017), serving as a catalyst for broader instructional innovation and the transformation of teaching practices (Berkovich and Becher, 2025). Compared to traditional offline training, which is often constrained by temporal and spatial rigidities, online training offers a more flexible, efficient, and cost-effective avenue for development (McCoy and McNaughton, 2021). Furthermore, it provides the necessary infrastructure to foster the digital skills required for cutting-edge educational technologies, such as generative AI (Lan *et al.*, 2025).

Against this backdrop, Expert Teacher Workshops (ETWs), commonly rendered in English-language scholarship as Master Teacher Studios (e.g., Zheng *et al.*, 2019), have become a distinctive form of blended teacher development in China since the early twenty-first century. Within China's public teacher-development system, ETWs are commonly organized through education authorities, schools, and platform-supported professional networks. Rather than operating as stand-alone short courses, ETWs are policy-supported, cross-school professional learning arrangements organized around a recognized master teacher and sustained through ongoing online and offline activities. In Zhejiang, this model is platform-mediated through *Zhijianghui*, which hosts named-teacher studio functions and related teacher-development applications. Participation is typically organized through role-differentiated arrangements and platform-based entry procedures rather than through open, one-off enrollment. ETWs should therefore be understood less as temporary training events than as institutionally organized infrastructures for sustained teacher learning, expert mentoring, and regional teacher development.

Conceptually, ETWs overlap with professional learning communities (PLCs) and communities of practice (CoPs) because they rely on sustained interaction, shared inquiry, and the improvement of teaching practice (Stoll *et al.*, 2006). They also resemble online teacher professional development programs in that participation is mediated by digital platforms, curated resources, and organized learning tasks. Yet ETWs are not a direct equivalent of any single international form. Their distinctive features lie in named expert leadership, formal role differentiation, cross-school recruitment, and platform-based governance backed by educational authorities. ETWs may therefore be understood as a Chinese, institutionalized variant of online professional learning community design—one that combines vertical mentoring with horizontal peer exchange and connects professional learning to an explicit regional development agenda.

Nevertheless, the transition to online environments is not without friction. Online training frequently presents challenges, including difficulties in technology adaptation, insufficient depth of interaction, and alarmingly low rates of sustained participation (Mulaimović *et al.*, 2025). Research underscores that a teacher's intention to continue with online training is not merely a reflection of their initial evaluation but is a decisive factor in whether professional development programs can achieve a long-term, transformative impact on the profession (Garet *et al.*, 2001).

In the domain of educational technology, most prior studies have focused on the initial adoption of specific technologies by students or teachers (W. Zheng *et al.*, 2024). While international research on online teacher professional development communities, professional learning networks, and networked communities of practice establishes that long-term involvement depends on the quality of interaction, the usability of shared resources, and the strength of collegial exchange (Lantz-Andersson *et al.*, 2018; Lu *et al.*, 2024), the existing literature primarily emphasizes early participation patterns, policy frameworks, or basic platform design (Takizawa *et al.*, 2024), and has often conceptualized professional networks as relatively decentralized or voluntary affiliations. Far fewer studies have examined continuance intention within institutionally organized, expert-led, and role-differentiated blended

communities such as Chinese ETWs, or theorized how structured external social support is translated into teachers' cognitive and affective evaluations over time. To address this gap, this study treats ETWs as socially organized professional learning environments and investigates how sustained engagement is shaped jointly by platform experience and the multidimensional support structures surrounding participation.

To rigorously examine these operational influences, this study introduces Social Support Theory to capture the multifaceted external assistance teachers receive during training. Building upon this foundation, we employ the Technology Acceptance Model (TAM), specifically its constructs of perceived ease of use (PEOU) and perceived usefulness (PU), to understand how such social support translates into rational evaluations of the platform's value (Almenara *et al.*, 2021). However, recognizing that TAM alone may not fully account for the emotional experiences and expectation assessments that emerge during long-term usage (Teo, 2015), we incorporate Expectation Confirmation Theory (ECT). This allows for the integration of satisfaction and continuance intention at the behavioral level, ultimately forming a comprehensive three-tier framework of external support, perception, and behavior.

Guided by this integrated perspective, the study addresses three core questions:

- RQ1. How does social support shape teachers' continuance intention via their perceived experiences of the platform?
- RQ2. What is the relative significance of rational evaluation (e.g. perceived usefulness) versus emotional experience (e.g., satisfaction) in determining the decision to maintain engagement?
- RQ3. Does merging Social Support Theory with the TAM-ECT framework provide a more comprehensive explanation for teachers' continued engagement in Expert Teacher Workshops?

A conceptual clarification is necessary here. Although the title refers to professional capital, the empirical model does not directly measure human, social, or decisional capital as outcomes. Instead, it examines continuance intention as a proximal mechanism through which professional capital may be developed in digitally mediated professional communities. Sustained participation is analytically important because it creates conditions for expert knowledge to circulate, collegial trust to deepen, and professional judgment to be refined through repeated interaction.

By empirically testing a Partial Least Squares Structural Equation Model (PLS-SEM), this research elucidates how instrumental, emotional, and informational support collectively fortify teacher training. We verify the parallel influence of rational evaluation and emotional experience on continued use, demonstrating that teachers exhibit notably stable continuance intentions under the dual influence of perceived usefulness and satisfaction. These findings offer a nuanced model for designing effective and enduring online TPD, highlighting the necessity of cultivating a socially supportive ecosystem that addresses both the practical and emotional needs of the professional community.

2. Theoretical framework and hypotheses development

Against the backdrop of increasingly close integration between digital technologies and educational concepts, teacher professional development has gradually shifted from traditional offline approaches to blended formats that combine real-time interaction with self-directed learning (Lu *et al.*, 2024; Xu *et al.*, 2026). In this context, Expert Teacher Workshops (ETWs) offer a more flexible and personalized environment for professional growth by incorporating expert guidance, online resources, and opportunities for independent study (Liu, 2022). In Zhejiang Province, ETWs are typically organized as provincially or locally recognized workshops led by a master teacher and structured through differentiated roles—such as

JPCC

workshop leaders, subject leaders, core trainees, and wider network participants. Participation is commonly mediated through platform-based review procedures on systems like Zhijianghui, rather than open, one-off enrollment. This organizational design matters analytically because it makes ETWs simultaneously a policy-supported training arrangement, a professional community, and a platform-mediated network. Ensuring sustained and in-depth teacher engagement in such extensive and cross-regional blended training platforms is therefore crucial for maximizing their impact.

To systematically examine teachers' continuance intention within these blended training platforms and its associated mechanisms, this study draws upon three theoretical perspectives: social support theory, the TAM, and the ECT. These perspectives are combined sequentially rather than additively. First, at the external level, Social Support Theory captures the relational and organizational conditions under which teachers receive instrumental, informational, and emotional assistance from experts, peers, and the broader workshop structure. Second, at the perceptual level, we investigate how TAM concepts (i.e. perceived ease of use and perceived usefulness) shape teachers' evaluations of the platform. In this model, TAM explains how external support is translated into rational appraisals of usability and value. Finally, at the level of continuous motivation, we employ ECT to elucidate the relationships among satisfaction, expectation confirmation, and continuance intention.

ECT is therefore used to explain how post-adoption experience confirms or disconfirms initial expectations and whether those experiences are strong enough to sustain continued engagement. This sequencing allows the integrated model to answer the three research questions more fully than any single theory alone. The subsections below detail the core concepts of social support theory, TAM, and ECT, illustrating their relevance to this study and presenting the hypotheses and structural equation models that guide our empirical work.

2.1 Social support theory

Social Support Theory provides a critical lens for understanding the relational dynamics within online teacher training, moving beyond individual coping mechanisms to the collective construction of professional capital. Cobb (1976) originally conceptualized social support as information leading individuals to believe they are cared for and valued, a notion that remains pivotal in digital environments where isolation can impede learning. In the context of TPD, social support acts as a buffer against the anxieties of technological disruption while simultaneously fueling self-efficacy and resilience (Kärner *et al.*, 2021).

Adopting the framework by Jolly *et al.* (2021), we categorize this support into three distinct dimensions that collectively foster a supportive ecosystem: (1) *Instrumental Support* serves as the logistical foundation, offering technical aid and operational guidance to lower the barriers to digital entry (Zhang *et al.*, 2023). (2) *Informational Support* functions as the intellectual currency of the community, providing instructional feedback and facilitating knowledge co-creation across regional boundaries (Karatay and Hegelheimer, 2021). (3) *Emotional Support* offers the necessary psychological validation through peer recognition and encouragement, crucial for sustaining motivation amidst professional pressures (Peng *et al.*, 2024).

By integrating these dimensions with TAM and ECT, this study treats social support not simply as an external input but as a condition that helps transform a digital platform from a technical tool into a professional learning environment. This framing is consistent with Hargreaves and Fullan's (2012) conception of professional capital as the interrelationship of human, social, and decisional capital. In the present model, instrumental support helps teachers participate with confidence, informational support helps professional knowledge circulate and become usable, and emotional support reinforces the trust and reciprocity needed for collective work. The model therefore does not claim to measure professional capital directly; rather, it examines a participation mechanism through which these forms of capital may gradually be developed over time.

2.2 Technology acceptance model

Davis (1989) proposed the TAM, drawing on the theory of reasoned action, to elucidate the key factors influencing users’ acceptance and utilization of new technologies. The model primarily revolves around two constructs: perceived usefulness (PU) and perceived ease of use (PEOU). Perceived usefulness refers to an individual’s belief that using a particular technology will enhance work or learning performance, whereas perceived ease of use denotes one’s perception of how effortless it is to operate the technology, often indirectly shaped by factors such as social influence (Kemp et al., 2024).

Within online teacher training, if teachers perceive the platform as easy to use and see its genuine value in advancing their professional competencies, they tend to develop favorable attitudes toward the system and exhibit higher intentions to continue using it (Almenara et al., 2021). Importantly, however, technology adoption among teachers is not solely an individual cognitive phenomenon; it is also significantly affected by external social support (Ifinedo et al., 2018).

As shown in Figure 1, instrumental support, information support, and emotional support each enhance teachers’ perceptions of platform ease of use and usefulness through distinct channels:

- (1) Emotional support (ES) fosters a climate of trust and belonging, thereby reinforcing professional identity and self-efficacy (Pozo-Rico et al., 2023).
- (2) Information support (INS) reduces uncertainty about the platform’s value by supplying high-quality teaching materials and timely feedback, thereby cultivating more positive expectations of platform functionality (Qin et al., 2017).
- (3) Instrumental support (IS) alleviates concerns about technological complexity through technical assistance and strategy training, ultimately boosting digital literacy and promoting sustained platform engagement (Yao and Wang, 2024).

Adopting social support theory as part of TAM’s conceptual framework, this study examines how these three forms of support feed into perceptions of ease of use and usefulness, thus shaping teachers’ attitudes toward platform utilization and their continuance intention. Based on the above discussion, we propose the following hypotheses:

- H1. Instrumental support positively influences perceived ease of use.
- H2. Emotional support positively influences perceived usefulness.
- H3. Information support positively influences perceived usefulness.

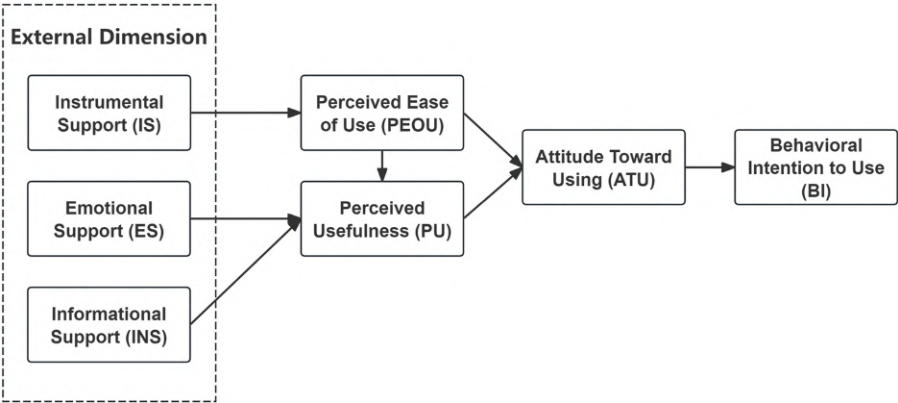


Figure 1. TAM model with external support. Source: Authors’ own work

2.3 Expectation confirmation theory

Originally introduced by [Oliver \(1977\)](#) and later extended to the context of information system continuance by [Bhattacharjee \(2001\)](#), the ECT elucidates how the alignment between users’ preliminary expectations and their actual experiences shapes satisfaction and subsequent usage intentions ([Cheng, 2022](#)). In ECT (see [Figure 2](#)), the central constructs include expectation confirmation, perceived usefulness, satisfaction, and continuance intention. When individuals’ real-world interactions with a system meet or exceed their prior expectations, the resulting elevated confirmation fosters higher satisfaction and ongoing motivation to use the system ([Bhattacharjee, 2001](#)).

In the context of online teacher training, teachers often begin with certain assumptions about platform functionality, interaction modes, and developmental benefits. If, during use, they experience meaningful emotional connections and gain access to the high-quality professional resources they anticipated (or better), they are inclined toward greater confirmation and satisfaction ([Pan et al., 2024](#)). Emotional support and information support are particularly salient in shaping both initial expectations and their eventual fulfillment: emotional support helps alleviate teachers’ anxiety regarding novel technologies or instructional approaches, and information support reinforces teachers’ perceptions of the platform’s utility by providing relevant resources and feedback. Accordingly, we hypothesize:

- H4. Emotional support positively influences expectation confirmation.
- H5. Information support positively influences expectation confirmation.

Moreover, perceived usefulness within ECT reflects teachers’ holistic judgment of whether the platform’s functions align with their professional needs—a notion consistent with the TAM constructs of PU and PEOU ([Obeid et al., 2024](#)). When teachers find the platform both easy to operate and effective, they become more likely to confirm their initial expectations and feel satisfied overall. Therefore, we propose that expectation confirmation, perceived ease of use, and perceived usefulness all serve as positive drivers of satisfaction:

- H6. Expectation confirmation positively influences perceived usefulness.
- H7. Expectation confirmation positively influences satisfaction.
- H8. Perceived ease of use positively influences satisfaction.
- H9. Perceived usefulness positively influences satisfaction.

Within the ECT framework, both expectation confirmation and perceived usefulness exert influence on continuance intention through the mediating role of satisfaction ([Han and Sa, 2022](#)). At the same time, perceived usefulness exerts a direct effect on continuance intention, reflecting teachers’ value-based assessment. As teachers’ comprehensive evaluation of the platform and their affective responses (satisfaction)

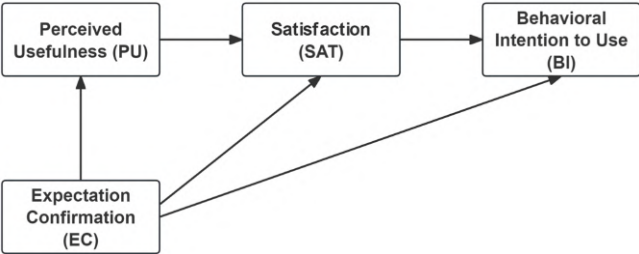


Figure 2. ECT original model. Source: Authors’ own work

improve, they develop a stronger inclination to keep using it, fostering a reinforcing loop of continued engagement. Consequently, we posit:

H10. Perceived usefulness positively influences continuance intention.

H11. Satisfaction positively influences continuance intention.

Collectively, as shown in Figure 3, the integration of ECT with TAM and social support theory enables a more nuanced exploration of teachers’ behavioral patterns in blended training environments. By analyzing how expectation confirmation, perceived usefulness, and satisfaction interact, we can better understand the mechanisms underpinning teachers’ decisions to continue or discontinue online training, thereby laying a robust theoretical foundation for subsequent empirical testing.

Furthermore, we propose that each construct’s effect on continuance intention may be mediated by specific intervening mechanisms. Drawing on the established direct relationships, we highlight three mediating pathways within the integrated TAM-ECT framework. First, perceived usefulness is posited to mediate the relationship between perceived ease of use and satisfaction (PEOU → PU → SAT), suggesting that perceptions of ease of use enhance satisfaction by augmenting functional value. Second, the influence of expectation confirmation on satisfaction may be partially mediated by perceived usefulness (EC → PU → SAT): when actual experience meets or surpasses initial expectations, it not only contributes directly to satisfaction but also promotes a more favorable reassessment of the platform’s practical benefits, thereby strengthening satisfaction. Finally, satisfaction is theorized to mediate the link between perceived usefulness and continuance intention (PU → SAT → CU): while rational judgments about a platform’s value can directly encourage continued usage, these assessments also indirectly foster continuance by increasing emotional satisfaction (Koutromanos *et al.*, 2024). Thus, we present the following mediating hypotheses:

H12. Perceived usefulness mediates the path from perceived ease of use to satisfaction.

H13. Perceived usefulness mediates the path from expectation confirmation to satisfaction.

H14. Satisfaction mediates the path from perceived usefulness to continuance intention.

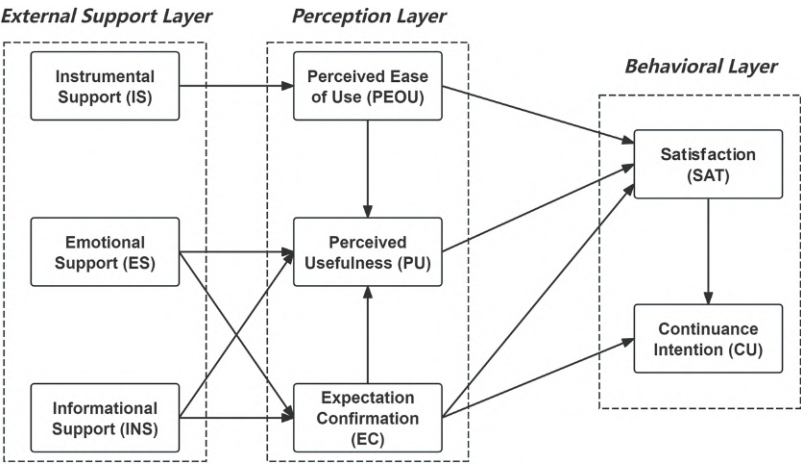


Figure 3. Research theoretical levels model. Source: Authors’ own work

3. Methodology and data collection

3.1 Measurement instrument development

This study developed a multidimensional measurement framework by combining social support theory with core constructs from the TAM and ECT. First, the research team adapted existing scales for PU, PEOU, EC, SAT, and CU from [Park \(2020\)](#), [So and Brush \(2008\)](#), and [Dai et al. \(2020\)](#). Subsequently, three dimensions of social support—IS, ES, and INS—were incorporated to capture the external support teachers receive during professional development ([Malecki et al., 2018](#); [Mathieu et al., 2019](#)).

To ensure measurement equivalence across different linguistic contexts, the questionnaire was prepared in Chinese because the participants were Chinese in-service teachers and Chinese was the working language of the workshops. Items adapted from English-language scales were translated into Chinese by the research team and then checked using the back-translation procedure recommended by [Ozolins et al. \(2020\)](#), verifying each item's semantic consistency through a "source language–target language–source language" cycle. Next, a focus group of educational technology experts, workshop facilitators, and participating teachers ($n = 6$) reviewed the initial 31 items, suggesting adjustments to enhance clarity and contextual fit. Following these refinements, a small-scale pilot study ($n = 45$) was conducted to verify scale reliability, validity, and item performance. One item was removed due to a factor loading under 0.60, resulting in a final instrument comprising 8 latent variables and 30 indicators. The instrument employed five-point Likert scales (1 = "strongly disagree," 5 = "strongly agree") and included control variables, such as demographic information (gender, age, education level, teaching experience) and workshop participation characteristics (role type, weekly usage time), at the end of the questionnaire.

3.2 Data collection

The study targeted in-service teachers from Expert Teacher Workshops in Zhejiang Province, China. To optimize the instrument and strengthen response quality, a two-stage questionnaire process was used. In the first stage, 45 workshop teachers participated in a pilot test assessing item content, expression, and psychometric properties, followed by minor word choice adjustments. After these revisions, the research team formally distributed the questionnaire via the Wenjuanxing platform. Consistent with [Singer and Ye's \(2013\)](#) guidance on incentivizing survey participation, each teacher received 5 RMB (around 0.70 USD) via WeChat Pay after consenting to participate. This approach yielded 557 initial responses over 14 days.

To ensure data integrity, a two-step standardized screening procedure was adopted. First, the team used Temporal Threshold Filtering to discard 18 surveys completed in under 100 s. Second, an attention-check item (requiring participants to select the option "1") excluded an additional 12 surveys. Ultimately, 527 valid questionnaires remained (a 94.61% effective response rate). The sample-to-indicator ratio was about 18:1, meeting established criteria for structural equation modeling and ensuring adequate statistical power for subsequent analyses.

3.3 Descriptive statistics

Using the final 527 valid responses, descriptive analysis showed that the sample covered a wide range of professional profiles and participation roles in Expert Teacher Workshops. As shown in [Table 1](#), gender was relatively balanced (47.40% male, 52.60% female), and most respondents were aged 31–40 (62.05%). The majority held a bachelor's degree or below (84.82%), had 10–20 years of teaching experience (57.90%), and held intermediate or senior professional titles (87.48% combined). In terms of studio role, network trainees constituted the largest group (52.94%), followed by core trainees (21.82%), subject leaders (12.71%), workshop leaders (5.12%), and other members (7.40%). Weekly online engagement was generally modest, with 56.93% reporting fewer than three hours and only 9.1% reporting more than six hours per week. Although the study does not claim province-wide statistical representativeness, these characteristics capture substantial heterogeneity. The sample

Table 1. Demographic information of participants (*N* = 527)

Variable	Category	Count	Percentage (%)
Gender	Male	250	47.40
	Female	277	52.60
Age Group	20–30 years	35	6.64
	31–40 years	327	62.05
	41–50 years	147	27.90
	51 years or older	18	3.40
Education	Bachelor’s degree or below	447	84.82
	Master’s degree or above	80	15.18
Teaching Experience	Less than 10 years	118	22.40
	10–20 years	305	57.90
	20–30 years	86	16.30
	30 years or more	18	3.40
Professional Title	Junior	66	12.52
	Intermediate	243	46.11
	Senior	218	41.37
Role in the Studio	Famous Teacher (Leader)	27	5.12
	Subject Leader	67	12.71
	Core Trainee	115	21.82
	Network Trainee	279	52.94
	Others	39	7.40
Weekly Engagement Time	<3 h	300	56.93
	3–6 h	179	33.98
	6–9 h	32	6.07
	≥9 h	16	3.04

Note(s): Percentages may not total 100% due to rounding

Source(s): Authors’ own work

represents a broad cross-section of active ETW participants rather than a single narrow subgroup, while still warranting caution in generalizing beyond Zhejiang.

Before constructing the structural model, extensive scale validation procedures were performed. Mean item scores (1–5 scale) ranged from 3.43 to 3.90 (SD 0.78–0.94), with coefficients of variation below 27.5%, indicating concentrated responses without extreme outliers. Skewness (0.02–0.68) and kurtosis (0.11–1.54) values were under [Kline’s \(2023\)](#) recommended cutoffs ($|\text{skewness}| < 2$, $|\text{kurtosis}| < 7$), implying mild deviations from normality. Shapiro–Wilk tests showed *p*-values below 0.05 for most variables, rejecting the hypothesis of a normal distribution ([Ghasemi and Zahediasl, 2012](#)). Following [Hair *et al.*’s \(2011\)](#) view that Partial Least Squares Structural Equation Modeling (PLS-SEM) is appropriate for such non-normal data, the study employed PLS-SEM to minimize potential biases that can arise with Covariance-Based Structural Equation Modeling (CB-SEM) under non-normal conditions, thus providing robust empirical support for the path analyses.

3.4 Post-survey qualitative follow-up

To contextualize the quantitative findings, we incorporated a concise post-survey qualitative follow-up within an explanatory sequential design. All interviewees were drawn from the survey sample and selected purposively by workshop role to capture perspectives from leadership, core participation, and wider network participation (2 workshop leaders, 2 core members, and 3 network participants). Semi-structured telephone interviews focused on participants’ experiences of platform support, usability, expectation confirmation, satisfaction, and reasons for continued participation. Each interview lasted approximately 20 min. The interview material was coded to identify recurrent explanations for the significant

JPCC PLS-SEM paths, and the themes retained in Table 8 were those that most consistently illuminated the key structural relationships.

4. Results

Following the PLS-SEM reporting guidelines proposed by Hair et al. (2019), this study conducted a hierarchical evaluation of both measurement and structural models. First, the reliability and validity of the measurement model were examined to ensure scale quality and effectiveness. Subsequently, predictive power and robustness indicators were reported at the structural model level, followed by mediation effect analysis to provide a more comprehensive explanation of the influence pathways for teachers' continuance intention.

4.1 Measurement model assessment

4.1.1 Reflective measurement model. This study employed PLS-SEM to investigate teachers' continuance intention in Expert Teacher Workshops. This approach offers advantages in predicting outcomes within complex online learning environments (Shi et al., 2024). To confirm that the measurement model objectively reflected the theoretical constructs, indicator loadings, internal consistency reliability, convergent validity, and discriminant validity were systematically examined.

As shown in Table 2, all latent variables demonstrated composite reliability (CR = 0.871–0.960) and Cronbach's α (α = 0.864–0.960) values above the recommended threshold of 0.70 (Hair et al., 2021; Li et al., 2024a). The initial internal consistency of Perceived Ease of Use (PEOU) was particularly high (CR = 0.960, α = 0.960), suggesting potential indicator redundancy. All observed indicators had factor loadings (0.717–0.945) exceeding the baseline of 0.70, and the Average Variance Extracted (AVE = 0.630–0.857) generally surpassed the threshold of 0.50, indicating good convergent validity (Fornell and Larcker, 1981). It should be noted that the initial CR value for PEOU was 0.960, suggesting potential indicator redundancy (Cepeda et al., 2024). Variance Inflation Factor (VIF) analysis revealed that PEOU4 had a VIF value of 5.684, exceeding the warning standard of 5. After removing PEOU4, the model indicators for PEOU were optimized (CR = 0.947, AVE = 0.866, α = 0.946), effectively reducing the risk of multicollinearity.

Regarding discriminant validity, as shown in Table 3, all constructs had Heterotrait-Monotrait (HTMT) ratios below 0.85 (Hair et al., 2019), and the confidence intervals obtained through bias-corrected bootstrapping did not include 1, further confirming discriminant validity between constructs. Additionally, based on the Fornell–Larcker criterion (Table 4), the square root of AVE for each latent variable was greater than its correlation coefficients with other latent variables, providing further evidence of good discriminant validity. Hence, the reflective measurement model exhibited sufficient reliability and validity, establishing a solid foundation for subsequent structural model analysis.

4.1.2 Collinearity assessment. To assess potential collinearity in the structural model, this study examined the inner VIF values for all predictor paths (see Table 5). All VIF values ranged from 1.000 to 2.364, below the recommended threshold of 3.0, indicating that multicollinearity did not introduce significant bias in parameter estimation (Hair et al., 2019).

4.2 Structural model assessment

In assessing the structural model, this study used the Bootstrapping procedure (5,000 resamples) in SmartPLS 4 to test path relationships and predictive power. Since PLS-SEM inherently emphasizes prediction, traditional goodness-of-fit indicators (such as GoF, SRMR) are more suitable for confirmatory research. Therefore, the focus here was on predictive indicators such as R^2 , Q^2 , and path coefficient effect sizes (Hair and Alamer, 2022; Hair et al., 2021).

Table 2. Summary of the results for the measurement model

Latent variable	Item	Internal consistency Composite reliability >0.70	Cronbach's alpha >0.70	Convergent validity Loadings >0.70	AVE >0.50
Instrumental support	IS1	0.881	0.883	0.843	0.711
	IS2			0.874	
	IS3			0.820	
Information Support	INS1	0.871	0.869	0.853	0.630
	INS2			0.744	
	INS3			0.717	
	INS4			0.852	
Emotional Support	ES1	0.885	0.929	0.825	0.657
	ES2			0.813	
	ES3			0.808	
	ES4			0.797	
Perceived Ease of Use	PEOU1	0.960	0.960	0.923	0.857
	PEOU2			0.923	
	PEOU3			0.929	
	PEOU4			0.945	
Perceived Usefulness	PU1	0.944	0.864	0.919	0.808
	PU2			0.885	
	PU3			0.890	
	PU4			0.910	
Expectation Confirmation	EC1	0.934	0.893	0.879	0.824
	EC2			0.927	
	EC3			0.917	
Satisfaction	SAT1	0.949	0.929	0.911	0.825
	SAT2			0.903	
	SAT3			0.908	
	SAT4			0.909	
Continued Use Intention	CU1	0.939	0.892	0.889	0.796
	CU2			0.890	
	CU3			0.908	
	CU4			0.881	

Source(s): Authors' own work

Table 3. HTMT values and discriminant validity analysis of constructs

	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS								
INS	0.699							
ES	0.630	0.689						
PEOU	0.655	0.677	0.490					
PU	0.699	0.748	0.673	0.675				
EC	0.630	0.655	0.599	0.630	0.749			
SAT	0.705	0.709	0.627	0.602	0.752	0.696		
CU	0.678	0.691	0.616	0.65	0.725	0.711	0.697	

Note(s) All HTMT values' confidence intervals do not include 1

Source(s): Authors' own work

4.2.1 Predictive power and robustness. Following the evaluation guidelines suggested by Hair et al. (2019), all endogenous variables achieved substantial or higher explanatory power ($R^2 > 0.360$): PEOU at 0.366, EC at 0.436, PU at 0.647, SAT at 0.554, and CU at 0.522.

JPCC

Table 4. Correlation matrix with AVE

	AVE	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS	0.807	0.898							
INS	0.805	0.629	0.897						
ES	0.769	0.561	0.628	0.877					
PEOU	0.866	0.605	0.639	0.458	0.931				
PU	0.833	0.634	0.693	0.617	0.639	0.913			
EC	0.823	0.580	0.625	0.558	0.604	0.710	0.907		
SAT	0.825	0.638	0.656	0.574	0.569	0.702	0.659	0.908	
CU	0.845	0.616	0.643	0.566	0.616	0.679	0.675	0.652	0.919

Note(s) Lower triangle shows the correlation coefficients
The diagonal values show the square root of AVE
The AVE values in Table 4 are based on the refined measurement model after excluding PEOU4. Table 2 reports the initial item-level measurement results before this exclusion
Source(s): Authors' own work

Table 5. Inner VIF for the predictors

	IS	INS	ES	PEOU	PU	EC	SAT	CU
IS				1.000				
INS					1.783	1.650		
ES					2.364	1.650		
PEOU					1.912		1.833	
PU							2.349	1.971
EC					1.989		2.186	
SAT								1.971
CU								

Note(s): VIF values from row to column variables
Source(s): Authors' own work

Notably, PU had the highest R^2 of 0.647 (see Table 6) suggesting that multiple paths in the integrated model comprehensively influenced teachers' functional perceptions. Furthermore, the Stone–Geisser Q^2 values (0.315–0.533), calculated via blindfolding, all significantly exceeded zero. With most endogenous variables showing Q^2 values above 0.35, the structural model demonstrated strong predictive relevance and excellent out-of-sample generalization capability (Cohen, 2013).

Table 6. R^2 and Q^2 values of latent variables

Latent variable	Q^2 predict	R^2
Instrumental support	–	–
Emotional Support	–	–
Information Support	–	–
Perceived Ease of Use	0.315	0.366
Perceived Usefulness	0.533	0.647
Expectation Confirmation	0.403	0.436
Satisfaction	0.452	0.554
Continuance Intention	0.436	0.522

Source(s): Authors' own work

4.2.2 Path coefficient analysis. After the overall structural model assessment, the path coefficients and their effect sizes were examined to confirm the strength and substantive contribution of exogenous variables to endogenous variables (see Table 7). Following Hu *et al.* (2025), this study considered f^2 values of 0.02, 0.15, and 0.35 as small, medium, and large effect sizes, respectively. All hypotheses (H1 to H11) were supported at the 0.05 significance level, and the distribution of effect sizes indicated substantial theoretical and practical significance.

From a latent variable perspective, Instrumental Support (IS) had the most pronounced influence on PEOU ($\beta = 0.605, f^2 = 0.578$). This path showed a large effect size, partly because only IS influenced PEOU in the model, resulting in relatively concentrated variance explanation. This finding aligns with Skulmowski and Xu’s (2022) research, which noted that intuitive operations and task-specific interface designs in digital learning environments can effectively reduce cognitive load and enhance users’ overall perception of system usability.

PU, a core variable in TAM theory, was influenced simultaneously by Expectation Confirmation (EC), Emotional Support (ES), and Information Support (INS) in the integrated model, forming a more complex path. Among these, EC had the strongest influence on PU ($\beta = 0.336, f^2 = 0.161, p < 0.001$), reaching a medium effect size; INS

Table 7. Results of significance tests of the path coefficients

Hypothesis	Relationship	Coefficient (β)	Std	t	95% CI	p	f-square
H1	IS $\rightarrow$ PEOU	0.605	0.037	16.323	[0.531, 0.677]	***	0.578
H2	ES $\rightarrow$ PU	0.190	0.053	3.563	[0.093, 0.300]	***	0.057
H3	INS $\rightarrow$ PU	0.238	0.056	4.248	[0.132, 0.348]	***	0.068
H4	ES $\rightarrow$ EC	0.274	0.062	4.439	[0.157, 0.397]	***	0.081
H5	INS $\rightarrow$ EC	0.453	0.063	7.200	[0.330, 0.576]	***	0.221
H6	EC $\rightarrow$ PU	0.336	0.049	6.803	[0.241, 0.434]	***	0.161
H7	EC $\rightarrow$ SAT	0.283	0.064	4.433	[0.162, 0.410]	***	0.082
H8	PEOU $\rightarrow$ SAT	0.132	0.052	2.531	[0.034, 0.237]	0.011	0.021
H9	PU $\rightarrow$ SAT	0.416	0.074	5.623	[0.274, 0.564]	***	0.165
H10	PU $\rightarrow$ CU	0.436	0.065	6.718	[0.307, 0.561]	***	0.228
H11	SAT $\rightarrow$ CU	0.346	0.062	5.582	[0.243, 0.486]	***	0.162

Note(s): *** $p < 0.001$
Source(s): Authors’ own work

Table 8. Qualitative explanations for key structural paths

Structural path	Qualitative theme	Representative quote
IS $\rightarrow$ PEOU	Alleviation of technological anxiety	“Whenever I struggled with uploading my video lessons, the workshop’s technical assistant provided step-by-step help. It made the platform feel effortless to use.” (Network Participant, 42)
INS $\rightarrow$ PU	Immediate pedagogical transferability	“The lesson plans and expert feedback I received were exactly what I needed for my next day’s class. This practical relevance proved the platform’s worth.” (Core Member, 35)
ES $\rightarrow$ EC	Affective validation and belonging	“Getting encouraged by a master teacher made me feel seen. It confirmed my hope that this wasn’t just a resource hub, but a real, supportive community.” (Network Participant, 29)
SAT $\rightarrow$ CU	Emotional attachment and habituation	“I am highly satisfied because it feels like a professional home. Logging in every week has become a natural habit rather than a forced administrative task.” (Workshop Leader, 48)

Source(s): Authors’ own work

JPCC

($\beta = 0.238, f^2 = 0.068, p < 0.001$) and ES ($\beta = 0.190, f^2 = 0.057, p < 0.001$) both showed small effect sizes. These findings align with [Xianhan et al. \(2022\)](#), who observed that teachers focus more on whether actual experiences match their initial expectations rather than depending solely on external support. The R^2 value of PU reached 0.647, the highest in the model, indicating that multiple paths comprehensively explained teachers' cognitive processes regarding platform functionality.

Satisfaction (SAT), as a comprehensive evaluation of individual perception and expectation fulfillment, was primarily influenced by PU, EC, and PEOU. PU had the most significant impact on SAT ($\beta = 0.416, f^2 = 0.165, p < 0.001$), indicating a medium effect size; EC followed ($\beta = 0.283, f^2 = 0.082, p < 0.001$), showing a small-to-medium effect; while PEOU exhibited a relatively weaker effect ($\beta = 0.132, f^2 = 0.021, p = 0.011$), corresponding to a small effect size. These results echo [Han and Sa's \(2022\)](#) study, suggesting that in teacher professional development contexts, the platform's functional value and fulfillment of expectations play more crucial roles in shaping satisfaction, whereas ease of use, though necessary, is less decisive. The R^2 value for SAT was 0.554, indicating that the model explained about 55.4% of the variance in satisfaction, thereby playing an important mediating role in revealing how continuance intention forms.

Continuance Intention (CU), the ultimate outcome variable, was influenced by both PU and SAT. The direct effect of PU on CU ($\beta = 0.436, f^2 = 0.228, p < 0.001$) was slightly stronger than that of SAT on CU ($\beta = 0.346, f^2 = 0.162, p < 0.001$), and both had medium effect sizes. This is consistent with [Salmerón et al. \(2020\)](#), who argued that in online learning environments, users' rational assessments of a system's value have a more direct impact on usage decisions, though emotional experiences (such as satisfaction) also remain vital. The R^2 value for CU was 0.522, indicating that over half of the variance in teachers' continuance intention could be explained, which is notably persuasive in social science research and demonstrates the integrated model's effectiveness in predicting teachers' continued use of online professional development.

4.2.3 Mediation path verification. To clarify the model's internal mechanisms, three theoretically meaningful indirect paths were tested. Following [Hair et al. \(2014\)](#), this study used Variance Accounted For (VAF) values to assess mediation effects: VAF >80% indicates full mediation, $20\% \leq \text{VAF} \leq 80\%$ indicates partial mediation, and VAF <20% indicates no mediation. The results showed that in the PEOU→PU→SAT path, the indirect effect coefficient was $\beta = 0.082$ ($p = 0.001$) with VAF = 0.383, indicating a significant partial mediating role of PU between PEOU and SAT. In the EC→PU→SAT path, the indirect effect coefficient was $\beta = 0.140$ ($p < 0.001$) with VAF = 0.331, again reflecting a strong partial mediation effect. Finally, in the PU→SAT→CU path, the indirect effect coefficient was $\beta = 0.144$ ($p < 0.001$) with VAF = 0.249, demonstrating another robust partial mediation effect. These findings confirm the core hypotheses of this study: teachers' rational cognition and emotional experience jointly shape their decisions to continue using online training platforms, with rational cognition generally holding a more prominent role, while emotional factors play a key complementary function.

In summary, as illustrated in [Figure 4](#), the structural model integrates the main variables and pathways from social support theory, the technology acceptance model, and the expectation confirmation model. All paths reached significance ($p < 0.05$), further validating the TAM-ECT integrated model grounded in social support for explaining teachers' continuance intention in online professional development. This analysis underscores how both rational and emotional drivers are essential in digital teacher training, while also offering theoretical and practical implications for further investigations into the complex relationships among latent variables.

5. Discussion and implications

5.1 Qualitative insights into path significance

While the PLS-SEM results confirm the structural relationships, the post-survey interviews provide concise explanatory evidence for why these paths emerged in practice (see [Table 8](#)).

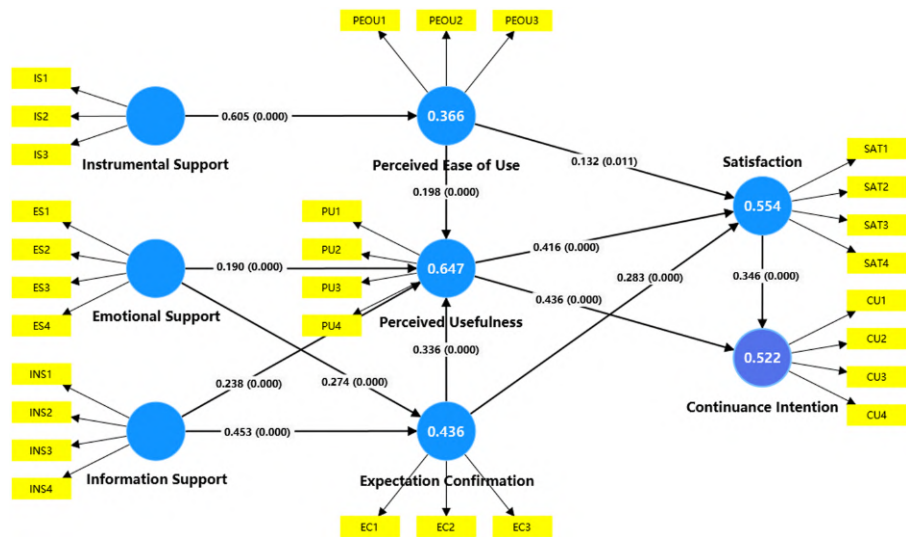


Figure 4. Teachers' continuance intention structural equation model. Source: Authors' own work

The interview material was coded in relation to the significant structural paths and then grouped into higher-order themes that captured recurrent teacher explanations. Table 8 therefore reports path-linked themes rather than stand-alone qualitative categories. For example, instrumental support was repeatedly associated with the alleviation of technological anxiety, information support with the immediate pedagogical transferability of resources and feedback, emotional support with affective validation and belonging, and satisfaction with emotional attachment to continued participation. These qualitative insights do not alter the model, but they make the statistical relationships more interpretable from teachers' lived experience.

5.2 The architecture of engagement: the triadic role of social support

Our findings delineate a hierarchical structure of support required to sustain online professional communities. Rather than operating in isolation, the three forms of social support create a compounded effect that bridges the gap between mere access and deep adherence. First, Instrumental Support acts as the gateway enabler. By significantly influencing perceived ease of use, it reduces the cognitive load associated with technical friction, ensuring that the technology itself does not become an impediment to professional exchange (Skulmowski and Xu, 2022).

Once access is secured, Informational Support serves as the value anchor. It directly enhances teachers' rational assessment of the platform's utility and confirms their professional expectations. This aligns with the view that for knowledge workers, the tangible exchange of pedagogical strategies and feedback constitutes the core value of any professional network (Karatay and Hegelheimer, 2021). Finally, Emotional Support functions as the relational sustenance mechanism. While its direct impact on perceived usefulness is more subtle, its strong connection to expectation confirmation suggests that the sense of belonging and peer validation is essential for converting a transactional training experience into a transformative professional journey (Darling-Hammond et al., 2017; Johnsen, 2001). Collectively, these findings imply that an effective online TPD ecosystem must be architected to simultaneously remove technical barriers, provide rich professional content, and nurture a supportive community culture.

JPCC

5.3 Dual-track drivers of rational evaluation and emotional experience in online teacher training

Structural equation analysis reveals that teachers' continuance intention in online training is driven by two parallel pathways: the rational evaluation pathway represented by perceived usefulness (PU) and the emotional experience pathway centered on satisfaction (SAT). Specifically, the direct effect of perceived usefulness on continuance intention ($\beta = 0.436$, $f^2 = 0.228$) is slightly stronger than that of satisfaction ($\beta = 0.346$, $f^2 = 0.162$), and perceived usefulness also exerts a partial mediating effect on continuance intention through satisfaction ($\beta = 0.144$, VAF = 0.249). This parallel influence of rational and emotional factors is consistent with the technology adoption model proposed by Bettiga and Lamberti (2017) and supports Li et al. (2024b) finding that knowledge-based professionals prioritize the platform's knowledge value.

The integrated TAM and ECT model offers a targeted framework for actionable interventions in online teacher training, suggesting that administrators should design a layered community architecture aligned with specific process needs rather than merely mandating participation (Xu et al., 2025). First, developers must lower initial access barriers through technical onboarding and just-in-time support mechanisms, including interface simplification and real-time help desks. These provisions significantly enhance perceived ease of use ($\beta = 0.605$, IS $\rightarrow$ PEOU), particularly for educators navigating digital literacy challenges (Lane and Lyle, 2011). Second, facilitators can strengthen the platform's functional value by establishing recurring pedagogical routines. Integrating curated resource pathways and expert feedback cycles delivers the high-quality informational support ($\beta = 0.238$, INS $\rightarrow$ PU) necessary to transform static repositories into interactive professional communities (Lantz-Andersson et al., 2018). Third, sustaining affective engagement demands role-based mentoring and differentiated participation pathways. Tailoring recognition and encouragement to specific roles, such as distinguishing between core trainees and wider network participants, provides essential emotional support ($\beta = 0.274$, ES $\rightarrow$ EC) that realistically confirms diverse professional expectations ($\beta = 0.336$, EC $\rightarrow$ PU). Ultimately, while derived from Chinese ETWs, this multifaceted alignment of technical assistance, informational routines, and role-based emotional support provides potentially transferable design principles for comparable expert-led, platform-mediated professional development settings, while remaining grounded in the Chinese ETW context.

5.4 Theoretical contributions: from continuance mechanisms to conditions for professional capital

This study extends traditional technology acceptance research by situating teachers' sustained engagement within the collaborative fabric of the teaching profession. While the classic TAM framework robustly explains individual cognitive processing, it is less well suited to capturing the relational dynamics inherent in organized professional learning environments (Mogaji et al., 2024). By synthesizing Social Support Theory with TAM and ECT, our integrated model shows that continuance intention is shaped not only by functional utility but also by the social conditions of participation within professional networks. This, in turn, clarifies the study's bounded contribution to the literature on professional capital. Following Hargreaves and Fullan (2012), professional capital arises through the interaction of human, social, and decisional capital rather than from isolated individual competence alone. From this perspective, our findings do not demonstrate professional capital directly; rather, they suggest that ETWs may create conditions supportive of its development by reducing participation barriers, facilitating professional knowledge exchange, and sustaining collegial trust over time. Recent work on organizational social capital further reinforces the importance of relational trust, shared understandings, and institutional memory for collective problem-solving and instructional improvement (Jabbar and Holme, 2026), but these are treated here as supportive conditions rather than directly observed outcomes.

The high explanatory power of the model (Continuance Intention $R^2 = 0.522$) validates the premise that “trust” and “community” are as critical as “interface” and “content” in digital learning environments (AlSokkar *et al.*, 2024; Tawafak *et al.*, 2023). Accordingly, the study does not claim to measure professional capital directly. Rather, it identifies continuance intention as a theoretically important bridge between platform participation and the longer-term accumulation of professional capital within expert-led professional communities. This contributes to the literature on professional capital by quantifying how external support networks—comprising peers, mentors, and administrators—can be operationalized to drive behavioral persistence in online settings. Over time, such persistence increases the likelihood that expertise will circulate, collegial relations will deepen, and professional judgment will be refined through repeated interaction. Future research can build upon this “socially embedded” perspective to explore how specific community structures, such as interdisciplinary alliances or AI-augmented tutoring groups, further modulate the relationship between professional interaction and continuous learning.

6. Conclusion and limitations

This study investigated teachers’ continuance intention in online training contexts by integrating TAM and ECT within a social support framework. The findings show that instrumental, informational, and emotional support jointly shape teachers’ cognitive and affective evaluations, thereby encouraging continued participation in Expert Teacher Workshops. Perceived usefulness occupied the core position in teachers’ rational evaluation of platform value, while satisfaction provided affective reinforcement; together, they formed a dual pathway to continuance intention. In substantive terms, these findings do not demonstrate professional capital directly. Rather, they indicate that socially supported and sustained participation in ETWs may contribute to professional capital by creating conditions for professional knowledge sharing, collegial trust, and the refinement of professional judgment over time. The study therefore offers a bounded explanation of how digital engagement can support, rather than substitute for, the longer-term development of professional capital in expert-led professional communities.

Despite this study’s theoretical and practical contributions, several limitations remain. First, because the questionnaire survey was cross-sectional, it is challenging to capture dynamic changes in teachers’ training status, and reliance on self-report may introduce bias. Second, the research sample was largely drawn from a specific type of Expert Teacher Workshop in a single region, which may limit generalizability. To address these constraints, future research should examine whether reported continuance intention is subsequently enacted in practice, rather than assuming a perfect correspondence between intention and sustained participation. This can be effectively achieved through longitudinal or multi-site comparative designs. Furthermore, while this study incorporated a preliminary qualitative follow-up, future investigations could employ more extensive ethnographic observations or in-depth interviews to capture the multi-dimensional evolution of teachers’ training experiences over time. Finally, integrating additional external support elements and moderating factors, such as teachers’ technology adaptability and subject backgrounds, would further validate the TAM-ECT approach across diverse educational contexts. Through continued expansion, the analytical framework proposed here holds promise for providing solid theoretical foundations and actionable guidance to broader online professional development initiatives.

Data availability statement

All data presented in this table were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Author contributions

Haojun Li and Yuying Luo contributed equally and share first authorship.

References

- Almenara, J., Tena, R., Llorente-Cejudo, C. and Palacios-Rodríguez, A. (2021), “Academic performance and technology acceptance model (TAM) through a flipped classroom experience: training of future teachers of primary education”, *Contemporary Educational Technology*, Vol. 13, p. 305, doi: [10.30935/cedtech/10874](https://doi.org/10.30935/cedtech/10874)
- AlSokkar, A.A.M., Law, E.L.C., AlMajali, D.A., Al-Gasawneh, J.A. and Alshinwan, M. (2024), “An indexed approach for expectation-confirmation theory: a trust-based model”, *Electronic Markets*, Vol. 34 No. 1, p. 12, doi: [10.1007/s12525-024-00694-3](https://doi.org/10.1007/s12525-024-00694-3).
- Berkovich, I. and Becher, A. (2025), “Historical changes in teacher training in Israel: an analysis of professional perceptions from 1960 to 2020”, *Journal of Education for Teaching: International Research and Pedagogy*, Vol. 51 No. 2, pp. 1-16, doi: [10.1080/02607476.2025.2449682](https://doi.org/10.1080/02607476.2025.2449682).
- Bettiga, D. and Lamberti, L. (2017), “Exploring the adoption process of personal technologies: a cognitive-affective approach”, *The Journal of High Technology Management Research*, Vol. 28 No. 2, pp. 179-187, doi: [10.1016/j.hitech.2017.10.002](https://doi.org/10.1016/j.hitech.2017.10.002).
- Bhattacharjee, A. (2001), “Understanding information systems continuance: an expectation-confirmation model”, *MIS Quarterly*, Vol. 25 No. 3, pp. 351-370, doi: [10.2307/3250921](https://doi.org/10.2307/3250921).
- Brandão, A., Pedro, L. and Zagalo, N. (2024), “Teacher professional development for a future with generative artificial intelligence – an integrative literature review”, *Digital Education Review*, No. 45, pp. 45-157, doi: [10.1344/der.2024.45.151-157](https://doi.org/10.1344/der.2024.45.151-157).
- Cepeda, G., Roldán, J.L., Sabol, M., Hair, J. and Chong, A.Y.L. (2024), “Emerging opportunities for information systems researchers to expand their PLS-SEM analytical toolbox”, *Industrial Management & Data Systems*, Vol. 124 No. 6, pp. 2230-2250, (world), doi: [10.1108/IMDS-08-2023-0580](https://doi.org/10.1108/IMDS-08-2023-0580).
- Chen, R., Lee, V.R. and Lee, M.G. (2025), “A cross-sectional look at teacher reactions, worries, and professional development needs related to generative AI in an urban school district”, *Education and Information Technologies*, Vol. 30 No. 11, pp. 16045-16082, doi: [10.1007/s10639-025-13350-w](https://doi.org/10.1007/s10639-025-13350-w).
- Cheng, Y.-M. (2022), “Which quality determinants cause MOOCs continuance intention? A hybrid extending the expectation-confirmation model with learning engagement and information systems success”, *Library Hi Tech*, Vol. 41 No. 6, pp. 1748-1780, (world), doi: [10.1108/LHT-11-2021-0391](https://doi.org/10.1108/LHT-11-2021-0391).
- Cobb, S. (1976), “Social support as a moderator of life stress”, *Psychosomatic Medicine*, Vol. 38 No. 5, pp. 300-314, doi: [10.1097/00006842-197609000-00003](https://doi.org/10.1097/00006842-197609000-00003).
- Cohen, J. (2013), *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Routledge, doi: [10.4324/9780203771587](https://doi.org/10.4324/9780203771587).
- Dai, H.M., Teo, T., Rappa, N.A. and Huang, F. (2020), “Explaining Chinese university students’ continuance learning intention in the MOOC setting: a modified expectation confirmation model perspective”, *Computers & Education*, Vol. 150, 103850, doi: [10.1016/j.compedu.2020.103850](https://doi.org/10.1016/j.compedu.2020.103850).
- Darling-Hammond, L., Hyler, M.E. and Gardner, M. (2017), “Effective teacher professional development”, *Learning Policy Institute*, available at: <https://eric.ed.gov/?id=ED606743>
- Davis, F.D. (1989), “Perceived usefulness, perceived ease of use, and user acceptance of information technology”, *MIS Quarterly*, Vol. 13 No. 3, pp. 319-340, doi: [10.2307/249008](https://doi.org/10.2307/249008).
- Fornell, C. and Larcker, D.F. (1981), “Evaluating structural equation models with unobservable variables and measurement error”, *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50, doi: [10.1177/002224378101800104](https://doi.org/10.1177/002224378101800104).
- Garet, M., Porter, A., Desimone, L., Birman, B. and Yoon, K.S. (2001), “What makes professional development effective? Results from a national sample of teachers”, *American Educational Research Journal*, Vol. 38 No. 4, pp. 915-945, doi: [10.3102/00028312038004915](https://doi.org/10.3102/00028312038004915).
- Ghasemi, A. and Zahediasl, S. (2012), “Normality tests for statistical analysis: a guide for non-statisticians”, *International Journal of Endocrinology and Metabolism*, Vol. 10 No. 2, pp. 486-489, doi: [10.5812/ijem.3505](https://doi.org/10.5812/ijem.3505).

- Hair, J. and Alamer, A. (2022), "Partial Least Squares Structural Equation Modeling (PLS-SEM) in second language and education research: guidelines using an applied example", *Research Methods in Applied Linguistics*, Vol. 1 No. 3, 100027, doi: [10.1016/j.rmal.2022.100027](https://doi.org/10.1016/j.rmal.2022.100027).
- Hair, J.F., Ringle, C.M. and Sarstedt, M. (2011), "PLS-SEM: indeed a silver bullet", *Journal of Marketing Theory and Practice*, Vol. 19, world, pp. 139-152, doi: [10.2753/MTP1069-6679190202](https://doi.org/10.2753/MTP1069-6679190202).
- Hair, J.F. Jr, Sarstedt, M., Hopkins, L. and Kuppelwieser, V.G. (2014), "Partial least squares structural equation modeling (PLS-SEM)", *European Business Review*, Vol. 26 No. 2, pp. 106-121, doi: [10.1108/EBR-10-2013-0128](https://doi.org/10.1108/EBR-10-2013-0128).
- Hair, J.F., Risher, J.J., Sarstedt, M. and Ringle, C.M. (2019), "When to use and how to report the results of PLS-SEM", *European Business Review*, Vol. 31 No. 1, pp. 2-24, doi: [10.1108/EBR-11-2018-0203](https://doi.org/10.1108/EBR-11-2018-0203).
- Hair, J.F. Jr, Hult, G.T.M., Ringle, C.M., Sarstedt, M., Danks, N.P. and Ray, S. (2021), *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R: A Workbook*, Springer Nature, doi: [10.1007/978-3-030-80519-7](https://doi.org/10.1007/978-3-030-80519-7).
- Han, J.-H. and Sa, H.J. (2022), "Acceptance of and satisfaction with online educational classes through the technology acceptance model (TAM): the COVID-19 situation in Korea", *Asia Pacific Education Review*, Vol. 23 No. 3, pp. 403-415, doi: [10.1007/s12564-021-09716-7](https://doi.org/10.1007/s12564-021-09716-7).
- Hargreaves, A. and Fullan, M. (2012), *Professional Capital: Transforming Teaching in Every School*, Teachers College Press, New York, NY.
- Hu, L., Wang, H. and Xin, Y. (2025), "Factors influencing Chinese pre-service teachers' adoption of generative AI in teaching: an empirical study based on UTAUT2 and PLS-SEM", *Education and Information Technologies*, Vol. 30 No. 9, pp. 12609-12631, doi: [10.1007/s10639-025-13353-7](https://doi.org/10.1007/s10639-025-13353-7).
- Ifinedo, P., Pyke, J. and Anwar, A. (2018), "Business undergraduates' perceived use outcomes of Moodle in a blended learning environment: the roles of usability factors and external support", *Telematics and Informatics*, Vol. 35 No. 1, pp. 93-102, doi: [10.1016/j.tele.2017.10.001](https://doi.org/10.1016/j.tele.2017.10.001).
- Jabbar, H. and Holme, J.J. (2026), "Teacher turnover, social capital, and improvement: how instability disrupts schools", *Educational Evaluation and Policy Analysis*, Vol. 48 No. 1, pp. 268-292, doi: [10.3102/01623737241311304](https://doi.org/10.3102/01623737241311304).
- Johnsen, M. (2001), "Social support measurement and intervention: a guide for health and social scientists", *Psychiatric Services - PSYCHIATR SERVICES*, Vol. 52 No. 10, p. 1404, doi: [10.1176/appi.ps.52.10.1404](https://doi.org/10.1176/appi.ps.52.10.1404).
- Jolly, P.M., Kong, D.T. and Kim, K.Y. (2021), "Social support at work: an integrative review", *Journal of Organizational Behavior*, Vol. 42 No. 2, pp. 229-251, doi: [10.1002/job.2485](https://doi.org/10.1002/job.2485).
- Karatay, Y. and Hegelheimer, V. (2021), "CALL teacher training—considerations for low-resource environments: overview of CALL teacher training", *CALICO Journal*, Vol. 38 No. 3, available at: <https://search.ebscohost.com/login.aspx?direct=true&profile=ehost&scope=site&authtype=crawler&jrnl=07427778&AN=153307566&h=uRIfiWGpYzkBUe9UWV7jZsI9RTC%2FNYbpuf0ep%2Fi3%2B0fYzNqGHYDBOPxDlqgFK4xH3VWtOuBsJqMYJz2dUdVMQg%3D%3D&cr1=c>
- Kärner, T., Weiß, J.K. and Heinrichs, K. (2021), "A social perspective on resilience: social support and dyadic coping in teacher training", *Empirical Research in Vocational Education and Training*, Vol. 13 No. 1, p. 24, doi: [10.1186/s40461-021-00126-y](https://doi.org/10.1186/s40461-021-00126-y).
- Kemp, A., Palmer, E., Strelan, P. and Thompson, H. and Mery (2024), "Testing a novel extended educational technology acceptance model using student attitudes towards virtual classrooms", *British Journal of Educational Technology*, Vol. 55 No. 5, pp. 2110-2131, doi: [10.1111/bjet.13440](https://doi.org/10.1111/bjet.13440).
- Kline, R.B. (2023), *Principles and Practice of Structural Equation Modeling*, Guilford publications, available at: https://books.google.com/books?hl=zh-CN&lr=&id=t2CvEAAAQBAJ&oi=fnd&pg=PP1&dq=Principles+and+practice+of+structural+equation+modeling&ots=sWUfY281IO&sig=pyEdbPQrs_xZ-itl3tHX8MR7a2w

JPCC

- Koutromanos, G., Mikropoulos, A.T., Mavridis, D. and Christogiannis, C. (2024), "The mobile augmented reality acceptance model for teachers and future teachers", *Education and Information Technologies*, Vol. 29 No. 7, pp. 7855-7893, doi: [10.1007/s10639-023-12116-6](https://doi.org/10.1007/s10639-023-12116-6).
- Lan, G., Feng, X., Du, S., Song, F. and Xiao, Q. (2025), "Integrating ethical knowledge in generative AI education: constructing the GenAI-TPACK framework for university teachers' professional development", *Education and Information Technologies*, Vol. 30 No. 11, pp. 15621-15644, doi: [10.1007/s10639-025-13427-6](https://doi.org/10.1007/s10639-025-13427-6).
- Lane, C.A. and Lyle, H.F. (2011), "Obstacles and supports related to the use of educational technologies: the role of technological expertise, gender, and age", *Journal of Computing in Higher Education*, Vol. 23 No. 1, pp. 38-59, doi: [10.1007/s12528-010-9034-3](https://doi.org/10.1007/s12528-010-9034-3).
- Lantz-Andersson, A., Lundin, M. and Selwyn, N. (2018), "Twenty years of online teacher communities: a systematic review of formally-organized and informally-developed professional learning groups", *Teaching and Teacher Education*, Vol. 75, pp. 302-315, doi: [10.1016/j.tate.2018.07.008](https://doi.org/10.1016/j.tate.2018.07.008).
- Li, H., Xu, J. and Luo, Y. (2024a), "The impact of mobile social media on knowledge sharing among vocational school teachers: a social cognitive career perspective", *Journal of Information Science*, 01655515241293754, doi: [10.1177/01655515241293754](https://doi.org/10.1177/01655515241293754).
- Li, H., Xu, J., Luo, Y. and Wang, C. (2024b), "The role of teachers' direct and emotional mentoring in shaping undergraduates' research aspirations: a social cognitive career theory perspective", *International Journal of Mentoring and Coaching in Education*, Vol. 14 No. 2, pp. 123-142, doi: [10.1108/IJMCE-07-2023-0064](https://doi.org/10.1108/IJMCE-07-2023-0064).
- Liu, P. (2022), "Understanding the roles of expert teacher workshops in building teachers' capacity in Shanghai turnaround primary schools: a Teacher's perspective", *Teaching and Teacher Education*, Vol. 110, 103574, doi: [10.1016/j.tate.2021.103574](https://doi.org/10.1016/j.tate.2021.103574).
- Lu, Y., Ma, N. and Yan, W.-Y. (2024), "Social comparison feedback in online teacher training and its impact on asynchronous collaboration", *International Journal of Educational Technology in Higher Education*, Vol. 21 No. 1, p. 55, doi: [10.1186/s41239-024-00486-x](https://doi.org/10.1186/s41239-024-00486-x).
- Malecki, C.K., Demaray, M.K., Elliott, S.N. and Nolten, P.W. (2018), "Child and adolescent social support scale", [Data set], doi: [10.1037/t57891-000](https://doi.org/10.1037/t57891-000).
- Mathieu, M., Eschleman, K.J. and Cheng, D. (2019), "Meta-analytic and multiwave comparison of emotional support and instrumental support in the workplace", *Journal of Occupational Health Psychology*, Vol. 24 No. 3, pp. 387-409, doi: [10.1037/ocp0000135](https://doi.org/10.1037/ocp0000135).
- McCoy, A. and McNaughton, D. (2021), "Effects of online training on educators' knowledge and use of system of least prompts to support augmentative and alternative communication", *Journal of Behavioral Education*, Vol. 30 No. 6, doi: [10.1007/s10864-020-09374-09376](https://doi.org/10.1007/s10864-020-09374-09376).
- Mogaji, E., Viglia, G., Srivastava, P. and Dwivedi, Y.K. (2024), "Is it the end of the technology acceptance model in the era of generative artificial intelligence?", *International Journal of Contemporary Hospitality Management*, Vol. 36 No. 10, pp. 3324-3339, world doi: [10.1108/IJCHM-08-2023-1271](https://doi.org/10.1108/IJCHM-08-2023-1271).
- Mulajmović, N., Richter, E., Lazarides, R. and Richter, D. (2025), "Comparing quality and engagement in face-to-face and online teacher professional development", *British Journal of Educational Technology*, Vol. 56 No. 1, pp. 61-79, doi: [10.1111/bjet.13480](https://doi.org/10.1111/bjet.13480).
- Obeid, A., Ibrahim, R. and Fadhil, A. (2024), "Extended model of expectation confirmation model to examine users' continuous intention toward the utilization of E-learning platforms", *IEEE Access*, Vol. 12, pp. 40752-40764, doi: [10.1109/ACCESS.2024.3373190](https://doi.org/10.1109/ACCESS.2024.3373190).
- Oliver, R. (1977), "Effect of expectation and disconfirmation on postexposure product evaluations: an alternative interpretation", *Journal of Applied Psychology*, Vol. 62 No. 4, pp. 480-486, doi: [10.1037/0021-9010.62.4.480](https://doi.org/10.1037/0021-9010.62.4.480).
- Ozolins, U., Hale, S., Cheng, X., Hyatt, A. and Schofield, P. (2020), "Translation and back-translation methodology in health research – a critique", *Expert Review of Pharmacoeconomics & Outcomes Research*, Vol. 20 No. 1, pp. 69-77, doi: [10.1080/14737167.2020.1734453](https://doi.org/10.1080/14737167.2020.1734453).

- Pan, G., Mao, Y., Song, Z. and Nie, H. (2024), "Research on the influencing factors of adult learners' intent to use online education platforms based on expectation confirmation theory", *Scientific Reports*, Vol. 14 No. 1, 12762, doi: [10.1038/s41598-024-63747-9](https://doi.org/10.1038/s41598-024-63747-9).
- Park, E. (2020), "User acceptance of smart wearable devices: an expectation-confirmation model approach", *Telematics and Informatics*, Vol. 47, 101318, doi: [10.1016/j.tele.2019.101318](https://doi.org/10.1016/j.tele.2019.101318).
- Peng, T., Wang, C., Xu, J., Dai, J. and Yu, T. (2024), "Evolution and current research status of educational leadership theory: a content analysis-based study", *Sage Open*, Vol. 14 No. 3, 21582440241285763, doi: [10.1177/21582440241285763](https://doi.org/10.1177/21582440241285763).
- Pozo-Rico, T., Poveda, R., Gutiérrez-Fresneda, R., Castejón, J.-L. and Gilar-Corbi, R. (2023), "Revamping teacher training for challenging times: teachers' well-being, resilience, emotional intelligence, and innovative methodologies as key teaching competencies", *Psychology Research and Behavior Management*, world, available at: <https://www.tandfonline.com/doi/abs/10.2147/PRBM.S382572>
- Qin, L., Li, N., Zha, S. and He, W. (2017), "Research on factors influencing perceived usefulness of a virtual teacher community: a case study of rural teachers in Inner Mongolia, China", *Telematics and Informatics*, Vol. 34 No. 5, pp. 463-471, doi: [10.1016/j.tele.2016.09.008](https://doi.org/10.1016/j.tele.2016.09.008).
- Salmerón, L., Delgado, P. and Mason, L. (2020), "Using eye-movement modelling examples to improve critical reading of multiple webpages on a conflicting topic", *Journal of Computer Assisted Learning*, Vol. 36 No. 6, pp. 1038-1051, doi: [10.1111/jcal.12458](https://doi.org/10.1111/jcal.12458).
- Shi, G., Li, J. and Yang, J. (2024), "A study on the influencing factors of university students' online persistent learning supported by intelligent technology in the post-pandemic era: an empirical study with PLS-SEM", *Interactive Learning Environments*, (world), available at: <https://www.tandfonline.com/doi/abs/10.1080/10494820.2023.2205901>
- Singer, E. and Ye, C. (2013), "The use and effects of incentives in surveys", *The Annals of the American Academy of Political and Social Science*, Vol. 645 No. 1, pp. 112-141, doi: [10.1177/0002716212458082](https://doi.org/10.1177/0002716212458082).
- Skulmowski, A. and Xu, K.M. (2022), "Understanding cognitive load in digital and online learning: a new perspective on extraneous cognitive load", *Educational Psychology Review*, Vol. 34 No. 1, pp. 171-196, doi: [10.1007/s10648-021-09624-7](https://doi.org/10.1007/s10648-021-09624-7).
- So, H.-J. and Brush, T.A. (2008), "Student perceptions of collaborative learning, social presence and satisfaction in a blended learning environment: relationships and critical factors", *Computers & Education*, Vol. 51 No. 1, pp. 318-336, doi: [10.1016/j.compedu.2007.05.009](https://doi.org/10.1016/j.compedu.2007.05.009).
- Stoll, L., Bolam, R., McMahon, A., Wallace, M. and Thomas, S. (2006), "Professional learning communities: a review of the literature", *Journal of Educational Change*, Vol. 7 No. 4, pp. 221-258, doi: [10.1007/s10833-006-0001-8](https://doi.org/10.1007/s10833-006-0001-8).
- Takizawa, Y., Bambling, M., Teoh, H.-J. and Edirippulige, S. (2024), "Evaluation of co-designed online teacher training program for providing neuroscience-informed mental health intervention to Japanese children", *Frontiers in Education*, Vol. 9, 1396271, doi: [10.3389/feduc.2024.1396271](https://doi.org/10.3389/feduc.2024.1396271).
- Tawafak, R.M., Al-Rahmi, W.M., Almogren, A.S., Al Adwan, M.N., Safori, A., Attar, R.W. and Habes, M. (2023), "Analysis of E-learning system use using combined TAM and ECT factors", *Sustainability*, Vol. 15 No. 14, p. 11100, doi: [10.3390/su151411100](https://doi.org/10.3390/su151411100).
- Teo, T. (2015), "Comparing pre-service and in-service teachers' acceptance of technology: assessment of measurement invariance and latent mean differences", *Computers & Education*, Vol. 83, pp. 22-31, doi: [10.1016/j.compedu.2014.11.015](https://doi.org/10.1016/j.compedu.2014.11.015).
- Teo, T., Unwin, S., Scherer, R. and Gardiner, V. (2021), "Initial teacher training for twenty-first century skills in the Fourth Industrial Revolution (IR 4.0): a scoping review", *Computers & Education*, Vol. 170, 104223, doi: [10.1016/j.compedu.2021.104223](https://doi.org/10.1016/j.compedu.2021.104223).
- Xianhan, H., Chun, L., Mingyao, S. and Caixia, S. (2022), "Associations of different types of informal teacher learning with teachers' technology integration intention", *Computers & Education*, Vol. 190, 104604, doi: [10.1016/j.compedu.2022.104604](https://doi.org/10.1016/j.compedu.2022.104604).

JPCC

Xu, J., Liu, F., Luo, Y. and Xu, X. (2025), "Stage-specific impacts of digital technologies on supply-chain resilience: meta-analytic evidence for continuous process improvement", *Business Process Management Journal*, pp. 1-22, doi: [10.1108/BPMJ-04-2025-0550](https://doi.org/10.1108/BPMJ-04-2025-0550).

Xu, J., Luo, Y., Wang, C., Wang, M. and Wu, Y. (2026), "AI support in self-regulated learning: a decade of technological evolution and meta-analysis", *British Journal of Educational Technolog*, 70058, doi: [10.1111/bjet.70058](https://doi.org/10.1111/bjet.70058)

Yao, N. and Wang, Q. (2024), "Factors influencing pre-service special education teachers' intention toward AI in education: digital literacy, teacher self-efficacy, perceived ease of use, and perceived usefulness", *Heliyon*, Vol. 10 No. 14, e34894, doi: [10.1016/j.heliyon.2024.e34894](https://doi.org/10.1016/j.heliyon.2024.e34894).

Zhang, D., Gao, S. and Ren, L. (2023), "A study on the mechanisms of teachers' academic emotions and motivational beliefs on learning engagement in the context of online training", *Frontiers in Psychology*, Vol. 14, 1255660, doi: [10.3389/fpsyg.2023.1255660](https://doi.org/10.3389/fpsyg.2023.1255660).

Zheng, X., Zhang, J. and Wang, W. (2019), "Teacher learning as boundary crossing: a case study of Master Teacher Studios in China", *Teachers and Teaching*, Vol. 25 No. 7, pp. 837-854, doi: [10.1080/13540602.2019.1673358](https://doi.org/10.1080/13540602.2019.1673358).

Zheng, W., Ma, Z., Sun, J., Wu, Q. and Hu, Y. (2024), "Exploring factors influencing continuance intention of pre-service teachers in using generative artificial intelligence", *International Journal of Human-Computer Interaction*, Vol. 41 No. 16, pp. 1-14, doi: [10.1080/10447318.2024.2433300](https://doi.org/10.1080/10447318.2024.2433300).

Corresponding author

Jun Xu can be contacted at: xujun1569@gmail.com

Stage-specific impacts of digital technologies on supply chain resilience: meta-analytic evidence for continuous process improvement

Business Process
Management
Journal

Jun Xu

*Department of Educational Information Technology, East China Normal University,
Shanghai, China, and*

Fengmei Liu, Yuying Luo and Xueguo Xu

School of Management, Shanghai University, Shanghai, China

Received 23 April 2025
Revised 29 July 2025
Accepted 8 September 2025

Abstract

Purpose – This study quantifies how digital technologies enhance business process resilience across supply chain stages, addressing a critical gap in process management literature on technology-enabled continuous improvement.

Design/methodology/approach – A random-effects meta-analysis of 70 studies (218 effect sizes) maps digital technologies (sensing, analytics, collaboration) to three supply chain resilience process stages (preparedness, response and recovery), with moderator analysis of economic and cultural factors.

Findings – Digital technologies are positively associated with resilience ($r = 0.331$), preparedness ($r = 0.332$) and response ($r = 0.322$) gain more than recovery ($r = 0.273$). Analytics tools yield largest benefits in early stages ($r = 0.36$); recovery processes rely more on sensing technologies ($r = 0.299$). Effects are stronger in developing economies ($r = 0.350$) than developed ones ($r = 0.267$) and diminish as digital maturity rises ($r = -0.007$). Notably, long-term orientation culture ($r = 0.003$) positively moderates technology effectiveness, demonstrating how cultural factors shape digital transformation trajectories in business processes.

Practical implications – Organizations should adopt stage-specific technology deployment strategies while balancing technological implementation with governance mechanisms. In digitally mature environments, process managers should prioritize institutional improvements over additional technology investments to overcome diminishing returns and achieve sustainable resilience.

Originality/value – Our analysis reveals digital technologies more effectively support the “risk perception–dynamic response” process loop than “post-disruption reconfiguration–adaptive learning” processes. The identified “diminishing digital dividend” phenomenon challenges assumptions that technology alone ensures process resilience.

Keywords Supply chain resilience, Digital technologies, Meta-analysis, Business process management, Continuous improvement, Cultural moderation

Paper type Literature review

1. Introduction

In the face of unprecedented global uncertainty, managing business processes across supply chains has become a critical strategic imperative. Recent disruptions – including the COVID-19 pandemic, geopolitical conflicts and climate-related disasters – have exposed fundamental vulnerabilities in cross-organizational business process integration, highlighting the urgent need for resilience-building strategies (Zavala-Alcívar *et al.*, 2023). More recently, escalating China–US trade tensions have triggered profound reconfigurations of business processes, as rising protectionism and regionalization trends necessitate a reassessment of outsourcing and network design (Kazancoglu *et al.*, 2024). Within this challenging landscape, digital transformation offers promising pathways for business process managers. Research indicates that digital technologies can strengthen organizational responsiveness through advanced risk forecasting, real-time information processing and adaptive resource allocation (Ghobakhloo *et al.*, 2025).

As organizations embed digital tools into their process management systems (Ho *et al.*, 2023), these technologies have become recognized as essential enablers for improving how



Business Process Management Journal
© Emerald Publishing Limited
e-ISSN: 1758-4116
p-ISSN: 1463-7154
DOI 10.1108/BPMJ-04-2025-0550

BPMJ

supply chains respond to, recover from and adapt to disruptions (Li *et al.*, 2025). However, the business process management literature lacks a systematic, quantitative synthesis of how specific digital technologies differentially enhance supply chain resilience (SCR) across various operational contexts. Three critical gaps remain unaddressed:

First, resilience is often treated as a single, uniform concept, leading to the misleading assumption that technology enhances it in the same way across all situations. This view is problematic because SCR comprises distinct capabilities activated at different stages of a disruption (Stadtfeld and Gruchmann, 2024). For instance, the capacity to absorb a shock is different from the capacity to adapt and restore operations. The value of digital technologies, therefore, depends on the specific functions they support, from basic data visibility to more advanced capabilities like strategic flexibility and adaptive learning (Ghobakhloo *et al.*, 2025). A “one-size-fits-all” approach to technology thus fails to capture the distinct value that digital tools provide at each process stage (Al-Swidi *et al.*, 2024). Second, as the technological landscape evolves, research typically investigates either individual technologies or specific business contexts, limiting our comprehensive understanding of how diverse digital solutions contribute to resilience-building across different process dimensions (Tortorella *et al.*, 2024). For instance, some researchers have stratified Industry 4.0 technologies into foundational, advanced and adaptive layers using the Delphi method (Spieske *et al.*, 2023), highlighting the multifaceted nature of the digital technology spectrum in business process environments. Third, from a process management perspective, the relationship between digital technologies and SCR exhibits nonlinear characteristics embedded within complex socioeconomic and cultural contexts (Zhao *et al.*, 2023). Research indicates that macro-level factors – including economic development indicators and cultural dimensions – exert significant moderating effects on organizations’ digital adoption patterns and business process risk management approaches (Rodríguez-González *et al.*, 2023).

While existing reviews have confirmed a generally positive technology–resilience relationship (e.g. Iftikhar *et al.*, 2024; Spieske and Birkel, 2021), none have quantitatively deconstructed this relationship to reveal stage-specific effect sizes, leaving practitioners without empirical guidance for targeted technology deployment. This study is the first meta-analysis to provide comprehensive quantitative evidence of how the impact of digital technologies varies systematically across the different stages of resilience. In doing so, it challenges the common but untested assumption that technology provides uniform benefits. This oversight has left managers and scholars without stage-specific evidence, a critical gap this research aims to fill.

To address these research gaps – specifically, the need to (1) disaggregate the unitary construct of resilience into distinct process stages, (2) synthesize fragmented evidence on how different digital technologies perform across these stages and (3) quantitatively assess the influence of macro-level contextual factors – this meta-analysis is guided by four fundamental questions. RQ1 first maps the technological landscape. Subsequently, RQ2 and RQ3 directly tackle the stage-specific and technology-specific effects, while RQ4 examines the critical economic and cultural moderators. Specifically, we ask:

- RQ1. Which digital technologies feature prominently in SCR-related business process research?
- RQ2. What are the quantitative effect sizes of digital technologies on SCR overall, and on each of its three operational stages in organizational contexts?
- RQ3. Which moderating variables influence how digital technologies enable SCR in business process systems?
- RQ4. From a global business perspective, how do economic and cultural factors shape this technological relationship?

This meta-analysis makes several contributions to business process management research. Theoretically, by examining the three critical SCR phases – preparedness, response and recovery – and systematically cross-analyzing them with different categories of digital technologies, our work transcends the limitations of qualitative studies and single-dimensional empirical analyses. The resulting framework offers practical, stage-specific guidance for applying digital technologies, supporting a shift from isolated process reengineering to continuous improvement. From a practical standpoint, our findings deliver evidence-based decision support for business process managers and policymakers, enabling organizations to identify optimal digital approaches for enhancing process resilience at different operational stages and offering targeted strategies for managing cross-organizational processes under varying economic and cultural contexts.

2. Literature review and early work

2.1 Resilience and technology in supply chains

SCR – a network’s dynamic capability to rebound from disruptions – has been a central theme in operations research since Christopher and Peck (2004) introduced the concept. Drawing on publication data from the Web of Science SSCI and SCI databases, Figure 1 confirms scholarly interest in SCR has grown exponentially since 2015. This surge mirrors both the increasing frequency of global shocks and the rapid diffusion of digital technologies (Stadtfeld and Gruchmann, 2024).

Digitalization has advanced in two discernible waves. During 2014–2020 firms experimented with isolated tools, gradually aligning them with resilience routines. Since 2021 – accelerated by COVID-19 – IoT sensors, big-data analytics, blockchain, digital twins and, from 2022, advanced AI have converged into an end-to-end ecosystem that supports real-time monitoring, risk assessment and adaptive decision-making (Li et al., 2025; Spieske and Birkel, 2021). Collectively these tools heighten visibility, collaboration, agility and responsiveness, enabling earlier disruption detection and faster recovery (Iftikhar et al., 2024). By dissolving conventional boundaries among information, talent and capital flows, they embody Industry 4.0 principles of modularity, interoperability and decentralization (Büyükoğkan and Göçer, 2018; Ghobakhloo et al., 2025).

Regarding how SCR is segmented, multiple research perspectives exist: Wieland and Wallenburg (2013) classify resilience from an attribute perspective into passive resilience (agility) and active resilience (robustness); Pillay (2017), approaching from capability

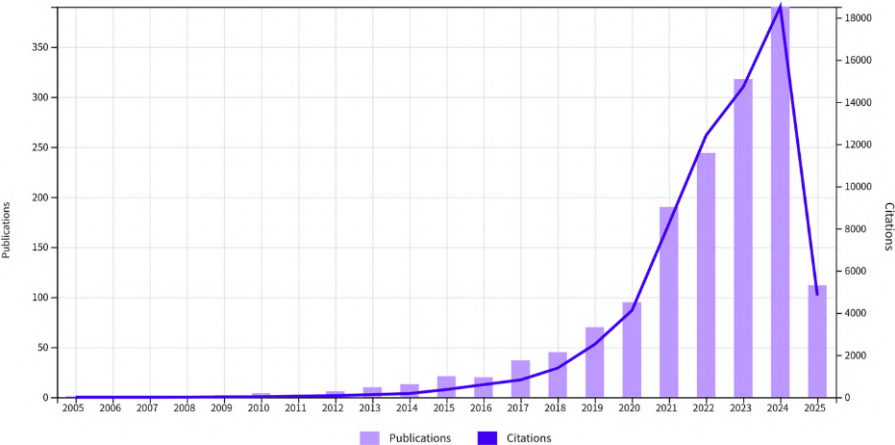


Figure 1. Publication and citation trends in supply chain resilience research. Source: Authors’ own work

BPMJ

building, proposes four key competencies – response, monitoring, learning and foresight. Other scholars have mapped SCR to specific functional elements such as production/manufacturing, logistics/distribution, demand management and collaboration and returns/remanufacturing (Akçali and Çetinkaya, 2011; Jung *et al.*, 2014).

While various frameworks exist, this study adopts the process-oriented three-stage classification from Ponomarov and Holcomb (2009), as it aligns with a business process lifecycle view of disruption management. This model conceptualizes resilience not as a single outcome, but as a dynamic capability unfolding across distinct phases. The first is preparedness, which involves proactive capabilities for “event readiness”, such as developing risk warning systems and establishing resource redundancy. This is followed by response, centered on executing an “efficient response” to a disruption through actions like flexible resource deployment and innovative decision-making. The final stage is recovery, which extends beyond simple restoration to include system reconfiguration and adaptive learning, enabling the supply chain to return to its original state or evolve to a better one. This granular, stage-based operationalization, detailed in Table 1, forms the analytical foundation for quantifying the differentiated impact of digital technologies.

2.2 Overview of prior review literature

Early reviews focused on defining resilience and its metrics (Hosseini *et al.*, 2019). More recently, attention has shifted to the digitalization–resilience nexus, with reviews showing that Industry 4.0 enablers elevate disruption tolerance (Spieske and Birkel, 2021) and that technologies like digital twins can enhance real-time scheduling (Golan *et al.*, 2021). However, it is also argued that genuine resilience gains require cross-organizational collaboration, not just technology (Ghobakhloo *et al.*, 2025).

Despite this progress, critical gaps remain, which this study addresses. Most reviews treat SCR as a unitary construct, obscuring how digital tools perform across preparedness, response and recovery (Spieske and Birkel, 2021). Furthermore, evidence is often fragmented, and large-sample quantitative validation is scarce.

Our study overcomes these limitations with a random-effects meta-analysis that (1) maps digital technologies to the three SCR stages, (2) quantifies their stage-specific effect sizes and (3) tests macro-level moderators. This approach advances SCR theory with robust statistical evidence and offers managers stage-aware guidance for technology deployment, reinforcing the strategic value of digital transformation.

3. Research design

Following Ponomarov and Holcomb’s (2009) three-stage model categorizing SCR into preparedness, response and recovery, this study employs meta-analysis guidelines by Pigott and Polanin (2020) to assess how digital technologies impact these stages differently.

3.1 Literature search and inclusion criteria

Adhering to PRISMA guidelines, our systematic literature search queried four major databases (Web of Science, Scopus, ProQuest and Emerald) to identify all relevant studies published, or made available via early access, between January 2004 and April 2025. The search strategy intentionally focused on resilience concepts – using a representative query such as TS= (“supply chain resilience” OR “supply chain robustness” OR “supply chain adaptation”) – rather than specific technology keywords. This approach was designed to ensure a broad and unbiased capture of diverse and evolving digital solutions.

The search and screening protocol is detailed in Figure 2. The initial search yielded 8,427 records. After removing 5,371 duplicates, we screened the titles and abstracts of the remaining 3,056 articles, excluding 2,057 that did not meet the research topic (e.g., they were non-English, review articles or outside the business process scope). A full-text review of the 999

Table 1. SCR four-stage coding

Stage	Indicator name	Definition/explanation
Preparedness	Risk Warning	Systematically identifies, evaluates and flags potential risks, establishing a forward-looking prevention foundation for the supply chain
	Information Coordination	Achieves full-chain information transparency and integration, improving communication and decision-making efficiency across nodes
	Resource Redundancy	Builds buffers through redundant resources and robust infrastructure to safeguard against external shocks
	Organizational Learning	Enhances overall risk response and prevention capabilities through skill development, network governance and the accumulation of experience
Response	Flexible Response	Rapidly reallocates and integrates both internal and external resources to enable swift adjustments and responses under disruption conditions
	Innovative Decision-Making Disruption Containment	Leverages improvisational innovation and contextual judgment in dynamic environments to drive precise decisions and plan execution Implements effective measures to isolate disruptions and prevent their spread, ensuring core operations and services remain uninterrupted
Recovery	Speed of Restoration	Measures how quickly key functions return to normal operation post-disruption, reflecting the system's self-repair capacity
	Reconfiguration	Undertakes structural reorganization and process redesign based on disruption experiences, promoting optimized realignment of the supply chain as a whole
	Resource Regeneration	Enables swift replenishment and regeneration of strategic resources, restoring supply and production capacity balances disrupted by interruptions
	Evolution and Upgrading	Propels strategic adaptation and capability enhancements through experiential learning and continuous improvement, fostering long-term competitive advantages
Combined	Systemic Resilience	Integrates resilience-building achievements from all stages, reflecting the overall capacity of the supply chain to withstand shocks and maintain continuous operations
	Collaborative Integration	Reinforces cross-organizational and cross-boundary synergy, achieving resource sharing and risk-sharing to strengthen overall resilience
	Strategic Adaptation	Relies on dynamic adaptation and innovation, transforming resilience into sustainable competitive advantages, serving as a long-term market strategy foundation
	Benefit Realization	Maximizes the comprehensive economic and operational return on investments in resilience and governance optimization through strategic resource allocation

Source(s): Authors' own work

eligible articles led to the exclusion of 450 papers for lacking empirical research design. For final inclusion, studies had to provide sufficient statistical data for effect-size calculation. This critical step led to the exclusion of 424 articles with no available data and a further 55 with insufficient data, yielding the final sample of 70 studies for meta-analysis.

3.2 Study characteristics and intervention coding

We established a multidimensional coding framework capturing:

- (1) Study background (publication year, journal type, research method, economic development level, firm size, industry)

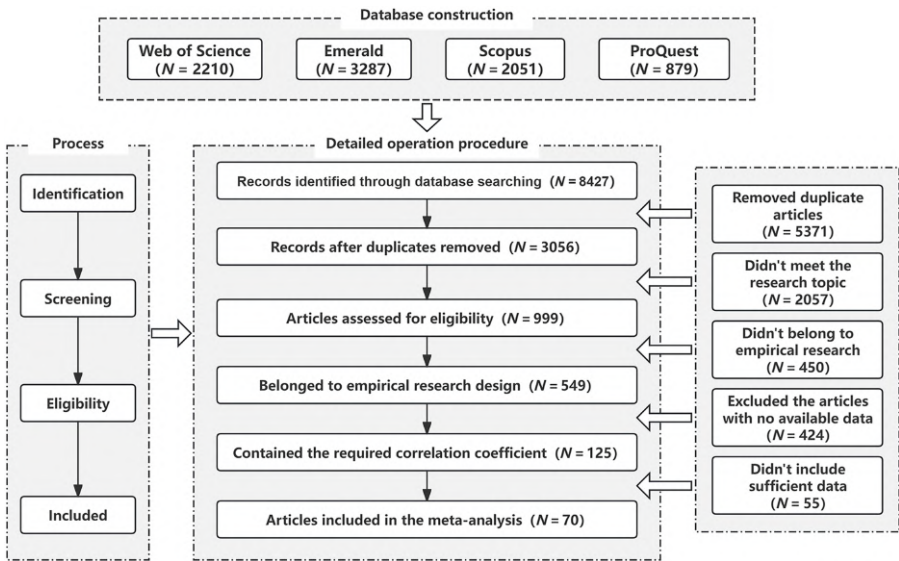


Figure 2. Search flow diagram. Source: Authors' own work

- (2) SCR stage was coded into preparedness, response and recovery, plus a combined category for studies addressing multiple phases non-distinctly, based on the specific indicators in Table 1 which were refined through iterative discussion.
- (3) Digital technology type (sensing/perception, analytics/decision-making, collaboration/integration, multiple/comprehensive)

A complete list of the 70 primary studies included in the meta-analysis is presented in Appendix A. Two researchers independently coded the studies; Cohen's kappa values for both SCR stage and technology classification exceeded 0.80, indicating high inter-rater reliability. Any disagreements were resolved through discussion until consensus was reached.

3.3 Sensitivity analysis and heterogeneity tests

Sensitivity analysis using a leave-one-out approach confirmed minimal fluctuation in effect sizes (interval [0.326, 0.334]), indicating resistance to outliers. Heterogeneity assessment via Cochran's Q test and the I² statistic showed significant heterogeneity across all subsamples (I² > 90%), justifying our random-effects model and the reporting of the variance of true effect sizes (τ^2) (Table 2).

3.4 Publication bias

We assessed publication bias following Oesterreich et al. (2025) by examining funnel-plot symmetry, Egger's regression test and fail-safe analyses. The overall funnel plot (Figure A1) was essentially symmetrical, and Egger's test was non-significant ($p > 0.05$), indicating minimal small-study effects. The classic Rosenthal's (1979) fail-safe N reached 335,369 – far above the 5k + 10 threshold of 525 – and Orwin's (1983) fail-safe N of 6,430 to reduce the standardized mean difference to 0.01 further confirms the stability of our overall effect-size estimate.

Stage-specific funnel plots (Figure A2) appeared largely symmetrical; however, the more sensitive Egger's regression test revealed a significant publication bias for the response stage

Table 2. Publication bias

Resilience phase	Egger's					Fail-safe <i>N</i>	
	K	Score	SD	<i>z</i>	<i>p</i>	Classic	Orwin– 0.01
Readiness	69	−0.80	1.950	−0.41	0.6807	33,215	2,041
Response	55	−5.89	2.554	−2.30	0.0212	21,911	1,556
Recovery	42	−0.34	2.465	−0.14	0.8902	8,796	1,040
Comprehensive	52	−1.38	1.567	−0.88	0.3788	23,864	1,795
Overall	218	−1.53	1.028	−1.49	0.1372	335,369	6,430

Source(s): Authors' own work

($p < 0.05$). This suggests the reported effect size for this stage ($r = 0.322$) is best interpreted as a potential upper bound, as the bias likely inflates the estimate by under-representing studies with smaller or null findings. Nevertheless, the finding's statistical significance remains highly robust: the fail-safe N of 21,911 confirms that the positive relationship is highly unlikely to be nullified by this potential bias.

4. Results

4.1 Review of digital technologies in supply chain resilience (for RQ1)

Based on analysis of 70 empirical studies, this research categorizes digital technologies into three functional dimensions that form an integrated ecosystem for SCR development. Rather than classifying by chronology or type, we organize technologies by functionality:

4.1.1 *Information acquisition and sensing technologies.* Technologies such as the Internet of Things, radio frequency identification and sensors constitute the foundational layer for SCR, enabling continuous monitoring and transmission of supply chain status data. These technologies heighten visibility, allowing firms to identify potential risks and reduce reaction times to disruptions. As Huang *et al.* (2023) demonstrate, supply chain visibility mediates between information acquisition technologies and SCR, helping organizations recover faster. When paired with analytics, these technologies transition supply chains from passive defense to proactive warning systems.

4.1.2 *Data analytics and decision-making technologies.* Tools such as artificial intelligence (AI), machine learning, predictive analytics and digital twins transform data into actionable insights, which can enhance risk management efficiency. Belhadi *et al.* (2024) show that AI-driven innovations are linked to a boost in risk prediction capabilities in dynamic environments. These technologies provide forward-looking solutions through contingency optimization and adaptive supply network configurations. However, as Xu and Liu (2024) note, excessive reliance on analytics may diminish intuitive judgment, suggesting the need for balance between technological inputs and human insight.

4.1.3 *Collaboration, integration and synchronization technologies.* Blockchain, cloud computing, collaborative platforms and supply chain control towers represent the networked collaboration layer, dismantle information silos and facilitate connectivity among supply chain partners. These technologies are associated with enhanced SCR, partly by establishing unified information-sharing platforms, standardizing cross-organizational processes and increasing transparency. Harju *et al.* (2023) demonstrate that collaborative synergy and compatible standards are necessary to convert digital investments into resilience capabilities.

4.2 Primary effect-size calculations (for RQ2)

Using correlation coefficients (r) to measure effect sizes, this study employs the approach of Wang *et al.* (2024) to standardize other metrics into correlation coefficients, followed by

BPMJ

Fisher’s z-transformation via Equation (1), ensuring the accuracy and comparability of the estimated effect sizes. To enhance methodological transparency, Appendix B provides a detailed guide with concrete examples illustrating the conversion of commonly reported statistics (e.g. standardized regression coefficients) into Pearson’s *r*. Following Cohen’s (2013) guidelines, an effect size below 0.10 indicates a weak correlation, 0.10–0.30 indicates a moderate correlation and above 0.30 indicates a strong correlation.

$$z' = 0.5 \times \ln \left(\frac{1 + r}{1 - r} \right) \tag{1}$$

Meta-analysis results show that digital technologies have a strong positive association with SCR (*r* = 0.331, 95% CI [0.300, 0.363], *p* < 0.001) but display significant disparities among the three SCR stages – preparedness, response and recovery (Table 3). Specifically, the association is strongest for preparedness (*r* = 0.332), followed closely by response (*r* = 0.322), while the relationship is more moderate for recovery (*r* = 0.273). Of note, the combined stage exhibits a markedly higher effect size (*r* = 0.388), potentially stemming from synergistic effects when multiple technologies are applied jointly or because some studies did not isolate the recovery phase in their analyses.

4.3 Stage-specific impacts of digital technologies (for RQ3)

Further subgroup analysis reveals the interaction effects between technology type and resilience stage. Figure 3 provides a heatmap of these technology-stage synergies – where color intensity represents the effect size (*r*) – visually complementing the detailed statistical results in Table 4.

In the preparedness stage, data analytics and decision-making technologies are associated with the strongest effect size (*r* = 0.361), slightly surpassing that of collaboration technologies (*r* = 0.355), while sensing technologies show a lower but still significant effect (*r* = 0.263). This finding points to the importance of analytical capabilities in the processes of identifying latent threats and formulating preventive strategies.

For the response stage, analytics technologies maintain the strongest association (*r* = 0.366), followed by collaboration technologies (*r* = 0.348) and sensing technologies (*r* = 0.314). This distribution reflects organizations’ reliance on data-driven decisions and resource deployment during disruptions.

In the recovery stage, sensing technologies (*r* = 0.299) are linked to a slightly stronger effect size than collaboration technologies (*r* = 0.290), while analytics technologies show a relatively lower effect (*r* = 0.256). This pattern suggests that processes related to real-time monitoring are particularly salient for reconfiguring supply chains and repairing damaged nodes.

Table 3. Random effects model for the influence of digital technologies on SCR across stages

Variable	k	g	SE	95% CI	Q	df	p	I ² (%)	Tau-2	Tau
Readiness	69	0.332	0.029	[0.275, 0.389]	1099.31	68	0.000	93.21	0.054	0.232
Response	55	0.322	0.035	[0.253, 0.391]	879.52	54	0.000	94.26	0.063	0.251
Recovery	42	0.273	0.036	[0.202, 0.344]	715.79	41	0.000	94.90	0.050	0.224
Comprehensive	52	0.388	0.030	[0.330, 0.446]	568.57	51	0.000	90.24	0.040	0.200
Overall	218	0.331	0.016	[0.300, 0.363]	3804.92	217	0.000	93.94	0.053	0.230

Source(s): Authors’ own work

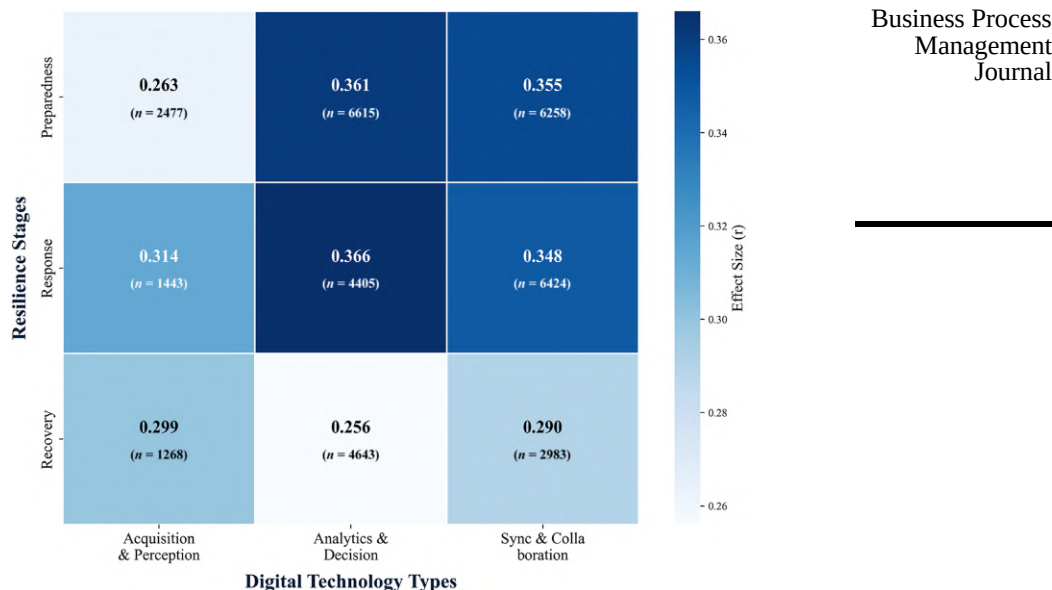


Figure 3. Heatmap of technology–stage synergies. Source: Authors’ own work

4.4 Analysis of the moderating effects of technology’s support for SCR (for RQ4)

Under the meta-analysis framework adopted in this study, subgroup analysis and meta-regression were conducted to examine whether the relationship between digital technologies and SCR varies across different contexts. An overview of both analyses is presented in Figure 4.

4.4.1 Subgroup analysis. As shown in Table 5, subgroup analyses were performed according to industry type, economic development level and firm size.

Industry type: The effect size for manufacturing ($g = 0.350$, 95% CI [0.300, 0.401]) is slightly higher than for services ($g = 0.322$, 95% CI [0.250, 0.394]), but this difference is not statistically significant ($p = 0.525$), indicating digital technologies’ broad applicability across these categories. This broad classification, however, masks the nuanced role of technology in diverse sectoral contexts (e.g., humanitarian aid, construction) which, while having sample sizes too small for separate estimation, offer key insights. For instance, in humanitarian logistics, digital twin technology is shown to be a critical contributor to SCR (Singh, 2025), highlighting its value in highly volatile environments.

Economic development level: Developing economies show a significantly higher effect size ($g = 0.350$, 95% CI [0.314, 0.386]) than developed economies ($g = 0.267$, 95% CI [0.197, 0.337]) ($p = 0.039$). Regions with fewer legacy systems may find it easier to adopt emerging digital technologies and quickly strengthen SCR.

Firm size: Large enterprises ($g = 0.353$, 95% CI [0.305, 0.401]) and SMEs ($g = 0.300$, 95% CI [0.231, 0.370]) show no significant difference ($p = 0.219$). While larger organizations typically have more established resources, SMEs can leverage external partnerships and flexible digital tools to compensate for resource constraints.

4.4.2 Meta-regression. This study extends the subgroup analysis by using meta-regression to examine the macro-level moderating factors, specifically national economic development and cultural influences, that contextualize the relationship between digital technologies and SCR (see Equation (2)). The Digital Competitiveness Score is employed to measure a country’s digital infrastructure maturity, while GDP per capita serves as a proxy for economic development. Additionally, several of Hofstede’s cultural dimensions – such as long-term orientation and uncertainty avoidance, which are related to organizational practices – are

Table 4. Effect sizes for digital technology–stage combinations (Fisher’s z)

Resilience phase	Subsamples	k	n	Effect size (r)	95% CI	Q (p, χ^2)	I ² (%)
Readiness	Overall	69	17,901	0.332	[0.275, 0.389]	1099.31 (p = 0.000)	93.21
	Acquisition and Perception	8	2,477	0.263	[0.088, 0.438]	182.98 (p = 0.000)	94.75
	Analytics and Decision	29	6,615	0.361	[0.295, 0.428]	190.19 (p = 0.000)	86.28
	Sync and Collaboration	20	6,258	0.355	[0.205, 0.506]	605.76 (p = 0.000)	97.19
	Comprehensive	12	2,551	0.265	[0.204, 0.325]	24.46 (p = 0.011)	54.99
Response	Overall	55	14,585	0.322	[0.253, 0.391]	879.52 (p = 0.000)	94.26
	Acquisition and Perception	5	1,443	0.314	[0.179, 0.449]	27.08 (p = 0.000)	84.91
	Analytics and Decision	19	4,405	0.366	[0.289, 0.444]	114.53 (p = 0.000)	85.07
	Sync and Collaboration	22	6,424	0.348	[0.206, 0.491]	601.10 (p = 0.000)	96.96
	Comprehensive	9	2,323	0.170	[0.047, 0.293]	65.71 (p = 0.000)	88.75
Recovery	Overall	42	22,277	0.273	[0.202, 0.344]	715.79 (p = 0.000)	94.90
	Acquisition and Perception	5	1,268	0.299	[0.151, 0.448]	29.83 (p = 0.000)	85.09
	Analytics and Decision	22	4,643	0.256	[0.148, 0.364]	290.26 (p = 0.000)	92.71
	Sync and Collaboration	11	2,983	0.290	[0.135, 0.444]	151.73 (p = 0.000)	94.48
	Comprehensive	4	13,383	0.275	[0.126, 0.424]	69.83 (p = 0.000)	92.91
Comprehensive stage	Overall	52	12,143	0.388	[0.330, 0.446]	568.57 (p = 0.000)	90.24
	Acquisition and Perception	11	2,729	0.361	[0.264, 0.457]	57.34 (p = 0.000)	83.24
	Analytics and Decision	17	3,464	0.361	[0.288, 0.435]	70.82 (p = 0.000)	78.55
	Sync and Collaboration	12	2,795	0.498	[0.317, 0.679]	313.35 (p = 0.000)	95.71
	Comprehensive	12	3,155	0.341	[0.252, 0.430]	68.65 (p = 0.000)	83.95
Grand mean (overall sample)	Overall Sample	218	66,906	0.331	[0.300, 0.363]	3804.92 (p = 0.000)	93.94
	Acquisition and Perception	29	7,917	0.313	[0.244, 0.382]	352.67 (p = 0.000)	89.3
	Analytics and Decision	87	19,127	0.366	[0.294, 0.378]	701.13 (p = 0.000)	88.36
	Sync and Collaboration	65	18,460	0.368	[0.289, 0.447]	1749.62 (p = 0.000)	96.60
	Comprehensive	37	21,412	0.269	[0.217, 0.320]	407.67 (p = 0.000)	89.24

Source(s): Authors’ own work

incorporated. Previous cross-national research and meta-analytic studies underscore the significance of these macro and cultural variables in moderating technology adoption and risk

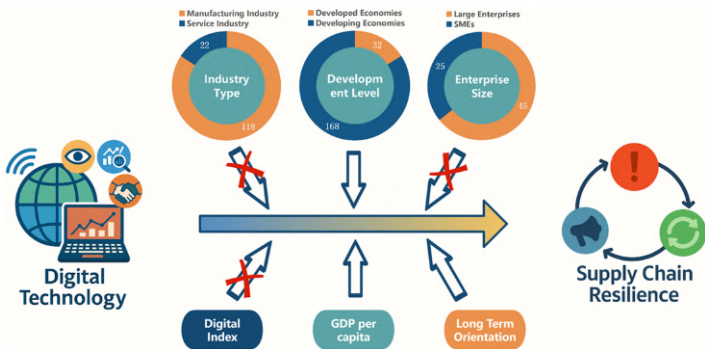


Figure 4. Moderating effects. Source: Authors’ own work

Table 5. Subgroup analysis of moderating effects on supply chain resilience

Moderator	Subgroup	k	r	95% CI	p	I ² (%)	Between p (χ ²)
Industry type (excluding mixed/ other)	Manufacturing Industry	118	0.350	[0.300, 0.401]	<0.001	94.79	0.525 (0.40)
	Service Industry	22	0.322	[0.250, 0.394]	<0.001	83.75	
	Overall	140	0.346	[0.302, 0.390]	<0.001	94.11	
Economic development level (excluding unreported)	Developed Economies	32	0.267	[0.197, 0.337]	<0.001	83.89	0.039 (4.24)
	Developing Economies	168	0.350	[0.314, 0.386]	<0.001	94.34	
	Overall	200	0.337	[0.305, 0.370]	<0.001	93.66	
Enterprise size (excluding mixed/ other)	Large Enterprises	45	0.353	[0.305, 0.401]	<0.001	87.42	0.219 (1.51)
	Small- and Medium-sized Enterprises (SMEs)	25	0.300	[0.231, 0.370]	<0.001	88.18	
	Overall	70	0.334	[0.295, 0.373]	<0.001	88.65	

Source(s): Authors’ own work

management outcomes (Hofstede, 2001; Kumar and Raj, 2025). This approach thus examines how institutional, economic and cultural factors contribute to cross-national variations and assesses their role in moderating the incremental effects of digital technologies on SCR.

$$ES_{ij} = \beta_{0j} + r_{ij} \tag{2}$$

$$\begin{aligned} \beta_{0j} = & \gamma_{00} + \gamma_{01}n_j + \gamma_{02} \text{Year}_j + \gamma_{03} \text{DigitalIndex}_j + \gamma_{04} \text{GDPpc}_j \\ & + \gamma_{05} \text{PowerDistance}_j + \gamma_{06} \text{Individualism}_j + \gamma_{07} \text{Masculinity}_j \\ & + \gamma_{08} \text{UncertaintyAvoidance}_j + \gamma_{09} \text{LongTermOrientation}_j + \gamma_{010} \text{Indulgence}_j + u_{0j} \end{aligned} \tag{3}$$

where ES_{ij} is the i^{th} effect size describing the relationship reported within the j^{th} sample. Equation (3) describes the effect of all identified moderators that vary between studies. Here, u_j

BPMJ

is the term for the study-level residual error. A fully nested model (Table 6) illustrates the meta-regression results, which provide additional insights.

Table 6 presents the fully nested meta-regression. The model explains 6.34% of the between-study variance in effect sizes ($R^2 = 0.0634$). Even after all moderators are included, the intercept remains positive and significant ($\gamma = 0.699$, $p < 0.01$), confirming a stable, positive link between digital technologies and SCR across contexts.

Moderator analysis, reported with standardized beta (β) coefficients, highlights three notable patterns. First, the digitalization index exerts a significant negative moderating effect ($\gamma = -0.007$, $p < 0.10$; $\beta = -0.1935$), indicating that the “digital dividend” diminishes as national digital infrastructure matures. Second, a culture’s long-term orientation has a positive influence ($\gamma = 0.003$, $p < 0.10$; $\beta = 0.0709$), suggesting that forward-looking societies strengthen the technology–resilience relationship. Third, publication year shows a marginally negative coefficient ($\gamma = -0.020$, $p < 0.10$), hinting that reported effect sizes may be converging over time. No other cultural or economic indicators emerge as significant moderators.

Table 6. Meta-regression results (digital technology and supply chain resilience)

Parameter	Unstandardized Coeff. (γ)	SE	z- value	95% CI	Standardized (β)
<i>Constants and controls</i>					
Intercept (_cons)	0.6994***	0.230941	3.03	[0.2467147, 1.151987]	0.2955***
n (γ_{01})	−0.000013	0.0000155	−0.84	[−0.0000434, 0.0000173]	0.0048
year (γ_{02})	−0.0198*	0.0109648	−1.81	[−0.0413180, 0.0016631]	−0.0314*
<i>Infrastructure – economy</i>					
Digitalization Index (γ_{03})	−0.0073*	0.0038931	−1.88	[−0.0149429, 0.0003178]	−0.1935*
GDP per Capita (γ_{04})	3.81e−06	4.29e−06	0.89	[−4.60e−06, 0.0000122]	0.1448
<i>Culture</i>					
PowerDistance (γ_{05})	−0.0016	0.0016104	−1.00	[−0.0047698, 0.0015429]	−0.0381
Individualism (γ_{06})	−0.0018	0.0015728	−1.15	[−0.0023443, 0.0038209]	0.0132
Masculinity (γ_{07})	−0.0004	0.0033915	−0.20	[−0.0073221, 0.0059724]	0.0037
UncertaintyAvoidance (γ_{08})	−0.0016	0.002071	−1.03	[−0.0061873, 0.0019307]	−0.0548
LongTermOrientation (γ_{09})	0.0029*	0.0015448	1.75	[−0.0003267, 0.0057288]	0.0709*
Indulgence (γ_{10})	0.0007	0.0013702	0.49	[−0.0020084, 0.0033626]	−0.0196
<i>Number of effects</i>		199			
<i>Residual heterogeneity (τ^2)</i>		0.0346			
<i>R-squared (%)</i>		6.34%			

Note(s): ***($p < 0.01$); **($p < 0.05$); *($p < 0.1$); ns ($p > 0.1$)
All other statistics (z-value, 95% CI, τ^2 , R^2) correspond to the unstandardized model
To account for skewness, n (Sample Size) and GDP per capita were log-transformed prior to z-score standardization; all other moderators were standardized using the z-score method
Source(s): Authors’ own work

5. Discussion

5.1 *The differentiated effects of various digital technology types on three-stage resilience*

Drawing on the interaction between digital technologies and SCR, this research investigates both the “horizontal” and “vertical” dimensions: the former concerns how functional layers such as sensing, analytics and collaboration can be integrated, whereas the latter examines how these functions may yield different outcomes across the “preparedness, response and recovery” stages. Our central theoretical contribution lies in revealing a fundamental asymmetry in the value of digitalization for business process resilience. The findings demonstrate that digital technologies are highly effective at enhancing the “risk perception–dynamic response” loop but are significantly less associated with the more complex “post-disruption reconfiguration–adaptive learning” process. This distinction, quantitatively validated by our meta-analysis, challenges the prevailing view of technology as a uniform solution and instead posits that its value is contingent on the specific process requirements of each resilience stage.

5.1.1 Horizontal dimension of technology. From the “horizontal” perspective, this study classifies digital technologies into information acquisition and sensing technologies, data analytics and decision-making technologies, and collaboration and synchronization technologies, along with multiple/comprehensive technologies. While sensing-layer tools (e.g., RFID, NFC, sensor systems) show relatively modest effect sizes in isolation, their core value lies in providing a front-end interface for data collection, thereby creating foundational visibility throughout the entire supply chain (Taj *et al.*, 2023). When sensing technologies combine with higher-level analysis and coordination tools such as AI, cloud computing or blockchain, raw monitoring data can fuel highly automated and intelligent operational decisions. For instance, AI models can leverage large-scale real-time data gathered at the sensing end to quickly identify abnormal demand or potential disruption risks (Jauhar *et al.*, 2024), significantly boosting the speed and accuracy of organizational responses to disturbances.

Data analytics and decision-making technologies, by harnessing big data and machine learning algorithms for forecasting, scenario simulation and resource allocation, enable organizations to swiftly implement targeted responses. While these intelligent decision systems perform especially well in the “risk perception–dynamic response” domain, they may not comprehensively address the broader human–machine collaboration and flexible management elements needed during post-disruption organizational learning and complex situational assessments.

Collaboration and synchronization technologies further eliminate information silos among enterprises by using distributed mechanisms such as cloud platforms or blockchain to strengthen cross-node process integration (Alikhani *et al.*, 2023). In times of unexpected incidents, such solutions can support synchronized response and resource-sharing among multiple parties, while also serving as a data exchange and business integration foundation in subsequent recovery stages. Particularly in large-scale, networked supply chains, these collaborative platforms – combining real-time communication with traceability functions – can support a more resilient systemic linkage across diverse supply chain partners.

The synergistic benefits of deploying multiple digital technologies arise from structured layering, not simple addition. Industry 4.0, for instance, involves using various technologies together across the value network (Ghobakhloo *et al.*, 2025). Foundational sensing technologies like IoT create a data-rich environment. Analytics technologies like AI use this data to generate predictive insights. Finally, collaboration technologies like blockchain provide a secure mechanism for sharing information between firms, which is essential for coordinated action (Gozali *et al.*, 2024). This layering – where sensing, analytics and collaboration are complementary – builds comprehensive resilience in a way that no single technology can achieve alone.

5.1.2 Vertical dimension of the supply chain. From the “vertical” perspective, this study explores how digital technologies differ in the three major supply chain stages of

BPMJ

preparedness, response and recovery. First, the preparedness phase is highly reliant on capabilities for risk forecasting and prevention, reflected by its pronounced correlation with data analytics and decision-making technologies (Dai *et al.*, 2025). Early identification of potential threats and the establishment of redundancy strategies not only draw on data from the sensing layer but also rely on intelligent algorithms to support scenario simulation and emergency planning (Ju *et al.*, 2021). If combined with collaboration and synchronization technologies, these capabilities can offer visualized decision-making references for cross-firm or cross-departmental strategic coordination, thus enhancing supply chain stability in the face of risks.

In the response phase when a disruption occurs, platform-based collaboration and real-time data analysis have heightened importance: visibility and instantaneous decision-making allow each link in the chain to adjust resources in tandem, dynamically revise production plans or logistics schemes (Ali *et al.*, 2024). Information acquisition and sensing technologies gather critical inputs at this point, relaying them via collaborative platforms to the decision-making level, facilitating rapid iteration and collective action. Yet as noted regarding the inherent shortfall in “post-disruption reconfiguration–adaptive learning,” if algorithms or data analytics focus exclusively on short-term interventions while neglecting deeper analysis of disruption causes or long-range developments, organizations may handle the initial impact effectively but struggle later to advance capability evolution and systemic reconfiguration.

The weaker link between digital technologies and the recovery stage ($r = 0.273$) requires careful interpretation. Our data reveal a critical misalignment: the analytics tools most effective in the preparedness and response stages ($r \approx 0.36$) are the least effective during recovery ($r = 0.256$). This gap is not just about technology; it reflects a fundamental difference in the business processes themselves. Preparedness and response are largely about optimizing activities within established frameworks. In contrast, recovery is about redesign and learning, requiring new routines and models to be created under ambiguous conditions.

This suggests that current data-driven tools, while excellent for optimizing known processes, are less suited for the unstructured problem-solving and organizational learning needed to reconfigure the supply chain after a major disruption. This finding highlights a key limit to the value of current AI and analytics in business process management: they are powerful for executing existing processes but offer limited support for evolving and re-learning them. Instead, blending digital solutions with experiential insights, team-based learning, and supply chain memory can progressively refine recovery strategies and bolster defenses against future similar risks.

The synergistic effect of comprehensive technologies is vividly illustrated in applications such as pandemic vaccine distribution, where integrating IoT for demand sensing with drones for rapid logistics creates resilience across the entire process (Ansari *et al.*, 2025). This moves beyond a simple, stage-isolated view of technology’s association with resilience. From a theoretical standpoint, our findings suggest that digitalization is associated with a dynamic, cross-stage process of capability-building. This process can be interpreted through the lens of dynamic capabilities, potentially manifesting as a sequential pathway where digitalization is first linked to supply chain adaptability, which in turn may provide a platform for effective supply chain innovation, ultimately contributing to resilience (Aslam *et al.*, 2025).

Yet precisely because algorithms are relatively more effective for short-term disruption management, deeper systemic transformations required for “post-disruption reconfiguration–adaptive learning” still rely on organizational capabilities and resource alignment to ensure that the resilience advantages unleashed by digital technologies persist and become internalized.

5.2 Influence of moderators and contextual characteristics

This section discusses how the observed relationship between digital technologies and SCR is contextualized by various factors – firm size, industry type, economic development and

institutional and cultural environments – and explores the potential mechanisms behind the observed moderating effects.

5.2.1 Consistency across firm size and industry characteristics. Regarding differences in firm size and industry, subgroup analysis reveals that although digital technology usage patterns vary across contexts, there is no statistically significant difference in overall effect sizes between the broad categories of manufacturing and services ($p = 0.525$). The underlying reason for this cross-sectoral consistency is that digital technologies primarily augment fundamental business processes – such as demand forecasting, logistics coordination and partner communication – that are ubiquitous across both manufacturing and service supply chains. In high-uncertainty settings like healthcare, for example, data analytics becomes pivotal for managing disruptions (Bag *et al.*, 2024), while in project-based sectors like construction, blockchain is instrumental in fostering the supply chain integration that underpins resilience (Wang *et al.*, 2025). The aggregate non-significant finding therefore reinforces our core thesis: technology's value is realized when it addresses specific, context-dependent vulnerabilities. Similarly, no notable difference emerges between large enterprises and small-to-medium enterprises (SMEs) ($p = 0.219$). These findings challenge earlier assumptions that large firms enjoy a “natural advantage” in technological resources and specialized teams (Qi and Zhang, 2024).

The growing trend of “digital democratization” does not imply that SMEs face no obstacles in digital transformation. On the contrary, implementing digital systems often demands specialized training and upfront infrastructure investments (Jauhar *et al.*, 2024). However, our finding of performance parity suggests that SMEs can compensate for resource limitations with structural agility, enabling a more targeted and rapid integration of select digital tools (Ghobakhloo *et al.*, 2025). This allows them to master internal processes – a key antecedent of SCR (Lin *et al.*, 2025) – by focusing on accessible solutions like No-Code AI rather than large-scale capital projects (Jauhar *et al.*, 2024).

5.2.2 Economic development level and the marginal benefits of digital transformation. Contrasting with the non-significant findings regarding firm size and industry attributes, economic development level exhibits a more pronounced moderating effect. The overall effect size for digital technologies in developing economies ($r = 0.350$) significantly exceeds that of developed economies ($r = 0.267$), and the difference is statistically meaningful ($p = 0.039$) – consistent with findings by He *et al.* (2023) concerning port enterprises. This result suggests that when a country or region has not yet reached the “saturation threshold” of digital infrastructure, digital transformation is associated with more significant marginal returns. For developing economies lacking established supply chain architecture, emerging digital tools frequently provide “latecomer advantages,” allowing rapid resource realignment and risk management in the face of extensive disruptions such as trade barriers or geopolitical conflict (Mandal, 2017).

By contrast, developed economies with relatively mature digital ecosystems already incorporate automation and data analytics in routine operations, reducing the “incremental space” in which digital technologies can further improve SCR (Bown, 2023). However, this does not imply a weakness in the digital capabilities of developed countries; rather, it indicates that the next step must involve deeply integrating digital solutions with organizational competencies and pursuing refined management and innovative value-chain development to achieve lasting competitive differentiation (Xue *et al.*, 2024). In this sense, economic development level not only corresponds with the immediate gains from digital transformation but also appears related to how different regions rebuild capabilities and pursue sustainable pathways during a crisis.

5.2.3 Institutional environment: balancing digital dividends with governance mechanisms. The negative moderating effect of the digitalization index ($\gamma = -0.007$, $p < 0.1$) supports our “diminishing digital dividend” hypothesis and signals a critical shift in what drives process resilience. In digitally advanced regions, further technology investments may yield smaller gains. As organizations adopt similar foundational technologies, simply investing more

BPMJ

without improving management practices can lead to a “digital trap” (Dai *et al.*, 2024). In other words, when technology becomes commonplace, competitive advantage shifts toward better institutional and governance capabilities (Lin *et al.*, 2025).

From a governance perspective, this statistical trend can be interpreted as evidence of a growing imbalance: a gap between “technology excess” and “inadequate governance,” where digital expansion outpaces the cross-organizational coordination and risk-control mechanisms needed to harness its full potential (Li *et al.*, 2024; Zhu *et al.*, 2023). This suggests that to counteract diminishing returns, the focus of process improvement may shift from firm-level technology to inter-firm governance. This requires a synergy where digital contractual governance (e.g., via blockchain) creates a transparent, enforceable foundation of rules (Wang *et al.*, 2025), which in turn enables the trust-based relational governance needed for adaptive, collaborative problem-solving across the ecosystem (Lu *et al.*, 2023; Peng *et al.*, 2024).

5.2.4 Cultural perspective: cross-national variations and trust mechanisms. Macro-level cultural differences add further explanatory depth to how digital technologies enable SCR (Hofstede, 2001). Although only long-term orientation (LTO) exhibits a significant effect in this study, while other cultural measures such as power distance or uncertainty avoidance do not, this does not necessarily mean they lack relevance. Instead, more complex or nonlinear interplays may be at work (Dubey *et al.*, 2020). In this study, higher LTO tendencies encourage sustained investments in digital infrastructure and risk management, leading to quicker recovery from disruptions. In contrast, cultural traits such as high power distance or strong uncertainty avoidance may support rapid centralization and consistent directives in the initial stages of a crisis but can create organizational rigidity during subsequent collaboration and innovation.

Trust is another critical factor. Cross-national supply chains routinely involve diverse cultural settings. In environments marked by higher trust levels and stronger collaborative norms, organizations are typically more inclined to use digital platforms for information-sharing and collective decision-making, thereby mitigating asymmetrical information risks. Conversely, in contexts emphasizing individualism or lacking trust in collaboration, more stringent governance mechanisms (e.g. blockchain or smart contracts) may be required to prevent moral hazard and alleviate anxieties about partnership (Hofstede, 2001). Notably, lower power distance and greater indulgence can enhance the fluidity of internal communication, encourage team collaboration and increase risk tolerance (Alsanoosy *et al.*, 2019). This cultural backdrop enables digital technology to permeate all supply chain stages more freely, bolstering overall agility and synchronization across the network.

In summary, our analysis suggests that cultural factors are profoundly intertwined with how organizations perceive and adopt digital solutions, as well as with the governance frameworks and resource allocation decisions for handling uncertainty. For future research, applying cross-cultural perspectives and multi-case longitudinal studies could yield a deeper understanding of how cultural dimensions interact with digital technologies to augment SCR.

6. Conclusion, implications and limitations

6.1 Main conclusions

This study addresses the four key research questions raised in the introduction. First, through an extensive literature review, digital technologies were categorized into three core functional categories. Then, employing a meta-analysis approach, we systematically examined the differentiated associations between digital technologies and the three phases of SCR. The results indicate that digital technologies are significantly associated with resilience in the preparedness ($r = 0.332$), response ($r = 0.322$) and recovery ($r = 0.273$) stages, yet these associations are uneven across the phases. Data analytics and decision-making technologies show the strongest association during preparedness ($r = 0.361$) and response ($r = 0.366$), whereas the recovery stage is more strongly linked to information acquisition and sensing technologies ($r = 0.299$). This finding suggests that digital technologies are more strongly

associated with the “risk perception–dynamic response” loop than with the “post-disruption reconfiguration–adaptive learning” process. Additionally, the study confirms the significant moderating effects of macro-level variables: the association is stronger in developing economies ($r = 0.350$) than in developed economies ($r = 0.267$), while the digitalization index ($\gamma = -0.007$) acts as a negative moderator and a long-term orientation culture ($\gamma = 0.003$) serves as a positive moderator. These findings highlight that the phenomenon of diminishing digital dividends and cultural values are strongly intertwined with digitalized pathways toward building SCR.

6.2 Theoretical contributions

This study makes three primary contributions to the literature on SCR and digitalization. First, it develops an innovative cross-stage analytical framework that integrates the dynamic process view of SCR with the multilayer functional perspective of digital technologies, thus answering [Ponomarov and Holcomb's \(2009\)](#) call to examine the multi-stage dynamics of resilience. Second, our large-scale quantitative analysis uncovers the “uneven effect” of digital technologies. We show their effectiveness is significantly higher for risk prevention and immediate response than it is for rebuilding systems and evolving capabilities after a disruption. This finding extends the research boundary of [Spieske and Birkel \(2021\)](#). Third, by incorporating macro-level moderating variables, the study demonstrates that economic development levels, digital maturity and cultural values collectively shape how digital technologies enable resilience, notably revealing the phenomenon of “diminishing digital dividends” and the positive influence of long-term orientation culture on digital transformation. These findings enrich social and cultural considerations in research exploring digital technologies’ support for supply chains ([Kumar and Raj, 2025](#)).

6.3 Managerial implications

The study offers insights on three levels for enterprise managers and policymakers. At the technology planning level, firms should adopt differentiated, evidence-based deployment strategies aligned with our stage-specific findings. For the preparedness ($r = 0.332$) and response ($r = 0.322$) stages, investments should prioritize analytics and decision-making technologies, which demonstrate the strongest effects ($r \approx 0.36$). Conversely, for the recovery stage – where our findings reveal analytics are less effective ($r = 0.256$) – the strategy must pivot to integrating sensing technologies ($r = 0.299$) with organizational learning to address the critical “post-disruption reconfiguration–adaptive learning” gap.

At the organizational structure level, firms must build decision-making frameworks that align with digital technologies. For SMEs, our finding of performance parity with large firms ($p = 0.219$) suggests a strategy of prioritizing mastery over breadth: achieving deep process integration with a few targeted, low-cost digital tools (e.g. a cloud-based logistics platform) rather than attempting a large-scale, resource-intensive transformation.

Finally, at the ecosystem level, our moderator analysis provides targeted policy guidance. For developing economies, where technology yields higher returns ($r = 0.350$), infrastructure investment remains the priority. In contrast, developed economies experiencing diminishing returns ($\gamma = -0.007$) should redirect resources toward establishing governance frameworks and institutional standards for data sharing and automated decision-making. Additionally, societies with strong long-term orientation ($\gamma = 0.003$) are better positioned to frame digital investments as strategic foundations for future competitiveness, suggesting more favorable conditions for sustained technology adoption policies.

6.4 Limitations and future research directions

Although this study broadens the understanding of the relationship between digital technologies and SCR, several limitations remain. First, due to the present sample

BPMJ

conditions, it is not possible to measure correlations among the three stages of SCR within the current research design. Second, the study falls short of fully capturing the dynamic interactions among different types of digital technologies.

This paper's findings open several promising avenues for future research that can build directly upon our contributions. First, our discovery of an "uneven effect" across resilience stages calls for mixed-methods studies to uncover the micro-foundations of this quantitative pattern; specifically, longitudinal case research is needed to explore how business processes and digital tools co-evolve through the preparedness, response and recovery phases. Second, our results reveal a critical "capability misalignment": the analytics tools effective in early stages are weakest during recovery. Future inquiry should therefore investigate how to design systems that effectively blend technology with organizational routines and human skills to support the unstructured, learning-intensive processes that define recovery. Finally, our findings on the "diminishing digital dividend" in mature economies and the retrospective nature of our analysis highlight the need for future-oriented research. The next step is to evaluate whether emerging technologies like generative AI can overcome the capability misalignment we identified and offer new pathways to resilience in an increasingly complex global landscape.

About the authors

Jun Xu is a PhD candidate in the Department of Education Information Technology, Faculty of Education, East China Normal University. His research interests focus on large model research in educational technology, big data analysis in online education and learning analytics.

Fengmei Liu is a PhD candidate in the Department of Management Science and Engineering, School of Management, Shanghai University, Shanghai, China. Her research interests include operations management, data-driven decision-making and optimization techniques in management engineering.

Yuying Luo is a postgraduate in the Department of Management Science and Engineering, School of Management, Shanghai University, Shanghai, China. Her research interests include supply chain resilience, digital transformation in operations and quantitative methods for risk assessment in global production networks.

Xueguo Xu is a professor in the Department of Management Science and Engineering, School of Management, Shanghai University, Shanghai, China. His research focuses on production system optimization, supply chain analytics, digital technology adoption in manufacturing and decision support frameworks for operational resilience in complex industrial environments.

Supplementary material

The supplementary material for this article can be found online.

References

- Akçalı, E. and Çetinkaya, S. (2011), "Quantitative models for inventory and production planning in closed-loop supply chains", *International Journal of Production Research*, Vol. 49 No. 8, pp. 2373-2407, doi: [10.1080/00207541003692021](https://doi.org/10.1080/00207541003692021).
- Al-Swidi, A.K., Al-Hakimi, M.A., Al Halbusi, H., Al Harbi, J.A. and Al-Hattami, H.M. (2024), "Does blockchain technology matter for supply chain resilience in dynamic environments? The role of supply chain integration", *Plos One, Public Library of Science*, Vol. 19 No. 1, e0295452, doi: [10.1371/journal.pone.0295452](https://doi.org/10.1371/journal.pone.0295452).
- Ali, A.A.A., Sharabati, A.-A.A., Alqurashi, D.R., Shkeer, A.S. and Allahham, M. (2024), "The impact of artificial intelligence and supply chain collaboration on supply chain resilience: mediating the effects of information sharing", *Uncertain Supply Chain Management*, Vol. 12 No. 3, pp. 1801-1812, doi: [10.5267/j.uscm.2024.3.002](https://doi.org/10.5267/j.uscm.2024.3.002).
- Alikhani, R., Ranjbar, A., Jamali, A., Torabi, S.A. and Zobel, C.W. (2023), "Towards increasing synergistic effects of resilience strategies in supply chain network design", *Omega*, Vol. 116, 102819, doi: [10.1016/j.omega.2022.102819](https://doi.org/10.1016/j.omega.2022.102819).

- Alsanoosy, T., Spichkova, M. and Harland, J. (2019), "The influence of power distance on requirements engineering activities", *Procedia Computer Science*, Vol. 159, pp. 2394-2403, doi: [10.1016/j.procs.2019.09.414](https://doi.org/10.1016/j.procs.2019.09.414).
- Ansari, N., Fattahi, P. and Shiri, M. (2025), "The role of drone technology and application of IoT on vaccine supply chain during a pandemic under uncertain environment: a real case study of COVID-19 in Iran", *Transportation Research Part E: Logistics and Transportation Review*, Vol. 193, 103831, doi: [10.1016/j.tre.2024.103831](https://doi.org/10.1016/j.tre.2024.103831).
- Aslam, H., ur Rehman, A., Iftikhar, A., ul Haq, M.Z., Akbar, U. and Kamal, M.M. (2025), "Digital transformation: unlocking supply chain resilience through adaptability and innovation", *Technological Forecasting and Social Change*, Vol. 219, 124234, doi: [10.1016/j.techfore.2025.124234](https://doi.org/10.1016/j.techfore.2025.124234).
- Bag, S., Gupta, S., Choi, T.-M. and Kumar, A. (2024), "Roles of innovation leadership on using big data analytics to establish resilient healthcare supply chains to combat the COVID-19 pandemic: a multimethodological study", *IEEE Transactions on Engineering Management, IEEE-Inst Electrical Electronics Engineers*, Vol. 71, pp. 13213-13226, doi: [10.1109/TEM.2021.3101590](https://doi.org/10.1109/TEM.2021.3101590).
- Belhadi, A., Mani, V., Kamble, S.S., Khan, S.A.R. and Verma, S. (2024), "Artificial intelligence-driven innovation for enhancing supply chain resilience and performance under the effect of supply chain dynamism: an empirical investigation", *Annals of Operations Research*, Vol. 333 Nos 2-3, pp. 627-652, doi: [10.1007/s10479-021-03956-x](https://doi.org/10.1007/s10479-021-03956-x).
- Bown, C.P. (2023), "The WTO and vaccine supply chain resilience during a pandemic", *Journal of International Economic Law*, Vol. 26 No. 2, pp. 343-362, doi: [10.1093/jiel/jgad007](https://doi.org/10.1093/jiel/jgad007).
- Büyüközkan, G. and Göçer, F. (2018), "Digital supply chain: literature review and a proposed framework for future research", *Computers in Industry*, Vol. 97, pp. 157-177, doi: [10.1016/j.compind.2018.02.010](https://doi.org/10.1016/j.compind.2018.02.010).
- Christopher, M. and Peck, H. (2004), "Building the resilient supply chain", *The International Journal of Logistics Management*, Vol. 15 No. 2, pp. 1-14, doi: [10.1108/09574090410700275](https://doi.org/10.1108/09574090410700275).
- Cohen, J. (2013), *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Routledge, New York, doi: [10.4324/9780203771587](https://doi.org/10.4324/9780203771587).
- Dai, J., Geng, R., Xu, D., Shangguan, W. and Shao, J. (2024), "Unveiling the impact of the congruence between artificial intelligence and explorative learning on supply chain resilience", *International Journal of Operations and Production Management*, Vol. 45 No. 2, pp. 570-593, doi: [10.1108/IJOPM-12-2023-0990](https://doi.org/10.1108/IJOPM-12-2023-0990).
- Dai, J., Geng, R., Xu, D., Shangguan, W. and Shao, J. (2025), "Unveiling the impact of the congruence between artificial intelligence and explorative learning on supply chain resilience", *International Journal of Operations and Production Management*, Vol. 45 No. 2, pp. 570-593, doi: [10.1108/IJOPM-12-2023-0990](https://doi.org/10.1108/IJOPM-12-2023-0990).
- Dubey, R., Gunasekaran, A., Bryde, D.J., Dwivedi, Y.K. and Papadopoulos, T. (2020), "Blockchain technology for enhancing swift-trust, collaboration and resilience within a humanitarian supply chain setting", *International Journal of Production Research*, Vol. 58 No. 11, pp. 3381-3398, doi: [10.1080/00207543.2020.1722860](https://doi.org/10.1080/00207543.2020.1722860).
- Ghobakhloo, M., Iranmanesh, M., Foroughi, B., Tseng, M.-L., Nikbin, D. and Khanfar, A.A.A. (2025), "Industry 4.0 digital transformation and opportunities for supply chain resilience: a comprehensive review and a strategic roadmap", *Production Planning and Control*, Vol. 36 No. 1, pp. 61-91, doi: [10.1080/09537287.2023.2252376](https://doi.org/10.1080/09537287.2023.2252376).
- Golan, M.S., Trump, B.D., Cegan, J.C. and Linkov, I. (2021), "Supply chain resilience for vaccines: review of modeling approaches in the context of the COVID-19 pandemic", *Industrial Management and Data Systems*, Vol. 121 No. 7, pp. 1723-1748, doi: [10.1108/imds-01-2021-0022](https://doi.org/10.1108/imds-01-2021-0022).
- Gozali, L., Kristina, H.J., Yosua, A., Zagloel, T.Y.M., Masrom, M., Susanto, S., Tanujaya, H., Irawan, A.P., Gunadi, A., Kumar, V., Garza-Reyes, J.A., Jap, T.B. and Daywin, F.J. (2024), "The improvement of block chain technology simulation in supply chain management (case study: pesticide company)", *Scientific Reports*, Vol. 14 No. 1, 3784, doi: [10.1038/s41598-024-53694-w](https://doi.org/10.1038/s41598-024-53694-w).

BPMJ

- Harju, A., Hallikas, J., Immonen, M. and Lintukangas, K. (2023), "The impact of procurement digitalization on supply chain resilience: empirical evidence from Finland", *Supply Chain Management-An International Journal*, Vol. 28 No. 7, pp. 62-76, doi: [10.1108/SCM-08-2022-0312](https://doi.org/10.1108/SCM-08-2022-0312).
- He, X., Hu, W., Li, W. and Hu, R. (2023), "Digital transformation, technological innovation, and operational resilience of port firms in case of supply chain disruption", *Marine Pollution Bulletin*, Vol. 190, 114811, doi: [10.1016/j.marpolbul.2023.114811](https://doi.org/10.1016/j.marpolbul.2023.114811).
- Ho, W.R., Tsolakis, N., Dawes, T., Dora, M. and Kumar, M. (2023), "A digital strategy development framework for supply chains", *IEEE Transactions on Engineering Management*, Vol. 70 No. 7, pp. 2493-2506, doi: [10.1109/TEM.2021.3131605](https://doi.org/10.1109/TEM.2021.3131605).
- Hofstede, G. (2001), *Culture's Consequences: Comparing Values, Behaviors, Institutions and Organizations Across Nations*, SAGE Publications, Thousand Oaks, CA.
- Hosseini, S., Ivanov, D. and Dolgui, A. (2019), "Review of quantitative methods for supply chain resilience analysis", *Transportation Research Part E: Logistics and Transportation Review*, Vol. 125, pp. 285-307, doi: [10.1016/j.tre.2019.03.001](https://doi.org/10.1016/j.tre.2019.03.001).
- Huang, K., Wang, K., Lee, P.K.C. and Yeung, A.C.L. (2023), "The impact of Industry 4.0 on supply chain capability and supply chain resilience: a dynamic resource-based view", *International Journal of Production Economics*, Vol. 262, 108913, doi: [10.1016/j.ijpe.2023.108913](https://doi.org/10.1016/j.ijpe.2023.108913).
- Iftikhar, A., Ali, I., Arslan, A. and Tarba, S. (2024), "Digital innovation, data analytics, and supply chain resiliency: a bibliometric-based systematic literature review", *Annals of Operations Research*, Vol. 333 No. 2, pp. 825-848, doi: [10.1007/s10479-022-04765-6](https://doi.org/10.1007/s10479-022-04765-6).
- Jauhar, S.K., Jani, S.M., Kamble, S.S., Pratap, S., Belhadi, A. and Gupta, S. (2024), "How to use no-code artificial intelligence to predict and minimize the inventory distortions for resilient supply chains", *International Journal of Production Research*, Vol. 62 No. 15, pp. 5510-5534, doi: [10.1080/00207543.2023.2166139](https://doi.org/10.1080/00207543.2023.2166139).
- Ju, Y., Hou, H. and Yang, J. (2021), "Integration quality, value co-creation and resilience in logistics service supply chains: moderating role of digital technology", *Industrial Management and Data Systems*, Vol. 121 No. 2, pp. 364-380, doi: [10.1108/IMDS-08-2020-0445](https://doi.org/10.1108/IMDS-08-2020-0445).
- Jung, K.S., Dawande, M., Geismar, N., Guide, V.D.R. and Sriskandarajah, C. (2014), "Supply planning models for a remanufacturer under just-in-time manufacturing environment with reverse logistics", SSRN Scholarly Paper, Social Science Research Network, Rochester, NY, 21 March, doi: [10.2139/ssrn.2427326](https://doi.org/10.2139/ssrn.2427326).
- Kazancoglu, Y., Lafci, C., Berberoglu, Y., Upadhyay, A., Rocha-Lona, L. and Kumar, V. (2024), "The effects of globalization on supply chain resilience: outsourcing techniques as interventionism, protectionism, and regionalization strategies", *Operations Management Research*, Vol. 17 No. 2, pp. 505-522, doi: [10.1007/s12063-023-00429-1](https://doi.org/10.1007/s12063-023-00429-1).
- Kumar, R.R. and Raj, A. (2025), "Big data adoption and performance: mediating mechanisms of innovation, supply chain integration and resilience", *Supply Chain Management-An International Journal*, Vol. 30 No. 1, pp. 67-85, doi: [10.1108/SCM-03-2024-0186](https://doi.org/10.1108/SCM-03-2024-0186).
- Li, H., Xu, J. and Luo, Y. (2024), "The impact of mobile social media on knowledge sharing among vocational school teachers: a social cognitive career perspective", *Journal of Information Science*. Online First, doi: [10.1177/01655515241293754](https://doi.org/10.1177/01655515241293754).
- Li, P., Chen, Y. and Guo, X. (2025), "Digital transformation and supply chain resilience", *International Review of Economics and Finance*, Vol. 99, 104033, doi: [10.1016/j.iref.2025.104033](https://doi.org/10.1016/j.iref.2025.104033).
- Lin, Y., Pang, G., Duan, K., Luo, J., Wang, S. and Qu, J. (2025), "The impacts of digital and learning orientations on supply chain resilience", *Industrial Management and Data Systems*, Vol. 125 No. 2, pp. 535-574, doi: [10.1108/IMDS-04-2024-0379](https://doi.org/10.1108/IMDS-04-2024-0379).
- Lu, Q., Wang, X. and Wang, Y. (2023), "Enhancing supply chain resilience with supply chain governance and finance: the enabling role of digital technology adoption", *Business Process Management Journal*, Vol. 29 No. 4, pp. 944-964, doi: [10.1108/BPMJ-11-2022-0601](https://doi.org/10.1108/BPMJ-11-2022-0601).

- Mandal, S. (2017), "The influence of organizational culture on healthcare supply chain resilience: moderating role of technology orientation", *Journal of Business and Industrial Marketing*, Vol. 32 No. 8, pp. 1021-1037, doi: [10.1108/JBIM-08-2016-0187](https://doi.org/10.1108/JBIM-08-2016-0187).
- Oesterreich, T.D., Anton, E., Schuir, J., Teuteberg, F., Al-Busaidi, A.S. and Dwivedi, Y.K. (2025), "Learning from the past to advance the future: synthesizing the meta-analysis landscape in information systems research", *Information Systems Frontiers*. Online First, doi: [10.1007/s10796-025-10598-y](https://doi.org/10.1007/s10796-025-10598-y).
- Orwin, R.G. (1983), "A fail-safe N for effect size in meta-analysis", *Journal of Educational Statistics*, Vol. 8 No. 2, pp. 157-159, doi: [10.3102/10769986008002157](https://doi.org/10.3102/10769986008002157).
- Peng, T., Wang, C., Xu, J., Dai, J. and Yu, T. (2024), "Evolution and current research status of educational leadership theory: a content analysis-based study", *Sage Open*, Vol. 14 No. 3, doi: [10.1177/21582440241285763](https://doi.org/10.1177/21582440241285763).
- Pigott, T.D. and Polanin, J.R. (2020), "Methodological guidance paper: high-quality meta-analysis in a systematic review", *Review of Educational Research*, Vol. 90 No. 1, pp. 24-46, doi: [10.3102/0034654319877153](https://doi.org/10.3102/0034654319877153).
- Pillay, M. (2017), "Resilience engineering: an integrative review of fundamental concepts and directions for future research in safety management", *Open Journal of Safety Science and Technology*, Vol. 7 No. 04, pp. 129-160, doi: [10.4236/ojsst.2017.74012](https://doi.org/10.4236/ojsst.2017.74012).
- Ponomarov, S.Y. and Holcomb, M.C. (2009), "Understanding the concept of supply chain resilience", *The International Journal of Logistics Management*, Vol. 20 No. 1, pp. 124-143, doi: [10.1108/09574090910954873](https://doi.org/10.1108/09574090910954873).
- Qi, N. and Zhang, A. (2024), "Technological cooperation relationships among supply chain enterprises under the perspective of technological blockade: a tripartite evolutionary game model", *Journal of Cleaner Production*, Vol. 447, 141240, doi: [10.1016/j.jclepro.2024.141240](https://doi.org/10.1016/j.jclepro.2024.141240).
- Rodríguez-González, R.M., Madrid-Guijarro, A. and Maldonado-Guzmán, G. (2023), "Digital organizational culture and absorptive capacity as precursors to supply chain resilience and sustainable performance", *Journal of Cleaner Production*, Vol. 420, 138411, doi: [10.1016/j.jclepro.2023.138411](https://doi.org/10.1016/j.jclepro.2023.138411).
- Rosenthal, R. (1979), "The file drawer problem and tolerance for null results", *Psychological Bulletin*, Vol. 86 No. 3, pp. 638-641, doi: [10.1037/0033-2909.86.3.638](https://doi.org/10.1037/0033-2909.86.3.638).
- Singh, R.K. (2025), "Transforming humanitarian supply chains with digital twin technology: a study on resilience and agility", *International Journal of Logistics Management*, Vol. 36 No. 4, pp. 1119-1135, doi: [10.1108/IJLM-08-2024-0504](https://doi.org/10.1108/IJLM-08-2024-0504).
- Spieske, A. and Birkel, H. (2021), "Improving supply chain resilience through Industry 4.0: a systematic literature review under the impressions of the COVID-19 pandemic", *Computers and Industrial Engineering*, Vol. 158, 107452, doi: [10.1016/j.cie.2021.107452](https://doi.org/10.1016/j.cie.2021.107452).
- Spieske, A., Gebhardt, M., Kopyto, M., Birkel, H. and Hartmann, E. (2023), "The future of Industry 4.0 and supply chain resilience after the COVID-19 pandemic: empirical evidence from a Delphi study", *Computers and Industrial Engineering*, Vol. 181, 109344, doi: [10.1016/j.cie.2023.109344](https://doi.org/10.1016/j.cie.2023.109344).
- Stadtfeld, G.M. and Gruchmann, T. (2024), "Dynamic capabilities for supply chain resilience: a meta-review", *International Journal of Logistics Management*, Vol. 35 No. 2, pp. 623-648, doi: [10.1108/IJLM-09-2022-0373](https://doi.org/10.1108/IJLM-09-2022-0373).
- Taj, S., Imran, A.S., Kastrati, Z., Daudpota, S.M., Memon, R.A. and Ahmed, J. (2023), "IoT-based supply chain management: a systematic literature review", *Internet of Things*, Vol. 24, 100982, doi: [10.1016/j.iot.2023.100982](https://doi.org/10.1016/j.iot.2023.100982).
- Tortorella, G., Gloet, M., Samson, D., Kurnia, S., Fogliatto, F.S. and Anzanello, M.J. (2024), "Food supply chain resilience through digital transformation: a mixed-method approach", *The International Journal of Logistics Management*, Vol. 36 No. 2, pp. 381-412, doi: [10.1108/IJLM-01-2024-0030](https://doi.org/10.1108/IJLM-01-2024-0030).
- Wang, Q., Chen, L., Jia, F., Luo, Y. and Zhang, Z. (2024), "The relationship between supply chain integration and sustainability performance: a meta-analysis", *International Journal of Logistics*

BPMJ

Research and Applications, Vol. 27 No. 8, pp. 1388-1409, doi: [10.1080/13675567.2022.2144812](https://doi.org/10.1080/13675567.2022.2144812).

- Wang, L., Zeng, Y., Xu, Y., Chi, M. and Wu, H. (2025), "The effect of blockchain on construction supply chain resilience: a mediated moderation model", *Ain Shams Engineering Journal*, Vol. 16 No. 2, 103253, doi: [10.1016/j.asej.2024.103253](https://doi.org/10.1016/j.asej.2024.103253).
- Wieland, A. and Wallenburg, C.M. (2013), "The influence of relational competencies on supply chain resilience: a relational view", *International Journal of Physical Distribution and Logistics Management*, Vol. 43 No. 4, pp. 300-320, doi: [10.1108/IJPDLM-08-2012-0243](https://doi.org/10.1108/IJPDLM-08-2012-0243).
- Xu, T. and Liu, X. (2024), "Achieving manufacturing supply chain resilience: the role of paradoxical leadership and big data analytics capability", *Journal of Manufacturing Technology Management*, Vol. 35 No. 2, pp. 205-225, doi: [10.1108/JMTM-05-2023-0206](https://doi.org/10.1108/JMTM-05-2023-0206).
- Xue, L., Gong, Y., Yang, B. and Xu, X. (2024), "Resilience, efficiency fluctuations, and regional heterogeneity in disaster: an empirical study on logistics", *Socio-Economic Planning Sciences*, Vol. 93, 101854, doi: [10.1016/j.seps.2024.101854](https://doi.org/10.1016/j.seps.2024.101854).
- Zavala-Alcívar, A., Verdecho, M.-J., Alfaro-Saiz, J.-J. and González Ortega, R. (2023), "The impact of Industry 4.0 on supply chain resilience management", in Camarinha-Matos, L.M., Boucher, X. and Ortiz, A. (Eds), *Collaborative Networks in Digitalization and Society 5.0*, Springer Nature Switzerland, Cham, pp. 107-120, doi: [10.1007/978-3-031-42622-3_8](https://doi.org/10.1007/978-3-031-42622-3_8).
- Zhao, N., Hong, J. and Lau, K.H. (2023), "Impact of supply chain digitalization on supply chain resilience and performance: a multi-mediation model", *International Journal of Production Economics*, Vol. 259, 108817, doi: [10.1016/j.ijpe.2023.108817](https://doi.org/10.1016/j.ijpe.2023.108817).
- Zhu, T., Li, X., Wu, H. and Chu, Z. (2023), "Does the collaboration of digitalization foster regional green development?", *Sustainability, Multidisciplinary Digital Publishing Institute*, Vol. 15 No. 20, 14799, doi: [10.3390/su152014799](https://doi.org/10.3390/su152014799).

Corresponding author

Yuying Luo can be contacted at: lyy24720692@shu.edu.cn

Trust the messenger, then the message: unpacking the cross-route logic of danmaku engagement

Aslib Journal of
Information
Management

Jun Xu

*Department of Education Information Technology, East China Normal University,
Shanghai, China*

Yuying Luo

School of Management, Shanghai University, Shanghai, China

Chengliang Wang

*Institute for Positive Psychology and Education, Australian Catholic University,
North Sydney, Australia, and*

Huina Wu

*Department of Education Information Technology, East China Normal University,
Shanghai, China*

Received 31 January 2026
Revised 19 May 2026
Accepted 12 July 2026

Abstract

Purpose – Contemporary video repositories increasingly overlay real-time, user-generated metadata in the form of danmaku (bullet comments) onto linear content, creating a complex retrieval environment defined by the tension between collaborative sensemaking and cognitive overload. This study examines this human information interaction dynamic, analyzing how observable creator-related source cues, Content value (CV) signals and peer feedback are associated with engagement within information-dense streams.

Design/methodology/approach – Leveraging large-scale behavioral trace data from 261 informational videos and over 130,000 time-anchored comments on Bilibili, the study employs a multi-method approach combining partial least squares structural equation modeling, partial least squares multi-group analysis and BERTopic modeling to triangulate structural engagement associations with semantic discourse patterns.

Findings – The analysis identifies association patterns consistent with a sequential cross-route account in which peripheral source cues function as heuristic filters that are linked to CV signals and subsequent feedback interaction. Crucially, this pattern is context-contingent: high-efficacy information environments show stronger content-oriented engagement, whereas low-efficacy contexts show stronger associations with peripheral source heuristics and social signals.

Originality/value – The research refines dual-process theories within digital information management by specifying a measurable, sequential association pattern for observable engagement under information density. It distinguishes Audience Conversion Efficacy as a critical moderator, offering evidence-based strategies for adaptive interface design that align information presentation with content utility.

Keywords Elaboration likelihood model, Human information interaction, Behavioral trace data, User engagement, Social video platforms

Paper type Research article

1. Introduction

Video-sharing platforms increasingly function as information environments in which users evaluate content, interpret platform-visible cues and respond to peer reactions within the same viewing interface (Amaldoss and Shin, 2026). Bilibili provides a relevant setting for examining this process because its danmaku (bullet comments) system overlays time-synchronized viewer messages directly onto the video stream (Li and Tao, 2026). This feature turns video viewing into a layered information process in which the primary video content is accompanied by a continuous stream of social annotation. At the behavioral level, danmaku



Aslib Journal of Information
Management
© Emerald Publishing Limited
e-ISSN: 2050-3814
p-ISSN: 2050-3806
DOI 10.1108/AJIM-01-2026-0111

Declaration of conflicting interest: The authors declare that they have no conflict of interest with respect to the research, authorship and publication of this article.

AJIM

comment density on Bilibili is associated with users' viewing and awarding behavior and this association may become non-linear above a density threshold (Li *et al.*, 2025). At the socio-cognitive level, viewer cues and danmaku timing are linked to social cognition and narrative engagement on online video platforms (Zhang *et al.*, 2025b). Taken together, this evidence positions danmaku not merely as an interaction feature but as part of the information environment through which users evaluate whether and how to engage.

This information layer is especially consequential in educational video contexts, where engagement should support effortful information processing rather than only social co-presence. Danmaku can enhance perceived togetherness and prompt peer exchange (Xiang and Chae, 2022), functioning as a social media affordance that facilitates knowledge sharing within the learning community (Li *et al.*, 2024a), but it can also introduce competing information that fragments attention and intensifies cognitive overload (Peng *et al.*, 2025). In instructional video learning, danmaku interruptions can distract learners and increase cognitive load for some cognitive styles (Mu *et al.*, 2025). This tension is central to human information interaction (HII) because viewers encounter competing creator-related source cues, content value (CV) signals and peer feedback while deciding whether to remain passive viewers or become real-time contributors. Although prior work offers valuable insights, two gaps remain. First, existing studies often examine these cues in isolation or as parallel predictors, which provides limited explanation of how users combine multiple signals when navigating high-density environments (Dai *et al.*, 2026). Second, research based mainly on self-reported perceptions may diverge from observable behavior in naturalistic platform settings (Costa-Feito *et al.*, 2025). Together, these limitations motivate an association-based account of danmaku engagement as a sequential HII process in which users' source screening, content evaluation, peer feedback and real-time participation are modeled as linked components of engagement.

We ground the model in the elaboration likelihood model (ELM) (Petty and Cacioppo, 1986) and use the community of inquiry (COI) framework (Garrison *et al.*, 2001) as a contextual lens to interpret learning-oriented participation. Methodologically, we analyze a large-scale archival dataset of 261 educational videos and integrate partial least squares structural equation modeling (PLS-SEM), multi-group analysis (MGA) and BERTopic modeling of more than 130,000 real-time comments to triangulate structural relations with discourse patterns.

This study therefore addresses the following central research questions: (1) How do peripheral cues (source credibility (SC)) and central cues (CV and peer feedback) interact to shape user engagement in user-generated content (UGC) video environments, and can a sequential, cross-route model better explain this dynamic than traditional parallel-processing frameworks? (2) Under what specific conditions does this engagement pattern emerge, and how is the intrinsic efficacy of the video content associated with differences in these structural paths? (3) What underlying behavioral signature, as revealed through users' real-time discourse, is associated with this moderation?

In answering these questions, this paper makes three contributions to information management and HII research. First, it specifies and tests a sequential cross-route association pattern showing how source-related cues, content evaluation signals, and peer feedback jointly relate to real-time engagement (RTE) in social video environments. Second, it specifies boundary conditions by showing that key path strengths differ across content contexts with high versus low audience conversion efficacy (CE), clarifying when observable engagement is more closely aligned with source cues versus content-centric interaction. Third, it triangulates structural results with discourse evidence by linking moderation patterns to thematic differences in danmaku, offering an ecologically valid account of how engagement patterns vary under information density.

2. Theoretical background and hypotheses

2.1 *Danmaku as a dynamic information layer*

Danmaku refers to time-anchored viewer messages that overlay the video screen, functioning as a distinct secondary information layer during content consumption (Li, 2017). Unlike traditional asynchronous comments, this feature externalizes viewers' in-the-moment interpretations and exposes them to continuous peer-produced cues that can redirect attention and reshape ongoing information processing (Yuan *et al.*, 2025). This dual character makes danmaku a useful behavioral trace for information management research, as it records time-stamped moments of active contribution under conditions of high information density. Consequently, rather than viewing engagement solely as social participation, we conceptualize it as a sequence of information filtering and processing decisions in which creator-related source cues, CV signals and peer feedback are jointly present.

2.2 *Elaboration likelihood model and conceptual framework*

To explain engagement in this cue-rich environment, we situate the ELM within the broader tradition of dual-process theories of information processing. Dual-process accounts distinguish relatively fast, heuristic processing from more effortful, analytic processing (Evans, 2008). The ELM applies this distinction to persuasion and information evaluation by proposing that individuals rely on a central route when they scrutinize message or content arguments, and on a peripheral route when they use cues such as SC, popularity, or social endorsement under limited motivation or ability (Petty and Cacioppo, 1986).

In danmaku-enabled educational videos, these two routes are unlikely to operate only as independent predictors. Creator-related source cues, CV signals and peer feedback appear together in a dense and time-sensitive information environment. Users may rely on creator-related cues to reduce uncertainty about whether a video warrants attention. They may also use CV signals and peer feedback to assess whether deeper processing and real-time participation are worthwhile. Accordingly, this study advances an association-based sequential cross-route interpretation of ELM. Within the specified model sequence, creator-related SC cues are modeled as being linked with CV signals and user feedback interaction (UFI), which are then associated with RTE observed in danmaku traces. The COI framework (Garrison *et al.*, 2001) is used as a contextual lens for interpreting why engagement traces in learning-oriented environments often combine social and cognitive dimensions. It is not treated as a measurement model, but as an interpretive framework for understanding the nature of peer discourse.

2.3 *Research hypotheses*

This study operationalizes its constructs using publicly observable platform traces at the video level. Creator-related SC cues, CV signals, peer feedback interaction and RTE are derived from standard counters and text-derived indicators (comments and danmaku). Figures 1–3 provide interface context and an overview of key platform cues and conceptual routes; the hypotheses below specify the expected associations among constructs under a sequential cross-route logic rooted in ELM (Petty and Cacioppo, 1986).

2.3.1 Source credibility (SC). SC encompasses observable creator-level prominence signals, such as follower counts and historical performance metrics. In digital information environments, users often rely on these aggregated indicators as a heuristic proxy for authority and potential utility, rather than assessing credibility through direct expertise verification. Accordingly, we conceptualize SC as a peripheral cue that may reduce uncertainty and inform users' initial expectations and is therefore expected to be associated with subsequent engagement patterns (Li and See-To, 2024; Muda and Hamzah, 2021).

2.3.2 Content value (CV). Because this study utilizes archival platform traces, CV is captured through post-consumption appreciation metrics (e.g. likes, collections and coins). These indicators are treated as visible collective appreciation signals within the platform



Figure 1. Video playback interface

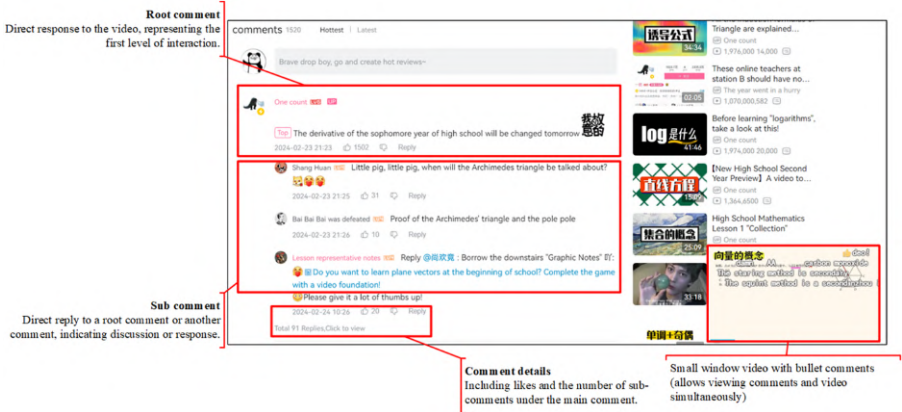


Figure 2. Video comment interface

environment. More broadly, perceived usefulness has been identified as a central rational driver of continued engagement in digitally mediated learning environments (Ke et al., 2026; Li et al., 2026). While these metrics are accumulated post-viewing, they serve as a visible quality signal within the information environment that users process in conjunction with source cues. Therefore, we interpret CV not as a temporal antecedent to the viewing act itself, but as a proxy for the intrinsic “informational currency” of the content that is visible within the user’s engagement environment.

H1 is stated accordingly:

H1. Source Credibility (SC) is positively associated with content value signals (CV).

2.3.3 User feedback interaction (UFI). UFI reflects users’ observable participation in the shared evaluation space around a video. In this study, UFI is captured via comment volume and related interaction signals (e.g. root comments, comment likes and sentiment-derived counts). These indicators represent the extent to which viewers externalize evaluations, request clarification, or respond to others, thereby making feedback an explicit part of the information environment (Phua et al., 2020).

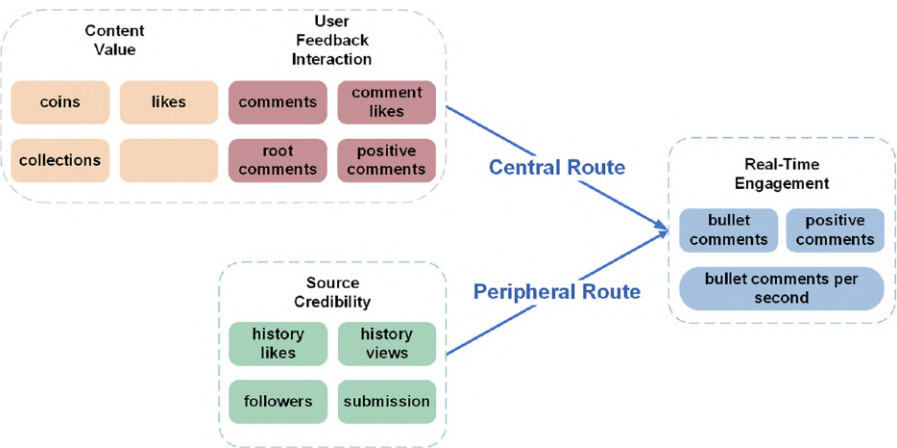


Figure 3. Conceptual factor and route diagram

Under the cross-route logic, SC may be associated with lower perceived risk of engaging and with entry into the feedback space, while stronger CV signals may be associated with user evaluations and questions (Oh *et al.*, 2018). Therefore, we state the following hypotheses in associational terms consistent with archival data:

- H2. Source credibility (SC) is positively associated with user feedback interaction (UFI).
- H3. Content value signals (CV) are positively associated with user feedback interaction (UFI).

2.3.4 Real-time engagement (RTE). RTE refers to users' during-view participation that is temporally anchored to the video stream. On danmaku-enabled platforms, RTE is reflected in the volume and density of danmaku as well as sentiment-derived counts, capturing the immediacy and intensity of real-time reactions (Zeng *et al.*, 2024). As a behavioral trace, RTE marks observable moments of active, time-sensitive expression while information is still unfolding.

Based on ELM, both peripheral cues, represented by SC and central cues, represented by CV signals and peer feedback interaction, can relate to real-time participation. Using archival data, we formulate the following hypotheses as associations rather than time-ordered causal claims:

- H4. Source Credibility (SC) is positively associated with real-time engagement (RTE).
- H5. Content value signals (CV) are positively associated with real-time engagement (RTE).
- H6. User feedback interaction (UFI) is positively associated with real-time engagement (RTE).

2.3.5 Mediating hypothesis. To examine whether the observed associations are consistent with a *sequential cross-route* mechanism, we test a set of indirect-effect hypotheses among SC, CV, UFI and RTE. Conceptually, creator-related SC cues may be linked to initial expectations; this can be reflected in stronger CV signals and greater entry into the peer feedback space, which may in turn relate to more frequent real-time participation.

Because our indicators are archival behavioral traces aggregated at the video level, the indirect paths reported below should be interpreted as statistical decompositions of associations rather than causal mediation in the strict time-ordered sense. We therefore use

cautious language and treat the results as evidence of pattern consistency with the proposed sequential mechanism.

- H7. The associations among SC, CV, UFI and RTE exhibit indirect-effect patterns consistent with a sequential cross-route mechanism.
- H7a. SC is indirectly associated with RTE through CV and UFI (SC → CV → UFI → RTE).
- H7b. SC is indirectly associated with RTE through CV (SC → CV → RTE).
- H7c. SC is indirectly associated with UFI through CV (SC → CV → UFI).
- H7d. SC is indirectly associated with RTE through UFI (SC → UFI → RTE).
- H7e. CV is indirectly associated with RTE through UFI (CV → UFI → RTE).

The hypothesized model of this study is illustrated in Figure 4.

2.3.6 *The moderating role of contextual factors.* While the ELM explains how users rely on central and peripheral cues, the relative salience of these cues is unlikely to remain constant across all video contexts. User-generated content platforms vary substantially in content performance, creator visibility and peer interaction dynamics. In danmaku-enabled educational videos, users encounter creator-related cues, CV signals, peer feedback and real-time commentary within the same information environment. These contextual conditions may alter the strength of the structural paths proposed in the engagement model. More broadly, evidence from digital process research indicates that the effects of digital technologies are not uniform but vary across functional stages and contextual conditions (Xu et al., 2025). We therefore examine three moderators that correspond to content context, source context and peer discourse context: Audience CE, Creator Standing (CS) and Danmaku Climate (DC).

Audience CE captures the extent to which a video context is accompanied by deliberate engagement. Conceptually, CE differs from CV. CV represents accumulated appreciation attached to a video, whereas CE reflects the contextual efficiency with which viewed content is associated with active responses. Under ELM, higher CE contexts are expected to show greater salience of substantive content cues because viewers are more likely to treat the video as worthy of continued attention and interaction. Lower CE contexts, in contrast, may show stronger alignment with SC and social heuristics in observable engagement patterns. Accordingly, we propose:

- H8. The strengths of the path relationships in the engagement model differ significantly between videos with high and low Audience Conversion Efficacy.

CS represents the broader platform position of the uploader. It extends beyond a single reputation cue by reflecting whether the creator has an established presence and historical

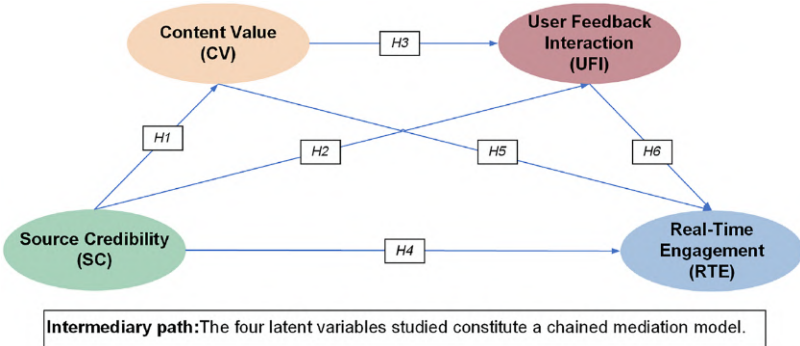


Figure 4. Hypothesized structural model

footprint in the platform community. As a source-context condition, high CS may correspond to differences in how viewers interpret SC and to lower uncertainty around entry into the feedback and danmaku spaces. Accordingly, we propose:

- H9. The strengths of the path relationships in the engagement model differ significantly between videos from creators with high and low Creator Standing.

DC represents the affective tone of the real-time peer information environment. A constructive climate may be associated with peer feedback appearing more useful and socially safe, whereas a passive or mixed climate may weaken the informational value of the surrounding commentary (Wang and Chen, 2026). Because danmaku is processed alongside the focal video, this climate may condition how feedback and CV cues are associated with real-time participation. Accordingly, we propose:

- H10. The strengths of the path relationships in the engagement model differ significantly between videos with a constructive and a passive or mixed Danmaku Climate.

For empirical testing, each moderator hypothesis is examined across the same six structural paths in the model. The suffixes a through f correspond respectively to SC → CV, SC → UFI, SC → RTE, CV → UFI, CV → RTE and UFI → RTE. Accordingly, the MGA results report H8a through H8f for CE, H9a through H9f for CS and H10a through H10f for DC.

3. Research methodology

3.1 Sample selection

Bilibili serves as a prominent ecosystem for examining HII, particularly within the dissemination of educational content. The platform is distinguished by its integration of rich learning resources with a synchronous “danmaku” comment mechanism, creating a dense social information layer (Wei *et al.*, 2023). This study targets the “Campus Learning” section under the “Knowledge” category, which hosts a spectrum of content ranging from formal instructional videos (e.g. exam analyses) to informal knowledge sharing (e.g. interview strategies). In this context, the interaction between creators and viewers extends beyond passive consumption and is associated with a collaborative learning dynamic linked to stronger engagement outcomes (Leem, 2023).

To ensure the sample represents high-utility information artifacts rather than merely popular entertainment, data collection was strictly prioritized based on “favorites” (collections) rather than view counts or likes. Within the framework of digital repositories, the “favorite” function operates as a personal knowledge management tool, explicitly signaling a user’s intent to archive, retrieve and reuse information (Wei *et al.*, 2023). In contrast to “likes”, which often capture fleeting affective responses, “favorites” indicate a deliberate cognitive judgment regarding the content’s long-term educational value and the user’s consolidation needs. Consequently, selecting samples via this archival metric helps focus the analysis on videos likely to have functioned as knowledge carriers, thereby supporting the validity of investigating deep information adoption behaviors.

3.2 Data collection

We curated a dataset of the 300 most-favorited videos within the “Campus Learning” section, identified by their unique BV numbers through the platform’s native filtering interface. The collection window was strategically limited to content published between December 10, 2023 and March 10, 2024. This specific three-month timeframe was selected to increase the likelihood that each video had accrued a stable viewership baseline post-publication, thereby reducing the transient inflation of view counts often observed in newly released content (Dong *et al.*, 2022), while simultaneously minimizing potential data biases arising from significant temporal discrepancies.

AJIM

Following sample identification, comprehensive behavioral traces – including metadata for the 300 videos and their respective uploaders – were retrieved via Bilibili’s open API. All collected data are publicly accessible, ensuring full compliance with ethical and legal standards for digital research. The final dataset presents a broad spectrum of engagement, with favorite counts ranging from 6,579 to 301,343, providing a robust empirical foundation for the subsequent structural analysis.

3.3 Data filtering and processing

Following initial data collection, a notable observation was made regarding videos that, while possessing a high number of favorites, lacked interaction. This suggests that some videos conveyed information but showed limited interactivity relative to the focus of this study. To align the sample with the study’s focus on observable interaction, videos with fewer than 10 comments or danmaku were filtered out, resulting in a final selection of 261 videos for analysis.

The data processing for this study primarily involved sentiment analysis of the comments and danmaku texts. To ensure accurate text sentiment classification, a Chinese sentiment analysis model, RoBERTa-JD-Binary, fine-tuned on the JD review corpus, was utilized. This model has been reported to achieve high accuracy in Chinese sentiment classification tasks (Fei, 2024). The model was applied to perform binary classification of the sentiments (positive or negative) for 118,380 comments and 130,877 danmaku entries from the 261 video samples.

3.4 Construct operationalization and grouping variables

The constructs were operationalized using publicly observable video-level platform traces. SC was measured using uploader followers, historical likes, historical views and submissions. CV was measured using coins, collections and likes, which capture collective appreciation attached to the video. UFI was measured using comments, positive comments, root comments and total comment likes. RTE was measured using danmaku comments, positive danmaku comments and danmaku comments per second. These indicators were log-transformed before PLS-SEM estimation, as reported in the descriptive statistics section.

The three contextual grouping variables used in the MGA were also constructed at the video level. Audience CE was operationalized as the ratio of observable engagement actions to views and was calculated as follows: $CE = \frac{(Likes + Collections + Coins + Shares)}{Views}$. This grouping variable differs from CV because it normalizes engagement actions by exposure rather than representing the absolute level of accumulated appreciation. CS was operationalized as a composite standing index and was calculated as follows, where z denotes a standardized score: $CS = z(followers) + z(submissions) + z(historical\ views)$. DC was operationalized as the proportion of positive danmaku comments and was calculated as positive danmaku comments divided by total danmaku comments.

For the multi-group analysis, the sample was split into high and low groups using median-based cutoffs for each moderator. Because the sample included 261 videos, observations equal to or above the median were assigned to the high group, producing 131 videos in the high group and 130 videos in the low group for each comparison. Videos with $CE \geq 0.157$ were assigned to the high-CE group, and the remaining videos were assigned to the low-CE group. Videos with $CS \geq -0.837$ were assigned to the high CS group, and the remaining videos were assigned to the low CS group. Videos with $DC \geq 0.741$ were assigned to the constructive DC group, and the remaining videos were assigned to the passive or mixed DC group.

4. Results

4.1 Descriptive statistics

This study utilized archival data from Bilibili, drawing on publicly available video-level platform indicators and uploader information rather than survey responses. Archival data refers to data that has already been collected and preserved to support new research or analysis

(Fisher and Chaffee, 2018). Table 1 reports descriptive statistics for the original, untransformed platform indicators so that the empirical scale of the dataset remains transparent. The log-transformed indicators used for PLS-SEM estimation are reported separately in Appendix Table A1.

Prior to model estimation, the distributional properties of the indicators were assessed using the Shapiro–Wilk test. As shown in Table 1, the original platform indicators did not satisfy normality assumptions. Given this non-normality, PLS-SEM was selected as the primary analytical method. Unlike its covariance-based counterpart (CB-SEM), PLS-SEM is a variance-based, non-parametric approach that does not require normally distributed data, making it suitable for predictive modeling with non-normal behavioral trace data (Hair *et al.*, 2011).

For model estimation, natural logarithmic transformations were applied to reduce distributional skewness, alleviate scale differences and improve numerical stability. Where required for variables with possible zero values, a small pseudocount was used to ensure the feasibility of the logarithmic transformation, following the data-processing logic suggested by Biswas (2016). The transformed indicators were used for PLS-SEM estimation, whereas Table 1 preserves the original measurement scale.

4.2 Reflective measurement model assessment

This study employed PLS-SEM to evaluate real-time interactive behaviors in danmaku videos. This methodology is widely applied in user behavior research on online platforms, particularly excelling in predicting and explaining user behaviors (Klein *et al.*, 2015; Moghadamzadeh *et al.*, 2020). To validate the proposed model's effectiveness and reliability, an assessment of the measurement model was conducted.

The evaluation of the measurement model was based on three criteria: internal consistency, convergent validity and discriminant validity. As shown in Table 2, the Composite Reliability and Cronbach's Alpha coefficients for all latent variables exceeded 0.70, indicating good internal consistency (Hair *et al.*, 2021; Li *et al.*, 2024b; Xia *et al.*, 2024). Additionally, with the exception of the natural logarithm of submissions, all factor loadings surpassed 0.70 and the Average Variance Extracted (AVE) values exceeded 0.50, confirming the measurement model's convergent validity and indicating strong associations between observed variables and their corresponding latent variables (Fornell and Larcker, 1981; Wang *et al.*, 2024b, 2025).

For discriminant validity, the AVE square roots of each latent variable were compared against their correlation coefficients, following Fornell and Larcker's (1981) criteria (Fornell and Larcker, 1981; Wang *et al.*, 2024a). As shown in Table 3, all AVE square roots exceeded their respective correlation coefficients, demonstrating good discriminant validity. Furthermore, the Heterotrait-Monotrait Ratio (HTMT) and bootstrapping statistical tests further supported discriminant validity, with all HTMT values below the 0.85 threshold (Hair *et al.*, 2019). Confidence interval analyses indicated that none of the intervals included 1, further reinforcing the model's discriminant validity. These results provide a solid foundation for subsequent structural model analyses.

4.3 Structural model assessment

In the structural model assessment, SmartPLS 4 was utilized for analysis via PLS-SEM. Unlike CB-SEM, PLS-SEM is a predictive-oriented modeling approach that does not depend on model fit indices (Shmueli *et al.*, 2019). Therefore, fit indices such as SRMR and NFI require cautious application in PLS-SEM, and there is a lack of direct evidence supporting their use for model evaluation based on archival data. Consequently, model evaluation and selection should primarily rely on predictive indicators rather than traditional fit indices (Hair *et al.*, 2019, 2021). Accordingly, the assessment of the structural model focused primarily on predictive capability and the significance of path coefficients.

Original untransformed platform indicators

n	Min	Q1	Median	Q3	Max	Mean	SD	SW <i>p</i> > 0.05
of uploader	856	33,233	169,569	483,478	14,138,613	834,571.76	2,420,107.82	No
likes	2,460	172,854	596,663	1,482,732	33,941,536	2,293,136.71	5,739,761.82	No
views	57,946	2,924,960	8,655,928	26,905,507	782,217,907	47,411,076.97	131,128,292.60	No
ns	1	59	141	343	2,135	257.90	340.12	No
s	226	973	1,834	3,838	46,660	3,981.06	6,456.35	No
s	6,579	8,720	11,957	23,405	301,343	23,023.68	36,241.29	No
s	1,494	7,803	18,148	42,431	444,991	38,784.09	58,185.24	No
s	14	115	268	485	5,382	453.56	657.14	No
omments	9	89	200	383	3,867	348.92	490.94	No
ments	9	81	177	350	4,436	349.14	555.47	No
ment likes	19	789	2,340	8,137	114,001	8,754.23	16,262.40	No
comments	10	49	144	428	7,200	501.44	1,022.80	No
annaku comments	1	39	106	331	5,518	373.66	777.19	No
comments per	<0.01	0.13	0.66	2.20	9.52	1.62	2.12	No

D = standard deviation; SW = Shapiro–Wilk. Yes indicates *p* > 0.05; No indicates that normality was not supported

Table 2. Summary of the results for the measurement model

Latent variable	Item	Internal consistency Composite reliability	Cronbach's alpha	Convergent validity		Discriminant validity
		>0.70	>0.70	Loadings	AVE	HTMT CI < 1 (upper <0.85)
Source credibility	The natural logarithm of the number of followers of the uploader	0.937	0.916	0.882	0.790	Yes
	The natural logarithm of historical likes			0.973		
	The natural logarithm of historical views			0.978		
	The natural logarithm of submissions			0.691		
Content value	The natural logarithm of coins	0.822	0.716	0.785	0.608	Yes
	The natural logarithm of collections			0.703		
	The natural logarithm of likes			0.844		
User feedback interaction	The natural logarithm of comments	0.969	0.956	0.983	0.887	Yes
	The natural logarithm of positive comments			0.968		
	The natural logarithm of root comments			0.979		
	The natural logarithm of total comment likes			0.828		
Real-time engagement	The natural logarithm of bullet comments	0.902	0.829	0.943	0.757	Yes
	The natural logarithm of positive bullet comments			0.937		
	The natural logarithm of bullet comments per second			0.710		

Table 3. Correlation matrix with AVE

	AVE	Source credibility	Content value	User feedback interaction	Real-time engagement
Source credibility	0.790	0.889	0.086	0.082	0.287
Content value	0.608	0.294	0.780	0.362	0.179
User feedback interaction	0.887	0.287	0.602	0.942	0.461
Real-time engagement	0.757	0.423	0.536	0.679	0.870

Note(s): 1) Lower triangle shows the correlation coefficients
2) Upper triangle shows the squared values of the correlation coefficients
3) Diagonal elements show the square roots of the average variance extracted (AVE)
4) The diagonal elements (square roots of AVE) should be greater than the corresponding inter-construct correlation coefficients

AJIM

The structural model was first evaluated in terms of collinearity and explanatory power. Inner Variance Inflation Factor (VIF) values were used to assess potential multicollinearity among structural predictors, with values below 5 indicating no serious concern (Hair *et al.*, 2017). As shown in Table 4, all inner VIF values ranged from 1.000 to 1.610. The same table also reports R^2 and adjusted R^2 values for the endogenous constructs: CV ($R^2 = 0.087$, adjusted $R^2 = 0.083$), UFI ($R^2 = 0.376$, adjusted $R^2 = 0.371$) and RTE ($R^2 = 0.533$, adjusted $R^2 = 0.528$).

Because the structural model was estimated using standardized latent construct scores derived from log-transformed indicators, the path coefficients should be interpreted as standardized conditional associations among constructs. For example, the coefficient for CV → UFI indicates the expected standard-deviation change in UFI associated with a one-standard-deviation increase in CV, conditional on the other predictors in the model. These coefficients should not be read as raw-count changes in platform metrics or as direct elasticity estimates.

The bootstrapping procedure in SmartPLS 4 was then used to estimate the significance of the structural path coefficients. As reported in Table 5, all six hypothesized direct associations were statistically significant and the corresponding 95% confidence intervals did not include zero. Therefore, H1 through H6 are supported.

The structural model for this study is illustrated in Figure 5.

The bootstrapping procedure in SmartPLS 4 was also used to estimate the total, direct and indirect associations among SC, CV, UFI and RTE. Because the data are observational and aggregated at the video level, these results are interpreted as statistical decompositions of associations rather than as evidence of causal mediation.

As shown in Table 6, the total indirect association of SC with RTE was statistically significant ($\beta = 0.195$, $p < 0.001$, 95% CI [0.139, 0.260]), with VAF = 0.461, which is consistent with partial statistical mediation. At the specific-path level, H7a, H7c and H7e are supported. H7b and H7d show statistically significant indirect associations, but they are not interpreted as substantive mediation paths under the VAF threshold because their VAF values are <0.20.

Table 4. Inner VIF and explanatory power of the structural model

	Content value $R^2 = 0.087$; adj. $R^2 = 0.083$	User feedback interaction $R^2 = 0.376$; adj. $R^2 = 0.371$	Real-time engagement $R^2 = 0.533$; adj. $R^2 = 0.528$
Source credibility	1.000	1.095	1.118
Content value	–	1.095	1.610
User feedback interaction	–	–	1.602

Note(s): Cells report inner VIF values; column headings report R^2 and adjusted R^2 for endogenous constructs

Table 5. Bootstrapped significance tests of standardized path coefficients

Hypothesis	Relationship	β	STDEV	t value	95% CI	p value	Decision
H1	SC → CV	0.294	0.049	5.988	[0.206, 0.400]	<0.001	Supported
H2	SC → UFI	0.120	0.044	2.732	[0.032, 0.202]	0.006	Supported
H3	CV → UFI	0.567	0.046	12.416	[0.478, 0.658]	<0.001	Supported
H4	SC → RTE	0.228	0.050	4.601	[0.128, 0.323]	<0.001	Supported
H5	CV → RTE	0.155	0.049	3.143	[0.058, 0.252]	0.002	Supported
H6	UFI → RTE	0.521	0.050	10.465	[0.416, 0.612]	<0.001	Supported

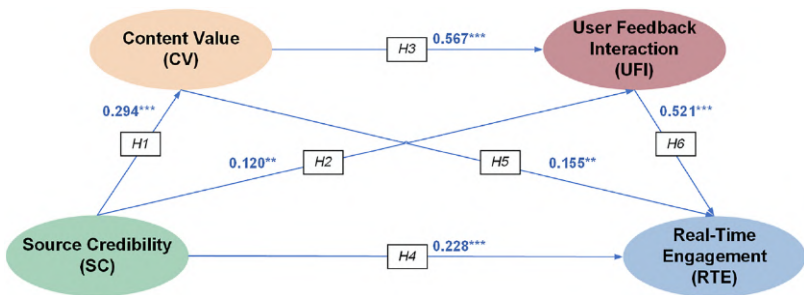


Figure 5. Structural model with standardized path coefficients. **Note.** Two asterisks indicate $p < 0.01$; three asterisks indicate $p < 0.001$

4.4 Multi-group analysis

To advance beyond a static model of engagement and to test H8, H9 and H10, a multi-group analysis (MGA) was conducted. The analysis examined whether the structural path relationships differed across subgroups defined by three contextual moderators: Audience CE, CS and DC. Before comparing path coefficients, the Measurement Invariance of Composite Models (MICOM) procedure was conducted to assess measurement equivalence across the subgroups (Henseler et al., 2016). As shown in Table 7, the three-step MICOM procedure indicated that at least partial measurement invariance was achieved across the CS and DC groupings. For the CE grouping, full or partial invariance was established for SC, CV and UFI. However, the RTE construct did not meet the compositional invariance criterion in the CE comparison. Therefore, CE-related path comparisons involving RTE should be interpreted with caution.

Following the measurement invariance assessment, MGA was conducted using the permutation test procedure in SmartPLS 4.1 to examine path-coefficient differences between groups. To align the results with the hypothesis numbering, each moderator was tested across the same six structural paths: a = SC → CV, b = SC → UFI, c = SC → RTE, d = CV → UFI, e = CV → RTE and f = UFI → RTE. Thus, Table 8 reports H8a through H8f for CE, H9a through H9f for CS and H10a through H10f for DC.

The results show that significant group differences emerged only for Audience CE. H8b was supported because the SC → UFI path was significant in the low-CE group ($\beta = 0.172$, $p < 0.05$) but not in the high-CE group ($\beta = -0.009$), with a significant difference between groups ($p = 0.037$). The H8c comparison also showed a significant between-group difference because the SC → RTE path was stronger in the low-CE group ($\beta = 0.316$, $p < 0.001$) than in the high-CE group ($\beta = 0.088$), with a significant difference between groups ($p = 0.017$). However, this RTE-related CE comparison should be interpreted cautiously in light of the MICOM result reported above. H8d was supported because the CV → UFI path was stronger in the high-CE group ($\beta = 0.684$, $p < 0.001$) than in the low-CE group ($\beta = 0.490$, $p < 0.001$), with a significant difference between groups ($p = 0.030$). No other path coefficient differed significantly across the CE, CS, or DC groupings.

4.5 Thematic analysis of danmaku content

While the multi-group analysis shows that some model path relationships differ across contextual factors, it does not explain the underlying discursive patterns associated with these variations. To provide a complementary thematic account, we conducted BERTopic-based topic modeling on the danmaku corpus. The danmaku texts were aggregated and cleaned before topic modeling. Sentence embeddings were generated using the multilingual SentenceTransformer model paraphrase-multilingual-MiniLM-L12-v2. Chinese tokenization was implemented through a customized Jieba tokenizer within CountVectorizer, together with predefined Chinese stop-word filtering. BERTopic was configured with multilingual language

AJIM

ons and mediation assessment

	Estimate	STDEV	t value	95% CI	p value	VAF	Interpretation
I → RTE	0.423	0.049	8.714	[0.329, 0.521]	<0.001	N/A	N/A
	0.287	0.045	6.436	[0.204, 0.376]	<0.001	N/A	N/A
	0.450	0.045	9.986	[0.362, 0.538]	<0.001	N/A	N/A
	0.228	0.050	4.601	[0.128, 0.323]	<0.001	N/A	N/A
	0.195	0.031	6.317	[0.139, 0.260]	<0.001	0.461	Partial statistical mediation
I → TE	0.087	0.017	4.997	[0.057, 0.126]	<0.001	0.206	Supported; partial statistical mediation
TE → I	0.046	0.018	2.592	[0.016, 0.086]	0.010	0.109	Significant indirect association, but not substantive mediation because VAF <0.20
I → TE	0.167	0.032	5.290	[0.113, 0.238]	<0.001	0.582	Supported; partial statistical mediation
TE → I	0.062	0.024	2.607	[0.017, 0.109]	0.009	0.147	Significant indirect association, but not substantive mediation because VAF <0.20
TE → I	0.295	0.035	8.470	[0.229, 0.368]	<0.001	0.656	Supported; partial statistical mediation

eviation; VAF = variance accounted for; N/A = not applicable

g results using MICOM

1	Step 2	Step 3(a)		Step 3(b)		Final invariance status	
Figurational invariance	Correlation c-value	Comp. inv. (<i>p</i> -value)	Difference (mean)	95% CI (mean)	Difference (variance)		95% CI (variance)
blished	0.976	0.065	-0.041	-0.248, 0.236	-0.166	-0.388, 0.375	Full Invariance
blished	0.992	0.013	0.403	-0.248, 0.242	0.193	-0.295, 0.273	Invariance Not Established
blished	0.991	0.121	0.454	-0.242, 0.245	0.202	-0.335, 0.362	Partial Invariance
blished	1.000	0.894	0.232	-0.244, 0.231	0.070	-0.342, 0.340	Full Invariance
blished	0.990	0.264	-0.144	-0.232, 0.242	0.080	-0.394, 0.392	Full Invariance
blished	0.996	0.102	-0.387	-0.239, 0.228	-0.382	-0.297, 0.289	Partial Invariance
blished	0.998	0.732	-1.393	-0.250, 0.239	0.080	-0.360, 0.337	Partial Invariance
blished	1.000	0.079	-0.261	-0.248, 0.251	0.123	-0.324, 0.335	Partial Invariance
blished	0.976	0.074	0.002	-0.255, 0.245	-0.145	-0.356, 0.365	Full Invariance
blished	1.000	0.804	-0.103	-0.241, 0.241	-0.119	-0.294, 0.284	Full Invariance
blished	0.999	0.730	0.048	-0.228, 0.234	-0.165	-0.368, 0.371	Full Invariance
blished	1.000	0.712	0.076	-0.233, 0.252	0.104	-0.343, 0.353	Full Invariance

0.05 to establish compositional invariance. This criterion is decisive even in cases where the 95% confidence interval may contain zero

0.05 to establish compositional invariance. This criterion is decisive even in cases where the 95% confidence interval may contain zero

i-group analysis (MGA) on path coefficients

grouping variable	Low group coefficient	Low group 95% CI	High group coefficient	High group 95% CI	Coefficient differences	p-value difference	Difference supported
	0.372***	0.214, 0.532	0.300**	0.188, 0.448	-0.072	0.456	No
	0.417***	0.323, 0.533	0.373***	0.259, 0.498	-0.044	0.652	No
	0.339***	0.243, 0.477	0.261**	0.090, 0.423	-0.078	0.420	No
	0.172*	0.026, 0.299	-0.009	-0.137, 0.119	-0.181	0.037	Yes
	0.007	-0.146, 0.148	0.150*	0.025, 0.268	0.143	0.110	No
	0.099	-0.032, 0.211	0.128*	0.000, 0.257	0.029	0.741	No
	0.316***	0.171, 0.456	0.088	-0.054, 0.214	-0.228	0.017	Yes
	0.210***	0.082, 0.337	0.232***	0.098, 0.367	0.022	0.825	No
	0.148*	0.002, 0.278	0.292***	0.162, 0.424	0.144	0.147	No
	0.490***	0.344, 0.654	0.684***	0.596, 0.763	0.194	0.030	Yes
	0.513***	0.353, 0.673	0.617***	0.525, 0.707	0.104	0.266	No
	0.587***	0.496, 0.701	0.584***	0.450, 0.708	-0.003	0.981	No
	0.121†	-0.017, 0.253	0.272**	0.087, 0.436	0.151	0.126	No
	0.111	-0.036, 0.281	0.157*	0.012, 0.296	0.046	0.685	No
	0.248***	0.127, 0.368	0.095	-0.057, 0.250	-0.153	0.120	No
	0.476***	0.327, 0.614	0.497***	0.347, 0.640	0.021	0.826	No
	0.562***	0.400, 0.693	0.501***	0.365, 0.628	-0.061	0.583	No
	0.552***	0.424, 0.671	0.479***	0.308, 0.630	-0.073	0.479	No

acy; CS, Creator Standing; DC, Danmaku Climate
0.01, * $p < 0.05$, † $p < 0.10$. Italic p -values indicate a significant difference between groups

processing, the customized vectorizer, a minimum topic size of 50 and verbose output enabled. No custom UMAP or HDBSCAN settings were specified in the archived workflow; therefore, the dimensionality-reduction and density-based clustering stages followed BERTopic’s default pipeline. These settings make the topic modeling procedure traceable while avoiding the introduction of unsupported hyperparameters.

The exported topic summary contained 34 rows, including the noise topic (–1) and 33 non-noise topics, with topic IDs, topic sizes, model-generated topic labels, representative keywords and representative documents. Intertopic distance and topic word-score visualizations were also generated to support topic inspection. To assess topic interpretability, overall topic coherence was calculated as $C_v = 0.364$ using a C_v -style indirect confirmation procedure across the eligible topic set. Given the short, colloquial and highly context-dependent nature of danmaku comments, this coherence value was treated as a diagnostic indicator of topic interpretability rather than as standalone validation (Wang *et al.*, 2024b, 2025). The model-generated topics were subsequently inspected and aggregated through inductive coding into five overarching theme categories: cognitive and content, metacognitive and navigational, affective and motivational, social and community and off-topic and photic.

The thematic distribution across the moderating conditions is presented in Figure 6. As depicted in Figure 6(a), the analysis shows pronounced thematic differences across CE groups. In the high-CE group, danmaku discourse was characterized by Cognitive and Content themes (26.7%), a proportion more than double that of the low-CE group (12.5%). Conversely, Social and Community themes were more prevalent in the low-CE group (40.0%) than in the high-CE group (26.7%). A similar pattern was observed for Affective & Motivational content, which was more common in low-efficacy contexts (20.0% vs. 16.7%). This thematic divergence provides an association-based interpretation of the MGA findings. The stronger presence of substantive content in high-efficacy videos is consistent with the stronger CV → UFI association (H8d) in this group; when danmaku discourse is more oriented toward the material itself, the perceived value of that content appears more closely linked with further interaction. In contrast, the prevalence of social and affective commentary in low-efficacy videos is consistent with the stronger associations involving the peripheral cue of SC with both UFI (H8b) and RTE (H8c). When the content itself shows less content-oriented engagement, discourse places greater emphasis on the social environment and heuristic cues like creator reputation as part of the observed interaction pattern. While thematic differences were also observed across CS and DC groups (Figure 6b and 6c), they did not correspond to statistically significant path differences in the MGA, supporting CE as the most evident contextual moderator in this dataset.

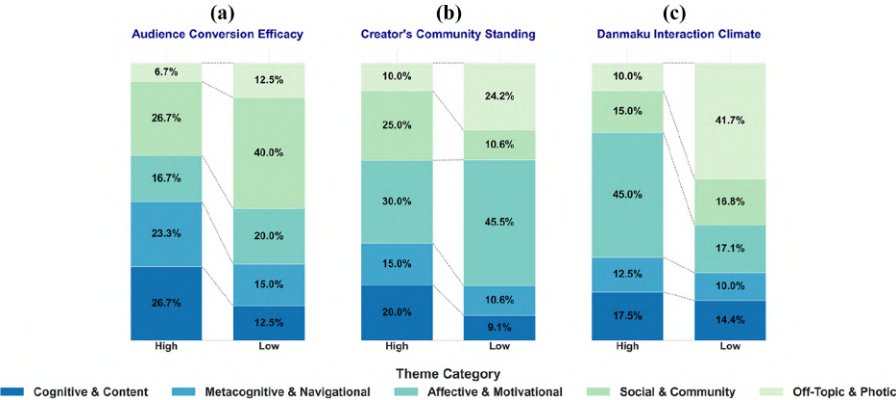


Figure 6. Thematic distribution of danmaku content across contextual moderators

5. Discussion

5.1 Modeling observed engagement patterns: from peripheral heuristics to central engagement

This study employed the ELM to examine user engagement patterns in information-rich, user-generated content (UGC) environments. The central design tension in such systems is a core Human-Computer Interaction (HCI) challenge: socio-technical features like *danmaku* commentary, intended to enhance social presence, risk inducing cognitive overload and a split-attention effect by presenting a constant stream of peripheral information (Peng *et al.*, 2025; Zhang *et al.*, 2025a).

The ELM's dual-route framework provides a useful lens for interpreting observable engagement in environments saturated with competing signals. The findings suggest that SC should not be treated merely as a parallel variable, but as an orientation cue or usability scaffold (Wang, 2015). In high-density information streams, SC may serve as a heuristic filter through which viewers assess whether an information source warrants attention. When higher SC is accompanied by stronger CV, content quality can be interpreted as a value signal associated with active processing (Shoufan, 2019), while peer feedback (UFI) is associated with the social validation surrounding engagement (Lee *et al.*, 2024). This pattern is consistent with a sequential cross-route account (SC → CV → UFI → RTE): creator reputation is linked with CV, which is then associated with social interaction and RTE. This refines prior ELM research by suggesting that, in goal-oriented information seeking, peripheral cues may need to be accompanied by perceived content efficacy to be associated with meaningful behavioral participation.

Beyond estimating the direct associations of both peripheral and central cues, the findings indicate a cross-route pattern that can inform the design of user journeys in complex interactive systems. This pattern suggests a refinement of the ELM in this context (Beautemps and Bresges, 2021) and offers a granular, association-based account for linking the COI framework with HCI principles. The analysis suggests that SC, a peripheral cue, does not function merely as a simple heuristic; instead, it is associated with an initial usability-scaffold role that may help viewers orient themselves before engaging in more effortful central processing. In a UGC learning context, a credible creator may be perceived as a trustworthy instructor (Zha *et al.*, 2018) and this perception can be associated with a psychologically safe and cohesive group climate – a socio-emotional foundation that may make deeper cognitive engagement more accessible (Maré and Mutezo, 2025). This association pattern is further clarified by the chain mediation analysis, which provides an empirical basis for adaptive interface design. The full mediation path, SC → CV → UFI → RTE, was statistically significant (H7a), representing an observed association pattern from peripheral UI cues to CV and feedback interaction. This pathway can be interpreted as an observable learning-oriented sequence: source-related interface cues are associated with CV, which is then linked with sustained UFI and observable RTE (Garrison *et al.*, 2001).

Crucially, the weaker substantive support for the simpler indirect paths, namely SC → CV → RTE and SC → UFI → RTE, helps clarify how the present findings both align with and differ from prior work. Existing studies on *danmaku* and online video interaction have shown that real-time comments, viewer cues and social presence can shape engagement and narrative involvement (Zhang *et al.*, 2025a, b). Studies of SC also suggest that credible sources can be associated with stronger engagement (Li and See-To, 2024; Yu *et al.*, 2026), while ELM research has long recognized that central and peripheral routes may influence one another (Lee, 2012). The present results extend this literature by suggesting that, in *danmaku*-enabled educational videos, SC is not best interpreted as an isolated engagement cue. Its association with RTE appears to be embedded in a broader sequence in which source cues are linked with CV and UFI. Social influence therefore operates less as a simple volume effect and more as a context-sensitive information layer through which users evaluate whether participation is worth sustaining. This interpretation remains associational because the archival data cannot establish the temporal order of users' internal processing.

5.2 An evidence-based principle for adaptive interface design

The study's primary contribution to HCI and information management is the identification of Audience CE not merely as a performance metric, but as a moderator of observed information-processing-related engagement patterns. As visualized in [Figure 7](#), the associations among source cues, CV, feedback interaction and RTE vary across content-efficacy contexts, offering an empirical basis for context-aware interface adaptation ([Keehr and Reardon, 2025](#)).

Specifically, the CE comparison indicates a context-contingent association pattern rather than evidence of a universal difference in processing routes. In lower-CE contexts, the stronger SC → UFI and SC → RTE associations suggest that SC and social signals may function as uncertainty-reduction cues when content efficacy is less evident. This pattern is consistent with the role of the social layer as an orientation aid in dense information streams, where affective and community cues can help users decide whether further participation is worthwhile ([Du and Huang, 2025](#)). However, it should not be interpreted as direct evidence that users abandon content processing.

Conversely, in higher-CE contexts, the stronger CV → UFI association suggests that CV is more closely linked with feedback interaction. This result is consistent with a content-oriented interpretation of engagement, where peer exchange appears more connected with semantic discussion than with social connectedness alone ([Zhu et al., 2025](#)). It does not establish that users directly invest cognitive effort or that content alone accounts for engagement. Rather, high-CE conditions appear to make CV more salient in the observed relationship between content evaluation and feedback interaction.

This structural divergence is further supported by the thematic analysis of the danmaku metadata. In high-efficacy videos, the stronger presence of Cognitive and Content themes is consistent with the stronger CV → UFI association, suggesting that peer exchange may support content-oriented engagement. In low-efficacy videos, the greater presence of Social and Community and Affective themes is consistent with the stronger role of source and social cues as heuristic anchors. This pattern cautions against treating denser danmaku or higher interaction volume as inherently better ([Zhao et al., 2026](#)). In educational settings, engagement metrics should be interpreted alongside whether interaction supports comprehension, retention, knowledge organization, or self-regulated learning, because learning analytics research shows that digital trace indicators are most meaningful when related to learners' goals, cognitive engagement and learning outcomes ([Xu et al., 2026](#)). From an attention economy perspective, interface optimization should avoid turning learning spaces into systems that maximize attention capture at the expense of learner welfare ([De La Torre et al., 2026](#)). More broadly, adaptive designs should be evaluated for whether they distribute learning support equitably rather than reproducing visibility bias or unequal access to useful guidance ([Albuquerque et al., 2026](#)).

Accordingly, this study proposes a principle of Dynamic Information Presentation. For lower-CE content, interfaces may foreground peripheral metadata, such as creator credentials, historical performance indicators and social proof elements, to support orientation and discovery. For higher-CE content, interfaces may reduce peripheral distractions and prioritize features that support semantic analysis and knowledge organization, such as threaded discussions or content-anchored annotations, so that interaction is more likely to support learning-oriented engagement. This design logic is consistent with recent human-centered adaptive learning research, which suggests that adaptive systems can adjust learning strategies based on behavioral indicators and learner-related psychological states while still requiring controlled validation before causal effectiveness claims can be made ([Lin et al., 2025](#)).

Finally, the non-significant moderation by CS and DC defines the boundary of this design principle. The results do not imply that creator position or DC are irrelevant. Rather, within this dataset, CE is the contextual condition most clearly associated with differences in the structural paths. Prior work suggests that creator type, influencer context ([Dong et al., 2025](#)) and real-time information flow can shape digital engagement in ways that may not be fully captured by aggregate standing or sentiment-ratio indicators ([Wu, 2025](#); [Yuan et al., 2025](#)).

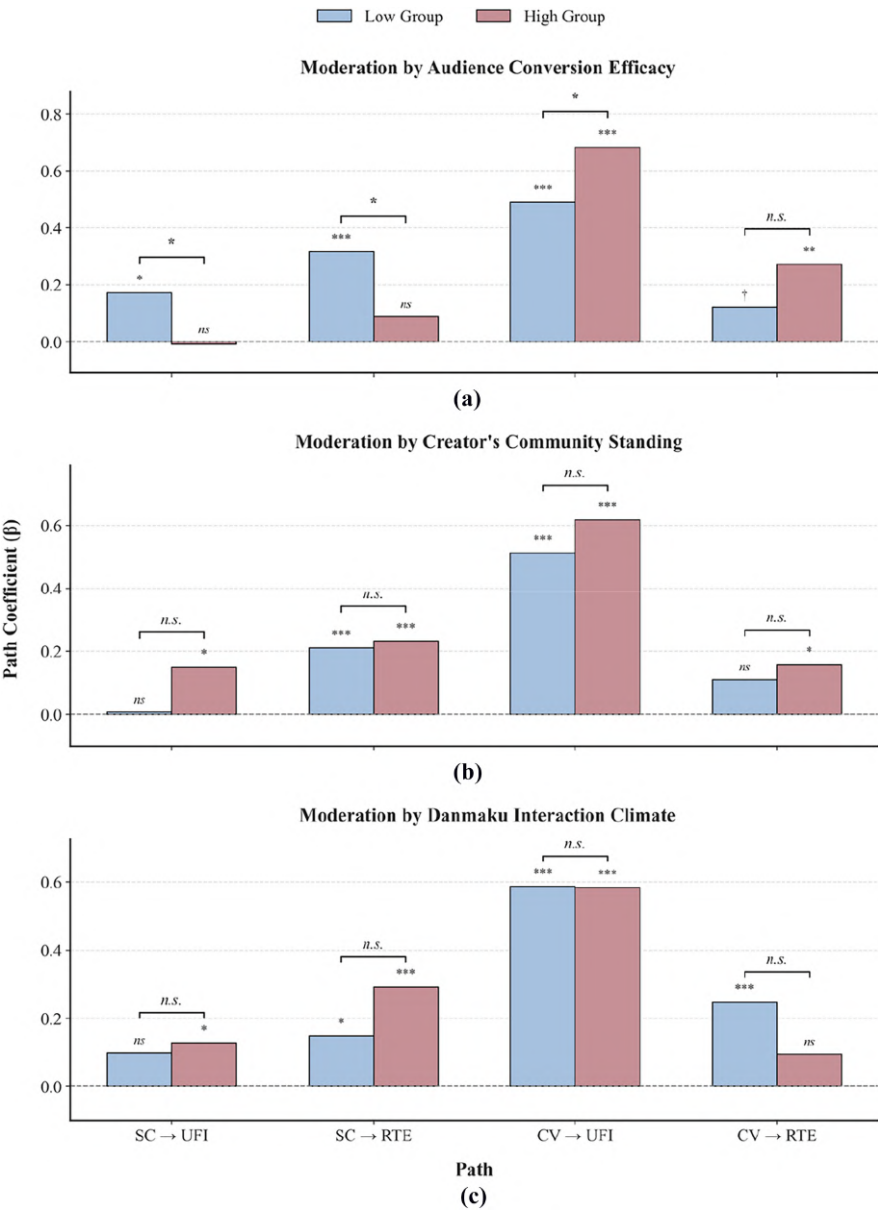


Figure 7. Multi-group analysis results for the three contextual moderators

Future research should therefore use richer measures of creator reputation and peer climate to examine whether these social-context variables condition engagement beyond the operationalization used in this study.

6. Conclusion and limitations

This study examined danmaku engagement as a form of HII in educational videos. Using behavioral trace data from Bilibili, it modeled how SC, CV, UFI and RTE are associated within a sequential cross-route pattern. The findings suggest that engagement in danmaku-enabled educational videos is not only a matter of isolated source, content, or peer cues. Rather, observable participation is more coherently explained as a linked process in which creator-related cues, content evaluation, feedback interaction and real-time expression operate together within an information-dense viewing environment.

The study also clarifies the role of Audience CE as a contextual condition for this process. In lower-CE contexts, the stronger associations involving SC suggest that source and social cues may play a greater role when the informational value of the content is less immediately evident. In higher-CE contexts, the stronger CV → UFI association suggests that peer interaction is more closely connected with content-oriented engagement. This distinction refines the interpretation of social influence in danmaku environments. Social cues are not merely associated with greater engagement volume; their role appears to depend on whether the surrounding content context supports heuristic orientation or content-focused exchange.

These findings have implications for platform and interface design, but they should be interpreted cautiously. Adaptive educational-video systems should not optimize only for visible engagement indicators such as danmaku volume, comments, clicks, or other activity counts. Such indicators may reflect meaningful knowledge exchange, but they may also reflect distraction, social following, or attention fragmentation. A more appropriate design principle is to align interface affordances with informational efficacy. Source and peer cues may be useful when learners need orientation and uncertainty reduction, whereas high-efficacy educational content may benefit from reduced peripheral noise and stronger support for semantic discussion, annotation and knowledge organization.

For educational practice, the results point to a distinction between behavioral engagement and learning-oriented engagement. This study did not directly measure learning outcomes, so it cannot claim that the observed engagement pathways improve comprehension, retention, transfer, or self-regulated learning. However, the model can inform future evaluations of educational-video interfaces by identifying when engagement is more likely to reflect content-oriented interaction rather than activity volume alone. From an equity perspective, adaptive systems should also avoid privileging only already visible creators, highly active users, or high-performing videos. They should support learners with different cognitive styles, attention capacities, prior knowledge, emotional responses and resource needs by providing manageable information environments.

Several limitations define the boundaries of these conclusions. First, the study uses archival behavioral trace data at the video level, so the indicators are indirect proxies for internal cognitive states rather than direct measures of attention, comprehension, or learning. Second, the cross-sectional and observational design limits causal and temporal inference. The reported paths should therefore be interpreted as statistical associations consistent with the proposed cross-route logic, not as evidence of a causal processing sequence. Third, the empirical setting is limited to Bilibili educational videos, and the generalizability of the model should be examined across other platforms, cultural contexts and instructional formats. Future research should combine trace data with learning assessments, experiments, longitudinal designs, surveys, eye-tracking, cognitive load measures, or other process-level evidence and should evaluate whether adaptive interface designs improve learning quality rather than only increasing observable engagement.

Data availability statement

All data used in this study were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

AJIM

Appendix

log-transformed PLS-SEM indicators									
description	Min	Q1	Median	Q3	Max	Mean	SD	SW <i>p</i> > 0.05	
g followers of uploader	6.75	10.41	12.04	13.09	16.46	11.76	2.04	No	
g historical likes	7.81	12.06	13.30	14.21	17.34	13.12	1.88	No	
g historical views	10.97	14.89	15.97	17.11	20.48	16.04	1.82	Yes	
g submissions	0.00	4.08	4.95	5.84	7.67	4.84	1.34	No	
g coins	5.42	6.88	7.51	8.25	10.75	7.62	1.07	No	
g collections	8.79	9.07	9.39	10.06	12.62	9.62	0.76	No	
g likes	7.31	8.96	9.81	10.66	13.01	9.85	1.18	No	
g comments	2.64	4.74	5.59	6.18	8.59	5.50	1.11	Yes	
g positive comments	2.20	4.49	5.30	5.95	8.26	5.24	1.13	Yes	
g root comments	2.20	4.39	5.18	5.86	8.40	5.18	1.15	Yes	
g total comment likes	2.94	6.67	7.76	9.00	11.64	7.76	1.79	Yes	
g danmaku comments	2.30	3.89	4.97	6.06	8.88	5.09	1.48	No	
g positive danmaku comments	0.69	3.69	4.67	5.81	8.62	4.78	1.52	Yes	
g danmaku comments per second	-8.36	-2.06	-0.42	0.79	2.25	-0.87	2.29	No	

e used for PLS-SEM estimation; original-scale statistics are reported in [Table 1](#)

References

- Albuquerque, J., Rienties, B., Hlosta, M. and Holmes, W. (2026), "Learning analytics to uncover ethnic bias in educational texts: an ensemble learning approach", *Journal of Learning Analytics*, Vol. 13 No. 1, pp. 143-162, doi: [10.18608/jla.2026.8905](https://doi.org/10.18608/jla.2026.8905).
- Amaldoss, W. and Shin, W. (2026), "Content provision on UGC platforms", *Management Science*, Vol. 72 No. 2, pp. 1489-1508, doi: [10.1287/mnsc.2022.01066](https://doi.org/10.1287/mnsc.2022.01066).
- Beautemps, J. and Bresges, A. (2021), "What comprises a successful educational science YouTube video? A five-thousand user survey on viewing behaviors and self-perceived importance of various variables controlled by content creators", *Frontiers in Communication*, Vol. 5, 600595, doi: [10.3389/fcomm.2020.600595](https://doi.org/10.3389/fcomm.2020.600595).
- Biswas, S. (2016), "The latent logarithm", *arXiv preprint*, arXiv: 1605.06064.
- Costa-Feito, A., Saavedra, A. and Barta, S. (2025), "Enhancing student memorisation through teacher webcam usage and the interplay of social presence", *International Journal of Educational Technology in Higher Education*, Vol. 22 No. 1, p. 58, doi: [10.1186/s41239-025-00554-w](https://doi.org/10.1186/s41239-025-00554-w).
- Dai, H., Liu, J., Chen, Y., Yang, Y., Zhou, Y., Zheng, Z., Gong, Z., Ma, X., Zhang, X. and Zhu, J. (2026), "Do likes reflect quality? Engagement metrics and information reliability of robotic surgery short videos", *Journal of Robotic Surgery*, Vol. 20 No. 1, p. 187, doi: [10.1007/s11701-026-03144-8](https://doi.org/10.1007/s11701-026-03144-8).
- De La Torre, P.G., Pérez-Verdugo, M. and Barandiaran, X.E. (2026), "Attention is all they need: cognitive science and the (techno)political economy of attention in humans and machines", *AI and Society*, Vol. 41 No. 1, pp. 5-21, doi: [10.1007/s00146-025-02400-z](https://doi.org/10.1007/s00146-025-02400-z).
- Dong, J., He, Y., Song, J., Ding, H. and Kong, Y. (2022), "Universal scaling behavior and Hawkes process of videos' views on Bilibili.com", *Frontiers in Physics*, Vol. 10, 1018704, doi: [10.3389/fphy.2022.1018704](https://doi.org/10.3389/fphy.2022.1018704).
- Dong, X., Wang, B., Chu, W., Filieri, R. and Liao, J. (2025), "The impact of danmaku ritual types on user digital engagement in video-based social media: the moderating role of influencer types and domains", *Psychology and Marketing*, Vol. 42 No. 10, pp. 2650-2672, doi: [10.1002/mar.70003](https://doi.org/10.1002/mar.70003).
- Du, P. and Huang, Z. (2025), "Happiness backfires: emotion and sales in live streaming", *Electronic Commerce Research*, Vol. 25 No. 4, pp. 2641-2672, doi: [10.1007/s10660-023-09760-y](https://doi.org/10.1007/s10660-023-09760-y).
- Evans, J.S.B.T. (2008), "Dual-processing accounts of reasoning, judgment, and social cognition", *Annual Review of Psychology*, Vol. 59 No. 1, pp. 255-278, doi: [10.1146/annurev.psych.59.103006.093629](https://doi.org/10.1146/annurev.psych.59.103006.093629).
- Fei, D. (2024), "Research on interpretable text sentiment analysis", PhD Thesis, Tokushima University, Japan.
- Fisher, G.G. and Chaffee, D.S. (2018), "Research using archival data", in *Advanced Research Methods for Applied Psychology*, Routledge.
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50, doi: [10.2307/3151312](https://doi.org/10.2307/3151312).
- Garrison, D.R., Anderson, T. and Archer, W. (2001), "Critical thinking, cognitive presence, and computer conferencing in distance education", *American Journal of Distance Education*, Vol. 15 No. 1, pp. 7-23, doi: [10.1080/08923640109527071](https://doi.org/10.1080/08923640109527071).
- Hair, J.F., Ringle, C.M. and Sarstedt, M. (2011), "PLS-SEM: indeed a silver bullet", *Journal of Marketing Theory and Practice*, Vol. 19 No. 2, pp. 139-152, doi: [10.2753/MTP1069-6679190202](https://doi.org/10.2753/MTP1069-6679190202).
- Hair, J., Hollingsworth, C.L., Randolph, A.B. and Chong, A.Y.L. (2017), "An updated and expanded assessment of PLS-SEM in information systems research", *Industrial Management and Data Systems*, Vol. 117 No. 3, pp. 442-458.
- Hair, J.F., Risher, J.J., Sarstedt, M. and Ringle, C.M. (2019), "When to use and how to report the results of PLS-SEM", *European Business Review*, Vol. 31 No. 1, pp. 2-24, doi: [10.1108/EBR-11-2018-0203](https://doi.org/10.1108/EBR-11-2018-0203).

AJIM

- Hair, J.F. Jr, Hult, G.T.M., Ringle, C.M., Sarstedt, M., Danks, N.P. and Ray, S. (2021), *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R: A Workbook*, Springer Nature, doi: [10.1007/978-3-030-80519-7](https://doi.org/10.1007/978-3-030-80519-7).
- Henseler, J., Ringle, C.M. and Sarstedt, M. (2016), "Testing measurement invariance of composites using partial least squares", in Sinkovics, R., "Bryan" J, R.-J. and Rudolf, D, K. (Eds), *International Marketing Review*, Vol. 33 No. 3, pp. 405-431, doi: [10.1108/IMR-09-2014-0304](https://doi.org/10.1108/IMR-09-2014-0304).
- Ke, Q., Zhang, S., Xu, J. and Cao, Y. (2026), "Beyond content: enhancing misinformation detection using stance and sentiment clues from online comments", *Aslib Journal of Information Management*, pp. 1-20, doi: [10.1108/AJIM-04-2025-0206](https://doi.org/10.1108/AJIM-04-2025-0206).
- Keehr, J. and Reardon, R. (2025), "Is speed-watching a good idea? Potential metacognitive complications of watching instructional videos at faster speeds", *Computers in Human Behavior*, Vol. 165, 108556, doi: [10.1016/j.chb.2025.108556](https://doi.org/10.1016/j.chb.2025.108556).
- Klein, A., Ahlf, H. and Sharma, V. (2015), "Social activity and structural centrality in online social networks", *Telematics and Informatics*, Vol. 32 No. 2, pp. 321-332, doi: [10.1016/j.tele.2014.09.008](https://doi.org/10.1016/j.tele.2014.09.008).
- Lee, W.-K. (2012), "An elaboration likelihood model based longitudinal analysis of attitude change during the process of IT acceptance via education program", *Behaviour and Information Technology*, Vol. 31 No. 12, pp. 1161-1171.
- Lee, E., Shin, S., Joo, H. and Koo, C. (2024), "The cognitive effect of YouTube video and user-generated content: a preliminary study", in Berezina, K., Nixon, L. and Tuomi, A. (Eds), *Information and Communication Technologies in Tourism 2024*, Springer Nature Switzerland, Cham, pp. 440-445, doi: [10.1007/978-3-031-58839-6_45](https://doi.org/10.1007/978-3-031-58839-6_45).
- Leem, B.-H. (2023), "Impact of interactivity on learning outcome in online learning settings: ordinal logit model", *International Journal of Engineering Business Management*, Vol. 15, doi: [10.1177/18479790231203107](https://doi.org/10.1177/18479790231203107).
- Li, J. (2017), *The Interface Affect of a Contact Zone: Danmaku on Video-Streaming Platforms*, Brill, doi: [10.1163/22142312-12340079](https://doi.org/10.1163/22142312-12340079).
- Li, H., Xu, J. and Luo, Y. (2024a), "The impact of mobile social media on knowledge sharing among vocational school teachers: a social cognitive career perspective", *Journal of Information Science*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1177/01655515241293754](https://doi.org/10.1177/01655515241293754).
- Li, H., Xu, J., Luo, Y. and Wang, C. (2024b), "The role of teachers' direct and emotional mentoring in shaping undergraduates' research aspirations: a social cognitive career theory perspective", *International Journal of Mentoring and Coaching in Education*, Vol. ahead-of-print No. ahead-of-print, pp. 123-142, doi: [10.1108/IJMCE-07-2023-0064](https://doi.org/10.1108/IJMCE-07-2023-0064).
- Li, H. and See-To, E.W.K. (2024), "Source credibility plays the central route: an elaboration likelihood model exploration in social media environment with demographic profile analysis", *Journal of Electronic Business and Digital Economics*, Vol. 3 No. 1, pp. 36-60, doi: [10.1108/JEBDE-10-2022-0038](https://doi.org/10.1108/JEBDE-10-2022-0038).
- Li, D. and Tao, Y. (2026), "Translanguaging space construction in a Chinese-Russian border city: a multimodal perspective from vlogs", *Journal of Multilingual and Multicultural Development*, pp. 1-16, doi: [10.1080/01434632.2026.2613676](https://doi.org/10.1080/01434632.2026.2613676).
- Li, N., Li, F. and Lai, W. (2025), "Exploring the impacts of Danmu on online video consumption behaviour: a data-driven method", *Journal of Information Science*, 01655515251362299, doi: [10.1177/01655515251362299](https://doi.org/10.1177/01655515251362299).
- Li, H., Luo, Y., Wang, J., Ye, Y. and Xu, J. (2026), "From access to adherence: fostering professional capital and continued engagement in blended teacher workshops", *Journal of Professional Capital and Community*, pp. 1-22, doi: [10.1108/JPCCE-12-2025-0145](https://doi.org/10.1108/JPCCE-12-2025-0145).
- Lin, K.-Y., Li, M.-H., Lo, F., Huang, H.-C., Matsuno, K. and Watanabe, R. (2025), "Adaptive learning with human factors and Artificial Intelligence: associations with training effectiveness in programming education", *International Journal of Industrial Ergonomics*, Vol. 110, 103834, doi: [10.1016/j.ergon.2025.103834](https://doi.org/10.1016/j.ergon.2025.103834).

- Maré, S. and Mutezo, A.T. (2025), "The community of inquiry, shared metacognition, and student engagement in online learning: a systematic review", *Interactive Learning Environments*, Vol. 33 No. 9, pp. 5213-5234.
- Moghadamzadeh, A., Ebrahimi, P., Radfard, S., Salamzadeh, A. and Khajeheian, D. (2020), "Investigating the role of customer Co-creation behavior on social media platforms in rendering innovative services", *Sustainability*, Vol. 12 No. 17, p. 6926, doi: [10.3390/su12176926](https://doi.org/10.3390/su12176926).
- Mu, D., Zou, M. and Chen, Y. (2025), "How do different cognitive styles learners deal with the bullet screen interruption in instructional videos? An eye-tracking study", *BMC Psychology*, Vol. 13 No. 1, p. 961, doi: [10.1186/s40359-025-03182-1](https://doi.org/10.1186/s40359-025-03182-1).
- Muda, M. and Hamzah, M.I. (2021), "Should I suggest this YouTube clip? The impact of UGC source credibility on eWOM and purchase intention", *Journal of Research in Interactive Marketing*, Vol. 15 No. 3, pp. 441-459, doi: [10.1108/JRIM-04-2020-0072](https://doi.org/10.1108/JRIM-04-2020-0072).
- Oh, J., Bellur, S., Sundar, S. and Morris, R. (2018), "Clicking, assessing, immersing, and sharing: an empirical model of user engagement with interactive media", *Communication Research*, Vol. 45 No. 5, pp. 737-763, doi: [10.1177/0093650215600493](https://doi.org/10.1177/0093650215600493).
- Peng, G., Wang, X., Li, J. and Wu, J. (2025), "Is it beneficial for consumers to ask more questions in Danmaku? The inverted U-shaped effect of information-seeking danmaku density on e-commerce livestream sales", *Electronic Commerce Research*, Vol. 26 No. 2, pp. 1745-1782, doi: [10.1007/s10660-025-09959-1](https://doi.org/10.1007/s10660-025-09959-1).
- Petty, R.E. and Cacioppo, J.T. (1986), "The elaboration likelihood model of persuasion", in Berkowitz, L. (Ed.), *Advances in Experimental Social Psychology*, Academic Press, Vol. 19, pp. 123-205, doi: [10.1016/S0065-2601\(08\)60214-2](https://doi.org/10.1016/S0065-2601(08)60214-2).
- Phua, J., Jin, S.V., Kim, J. and Jay (2020), "Pro-veganism on Instagram: effects of user-generated content (UGC) types and content generator types in Instagram-based health marketing communication about veganism", *Online Information Review*, Vol. 44 No. 3, pp. 685-704, doi: [10.1108/OIR-06-2019-0213](https://doi.org/10.1108/OIR-06-2019-0213).
- Shmueli, G., Sarstedt, M., Hair, J.F., Cheah, J.-H., Ting, H., Vaithilingam, S. and Ringle, C.M. (2019), "Predictive model assessment in PLS-SEM: guidelines for using PLSpredict", *European Journal of Marketing*, Vol. 53 No. 11, pp. 2322-2347, doi: [10.1108/EJM-02-2019-0189](https://doi.org/10.1108/EJM-02-2019-0189).
- Shoufan, A. (2019), "Estimating the cognitive value of YouTube's educational videos: a learning analytics approach", *Computer Human Behaviour*, Vol. 92, pp. 450-458, doi: [10.1016/j.chb.2018.03.036](https://doi.org/10.1016/j.chb.2018.03.036).
- Wang, P. (2015), "Exploring the influence of electronic word-of-mouth on tourists' visit intention: a dual process approach", *Journal of Systems and Information Technology*, Vol. 17 No. 4, pp. 381-395, doi: [10.1108/JSIT-04-2015-0027](https://doi.org/10.1108/JSIT-04-2015-0027).
- Wang, C., Dai, J., Zhu, K., Yu, T. and Gu, X. (2024a), "Understanding the continuance intention of college students toward new E-learning spaces based on an integrated model of the TAM and TTF", *International Journal of Human-Computer Interaction*, Vol. 40 No. 24, pp. 8419-8432, doi: [10.1080/10447318.2023.2291609](https://doi.org/10.1080/10447318.2023.2291609).
- Wang, Z., Chen, J., Chen, J. and Chen, H. (2024b), "Identifying interdisciplinary topics and their evolution based on BERTopic", *Scientometrics*, Vol. 129 No. 11, pp. 7359-7384, doi: [10.1007/s11192-023-04776-5](https://doi.org/10.1007/s11192-023-04776-5).
- Wang, C., Wang, H., Li, Y., Dai, J., Gu, X. and Yu, T. (2025), "Factors influencing university students' behavioral intention to use generative artificial intelligence: integrating the theory of planned behavior and AI literacy", *International Journal of Human-Computer Interaction*, Vol. 41 No. 11, pp. 6649-6671.
- Wang, Y. and Chen, C. (2026), "Support or disrupt? The impact of danmaku on learners' continuous learning intentions in online learning environment", *Interactive Learning Environments*, Vol. 34 No. 2, pp. 718-733.
- Wei, X., Lv, S. and Jia, M. (2023), "Research on English autonomous learning model of college students in baoding: taking the Bilibili bullet screen network as an example", *International Journal of Education and Humanities*, Vol. 7 No. 2, pp. 167-173, doi: [10.54097/ijeh.v7i2.5571](https://doi.org/10.54097/ijeh.v7i2.5571).

AJIM

- Wu, J. (2025), “‘High energy ahead!’: exploring Chinese adolescents’ pragmatic identities in bullet-screen discourse”, *Humanities and Social Sciences Communications*, Vol. 12 No. 1, p. 167, doi: [10.1057/s41599-025-04475-x](https://doi.org/10.1057/s41599-025-04475-x).
- Xia, Y., Deng, Y., Tao, X., Zhang, S. and Wang, C. (2024), “Digital art exhibitions and psychological well-being in Chinese Generation Z: an analysis based on the S-O-R framework”, *Humanities and Social Sciences Communications*, Vol. 11 No. 1, pp. 1-15, doi: [10.1057/s41599-024-02718-x](https://doi.org/10.1057/s41599-024-02718-x).
- Xiang, Y. and Chae, S.W. (2022), “Influence of perceived interactivity on continuous use intentions on the danmaku video sharing platform: belongingness perspective”, *International Journal of Human-Computer Interaction*, Vol. 38 No. 6, pp. 573-593.
- Xu, J., Liu, F., Luo, Y. and Xu, X. (2025), “Stage-specific impacts of digital technologies on supply-chain resilience: meta-analytic evidence for continuous process improvement”, *Business Process Management Journal*, pp. 1-22, doi: [10.1108/BPMJ-04-2025-0550](https://doi.org/10.1108/BPMJ-04-2025-0550).
- Xu, J., Luo, Y., Wang, C., Wang, M. and Wu, Y. (2026), “AI support in self-regulated learning: a decade of technological evolution and meta-analysis”, *British Journal of Educational Technology*, 70058, doi: [10.1111/bjet.70058](https://doi.org/10.1111/bjet.70058).
- Yu, T., Dai, J., Chen, X. and Wang, C. (2026), “From information evaluation to technology use: generative AI adoption in Chinese business schools”, *The International Journal of Management Education*, Vol. 24 No. 3, 101430, doi: [10.1016/j.ijme.2026.101430](https://doi.org/10.1016/j.ijme.2026.101430).
- Yuan, H., Lu, K., Ausaf, A. and Zhu, M. (2025), “Constant or inconstant? The time-varying effect of danmaku on user engagement in online video platforms”, *Internet Research*, Vol. 35 No. 2, pp. 771-797, doi: [10.1108/intr-06-2023-0479](https://doi.org/10.1108/intr-06-2023-0479).
- Zeng, L., Tan, Z., Ke, Y. and Xia, L. (2024), “Danmaku-based automatic analysis of real-time online learning engagement”, *International Journal of Interactive Mobile Technologies (iJIM)*, Vol. 18 No. 08, pp. 127-139, doi: [10.3991/ijim.v18i08.48025](https://doi.org/10.3991/ijim.v18i08.48025).
- Zha, X., Yang, H., Yan, Y., Liu, K. and Huang, C. (2018), “Exploring the effect of social media information quality, source credibility and reputation on informational fit-to-task: moderating role of focused immersion”, *Computers in Human Behavior*, Vol. 79, pp. 227-237, doi: [10.1016/j.chb.2017.10.038](https://doi.org/10.1016/j.chb.2017.10.038).
- Zhao, X., Zhang, Y., Yue, S., Wang, X., Qin, C. and Xia, X. (2026), “Understanding continuance learning intention in online educational videos: the role of danmaku (real-time interactive comments) in Informal learning”, *Education and Information Technologies*, doi: [10.1007/s10639-026-14008-x](https://doi.org/10.1007/s10639-026-14008-x).
- Zhang, M., Jiang, Q., Xiong, W., Li, Q. and Zhao, W. (2025a), “Effects of real-time danmaku interaction on student engagement in live video-streaming teaching: analyzing the mediating roles of social, teaching, and cognitive presences”, *Interactive Learning Environments*, Vol. 33 No. 3, pp. 2023-2043, doi: [10.1080/10494820.2024.2388766](https://doi.org/10.1080/10494820.2024.2388766).
- Zhang, X., Kim, H.K. and Zhou, S. (2025b), “Time versus timing in social cognition: how concurrent viewer cues and plot-aligned Danmaku affect narrative outcomes on online video platforms”, *Computers in Human Behavior*, Vol. 172, 108748, doi: [10.1016/j.chb.2025.108748](https://doi.org/10.1016/j.chb.2025.108748).
- Zhu, Y., Lin, X., Kim, J., Al-Adwan, A.S. and Li, N. (2025), “Exploring human interaction in online self-regulated learning through danmaku comments”, *International Journal of Human-Computer Interaction*, Vol. 41 No. 22, pp. 1-14, doi: [10.1080/10447318.2025.2480826](https://doi.org/10.1080/10447318.2025.2480826).

Corresponding author

Yuying Luo can be contacted at: lyy24720692@shu.edu.cn

基于文献计量学的住房不平等研究
Research of Housing Inequality Based on Bibliometrics

作者简介

陈多长 Chen Duo Zhang 浙江工业大学管理学院 教授 博士
罗玉莹 Luo Yuying (通讯作者) 浙江工业大学管理学院 本科生

摘要: 住房不平等是全球性热点问题。运用 VOSviewer 与 CiteSpace 软件对 WoS 数据库中 2000-2022 年的相关文献进行量化分析和可视化分析,通过绘制高频关键词关系网络和关键词时区图谱,梳理住房不平等研究的文献特点。主要结论如下:一是该议题研究已形成了较稳定的作者合作群体;二是美国和中国是该议题研究最活跃的国家;三是研究住房和社会问题的期刊占比极高,表明该议题有较强专业性;四是不同时代背景下住房不平等研究的关注点不同,而贫富差距和种族歧视引致住房不平等始终是研究焦点。

关键词: 住房不平等; 文献计量学; 知识图谱; 研究进展

Abstract: Housing inequality is a global hot issue. It uses VOSviewer and CiteSpace software to conduct quantitative analysis and visual analysis of relevant literature from 2000 to 2022 in WoS database, and analyzes the research topic of housing inequality by drawing high-frequency keyword relationship network and keyword time zone map; The research found that: first, the research on this topic has formed a relatively stable cooperative group of authors. Second, the United States and China are the most active countries on this issue. Third, the proportion of journals studying housing and social issues is very high, indicating that the topic is highly professional. Fourth, the housing inequality caused by the gap between the rich and the poor and racial discrimination has always been the focus of this study.

Keywords: housing inequality; bibliometrics; knowledge map; research progress

前言

住房是家庭经济安全和财富战略的核心。在许多发达经济体中,国家福利的不平衡以及劳动力市场重组导致住房资产的价值波动对家庭经济安全有着极大的影响^[1,2]。获得住房所有权和财富积累的差异化似乎是形成社会不平等的核心。在过去半个世纪里,房价不断升值。成为了 21 世纪私人财产增长的主力军,并被认为是有史以来最好的长期投资^[3]。也因此住房从消费品逐渐演变为投资品和金融品,并成为城市家庭财富和社会分化的重要机制^[4]。有研究指出近年来许多国家的住房不平等现象明显加剧^[5,6],史密斯 (Smith)^{[7][8]}等人的调查显示,如今自住户之间住宅物业财富(扣除抵押贷款债务)的分配不平等正不断扩大。

导致住房不平等的因素颇多,如调查显示新一代年轻人由于可支配收入的增加无法企及房价的增长,使得 25-34 岁人群自购住房拥有率较以低,而 50 岁以上人均房产往往远高于一个国家的平均值,在住房上形成巨大的代际差异^{[8][7]}。房产继承也是导致住房不平等的一个重要因素,而近年来生育率的下降使得这种继承导致的不平等更加突出。同时住房资产丰富者,通过出租房屋获得大量收入,进一步加剧了住房财富的不平等。另外空间两极分化和差异化准入也是导致住房财富积累不平等的重要原因,有学者^[9]在获取完整的人口登记数据后,运用 GIS 制图揭示了房屋价值开发中的空间极化。并用通过在家庭层面建模,证明了收入、就业状况和父母财富会影响住房子市场的分割。通过该研究还可以发现住房不平等推动了地区之间高低收益子市场之间的分歧,由此拉大贫富差距和加剧发展不平衡。除此之外,由新自由主义政治经济关系——放松管制、倒退的税收制度——推动的住房自有制的增长,也导致房地产市场的不平等发展。

住房不平等会带来许多不利影响,如有些家庭拥有多套住宅,可能导致房屋空置率上升,

土地利用降低，而部分贫困人口却难以承担房价，会成为影响社会和谐稳定发展的不利因素。由此，住房财富不平等加剧将日益影响代际以及相关的基于阶级的政治冲突^{[8][7]}。另有研究表明，居民的生活满意度主要受住房价值和住房面积区域不平等的影响^[10]。尤其在 Covid-19 疫情期间，人们居家时长增加暴露了全球住房质量和可用性的不平等。一些人缺乏足够的空间来开展转移到自己家中的活动，并且疫情带来的经济衰退增加了住房还贷压力。由此导致居民生活在负担不起或质量差的住房中，从而对心理健康产生负面影响^[11]。西方学术界早在 20 世纪 60 年代就已开始关注住房财富分配不平等问题^[12]。20 世纪 70 年代到 20 世纪末，直击“住房不平等”（housing inequality）这一主题的英文文献持续散见于社会学、经济学和政治学等领域。21 世纪以来，世界各地房地产市场屡创新高，住房购买愈加困难，住房不平等状况恶化，住房不平等再次受到西方学术界关注，相关文献也开始增加^[13]。国内住房不平等研究晚于西方，大致萌芽于 21 世纪初。2010 年之后相关文献快速增加，渐成社会学、城市经济学和地理学等学科的研究热点。由于住房不平等问题的研究涉及多学科、多领域，单一视角的案例分析和实证研究可能无法全面了解住房不平等产生的原因和造成的结果。因此，本文利用文献计量学方法对现有关于住房不平等的研究文献进行梳理，分析该领域的研究结构和量化信息，并提供了系统综述和可视化图谱，为该领域的学者勾勒出目前关于住房不平等研究现状，展现该领域研究聚焦点和近年来的发展趋势。

1 方法和数据

1.1 文献计量学与分析工具

文献计量是一种用于描述分析某一学科或研究领域的动态与进展的计量方法，受益于近年来计算机科学技术的发展，计量分析结合可视化分析成为了该研究领域的新态势，通过清晰明了的知识图谱来展示计量分析的结果，成功实现了“一图胜万言”的目标^{[14][12]}。本文利用 CiteSpace 和 VOSviewer 进行知识图谱的绘制。CiteSpace 采用基于集合论的数据标准化方法来进行知识单元的相似性测度，通过绘制时区视图（Timezone）在时间维度上清晰勾勒出研究热点演进的过程，从而呈现该领域的发展进程和趋势^{[15][13]}。VOSviewer 采用的是基于概率论的数据标准化方法，在关键词、共机构、共作者等领域提供了多种可视化视图，制图简易且图像美观^{[16][14]}，目前在文献计量的可视化领域受到了越来越多学者的关注。

1.2 数据检索与提取

本文选取 Web of Science（核心集合）作为数据来源，同时为了保证检索数据的全面且准确，索引选择为 SCI-EXPANDED、SSCI，最终使用的检索策略为 TS=（“housing inequality” OR “housing equality” OR （“housing” AND （“inequality” OR “equality”））），时间跨度从 2000 年 1 月开始到 2022 年 2 月，检索时间截止为 2022 年 2 月 4 日，文献类型选择为 Articles。检索结果去重后共得到期刊论文 2089 篇。利用 CiteSpace 对数据进行重复性审查，再根据标题和摘要去除和主题完全无关的文献，最后通过浏览全文筛除不符合要求的文献，保留文献 174 篇。整体来看，该领域的发文量在不断上升，并以 2011 年和 2018 年为界可以分成三个阶段：第一阶段 2000 年 2011 年，每年只发表少量文章；2011 年到 2018 年为第二阶段，发文量徘徊在每年 8 篇左右；2018 年开始进入第三阶段，每年的发文量显著增长，2020 年更是高达 41 篇，这表明该研究领域受到了越来越多学者的关注，进入了蓬勃发展的阶段。

2 数据的计量分析

2.1 对作者发文量的统计分析

对文献作者进行分析，可以获悉该研究领域的代表学者和核心研究力量。著名学者普赖斯（Price）^{[17][15]}指出，在同一主题中，半数论文为一群高生产力作者所撰，这一作者集合的数量上约等于全部作者总数的平方根，即：

$$\sum_{m+1}^I n(x) = \sqrt{N}$$

其中 $n(x)$ 表示撰写了 x 篇论文的作者数， $I=n_{\max}$ 为该领域内最高产作者的论文数（通过 VOSviewer 统计可知 $n_{\max}=6$ 篇）， N 为作者的总数， m 为核心作者的最低发文数。依据 Price's Law，某一领域的核心作者最低发文数 $m = 0.749 \times \sqrt{n_{\max}} \approx 1.83$ ，因此将发文量 2 篇以上（包含 2 篇）的作者定位为该领域的核心作者，共计 35 名核心作者，共计发文 92 篇，占总发文量的 52.9%，达到了普赖斯（Price）所提出的半数（50%）标准，将上述数值代入公式计算也基本符合 Price's Law 的计算公式。因此，可以认为住房不平等研究议题已形成了较为稳定的作者合作群体。展示了在该领域发文量大于 4 篇的高生产力作者相关信息（表 1）。

表 1 作者信息表（发文量大于 4 篇）

作者	发文量	被引次数	篇均被引次数
Huang, Y.	6	221	36.83
Arundel, R.	5	117	23.40
Wang, Y	5	15	3.00
He, S.	4	117	29.25
Yi, C.	4	67	16.75

由表 1 可以发现，学者黄（Huang）是近 20 余年来在住房不平等研究领域的最高生产力，共计发表了 6 篇论文，同时也是高被引作者，篇均被引次数高达 36.83 次，研究该作者关注的核心点，可知他主要关注中国在住房转型和过渡时期的住房财富不平等以及农民工住房条件相对较差的制度原因^{[18, 19][16-17]}。此外，近 20 余年住房不平等领域的高发文量和高被引作者大多来自中国，可见由于近年来中国经济高速增长下社会变革的快速推进和房地产领域的飞速发展，不仅加速了中国住房的商品化进程、居民住房自有率显著提高，也使得各种住房不平等也日益严重^{[20][18]}，该议题的研究亦受到了较多房地产领域学者的关注。

2.2 对期刊载文量的统计分析

对文献所属的期刊进行统计，发现近 20 余年来，刊载该议题论文的刊物除了少量综合性期刊外，大部分期刊都与房地产研究密切相关，这说明该议题的研究有较强的专业性和针对性。对载文量大于 4 篇的核心期刊进行了展示（表 2）。其中载文量最大的期刊是 *Housing Studies*，多达 25 篇。此外，篇均被引次数最多的期刊为 *Social Problems*，为 JCR1 区的社会学类顶尖期刊，共计 4 篇文章获得了 303 次引用，篇均被引次数达 75.25 次，这说明该期刊刊载的文章质量较高，在住房不平等研究议题上受到了诸多关注。对其刊载的文献进行分析发现，该期刊对种族问题较为关注，分析了种族歧视导致的住房不平等^{[21, 22][19-20]}，此外也关注了财富不平衡导致的居住隔离现象^{[23][21]}。

表 2 期刊的信息表（载文量大于 4 篇）

期刊	发文量	被引数	篇均被引数
Housing Studies	25	323	12.92
Housing Policy Debate	8	84	10.50
Urban Studies	7	107	15.29
Cities	7	88	12.57
Journal of Housing and the Built Environment	5	31	6.20
Land	5	9	1.80
Social Problems	4	303	75.75
China Economic Review	4	115	28.75

（三）对国家合作和发文量的统计分析

本研究对 32 个国家的发文量进行了分析。首先通过 VOSviewer 对所有在该领域有发文的国家进行了可视化,结果如图所 1 示。从图中可以看出,该领域不同的国家的发文量差异巨大,马太效应十分显著,大部分论文为少数几个国家的学者所著。

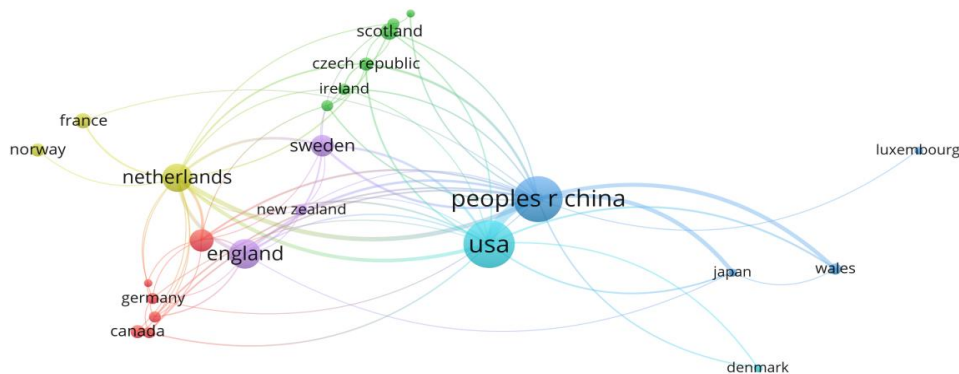


图 1 国家发文量与合作的知识图谱

进一步分析该领域的高生产力国家,发现该领域发文量前 5 的国家中,美国和中国学者贡献了最多的论文,共发表 107 篇论文,占该领域总发文量的 61.5%。其次是英国和新西兰,分别发表了 19 篇和 17 篇;从篇均被引量来看,美国学者的研究在该领域中受到了较多学者的关注,篇均被引次数达到了 26.95 次(表 3)。

表3 发文量大于10篇的国家一览表

国家	发文量	被引次数	篇均被引次数
美国	56	1509	26.95
中国	51	688	13.49
英国	19	191	10.05
荷兰	17	408	24.00
澳大利亚	10	46	4.60

3 关键词聚类分析与演进分析

3.1 关键词聚类分析

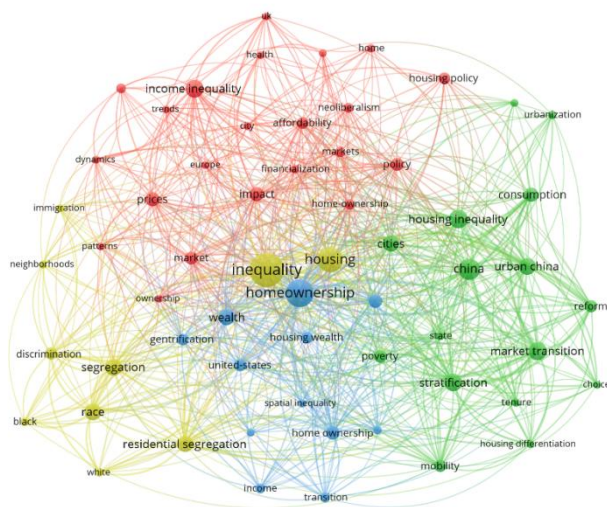


图 2 共现关键词聚类知识图谱

关键词凝练了一篇论文的核心与精髓,通过关键词共现分析可以发现某个领域的研究热

点，因此关键词已被广泛用于揭示研究领域的知识结构^{[24][22]}。使用 VOSviewer 对 167 篇文献绘制关键词共现网络视图，选择其中频次大于等于 5 的 60 个重点关键词进行可视化（图 2）。图中圆节点越大，表示关键词出现的次数越多，越能代表该领域的热点；节点连线代表关键词的关联强度，连线越粗表明连线两端的关键词共同出现在同一篇文献中的次数越多；不同的节点颜色代表不同的聚类，即研究主题。

探究关键词共现图可以发现，图中关键词可分为四个聚类，且蓝色聚类分别和黄色聚类绿色聚类关系较为紧密，而红色聚类相对较独立。为更清晰地了解关键词之间的关系，本研究对 4 个聚类的词汇进行了分析，逐一研究了关键词聚类所关联的研究内容。

红色聚类关键词围绕的核心是收入不平等（income inequality），而收入不平等导致的贫富差异很可能会影响整个社会在住房上的负担能力（affordability）差异，从而导致社会住房不平等加剧。如在加拿大，获得经济适用房的机会根据性别和移民身份进行了分化，随着时间推移，中低收入家庭的住房负担能力显著恶化^{[25][23]}。德维德（Dewilde）和兰斯（Lancee）^{[26][24]}通过研究 28 个国家的多层次模型得出，收入不平等会影响人们的期望从而加剧地位竞争，进而影响人们获得住房的机会；收入不平等加剧了低收入者按市场价租房或购房能力的问题；不平等与住房拥挤之间存在正相关；收入不平等程度越高，住房质量越低。但事实上收入平等与住房不平等之间的关系很复杂，如 1981 年至 2011 年的英格兰和威尔士就是例外，该地区的收入不平等程度较高，但住房不平等程度较低^{[27][25]}。

黄色聚类围绕的核心是居住隔离（residential segregation），居住隔离是指居民由于种族、宗教、职业、习俗、文化水准或财富差异等关系，特征相类似的集居于一特定地区，不相类似的团体间彼此分开产生隔离现象。贝布里斯（Besbris）和费伯（Faber）^{[28][26]}利用回归方法和定性访谈调查房地产经纪人，研究得出代理商集中和种族隔离与地方不平等之间存在显著关系。同时有学者通过实证分析，可知移民的房屋所有权相对较低，且家庭住房拥挤程度较高，但移民特征对住房成本的影响较小^[29]。另外，我们可以感受到中心城市和郊区之间存在居住隔离。且通过大量文献阅读后我们发现该现象普遍存在于全球各地，如旨在为陷入困境的房地产市场提供救济的美国房主贷款公司（HOLC），通过拒绝向有色人种社区提供抵押贷款将种族隔离主义制度化。在随后的几十年里，该做法从少数族裔社区转移了数十亿美元，并塑造了种族隔离模式。这种通过“空间标记”进行种族排斥的现象也体现在当代住房不平等中，住房不平等仍然是当代种族分层最明显的体现^{[30][28]}。学者将最新的数字化档案数据与近期抵押贷款结果的数据相结合，以调查制度性地方分化在住房金融不平等问题上的影响。结果表明 21 世纪的有色人种借款人更有可能被拒绝贷款，或获得次级贷款^[30]。包括中国的户籍隔离所导致的国家福利条款双重性，也会在一定程度上导致住房不平等。

绿色聚类关键词所关联的研究大多是在中国（China）社会背景下展开的。在文献查阅时，我们发现大量的研究聚焦于中国住房的城乡不平等。通过分析中国家庭金融调查（CHFS）的数据，发现城乡不平等是整体住房财富不平等的重要组成部分^[31]。本身城市和农村人口的收益和财富就存在差距，这直接导致了城乡住房不平等。且研究发现在过去几年里，中国城乡居民住房财富差距不断加大，并且大于收入差距^[32]。此外，中国城市住房市场化程度相对较高，而投资性住房财富在城镇居民住房财富中所占比重也越来越高^[32]，有助于城市家庭通过交易房产加速财富积累，从而拉大城乡贫富差距，这显著的促增了城乡居民住房财富的不平等。这些发现对进一步的住房改革以及城乡融合具有重要的政策意义。

蓝色聚类围绕的核心关键词是财富（wealth）。住房在很多家庭中都占有极大的财富比例，根据有关住房金融化的文献，可得出住房公平性是比较分析财富平等性的核心，且通过追踪不同财富组成的影响，财富不平等和集中度的跨国差异主要由住房权益的分布决定^[33]。同时，随着住房金融化，越来越多的人通过房地产交易来赚取收益，这使本就拥有大量财富的人获得了更多的收益，而炒房提高了市场房价，导致本就财富不足的人口，更加承担不起住房支出，由此加剧了住房不平等。

在图 3 中最大的两个节点为不平等（inequality）和房屋所有权（homeownership），也是

全图的核心。由此可以推知，住房不平等主要体现在房屋所有权的不平等。研究表明，房屋所有权在总体上具有均衡效应，并且近年来住房拥有率呈上升趋势，但住房拥有率的上升并不代表空间和财富的分配均衡，反而与空间隔离的加剧和财富在城市的集中化有关^[34]。需要专家进一步研究改进住房机制。

3.2 关键词演进分析

对关键词的共现分析可以发现研究领域的背景、热点和聚焦点，对关键词的演进分析则可以获悉研究领域的发展过程，因此本研究运用 CiteSpace 对关键词进行了演进分析，勾画关键词时区视图（图 3）。图 3 时区图中每一个背景条代表了一个年份，关键词大小代表了关键词的频次高低，连线代表了关键词间的共现关系。

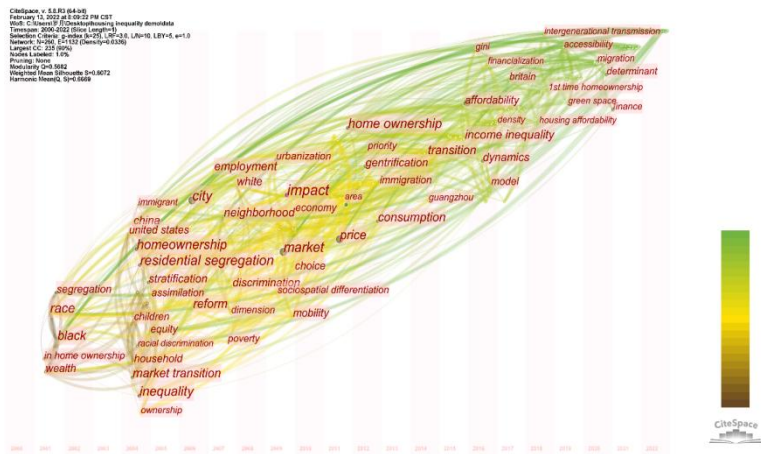


图 3 共现关键词时区演进知识图谱

从图 3 可发现，高频关键词主要出现在 2004 年到 2010 年左右，但在那期间关于住房不平等问题发文量较少，且在该领域最初主要关注于种族（race）和（对不同种族、宗教的人所采取的）隔离（segregation）问题。学者们认为，财富和房屋所有权方面的种族差异是导致美国大都市住房市场种族不平等和产生住宅隔离的关键因素^{[35][34]}。该现象受到越来越多的关注。有学者通过对 5 个纬度的隔离因素进行研究，证明种族隔离对少数族裔住房所有权有负面影响^{[36][35]}。该现象凸显了对财富分层采取多种族观点的必要性，突出了住房分层的背景性质以及城市人口动态对种族和民族住房不平等的影响。另一方面，由图可知，2004 年开始美国（the United State）和中国（China）对住房不平等问题关注和研究较多，本文的许多综述例子也来源于这两个国家。例如，中国有不少学者对流动人口的住房问题进行研究，发现在过去的 20 年里，中国经历了住房系统的急剧转型和快速的城市化，在城市地区住房过度拥挤现象加剧，基于回归的分解分析进一步表明，在其他条件相同的情况下，外来人员受到住房过度拥挤的风险影响远高于当地人口^{[37][37]}由此国内相关住房部门需要更加关注由流动人口构成的“漂泊社会结构”下的住房分化问题。另外，在中国户口也会影响流动人口的居住状况，非农业户口者在城市中更可能获得相对优越的居住条件^{[38][38]}。而美国住房的种族差异较大，有学者通过研究美国黑人，西班牙裔，亚洲人和非西班牙裔白人之间的住房公平差异，探索了住房和财富不平等的根源。查看美国住房调查的数据，学者发现黑人和西班牙裔的住房公平性与白人相比存在巨大的差距。这一发现在年龄，社会经济地位和住房市场价值的影响中也表现一致。黑人和西班牙裔人从抵押贷款和住房政策中获得的好处也比白人少。这些发现为关于财富和住房不平等的新兴分层观点提供了依据，该观点表明种族分层的社会和制度有利于白人，而拉大了有色人种和白人的住房差距^{[39][40]}。

此外，移民（immigrant）的住房不平等问题也开始受到关注，如塞米约诺夫（Semyonov）^{[40][41]}等人应用 Tobit 估算程序来描述以色列移民群体住房资产积累背后的社会机制。研究发现，移民的住房价值随着在新国家居住时间的增长而上升。住房价值也受到移民时期的影响，

但以以色列出生者和移民（即使在以色列居住了 30 年之久）之间的住房价值仍然差距显著。另有学者福克斯高谭（Fox Gotham）^{[35][34]}通过调查数据和访谈的形式，发现以市场为中心的联邦住房政策削弱了华裔美国人通过拥有房屋积累财富的能力，并加剧了住房的种族隔离。为应对在美国的移民住房不平等问题在 21 世纪初，也诞生了一个非盈利民权和住房组织——亚裔美国人争取平等组织（AAFE），为亚裔美国人提供房屋所有权和其他服务^{[41][42]}。

随着近 20 余年来社会经济的发展，住房不平等研究也与时俱进，从时区图中可看出该研究领域的演进趋势。首先从高频的词汇中可以发现有关房屋所有权（homeownership）和居住隔离（residential segregation）的研究是住房不平等议题中贯穿始终的热点。其次，近年来的研究呈现出两个大趋势，即研究内容不断细化和研究领域也在不断拓展，并开始呈现与其他研究领域交汇的态势。近些年的研究开始关注基尼（gini）系数与住房不平等之间的关系，也关注了部分地区出现的士绅化（gentrification）现象^{[42][43]}，士绅化是一个复杂的社会现象，具体是指一个旧社区从原本聚集低收入人士，到重建后地价及租金上升，引致较高收入人士迁入，并取代原有低收入者。士绅化的转变过程可能因为重建速度而需时多年，但引申的结果是本区生活指数提高。此外，住房金融化（financialization）成为了房地产业不可逆趋势^{[43][44]}。整体而言，结合关键词演进分析与发文量的相关信息，可以推知关于住房不平等的问题在近几年依旧是越来越受到重视的。

4、研究结论

基于文献计量学方法，本文对国内外住房不平等问题研究文献进行了量化分析和可视化分析，主要发现如下：第一，对住房不平等议题的研究已经形成了较为稳定的作者合作群体，形成了若干知名学者，且具有较高生产力的学者大多属于中国，这些学者所带领的学术团体在推动该研究议题发展的过程起到了举足轻重的作用。第二，美国和中国学者对住房不平等研究的贡献最大。刊载该议题论文的期刊主要为研究住房问题和社会问题的期刊，也有部分论文发表于一些综合性期刊如（如 *Sustainability*）；第三，基于关键词共现分析和演进分析，发现该研究领域在 20 余年间发展迅速，研究议题十分广泛且复杂。且很多研究都是在特定的社会背景下展开，演进分析较为清晰地呈现了不同时代背景下研究关注点的不同，但贫富差距和种族歧视引致的住房不平等始终是研究的焦点之一。

[参考文献]

[1] BELL D N, BLANCHFLOWER D G. Young people and the Great Recession [J]. Oxford Review of Economic Policy, 2011, 27(2): 241-67.

[2] STOCKHAMMER E. Why have wage shares fallen? An analysis of the determinants of functional income distribution [M]. Wage-led growth. Springer. 2013: 40-70.

[3] RENAUD B, KIM K. The global housing price boom and its aftermath [J]. Housing Finance International, 2007, 22(2): 3.

[4] 吴开泽. 住房市场化与住房不平等 [J]. 社会学研究, 2019: 6.

[5] ARUNDEL R. Equity inequity: Housing wealth inequality, inter and intra-generational divergences, and the rise of private landlordism [J]. Housing, Theory and Society, 2017, 34(2): 176-200.

[6] HOEKSTRA M S, HOCHSTENBACH C, BONTJE M A, et al. Shrinkage and housing inequality: Policy responses to population decline and class change [J]. Journal of Urban Affairs, 2020, 42(3): 333-50.

[7] SMITH S J, CLARK W A, ONG VIFORJ R, et al. Housing and economic inequality in the long run: The retreat of owner occupation [J]. Economy and Society, 2022, 51(2): 161-86.

[8] FULLER G W, JOHNSTON A, REGAN A. Housing prices and wealth inequality in Western Europe [J]. West European Politics, 2020, 43(2): 297-320.

[9] ARUNDEL R, HOCHSTENBACH C. Divided access and the spatial polarization of housing wealth [J]. Urban Geography, 2020, 41(4): 497-523.

[10] 王迪. 住房区域不平等对个体住房消费和生活满意度的影响研究 [D]; 吉林大学, 2020.

- [11] BOWER M, BUCKLE C, RUGEL E, et al. 'Trapped', 'anxious' and 'traumatised': COVID-19 intensified the impact of housing inequality on Australians' mental health [J]. *International Journal of Housing Policy*, 2021: 1-32.
- [12] LAMPMAN R J. The share of top wealth-holders in national wealth, 1922-56 [J]. NBER Books, 1962.
- [13] GHOSH P, ALAM A, GHOSAL N, et al. A geospatial analysis of temporary housing inequality among socially marginalized and privileged groups in India [J]. *Regional Science Policy & Practice*, 2021, 13(3): 798-819.
- [14] MERIGÓ J M, GIL-LAFUENTE A M, YAGER R R. An overview of fuzzy research with bibliometric indicators [J]. *Applied Soft Computing*, 2015, 27: 420-33.
- [15] CHEN C. CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature [J]. *Journal of the American Society for information Science and Technology*, 2006, 57(3): 359-77.
- [16] VAN ECK N, WALTMAN L. Software survey: VOSviewer, a computer program for bibliometric mapping [J]. *scientometrics*, 2010, 84(2): 523-38.
- [17] PRICE D. Little science, big science Columbia University Press New York [Z]. London. 1963
- [18] YI C, HUANG Y. Housing consumption and housing inequality in Chinese cities during the first decade of the twenty-first century [J]. *Housing Studies*, 2014, 29(2): 291-311.
- [19] HUANG Y, TAO R. Housing migrants in Chinese cities: Current status and policy design [J]. *Environment and Planning C: Government and Policy*, 2015, 33(3): 640-60.
- [20] 吴翔华, 赵亿. 住房基尼系数与我国住房不平等研究进展 [J]. *现代城市研究*, 2018, (8): 47-51.
- [21] SHARP G, HALL M. Emerging forms of racial inequality in homeownership exit, 1968–2009 [J]. *Social Problems*, 2014, 61(3): 427-47.
- [22] ROSCIGNO V J, KARAFIN D L, TESTER G. The complexities and processes of racial housing discrimination [J]. *Social Problems*, 2009, 56(1): 49-69.
- [23] DWYER R E. Expanding homes and increasing inequalities: US housing development and the residential segregation of the affluent [J]. *social problems*, 2007, 54(1): 23-46.
- [24] CHEN G, XIAO L. Selecting publication keywords for domain analysis in bibliometrics: A comparison of three methods [J]. *Journal of Informetrics*, 2016, 10(1): 212-23.
- [25] ZHU Y, YUAN Y, GU J, et al. Neoliberalization and inequality: disparities in access to affordable housing in urban Canada 1981–2016 [J]. *Housing Studies*, 2021: 1-28.
- [26] DEWILDE C, LANCEE B. Income inequality and access to housing in Europe [J]. *European sociological review*, 2013, 29(6): 1189-200.
- [27] TUNSTALL R. Is housing growth ever inclusive growth? Evidence from three decades of housing development in England and Wales, 1981–2011 [J]. *Social Inclusion*, 2020, 8(3): 16-27.
- [28] BESBRIS M, FABER J W. Investigating the relationship between real estate agents, segregation, and house prices: Steering and upselling in New York State; proceedings of the Sociological Forum, F, 2017 [C]. Wiley Online Library.
- [29] KRIVO L J. Immigrant characteristics and Hispanic-Anglo housing inequality [J]. *Demography*, 1995, 32(4): 599-615.
- [30] FABER J. Contemporary echoes of segregationist policy: Spatial marking and the persistence of inequality [J]. *Urban Studies*, 2021, 58(5): 1067-86.
- [31] WANG Y, LI Y, HUANG Y, et al. Housing wealth inequality in China: An urban-rural comparison [J]. *Cities*, 2020, 96: 102428.
- [32] 王婷婷. 我国城镇居民住房财富分化的程度, 特征及其影响研究 [D]; 上海师范大学, 2014.
- [33] PFEFFER F T, WAITKUS N. The wealth inequality of nations [J]. *American Sociological Review*, 2021, 86(4): 567-602.
- [34] WALKS A. Homeownership, asset-based welfare and the neighbourhood segregation of wealth [J]. *Housing Studies*, 2016, 31(7): 755-84.
- [35] FOX GOTHAM K. Separate and unequal: The Housing Act of 1968 and the Section 235 program; proceedings of the Sociological Forum, F, 2000 [C]. Springer.
- [36] FLIPPEN C A. Racial and ethnic inequality in homeownership and housing equity [J]. *Sociological Quarterly*, 2001, 42(2): 121-49.
- [37] ZHANG Y, CHEN J. The changing prevalence of housing overcrowding in post-reform China: The case of Shanghai, 2000–2010 [J]. *Habitat International*, 2014, 42: 214-23.
- [38] 胡陈冲, 朱宇, 林李月, et al. 流动人口的户籍迁移意愿及其影响因素分析——基于一项在福建省的问卷调查 [J]. *人口与发展*, 2011, (3): 2-10.

- [39] KRIVO L J, KAUFMAN R L. Housing and wealth inequality: Racial-ethnic differences in home equity in the United States [J]. *Demography*, 2004, 41(3): 585-605.
- [40] SEMYONOV M, LEWIN-EPSTEIN N, DAVIDOV E. Period and duration effects on the value of housing among immigrants [J]. *Social Science Research*, 2003, 32(1): 2-24.
- [41] LISTOKIN D, LISTOKIN B. Asian Americans for equality: A case study of strategies for expanding immigrant homeownership [J]. *Housing Policy Debate*, 2001, 12(1): 47-75.
- [42] TACH L, EMORY A D. Public housing redevelopment, neighborhood change, and the restructuring of urban inequality [J]. *American Journal of Sociology*, 2017, 123(3): 686-739.
- [43] DEWILDE C, DE DECKER P. Changing inequalities in housing outcomes across Western Europe [J]. *Housing, Theory and Society*, 2016, 33(2): 121-61.

Research Hotspot and Evolution Trend of REITs Since the New Century—from the Perspective of Bibliometrics

Yuying Luo¹, Xinghua Zhang², Peiqi Jin³

^{1,2}School of Management, Zhejiang University of Technology, Hangzhou, China
³School of Foreign Language, Zhejiang University of Technology, Hangzhou, China
¹2909435796@qq.com, ²wwew007@me.com, ³823893488@qq.com

Abstract: To understand the research status of REITs since the new century, this paper uses VOSviewer and CiteSpace software to conduct quantitative research and visual analysis on 515 relevant documents in the WOS core database from 2000 to 2022. It has found that: 1) the core author group of the research topic has been formed, the cluster structure of the cooperation network among the core authors is obvious, and the distribution of the cooperation groups is relatively clear; 2) Core journals and related journals in the research field have already been formed; 3) The research on this topic has close international cooperation. Generally speaking, as the country that first gave birth to REITs, the United States has the highest scientific productivity in this field; 4) The research presents an overview of this field through the co-occurrence analysis and evolution analysis of keywords. Combined with the analysis of mutant words, it can be considered that this research field has formed a relatively stable research trend. Through the combination of econometric analysis and visualization, the research grasps the development context of this field and provides a comprehensive and clear research framework and structure for REITs research topics.

Keywords: Knowledge map, Visual analysis, Bibliometrics, REITs.

1. Introduction

Real estate investment trusts (REITs) are trust or corporate funds that collect investors’ funds and are managed by professional institutions to invest in rental real estate assets that can generate stable cash flow, and distribute the generated cash income in a timely manner in the form of dividend distribution to provide investors with long-term and stable cash income. One of the most significant advantages of REITs is to avoid double taxation. Tax drive is the driving force for the emergence, development, and innovation of the world’s largest REITs market, that is, the United States REITs.

On June 21, 2020, marked by the official landing of 9 infrastructure REITs products on China’s Shanghai and Shenzhen Stock Exchange, the innovative development stage of China’s REITs in line with international standards has commenced, declaring the official departure of China’s public offering REITs. In this context, it is necessary to sort out and analyze the research on REITs from a global perspective, so as to provide reasonable suggestions for the management of domestic REITs and the implementation of relevant policies.

2. Research Method and Materials

2.1 Research Methods

Bibliometrics is a science that takes literature as the object and adopts the ideas and methods of Metrology. Since the British scholar Pritchard proposed “Bibliometrics” in 1969, bibliometrics, as an independent discipline, has emerged in scientific quantitative researches. Especially in recent years, thanks to the rapid development of computer technology and the combination of visualization technology and metrology principles, bibliometrics analysis has become more

three-dimensional and detailed, forming a new trend of thought on quantitative analysis.

This study mainly uses econometric methods to analyze the scientific productivity of countries and authors in the research field of REITs and the publication volume of journals, describing the laws and quasi laws of scientific activities with the help of Price’s law and Bradford’s law of scattering, and learns the tendencies and paths of research’s development through keyword cluster analysis and evolution analysis.

2.2 Research Materials

To ensure the coverage and authority of the analyzed data, this study selects the Web of Science (Core Collection) as the data source, and carries out the preliminary retrieval of the data according to the following standards:

Table 1: Summary of data source and selection

Category	Specific Standard Requirements
Research database	Web of Science core collection
Citation indexes	SSCI
Searching period	January 2000 to March 22 nd 2022
Language	“English”
Searching keywords	TS=(“REITs” or “REOCs” or “Real Estate Investment Trust”)
Subject categories	“Economics” or “Management” or “Sociology”
Document types	“Articles” or “Review Articles”
Data extraction	Export with full records and cited references in plain text format
Sample size	669

Because the data obtained directly through retrieval may have some problems, such as repetition or inconsistency with the subject, the data needs to be preprocessed before analysis, so as to avoid affecting the analysis results due to the quality of the data itself. According to the literature screening criteria

RESEARCH ARTICLE

Who Upgrades the Plant? A Tripartite Evolutionary Game Model for PPCPs Removal Technology Adoption

Xue Lei  | Yuying Luo  | Fengmei Liu  | Xueguo Xu 

School of Management, Shanghai University, Shanghai, China

Correspondence: Xueguo Xu (xu_xueguo_work@163.com)**Received:** 8 January 2026 | **Revised:** 20 January 2026 | **Accepted:** 3 February 2026**Keywords:** environmental governance | personal care products | pharmaceuticals | wastewater treatment**ABSTRACT**

Pharmaceuticals and personal care products (PPCPs) contamination in aquatic environments has attracted considerable research attention, yet most studies focus on treatment technologies while the governance mechanisms coordinating stakeholder behavior remain underexplored. This study develops a tripartite evolutionary game model to analyze strategic interactions among pharmaceutical enterprises, wastewater treatment plants, and government environmental agencies in PPCPs pollution control. Replicator dynamics and stability analysis identify six stable equilibrium configurations and reveal that complete cooperation equilibrium cannot achieve asymptotic stability due to government fiscal incentives for regulatory withdrawal. Results demonstrate that technology-upgrading decisions of treatment plants respond primarily to government subsidies and penalties rather than upstream enterprise behavior. The mathematical finding that upstream enterprise production choices do not directly alter treatment plant upgrading incentives through cost transmission channels reveals that market-based coordination cannot spontaneously emerge without regulatory intervention, thereby strengthening the theoretical case for sustained government engagement. Sensitivity analysis indicates that reducing clean production costs and enhancing green product revenues accelerate system convergence toward environmentally favorable equilibria. Additional sensitivity analysis of treatment plant subsidy and penalty parameters confirms that both positive incentives and negative sanctions effectively promote technology adoption when appropriately calibrated. These findings provide theoretical guidance for designing differentiated policy instruments that balance market incentives with sustained regulatory oversight.

1 | Introduction

Pharmaceuticals and personal care products (PPCPs) have become pervasive contaminants in aquatic ecosystems across the globe, raising significant concerns for both ecological integrity and human health (Kallenborn et al. 2018). These biologically active compounds are intentionally designed to elicit physiological responses at trace concentrations, a characteristic that renders them particularly problematic when they enter environmental compartments for which they were never intended. The primary pathway through which PPCPs reach surface waters is via wastewater treatment plant (WWTP) effluents, where conventional activated sludge processes often fail to achieve

adequate removal due to the recalcitrant molecular structures and low biodegradability of many pharmaceutical compounds (Tarpani and Azapagic 2018). This incomplete elimination transforms WWTPs from pollution control barriers into conduits for micropollutant discharge, creating a persistent source of contamination that affects receiving water bodies on a continuous basis.

The geographical scope of PPCP contamination spans from densely urbanized watersheds to remote polar regions. Monitoring campaigns conducted in the River Lambro basin, one of the most industrialized areas in Italy, detected pharmaceuticals and other emerging contaminants at concentrations

Summary

- Government subsidies drive treatment plant technology upgrading more than upstream pollution loads.
- Complete cooperation equilibrium remains structurally unstable without sustained regulatory oversight.
- Reducing clean production costs accelerates voluntary adoption of environmentally friendly strategies.
- Differentiated policy instruments outperform uniform penalties in PPCPs governance effectiveness.

reaching the microgram per liter range in untreated wastewater, with substantial mass loads persisting through treatment processes into final effluents (Castiglioni et al. 2018). Similar patterns emerge in mixed land use watersheds across North America, where contaminants including carbamazepine, sulfamethoxazole, and caffeine exhibit detection frequencies exceeding 70% in surface water samples, with concentrations influenced by proximity to WWTP discharge points and seasonal hydrological conditions (Fairbairn et al. 2016). Even the Arctic environment, long considered pristine, now harbors detectable residues of over 110 pharmaceutical substances in coastal seawater and biota, attributable largely to domestic wastewater discharges from settlements lacking modern treatment infrastructure (Kallenborn et al. 2018). This ubiquity underscores that PPCP pollution transcends local boundaries and demands coordinated responses across diverse governance contexts.

The ecological consequences of chronic PPCP exposure have attracted sustained investigative attention. Laboratory studies demonstrate that environmentally relevant concentrations of metformin, one of the most widely prescribed antidiabetic medications, induce endocrine disruption in adult male fathead minnows, manifesting as significant upregulation of vitellogenin messenger RNA, a biomarker typically associated with female reproductive physiology (Niemuth et al. 2015). Mesocosm experiments exposing three-spined stickleback populations to mixtures of commonly detected pharmaceuticals reveal impaired survival among female fish at higher treatment levels, alongside measurable impacts on condition factors at both individual and population scales (David et al. 2020). Marine organisms face analogous threats; tiered toxicity assessments of diclofenac exposure in the brown mussel *Perna perna* document decreased lysosomal membrane stability, DNA damage, and oxidative stress responses at concentrations as low as nanograms per liter, well within the range observed in coastal waters receiving submarine outfall discharges (Fontes et al. 2018). These findings collectively indicate that PPCPs exert subtle yet consequential effects on non-target organisms, with implications for ecosystem structure and function that may propagate through food webs.

The performance of treatment technologies in removing PPCPs varies considerably across compounds, processes, and operational conditions. Seasonal monitoring at Himalayan foothill WWTPs reveals negative removal rates for certain pharmaceuticals, meaning effluent concentrations exceed influent levels, a phenomenon attributed to deconjugation of metabolites and desorption from particulate matter during biological treatment (Silori et al. 2025). Comparative assessments of wastewater

treatment works demonstrate wide-ranging removal efficiencies for trace contaminants, with performance influenced by site-specific operational factors that remain incompletely characterized (Gardner et al. 2013). Advanced treatment technologies offer enhanced removal capabilities; membrane bioreactors achieve selective elimination of antibiotics through combined biodegradation and adsorption mechanisms (Zheng et al. 2019), while advanced oxidation processes (AOPs) and granular activated carbon filtration attain removal efficiencies exceeding 80% for many recalcitrant compounds (Risch et al. 2022). Yet life cycle assessments reveal that the direct water quality benefits obtained from advanced treatment are frequently outweighed by indirect environmental impacts stemming from elevated energy and resource demands, complicating straightforward recommendations for technology adoption (Risch et al. 2022). Multi-criteria decision-making frameworks have accordingly been developed to navigate these tradeoffs, incorporating factors such as hydraulic retention time and population equivalent alongside removal efficiency metrics to guide WWTP design decisions (Fernandez-Lopez et al. 2021).

Prioritization schemes represent another active research domain, seeking to identify which among thousands of pharmaceutical compounds warrant focused monitoring and regulatory attention. Approaches employing predicted environmental concentrations, persistence indicators, bioaccumulation potential, and ecotoxicological thresholds have been applied to rank pharmaceuticals according to their likelihood of causing environmental harm (Letsinger and Kay 2019). Risk-based prioritization of micropollutants detected in WWTP effluents across Chinese facilities identifies perfluorinated compounds, pesticide transformation products, and bisphenol A among the top-ranked substances, many of which are not currently subject to effluent discharge limits (Liu et al. 2024). The COVID-19 pandemic provided an unintended natural experiment illustrating how rapidly pharmaceutical consumption patterns translate into environmental contamination; wastewater surveillance documented dramatic surges in antipyretic concentrations, with ibuprofen levels increasing up to 67-fold and paracetamol up to 311-fold relative to pre-pandemic baselines, accompanied by elevated ecological risk quotients for aquatic organisms (Yu et al. 2025). Therapeutic agents specific to COVID-19 management, including remdesivir and dexamethasone, were detected persisting through treatment processes into effluents destined for beneficial reuse, introducing novel pharmaceutical compounds directly to agricultural soils and surface waters (Hayden et al. 2022). These observations highlight the dynamic coupling between human health interventions and environmental quality outcomes, a relationship mediated substantially by the decisions and capacities of wastewater treatment infrastructure.

Despite growing recognition of PPCPs as priority contaminants, existing governance frameworks exhibit structural deficiencies that constrain effective pollution control. Current regulatory architectures predominantly adopt end-of-pipe approaches that place remediation burdens exclusively on wastewater treatment infrastructure while neglecting upstream source reduction opportunities at pharmaceutical manufacturing facilities (Miarov et al. 2020). This fragmented regulatory design fails to internalize the externality costs that polluting enterprises impose on downstream treatment

operations, creating systematic misalignment between private incentives and social welfare objectives. Furthermore, the absence of PPCPs-specific effluent standards in most jurisdictions means that treatment plant operators lack clear compliance targets against which to evaluate technology investment decisions, while pharmaceutical manufacturers face no direct regulatory pressure to modify production processes that generate persistent micropollutants (Liu et al. 2024). The voluntary nature of many existing green manufacturing initiatives introduces collective action dilemmas wherein individual enterprises rationally defer costly process modifications while hoping competitors will bear the transition costs, resulting in suboptimal pollution reduction outcomes across the industry. Cross-jurisdictional regulatory heterogeneity compounds these challenges by enabling regulatory arbitrage, as pharmaceutical production may relocate to regions with less stringent environmental oversight, thereby displacing rather than eliminating contamination sources (El Hammoudani et al. 2024).

Understanding the strategic calculus of each stakeholder group proves essential for designing effective governance interventions. Pharmaceutical enterprises operate within competitive market environments where production cost minimization constitutes a primary operational objective, creating inherent tensions between environmental responsibility and profit maximization that shape clean production adoption decisions (Liang et al. 2023). These enterprises weigh the immediate costs of green manufacturing processes, including equipment upgrades, raw material substitutions, and process reengineering, against uncertain future benefits such as enhanced brand reputation, market access to environmentally conscious consumers, and reduced regulatory risk exposure. Wastewater treatment plants, whether publicly owned utilities or privately operated facilities, face capital constraints that limit technology upgrading capacity, particularly for AOPs and membrane separation systems requiring substantial initial investment and ongoing operational expenditures (Risch et al. 2022). Their technology adoption decisions respond to the interplay between regulatory compliance requirements, available fiscal support mechanisms, and the economic viability of reclaimed water utilization that could offset treatment costs. Government environmental protection agencies navigate complex political economy landscapes where regulatory stringency decisions must balance environmental protection mandates against industry competitiveness concerns, fiscal constraints on enforcement capacity, and accountability pressures that may favor visible short-term outcomes over sustained long-term environmental investments (Haoyang et al. 2022). The bounded rationality characterizing all three stakeholder groups means that strategic choices emerge through iterative learning and adaptation processes rather than instantaneous optimization, necessitating analytical frameworks capable of capturing such behavioral dynamics.

Public participation constitutes an increasingly salient dimension of environmental governance that shapes the strategic context within which enterprises, treatment facilities, and regulatory agencies operate. Household pharmaceutical disposal behaviors directly influence municipal wastewater PPCPs loading, while consumer-purchasing preferences can reward or

penalize manufacturers based on environmental performance credentials (Wang et al. 2015). Survey research documents significant public willingness to pay for PPCPs pollution control measures, indicating latent social demand for environmental protection that policymakers could potentially mobilize to support regulatory initiatives. Media attention to pharmaceutical contamination incidents amplifies reputational consequences for non-compliant enterprises and creates political pressure on government agencies to demonstrate regulatory effectiveness. While the present study focuses on the tripartite interactions among enterprises, treatment plants, and government agencies as the primary decision-makers in PPCPs governance, the broader social context within which these actors operate influences their strategic calculations and merits explicit acknowledgment as a boundary condition shaping model applicability.

Notwithstanding the extensive technical knowledge accumulated regarding PPCP occurrence, fate, and treatment, a critical dimension of micropollutant control remains underexplored: the governance mechanisms through which stakeholder behaviors are shaped and coordinated. Addressing PPCP pollution effectively requires concerted action among pharmaceutical manufacturers who generate contamination at source, wastewater treatment facilities charged with end-of-pipe removal, and government agencies possessing regulatory and fiscal instruments capable of influencing private sector decisions. Survey research among expert stakeholders drawn from government, academia, and industry across North America and Europe reveals broad consensus that pharmaceuticals in the environment constitute a legitimate concern for both ecosystem and human health, with respondents identifying advanced wastewater treatment, pharmaceutical return programs, and minimum secondary treatment requirements as promising management strategies (Doerr-MacEwen and Haight 2006). Public willingness to pay for measures aimed at reducing PPCP pollution has been documented in contingent valuation experiments, suggesting that societal demand for environmental protection exists and could potentially support policy interventions (Wang et al. 2015). Comparative analyses of regulatory frameworks across jurisdictions reveal substantial heterogeneity in approaches to pharmaceutical monitoring and standard-setting, with most countries lacking comprehensive regulatory oversight for these compounds in water resources (Miarov et al. 2020). This regulatory patchwork reflects underlying uncertainties regarding appropriate policy instruments and implementation strategies, uncertainties that cannot be resolved through technical analysis alone but require systematic examination of stakeholder incentives and behavioral dynamics.

The strategic interactions among actors in environmental governance systems can be productively analyzed using game-theoretic frameworks, which formalize how rational agents with potentially conflicting interests make interdependent decisions. Evolutionary game theory extends classical game theory by relaxing the assumption of perfect rationality, instead modeling strategy adoption as a population-level learning process wherein agents gradually shift toward behaviors that yield higher payoffs relative to alternatives. This approach proves particularly apt for environmental policy contexts where stakeholders operate under bounded rationality, possess incomplete information about others' strategies, and

adjust their decisions iteratively based on observed outcomes. Bibliometric analysis of micropollutant research demonstrates that technological innovation has dominated the field, with comparatively sparse investigation into policy implementation dynamics and stakeholder decision-making processes (El Hammoudani et al. 2024). The behavioral dimension of PPCP governance thus constitutes a notable gap; while we understand increasingly well how to remove these compounds technically, we understand far less about the conditions under which pharmaceutical enterprises will voluntarily adopt cleaner production practices, treatment plants will invest in technology upgrading, and government agencies will commit resources to regulatory enforcement.

This gap assumes particular significance because the decisions of these three stakeholder groups are inherently interdependent. Pharmaceutical enterprises choosing between clean production and traditional manufacturing impose differential pollution loads on downstream treatment infrastructure, affecting the costs and feasibility of achieving effluent quality targets. Wastewater treatment plants deciding whether to upgrade their technology do so within economic constraints shaped partly by influent characteristics determined by upstream enterprise behavior. Government agencies contemplating regulatory stringency must weigh enforcement costs against environmental benefits, with both influenced by the prevailing distribution of strategies among regulated entities. These feedback relationships create complex dynamics that single-actor analyses cannot capture; understanding system behavior requires examining how strategies co-evolve across all three populations simultaneously.

Against this background, several questions emerge as deserving systematic investigation. How do pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies interact strategically within the PPCP pollution control system, and what behavioral dynamics arise from their interdependent decision-making? Under what combinations of cost structures, benefit distributions, and policy instrument intensities will stakeholders voluntarily converge toward environmentally beneficial strategies without regulatory compulsion? Conversely, under what conditions does sustained government intervention become necessary to induce and maintain cooperative behavior among private actors? Which policy levers exert the greatest influence on system trajectories, and how should these instruments be calibrated to maximize governance effectiveness?

This study addresses these questions through the development and analysis of a tripartite evolutionary game model capturing the strategic interactions among pharmaceutical enterprises, wastewater treatment plants, and government environmental agencies in the context of PPCP pollution control. The model specifies binary strategy spaces for each actor type, derives payoff functions reflecting the economic incentives and cost structures characterizing real-world decision contexts, and employs replicator dynamics to trace the evolution of strategy frequencies within each population over time. Stability analysis of the resulting nonlinear dynamical system identifies equilibrium configurations and characterizes the parameter conditions under which each equilibrium emerges as asymptotically stable.

Numerical simulations validate theoretical predictions and explore sensitivity to key parameters.

The contributions of this research are threefold. First, this study extends the application of evolutionary game theory to the domain of PPCP governance, offering a novel analytical framework for examining stakeholder interactions in micropollutant control that complements the predominantly technical orientation of prior work. Second, the identification of stable equilibrium configurations and their associated parameter conditions provides actionable guidance for policy design, revealing circumstances under which market mechanisms suffice to achieve environmental objectives versus circumstances requiring sustained regulatory engagement. Third, sensitivity analysis pinpoints leverage points where policy interventions can most effectively redirect system dynamics toward socially desirable outcomes, informing the calibration of subsidy magnitudes, penalty intensities, and enforcement probabilities in practical regulatory contexts. These findings contribute to the theoretical understanding of environmental governance mechanisms while offering practical insights for addressing pharmaceutical pollution in aquatic environments, a challenge of growing urgency as global pharmaceutical consumption continues to expand and climate change intensifies pressures on water resources.

2 | Methodology

2.1 | Model Assumptions

The effective management of PPCPs in aquatic environments requires coordinated action among multiple stakeholders, each of whom possesses distinct interests, resource constraints, and strategic considerations. To systematically analyze the behavioral interactions and strategy evolution among these stakeholders during the adoption process of advanced treatment technologies, this study develops a tripartite evolutionary game model that captures the dynamic decision-making processes of pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies. Because each participant operates under conditions of bounded rationality rather than perfect information, their strategic choices emerge through iterative learning and adaptation rather than instantaneous optimization, which necessitates an evolutionary game-theoretic approach that can accommodate such behavioral complexity.

The first assumption (H1) establishes the identity and role of each participant within the PPCPs pollution control system. Pharmaceutical enterprises constitute the primary point-source emitters of PPCPs, as their manufacturing processes inevitably generate wastewater streams containing active pharmaceutical ingredients, intermediates, and various organic compounds that exhibit persistence and potential ecotoxicological effects in receiving waters. Wastewater treatment plants serve as the critical end-of-pipe barriers responsible for removing these micropollutants before treated effluent enters natural water bodies, yet conventional biological treatment processes demonstrate limited efficacy against many pharmaceutical compounds due to their recalcitrant

molecular structures and low biodegradability. Government environmental protection agencies occupy the regulatory position within this tripartite system, wielding policy instruments such as emission standards, financial incentives, and enforcement mechanisms that can significantly influence the strategic calculus of both enterprises and treatment facilities. The interactions among these three bounded-rational groups give rise to complex behavioral dynamics that this evolutionary game model seeks to elucidate.

The second assumption (H2) specifies the binary strategy space available to each participant, which, while representing a simplification of real-world decision complexity, enables tractable mathematical analysis while preserving the essential strategic tensions inherent in PPCPs governance. Pharmaceutical enterprises face a choice between clean production strategies and traditional production strategies, where the former involves adopting green manufacturing processes, optimizing product formulations, implementing source reduction measures, and deploying on-site pretreatment systems to minimize PPCPs discharge into municipal sewerage systems, whereas the latter maintains conventional production methods that externalize pollution costs onto downstream treatment infrastructure and the broader environment. Wastewater treatment plants must decide between technology upgrading strategies and conventional treatment strategies, with the upgrading option requiring substantial capital investment in AOPs, granular activated carbon adsorption systems, membrane separation technologies, or hybrid treatment trains capable of achieving high removal efficiencies for diverse pharmaceutical compounds, while the conventional option relies on existing activated sludge processes that, despite their effectiveness for traditional pollutant parameters, achieve only incidental and often inadequate removal of many PPCPs. Government regulatory agencies choose between active regulation strategies and passive regulation strategies, where active regulation encompasses the promulgation of PPCPs-specific discharge limits, provision of technology upgrading subsidies, implementation of rigorous monitoring programs, and imposition of penalties for non-compliance, whereas passive regulation implies regulatory forbearance that shifts environmental and public health costs onto society while avoiding immediate fiscal expenditures on enforcement infrastructure.

The third assumption (H3) introduces the probability parameters that characterize the population-level distribution of strategies at any given time and whose temporal evolution the model seeks to track. Let x denote the probability that a representative pharmaceutical enterprise adopts clean production, with $1 - x$ representing the probability of traditional production; let y denote the probability that a representative wastewater treatment plant chooses technology upgrading, with $1 - y$ representing conventional treatment; and let z denote the probability that the government implements active regulation, with $1 - z$ corresponding to passive regulation. These probabilities satisfy the constraints $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, and their dynamics over time reflect the aggregate outcome of individual learning processes as participants observe payoff differences and adjust their strategies accordingly. The complete set of model parameters and their economic interpretations are formally defined in Table 1.

2.2 | Game Strategies and Payoff Matrix

Building upon the assumptions and parameter definitions established above, this section constructs the complete payoff matrix that governs the evolutionary dynamics of the tripartite game system. The matrix encompasses all eight possible strategy combinations arising from the binary choices available to each participant, and careful specification of the payoff functions ensures that the model captures the essential economic incentives, cost structures, and externality relationships that shape stakeholder behavior in PPCPs governance contexts. Each cell of the payoff matrix reflects the net benefit or loss that each participant realizes under a specific strategy profile, accounting for direct revenues and costs as well as the interactive effects that one party's choices impose on others. The complete payoff structure for all eight strategy combinations is presented in Table 2.

Consider first the strategy combination in which pharmaceutical enterprises adopt clean production, wastewater treatment plants implement technology upgrading, and government pursues active regulation. Under this configuration, pharmaceutical enterprises realize revenue R_1 from their environmentally responsible market positioning while incurring clean production costs C_1 and receiving government subsidies S_1 , yielding a net payoff of $R_1 - C_1 + S_1$. Wastewater treatment plants obtain technology upgrading revenue R_3 , bear the substantial capital and operational costs C_3 associated with advanced treatment systems, and receive government financial support S_2 , producing a net payoff of $R_3 - C_3 + S_2$. Government captures the societal benefits R_5 from effective PPCPs control, expends regulatory resources C_6 on monitoring and enforcement, and disburses subsidies S_1 and S_2 to incentivize desirable behavior, resulting in a net payoff of $R_5 - C_6 - S_1 - S_2$. The positive-sum character of this outcome, wherein all parties can potentially benefit from cooperative behavior when supported by appropriate policy instruments, represents the socially optimal equilibrium toward which policy interventions should aim to steer system evolution.

When pharmaceutical enterprises maintain clean production and wastewater treatment plants upgrade their technology but government adopts passive regulation, the absence of both subsidies and penalties alters the payoff structure considerably. Pharmaceutical enterprises still earn R_1 and pay C_1 but receive no subsidy, yielding $R_1 - C_1$; wastewater treatment plants similarly obtain $R_3 - C_3$ without government financial support; and government enjoys the environmental benefits R_5 without incurring regulatory costs, which might appear advantageous in the short term but ignores the fragility of voluntary compliance in the absence of enforcement credibility. The situation changes markedly when pharmaceutical enterprises choose clean production and government regulates actively, but wastewater treatment plants opt for conventional treatment, in which case the treatment facilities face penalties F_2 and reputational damages L_2 for failing to meet effluent PPCPs standards despite receiving influent with relatively low pharmaceutical loading, producing a payoff of $R_4 - C_4 - F_2 - L_2$ that creates strong incentives for technology adoption.

The payoff implications become particularly consequential when pharmaceutical enterprises select traditional production,

TABLE 1 | Model parameters and definitions.

Parameter	Definition
R_1	Revenue accruing to pharmaceutical enterprises from clean production, including green brand premiums, enhanced market competitiveness, and improved corporate social responsibility reputation
R_2	Revenue accruing to pharmaceutical enterprises from traditional production
R_3	Revenue accruing to wastewater treatment plants from technology upgrading, encompassing improved effluent quality, reclaimed water utilization benefits, and enhanced social reputation
R_4	Baseline operational revenue accruing to wastewater treatment plants from conventional treatment
R_5	Potential benefits accruing to government from active regulation, including public health protection, ecological environment improvement, and enhanced governance performance
C_1	Costs incurred by pharmaceutical enterprises for clean production implementation, including process modification, green raw material substitution, and on-site pretreatment investment
C_2	Costs incurred by pharmaceutical enterprises for traditional production
C_3	Costs incurred by wastewater treatment plants for technology upgrading, encompassing capital investment and operational expenditure for AOPs, activated carbon adsorption, and membrane technologies
C_4	Operational costs incurred by wastewater treatment plants for conventional treatment
C_5	Additional treatment costs imposed on wastewater treatment plants when pharmaceutical enterprises employ traditional production, arising from elevated influent PPCPs concentrations
C_6	Costs incurred by government for active regulation implementation, including monitoring equipment, enforcement personnel, and analytical testing expenses
C_7	Environmental health damages and social costs borne by government under passive regulation
S_1	Subsidies provided by government to pharmaceutical enterprises adopting clean production under active regulation
S_2	Subsidies provided by government to wastewater treatment plants undertaking technology upgrading under active regulation
F_1	Fines imposed by government on pharmaceutical enterprises maintaining traditional production under active regulation
F_2	Fines imposed by government on wastewater treatment plants with non-compliant effluent PPCPs concentrations under active regulation
L_1	Reputational losses suffered by pharmaceutical enterprises from regulatory penalties for traditional production
L_2	Reputational losses suffered by wastewater treatment plants from effluent quality noncompliance

as this choice imposes negative externalities on downstream wastewater treatment infrastructure regardless of the treatment strategy employed. Should treatment plants upgrade their technology while enterprises pollute and government regulates, the plants must absorb additional treatment costs C_5 attributable to elevated influent PPCPs concentrations even as they receive subsidies S_2 , yielding $R_3 - C_3 - C_5 + S_2$, while enterprises suffer fines F_1 and reputational losses L_1 that reduce their payoff to $R_2 - C_2 - F_1 - L_1$. The most problematic scenario arises when all three parties defect from socially beneficial strategies: enterprises pursue traditional production, treatment plants maintain conventional processes, and government remains passive, whereupon enterprises capture $R_2 - C_2$ without penalty, treatment plants obtain $R_4 - C_4 - C_5$ as they struggle with high PPCPs loading using inadequate technology, and government bears environmental damage costs $-C_7$ that ultimately manifest as degraded water quality, ecosystem impairment, and public health risks.

3 | Replicator Dynamics and Equilibrium Analysis

3.1 | Replicator Dynamic Equations

The replicator dynamics framework provides the mathematical apparatus for analyzing how strategy frequencies evolve within each population as participants observe realized payoffs and gradually shift toward more successful behavioral patterns. An evolutionarily stable strategy (ESS) represents a strategy configuration that, once established in a population, resists invasion by alternative strategies because any small group of mutants adopting different behavior would obtain lower average payoffs than the incumbent population. Deriving the replicator dynamic equations for each party requires computing expected payoffs under each pure strategy, calculating the population average payoff, and specifying the rate of strategy adjustment as proportional to the payoff differential, which captures the intuition that strategies outperforming the

TABLE 2 | Strategy combinations and payoff matrix for three-party evolutionary game.

Strategy combination	Pharmaceutical enterprises	Wastewater treatment plants	Government environmental agency
(Clean production, technology upgrading, active regulation)	$R_1 - C_1 + S_1$	$R_3 - C_3 + S_2$	$R_5 - C_6 - S_1 - S_2$
(Clean production, technology upgrading, passive regulation)	$R_1 - C_1$	$R_3 - C_3$	R_5
(Clean production, conventional treatment, active regulation)	$R_1 - C_1 + S_1$	$R_4 - C_4 - F_2 - L_2$	$R_5 - C_6 - S_1 + F_2$
(Clean production, conventional treatment, Passive regulation)	$R_1 - C_1$	$R_4 - C_4$	$R_5 - C_7$
(Traditional production, technology upgrading, active regulation)	$R_2 - C_2 - F_1 - L_1$	$R_3 - C_3 - C_5 + S_2$	$-C_6 - S_2 + F_1$
(Traditional production, technology upgrading, passive regulation)	$R_2 - C_2$	$R_3 - C_3 - C_5$	$-C_7$
(Traditional production, conventional treatment, active regulation)	$R_2 - C_2 - F_1 - L_1$	$R_4 - C_4 - C_5 - F_2 - L_2$	$-C_6 + F_1 + F_2$
(Traditional production, conventional treatment, passive regulation)	$R_2 - C_2$	$R_4 - C_4 - C_5$	$-C_7$

population average will spread while underperforming strategies will decline.

For pharmaceutical enterprises, the expected payoff from clean production U_{x1} aggregates the payoffs realized across all possible strategy combinations of the other two parties, weighted by the respective probabilities of those combinations occurring. When treatment plants upgrade with probability y and government regulates actively with probability z , enterprises choosing clean production obtain the highest payoff component $(R_1 - C_1 + S_1)$ with probability yz ; when treatment plants upgrade but government remains passive, the payoff $(R_1 - C_1)$ occurs with probability $y(1 - z)$; analogous reasoning applies to the remaining combinations, yielding:

$$U_{x1} = (R_1 - C_1 + S_1)yz + (R_1 - C_1)y(1 - z) + (R_1 - C_1 + S_1)(1 - y)z + (R_1 - C_1)(1 - y)(1 - z)$$

Algebraic simplification reveals that the wastewater treatment plant's strategy choice does not directly affect the pharmaceutical enterprise's payoff from clean production, because clean production eliminates the externality channel through which enterprise behavior influences treatment costs, so the expected payoff reduces to:

$$U_{x1} = R_1 - C_1 + S_1z$$

The expected payoff from traditional production U_{x2} follows similar logic but incorporates the penalties and reputational losses that enterprises suffer when government regulates actively:

$$U_{x2} = (R_2 - C_2 - F_1 - L_1)yz + (R_2 - C_2)y(1 - z) + (R_2 - C_2 - F_1 - L_1)(1 - y)z + (R_2 - C_2)(1 - y)(1 - z)$$

This expression simplifies to:

$$U_{x2} = R_2 - C_2 - (F_1 + L_1)z$$

The average expected payoff for the enterprise population equals $\bar{U}_x = xU_{x1} + (1 - x)U_{x2}$, and the replicator dynamic equation governing the evolution of clean production adoption takes the standard form:

$$H(x) = \frac{dx}{dt} = x(U_{x1} - \bar{U}_x) = x(1 - x)(U_{x1} - U_{x2})$$

Substituting the simplified expected payoff expressions yields the enterprise replicator dynamics:

$$H(x) = x(1 - x)[(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)z]$$

This equation reveals that the direction of enterprise strategy evolution depends critically on the comparison between the net benefit differential of clean versus traditional production $(R_1 - C_1) - (R_2 - C_2)$ and the policy-induced incentive term $(S_1 + F_1 + L_1)z$ that government regulation activates. When the intrinsic profitability advantage of traditional production exceeds the expected regulatory penalty, enterprises will drift toward polluting behavior unless government increases either the probability or intensity of enforcement.

For wastewater treatment plants, the expected payoff from technology upgrading U_{y1} must account for the additional treatment burden C_5 imposed when upstream pharmaceutical enterprises employ traditional production:

$$U_{y1} = (R_3 - C_3 + S_2)xz + (R_3 - C_3)x(1 - z) + (R_3 - C_3 - C_5 + S_2)(1 - x)z + (R_3 - C_3 - C_5)(1 - x)(1 - z)$$

Simplification yields:

$$U_{y1} = R_3 - C_3 - C_5(1-x) + S_2z$$

The expected payoff from conventional treatment U_{y2} incorporates both the elevated treatment costs from enterprise pollution and the penalties for effluent non-compliance under active regulation:

$$U_{y2} = (R_4 - C_4 - F_2 - L_2)xz + (R_4 - C_4)x(1-z) + (R_4 - C_4 - C_5 - F_2 - L_2)(1-x)z + (R_4 - C_4 - C_5)(1-x)(1-z)$$

Which reduces to:

$$U_{y2} = R_4 - C_4 - C_5(1-x) - (F_2 + L_2)z$$

The treatment plant replicator dynamics follow as:

$$H(y) = \frac{dy}{dt} = y(1-y)(U_{y1} - U_{y2})$$

$$H(y) = y(1-y)[(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)z]$$

Notably, the enterprise clean production probability x affects both U_{y1} and U_{y2} equally through the externality cost term $C_5(1-x)$, which cancels in the payoff differential and therefore does not appear in the simplified replicator equation. This mathematical result merits careful interpretation rather than dismissal as a modeling limitation. The cancellation reflects a substantive economic reality: both technology upgrading and conventional treatment options face identical baseline cost increases when upstream pharmaceutical enterprises employ traditional production methods, because elevated influent PPCPs concentrations impose additional operational burdens regardless of the treatment technology employed. The critical distinction lies not in differential exposure to upstream pollution costs, but in the capacity of upgraded technologies to achieve regulatory compliance and capture associated benefits including government subsidies, avoided penalties, and enhanced social reputation. This finding carries important policy implications: market-based coordination between upstream and downstream actors cannot spontaneously emerge through cost transmission alone, necessitating government intervention through subsidy and penalty instruments to create the incentive structures that induce treatment plant technology adoption. The model thus reveals that PPCPs governance systems exhibit a form of strategic independence at the private sector level that only regulatory oversight can overcome, a conclusion that strengthens rather than undermines the case for active government engagement in micropollutant control. This mathematical result carries the important policy implication that upstream source control, while reducing absolute treatment costs, does not directly alter the relative attractiveness of technology upgrading versus conventional treatment; instead, government policy instruments $(S_2 + F_2 + L_2)z$ drive the treatment plant's technology adoption decision.

For government environmental protection agencies, the expected payoff from active regulation U_{z1} reflects the heterogeneous

outcomes that arise depending on enterprise and treatment plant behavior:

$$U_{z1} = (R_5 - C_6 - S_1 - S_2)xy + (R_5 - C_6 - S_1 + F_2)x(1-y) + (-C_6 - S_2 + F_1)(1-x)y + (-C_6 + F_1 + F_2)(1-x)(1-y)$$

The expected payoff from passive regulation U_{z2} captures the environmental benefits when enterprises comply voluntarily and the social costs when they do not:

$$U_{z2} = R_5xy + (R_5 - C_7)x(1-y) + (-C_7)(1-x)y + (-C_7)(1-x)(1-y)$$

Which simplifies to:

$$U_{z2} = R_5x - C_7(1-x)$$

The government replicator dynamics take the form:

$$H(z) = \frac{dz}{dt} = z(1-z)(U_{z1} - U_{z2})$$

The complexity of U_{z1} precludes simple closed-form simplification, but the essential insight remains that government's optimal regulatory intensity depends on the current distribution of strategies in the enterprise and treatment plant populations, creating feedback loops that can generate rich evolutionary dynamics including limit cycles and path-dependent equilibrium selection.

3.2 | Tripartite Evolutionary Game Replicator Dynamic System

Consolidating the replicator dynamic equations derived above produces the coupled nonlinear differential equation system that governs the co-evolution of strategies across all three populations:

$$\begin{aligned} \frac{dx}{dt} &= x(1-x)[(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)z] \\ \frac{dy}{dt} &= y(1-y)[(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)z] \\ \frac{dz}{dt} &= z(1-z)(U_{z1} - U_{z2}) \end{aligned}$$

This system of equations encapsulates the interdependent evolutionary trajectories of clean production adoption among pharmaceutical enterprises, technology upgrading among wastewater treatment plants, and regulatory activism among government agencies. The nonlinear coupling among the three equations implies that small perturbations in one population's strategy distribution can propagate through the system to influence equilibrium outcomes for all parties, which underscores the importance of coordinated policy design that accounts for these cross-population feedback effects. Subsequent analysis will identify the equilibrium points of this system by setting all three time derivatives equal to zero, assess the local stability properties of each equilibrium through linearization and eigenvalue analysis of the Jacobian matrix, and employ numerical

simulation to explore how variations in key parameters affect the basins of attraction and evolutionary paths connecting initial conditions to long-run outcomes.

4 | Stability Analysis of Equilibrium Points

4.1 | Jacobian Matrix and Eigenvalue Computation

To determine the local stability properties of equilibrium points within the tripartite evolutionary game system, this study employs the Jacobian matrix approach, which involves linearizing the nonlinear replicator dynamic system around each equilibrium and examining the signs of the resulting eigenvalues. At pure strategy equilibrium points where each probability variable assumes a boundary value of either 0 or 1, the Jacobian matrix becomes diagonal because cross-partial derivative terms containing products such as $x(1-x)$, $y(1-y)$, and $z(1-z)$ evaluate to zero. This diagonalization property allows the eigenvalues to be extracted directly from the diagonal entries, thereby circumventing the computational complexity associated with characteristic polynomial factorization.

Table 3 summarizes the eigenvalue expressions evaluated at each of the eight pure strategy equilibria. Evaluating the Jacobian matrix at each of the eight pure strategy equilibria yields the eigenvalue expressions summarized in the following table:

4.2 | Rationale for Excluding Mixed Strategy Equilibria

This analysis restricts attention to pure strategy equilibria for three reasons. First, pure strategy configurations possess

direct practical interpretability because they correspond to scenarios where all members of a stakeholder group adopt identical behavioral patterns, whereas mixed equilibria would require individual actors to randomize decisions according to specific probability weights. Second, mixed strategy equilibria in asymmetric multipopulational evolutionary games typically exhibit structural instability, meaning arbitrarily small parameter perturbations can eliminate such equilibria entirely. Third, interior equilibria generically function as saddle points rather than attractors in tripartite game systems, failing to satisfy asymptotic stability conditions even when they exist mathematically.

4.3 | Stability Conditions and Impossibility Results

An equilibrium achieves asymptotic stability if and only if all three eigenvalues possess strictly negative real parts. Systematic examination of the eigenvalue expressions reveals distinct stability requirements for each equilibrium, along with two configurations that cannot achieve stability under any parameter specification.

E1(0,0,0) achieves stability when $(R_2 - C_2) > (R_1 - C_1)$, $(R_4 - C_4) > (R_3 - C_3)$, and $(C_6 - F_1 - F_2) > C_7$. These conditions characterize a scenario where traditional production and conventional treatment generate higher net profits than their environmentally beneficial alternatives, while government perceives regulatory implementation costs as exceeding the sum of environmental damage costs and potential fine revenues.

E2(1,0,0) achieves stability when $(R_1 - C_1) > (R_2 - C_2)$, $(R_4 - C_4) > (R_3 - C_3)$, and $(C_6 + S_1 - F_2) > C_7$. Clean production must be intrinsically profitable, technology upgrading must

TABLE 3 | Eigenvalues of Jacobian matrix at pure strategy equilibria.

Equilibrium	λ_1	λ_2	λ_3
E1(0,0,0)	$(R_1 - C_1) - (R_2 - C_2)$	$(R_3 - C_3) - (R_4 - C_4)$	$(C_7 - C_6) + F_1 + F_2$
E2(1,0,0)	$(R_2 - C_2) - (R_1 - C_1)$	$(R_3 - C_3) - (R_4 - C_4)$	$(C_7 - C_6) + F_2 - S_1$
E3(0,1,0)	$(R_1 - C_1) - (R_2 - C_2)$	$(R_4 - C_4) - (R_3 - C_3)$	$(C_7 - C_6) + F_1 - S_2$
E4(0,0,1)	$(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)$	$(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_1 - F_2$
E5(1,1,0)	$(R_2 - C_2) - (R_1 - C_1)$	$(R_4 - C_4) - (R_3 - C_3)$	$-C_6 - S_1 - S_2$
E6(1,0,1)	$(R_2 - C_2) - (R_1 - C_1) - (S_1 + F_1 + L_1)$	$(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_2 + S_1$
E7(0,1,1)	$(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)$	$(R_4 - C_4) - (R_3 - C_3) - (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_1 + S_2$
E8(1,1,1)	$(R_2 - C_2) - (R_1 - C_1) - (S_1 + F_1 + L_1)$	$(R_4 - C_4) - (R_3 - C_3) - (S_2 + F_2 + L_2)$	$C_6 + S_1 + S_2$

remain economically unattractive, and government must find passive regulation preferable even after forgoing enterprise subsidy expenditures.

E3(0,1,0) achieves stability when $(R_2 - C_2) > (R_1 - C_1)$, $(R_3 - C_3) > (R_4 - C_4)$, and $(C_6 + S_2 - F_1) > C_7$. Technology upgrading must generate positive net returns while clean production remains unprofitable, with government preferring regulatory passivity.

E4(0,0,1) requires that $\lambda_1 < 0$ and $\lambda_2 < 0$ hold simultaneously with $\lambda_3 < 0$. The first two conditions demand that policy instruments fail to induce private sector cooperation, while the third requires government to find active regulation worthwhile. These requirements prove mutually inconsistent: if subsidies and penalties cannot shift enterprise and treatment plant behavior, the fiscal burden of regulation lacks justification. E4 therefore cannot constitute a stable equilibrium.

E5(1,1,0) has $\lambda_3 = -C_6 - S_1 - S_2$, which is automatically negative since regulatory costs and subsidy amounts are positive by definition. Stability additionally requires $(R_1 - C_1) > (R_2 - C_2)$ and $(R_3 - C_3) > (R_4 - C_4)$, meaning both clean production and technology upgrading must be intrinsically profitable without regulatory support.

E6(1,0,1) achieves stability when regulatory instruments effectively maintain enterprise compliance but fail to induce treatment plant upgrading, requiring $(R_1 - C_1) + (S_1 + F_1 + L_1) > (R_2 - C_2)$, $(R_4 - C_4) > (R_3 - C_3) + (S_2 + F_2 + L_2)$, and $(C_7 + F_2) > (C_6 + S_1)$.

E7(0,1,1) achieves stability under the converse conditions where regulatory pressure induces treatment plant upgrading but fails to shift enterprise behavior, requiring $R_2 - (C_2) > (R_1 - C_1) + (S_1 + F_1 + L_1)$, $(R_3 - C_3) + (S_2 + F_2 + L_2) > (R_4 - C_4)$, and $(C_7 + F_1) > (C_6 + S_2)$.

E8(1,1,1) has $\lambda_3 = C_6 + S_1 + S_2$, which is necessarily positive because regulatory costs and subsidy expenditures are positive quantities. When private actors already exhibit full cooperation, government faces fiscal incentives to discontinue costly regulatory programs, as environmental benefits accrue regardless of enforcement effort. E8 therefore cannot be asymptotically stable, implying that sustaining universal cooperation requires ongoing regulatory commitment to maintain enforcement credibility rather than representing a self-sustaining equilibrium configuration.

5 | Numerical Simulation and Stability Verification

5.1 | Parameter Configuration and Simulation Design

To validate the theoretical stability conditions derived from eigenvalue analysis and to visualize the evolutionary dynamics of the tripartite game system, this section presents numerical simulations for each potentially stable equilibrium configuration. The numerical simulations employ stylized

parameter values designed to satisfy the stability conditions derived from theoretical analysis and to illustrate the qualitative dynamic behaviors predicted by the model. Following established conventions in evolutionary game theory literature, parameter magnitudes are assigned to preserve economically meaningful relationships among cost, benefit, and policy instrument variables rather than to replicate specific empirical contexts. The ratio of clean production costs to traditional production costs reflects documented ranges for green manufacturing process adoption in pharmaceutical industries, while subsidy and penalty magnitudes correspond to typical policy instrument intensities observed in environmental governance programs (Haoyang et al. 2022; Liang et al. 2023). Life cycle assessment studies of advanced wastewater treatment technologies report that ozonation and granular activated carbon systems incur capital costs approximately 2–3 times higher than conventional activated sludge processes, with operational expenditures elevated by factors of 1.5–2.5 depending on target removal efficiencies and influent characteristics (Amaya-Santos et al. 2025; Rashid et al. 2023; Risch et al. 2022). These empirical ratios inform the specification of C_3 relative to C_4 in the simulation scenarios. The primary purpose of simulation analysis is to validate theoretical stability predictions and to explore how parameter variations influence convergence trajectories and equilibrium selection, providing qualitative guidance for policy design rather than quantitative forecasting of specific governance outcomes. The parameter values assigned to each scenario satisfy the corresponding stability conditions identified in Section 4.3, thereby ensuring that the simulated trajectories converge to the predicted equilibrium points. All simulations employ identical initial conditions with $x_0 = 0.2$, $y_0 = 0.3$, and $z_0 = 0.4$, representing a baseline scenario where pharmaceutical enterprises exhibit low clean production adoption, wastewater treatment plants demonstrate modest technology upgrading propensity, and government maintains moderate regulatory engagement. The simulation horizon extends to $T = 100$ time units, which proves sufficient for observing convergence behavior across all parameter configurations examined.

5.2 | Scenario Analysis for Stable Equilibria

The first simulation scenario examines conditions under which the system converges to the universal defection equilibrium E1(0,0,0), where pharmaceutical enterprises maintain traditional production, wastewater treatment plants operate conventional treatment processes, and government adopts passive regulation. The parameter configuration establishes $(R_2 - C_2) = 70 > (R_1 - C_1) = 30$, $(R_4 - C_4) = 50 > (R_3 - C_3) = 10$, and $(C_6 - F_1 - F_2) = 30 > C_7 = 20$, which collectively satisfy all three stability conditions for E1. These values reflect a market environment where traditional production yields substantially higher net profits than clean production due to elevated green technology costs, where conventional treatment remains economically attractive relative to capital-intensive advanced oxidation systems, and where government perceives regulatory implementation costs as exceeding environmental damage costs. Figure 1 displays the temporal evolution and three-dimensional phase trajectory for this scenario. The left panel reveals that all three strategy probabilities decline monotonically from their

initial values toward zero, with the enterprise clean production probability x and treatment plant upgrading probability y approaching the lower boundary within approximately 20-time units, while the government active regulation probability z exhibits slightly slower convergence. The right panel presents the corresponding stem plot in (x, y, z) phase space, illustrating how the system trajectory spirals downward from the initial condition $(0.2, 0.3, 0.4)$ toward the origin. This convergence pattern confirms that when environmentally beneficial strategies impose net costs on private actors and regulatory intervention appears fiscally unjustifiable, the evolutionary dynamics inevitably drive the system toward complete environmental governance failure.

The second scenario investigates the equilibrium configuration $E_2(1, 0, 0)$, characterized by universal enterprise clean production adoption coupled with treatment plant and government defection. Parameter values satisfy $(R_1 - C_1) = 75 > (R_2 - C_2) = 50$, $(R_4 - C_4) = 45 > (R_3 - C_3) = 0$, and $(C_6 + S_1 - F_2) = 45 > C_7 = 25$. This configuration represents market conditions where green

brand premiums and consumer environmental preferences render clean production intrinsically profitable, yet advanced treatment technologies remain economically unattractive due to high capital requirements, and government finds passive regulation optimal because enterprise compliance occurs voluntarily without regulatory expenditure. Figure 2 demonstrates that the enterprise strategy probability x increases rapidly from 0.2 toward unity while both y and z decline toward zero. The three-dimensional trajectory in the right panel traces a path from the initial condition toward the point $(1, 0, 0)$ on the positive x -axis, confirming theoretical predictions. This equilibrium possesses practical significance because it suggests that sufficiently strong market incentives for green products can induce enterprise compliance without government intervention, although the failure of treatment plant upgrading implies that end-of-pipe PPCPs removal remains inadequate even when source pollution decreases.

The third scenario examines the configuration $E_3(0, 1, 0)$, where treatment plants universally upgrade their

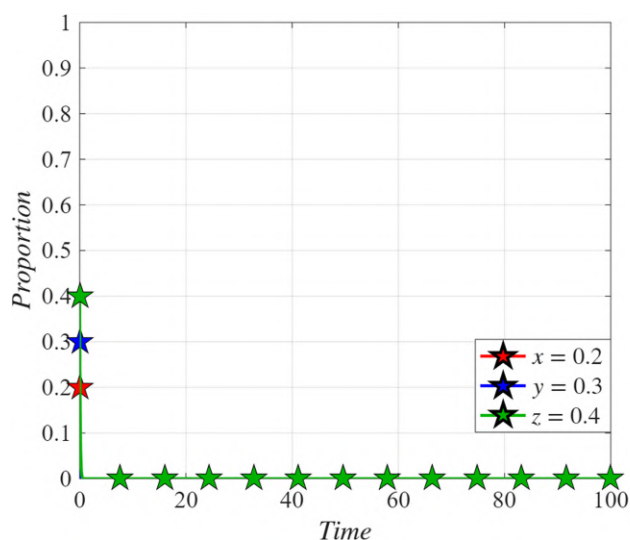


FIGURE 1 | Evolutionary dynamics converging to $E_1(0,0,0)$.

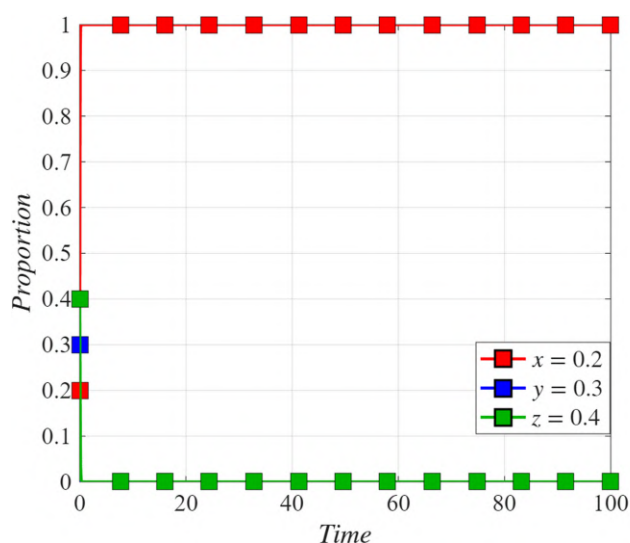
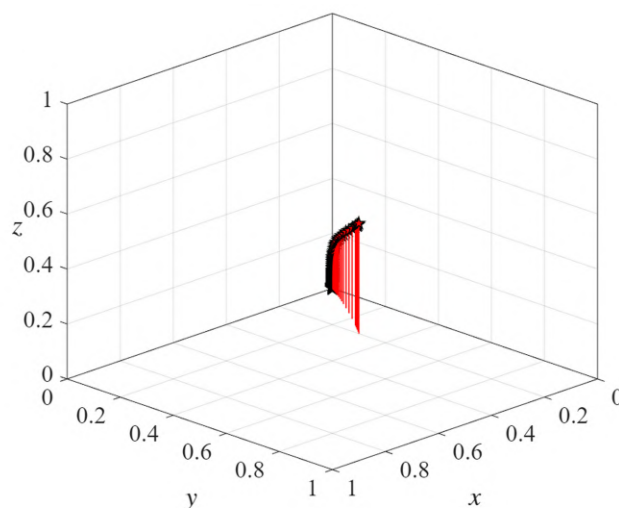
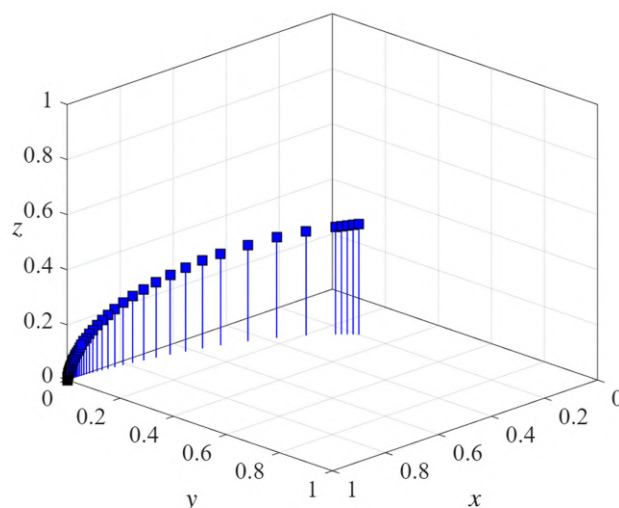


FIGURE 2 | Evolutionary dynamics converging to $E_2(1,0,0)$.



technology while enterprises and government select non-cooperative strategies. The stability conditions require $(R_2 - C_2) = 70 > (R_1 - C_1) = 20$, $(R_3 - C_3) = 55 > (R_4 - C_4) = 30$, and $(C_6 + S_2 - F_1) = 51 > C_7 = 22$. Such conditions might arise in jurisdictions where reclaimed water markets create strong revenue incentives for treatment plants to improve effluent quality, where clean production imposes prohibitive costs on pharmaceutical manufacturers, and where government lacks political will or fiscal capacity for regulatory enforcement. The simulation results in Figure 3 confirm that y converges to unity while x and z approach zero. The phase trajectory demonstrates movement toward the point $(0, 1, 0)$ along the positive y -axis. Although treatment plant upgrading can achieve partial PPCPs removal, the persistence of traditional pharmaceutical production means that influent concentrations remain elevated, imposing additional treatment costs C_3 on upgraded facilities and potentially compromising removal efficiency for the most recalcitrant compounds.

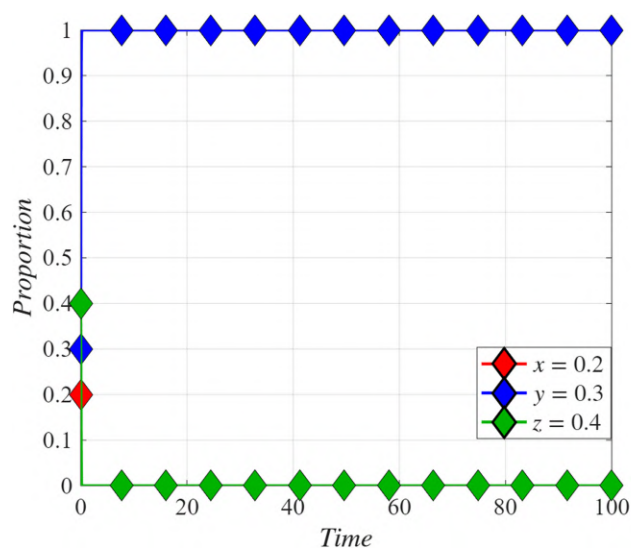


FIGURE 3 | Evolutionary dynamics converging to E3(0, 1, 0).

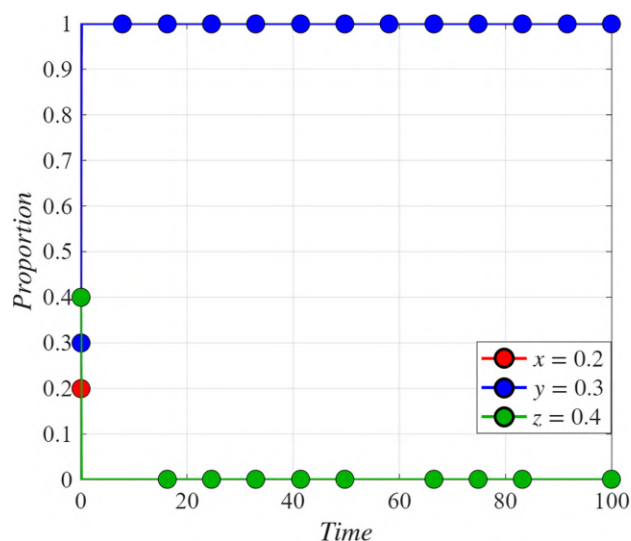
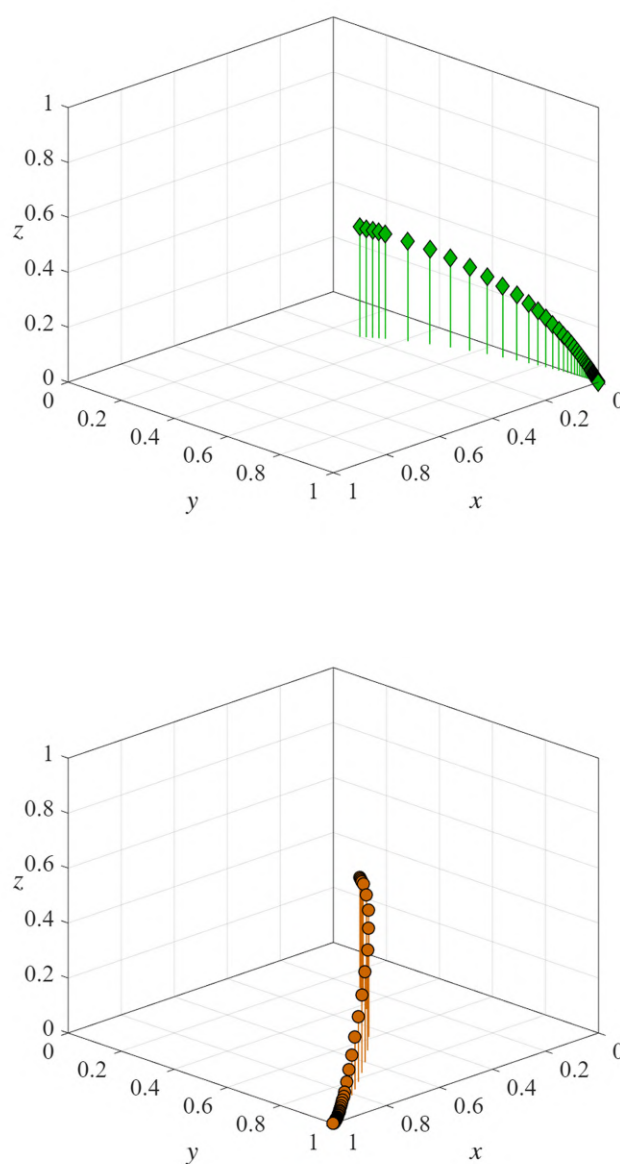


FIGURE 4 | Evolutionary dynamics converging to E5(1, 1, 0).

The fourth scenario represents arguably the most desirable stable configuration achievable within the model framework: voluntary private sector cooperation at E5(1, 1, 0), where both enterprises and treatment plants adopt environmentally beneficial strategies without government regulatory pressure. Stability requires $(R_1 - C_1) = 80 > (R_2 - C_2) = 50$ and $(R_3 - C_3) = 55 > (R_4 - C_4) = 25$, ensuring that clean production and technology upgrading are intrinsically profitable. The third eigenvalue $\lambda_3 = -C_6 - S_1 - S_2 = -70$ is automatically negative, guaranteeing government convergence toward passive regulation once private cooperation becomes established. Figure 4 illustrates the convergence dynamics toward E5. Both x and y increase rapidly toward unity as enterprises and treatment plants recognize the profitability advantages of their cooperative strategies, while z declines toward zero as government withdraws from costly regulatory activities that have become redundant. The stem plot trajectory moves toward the corner $(1, 1, 0)$ of the unit cube, representing full private sector cooperation with



government passivity. This equilibrium demonstrates that market-based mechanisms can achieve environmental objectives when technological progress or consumer preferences align private incentives with social welfare, although its stability depends critically on maintaining the profitability conditions that currently favor green strategies.

The fifth scenario explores the partial cooperation equilibrium $E6(1, 0, 1)$, where enterprises comply with clean production requirements under active government regulation, but treatment plants maintain conventional processes. The parameter configuration satisfies $(R_1 - C_1) + (S_1 + F_1 + L_1) = 125 > (R_2 - C_2) = 55$, $(R_4 - C_4) = 40 > (R_3 - C_3) + (S_2 + F_2 + L_2) = 28$, and $(C_7 + F_2) = 70 > (C_6 + S_1) = 55$. These values characterize situations where regulatory penalties and subsidies successfully induce enterprise compliance, yet technology upgrading costs remain prohibitively high for treatment plants even with government support, and environmental damages justify continued regulatory engagement. Figure 5 shows that x and z converge to unity while y approaches zero. The phase trajectory terminates at the edge $(1, 0, 1)$ of the

unit cube, representing an equilibrium where enterprises practice clean production under government oversight but treatment plants lack sufficient economic motivation to upgrade despite regulatory pressure. This configuration may reflect realistic constraints in developing economies where pharmaceutical manufacturers can feasibly adopt cleaner processes but municipal wastewater infrastructure faces severe capital constraints.

The sixth stable configuration examined is $E7(0, 1, 1)$, representing the mirror image of E 6 wherein treatment plants cooperate with government while enterprises defect. Stability conditions require $(R_2 - C_2) = 70 > (R_1 - C_1) + (S_1 + F_1 + L_1) = 43$, $(R_3 - C_3) + (S_2 + F_2 + L_2) = 125 > (R_4 - C_4) = 25$, and $(C_7 + F_1) = 65 > (C_6 + S_2) = 60$. This scenario might characterize industries where pharmaceutical production processes exhibit limited flexibility for green transformation, where treatment plant upgrading generates substantial ancillary benefits through reclaimed water utilization, and where government prioritizes end-of-pipe control over source reduction. Figure 6 confirms convergence to E7, with y and z increasing toward unity while x declines toward zero. The trajectory

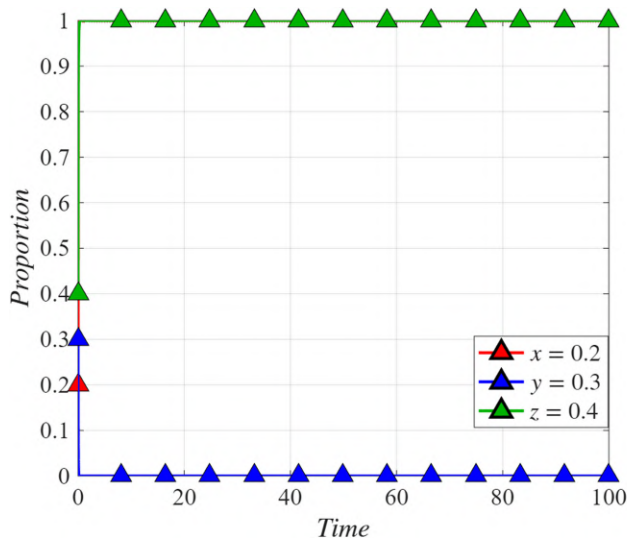


FIGURE 5 | Evolutionary dynamics converging to $E6(1, 0, 1)$.

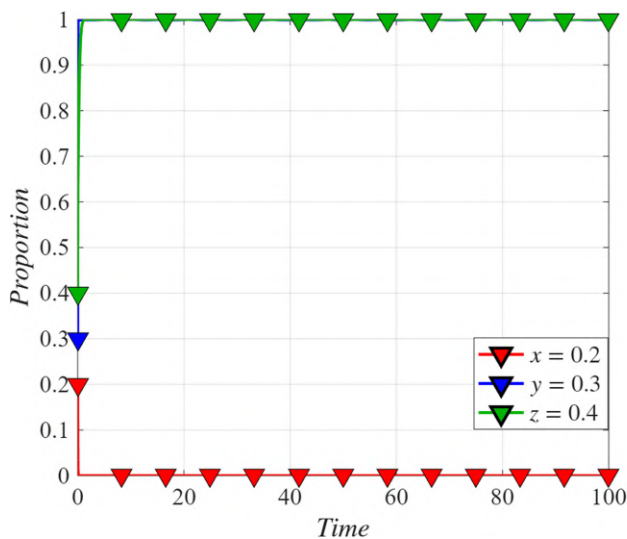
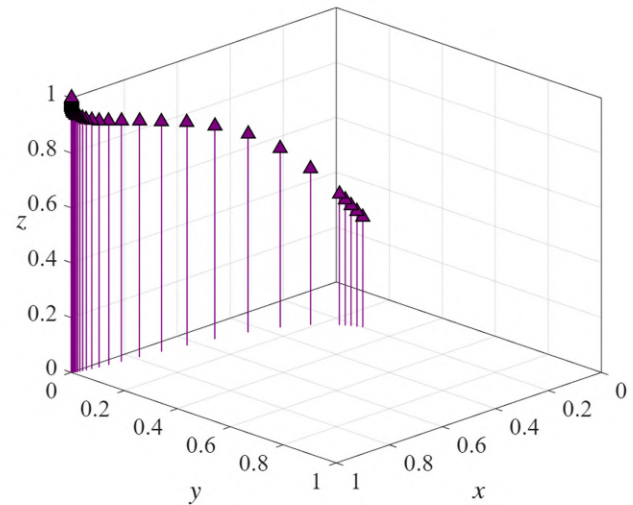
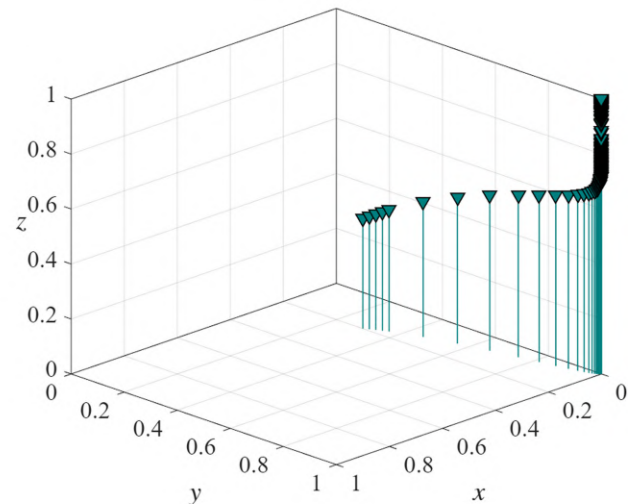


FIGURE 6 | Evolutionary dynamics converging to $E7(0, 1, 1)$.



approaches the edge (0, 1, 1) of the unit cube. Under this equilibrium, treatment plants bear the full burden of PPCPs removal from heavily contaminated influent, which elevates operational costs and may reduce removal efficiency compared to scenarios with concurrent source control.

5.3 | Sensitivity Analysis

To explore how variations in key parameters influence system dynamics and equilibrium selection, this section presents sensitivity analysis focusing on the voluntary cooperation scenario **E5(1,1,0)**, which represents an attractive policy target given its achievement of environmental objectives through market mechanisms rather than regulatory coercion. Two parameters with substantial practical and policy significance receive detailed examination: the clean production cost C_1 and the clean production revenue R_1 .

Figure 7 illustrates how different values of C_1 affect evolutionary trajectories in phase space, with C_1 varying across 0.3, 0.4 and 0.5 (scaled by factor 100). When clean production costs remain low ($C_1 = 30$), the profitability advantage $(R_1 - C_1) - (R_2 - C_2)$ increases, accelerating enterprise convergence toward clean production and hastening system arrival at E 5. As C_1 rises to 40 and subsequently to 50, the net benefit differential narrows, slowing the rate of enterprise strategy evolution and potentially threatening equilibrium stability if costs rise sufficiently to reverse the profitability ranking. The three trajectories in Figure 7 demonstrate progressively slower convergence as C_1 increases, with the highest cost scenario exhibiting the most gradual approach to the stable equilibrium. This sensitivity pattern carries important policy implications: government efforts to reduce clean production costs through technology transfer programs, green manufacturing subsidies, or research and development support can accelerate voluntary cooperation emergence and strengthen equilibrium stability against parameter fluctuations.

Figure 8 examines the impact of varying clean production revenue R_1 across values of 1.0, 1.2, and 1.4 (scaled by factor 100). Higher values of R_1 correspond to stronger green brand

premiums, more pronounced consumer preferences for environmentally responsible products, or enhanced market access for certified sustainable manufacturers. The simulation results reveal that elevated R_1 values substantially accelerate enterprise convergence toward clean production, as the widening profitability gap $(R_1 - C_1)(R_2 - C_2)$ creates stronger evolutionary pressure favoring cooperative strategy adoption. When $R_1 = 100$ (the lowest value examined), enterprise clean production remains profitable but the convergence trajectory exhibits relatively gradual dynamics. Increasing R_1 to 120 and 140 progressively steepens the trajectory slope, indicating faster strategy adjustment rates and more robust equilibrium stability. These findings suggest that policies enhancing green product market value, such as eco-labeling programs, green public procurement preferences, or environmental certification standards, can effectively promote voluntary private sector cooperation in PPCPs governance without requiring costly regulatory enforcement infrastructure.

5.4 | Sensitivity Analysis of Treatment Plant Parameters

The sensitivity analysis of treatment plant parameters employs equilibrium $E_7(0,1,1)$ as the convergence target, representing a governance configuration wherein pharmaceutical enterprises persist with traditional production methods while wastewater treatment plants universally adopt upgraded technologies under sustained government regulatory oversight. This equilibrium selection reflects realistic scenarios where enterprise clean production transition faces substantial economic barriers that render voluntary source reduction unattainable within policy-relevant time horizons, yet environmental protection objectives can nonetheless be achieved through intensified end-of-pipe treatment coupled with persistent regulatory engagement. From a stability analysis perspective, E_7 achieves asymptotic stability when the combined effect of treatment plant subsidies, penalties, and reputational losses exceeds the intrinsic cost disadvantage of technology upgrading, while simultaneously the enterprise clean production incentive remains insufficient to induce behavioral transition away from traditional production. The theoretical significance of E_7 lies in its characterization of

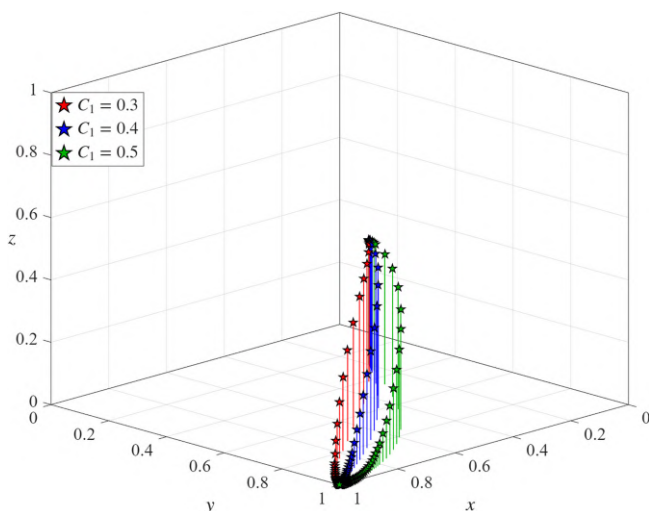


FIGURE 7 | Sensitivity analysis of clean production cost C_1 .

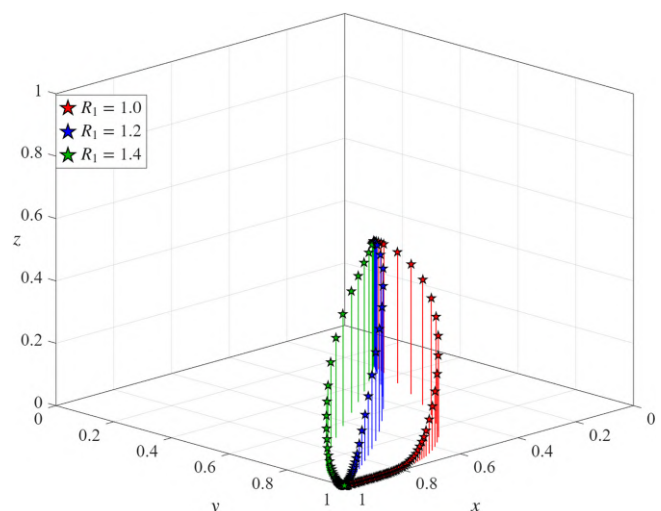


FIGURE 8 | Sensitivity analysis of clean production revenue R_1 .

governance systems where downstream remediation compensates for upstream pollution generation, constituting a second-best outcome that accepts continued enterprise emissions while ensuring adequate treatment capacity to protect receiving water quality. This equilibrium configuration carries particular policy relevance for jurisdictions where pharmaceutical industry competitiveness concerns constrain regulatory stringency on production processes, necessitating alternative governance pathways that achieve environmental objectives through treatment infrastructure investment rather than source control mandates. Examining parameter sensitivity around E_7 therefore reveals the instrument magnitudes required to ensure treatment plant cooperation even when upstream source control fails, providing critical guidance for designing resilient governance frameworks that achieve environmental protection despite incomplete stakeholder coordination.

Figure 9 depicts the evolutionary trajectories under three treatment plant subsidy scenarios converging toward equilibrium $E_7(0,1,1)$: $S_2 = 10$, $S_2 = 25$, and $S_2 = 40$, representing low, moderate, and high fiscal support intensities respectively. The three-dimensional phase space visualization tracks the joint evolution of enterprise clean production probability, treatment plant technology upgrading probability, and government active regulation probability from identical initial conditions across subsidy variations. As illustrated, all three trajectories converge toward the region characterized by low enterprise cooperation, high treatment plant upgrading, and sustained government regulation, confirming the asymptotic stability of equilibrium E_7 across the examined subsidy range. The enterprise clean production probability declines toward zero because the parameterization reflects scenarios where traditional production remains more profitable than clean alternatives even under regulatory pressure, causing the enterprise population to evolutionarily drift toward non-cooperative strategies regardless of treatment plant subsidy variations. However, the treatment plant dimension exhibits markedly different dynamics across subsidy treatments: the low subsidy scenario ($S_2 = 10$, red markers) produces a trajectory with more dispersed vertical distribution in the phase space, indicating that treatment plant upgrading probability requires extended evolutionary periods to approach unity when

fiscal incentives are modest. The high subsidy scenario ($S_2 = 40$, green markers) generates a more concentrated trajectory cluster with substantially higher point density, demonstrating that generous fiscal support accelerates treatment plant behavioral convergence even when upstream enterprises fail to cooperate. The moderate subsidy scenario ($S_2 = 25$, blue markers) displays intermediate characteristics, with trajectory density and convergence speed falling between the low and high subsidy cases, suggesting a monotonic relationship between subsidy intensity and convergence acceleration within this parameter range. The government regulation probability maintains elevated values throughout the evolutionary process across all subsidy treatments, reflecting the theoretical prediction that regulatory persistence becomes necessary when enterprise cooperation cannot be achieved, as the penalty threats and subsidy disbursements inducing treatment plant cooperation require continued government engagement to remain credible and effective.

Figure 10 examines the sensitivity of evolutionary dynamics to the penalty intensity F_2 imposed on treatment plants maintaining conventional technologies under active regulatory oversight, with trajectories again converging toward equilibrium $E_7(0,1,1)$. Three penalty magnitudes are analyzed: $F_2 = 5$, $F_2 = 20$, and $F_2 = 35$, spanning the range from nominal enforcement consequences to stringent regulatory sanctions. The trajectory patterns reveal that penalty intensity exerts substantial influence on treatment plant behavioral dynamics, complementing the subsidy effects documented in Figure 9 through a distinct incentive mechanism that increases the expected cost of non-compliance rather than reducing the cost of cooperation. The low penalty scenario ($F_2 = 5$, red markers) generates a trajectory characterized by gradual ascent in the treatment plant upgrading dimension and wider spatial dispersion, indicating that weak enforcement consequences provide insufficient deterrent pressure to accelerate technology adoption beyond baseline adjustment rates. Treatment plants facing minimal penalty exposure rationally delay costly technology investments, producing slower convergence dynamics even when government maintains active regulatory presence. The moderate penalty scenario ($F_2 = 20$, blue markers) demonstrates noticeably faster convergence with trajectory points

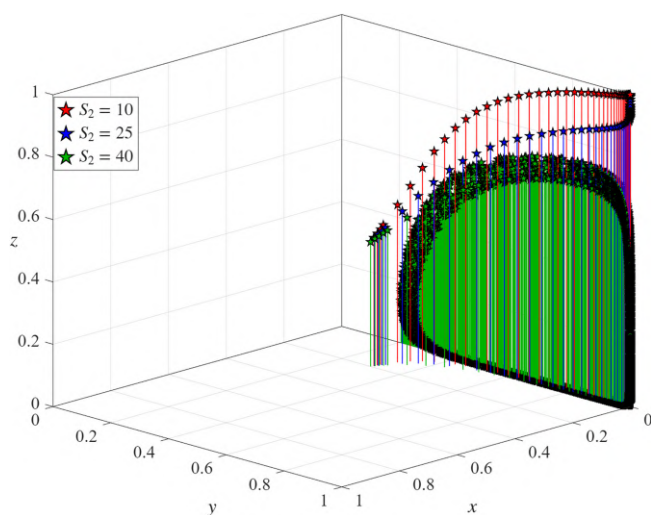


FIGURE 9 | Sensitivity analysis of treatment plant subsidy S_2 .

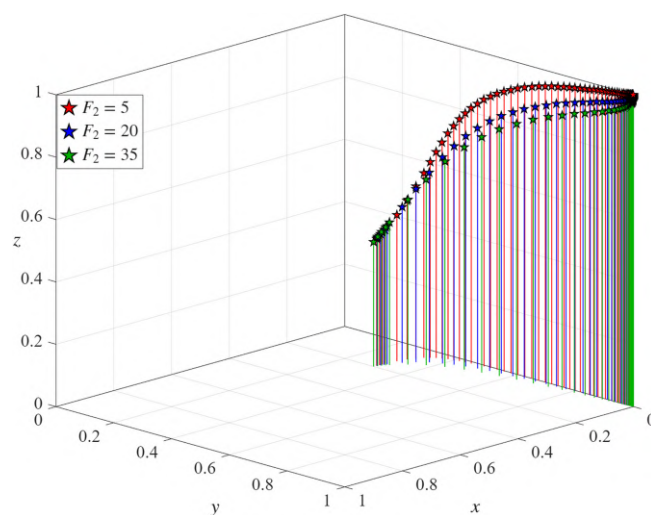


FIGURE 10 | Sensitivity analysis of treatment plant penalty F_2 .

concentrated at higher upgrading probability values, confirming that meaningful enforcement consequences can effectively modify treatment plant behavioral dynamics when calibrated at sufficient magnitudes. The high penalty scenario ($F_2=35$, green markers) produces the most rapid and concentrated convergence trajectory, with treatment plant upgrading probability reaching elevated values within fewer evolutionary periods, as the substantial penalty exposure renders conventional technology maintenance economically untenable under regulatory scrutiny. The comparative analysis across penalty treatments reveals behavioral mechanisms that complement subsidy effects documented in Figure 9: whereas subsidies reduce the absolute cost of technology upgrading by providing fiscal support for capital investment, penalties increase the expected cost of maintaining conventional treatment operations through regulatory sanctions, and both instruments shift the payoff differential in favor of technology upgrading adoption. The combined evidence from Figures 9 and 10 supports policy recommendations emphasizing coordinated deployment of both positive and negative incentive instruments, with subsidies potentially more effective during initial adoption phases when capital barriers constitute the primary obstacle, while penalties become increasingly potent for inducing behavioral change among recalcitrant operators who have not responded to fiscal incentives alone.

6 | Discussion

The tripartite evolutionary game model developed in this study reveals the strategic interaction mechanisms among pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies in PPCPs pollution control, and this theoretical contribution engages productively with empirical findings from the emerging contaminant governance literature. Model analysis demonstrates that government subsidies and penalty intensities exert decisive influence over wastewater treatment plant technology upgrading decisions, rather than upstream pharmaceutical enterprise clean production behavior alone determining such choices. Empirical evidence lends strong support to this conclusion: membrane bioreactor and other advanced treatment processes achieve markedly different antibiotic removal efficiencies across seasons, with biodegradation efficiency reaching 64.79% under elevated summer temperatures while winter microfiltration retention of antibiotics remains limited, and these observations indicate that wastewater treatment plants must weigh operational costs against actual removal performance when selecting treatment technologies (Zheng et al. 2019). Ecological risk assessments of micropollutants in wastewater treatment plant effluents have simultaneously revealed that perfluorooctanoic acid, bisphenol A, and other emerging contaminants remain excluded from current effluent regulatory frameworks, a finding that underscores the necessity of proactive government regulatory strategies (Liu et al. 2024). Monitoring data from Himalayan foothill wastewater treatment plants deserve particular attention because they document negative removal phenomena for certain PPCPs, wherein effluent concentrations exceed influent levels, and this pattern proves especially pronounced for ciprofloxacin, carbamazepine, and caffeine, thereby corroborating this model's core assumption that conventional treatment processes cannot

effectively address PPCPs contamination (Silori et al. 2025). Life cycle assessment studies of advanced treatment technologies have further demonstrated that ozone oxidation and granular activated carbon adsorption can substantially improve PPCPs removal rates, yet the indirect environmental impacts arising from energy consumption and resource demands may exceed direct water quality improvement benefits, and this evidence provides a scientific basis for governments to formulate differentiated subsidy policies (Risch et al. 2022).

Stability analysis of the model reveals another important finding: the complete cooperation equilibrium $E_8(1,1,1)$ cannot theoretically achieve asymptotic stability, which implies that even after private sector actors have fully adopted environmentally friendly strategies, governments still face fiscal incentives to withdraw from regulatory engagement. This theoretical finding warrants contextualization within broader governance realities. The model specification captures fiscal incentives that induce regulatory withdrawal when compliance objectives have been achieved, a phenomenon documented in environmental policy implementation where agencies reallocate enforcement resources toward emerging priorities once established programs demonstrate success. However, real-world regulatory systems operate under additional constraints that the stylized model does not fully incorporate. Legal mandates embedded in environmental legislation create binding obligations for continued monitoring and standard enforcement regardless of current compliance rates, insulating regulatory agencies from purely fiscal optimization pressures. Bureaucratic reputation concerns motivate sustained regulatory presence because agency credibility and future budget allocations depend on demonstrated ongoing engagement rather than historical accomplishments. Political accountability mechanisms, including legislative oversight and public expectations shaped by environmental advocacy organizations, constrain government discretion to reduce regulatory intensity even when immediate fiscal calculations might favor withdrawal. These institutional factors suggest that E_8 instability represents a theoretical limiting case that identifies governance vulnerabilities requiring institutional safeguards, rather than an inevitable empirical prediction. Policy designs incorporating statutory monitoring requirements, earmarked enforcement funding, and transparent compliance reporting can mitigate the regulatory withdrawal incentives that the model identifies, thereby approximating stable complete cooperation outcomes in practice even if pure asymptotic stability remains theoretically unattainable. This theoretical prediction aligns closely with stakeholder survey research conclusions: experts drawn from government, industry, and research institutions generally identify advanced wastewater treatment technologies, healthcare professional education to reduce overprescription, pharmaceutical return programs coupled with public education, and mandatory secondary treatment capacity requirements for all municipal authorities as the most effective management strategies, and the successful implementation of these strategies depends on sustained government regulatory commitment rather than one-time policy interventions (Doerr-MacEwen and Haight 2006). Regarding public participation dimensions, survey research among residents in Xiamen, China has found significant willingness to pay for PPCPs pollution control policies, with income level, environmental concern, and subjective financial health status serving as important predictors of payment

willingness, and these findings suggest that governments can draw upon a foundation of social support when implementing active regulatory strategies (Wang et al. 2015). Cross-national comparative research on pharmaceutical regulation in reclaimed water has noted that despite ongoing monitoring efforts in multiple developed countries, pharmaceutical contaminants have generally not been incorporated into water quality regulatory systems, and Israel, the country with the highest reclaimed water utilization rate globally, similarly lacks systematic regulation of pharmaceutical residues in its wastewater, a situation that effectively constitutes an epidemiological experiment conducted on citizens and the environment (Miarov et al. 2020). Urban wastewater monitoring data from China during the COVID-19 pandemic further confirm this study's theoretical inference regarding the impacts of public health emergencies on PPCPs discharge: following policy adjustments, ibuprofen and paracetamol concentrations surged to 67-fold and 311-fold pre-pandemic levels respectively; direct discharge of high pharmaceutical-load wastewater can pose significant ecological risks to aquatic organisms, and establishing comprehensive wastewater treatment systems proves particularly critical for remote and underdeveloped regions (Yu et al. 2025).

The policy instruments identified as effective levers for promoting cooperative behaviors nonetheless carry implementation risks and may trigger unintended stakeholder conflicts that warrant explicit consideration. Differentiated subsidy programs targeting treatment plant technology upgrading face administrative challenges in establishing credible eligibility criteria, monitoring actual technology deployment, and preventing strategic gaming wherein facilities claim subsidies without achieving intended pollution reduction outcomes. Fiscal sustainability concerns arise when subsidy magnitudes sufficient to induce widespread technology adoption impose significant burdens on government budgets, potentially competing with other environmental protection priorities or triggering taxpayer resistance that undermines political support for continued program funding. Penalty enforcement mechanisms require robust monitoring capacity and political willingness to impose consequences on non-compliant actors, conditions that may prove difficult to sustain when regulated enterprises possess significant economic or political influence. The pharmaceutical industry in particular may mobilize lobbying resources to resist stringent clean production requirements, arguing that elevated compliance costs threaten domestic production competitiveness and employment, thereby creating political economy dynamics that constrain regulatory ambition. Treatment plants facing technology upgrading mandates without adequate fiscal support may experience financial distress that compromises service quality for conventional pollutant removal, potentially creating unintended trade-offs between PPCPs control and traditional wastewater treatment performance. Small and medium-sized enterprises with limited capital access may disproportionately bear clean production transition burdens, raising equity concerns that complicate policy design and implementation. These challenges suggest that effective PPCPs governance requires not only well-calibrated economic instruments but also attention to stakeholder consultation processes, phased implementation schedules, and complementary support mechanisms that facilitate behavioral transitions while managing distributional consequences.

7 | Conclusion

This study constructs a tripartite evolutionary game model encompassing pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies, thereby systematically revealing the internal mechanisms and evolutionary patterns of multi-stakeholder strategic interactions in PPCPs pollution control. Research findings demonstrate that when clean production and technology upgrading possess inherent profitability advantages, market mechanisms can drive private sector actors to spontaneously converge toward the environmentally friendly equilibrium $E5(1,1,0)$ in the absence of government regulatory intervention, and under such circumstances governments may appropriately withdraw from direct regulation to conserve administrative resources. The complete cooperation equilibrium $E8(1,1,1)$, however, cannot achieve asymptotic stability due to the presence of a positive eigenvalue, which means that once private sector actors have fully adopted cooperative strategies, governments will face fiscal incentives to reduce regulatory investment, thereby threatening the long-term maintenance of cooperative configurations. The model further reveals that government subsidies and penalty intensities primarily govern wastewater treatment plant technology upgrading decisions, rather than upstream enterprise clean production behavior directly determining such choices, and this finding provides a theoretical basis for precisely targeted policy interventions. The mathematical finding that upstream enterprise production choices do not directly alter treatment plant upgrading incentives through cost transmission channels reflects the substantive economic reality that both technology options face equivalent pollution burden increases, revealing that market-based coordination cannot spontaneously emerge without regulatory intervention. Sensitivity analysis confirms that reductions in clean production costs and increases in green product revenues can significantly accelerate system convergence toward ideal equilibria, indicating that technological progress and market incentives serve as critical leverage points in PPCPs governance. Additional sensitivity analysis of treatment plant subsidy and penalty parameters demonstrates that both positive fiscal incentives and negative regulatory sanctions effectively promote technology adoption, with policy designers possessing flexibility to select instrument combinations based on fiscal constraints and political feasibility considerations.

Building upon these theoretical findings, this study advances the following policy recommendations. Governments should first establish differentiated fiscal incentive systems that provide targeted subsidies to wastewater treatment plants adopting advanced treatment technologies such as ozone oxidation, granular activated carbon adsorption, or membrane separation, while simultaneously linking subsidy magnitudes to actual PPCPs removal efficiencies, because this approach can overcome the inhibitory effects of high capital investment requirements on technology upgrading. For pharmaceutical enterprises, this study recommends enhancing the economic returns from clean production through market-based instruments including green certification labels and preferential government procurement of environmentally friendly products, rather than relying solely on end-of-pipe penalties, since this approach can stimulate intrinsic enterprise emission reduction motivation while reducing regulatory costs. Given the structural instability of complete

cooperation equilibrium, governments must maintain necessary monitoring capacity and enforcement deterrence even after widespread private sector compliance has been achieved, and adopting a low-cost regulatory model that combines random inspection with information disclosure can prevent system regression toward non-cooperative states triggered by regulatory relaxation. Finally, considering that public health emergencies may cause dramatic fluctuations in specific pharmaceutical discharges, this study recommends reserving emergency treatment redundancy in wastewater treatment facility planning and incorporating wastewater pharmaceutical monitoring into routine public health surveillance systems, as these measures can enhance the resilience and adaptive capacity of water environment governance systems.

Several limitations of this study merit acknowledgment and suggest directions for future research. The binary strategy specification, while enabling tractable mathematical analysis, simplifies the continuous spectrum of clean production intensity and technology upgrading options available to real-world decision-makers; future extensions could incorporate multi-level strategy spaces or continuous investment variables that better capture the incremental nature of environmental improvement decisions. The tripartite model structure excludes potentially important actors including pharmaceutical consumers whose purchasing behaviors shape market incentives for green products, health-care institutions whose prescribing practices influence total pharmaceutical consumption and environmental loading, and civil society organizations whose advocacy activities affect regulatory stringency and public awareness. Parameter calibration relies on stylized approximations drawn from diverse empirical sources rather than systematic data collection targeting the specific PPCPs governance context, limiting the precision of quantitative predictions regarding convergence speeds and equilibrium selection probabilities. The static payoff specification assumes fixed cost and benefit parameters throughout the evolutionary process, whereas real-world technological learning curves, market development trajectories, and regulatory capacity building would introduce parameter dynamics that could substantially alter long-run system behavior. The model treats each stakeholder group as homogeneous populations, neglecting heterogeneity in firm size, technology vintage, regulatory jurisdiction, and other factors that generate differential responses to policy instruments within each category. These limitations notwithstanding, the analytical framework developed herein provides valuable qualitative insights into the strategic interdependencies characterizing PPCPs governance systems and offers a foundation for more detailed empirical and computational investigations.

Author Contributions

Xue Lei: conceptualization, writing – original draft, formal analysis, methodology. **Yuying Luo:** writing – review and editing, methodology. **Fengmei Liu:** formal analysis, writing – review and editing. **Xueguo Xu:** conceptualization, writing – review and editing, project administration, supervision.

Funding

The authors have nothing to report.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The numerical simulation results, including all figures presented in this manuscript, were generated using MATLAB R2021a based on the theoretical model parameters specified in Tables 1 and 2. The simulation code and parameter configurations used to generate Figures 1–10 can be made available upon request to facilitate reproduction of results.

References

- Amaya-Santos, G., N. Boelee, A. Paulillo, and P. Lettieri. 2025. “Life Cycle Assessment and Life Cycle Costing of Full-Scale Ozonation for Micropollutants Removal From Wastewater. Case Study in the Netherlands.” *Science of the Total Environment* 961: 178259. <https://doi.org/10.1016/j.scitotenv.2024.178259>.
- Castiglioni, S., E. Davoli, F. Riva, et al. 2018. “Mass Balance of Emerging Contaminants in the Water Cycle of a Highly Urbanized and Industrialized Area of Italy.” *Water Research* 131: 287–298.
- David, V., S. Joachim, A. Catteau, et al. 2020. “Effects of Chronic Exposure to a Pharmaceutical Mixture on the Three-Spined Stickleback (*Gasterosteus aculeatus*) Population Dynamics in Lotic Mesocosms.” *Aquatic Toxicology* 224: 105499.
- Doerr-MacEwen, N. A., and M. E. Haight. 2006. “Expert Stakeholders’ Views on the Management of Human Pharmaceuticals in the Environment.” *Environmental Management* 38, no. 5: 853–866.
- El Hammoudani, Y., F. Dimane, K. Haboubi, et al. 2024. “Micropollutants in Wastewater Treatment Plants: A Bibliometric-Bibliographic Study.” *Desalination and Water Treatment* 317: 100190.
- Fairbairn, D. J., M. E. Karpuzcu, W. A. Arnold, et al. 2016. “Sources and Transport of Contaminants of Emerging Concern: A Two-Year Study of Occurrence and Spatiotemporal Variation in a Mixed Land Use Watershed.” *Science of the Total Environment* 551: 605–613.
- Fernandez-Lopez, C., M. G. García, and J. M. Sánchez-Lozano. 2021. “Analysis of WWTPs Technologies Based on the Removal Efficiency of Pharmaceutical Activated Compounds for Water Reuse Purposes. A Fuzzy Multi-Criteria Decision Making Approach.” *Journal of Water Process Engineering* 42: 102098.
- Fontes, M. K., P. K. Gusso-Choueri, L. A. Maranhão, et al. 2018. “A Tiered Approach to Assess Effects of Diclofenac on the Brown Mussel *Perna perna*: A Contribution to Characterize the Hazard.” *Water Research* 132: 361–370.
- Gardner, M., V. Jones, S. Comber, et al. 2013. “Performance of UK Wastewater Treatment Works With Respect to Trace Contaminants.” *Science of the Total Environment* 456: 359–369.
- Haoyang, W., G. Lei, and J. Ying. 2022. “The Predicament of Clean Energy Technology Promotion in China in the Carbon Neutrality Context: Lessons From China’s Environmental Regulation Policies From the Perspective of the Evolutionary Game Theory.” *Energy Reports* 8: 4706–4723. <https://doi.org/10.1016/j.egy.2022.03.142>.
- Hayden, K. R., M. Jones, K. R. Elkin, et al. 2022. “Impacts of the COVID-19 Pandemic on Pharmaceuticals in Wastewater Treated for

Beneficial Reuse: Two Case Studies in Central Pennsylvania.” *Journal of Environmental Quality* 51, no. 5: 1066–1082. <https://doi.org/10.1002/jeq2.20398>.

Kallenborn, R., E. Brorström-Lundén, L.-O. Reiersen, and S. Wilson. 2018. “Pharmaceuticals and Personal Care Products (PPCPs) in Arctic Environments: Indicator Contaminants for Assessing Local and Remote Anthropogenic Sources in a Pristine Ecosystem in Change.” *Environmental Science and Pollution Research* 25, no. 33: 33001–33013.

Letsinger, S., and P. Kay. 2019. “Comparison of Prioritisation Schemes for Human Pharmaceuticals in the Aquatic Environment.” *Environmental Science and Pollution Research* 26, no. 4: 3479–3491.

Liang, J., M. Zhang, Z. Yin, et al. 2023. “Tripartite Evolutionary Game Analysis and Simulation Research on Zero-Carbon Production Supervision of Marine Ranching Against a Carbon-Neutral Background.” *Frontiers in Ecology and Evolution* 11: 1–12. <https://doi.org/10.3389/fevo.2023.1119048>.

Liu, J., T. Ouyang, G. Lu, et al. 2024. “Ecosystem Risk-Based Prioritization of Micropollutants in Wastewater Treatment Plant Effluents Across China.” *Water Research* 263: 122168.

Miarov, O., A. Tal, and D. Avisar. 2020. “A Critical Evaluation of Comparative Regulatory Strategies for Monitoring Pharmaceuticals in Recycled Wastewater.” *Journal of Environmental Management* 254: 109794.

Niemuth, N. J., R. Jordan, J. Crago, C. Blanksma, R. Johnson, and R. D. Klaper. 2015. “Metformin Exposure at Environmentally Relevant Concentrations Causes Potential Endocrine Disruption in Adult Male Fish.” *Environmental Toxicology and Chemistry* 34, no. 2: 291–296.

Rashid, S. S., S. N. Harun, M. M. Hanafiah, K. K. Razman, Y.-Q. Liu, and D. A. Tholibon. 2023. “Life Cycle Assessment and Its Application in Wastewater Treatment: A Brief Overview.” *PRO* 11, no. 1: 208. <https://www.mdpi.com/2227-9717/11/1/208>.

Risch, E., L. Jaumaux, C. Maesele, and J.-M. Choubert. 2022. “Comparative Life Cycle Assessment of Two Advanced Treatment Steps for Wastewater Micropollutants: How to Determine Whole-System Environmental Benefits?” *Science of the Total Environment* 805: 150300.

Silori, R., D. M. Mahapatra, S. M. Tauseef, et al. 2025. “Seasonality in Negative Removal of Pharmaceutical and Personal Care Products at the Himalayan Foothills Wastewater Treatment Plants.” *Journal of Environmental Chemical Engineering* 13, no. 3: 116803.

Tarpani, R. R. Z., and A. Azapagic. 2018. “A Methodology for Estimating Concentrations of Pharmaceuticals and Personal Care Products (PPCPs) in Wastewater Treatment Plants and in Freshwaters.” *Science of the Total Environment* 622: 1417–1430.

Wang, X., P. Howley, A. B. Boxall, and M. A. Rudd. 2015. “Behavior, Preferences, and Willingness to Pay for Measures Aimed at Preventing Pollution by Pharmaceuticals and Personal Care Products in China.” *Integrated Environmental Assessment and Management* 12, no. 4: 793–800.

Yu, C., J. Zhang, X. Fu, et al. 2025. “Wastewater-Based Monitoring of Antipyretics Use During COVID-19 Outbreak in China and Its Associated Ecological Risks.” *Environmental Research* 267: 120680.

Zheng, W., X. Wen, B. Zhang, and Y. Qiu. 2019. “Selective Effect and Elimination of Antibiotics in Membrane Bioreactor of Urban Wastewater Treatment Plant.” *Science of the Total Environment* 646: 1293–1303.



The impact of mobile social media on knowledge sharing among vocational school teachers: A social cognitive career perspective

Journal of Information Science
1–19

© The Author(s) 2024

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/01655515241293754

journals.sagepub.com/home/jis



Haojun Li

Study of Education, Zhejiang University of Technology, China

Jun Xu

Faculty of Education, East China Normal University, China

Yuying Luo

Study of Management, Shanghai University, China

Abstract

In the rapidly evolving field of information systems, the role of mobile-based social media as a platform for knowledge sharing among vocational schoolteachers presents both opportunities and challenges. This study addresses a critical gap in the understanding of how psychological factors (such as self-efficacy) and contextual factors (such as trust environments) influence knowledge-sharing behaviours in information systems. This study includes 332 vocational schoolteachers and employs structural equation modelling to examine how psychological and contextual factors enhance or inhibit sharing intentions. The results revealed that psychological factors significantly impact sharing intentions, whereas contextual factors bolster self-efficacy and behaviours. The findings offer valuable insights into the optimisation of information systems environments to facilitate effective knowledge sharing, thereby contributing to enhanced collaborative practices and technology adoption in educational settings. This research underscores the need to integrate social cognitive career theory into information systems to better understand the complex dynamics of knowledge sharing on mobile platforms.

Keywords

Knowledge sharing; mobile social media; social cognitive career theory; empirical research; structural equation model

1. Introduction

With the development of a knowledge society, teachers' professional development and knowledge updating have become important issues in the field of education. As disseminators of vocational skills and knowledge, vocational schoolteachers need to constantly improve their subject knowledge, teaching knowledge and innovation ability to adapt to the needs of society and students [1,2]. Knowledge sharing is an effective method for teaching professional development and knowledge updating that can promote communication, cooperation and learning among teachers [3,4]. However, teachers' knowledge sharing also faces many challenges, such as time, space, resource and culture constraints [5,6].

As a new knowledge sharing platform, mobile social media (MSM) provides teachers with convenient, flexible and diverse ways to communicate and create knowledge [7]. During the COVID-19 pandemic, MSM became an important tool for teachers to acquire and disseminate knowledge and has become a mainstream method of social teachers' knowledge sharing [8,9]. However, the knowledge sharing environment of MSM is not perfect, and there are many factors that

Corresponding author:

Jun Xu, Faculty of Education, East China Normal University, Shanghai 200062, China.

Email: xujun1569@gmail.com

affect teachers' knowledge sharing behaviour (KSB) intentions and effects, such as social, psychological, cultural and technical factors [10]. Therefore, this paper aims to explore how to optimise the knowledge sharing environment provided by MSM and stimulate vocational schoolteachers to share knowledge on mobile social media platforms, which is a widely discussed topic in the field of vocational education and knowledge sharing research and is the focus of this paper.

Although MSM provide teachers with a new knowledge sharing platform, there is still a lack of research on the impact of MSM on vocational school teachers' knowledge sharing intentions (KSIs). This study is based on the belief that there are two main problems in this area of research:

(1) The particularities and needs of vocational schoolteachers are ignored. Vocational education has many differences from other forms of education, such as teaching objectives, student characteristics, and curriculum settings. The goals of this education are to facilitate students' transition from school to work, cultivate their active participation, self-directed learning and critical skills, and provide the talents needed for social and economic development [11,12]. Therefore, vocational education teachers need to constantly update and expand their subject, teaching and experience knowledge that is consistent with social requirements [13]. However, previous studies have focused mainly on the KSB of higher education or all teacher groups and lack attention and empirical research on vocational education teachers.

(2) The diversity and complexity of mobile social media environmental factors are ignored. MSM provide teachers with more time, space and channels to communicate and share knowledge [14]. However, not all MSM are conducive to knowledge sharing but are influenced by multiple dimensions of environmental factors, such as social, psychological, cultural and technical factors [15]. These environmental factors not only directly affect KSB but also indirectly affect this behaviour by affecting teachers' subjective efficacy and intentions [16,17]. Therefore, it is necessary to use a more comprehensive and integrated model structure to consider the interaction between the external environment and internal factors and reveal the specific mechanism involved.

Building on previous research gaps, this study aims to delve deeper into the KSBs of teachers in vocational colleges and universities in the context of mobile social media, focusing on how external environmental factors and intrinsic motivation work together to shape teachers' KSBs. This article constructs a structural equation model based on social cognitive career theory (SCCT), which provides a framework for understanding how individuals form attitudes and behavioural intentions towards knowledge sharing through social cognitive processes under the influence of a specific environmental factor (mobile social media platform application), using the external environmental factors of the mobile social media environment and the intrinsic motivation of the individual's psychological factors as the key variables in the study [18].

Numerous studies based on SCCT provide rich references for our path design. For example, Huang and Hsieh's [19] study on career exploration intention used self-efficacy and outcome expectations to directly point to the path setting of behavioural intention, revealing the mediating effect of psychological factors on entrepreneurial intention in the context of socioeconomic status. Aure et al. [20] regarded self-efficacy and outcome expectations as individual cognitive factors and personality traits and perceived social support as contextual factors, analysing the complex role of individual and contextual influences on career exploration intentions. Furthermore, this study draws on the meta-analytical work of Brown and colleagues [21,22] to synthesise the three basic models of SCCT, uses the performance model to analyse KSBs as a potential driver of educational outcomes, and draws on the choice model to explore how individuals make career-related decisions based on internal beliefs and external conditions. Using Lent et al.'s [23] perspective on how situational factors such as support and barriers indirectly influence individuals' goal setting and behavioural choices during career development through their self-efficacy, this study aims to elucidate how knowledge sharing environments can influence teachers' willingness and behaviours to share knowledge by shaping their self-efficacy and, consequently, their KSIs.

In view of the specific challenges faced by the teacher community in using mobile social media for knowledge sharing stemming from their educational needs and the unique role of vocational college teachers in vocational skills education and knowledge dissemination, teachers in Chinese vocational colleges and universities were specifically selected for this study. By using Mplus 8.3 software to analyse the data, this study aimed to explore the factors influencing KSIs and behaviours in the mobile social media environment.

2. Theoretical basis and related research

2.1. Theoretical basis

In this study, SCCT is used as the theoretical framework to explore the KSI of vocational schoolteachers in the MSM environment. SCCT is a comprehensive vocational psychology theory that emphasises the mutual influence of individual

cognition, behaviour and the environment [24]. SCCT originates from Bandura's social cognitive theory and inherits its research on core concepts such as individual self-efficacy, outcome expectations and goal setting [25].

According to SCCT, individual self-efficacy and outcome expectations are divided into different dimensions and sources, which can better explain the behavioural differences of individuals in different situations [26]. SCCT can also be used to overcome the limitations of traditional theories that separate psychological and social factors and link individual KSB intentions with the external environment [27]. Therefore, SCCT is suitable for analysing the direct and indirect effects of environmental factors of MSM on KSI and behaviour, as well as their interaction with individual cognitive processes and personal characteristics. Previous studies have often used SCCT to explore the factors that affect knowledge sharing, such as self-efficacy, outcome expectations, and community atmosphere [28], but systematic research on the KSI of vocational schoolteachers in MSM environments is lacking, and the influence mechanism of environmental factors and psychological factors has not been examined in depth.

2.2. Vocational school teachers and knowledge sharing

Knowledge sharing refers to the process of the communication and dissemination of knowledge between individuals or groups, including the process of knowledge provision and reception [29]. For teacher groups, knowledge sharing is a beneficial learning activity that can promote teachers' personal and collective professional development, thereby improving teaching effectiveness, student learning outcomes and school reputation [30]. Compared with teachers in higher education institutions and other schools, vocational schoolteachers are more willing and motivated to share knowledge, which is influenced by two main dimensions: individual knowledge pursuit and organisational incentive pursuit.

2.2.1. Individual knowledge pursuit. Individual knowledge pursuit refers to the outcome expectation of vocational education teachers for knowledge acquisition before they engage in KSB. This expectation is influenced by vocational education teachers' own professional development needs and education quality improvement goals.

First, vocational education teachers' own professional development needs motivate them to constantly update their teaching and professional skills. Antera [31] noted that vocational education teachers have the characteristics of dual professionalism, which requires them to master professional knowledge in teaching, industry and experience, as well as to cope with the challenges of knowledge updating and innovation. Through interviews with Swedish vocational schoolteachers, Andersson and Köpsén [32] found that they need to learn new knowledge and skills constantly to improve their teaching quality and adapt to market demand, which motivates them to participate in knowledge sharing activities more actively. Therefore, vocational education teachers expect to gain more professional knowledge and skills through knowledge sharing, thus enhancing their professional competitiveness and satisfaction.

Second, vocational education teachers' education quality improvement goals inspire them to constantly enrich both their practical and innovation knowledge. Runhaar and Sanders [33] argued that knowledge sharing is a key factor for improving the quality of vocational education because it can promote mutual learning and reflection among teachers and enhance their adaptability to student needs and industry changes. Cornelius and Stevenson [34] evaluated the data of vocational school teacher trainees in the International Online Collaboration Project (COLIGE) and found that knowledge sharing contributes to vocational education innovation, enabling learners and educators to cross boundaries between different practices, cultures and backgrounds and promote their professional identity and ability reflection and transformation. Therefore, vocational education teachers expect to gain more practical experience and innovative ideas through knowledge sharing, which will improve their teaching effectiveness and student satisfaction.

2.2.2. Organisational incentive pursuits. Organisational incentive pursuit refers to the expectation of vocational education teachers for the support and rewards provided by the organisation before they engage in KSB. This expectation is influenced by two factors: one is the technical change speed of the industry in which vocational education teachers are located, and the other is the degree of education reform and innovation in the field of vocational education.

First, technical change in the industry in which vocational education teachers are located motivates them to acquire new knowledge through knowledge sharing to keep up with the development of their professional fields [35]. Livingstone and Guile [36] argued that vocational education teachers are in the knowledge economy and need to constantly learn new knowledge and skills to adapt to work requirements and student needs. Therefore, vocational education teachers expect to obtain professional development opportunities, technical update training, learning resources and other support provided by the organisation through knowledge sharing.

Second, the degree of education reform and innovation in the field of vocational education affects the necessity and urgency of knowledge sharing for vocational education teachers. Wijnia et al. [37] noted that with changes in social

development and student needs, there are often reforms and innovations in teaching modes, curriculum settings, evaluation methods and other aspects in the field of vocational education. These reforms or innovations require vocational education teachers to acquire the latest knowledge achievements, improve their teaching and research abilities, promote the transformation of personal knowledge into team public knowledge, and enrich the practical knowledge base of teacher groups in a timely manner [38]. Therefore, vocational education teachers will expect to obtain research projects, experience sharing, academic exchange and other rewards provided by the organisation through knowledge sharing.

2.3. Knowledge sharing environment of mobile Social Media

Knowledge sharing among MSM has the characteristics of convenience, timeliness, richness and personalisation, which provide users with an environment that is conducive to knowledge communication and sharing [39]. Environmental factors refer to external conditions that affect individual KSB. They can promote or inhibit individuals' self-efficacy and outcome expectations [23]. Different studies divide the dimensions of the knowledge sharing environment from different perspectives, such as the physical environment, social environment, cultural environment, psychological environment, and technical environment [40]. In this paper, 'trust atmosphere' and 'technical adaptation' are selected as the environmental factors of MSM, and the analysis is conducted from multiple perspectives: the atmosphere of trust represents the influence of the environment on individual KSIs from the perspectives of individuals and teams, and technical adaptation represents the influence of the environment on individual KSIs from the perspective of technology.

2.3.1. Social trust atmosphere. The social trust atmosphere (STM) refers to the atmosphere of mutual trust and reliance that is established among MSM users. It can reduce the risk and cost of knowledge sharing and enhance knowledge sharers' confidence and sense of belonging [15]. Lee et al. [41] concluded that the recognition of knowledge recipients in virtual academic social networks will have a positive impact on STM, which will motivate users to share knowledge. Based on the theory of planned behaviour, Feng et al. [42] constructed a model of user KSB in question-and-answer communities and argued that an STM would affect individuals' knowledge sharing efficacy and value judgement. Chow and Chan [43] constructed a structural model of Hong Kong managers' KSI and concluded that this social trust had no significant direct impact on sharing intention, which also confirmed the conjecture proposed in this study that the environment affects KSB through internal factors.

2.3.2. Social technology adaptation. Social technology adaptation (STA) refers to users' cognition and evaluation of the technology functions and characteristics of MSM. This adaptation can affect users' attitudes and willingness to use technology and subsequently affect their KSB [39]. Based on the grounded theory method, Ming et al. [44] conducted interviews, questionnaires and structural equation modelling and concluded that individual differences are the internal driving factors for user behaviour intention, while system characteristics and interface characteristics are the external driving factors for user behaviour intention. Chai and Kim [45] adopted a method that included social technology system theory, and by using a structural model, they concluded that social network website service providers and technical security measures have a positive impact on KSB. Ahmad et al. [46] used a partial least squares structural equation model to conclude that technical availability has a positive impact on knowledge sharing with students. These studies support adding the technical adaptability of MSM as a technical link to the present model.

3. Research hypotheses and model construction

3.1. Research hypotheses

Based on the core concepts of SCCT, a theoretical model with three levels – psychological, environmental and behavioural – is constructed in this study. The psychological level includes self-efficacy (SE, a person's belief in his or her ability to succeed in specific situations), knowledge acquisition expectations (KAE, the expectation of vocational education teachers to gain professional knowledge and skills through knowledge sharing), and organisational reward expectations (ORE, the anticipation of support and rewards provided by the organisation), which reflect the individual's cognitive evaluation of his or her own ability and outcomes. The environmental level includes the STM and STA, which reflect the characteristics and influence of MSM. The behavioural level includes KSI and KSB, which reflect the individual's actual participation in knowledge sharing attitudes and behaviours towards MSM.

3.1.1. Psychological factors. SE is a concept proposed by Bandura in 1977 that refers to an individual's confidence and ability to complete a certain behaviour or task [47]. SCCT builds upon this concept and argues that SE is an important factor affecting individual career development and choice [24].

KAE refers to an individual's belief or expectation that he or she can acquire new knowledge or improve his or her knowledge level in the learning process [48]. KAE is a manifestation of learning motivation and is related to an individual's interest, goals, SE and other factors [49]. Previous studies have shown that KAE is an important motivational factor affecting KSB and has a significant positive correlation with the attitudes, intentions and behaviours of knowledge sharers [17]. Moreover, KAE is influenced by various personal and environmental factors, such as knowledge sharers' SE, trust, risk, and reward, as well as the scene, culture, atmosphere of knowledge sharing, etc [50,51].

ORE is defined as the expectation or belief that organisational members can obtain various rewards and support provided by the organisation for their KSB. That is, from the perspective of expectancy theory, incentives from the organisation can affect organisational members' outcome expectations and value perceptions of knowledge sharing, thus affecting their intention and behaviour of knowledge sharing [52]. For example, if the organisation provides rewards or recognition related to knowledge sharing, employees will think that sharing will bring positive results and thus will be more willing to share knowledge.

Individual factors have an important impact on teachers' KSI for MSM, which is reflected in the following relationships: (1) Teachers with high SE are more likely to think that they can effectively use MSM for knowledge sharing, thus increasing their sharing intention [53]. (2) Teachers' KAE in MSM, such as increasing professional knowledge or skills through communication with other teachers, obtaining and feedback information, participating in and creating content, etc., will also affect their sharing intention [54]. For teachers, the greater their KAE is, the more likely they are to believe that sharing can promote personal growth and development, thus increasing their sharing intention [55]. (3) Teachers' expectations of organisational rewards will also affect their sharing intentions. ORE means that teachers can expect or believe that they can obtain organisational-level incentives or praise through cooperation and collaboration with other teachers, contribution and sharing of resources, support and promotion of learning, etc. This expectation can enhance an individual's sense of belonging, loyalty and sharing intention [56]. When teachers believe that their sharing activities are consistent with the organisation's goals and values, they are more likely to think that sharing can facilitate the achievement of these goals and values, thus increasing their sharing intention.

3.1.2. Environmental factors. The STM of MSM refers to the degree and quality of trust among members, reflecting the interpersonal relationships and interactions among members. This study is based on the belief that STM affects the process and outcome of members' knowledge sharing by affecting their motivation, attitude, behaviour and so on. From the perspective of social cognitive theory, STM can affect online learners' cognition and evaluation of themselves, others and the environment, thus affecting online learners' attitudes and behaviours towards knowledge sharing [57,58].

STA refers to the effectiveness and efficiency of technology for completing specific tasks. In terms of online social knowledge sharing, from the perspective of the technology acceptance model (TAM), technology adaptation can affect people's perceived usefulness and perceived ease of use of social technology, thus affecting their attitude, intention, behaviour and outcome of using technology. For example, if technology can help sharers and receivers complete knowledge sharing tasks more effectively and is easy to use, they will be more willing to use technology for knowledge sharing and be able to achieve knowledge sharing goals better [56,59].

Based on the existing research results, STM and STA can enhance teachers' SE in MSM by improving their social SE and professional development; STM can enhance teachers' social SE, that is, their confidence and ability to achieve goals in social situations; [60] and STA can promote teachers' professional development and information sharing [61]. The STM and STA can directly affect teachers' attitudes and emotions related to knowledge sharing to affect teachers' KSI and behaviour in MSM [62,63].

In summary, focusing on the relationships among environmental and psychological factors and the internal mechanism of action, the following hypotheses are proposed:

- H1: STM has a positive impact on teachers' SE in mobile social media;
- H2: STA has a positive impact on teachers' SE in mobile social media;
- H3: SE has a positive impact on teachers' KSI on mobile social media;
- H4: KAE has a positive impact on teachers' KSI on mobile social media;
- H5: ORE has a positive impact on teachers' KSI on mobile social media.

3.1.3. Behaviour factors. KSI refers to the psychological tendency of individuals to share their knowledge with others in specific ways or methods [64]. In previous studies, KSI was shown to be influenced by various personal, organisational and environmental factors, such as an individual's motivation, attitude, trust, subjective norms, SE, environmental atmosphere, organisational reward, and social identity [65].

KSB involves actively transferring knowledge to others to bridge knowledge gaps or foster innovation and can be influenced by external incentives or constraints. This study examines both KSI and KSB to provide a comprehensive view of knowledge-sharing dynamics within the MSM environment. The model is informed by the theory of planned behaviour [66], which posits that intentions directly lead to behaviours [67]. By examining both intended and actual knowledge-sharing practices, we offer a holistic perspective on these processes, thus reflecting the interplay between individual intentions and their real-world actions in MSM settings.

In summary, focusing on the relationship between environmental factors and behavioural factors, the following hypotheses are proposed:

H6: The social trust environment has a positive impact on teachers' KSI on mobile social media.

H7: STA has a positive impact on teachers' KSI on mobile social media;

H8: The social trust environment has a positive impact on teachers' KSB on mobile social media.

H9: STA has a positive impact on teachers' KSB on mobile social media.

3.1.4. Mediation hypotheses. According to the social cognitive career model and its internal influence mechanism, in addition to the direct path to benefits, SE indirectly affects an individual's goal selection behaviour through outcome expectations as a substantive path in each subject model [18]. In light of this, it can be inferred that there is a mediating effect of outcome expectations between SE and KSI.

H10: SE affects teachers' KSI on mobile social media through KAE and ORE; that is, there is a dual-factor mediation effect, which can be decomposed into the following two hypotheses:

H10: SE affects teachers' KSI on mobile social media through KAE and ORE; that is, there is a dual-factor mediation effect, which can be decomposed into the following two hypotheses:

H10-1: KAE plays a mediating role in the effect of SE on KSI.

H10-2: ORE plays a mediating role in the effect of SE on KSI.

3.2. Model construction

Based on the relationship between each construct and the hypotheses proposed above, a research model is drawn, as shown in Figure 1.

4. Research design

4.1. Questionnaire design

A 27-question questionnaire was used in this study, including six basic information questions and specialised items based on a five-point Likert-type scale ranging from 1 (strongly disagree) to 5 (strongly agree). According to the SCCT model, questions on the three dimensions of psychological factors, SE, KAEs, and OREs, were based on the literature on individual psychology in the field of knowledge sharing [56,68,69]; questions on the two dimensions of environmental factors, STM and STA, were based on the literature on community environments in the field of knowledge sharing [68,70,71] and the literature on community environments in the field of knowledge sharing. In the field of knowledge sharing [56,71,72], the questions on the 2 constructs of knowledge sharing willingness and KSB in the action factor refer to the literature in the field of knowledge sharing on the mechanism of knowledge sharing willingness [56,70]. Most of the questions used in this study were based on the established scales of previous researchers, and the content of the questionnaire was modified to some extent according to the differences between the region and the target group. The specific scales are shown in Exhibit 1.

Before the official distribution of the questionnaire, a pilot test was conducted with 50 vocational school teachers in Zhejiang Province, China. This pretest used paper questionnaires to gather initial feedback. The respondents provided insights that led to modifications ensuring the clarity and precision of the questionnaire items. An introduction to the concept of 'knowledge sharing' was also added to enhance respondents' understanding and contextualise the questions.

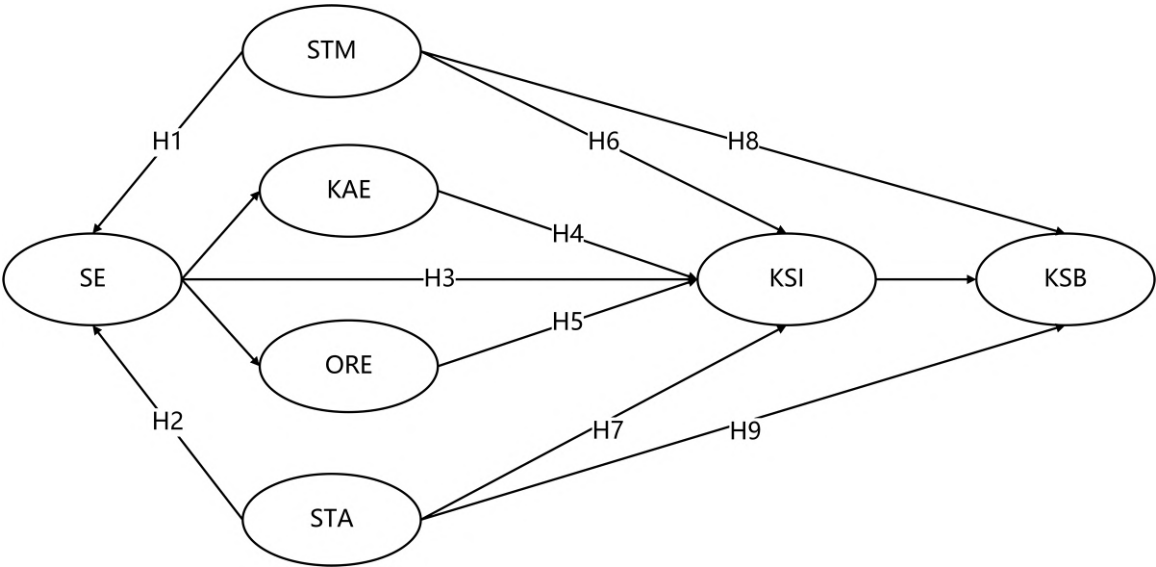


Figure 1. Research model.
Note: Paths SE → KAE, SE → ORE, and KSI → KSB are not hypothesised but are integral to the SCCT framework.

4.2. Data collection

To achieve comprehensive and accurate data collection, this study utilised the Questionnaire Star platform to design and release an online survey, which was distributed through mainstream social media platforms commonly used for educational purposes, such as WeChat and Nail. The target demographic comprised teachers from vocational colleges and universities in Zhejiang Province. A hierarchical distribution strategy was employed by collaborating with the heads of teaching and research groups at each institution to maximise survey participation and ensure data authenticity. In addition, detailed participant information was collected to analyse the factors influencing KSBs among teachers in these institutions and to develop a precise, empirically based research framework.

Building on the insights of Oldendick and Lambries [73], who highlighted the effectiveness of incentives in boosting survey participation, this study leveraged the host institution’s established network with vocational colleges to increase response rates. Close collaboration with the heads of teaching and research teams facilitated the distribution of the questionnaire while maintaining the voluntary nature and anonymity of the participants. A monetary incentive of RMB ten yuan was offered to respondents who completed the questionnaire via the Questionnaire Star platform. The collected data underwent ethical review by the corresponding author’s institution, thus ensuring adherence to ethical research standards.

Over three weeks, 387 responses were collected via online questionnaires. To ensure data quality, responses were screened via SPSS 27.0. Questionnaires that were completed in under 90 seconds or over 10 minutes were excluded to ensure the reliability of the data [74]. Therefore, 332 valid questionnaires were ultimately included, yielding a validity rate of 85.79%. Among the valid samples, 146 were male teachers and 186 were female teachers; 236 had bachelor’s degrees, 96 had master’s degrees, and there were no teachers with doctoral degrees; the age distribution was 47 under 25 years old, 157 between 26 and 35 years old, 105 between 36 and 45 years old, and 23 more than 46 years old; the distribution of the teaching experience was 87 less than 5 years old, 132 between 6 and 10 years old, and 113 more than 10 years old; and the type of teacher distribution was 85 teachers of cultural courses, 157 teachers of professional courses, 32 teachers of internship guidance, and 58 part-time teachers of professional courses and internship guidance. The normal distribution of the data suggests that the sample is suitable for studying teachers’ motivation for knowledge-sharing behaviours in the social media environment.

5. Data analysis

5.1. Reliability and validity tests

5.1.1. Reliability test. Reliability reflects the consistency or stability of the questionnaire measurement results. In this study, the path factor loadings obtained by Mplus 8.3 measurements were extracted. There is no unified standard for its

Table 1. Results of factor analysis.

Variable	Measurement	Factor Loading
SE	SE1	0.843
	SE2	0.779
	SE3	0.689
KAE	KAE1	0.817
	KAE2	0.819
	KAE3	0.722
ORE	ORE1	0.650
	ORE2	0.823
	ORE3	0.830
STM	STM1	0.744
	STM2	0.717
	STM3	0.852
	STM4	0.852
STA	STA1	0.698
	STA2	0.854
	STA3	0.720
KSI	KSI1	0.826
	KSI2	0.851
	KSI3	0.738
KSB	KSB1	0.758
	KSB2	0.714
	KSB3	0.712
	KSB4	0.785
	KSB5	0.748

acceptable range, but the guiding opinions from related studies suggest that a value higher than 0.5 is an acceptable standard and that a value higher than 0.7 is considered a clear structural target, indicating that there is a very strong correlation between the observed variables and the latent variables [75,76,77]. As shown in Table 1, most of the factor loadings are greater than 0.7, and all are greater than 0.5, which indicates that the setting of latent variables in each dimension of this study can explain each observation item under this dimension well.

SPSS 27.0 was used to calculate the composite reliability (CR) and average variance extracted (AVE) values. According to Fornell and Larcker's study [78], the CR should be greater than 0.7, and the AVE should be greater than 0.5. As shown in Table 2, all the AVE values and CR values in this study meet the requirements, which indicates that the internal consistency of the measurement items is good and that the reliability is acceptable.

5.1.2. Discriminant validity test. The criteria for the validity judgement were as follows: if a variable's correlation coefficient with other variables is smaller than the square root of the average variance extracted of that variable, it indicates that the variable has good discriminant validity [78]. As shown in Table 3, the diagonal line is the square root of the AVE, and the square root of the AVE value is greater than the correlation coefficient between structures, indicating that each variable has discriminant validity.

5.1.3. HTMT test. In addition, the heterogeneous-homogeneous ratio, which is the ratio of between-trait and within-trait correlations [79] and may affect the estimation of path coefficients and model fit, was used in this study to avoid potential multicollinearity problems among latent variables. In addition, the HTMT test can provide more accurate results than other traditional methods (such as the Fornell–Larcker criterion or cross loadings), especially when the measurement model is homogeneous [80]. Henseler et al.'s study [79] showed that an HTMT value of 0.85 is a conservative and strict threshold that can effectively detect discriminant validity problems. As shown in Table 4, the maximum HTMT value in this study is 0.831, and all the samples meet the more stringent HTMT test standard. In summary, the model has good reliability and validity.

Table 2. Reliability and validity analysis results – CR, Cronbach's alpha, and AVE.

Variable	Cronbach's alpha	CR	AVE
SE	0.821	0.816	0.597
KAE	0.827	0.830	0.620
ORE	0.811	0.814	0.596
STM	0.869	0.871	0.630
STA	0.799	0.803	0.578
KSI	0.851	0.848	0.650
KSB	0.865	0.861	0.553

Table 3. Discriminant validity.

	AVE	SE	KAE	ORE	STM	STA	KSI	KSB
SE	0.597	0.773						
KAE	0.620	0.349	0.787					
ORE	0.596	0.443	0.265	0.772				
STM	0.630	0.697	0.397	0.425	0.794			
STA	0.578	0.433	0.611	0.289	0.473	0.760		
KSI	0.650	0.610	0.621	0.548	0.606	0.571	0.806	
KSB	0.553	0.595	0.522	0.458	0.594	0.597	0.737	0.744

Table 4. Differential validity (HTMT).

	SE	KAE	ORE	STM	STA	KSI	KSB
SE							
KAE	0.296						
ORE	0.364	0.242					
STM	0.587	0.353	0.345				
STA	0.350	0.502	0.229	0.396			
KSI	0.520	0.533	0.459	0.528	0.493		
KSB	0.497	0.444	0.373	0.526	0.496	0.648	

Table 5. Fitting index of the structural equation model and results of this study.

Index	Acceptable standard	Results of this study
χ^2	N/A (the smaller, the better)	363.867
χ^2/df	≤ 3	1.849
CFI	≥ 0.8	0.968
TFI	≥ 0.8	0.962
RSMEA	≤ 0.08	0.042
SRMR	≤ 0.08	0.039

5.2. Structural equation modelling

5.2.1. Model fit test. Many indicators are used to measure model fit in structural equation modelling; these include the chi-square to degree of freedom ratio, CFI, TLI, RMSEA, etc. [81]. The chi-square degrees of freedom ratio is used to assess the deviation between model predictions and actual observed data, with lower values suggesting a better model fit. CFI and TLI are used as relative fit indices reflecting the improvement of the model with respect to the baseline model, with values converging to 1 indicating an excellent fit. The RMSEA evaluates the average magnitude of the model error, with smaller values suggesting greater consistency between the model and the data. In this study, as shown in Table 5, all

Table 6. Path coefficients.

Hypothesis	Dependent Variable	Independent Variable	Ustd	Std	S.E.	C.R.	<i>p</i>
H1	SE	STM	0.658	0.641	0.07	9.415	***
H2	SE	STA	0.179	0.175	0.063	2.845	**
H3	KSI	SE	0.234	0.221	0.089	2.632	**
H4	KSI	KAE	0.369	0.349	0.063	5.818	***
H5	KSI	ORE	0.351	0.272	0.075	4.694	***
H6	KSI	STM	0.163	0.150	0.082	1.976	*
H7	KSI	STA	0.160	0.147	2.409	0.016	*
H8	KSB	STM	0.197	0.187	0.067	2.918	**
H9	KSB	STA	0.222	0.210	0.068	3.261	**

Note: * is $p < 0.05$, ** is $p < 0.01$, *** is $p < 0.001$.

the fit indices are within the acceptable range, showing that the model of teachers' motivation for knowledge-sharing behaviour fits well with the actual data in the mobile social media environment, which provides a reliable basis for further exploring the relationship between the motivational factors of teachers' knowledge-sharing behaviour.

5.2.2. Hypothesis testing. In this study, the maximum likelihood method is used to estimate the coefficients of each path. The test results are shown in Table 6. Here, the estimated value represents the path coefficient, S.E. represents the standard error of the estimated difference value, and C.R. represents the ratio of the regression coefficient value to the estimated value standard error. The path coefficients of the knowledge-sharing model reach the $p < 0.001$ or $p < 0.05$ significance level.

5.2.3. Mediation test. This study used the self-help method (bootstrapping confidence interval method) and Mplus 8.3 software with 5000 resamples to construct and test a two-factor juxtaposed mediation model aiming to investigate the mediating roles of KAE and ORE between SE and KSI.

SE's total effect on KSI is 0.524, which has a significant positive impact ($Z = 4.637$, $p < 0.001$). SE's direct effect on the KSI is significant, with an effect size of 0.234, which indicates a significant positive impact ($Z = 2.095$, $p < 0.05$). The indirect effect can be decomposed into two parallel mediation paths: SE-KAE-KSI and SE-ORE-KSI. The confidence interval of the indirect path of SE-KAE-KSI is [0.089, 0.249], which does not include 0 and has a significant positive impact ($Z = 3.907$, $p < 0.001$), and the effect size is 0.068. Similarly, the indirect path of SE-ORE-KSI is also significant, with an effect size of 0.071. This shows that SE not only directly enhances the KSI but also indirectly enhances the KSI by enhancing the KAE and ORE. Therefore, this study supports the hypothesis of a dual-factor parallel mediation model.

The data show that KAE and ORE play partial mediating roles in the effect of SE on KSI, and the indirect effects account for 19.0% and 17.1%, respectively, of the total effects. The significance level of the difference between them is greater than 0.05, the confidence interval [−0.089, 0.119] also contains 0, and there is no significant difference in the effect size of the two.

5.3. Results analysis

This study assessed the model effects of teachers' KSB and its predictors through structural equation modelling (SEM) analysis, as shown in Figure 2. The results showed that the explanatory power (R^2) of the model for both KSB and KSI was 0.607, indicating that the present model was effective in predicting teachers' KSB. Meanwhile, the explanatory power of SE, knowledge harvest expectation, and organisational reward expectation were 0.54, 0.15, and 0.21, respectively, which were consistent with the benchmark values of Lent et al.'s [25] study on the application of the SCCT and confirmed the robustness of the model.

Based on the model analysis and hypothesis testing (see Table 7), the following key findings were derived from this study on teachers' knowledge sharing in the mobile social media environment:

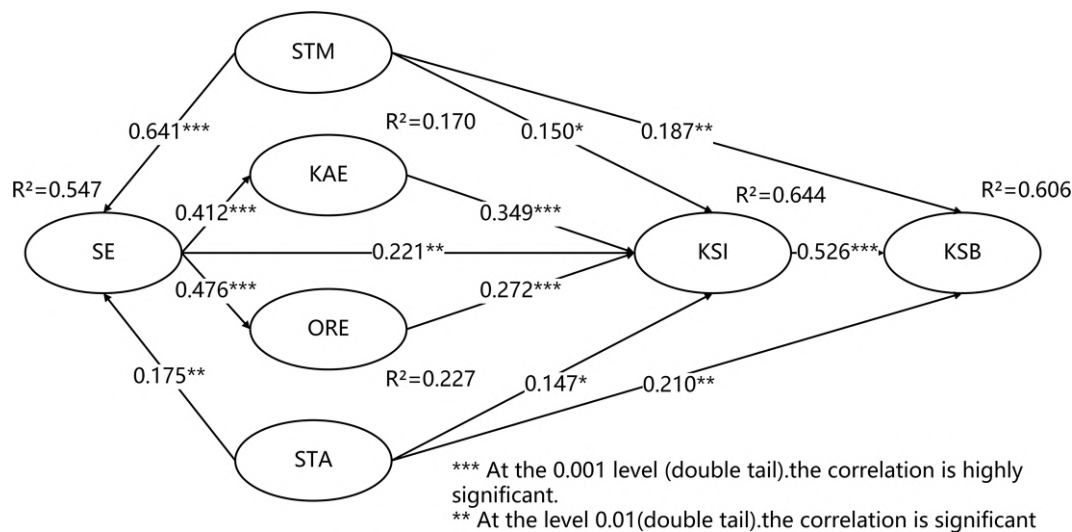


Figure 2. Structural model.

Table 7. Mediating effect test.

Path Effect	Size	Standard Error	Z value	p value	Bias-corrected 95% CI	
					lower limit	upper limit
SE-KSI	0.234	0.112	2.095	*	0.027	0.467
SE-KAE-KSI	0.152	0.039	3.907	***	0.089	0.249
SE-ORE-KSI	0.137	0.039	3.510	***	0.078	0.236
Total	0.524	0.112	4.637	***	0.318	0.764
Diff	0.015	0.052	0.291	0.771	-0.089	0.119

Note: * is $p < 0.05$, ** is $p < 0.01$, *** is $p < 0.001$.

1. Knowledge harvest expectations, as well as organisational reward expectations, positively influence KSIs (H4, H5).
2. Trust atmosphere and technology suitability positively affect teachers' SE (H1, H2).
3. Trust climate and technology suitability directly and positively affect teachers' willingness to share knowledge as well as their behaviour (H6–H9).
4. Teachers' SE, mediated through outcome expectations, affects willingness to share knowledge (H10).

These findings will be analysed in further depth in the discussion section.

6. Discussion and implications

6.1. Discussion

6.1.1. Direct and indirect effects of environmental factors on knowledge sharing. Environmental factors play crucial roles in shaping teachers' KSIs and behaviours within mobile social media contexts. According to SCCT, these factors serve as both direct incentives and indirect facilitators through their impact on SE. For example, social trust and platform technological fitness have been identified as key determinants of effective knowledge sharing [39]. This study extends these findings by emphasising how environmental contexts influence individual behaviour, aligning with the SCCT's assertion that such factors shape individual actions and beliefs [82].

The SCCT posits that environmental factors – such as family, social, cultural, economic, and political contexts – can either facilitate or impede individual agency [82]. These elements exert a direct influence on behavioural intentions, affecting how individuals perceive and act within their environments [23]. Furthermore, an individual's SE beliefs are continually shaped by these environmental factors, highlighting the dynamic interplay between the individual and his or

her surroundings [57]. This interaction underscores the complex process by which teachers adapt and respond to their environments, ultimately impacting their willingness and ability to share knowledge.

In the context of mobile social media, this study suggests that a climate of trust in social environments may positively influence KSIs and practices by reducing the perceived risk of sharing knowledge and enhancing psychological security and belonging. Studies by Hoseini et al. [39] and Kmiecik [83] confirm that trusting relationships within organisations mitigate the knowledge sharing process of psychological barriers and enhance knowledge sharing motivation. Furthermore, Hsu et al. [84] used a social cognitive theory model to validate how social trust can indirectly and positively promote KSBs by enhancing an individual's SE.

Technology suitability directly and indirectly reinforces users' SE and pleasure sharing by enhancing platform satisfaction and frequency of use, effectively promoting KSIs and practices. Wang and Xie's [85] study of online sharing behaviours was closely linked to perceived platform ease of use and user satisfaction, with the latter directly facilitating in-platform knowledge learning and sharing activities. Stibe et al. [86] used a model of differences in knowledge behaviours to illustrate how rich, convenient, and interactive sharing experiences can enhance audience breadth and platform recognition of social networking technologies, thereby stimulating KSBs. Alshahrani and Rasmussen [87] further suggest that optimising user experiences and mimicking experiences enhance SE and indirectly promote knowledge sharing activity.

In summary, social trust and technological appropriateness have important direct and indirect effects on knowledge sharing in mobile social media, highlighting the importance of optimising this environment to promote KSIs and behaviours among the teacher community. These findings provide a theoretical and empirical basis for strategically improving the mobile social media environment to promote knowledge sharing among teachers.

6.1.2. Influence of outcome expectations on knowledge sharing intentions. The analysis indicates that SE and outcome expectations have a significant positive effect on teachers' willingness to share knowledge via mobile social media. Teachers are motivated not only by their confidence in their ability to share knowledge but also by the potential gains and recognition they anticipate from this activity. This behaviour reflects a dual pursuit of personal empowerment and enhanced social status, thus highlighting both intrinsic and extrinsic motivations. These findings expand the theoretical understanding of knowledge sharing and provide practical insights into promoting this behaviour among teachers in digital environments.

The SCCT provides a robust framework for understanding these dynamics. According to Lent and Brown [18], there are strong correlations between outcome expectations, SE, and choice behaviours. SE enhances an individual's ability to set clear goals and develop positive knowledge-sharing beliefs, which subsequently increases their willingness to engage in knowledge sharing. Outcome expectations – defined as the anticipated results or rewards of a behaviour – serve as key determinants in deciding whether individuals adopt specific behaviours. This aligns with SCCT's assertion that both SE and outcome expectations are crucial in shaping individuals' actions and choices in professional contexts.

This study aimed to explore the effect of knowledge gain expectations on teachers' KSBs and found that teachers' KSBs or extent matched the expected knowledge gain on a given platform. According to Bock and Kim [88], KSBs are associated with costs, and individuals are willing to share their valuable knowledge resources only when the expected knowledge rewards exceed these costs. This view is corroborated by Thi and Duong's [63] study of KSBs on social networking sites in Vietnam, where an individual's willingness to share knowledge is only aroused when the expected benefits exceed costs. Tseng and Kuo [89] argued that teachers' knowledge gain expectations can enhance their positive attitudes and behavioural motivation towards knowledge sharing, thus promoting their KSIs.

This study reveals that teachers' expectations of organisational rewards – such as recognition, praise, career advancement and material rewards – can promote sharing motivation, which in turn significantly enhances their motivation to engage in knowledge sharing. Constant et al. [90] found that beliefs in organisational encouragement and status advancement can positively promote KSIs. Similarly, Zhang W. et al.'s [91] analysis of Chinese teachers confirmed that leadership originating from school administrators, as well as trust in the organisation, leads teachers to exhibit more positive KSBs.

Taken together, outcome expectations – that is, individuals' beliefs about the potential rewards of knowledge sharing – play an influential role at both the individual and organisational levels and are supported by both theoretical and empirical research. Heinz and Rice [92] suggested that by enhancing organisational status, increasing competence and experience, and realising social responsibility, teachers who are more likely to experience positive emotions in knowledge sharing are more likely to experience positive emotions and feel satisfied with the shared outcomes, which in turn strengthens their motivation to continue sharing knowledge in the future. This section summarises the role of outcome expectations in influencing willingness to promote knowledge sharing.

6.1.3. The partial mediating role of outcome expectations between SE and intention. The SCCT provides a valuable framework for understanding the complex relationship between SE and knowledge-sharing intentions. According to Lent and Brown [18], SE influences individuals' expectations of their ability to perform effectively in a given domain, thereby shaping their behavioural intentions and actions. In this study, outcome expectations were found to partially mediate the relationship between SE and the intention to share knowledge. This means that while SE directly enhances individuals' willingness to share, their expectations about the potential benefits and consequences of sharing also play a significant role in determining their intentions. These findings align with those of Safdar et al. [55], who demonstrated that outcome expectations are a crucial intermediary factor in the relationship between SE and intention. The partial mediation effect indicates that enhancing positive outcome expectations along with strengthening SE can effectively promote KSB, thus underscoring the need to integrate cognitive and motivational elements into interventions to foster collaboration among teachers.

Keren-Happuch and Moon-Ho [93] elaborated on the role of SE in shaping individual outcome expectations – the prediction or anticipation of the consequences of particular behaviours – and in setting personal goals, including goal choice and performance goals. The level of SE has an indirect effect on KSIs, a mechanism that can be explained by expectations about the effectiveness of sharing: lower SE leads individuals to be sceptical about their own ability to contribute to knowledge growth and thus show conservative sharing intentions; in contrast, higher SE prompts individuals to view knowledge sharing as a valuable contribution, which they perceive as contributing to organisational rewards and support [84,94].

6.2. Implications

6.2.1. Environmental optimisation strategy: building a good knowledge sharing network. This section explores strategies to optimise knowledge sharing in mobile social media from an environmental perspective, targeting a group of vocational schoolteachers. The findings suggest that STM and social technology appropriateness have a significant positive effect on teachers' KSI and behaviour.

First, it is crucial to shape a positive STM. The interactive and personalised features of mobile social platforms can be used to strengthen the trust relationship between users, which in turn facilitates knowledge sharing. Practical measures include using social features (e.g. likes, comments, and retweets) to recognise knowledge contributions, deepening connections between users through private interactions (e.g. private messages and group chats), and enhancing users' influence and credibility through personalised displays (e.g. personal homepages, hashtags, and topics). These strategies help to enhance community trust [87]. For professional teachers, it is crucial to create dedicated platforms for the exchange of pedagogical knowledge and skills, aiming to create a social environment of mutual trust and thus optimise knowledge sharing.

Second, improving the adaptability of social technologies is another key factor in optimising the knowledge sharing environment. The goal is to reduce the cost and risk of use and improve the user experience. Measures include improving the interface and functional design of social platforms and optimising search engines to reduce the cognitive burden and operational difficulty for users. At the same time, enhanced security measures (e.g. privacy protection, copyright management) are essential for safeguarding users' intellectual property and personal information and mitigating users' risk concerns [95,96].

6.2.2. Individual incentive strategy: cultivating teachers' knowledge-sharing beliefs. The discussion in this section aims to develop vocational school teachers' knowledge sharing beliefs through individual motivational strategies. The study revealed that teachers' positive expectations of knowledge sharing outcomes significantly contribute to their willingness to share. Therefore, this study proposes strategies to consolidate teachers' expectations of knowledge gains and organisational rewards while integrating the specific professional attributes and pedagogical needs of vocational teachers to provide customised support. This study demonstrates innovation in the field.

First, it increases teachers' expectations of knowledge gains. Strategies include the following: a. Implementing a knowledge product trading mechanism that encourages teachers to turn their teaching results (e.g. courseware, essays, cases) into traded products, which are rewarded in the form of points and prizes, aiming to broaden the impact and scope of knowledge sharing [38]. b. Enhancing teachers' self-directed research capacity and encouraging lifelong learning, including activities such as reading, reflecting, and writing, to continually update their professional knowledge and educational philosophy. Schools should provide learning resources and platforms, support teachers in taking appropriate professional development courses, and recognise and reward teachers' learning outcomes. c. Dedicate a knowledge-sharing section to facilitate interaction between teachers and industry experts, focusing on the sharing of practical experience

and keeping abreast of the latest industry news and technological advances, thereby stimulating pedagogical innovations and research collaborations within the field of vocational education. d. Promotion of the use of knowledge-sharing platforms to facilitate teachers' interaction with industry experts.

Second, teachers' expectations of organisational rewards should be increased. This study proposes two strategies to achieve this goal: a. Promote a learning organisational culture that encourages learning and knowledge sharing and provides regular training and learning opportunities to enhance teachers' sense of belonging, pride, job satisfaction and loyalty. b. Teachers should be provided with the necessary support and guidance in the process of knowledge sharing, including the establishment of mentorships and peer-to-peer support models, to promote their professional growth and enhance their trust and interaction. c. Teachers should be provided with the necessary support and guidance in the process of knowledge sharing, including the establishment of mentorships and peer-to-peer support models, to promote their professional growth and enhance their trust and interaction.

7. Limitations and future suggestions

This study has several limitations. First, the questionnaire survey was conducted on the Internet, and there may be cases of random completion. To reduce the impact of this situation, a larger sample size should be used in future studies. Second, due to the limitations of the research subjects, the sample of this study mainly consisted of teachers at secondary vocational schools in China, and the findings may not be generalisable to other vocational education teachers. Finally, in the mobile Internet environment, there are many complex factors that affect the knowledge sharing of vocational school teachers, and this study did not fully consider all the influencing factors, such as the social cultural atmosphere, which needs to be verified by follow-up studies.

8. Exhibit I. Survey questionnaire

Note: This survey was originally issued in Chinese on the Questionnaire Star platform to avoid ambiguity and enhance response accuracy. The following English version is a translation intended for broader accessibility.

8.1. Concept explanation

Mobile social media refers to the tools and platforms for individuals to share their opinions, insights, experiences and perspectives with each other in the mobile network environment, which mainly includes WeChat, QQ, Weibo, blogs, Baidu library, Zhihu, famous teachers' network studios, and all kinds of social networking sites.

Knowledge sharing of secondary teachers in the mobile Internet environment refers to teachers' sharing/sharing of knowledge information to others on mobile social media with the help of mobile devices, so as to realise the externalisation, dissemination, internalisation and innovation of teachers' knowledge.

8.2. Part I: your basic information (Single choice, mark '√' on your selected option)

1. Gender: ☐ Female ☐ Male
2. Age: ☐ 25 and below ☐ 26-35 years old ☐ 36-45 years old ☐ 46 years and above
3. Teaching experience: ☐ 5 years and below ☐ 6-10 years ☐ 11-15 years ☐ Over 16 years
4. Highest education level: ☐ Associate degree ☐ Bachelor's degree ☐ Master's degree ☐ Doctorate
5. Type of teacher: ☐ General subject teacher ☐ Professional subject teacher ☐ Internship guide teacher ☐ Professional subject and internship guide teacher
6. Mobile social media platforms you currently use (Multiple choices): _____
 A. WeChat B. QQ C. Weibo D. Blog E. Baidu Wenku F. Zhihu
 G. Famous Teacher Network Studios H. Others _____

8.3. Part II: psychological factors in knowledge sharing

Rate each statement on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree)

1. I can use mobile social media to share professional knowledge or teaching experiences with other teachers.
2. On mobile social media platforms, I can clearly express information or knowledge.

3. I am familiar with the operation and functions of various mobile social media platforms and can share knowledge smoothly.
4. The knowledge shared by teachers on mobile social media has improved my work performance.
5. The knowledge shared by teachers on mobile social media has provided me with new knowledge, information, skills, etc.
6. I think the knowledge shared by teachers on mobile social media is valuable.
7. My school will substantially reward knowledge-sharing behaviours
8. My coworkers will praise the act of sharing knowledge or experience through mobile social media.
9. In return for proactively sharing knowledge or experience, I will be given the opportunity to learn from others

8.4. Part III: environmental factors in knowledge sharing

Rate each statement on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree)

1. My mobile devices and mobile network support knowledge sharing.
2. The mobile social media platforms I use support knowledge sharing features.
3. The mobile social media platforms I use are versatile, supporting various forms of knowledge sharing.
4. If other members on mobile social media are trustworthy and friendly, I will actively share knowledge.
5. If most other members on mobile social media are colleagues, friends, or acquaintances, I will actively share knowledge.
6. By sharing my knowledge on mobile social media, I believe that when I need help, other members will also be willing to share their knowledge.
7. I believe that other members will share their knowledge, so I am willing to share mine.

8.5. Part IV: behavioural factors in knowledge sharing

Rate each statement on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree)

1. I am willing to share the knowledge, skills, or experiences I have gained in teaching or training through mobile social media.
2. I am willing to use mobile social media to share valuable information with colleagues.
3. I am willing to share my work experiences and knowledge through mobile social media in the future.
4. I share my teaching skills or work experiences with other members on mobile social media.
5. I often exchange experiences and share knowledge with colleagues through mobile social media.
6. I have contributed useful information to other teachers through mobile social media.
7. My school shares teaching skills or training experiences through mobile social media.
8. My school uses mobile social media for the transmission and sharing of teaching information, materials, and documents.

Author contributions

All authors contributed to the study conception and design. H.L. organized the entire project, while also making critical changes and reviews. J.X. conceived the initial idea and wrote the manuscript. Y.L. provide valuable advice and collate the literature. All authors read and approved the final version of the manuscript.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: This work was supported by the National Social Science Fund of China (Grant No. 16BTQ084) and the National College Students Innovation and Entrepreneurship Training Program (Grant No. GZ22251080006).

Ethical approval

This study was approved by Zhejiang University of Technology. The questionnaire respondents were informed about the purpose and scope of the study and gave their consent to participate.

ORCID iD

Jun Xu  <https://orcid.org/0000-0002-0557-2046>

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- [1] Costa RC. The place of the humanities in today's knowledge society. *Palgrave Commun* 2019; 5: 1–5.
- [2] van Bommel J, Randahl A-C, Liljekvist Y, et al. Tracing teachers' transformation of knowledge in social media. *Teach Teach Educ* 2020; 87: 102958.
- [3] Loo S. Vocational teachers' knowledge, experiences, and pedagogy. In: Mcgrath S, Mulder M, Papier J, et al. (eds) *Handbook of vocational education and training: developments in the changing world of work*. Cham: Springer, pp. 1611–1625.
- [4] Ahmad F and Karim M. Impacts of knowledge sharing: a review and directions for future research. *J Workplace Learn* 2019; 31: 207–230.
- [5] Kaiser G and König J. Competence measurement in (Mathematics) teacher education and beyond: implications for policy. *High Educ Policy* 2019; 32: 597–615.
- [6] Metsäpelto R-L, Poikkeus A-M, Heikkilä M, et al. A multidimensional adapted process model of teaching. *Educ Asse Eval Acc* 2022; 34: 143–172.
- [7] Zhang X, Tang J, Wei X, et al. How does mobile social media affect knowledge sharing under the 'Guanxi' system? *J Knowl Manag* 2020; 24: 1343–1367.
- [8] Hou H-T, Sung Y-T and Chang K-E. Exploring the behavioral patterns of an online knowledge-sharing discussion activity among teachers with problem-solving strategy. *Teach Teach Educ* 2009; 25: 101–108.
- [9] Khan MN, Ashraf MA, Seinen D, et al. Social media for knowledge acquisition and dissemination: the impact of the COVID-19 pandemic on collaborative learning driven social media adoption. *Front Psychol* 2021; 12: 648253.
- [10] Alsharo M, Gregg D and Ramirez R. Virtual team effectiveness: the role of knowledge sharing and trust. *Inform Manag* 2017; 54: 479–490.
- [11] OECD. *How do vocational education systems differ around the world?* Paris: OECD, 2020.
- [12] Placklé I, Könings KD, Struyven K, et al. Powerful learning environments in secondary vocational education: towards a shared understanding. *Europ J Teach Educ* 2020; 43: 224–242.
- [13] Charband Y and Jafari Navimipour N. Knowledge sharing mechanisms in the education. *Kybernetes* 2018; 47: 1456–1490.
- [14] Cai Y and Shi W. The influence of the community climate on users' knowledge-sharing intention: the social cognitive theory perspective. *Behav Inform Tech* 2022; 41: 307–323.
- [15] Renqiang X and Wende Z. An empirical study on the impact of platform environmental factors on knowledge sharing in virtual communities. *Tech Soc* 2022; 71: 102094.
- [16] Hao Q, Zhang B, Shi Y, et al. How trust in coworkers fosters knowledge sharing in virtual teams? A multilevel moderated mediation model of psychological safety, team virtuality, and self-efficacy. *Front Psychol* 2022; 13: 899142.
- [17] Lin H-F. Effects of extrinsic and intrinsic motivation on employee knowledge sharing intentions. *J Inform Sci* 2007; 33: 135–149.
- [18] Lent RW and Brown SD. Social cognitive career theory at 25: empirical status of the interest, choice, and performance models. *J Vocat Behav* 2019; 115: 103316.
- [19] Huang J-T and Hsieh H-H. Linking socioeconomic status to social cognitive career theory factors: a partial least squares path modeling analysis. *J Career Asses* 2011; 19: 452–461.
- [20] Aure PAP, Dui RP, Jimenez SV, et al. Understanding social entrepreneurial intention through social cognitive career theory: A partial least squares structural equation modelling approach. *Organiz Market Emerg Econ* 2019; 10: 92–110.
- [21] Brown SD, Lent RW, Telander K, et al. Social cognitive career theory, conscientiousness, and work performance: a meta-analytic path analysis. *J Vocat Behav* 2011; 79: 81–90.
- [22] Brown SD and Lent RW. Social cognitive career theory at 25: progress in studying the domain satisfaction and career self-management models. *J Career Asses* 2019; 27: 563–578.
- [23] Lent RW, Brown SD, Schmidt J, et al. Relation of contextual supports and barriers to choice behavior in engineering majors: test of alternative social cognitive models. *J Counsel Psychol* 2003; 50: 458.
- [24] Lent RW, Brown SD and Hackett G. Toward a unifying social cognitive theory of career and academic interest, choice, and performance. *J Vocat Behav* 1994; 45: 79–122.

- [25] Lent RW, Brown SD, Sheu H-B, et al. Social cognitive predictors of academic interests and goals in engineering: utility for women and students at historically black universities. *J Counsel Psychol* 2005; 52: 84.
- [26] Sheu H-B, Lent RW, Brown SD, et al. Testing the choice model of social cognitive career theory across Holland themes: a meta-analytic path analysis. *J Vocat Behav* 2010; 76: 252–264.
- [27] Wang D, Liu X and Deng H. The perspectives of social cognitive career theory approach in current times. *Front Psychol* 2022; 13: 1023994.
- [28] Almuqrin A and Mutambik I. The explanatory power of social cognitive theory in determining knowledge sharing among Saudi faculty. *PLoS ONE* 2021; 16: e0248275.
- [29] Ipe M. Knowledge sharing in organizations: a conceptual framework. *Human Res Develop Rev* 2003; 2: 337–359.
- [30] Zeinabadi H. Principals' role in teachers' knowledge-sharing beliefs, intention and behaviour in Iranian schools: exploring the impact of knowledge-sharing leadership. *J Educ Admin* 2022; 60: 493–510.
- [31] Antera S. Professional competence of vocational teachers: a conceptual review. *Vocat Learn* 2021; 14: 459–479.
- [32] Andersson P and Köpsén S. Continuing professional development of vocational teachers: participation in a Swedish national initiative. *Empiric Res Vocat Educ Train* 2015; 7: 7.
- [33] Runhaar P and Sanders K. Promoting teachers' knowledge sharing. The fostering roles of occupational self-efficacy and human resources management. *Educ Manag Admin Leader* 2015; 44: 64773.
- [34] Cornelius S and Stevenson B. International online collaboration as a boundary crossing activity for vocational educators. *J Vocat Educ Train* 2019; 71: 157–174.
- [35] OECD. Teachers and leaders in vocational education and training. OECD, https://www.oecd-ilibrary.org/education/teachers-and-leaders-in-vocational-education-and-training_59d4fbb1-en (2021, accessed 9 May 2023).
- [36] Livingstone DW and Guile D (eds) *The knowledge economy and lifelong learning*. Rotterdam: Sense Publishers, 2012.
- [37] Wijnia L, Kunst EM, van Woerkom M, et al. Team learning and its association with the implementation of competence-based education. *Teach Teach Educ* 2016; 56: 115–126.
- [38] Yu X, Hu D, Li N, et al. Comprehensive evaluation on teachers' knowledge sharing behavior based on the improved TOPSIS method. *Comput Intel Neurosci* 2022; 2022: e2563210.
- [39] Hoseini M, Saghaei F and Aghayi E. A multidimensional model of knowledge sharing behavior in mobile social networks. *Kybernetes* 2018; 48: 906–929.
- [40] Musfikar R. The best factors for influence knowledge sharing among teachers through website technology. *Cyberspace* 2017; 1: 101.
- [41] Lee J, Oh S, Dong H, et al. Motivations for self-archiving on an academic social networking site: a study on ResearchGate. *J Assoc Inform Sci Tech* 2019; 70: 563–574.
- [42] Feng X, Wang L, Yan Y, et al. The sustainability of knowledge-sharing behavior based on the theory of planned behavior in Q&A social network community. *Complex* 2021; 2021: 1526199.
- [43] Chow WS and Chan LS. Social network, social trust and shared goals in organizational knowledge sharing. *Inform Manag* 2008; 45: 458–465.
- [44] Ming J, Chen R and Tu R. Factors influencing user behavior intention to use mobile library application: a theoretical and empirical research based on grounded theory. *Data Inform Manag* 2021; 5: 131–146.
- [45] Chai S and Kim M. A socio-technical approach to knowledge contribution behavior: an empirical investigation of social networking sites users. *Int J Inform Manag* 2012; 32: 118–126.
- [46] Ahmad AR, Jameel A and Raewf M. Impact of social networking and technology on knowledge sharing among undergraduate students. *Int Busin Educ J* 2021; 14: 1–16.
- [47] Bandura A. Self-efficacy: toward a unifying theory of behavioral change. *Psychol Rev* 1997; 84: 191–215.
- [48] Wigfield A and Eccles JS. Expectancy–value theory of achievement motivation. *Contemp Educ Psych* 2000; 25: 68–81.
- [49] Pintrich PR and Schunk DH. *Motivation in education: theory, research, and applications*. Hoboken, NJ: Prentice Hall, 2002.
- [50] Wang S, Noe RA and Wang Z-M. Motivating knowledge sharing in knowledge management systems: a quasi-field experiment. *J Manag* 2014; 40: 978–1009.
- [51] Zhang X and Chen R. Examining the mechanism of the value co-creation with customers. *Int J Prod Econ* 2008; 116: 242–250.
- [52] Wang N, Yin J, Ma Z, et al. The influence mechanism of rewards on knowledge sharing behaviors in virtual communities. *J Knowl Manag* 2021; 26: 485–505.
- [53] Kim H, Lee J and Oh SE. Individual characteristics influencing the sharing of knowledge on social networking services: online identity, self-efficacy, and knowledge sharing intentions. *Behav Inform Tech* 2020; 39: 379–390.
- [54] Wasko MM and Faraj S. Why should I share? Examining social capital and knowledge contribution in electronic networks of practice. *MIS Quart* 2005; 29: 35–57.
- [55] Safdar M, Batool SH and Mahmood K. Relationship between self-efficacy and knowledge sharing: systematic review. *Global Knowl Memory Commun* 2020; 70: 254–271.
- [56] Bock G-W, Zmud RW, Kim Y-G, et al. Behavioral intention formation in knowledge sharing: examining the roles of extrinsic motivators, social-psychological forces, and organizational climate. *MIS Quart* 2005; 29: 87–111.
- [57] Savickas ML, Brown SD and Lent RW. Career development and counseling: putting theory and research to work. *Theory Pract Career Construct* 2005; 2005: 42–70.

- [58] Locke EA. Social foundations of thought and action: a social-cognitive view. *AMR* 1987; 12: 169–171.
- [59] Davis FD. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quart* 1989; 13: 319–340.
- [60] Vatou A, Gregoriadis A, Tsigilis N, et al. Teachers' social self-efficacy: development and validation of a new scale. *Cogent Educ* 2022; 9: 2093492.
- [61] Richter E, Carpenter JP, Meyer A, et al. Instagram as a platform for teacher collaboration and digital social support. *Comput Educ* 2022; 190: 104624.
- [62] Hsu P-C, Thuy TTH and Chen R-S. Female preschool teachers' perceptions of mobile communities and teacher self-efficacy for professional development: the mediating effects of trust and interaction via mobile apps. *Asia-Pacific Edu Res* 2022; 31: 147–154.
- [63] Pham Thi TD and Duong NT. Factors influencing knowledge-sharing intention on social network sites: an empirical study in Vietnam. *Soc Netw Anal Min* 2022; 12: 110.
- [64] Li X, Froese FJ and Pak YS. Promoting knowledge sharing in foreign subsidiaries through global talent management: the roles of local employees' identification and climate strength. *Int J Human Res Manag* 2022; 1: 1–28.
- [65] Chang Y-W, Hsu P-Y, Shiao W-L, et al. Knowledge sharing intention in the United States and China: a cross-cultural study. *Eur J Inf Syst* 2015; 24: 262–277.
- [66] Sheeran P. Intention – behavior relations: a conceptual and empirical review. *Europ Rev Social Psychol* 2002; 12: 1–36.
- [67] Dai J, Zhang X and Wang C. A meta-analysis of learners' continuance intention toward online education platforms. *Educ Inf Technol* 2024; 4: 126547.
- [68] Cao R. *Research on the influencing factors and patterns of information sharing of scientific research team in new media environment*. Changchun, China: Jilin University, 2017.
- [69] Chen C-J and Hung S-W. To give or to receive? Factors influencing members' knowledge sharing and community promotion in professional virtual communities. *Inform Manag* 2010; 47: 226–236.
- [70] Chung N, Nam K and Koo C. Examining information sharing in social networking communities: applying theories of social capital and attachment. *Telemat Inform* 2016; 33: 77–91.
- [71] Sun ZL and Li HQ. Research on the theory of knowledge ecological system. *Libr Inform* 2008; 5: 22–27.
- [72] Guo Y, Wang X, Li J, et al. A study on information sharing behaviors of mobile learning users in the new media environment. *Libr Inform Serv* 2017; 61: 34–42.
- [73] Oldendick RW and Lambries DN. Incentives for cell phone only users: what difference do they make? *Surv Pract* 2011; 4: 0002.
- [74] Wang C, Dai J, Zhu K, et al. Understanding the continuance intention of college students toward new E-learning spaces based on an integrated model of the TAM and TTF. *Int J Human Comput Inter* 2023; 1: 1–14.
- [75] Tabachnick BG and Fidell LS. *Using multivariate statistics*: Pearson new international edition. London: Pearson, 2013.
- [76] Asoka M. Structural equation modelling with AMOS. Unpublished Pridobljeno 2015; 15: 2019.
- [77] Hair J, Hollingsworth CL, Randolph AB, et al. An updated and expanded assessment of PLS-SEM in information systems research. *Ind Manag Data Syst* 2017; 117: 442–458.
- [78] Fornell C and Larcker DF. Evaluating structural equation models with unobservable variables and measurement error. *J Market Res* 1981; 18: 39–50.
- [79] Henseler J, Ringle CM and Sarstedt M. A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J Acad Market Sci* 2015; 43: 115–135.
- [80] Roemer E, Schuberth F and Henseler J. HTMT2—an improved criterion for assessing discriminant validity in structural equation modeling. *Ind Manag Data Syst* 2021; 121: 2637–2650.
- [81] Congjie Y and Qiaoling D. A study on the impact of characteristics of company recruitment we chat on employer attractiveness based on TAM. *Manag Rev* 2016; 28: 140.
- [82] Lent RW, Brown SD and Hackett G. Contextual supports and barriers to career choice: a social cognitive analysis. *J Counsel Psychol* 2000; 47: 36.
- [83] Kmiecik R. Trust, knowledge sharing, and innovative work behavior: empirical evidence from Poland. *Europ J Innovat Manag* 2020; 24: 1832–1859.
- [84] Hsu M-H, Ju TL, Yen C-H, et al. Knowledge sharing behavior in virtual communities: the relationship between trust, self-efficacy, and outcome expectations. *Int J Human Comput Stud* 2007; 65: 153–169.
- [85] Wang J and Xie J. Exploring the factors influencing users' learning and sharing behavior on social media platforms. *Library Hi Tech* 2022; 41: 1436–1455.
- [86] Stibe A, Krüger N and Behne A. Knowledge behavior gap model: an application for technology acceptance. In: Awan I, Younas M and Poniszewska-marañda A (eds) *Mobile Web and Intelligent Information Systems*. Cham: Springer, 2022, pp. 3–17.
- [87] Alshahrani H and Rasmussen PD. 'How to use it more?' Self-efficacy and its sources in the use of social media for knowledge sharing. *J Document* 2019; 76: 231–257.
- [88] Bock GW and Kim Y-G. Breaking the myths of rewards: an exploratory study of attitudes about knowledge sharing. *IRMJ* 2002; 15: 14–21.
- [89] Tseng F-C and Kuo F-Y. A study of social participation and knowledge sharing in the teachers' online professional community of practice. *Comput Educ* 2014; 72: 37–47.

- [90] Constant D, Kiesler S and Sproull L. What's mine is ours, or is it? A study of attitudes about information sharing. *Inform Syst Res* 1994; 5: 400–421.
- [91] Zhang W, Bowers AJ and Pang S. Promoting teacher knowledge sharing in China: the fostering roles of authentic leadership, reciprocity norms, and organizational trust. *Educ Manag Admin Leader* 2023; 2023: 17411432231199573.
- [92] Heinz M and Rice RE. An integrated model of knowledge sharing in contemporary communication environments. *Ann Intern Commun Assoc* 2009; 33: 135–175.
- [93] Keren-Happuch FFE and Moon-Ho RH. Examining the influence of individual, social cognitive and environmental factors on students' entrepreneurial intentions: application of the social cognitive career theory (SCCT) framework within a multidimensional career space. In: Ho MR, Kennedy JC, Uy MA, et al. (eds) *Entrepreneurship–professionalism–leadership: a multidimensional framework for human capital and career development in the 21st century*. Singapore: Springer, pp. 295–324.
- [94] Janssen O, van de, Vliert E and West M. The bright and dark sides of individual and group innovation: a Special Issue introduction. *J Organiz Behav* 2004; 25: 129–145.
- [95] Ansari JAN and Khan NA. Exploring the role of social media in collaborative learning the new domain of learning. *Smart Learn Environ* 2020; 7: 9.
- [96] Chen IYL, Chen N-S and Kinshuk. Examining the factors influencing participants' knowledge sharing behavior in virtual learning communities. *Educ Tech Soc* 2009; 12: 134–148.

The current issue and full text archive of this journal is available on Emerald Insight at:
<https://www.emerald.com/insight/2046-6854.htm>

The role of teachers' direct and emotional mentoring in shaping undergraduates' research aspirations: a social cognitive career theory perspective

International
Journal of
Mentoring and
Coaching in
Education

Received 25 July 2023
Revised 4 December 2023
19 March 2024
7 June 2024
23 August 2024
Accepted 25 August 2024

Haojun Li and Jun Xu

College of Education, Zhejiang University of Technology, Hangzhou, China

Yuying Luo

College of Management, Zhejiang University of Technology, Hangzhou, China, and

Chengliang Wang

*Department of Education Information Technology, Faculty of Education,
East China Normal University, Shanghai, China*

Abstract

Purpose – This study investigated the influence of teachers on undergraduate students' development of research aspirations and the mechanisms behind this process.

Design/methodology/approach – Employing social cognitive career theory, the study gathered data from 232 undergraduates, developed a structural equation model via the maximum likelihood method and executed empirical testing.

Findings – The findings reveal that neither direct nor emotional mentoring independently satisfies students' needs for self-efficacy and aspiration in research nor significantly influences research interest. Specifically, the study demonstrates that (1) research self-efficacy, outcome expectations and research interest significantly shape research aspirations; (2) an overemphasis on direct mentoring might impede research aspiration development and (3) a focus on emotional mentoring, while overlooking direct mentoring, could result in diminished research self-efficacy.

Originality/value – This research pioneers a comprehensive analysis of the role of teachers in shaping undergraduate research aspirations through the lens of social cognitive career theory. It underscores the critical need to both balance mentoring approaches and foster intrinsic research motivation.

Keywords Undergraduate mentorship, Social cognitive career theory, Mentor guidance, Research aspiration, Research self-efficacy, Mentoring relationships

Paper type Research paper

Declaration of conflicting interest: The authors declare that they have no conflict of interest with respect to the research, authorship, and publication of this article.

Ethical approval: Ethical clearance was granted by the corresponding author's institution, ensuring participant anonymity and confidentiality after informed consent was obtained.

Data availability statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Author contributions: Haojun Li has made critical revisions to important content. Jun Xu came up with the initial idea and wrote the manuscript. Yuying Luo conducted typesetting and document sorting. Chengliang Wang provides guidance on writing. All authors read and approved the final version of the manuscript.

Funding information: This work was supported by the National Natural Science Foundation of China (Grant No. 62077043); the Key Support Project of Philosophy and Social Science Planning Interdisciplinary in Zhejiang Province (Grant No. 22JCXK05Z); and the Zhejiang Provincial College Students' Innovation and Entrepreneurship Training Program (Grant No. GZ22251080006).



International Journal of Mentoring
and Coaching in Education
© Emerald Publishing Limited
2046-6854
DOI 10.1108/IJMC-07-2023-0064

Introduction

The global rise in undergraduate education and overall education levels underscores the growing importance of undergraduate involvement in scientific research. Research has consistently demonstrated that such participation offers valuable learning experiences (Seymour *et al.*, 2004). A pivotal element in this context is the undergraduate mentoring system, wherein university educators and professionals guide students across academic, career, and psychological aspects, fostering comprehensive development (Jung, 2018; Karaferye, 2018). In this study, the mentors are academic advisors assigned to undergraduate students in China, akin to graduate thesis advisors, responsible for guiding students in competitions, research projects, academic progress, and especially in degree thesis writing. Recognized for enhancing educational quality, stimulating learning motivation, and fostering innovative capabilities, this system plays a crucial role in shaping research aspirations (Leavitt *et al.*, 2022).

However, the efficacy of current undergraduate mentoring systems, particularly in China, faces critique due to over-reliance on personal relationships and an emphasis on mentor authority, potentially stifling innovation (Shi, 2016; Zhou *et al.*, 2019). Most existing studies on mentoring systems focus on systemic design, often overlooking the intricacies of mentor–student interactions and their impact on research aspirations.

The mechanisms through which mentors influence student research aspirations require clearer understanding. Pfund *et al.* (2006) highlighted the vital role of mentorship in enhancing research quality and student growth, yet there is a lack of comprehensive theoretical and empirical guidance for effective mentoring strategies aimed at boosting research aspirations. Moreover, the interaction between mentorship and student motivation for research remains underexplored. For instance, Overall *et al.* (2011) observed a complex, conditional mentor–student relationship, suggesting that fostering autonomy and setting clear expectations are key to effective mentorship.

Contrastingly, mentoring at the undergraduate level is an emerging area of study, with previous research primarily centered on graduate-level research behavior. For example, McDaniel *et al.* (2022) showed significant enhancements in doctoral students' research autonomy under effective mentoring systems. This highlights a gap in understanding the specific mentoring needs and mechanisms at the undergraduate level.

In summary, this study explores how mentors influence undergraduate students' research aspirations. We use social cognitive career theory (SCCT) as a theoretical framework and examine the effects of mentors' direct research mentoring and emotional mentoring on undergraduate students' research self-efficacy, research interests, and research aspirations. This study targeted undergraduate students and used structural equation modeling to analyze the data, aiming to understand the factors and mechanisms that affect students' willingness to participate in research from both theoretical and empirical perspectives. To enhance the stimulation of mentoring and cultivate students' research interests and aspirations, we propose suggestions and strategies with reference to value and enlightening significance.

This research focuses on the intricate dynamics of mentorship in undergraduate research, seeking to elucidate its multifaceted impacts. The study centers on key questions to understand the mentorship process better:

What is the influence of direct research mentoring by mentors on undergraduates' research aspirations, self-efficacy, and interests?

How does emotional mentoring by mentors shape undergraduates' research aspirations, self-efficacy, and interests?

In what ways can mentorship be optimized to bolster undergraduates' research aspirations and encourage their active participation in research?

When applied to student research, do the principles of self-efficacy and outcome expectation within the SCCT continue to hold a significant influence on students' research aspirations?

The study's primary contributions lie in reevaluating the commonly held belief that mentorship unilaterally enhances student research motivation. It enriches our understanding of the situational factors within SCCT, confirming its relevance in social sciences and specifically in the context of Chinese culture. By providing empirical insights into undergraduate research practices in China, this research offers valuable recommendations for enhancing mentorship approaches in Chinese universities, aiming to amplify the research competencies and aspirations of undergraduate students.

Literature review

The origin and mechanism of mentorship

The inception of the undergraduate mentorship framework is anchored in the collaborative research endeavors between faculty and students. This partnership, characterized by educators' deep engagement with students in both academic and extracurricular settings, contributes to a conducive atmosphere that positively influences students' scholarly achievements and career progression (Davis and Jones, 2020). The mentorship program in undergraduate education plays a supportive role in nurturing students' research skills and encouraging their pursuit of advanced academic qualifications (Eagan *et al.*, 2013). Crucial to this evolution is the recognition of diverse mentor backgrounds, experiences, and mentoring dispositions, which profoundly impact the mentor-mentee dynamic and, subsequently, the students' research directions and aspirations (Pfund *et al.*, 2022).

Different mentorship styles, such as direct research guidance and emotional support, are crucial in shaping students' self-efficacy, metacognition, and research intentions (Medakovich, 2019). Direct research guidance includes mentors providing hands-on assistance with research activities, expertise in research design, methodology, and data analysis, and guidance through academic publishing complexities, helping students develop essential critical thinking and technical skills (Leavitt *et al.*, 2022). Emotional support involves mentors offering encouragement, building confidence, and providing a safe space for discussing academic and personal challenges (Hudson, 2013).

The influencing factors of undergraduate research aspiration

Research aspiration among undergraduates, a reflection of their inherent motivation and desire to engage in scholarly research, is governed by a blend of internal and external forces. Central to these forces, within the ambit of SCCT, are internal factors which include individual attributes – demographics such as gender, age, and education level – and psychological constructs encompassing interest and outcome expectations (Butz *et al.*, 2018). These components not only forge research aspirations but also shape students' propensity for research participation (Lopatto, 2004; Mogk, 1993). Externally, mentorship emerges as a pivotal element, influencing students' trajectories toward graduate education and scientific careers. This influence was initially identified by Deemer *et al.* (2014) and subsequently expanded upon in the later works of Deemer *et al.* (2020), underscoring its significance within the context of academic and professional development. This notion of mentorship as a crucial environmental influence or source of social support aligns with the SCCT framework and echoes the principles of self-determination theory, underscoring the salience of external

IJMCE

factors in steering students' academic and research-related behaviors (Palmer *et al.*, 2015; Saini *et al.*, 2020).

The evolution of social cognitive career theory

SCCT, proposed by Lent and colleagues in 1994, elucidates the interplay between academic interests and career choices and performance (Lent *et al.*, 1994). Anchored in Bandura's social cognitive framework, the SCCT synthesizes principles of social cognition with empirical findings on academic self-efficacy (Lent *et al.*, 2005). SCCT posits that self-efficacy beliefs, when intertwined with anticipated outcomes, play a pivotal role not only in directing students' educational and vocational decisions but also in shaping the genesis of interests, which in turn profoundly influences their career paths and performance (Wu *et al.*, 2022). This study focuses on external factors, specifically the direct and emotional guidance received by students. According to Lent and Brown (2019), SCCT suggests that external influences, such as mentorship and social support, enhance self-efficacy and outcome expectations, subsequently shaping students' interests, goals, and actions in their academic and career development (Lent and Brown, 2019). Mentorship, in particular, is crucial for providing academic counsel and emotional encouragement, essential for educational and professional advancement (Mendez *et al.*, 2017; Young *et al.*, 2020).

Research model and hypotheses

Research hypotheses

Research self-efficacy. This study delves into the concept of research self-efficacy, defined as the confidence level of undergraduate students in their ability to successfully engage in academic research activities. Originating from the seminal work of Bandura, a renowned psychologist and the progenitor of social learning theory, self-efficacy is conceptualized as an individual's confidence in navigating challenges across various domains, including academic, social, and health-related spheres. In the realm of academia and scientific inquiry, research self-efficacy emerges as a critical predictor of success, underscoring the importance of this social cognitive concept in the academic and scientific research field (Livingi *et al.*, 2021). The study's scale for assessing research self-efficacy is based on, and linguistically refined from, the instrument developed by Bieschke *et al.* (1996).

Research interest. In educational research, interest is often explored as a mediating factor between self-efficacy and career intentions within the framework of SCCT. This concept, strongly influenced by individual learning experiences, has been consistently validated in prior studies (Lent *et al.*, 1994). Building on this, the current study adapts the SCCT model's concept of career interest to examine research interest in undergraduates. It is noteworthy that, unlike doctoral students who often require emotional rapport with their supervisors, undergraduates, due to their relative lack of research experience, are more reliant on direct guidance from their mentors (Kim *et al.*, 2018). This dynamic suggests that intimacy might play a lesser role in shaping their self-efficacy, and they might benefit more significantly from direct instruction and feedback from their supervisors (Zimmerman *et al.*, 1992). In this context, research interest is conceptualized as the individual's intrinsic curiosity and strong aspiration to engage in and further scientific activities.

Outcome expectation. Outcome expectation, a concept integral to understanding human behavior, is defined as the belief regarding the consequences or rewards ensuing from participation in specific activities or attainment of particular goals (Lent *et al.*, 1994). Golub *et al.* (2009) described it as a mental state of preparedness for outcomes, influencing individuals' emotional responses in task completion. Further exploring this concept, some scholars have investigated the impact of self-efficacy on interest development and

performance through experimental studies. Notably, they discovered that self-efficacy not only elevates the level of interest but also enhances its stability and persistence (Silvia, 2003). This relationship implies a positive correlation between self-efficacy and interest, a notion supported in the realm of research. For instance, Sawitri and Creed (2021) applied SCCT to analyze the interplay between self-efficacy, outcome expectation, and research interest. Their findings elucidated that research self-efficacy and outcome expectation can indirectly foster the development of research interest via research motivation.

Research aspiration. Research aspiration, a concept integral to understanding academic motivation, is defined as the intrinsic motivation or desire of undergraduates to engage in research projects. This concept is rooted in the principles of SCCT, which itself is an extension of Bandura's social cognitive theory. SCCT conceptualizes aspiration as the determination to undertake specific activities or achieve future outcomes, with a focus on the selection of career goals (Bandura, 1986). This study adapts this framework to examine the research aspirations of undergraduate students.

Within the SCCT framework, a critical interplay exists between self-efficacy, outcome expectation, and goal systems. When applied to goal selection, both self-efficacy and outcome expectation are seen to directly influence the choice of goals (Bandura, 1986; Lent *et al.*, 1994; Maddux *et al.*, 1982). Byars-Winston and Rogers (2019) research, employing structural equation modeling, indicated varying impacts of self-efficacy on research aspiration across different racial and ethnic groups, particularly in career and STEM fields. This variability underscores the need for further exploration into how these dynamics translate into the selection of research aspirations. Consequently, this study proposes the following hypotheses:

H1. Self-efficacy positively affects research aspiration.

H2. Outcome expectation positively affects research aspiration.

Furthermore, Lent's experimental findings suggest that self-efficacy indirectly influences goal selection through interest (Lent *et al.*, 1994). While these findings were initially applied to career choice, the applicability of this relationship in more nuanced academic contexts warrants further investigation. Therefore, this study also hypothesizes:

H3. Research interest positively affects research aspiration.

Mentor emotional mentoring and mentor direct mentoring. Within the framework of SCCT, contextual factors encompass a range of external influences that can either facilitate or impede individual agency, including family, societal, cultural, economic, and political elements. These factors, while impacting self-efficacy and interest, also interact with personal goals and behaviors. Yet, the specifics of these mechanisms remain underexplored (Lent *et al.*, 2005).

Mentoring, in the context of student research aspirations, is an influential environmental factor (Byars-Winston and Rogers, 2019). Easley and Wang (2017) highlighted the mentor's role as a critical social situational factor influencing student entrepreneurial behavior, even substituting parental influence in some cases. This study integrates mentorship – both direct research mentoring and emotional mentoring – into the SCCT's contextual factors, assessing their impact on student research aspirations.

Research on direct mentoring, such as Lopatto's (2007) survey of 2021 undergraduates, underscores its significance in enhancing the undergraduate research experience and fostering research interest. Similarly, Overall *et al.* (2011) found that academic mentors, through resource provision and process planning, can bolster students' research self-efficacy.

IJMCE

Regarding emotional mentoring, studies like [Armyanti's \(2020\)](#) emphasize mentors' role in boosting students' learning motivation and fostering independence. [Haeger and Fresquez's \(2016\)](#) research at California State University Monterey Bay further suggests the equal importance of emotional and cultural mentoring in developing research competencies.

However, mentorship as an external factor also has potential drawbacks. Excessive direct mentoring, detached mentor–student relationships, and untrained mentors might diminish its effectiveness or negatively impact research self-efficacy and interest ([Overall et al., 2011](#)).

Drawing from these findings, this study utilizes the SCCT model to probe into the mentor's influence as a contextual factor on students' research self-efficacy, interest, and aspiration. To address the research gap in this domain, the study posits the following hypotheses:

Impact of direct research mentoring

- H4.* Mentor direct research mentoring positively impacts students' research self-efficacy.
- H5.* Mentor direct research mentoring positively impacts students' research outcome expectation.
- H6.* Mentor direct research mentoring positively impacts students' research interest.

Impact of emotional mentoring

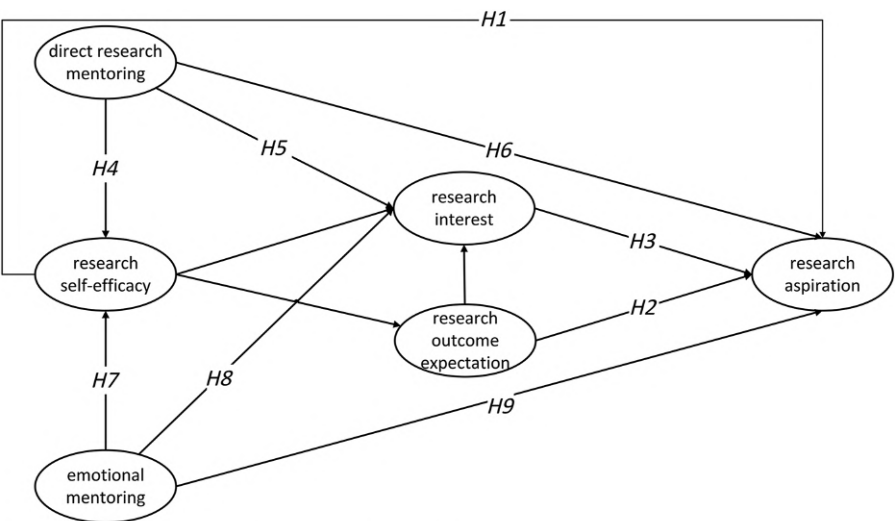
- H7.* Mentor emotional mentoring positively impacts students' research self-efficacy.
- H8.* Mentor emotional mentoring positively impacts students' research outcome expectation.
- H9.* Mentor emotional mentoring positively impacts students' research interest.

Conceptual model

This study constructs a conceptual model grounded in SCCT to investigate the influence of both direct and emotional mentoring on undergraduate research endeavors, as illustrated in [Figure 1](#). The model integrates four key internal factors – research self-efficacy, outcome expectations, research interest, and research aspiration – along with two pivotal contextual factors: mentor direct mentoring and emotional mentoring. It posits that these mentoring types exert direct effects on students' research self-efficacy, interest, and aspiration. Concurrently, the model hypothesizes that students' research self-efficacy and outcome expectations indirectly influence their research interest and aspiration. Furthermore, it identifies research interest as a critical mediating variable that bridges the gap between mentorship styles and the internal factors influencing student researchers.

Questionnaire design

In refining the research methodology, this study developed a structured questionnaire using a 7-point Likert scale, guided by established survey methodology practices ([Nworie et al., 2023](#)). This approach not only ensured accurate measurement of the constructs but also provided comprehensive data for structural equation modeling (SEM). The questionnaire



Source(s): Created by author Xu Jun 2024-08-24

Figure 1.
Initial structural
equation model

comprised 30 items: five items capturing basic demographic and academic information, 15 items focused on measuring aspects such as self-efficacy, outcome expectations, research interest, and research aspirations, and an additional 10 items specifically targeting direct research mentoring and emotional mentoring. This design was formulated to align with the SCCT framework. Survey participants were informed that in Chinese higher education, mentors are academic advisors guiding students in research projects, academic progress, and degree thesis writing throughout their undergraduate studies. The instrument’s development integrated insights from validated scales (Bieschke *et al.*, 1996; Gibbons and Borders, 2010; Overall *et al.*, 2011; Vaccaro, 2009; Zhang, 2013) and incorporated cultural considerations for enhanced contextual relevance and integrating cultural factors; the specific content of the scales can be found in the appendix. A pilot test of 102 undergraduates showed Cronbach’s alpha values between 0.871 and 0.939, exceeding 0.8 and indicating high reliability for formal distribution. The choice of a questionnaire-based method was driven by its proven efficacy in eliciting diverse and nuanced perspectives on the mentor–mentee relationship within the academic setting.

Data analysis

Data collection

To test the above hypotheses, this study conducted an online survey in 2023 through the Questionnaire Star platform, targeting students engaged in regular undergraduate education. A total of 242 valid questionnaires were collected, with an effective response rate of 88.6%. Following Höhne and Schlosser’s (2017) suggestions for online surveys, questionnaires that took less than 80 s or more than 10 min to complete, or that showed serious inconsistency, were excluded. Regarding the sample size, the selection of 242 respondents was deemed appropriate for SEM analysis, especially considering the complexity of the model with 24 test items. This sample size aligns with recommendations by Chigwende and Govender (2020) for SEM, where they emphasize the importance of a sufficient sample size to achieve reliable model estimations in complex

IJMCE

analyses. Furthermore, the reliability of the survey data was ensured through various tests, including internal consistency checks and exploratory factor analysis, supporting the validity of the SEM approach employed in this study.

Regarding the demographic characteristics of the respondents, this paper conducted a survey based on the gender, discipline category, learning stage, and research experience of undergraduate students. The results are shown in [Table 1](#).

Data inspection

AMOS 24.0 software was selected for its sophisticated SEM capabilities, enabling robust maximum likelihood estimation and confirmatory factor analysis. This choice is supported by academic research, such as [Shi et al.'s \(2021\)](#) study, demonstrating its efficacy in handling complex models and ensuring the reliability of SEM results.

Reliability test. Reliability reflects the consistency or stability of questionnaire measurement results. This study extracted the path factor loadings obtained by the AMOS 24.0 measurement. There is no unified standard for its acceptable range, but relevant research suggests that higher than 0.5 is an acceptable standard, and higher than 0.7 is considered a clear structural goal, indicating that there is a very strong correlation between the observed variables and the latent variables ([Hair et al., 2017](#)). [Table 2](#) displays factor loadings for each survey item, all exceeding 0.75, affirming the robust reliability of the individual test items in our research instrument. This high level of reliability indicates that the measured constructs are consistently represented across the sample.

The composite reliability (CR) value and average variance extracted (AVE) value were calculated by SPSS. According to [Fornell and Larcker's \(1981\)](#) study, the CR value should be greater than 0.7 and the AVE value should be greater than 0.5. [Table 3](#) presents the AVE and CR values for each construct, demonstrating that all exceed the thresholds of 0.5 and 0.7, respectively. These metrics affirm the questionnaire's strong internal consistency and the reliability of our constructs, underpinning the soundness of our research findings.

Table 1.
Summary of
respondents'
demographic
characteristics

Attribute	Group	Number of participants	Percentage of total sample
Gender	Male	124	53.4%
	Female	108	46.6%
Academic level	Freshman	50	21.6%
	Sophomore	76	32.8%
	Junior	62	26.7%
	Senior or above	44	19.0%
Discipline category	Philosophy	5	2.2%
	Economics	30	12.9%
	Engineering	37	15.9%
	Science	33	14.2%
	Education	33	14.2%
	Management	24	10.3%
	Jurisprudence	12	5.2%
	Medical science	12	5.2%
	Agronomy	15	6.5%
	Art	10	4.3%
	Literature	16	6.9%
	History	5	2.2%
Research experience	Yes or have	157	67.7%
	No or without	75	32.3%

Source(s): Authors' own creation

International
Journal of
Mentoring and
Coaching in
Education

Variable	Measurement	Factor loading
Research self-efficacy	RS1	0.822
	RS2	0.826
	RS3	0.836
	RS4	0.770
Research outcome expectation	RO1	0.773
	RO2	0.792
	RO3	0.834
Research interest	RI1	0.771
	RI2	0.824
	RI3	0.778
	RI4	0.795
Research aspiration	RA1	0.750
	RA2	0.759
	RA3	0.870
	RA4	0.781
Direct research mentoring	DM1	0.784
	DM2	0.809
	DM3	0.839
	DM4	0.817
	DM5	0.796
Emotional mentoring	EM1	0.850
	EM2	0.817
	EM3	0.828
	EM4	0.814
	EM5	0.843

Source(s): Authors' own creation

Table 2.
Results of factor
analysis

Variable	Cronbach's alpha	CR	AVE
Research self-efficacy	0.837	0.887	0.662
Research outcome expectation	0.727	0.842	0.640
Research interest	0.832	0.871	0.628
Research aspiration	0.820	0.870	0.626
Research direct mentorship	0.885	0.905	0.655
Emotional mentoring	0.902	0.905	0.655

Source(s): Authors' own creation

Table 3.
Reliability and validity
analysis results: CR,
Cronbach's alpha,
and AVE

Validity test. For the validity judgment, there are the following criteria: if a variable's correlation coefficient with other variables is less than the square root of its AVE, it indicates that the variable has good discriminant validity (Fornell and Larcker, 1981). As demonstrated in Table 4, the square roots of the AVE values, placed diagonally, surpass inter-construct correlation coefficients, validating the discriminant validity of our constructs. This ensures that each construct is distinctly measured and sufficiently independent from others within our model.

HTMT test. In addition, this study used heterotrait-monotrait (HTMT) ratio to avoid potential multicollinearity problems in latent variables, which is the ratio of between-construct and within-construct correlations (Henseler et al., 2015), which may affect the estimation of path coefficients and model fit. In addition, HTMT testing can provide more accurate results than other traditional methods (such as Fornell-Larcker criterion or cross-

IJMCE

	Research self-efficacy	Research outcome expectation	Research interest	Research aspiration	Research direct mentorship	Emotional mentoring
Research self-efficacy	0.831					
Research outcome expectation	0.744	0.810				
Research interest	0.423	0.370	0.792			
Research aspiration	0.328	0.410	0.534	0.793		
Research direct mentorship	0.186	0.214	0.443	0.503	0.800	
Emotional mentoring	0.296	0.453	0.691	0.471	0.353	0.815

Table 4. Discriminant validity

Source(s): Authors' own creation

loading), especially when the measurement model is homogeneous (Roemer *et al.*, 2021). Henseler *et al.*'s (2015) study showed that a HTMT value of 0.85 is a more conservative and strict threshold that can effectively detect discriminant validity problems. Table 5 elucidates that the highest HTMT value observed in our study is 0.831, aligning with stringent criteria for the HTMT assessment. This result underscores the model's reliability and the distinctiveness of the constructs, reinforcing the validity of our analytical framework.

Robustness test. Liu *et al.* (2021) verified the robustness of their model of facial recognition and privacy research in China by adding demographic control variables and comparing the changes in paths before and after. To test the robustness of the model in this study, we selected the grade, gender, major, institution, and past research experience of students as control variables and analyzed them using AMOS software. The robustness test outcomes in Table 6 indicate that after incorporating control variables, the path coefficients and fit

	Research self-efficacy	Research outcome expectation	Research interest	Research aspiration	Research direct mentorship	Emotional mentoring
Research self-efficacy						
Research outcome expectation	0.353					
Research interest	0.471	0.502				
Research aspiration	0.694	0.448	0.536			
Research direct mentorship	0.452	0.210	0.414	0.376		
Emotional mentoring	0.293	0.179	0.328	0.428	0.744	

Table 5. Differential validity (HTMT)

Source(s): Authors' own creation

Path/indicator	Before added	After added
DM → RS	0.522***	0.522***
DM → RI	0.184	0.184
DM → RA	−0.246***	−0.239***
EM → RS	−0.089	−0.089
EM → RI	0.056	0.056
EM → RA	0.348***	0.346***
RS → RA	0.568***	0.563***
RO → RA	0.138**	0.140**
RI → RA	0.186**	0.193**
RSMEA	0.410	0.410
χ^2/df	1.395	1.393
CFI	0.972	0.961
TLI	0.968	0.957

Note(s): *** At the 0.01 level (double tail), the correlation is highly significant

** At the level 0.05 (double tail), the correlation is significant

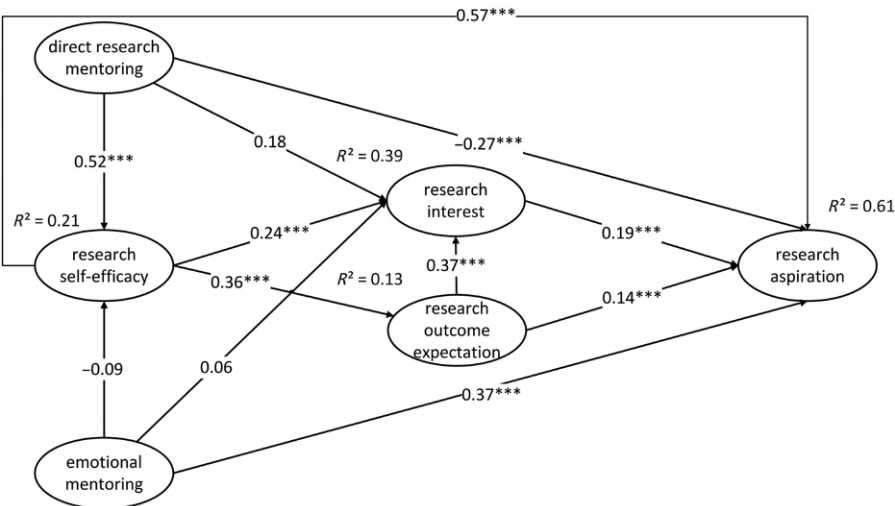
Source(s): Authors' own creation

Table 6.
Robustness test

indices remain substantially unchanged. This stability attests to the model's high resilience and reliability, suggesting that the identified relationships hold consistently across different contextual backgrounds.

Structural model

This study utilized the maximum likelihood estimation method to validate the structural model, delineating the explained variance (R2 values) for each construct within our model through AMOS 24.0 software, as depicted in Figure 2. Specifically, the R2 values for research



Note(s): ** At the 0.05 level (double-tailed), the correlation is significant

*** At the 0.01 level (double-tailed), the correlation is highly significant

Source(s): Created by author Xu Jun 2024-08-24

Figure 2.
Structural model

IJMCE

self-efficacy, outcome expectation, research interest, and research aspiration are 0.21, 0.13, 0.39, and 0.61, respectively, illustrating the substantial explanatory power of our model, particularly for research aspiration. The context of these R2 values suggests that our model effectively captures the key determinants influencing undergraduates' research aspirations and self-efficacy. Comparatively, these R2 figures align well with benchmarks from other studies applying SCCT in different contexts, such as [Lent et al. \(2005\)](#) and [Mohd Rasdi and Ahrari \(2020\)](#), thereby affirming our model's robustness. Notably, the variations in R2 for research self-efficacy and outcome expectation underscore the nuanced impact of individual and contextual factors within our theoretical framework, reinforcing the model's relevance and applicability in understanding the dynamics of research aspirations among undergraduates.

This study tested hypotheses [H1](#) to [H9](#) and found that [H1](#) to [H4](#) and [H9](#) were supported, while [H6](#) was contrary to expectations. Specifically, research self-efficacy ($\beta = 0.57$) and outcome expectation ($\beta = 0.14$) were significantly positively correlated with research aspiration ([H1–H2](#)), and research interest was also significantly positively correlated with research aspiration ($\beta = 0.19$) ([H3](#)).

For the situational perspective, direct research mentoring from mentors was significantly positively correlated with research self-efficacy ($\beta = 0.52$) ([H4](#)), but not related to research interest ([H5](#)), and was significantly negatively correlated with research aspiration ($\beta = -0.27$) ([H6](#)). Emotional mentoring from mentors was only significantly positively correlated with research aspiration ($\beta = 0.37$) ([H7](#)) but not related to research self-efficacy or research interest ([H8–H9](#)).

Discussion

Result discussion

The impact of internal factors on research aspiration. This investigation revealed a notable and substantial link between research self-efficacy, aspirations, and interests, substantiating the relevancy of SCCT's internal constructs within undergraduate research paradigms, predominantly concerning self-efficacy and outcome expectations. Notably, the influence of self-efficacy on research aspirations outstripped that of outcome expectations, likely attributable to self-efficacy's closer alignment with students' tangible behaviors. Conversely, outcome expectations might seem comparatively abstract or removed, potentially elucidating the observed direct and robust impact of self-efficacy on research aspirations within our SCCT-aligned structural model ([Lent et al., 2003](#)). This alignment underscores the imperative role of self-efficacy in fostering concrete research aspirations among undergraduates, as demonstrated in our findings.

The impact of direct mentoring on research aspiration and self-efficacy. This investigation delineates the nuanced influence of direct mentoring on undergraduates, identifying a nuanced dynamic where direct mentoring significantly augments research self-efficacy but, paradoxically, may attenuate research aspirations. These findings unfold through multiple dimensions.

Augmentation of self-efficacy: The study illustrates that direct mentoring enriches undergraduates' research self-efficacy via targeted feedback, methodical guidance, and demonstrative skill modeling, echoing findings from [Bieschke et al. \(1996\)](#) and [Chemers et al. \(2001\)](#). Such fortified self-efficacy potentially uplifts their research engagement, validated by established links to academic commitment ([Petko et al., 2020](#)).

Dampening of intrinsic motivation: Concurrently, our analysis suggests that while direct mentoring fosters competence, it might inadvertently diminish intrinsic motivation for research, a vital component elucidated by self-determination theory ([Ryan and Deci, 2000](#)).

Such a mentoring approach risks overlooking the essential elements of autonomy and connectivity, integral for nurturing enduring intrinsic motivation.

Risk of over-intervention: The findings further delineate risks associated with excessive mentor oversight, which could undermine students' autonomous drive, aligning with insights from [Pelletier et al. \(2002\)](#). This excessive guidance may inadvertently stifle long-term research engagement, manifesting a critical dichotomy within the mentoring paradigm.

The influence of emotional mentoring on research aspiration and research self-efficacy. This inquiry delineates the nuanced repercussions of emotional mentoring on undergraduates, spotlighting a substantial enhancement in research aspirations without a corresponding increase in self-efficacy. The bifurcated outcomes of this mentoring style are elaborated as follows.

Emotional support versus skill acquisition: Emotional mentoring's forte lies in sculpting a nurturing environment, akin to the solidarity observed in peer cohorts, which, while alleviating stress, falls short in fortifying undergraduates' self-assessment of their research competencies. This mentoring style, devoid of the direct skill transfer seen in direct mentoring, underscores the necessity for a balanced mentorship approach that integrates both emotional support and skill development to optimize research skill augmentation and passion ([Betz and Voyten, 1997](#); [Zimmerman et al., 1992](#)).

Challenges within the mentorship framework: Further scrutiny within this research reveals systemic hurdles embedded in the mentorship schema, like ill-defined goals and suboptimal student involvement ([Lambert et al., 2022](#)), that might attenuate the potency of emotional mentoring in bolstering self-efficacy.

Conversely, the pronounced enhancement of research aspirations through emotional mentoring underscores its indispensable role in engendering a conducive and engaging mentor-mentee nexus, pivotal for nurturing steadfast commitment and identity formation within the academic arena ([Hernandez et al., 2013](#); [Kram and Isabella, 1985](#)). This segment of mentorship emerges as a linchpin in sculpting students' professional identities and competencies, prerequisites for their enduring involvement in scholarly research ([Hudson, 2013](#)).

Assessment of mentorship's impact on research interest. This analysis delineates that neither direct nor emotional mentoring significantly alters undergraduates' research interest, reflecting its inherent stability as a deeply ingrained personal attribute. This study supports the idea that mentorship provides essential support but does not directly reshape the foundational aspects of research interest, which is consistent with [Deemer et al.'s \(2020\)](#) findings. Our investigation confirms that research self-efficacy is a crucial factor in maintaining research interest, which aligns with the views presented by [Renninger and Hidi \(2015\)](#). In alignment with the foundational principles of SCCT, the empirical evidence we garnered suggests that intrinsic determinants such as self-efficacy and outcome expectations predominantly orchestrate research interest. Concurrently, while mentorship contributes an ancillary and indirect role, it is the internal constructs that predominantly sculpt research interest, as posited by [Lent \(2004\)](#).

Revelation

Addressing challenges in China's undergraduate mentorship. In addressing the complexities of China's undergraduate mentorship system, significant challenges arise from resource constraints, cultural dynamics, systemic structures, and prevailing mindsets. Key issues identified include ineffective mentor-student communication, unrealistic mentor expectations, undue mentorship pressure, and students' diminished trust and respect towards mentors, coupled with their often vague or passive academic goals ([Zhou et al., 2019](#)). Navigating this landscape necessitates undergraduates to possess advanced cognitive

IJMCE

skills, robust emotional support, and a solid sense of research self-efficacy. Mentors, therefore, play pivotal roles, not only as academic facilitators providing guidance but also as emotional supporters enhancing student resilience (Medaille *et al.*, 2022). However, the research findings indicate that neither direct nor emotional mentoring alone suffices in meeting the comprehensive needs for students' research self-efficacy and aspirations.

In light of these insights, this study proposes several actionable recommendations:

- (1) Educational institutions should implement targeted mentor training programs, augment faculty resources, and guarantee substantial direct mentoring.
- (2) Mentors are encouraged to reevaluate and move beyond the preconception that nurturing undergraduates yields limited academic returns, instead fostering patience and encouragement to develop the potential academic leaders of tomorrow (Tsang, 2018).
- (3) There is a need for mentors to cultivate students' autonomy and sense of belonging within the academic community, which is instrumental in generating genuine motivation for research endeavors.

Balancing direct mentoring. The study's findings highlight the pivotal role of direct research mentoring in enhancing undergraduates' research self-efficacy. To optimize its benefits, several strategies are recommended:

- (1) Educational institutions should actively promote and value direct mentorship as a means to bolster students' research capabilities and confidence.
- (2) Mentors are advised to provide comprehensive guidance encompassing feedback on research methodologies, theoretical underpinnings, assistance in research planning and proposal development, and instruction in academic writing standards.

However, the study also indicates that excessive direct mentoring may inadvertently dampen students' research aspirations. In addressing this concern:

- (1) Educational institutions should establish clear guidelines and offer regular training to mentors, ensuring balanced involvement in student research projects.
- (2) Mentors should aim to achieve a delicate balance in their guidance, allowing sufficient intellectual space for students to engage in independent thought and exploration.

Emotional assistance cultivation. This study underscores the role of positive mentor–student relationships in fostering undergraduates' research engagement. To harness this potential, the following measures are advocated:

- (1) Educational institutions should formulate initiatives to nurture positive mentor–student dynamics, such as organizing seminars aimed at encouraging undergraduate research involvement.
- (2) Mentors should receive training to adopt a personalized approach for undergraduates, considering their backgrounds and specific needs. Balancing emotional support with academic guidance is crucial to avoid diminishing research self-efficacy, ensuring a distinct approach from that used with graduate students.
- (3) Providing ample research resources, alongside additional support such as mechanisms for course credit acquisition, can substantially lower students' opportunity costs and boost their research participation (Khoo, 2018).

- (4) Undergraduates should proactively engage in long-term research planning, recognizing the significance of their involvement, initiating projects independently, and considering mentors as facilitators rather than primary decision-makers in their research endeavors (Wang, 2019).

Limitations

The primary limitation identified in this study pertains to its sample concentration within South China, which potentially restricts the extrapolation of our results. Such geographical specificity raises concerns regarding the universality of our findings across disparate educational contexts, thus narrowing the scope of their interpretability. Moreover, the investigation's delineation of mentoring paradigms and the nuanced interplay between mentors and mentees – crucial for elucidating the mentorship's influence on students' research aspiration – lacks comprehensive detail. This omission restricts the breadth of our conclusions and fails to encompass the diversity of mentoring approaches documented within our sample. While the research offers substantive insights into the mentorship's role in bolstering undergraduate research motivation, its applicability remains confined to the investigated context, given the scant examination of mentor characteristics and practices. Future inquiries should endeavor to meticulously delineate these aspects, examining the spectrum of mentorship styles and their potential variations across educational landscapes. Augmenting the SCCT framework to incorporate a nuanced appraisal of mentor roles would significantly enrich our comprehension of mentorship efficacy and, in turn, fortify the external validity of this research, thereby amplifying its academic contribution to the educational domain.

References

- Armyanti, I. (2020), "The role of academic advisors in improving medical students' motivation: a case study", *Jurnal Pendidikan Kedokteran Indonesia: The Indonesian Journal of Medical Education*, Vol. 9 No. 1, p. 60, doi: [10.22146/jpki.45064](https://doi.org/10.22146/jpki.45064).
- Bandura, A. (1986), *Social Foundations of Thought and Action: A Social Cognitive Theory*, Prentice-Hall, Englewood Cliffs, NJ, pp. xiii-617.
- Betz, N.E. and Veyten, K.K. (1997), "Efficacy and outcome expectations influence career exploration and decidedness", *The Career Development Quarterly*, Vol. 46 No. 2, pp. 179-189, doi: [10.1002/j.2161-0045.1997.tb01004.x](https://doi.org/10.1002/j.2161-0045.1997.tb01004.x).
- Bieschke, K.J., Bishop, R.M. and Garcia, V.L. (1996), "The utility of the research self-efficacy scale", *Journal of Career Assessment*, Vol. 4 No. 1, pp. 59-75, doi: [10.1177/106907279600400104](https://doi.org/10.1177/106907279600400104).
- Butz, A.R., Byars-Winston, A., Leverett, P., Branchaw, J. and Pfund, C. (2018), "Promoting STEM trainee research self-efficacy: a mentor training intervention", *UI Journal*, Vol. 9 No. 1, available at: <https://www.understandinginterventionsjournal.org/article/3730-promoting-stem-trainee-research-self-efficacy-a-mentor-training-intervention>
- Byars-Winston, A. and Rogers, J.G. (2019), "Testing intersectionality of race/ethnicity $\times$ gender in a social-cognitive career theory model with science identity", *Journal of Counseling Psychology*, Vol. 66 No. 1, pp. 30-44, doi: [10.1037/cou0000309](https://doi.org/10.1037/cou0000309).
- Chemers, M.M., Hu, L. and Garcia, B.F. (2001), "Academic self-efficacy and first year college student performance and adjustment", *Journal of Educational Psychology*, Vol. 93 No. 1, pp. 55-64, doi: [10.1037/0022-0663.93.1.55](https://doi.org/10.1037/0022-0663.93.1.55).
- Chigwende, S. and Govender, K. (2020), "Corporate brand image and switching behavior: case of mobile telecommunications customers in Zimbabwe", *Innovative Marketing*, Vol. 16 No. 2, pp. 80-90, doi: [10.21511/im.16\(2\).2020.07](https://doi.org/10.21511/im.16(2).2020.07).

IJMCE

- Davis, S. and Jones, R. (2020), "The genesis, evolution, and influence of undergraduate research mentoring relationships", *International Journal for the Scholarship of Teaching and Learning*, Vol. 14 No. 1, doi: [10.20429/ijstol.2020.140106](https://doi.org/10.20429/ijstol.2020.140106).
- Deemer, E.D., Thoman, D.B., Chase, J.P. and Smith, J.L. (2014), "Feeling the threat: stereotype threat as a contextual barrier to women's science career choice intentions", *Journal of Career Development*, Vol. 41 No. 2, pp. 141-158, doi: [10.1177/0894845313483003](https://doi.org/10.1177/0894845313483003).
- Deemer, E.D., Navarro, R.L., Byars-Winston, A.M., Jensen, L.E. and Chen, C.P. (2020), "Investigating graduate education and undergraduate research intentions of college science students", *Journal of Career Assessment*, Vol. 28 No. 1, pp. 43-58, doi: [10.1177/1069072718823777](https://doi.org/10.1177/1069072718823777).
- Eagan, M.K. Jr, Hurtado, S., Chang, M.J., Garcia, G.A., Herrera, F.A. and Garibay, J.C. (2013), "Making a difference in science education: the impact of undergraduate research programs", *American Educational Research Journal*, Vol. 50 No. 4, pp. 683-713, doi: [10.3102/0002831213482038](https://doi.org/10.3102/0002831213482038).
- Eesley, C. and Wang, Y. (2017), "Social influence in career choice: evidence from a randomized field experiment on entrepreneurial mentorship", *Research Policy*, Vol. 46 No. 3, pp. 636-650, doi: [10.1016/j.respol.2017.01.010](https://doi.org/10.1016/j.respol.2017.01.010).
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50, doi: [10.2307/3151312](https://doi.org/10.2307/3151312).
- Gibbons, M.M. and Borders, L.D. (2010), "Prospective first-generation college students: a social-cognitive perspective", *The Career Development Quarterly*, Vol. 58 No. 3, pp. 194-208, doi: [10.1002/j.2161-0045.2010.tb00186.x](https://doi.org/10.1002/j.2161-0045.2010.tb00186.x).
- Golub, S.A., Gilbert, D.T. and Wilson, T.D. (2009), "Anticipating one's troubles: the costs and benefits of negative expectations", *Emotion*, Vol. 9 No. 2, pp. 277-281, doi: [10.1037/a0014716](https://doi.org/10.1037/a0014716).
- Haeger, H. and Fresquez, C. (2016), "Mentoring for inclusion: the impact of mentoring on undergraduate researchers in the sciences", *CBE-Life Sciences Education*, Vol. 15 No. 3, p. ar36, doi: [10.1187/cbe.16-01-0016](https://doi.org/10.1187/cbe.16-01-0016).
- Hair, J., Hollingsworth, C.L., Randolph, A.B. and Chong, A.Y.L. (2017), "An updated and expanded assessment of PLS-SEM in information systems research", *Industrial Management and Data Systems*, Vol. 117 No. 3, pp. 442-458, doi: [10.1108/IMDS-04-2016-0130](https://doi.org/10.1108/IMDS-04-2016-0130).
- Henseler, J., Ringle, C.M. and Sarstedt, M. (2015), "A new criterion for assessing discriminant validity in variance-based structural equation modeling", *Journal of the Academy of Marketing Science*, Vol. 43 No. 1, pp. 115-135, doi: [10.1007/s11747-014-0403-8](https://doi.org/10.1007/s11747-014-0403-8).
- Hernandez, P.R., Schultz, P., Estrada, M., Woodcock, A. and Chance, R.C. (2013), "Sustaining optimal motivation: a longitudinal analysis of interventions to broaden participation of underrepresented students in STEM", *Journal of Educational Psychology*, Vol. 105 No. 1, pp. 89-107, doi: [10.1037/a0029691.supp](https://doi.org/10.1037/a0029691.supp).
- Höhne, J.K. and Schlosser, S. (2017), "Investigating the adequacy of response time outlier definitions in computer-based web surveys using Paradata SurveyFocus", *Social Science Computer Review*, Vol. 36 No. 3, pp. 369-378, doi: [10.1037/a0029691.supp](https://doi.org/10.1037/a0029691.supp).
- Hudson, P. (2013), "Developing and sustaining successful mentoring relationships", *Journal of Relationships Research*, Vol. 4, p. e1, doi: [10.1017/jrr.2013.1](https://doi.org/10.1017/jrr.2013.1).
- Jung, J. (2018), "Learning experience and perceived competencies of doctoral students in Hong Kong", *Asia Pacific Education Review*, Vol. 19 No. 2, pp. 187-198, doi: [10.1007/s12564-018-9530-0](https://doi.org/10.1007/s12564-018-9530-0).
- Karaferye, F. (2018), "Student mentoring models: a systematic review", *Konferencia-előadás*, doi: [10.1145/3243157.3243170](https://doi.org/10.1145/3243157.3243170).
- Khoo, S.Y.-S. (2018), "How to make undergraduate research worthwhile", *Nature*, doi: [10.1038/d41586-018-07427-5](https://doi.org/10.1038/d41586-018-07427-5).
- Kim, E., Benson, S. and Alhaddab, T.A. (2018), "A career in academia? Determinants of academic career aspirations among PhD students in one research university in the US", *Asia Pacific Education Review*, Vol. 19 No. 2, pp. 273-283, doi: [10.1007/s12564-018-9537-6](https://doi.org/10.1007/s12564-018-9537-6).

-
- Kram, K.E. and Isabella, L.A. (1985), "Mentoring alternatives: the role of peer relationships in career development", *Academy of Management Journal*, Vol. 28 No. 1, pp. 110-132, doi: [10.2307/256064](https://doi.org/10.2307/256064).
- Lambert, W.M., Nana, N., Afonja, S., Saeed, A., Amado, A.C. and Golightly, L.M. (2022), "Addressing structural mentoring barriers in postdoctoral training: a qualitative study", *bioRxiv*, 23 August, doi: [10.1101/2022.08.20.504665](https://doi.org/10.1101/2022.08.20.504665).
- Leavitt, A.S., Nelson, K.L. and Cutucache, C.E. (2022), "The effect of mentoring on undergraduate mentors: a systematic review of the literature", *Frontiers in Education*, Vol. 6, doi: [10.3389/feduc.2021.731657](https://doi.org/10.3389/feduc.2021.731657).
- Lent, R.W. (2004), "Toward a unifying theoretical and practical perspective on well-being and psychosocial adjustment", *Journal of Counseling Psychology*, Vol. 51 No. 4, pp. 482-509, doi: [10.1037/0022-0167.51.4.482](https://doi.org/10.1037/0022-0167.51.4.482).
- Lent, R.W. and Brown, S.D. (2019), "Social cognitive career theory at 25: empirical status of the interest, choice, and performance models", *Journal of Vocational Behavior*, Vol. 115, 103316, doi: [10.1016/j.jvb.2019.06.004](https://doi.org/10.1016/j.jvb.2019.06.004).
- Lent, R.W., Brown, S.D. and Hackett, G. (1994), "Toward a unifying social cognitive theory of career and academic interest, choice, and performance", *Journal of Vocational Behavior*, Vol. 45 No. 1, pp. 79-122, doi: [10.1006/jvbe.1994.1027](https://doi.org/10.1006/jvbe.1994.1027).
- Lent, R.W., Brown, S.D., Schmidt, J., Brenner, B., Lyons, H. and Treistman, D. (2003), "Relation of contextual supports and barriers to choice behavior in engineering majors: test of alternative social cognitive models", *Journal of Counseling Psychology*, Vol. 50 No. 4, pp. 458-465, doi: [10.1037/0022-0167.50.4.458](https://doi.org/10.1037/0022-0167.50.4.458).
- Lent, R.W., Brown, S.D., Sheu, H.-B., Schmidt, J., Brenner, B.R., Gloster, C.S., Wilkins, G., Schmidt, L.C., Lyons, H. and Treistman, D. (2005), "Social cognitive predictors of academic interests and goals in engineering: utility for women and students at historically black universities", *Journal of Counseling Psychology*, Vol. 52 No. 1, pp. 84-92, doi: [10.1037/0022-0167.52.1.84](https://doi.org/10.1037/0022-0167.52.1.84).
- Liu, T., Yang, B., Geng, Y. and Du, S. (2021), "Research on face recognition and privacy in China—based on social cognition and cultural psychology", *Frontiers in Psychology*, Vol. 12, 809736, doi: [10.3389/fpsyg.2021.809736](https://doi.org/10.3389/fpsyg.2021.809736).
- Livinti, R., Gunnesch-Luca, G. and Ilescu, D. (2021), "Research self-efficacy: a meta-analysis", *Educational Psychologist*, Vol. 56 No. 3, pp. 215-242, doi: [10.1080/00461520.2021.1886103](https://doi.org/10.1080/00461520.2021.1886103).
- Lopatto, D. (2004), "Survey of undergraduate research experiences (SURE): first findings", *Cell Biology Education*, Vol. 3 No. 4, pp. 270-277, doi: [10.1187/cbe.04-07-0045](https://doi.org/10.1187/cbe.04-07-0045).
- Lopatto, D. (2007), "Undergraduate research experiences support science career decisions and active learning", *CBE-Life Sciences Education*, Vol. 6 No. 4, pp. 297-306, doi: [10.1187/cbe.07-06-0039](https://doi.org/10.1187/cbe.07-06-0039).
- Maddux, J.E., Sherer, M. and Rogers, R.W. (1982), "Self-efficacy expectancy and outcome expectancy: their relationship and their effects on behavioral intentions", *Cognitive Therapy and Research*, Vol. 6 No. 2, pp. 207-211, doi: [10.1007/BF01183893](https://doi.org/10.1007/BF01183893).
- McDaniel, J., Prael, A.H. and Schuele, C.M. (2022), "Rethinking research mentoring: a tutorial on how and why to implement a PhD student-mediated mentorship model", *Perspectives of the ASHA Special Interest Groups*, Vol. 7 No. 2, pp. 499-511, doi: [10.1044/2021_PERSP-21-00044](https://doi.org/10.1044/2021_PERSP-21-00044).
- Medaille, A., Beisler, M., Tokarz, R. and Bucy, R. (2022), "The role of self-efficacy in the thesis-writing experiences of undergraduate honors students", *Teaching and Learning Inquiry*, Vol. 10, doi: [10.20343/teachlearninqu.10.2](https://doi.org/10.20343/teachlearninqu.10.2).
- Medakovich, M. (2019), "Effectiveness of near-peer mentorship on self-efficacy and perceived stress during clinical transition for student nurse anesthetists", University at Buffalo, School of Nursing, doi: [10.1097/01.NPR.0000657320.35417.d2](https://doi.org/10.1097/01.NPR.0000657320.35417.d2).
- Mendez, S.L., Martin Conley, V., Keith, R.S., Haynes, C. and Gerhardt, R. (2017), "Mentorship in the engineering professoriate: exploring the role of social cognitive career theory", *International Journal of Mentoring and Coaching in Education*, Vol. 6 No. 4, pp. 302-316, doi: [10.1108/IJMCE-12-2016-0077](https://doi.org/10.1108/IJMCE-12-2016-0077).

IJMCE

- Mogk, D.W. (1993), "Undergraduate research experiences as preparation for graduate study in geology", *Journal of Geological Education*, Vol. 41 No. 2, pp. 126-128, doi: [10.5408/0022-1368-41.2.126](https://doi.org/10.5408/0022-1368-41.2.126).
- Mohd Rasdi, R. and Ahrari, S. (2020), "The applicability of social cognitive career theory in predicting life satisfaction of university students: a meta-analytic path analysis", *PLoS One*, Vol. 15 No. 8, e0237838, doi: [10.1371/journal.pone.0237838](https://doi.org/10.1371/journal.pone.0237838).
- Nworie, G.O., Gloria Okafor, T. and Christiana Igwebuike, C. (2023), "Performance of accounting students in Nigerian public universities: does accounting teaching methodology matter?", *International Journal of Research Publication and Reviews*, Vol. 04 No. 02, pp. 562-573, doi: [10.55248/gengpi.2023.42001](https://doi.org/10.55248/gengpi.2023.42001).
- Overall, N.C., Deane, K.L. and Peterson, E.R. (2011), "Promoting doctoral students' research self-efficacy: combining academic guidance with autonomy support", *Higher Education Research and Development*, Vol. 30 No. 6, pp. 791-805, doi: [10.1080/07294360.2010.535508](https://doi.org/10.1080/07294360.2010.535508).
- Palmer, R.J., Hunt, A.N., Neal, M. and Wuetherick, B. (2015), "Mentoring, undergraduate research, and identity development: a conceptual review and research agenda", *Mentoring and Tutoring: Partnership in Learning*, Vol. 23 No. 5, pp. 411-426, doi: [10.1080/13611267.2015.1126165](https://doi.org/10.1080/13611267.2015.1126165).
- Pelletier, L.G., Séguin-Lévesque, C. and Legault, L. (2002), "Pressure from above and pressure from below as determinants of teachers' motivation and teaching behaviors", *Journal of Educational Psychology*, Vol. 94 No. 1, pp. 186-196, doi: [10.1037/0022-0663.94.1.186](https://doi.org/10.1037/0022-0663.94.1.186).
- Petko, J.T., Sivo, S.A. and Lambie, G.W. (2020), "The research self-efficacy, interest in research, and research mentoring experiences of doctoral students in counselor education", *Journal of Counselor Preparation and Supervision*, Vol. 13 No. 1, p. 3, doi: [10.7729/12.0108](https://doi.org/10.7729/12.0108).
- Pfund, C., Maidl Pribbenow, C., Branchaw, J., Miller Lauffer, S. and Handelsman, J. (2006), "The merits of training mentors", *Science*, Vol. 311 No. 5760, pp. 473-474, doi: [10.1126/science.1123806](https://doi.org/10.1126/science.1123806).
- Pfund, C., Sancheznieto, F., Byars-Winston, A., Zárate, S., Black, S., Birren, B., Rogers, J. and Asai, D.J. (2022), "Evaluation of a culturally responsive mentorship education program for the advisers of howard hughes medical institute gilliam program graduate students", edited by Braun, D., *CBE-Life Sciences Education*, Vol. 21 No. 3, p. ar50, doi: [10.1187/cbe.21-11-0321](https://doi.org/10.1187/cbe.21-11-0321).
- Renninger, K.A. and Hidi, S. (2015), *The Power of Interest for Motivation and Engagement*, Routledge, New York.
- Roemer, E., Schuberth, F. and Henseler, J. (2021), "HTMT2—an improved criterion for assessing discriminant validity in structural equation modeling", *Industrial Management and Data Systems*, Vol. 121 No. 12, pp. 2637-2650, doi: [10.1108/IJMD-02-2021-0082](https://doi.org/10.1108/IJMD-02-2021-0082).
- Ryan, R.M. and Deci, E.L. (2000), "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being", *American Psychologist*, Vol. 55 No. 1, pp. 68-78, doi: [10.1037/0003-066X.55.1.68](https://doi.org/10.1037/0003-066X.55.1.68).
- Saini, M., Kumar, A. and Kaur, G. (2020), "Research perception, motivation and attitude among undergraduate students: a factor analysis approach", *Procedia Computer Science*, Vol. 167, pp. 185-192, doi: [10.1016/j.procs.2020.03.210](https://doi.org/10.1016/j.procs.2020.03.210).
- Sawitri, D.R. and Creed, P.A. (2021), "Perceived research environment, motivation, and academic interest in research: a social-cognitive perspective", *International Journal for Educational and Vocational Guidance*, Vol. 21 No. 3, pp. 489-506, doi: [10.1007/s10775-020-09449-w](https://doi.org/10.1007/s10775-020-09449-w).
- Seymour, E., Hunter, A.-B., Laursen, S.L. and DeAntoni, T. (2004), "Establishing the benefits of research experiences for undergraduates in the sciences: first findings from a three-year study", *Science Education*, Vol. 88 No. 4, pp. 493-534, doi: [10.1002/sc.10131](https://doi.org/10.1002/sc.10131).
- Shi, R.C. (2016), "Ben ke sheng dao shi zhi: Lei xing, shi shi xian zhuang ji wan shan dui ce [Undergraduate tutor system: types, implementation status and improvement measures]", *Da Xue Jiao Yu Ke Xue*, No. 03, pp. 70-73.
- Shi, J., Duan, K., Xu, Q. and Li, J. (2021), "Effect analysis of the driving factors of super-gentrification using structural equation modeling", *PLOS ONE, Public Library of Science*, Vol. 16 No. 3, e0248265, doi: [10.1371/journal.pone.0248265](https://doi.org/10.1371/journal.pone.0248265).

Silvia, P.J. (2003), "Self-efficacy and interest: experimental studies of optimal incompetence", *Journal of Vocational Behavior*, Vol. 62 No. 2, pp. 237-249, doi: [10.1016/S0001-8791\(02\)00013-1](#).

Tsang, B. (2018), "Undergraduate students can be a boon to your lab", *Natur*, doi: [10.1038/d41586-018-07665-7](#).

Vaccaro, N. (2009), "The relationship between research self-efficacy, perceptions of the research training environment and interest in research in co", Electronic Theses and Dissertations.

Wang, S. (2019), "A student's guide to undergraduate research", *Nature*. doi: [10.1038/d41586-019-00871-x](#).

Wu, J.R., Kundu, M., Iwanaga, K., Chan, F., Chen, X., Rumrill, P. and Wehman, P. (2022), "Social cognitive career theory predictors of goal persistence in African American college students with Disabilities", *Rehabilitation Counseling Bulletin*, Vol. 67 No. 2, 00343552221108407-93, doi: [10.1177/00343552221108407](#).

Young, J.Q., Sugarman, R., Schwartz, J., Thakker, K. and O'Sullivan, P.S. (2020), "Exploring residents' experience of career development scholarship tracks: a qualitative case study using social cognitive career theory", *Teaching and Learning in Medicine*, Vol. 32 No. 5, pp. 522-530, doi: [10.1080/10401334.2020.1751637](#).

Zhang, X. (2013), "Income disparity and digital divide: the Internet Consumption Model and cross-country empirical research", *Telecommunications Policy*, Vol. 37 No. 6, pp. 515-529, doi: [10.1016/j.telpol.2012.12.011](#).

Zhou, A.J., Lapointe, É. and Zhou, S.S. (2019), "Understanding mentoring relationships in China: towards a Confucian model", *Asia Pacific Journal of Management*, Vol. 36, pp. 415-444, doi: [10.1007/s10490-019-09653-5](#).

Zimmerman, B.J., Bandura, A. and Martinez-Pons, M. (1992), "Self-motivation for academic attainment: the role of self-efficacy beliefs and personal goal setting", *American Educational Research Journal*, Vol. 29 No. 3, pp. 663-676, doi: [10.3102/00028312029003663](#).

Appendix
Survey questionnaire

Part A: Demographic information

School: Discipline: Current Level of Education:

Participation status:

Have you ever had an academic advisor?

Have you had experience in academic conferences, paper publishing, research projects, or academic competitions?

Part B: Research self-efficacy, interests, outcome expectation and aspirations

Rate each statement on a scale from 1 (Extremely Inconsistent) to 7 (Extremely Consistent)

- (1) I am confident that I am competent for scientific research
- (2) I am confident that I can handle the difficulties in scientific research
- (3) I am good at scientific research analysis
- (4) I can express the research results very well
- (5) Willing to read research journal articles
- (6) Willing to be a member of the research team
- (7) Willing to lead the research team

IJMCE

- (8) Willing to develop data analysis strategies for research
- (9) Doing research will enhance my sense of self-worth
- (10) I believe that scientific research skills will be very helpful to my employment
- (11) I have a clear understanding of how to research
- (12) Hope to make achievements in professional fields
- (13) Hope to learn more data processing methods
- (14) Hope to further my study in my professional field
- (15) Hope to apply for scientific research project fund

Part C: Mentor support and relationship

Rate each statement on a scale from 1 (Extremely Inconsistent) to 7 (Extremely Consistent)

- (1) Clarify my expectations and goals in scientific research
- (2) Make institutional plans for me in scientific research
- (3) Provide me with pertinent suggestions on scientific research
- (4) Help me improve my scientific research writing
- (5) Teach me the knowledge and methods related to scientific research
- (6) When I encounter difficulties, he will help me
- (7) My mentor affirms my value and efforts when I am engaged in scientific research
- (8) When discussing my research with me, my mentor will be very enthusiastic and positive
- (9) Will listen kindly and respond to my concerns
- (10) Helped me and completed my scientific research task together

Corresponding author

Jun Xu can be contacted at: xujun1569@gmail.com

同行评审证明

论文标题	国家科技创新平台赋能关键核心技术创新质量的效应与机制		
投稿期刊	经济理论与经济管理		
作者位次	第三作者	当前状态	同行评审中

证明材料

国家科技创新平台赋能关键核心技术创新质量的效应与机制

提 要：关键核心技术创新质量是突破“卡脖子”技术封锁的基础，国家科技创新平台通过集聚创新资源、促进主体合作和优化生态环境，为其提升奠定了基础。本文以集成电路为例，采用渐进双重差分模型和模糊集定性比较分析法探讨了国家科技创新平台赋能关键核心技术创新质量的时空效应和组态机制。研究发现：国家科技创新平台对关键核心技术创新质量的赋能效应可持续 4 年，并通过空间溢出效应显著促进了其周边 100 千米内企业的技术发展。在时空交互的不同情景下，国家科技创新平台赋能关键核心技术创新质量的机制存在差异化：邻近-持续情景下，可采用设施-政策协同机制；远程-持续情景下，可采用政策-产业环境驱动和人员-原料-市场-产业协同两类机制；邻近-滞后情景下，可采用原料-政策协同和设施-原料-市场协同两类机制。研究结论为理解国家科技创新平台的复杂性和多维性提供了新视角，对加快国家科技自立自强，实现关键核心技术攻关具有重要意义。

关键词：国家科技创新平台；关键核心技术创新质量；时空交互效应；组态机制

一、引言

党的二十届三中全会明确提出“加强国家战略科技力量建设，完善国家实验室体系，优化国家科研机构、高水平研究型大学、科技领军企业定位和布局，推进科技创新央地协同，统筹各类科创平台建设”。国家科技创新平台是国家创新体系的重要组成，是国家战略科技力量的重要载体，也是实现高水平科技自立自强的重要基础。在关键核心技术攻关进程中，国家科技创新平台作为政府加强国家战略科技力量建设的统筹布局，为企业攻克“卡脖子”技术、掌握关键核心技术发挥了重大的引领和支持作用。国家科技创新平台包括（但不限于）国家重点实验室、国家技术创新中心、国家工程技术研究中心、国家工程研究中心、国家企业技术中心、国家产业创新中心和制造业创新中心等，通过资源聚合、主体合作、生态营造，构建“需求牵引-平台支撑-协同攻关-产业落地”的闭环体系，有利于企业“真需求”与科研单位的“好成果”有效对接（韩凤芹等，2023），对关键核心技术创新质量提升具有重要作用。

已有与本文主题相关研究主要包括：一是以国家科技创新平台为主题，现有研究主要围绕国际比较、分布特征与作用机制三个维度展开。国际比较方面，研究从平台发展、管理模式、经费配置等角度，对比分析了中美欧等主要科技强国在创新平台建设方面的差异（卞松保和柳卸林，2011；寇明婷等，2020；庄越和叶一军，2003）。分布特征方面，聚焦国家企业技术中心、国家重点实验室、国家实验室的空间分布格局与动态演化特征（敦帅和陈强，2023；庄德林等，2020）；作用机制方面，管理体制、开放与流动程度、国内外局势和科技发展趋势等内外驱动国家科技创新平台建设运行（李静等，2025；周岱等，2008）。二是以关键核心技术为主题，主要聚焦于概念特征界定、影响因素和突破路径。概念特征界定方面，现有研究基于国家安全观（胡旭博和原长弘，2022）、产业地位（郑刚等，2023）和技术差距（范书琴等，2023）等视角对其进行特征描述和概念界定；影响因素方面，宏观层面的政策引导、市场机制、中观层面的产业集聚、产学研合作以及微观层面的创新人才、资金投入等因素对关键核心技术创新具有影响（孙琴，2024）；突破路径方面，技术和制度升级与突破是现有研究关注的两大重点（吴晓波等，2025；Vivek & Khoo, 2021）。三是以国家科技创新平台与关键核心技术为主题，目前集中在影响效应与作用机制。影响效应方面，国

同行评审证明

论文标题	When posts outpace patents: Multimodal corporate AI washing and stakeholder verification frictions on digital platforms		
投稿期刊	Journal of Business Research		
作者位次	第一作者	当前状态	同行评审中

证明材料



Track your submission

This is a new submission-tracking service.

Is this helpful?

Yes

No

Peer review status

When posts outpace patents: Multimodal corporate AI washing and stakeholder verification frictions on digital platforms

• Reviews completed: 0

• Review invitations accepted: 0

• Review invitations sent: 2+

Under Review

Last review activity: 28th July 2026 ⓘ

Watch to learn what we're doing behind the scenes ↗

Journal:

Journal of Business Research

Corresponding author:

Jun Xu

First author:

Yuying Luo

Date of submission:

17th July 2026

Manuscript number:

JOB-26-07178

Need more help?

同行评审证明

论文标题	Beyond the Project Blueprint: How Regional Configurations Align Digital Development with Innovation Network Resilience for Infrastructure-Intensive Development		
投稿期刊	Engineering, Construction and Architectural Management		
作者位次	第一作者	当前状态	同行评审中

证明材料



Emerald Publishing

Home

My Dashboard

Journals

Author Guide

Reviewer Guide

Route Logic of Danmaku Engagement

Aslib Journal of Information Management

Status: Accepted Last Updated: Jul 14, 2026

Manuscript ID: AJIM-01-2026-0111.R1

Stage-Specific Impacts of Digital Technologies on Supply-Chain Resilience: Meta-Analytic Evidence for Continuous Process Improvement

Business Process Management Journal

Status: Accepted Last Updated: Sep 13, 2025

Manuscript ID: BPMJ-04-2025-0550.R1

Beyond the Project Blueprint: How Regional Configurations Align Digital Development with Innovation Network Resilience for Infrastructure-Intensive Development

Engineering, Construction and Architectural Management

Status: Awaiting Referee Selection Last Updated: Jul 10, 2026

Manuscript ID: ECAM-07-2026-1429

Submission Details

Beyond the Project Blueprint: How Regional Configurations Align Digital Development with Innovation Network Resilience for Infrastructure-Intensive Development

Manuscript submitted

Your manuscript has been submitted for peer review. You will be notified when the review process is complete.

Details

Abstract

Contacts

Submitting Author: Yuying Luo

Co-authors: Jun Xu, Fengmei Liu, Xueguo Xu

Journal: Engineering, Construction and Architectural Management

Document ID: ECAM-07-2026-1429

Status: Awaiting Referee Selection

View Proof

?

同行评审证明

论文标题	Seeing experience through multimodal traces: How practice auditability calibrates information processing and trust in the GenAI era		
投稿期刊	Information Processing & Management		
作者位次	通讯作者	当前状态	同行评审中

证明材料



Track your submission

This is a new submission-tracking service.

Is this helpful?

Yes

No

Peer review status

Seeing experience through multimodal traces:
How practice auditability calibrates information
processing and trust in the GenAI era

• Reviews completed: 0

• Review invitations accepted: 2+

• Review invitations sent: 2+

Under Review

Last review activity: 6th
August 2026 ⓘ

Watch to learn what we're doing behind
the scenes ↗

Journal:

Information Processing and Management

Corresponding author:

Yuying Luo

First author:

Jun Xu

Date of submission:

7th July 2026

Manuscript number:

IPM-D-26-04537

Need more help?

同行评审证明

论文标题	Frictionless tools, regulated minds: Designing GenAI diagnostic scaffolding for self-regulated learning in higher education		
投稿期刊	The Internet and Higher Education		
作者位次	共同通讯作者	当前状态	同行评审中

证明材料



Track your submission

This is a new submission-tracking service.

Is this helpful?

Yes

No

Peer review status

Frictionless tools, regulated minds: Designing GenAI diagnostic scaffolding for self-regulated learning in higher education

• Reviews completed: 0

• Review invitations accepted: 1

• Review invitations sent: 2+

Under Review

Last review activity: 29th July 2026 ⓘ

Watch to learn what we're doing behind the scenes ↗

Journal:

The Internet and Higher Education

Corresponding author:

Yonghe Wu

First author:

Jun Xu

Date of submission:

12th June 2026

Manuscript number:

INTHIG-D-26-00660

同行评审证明

论文标题	Recipes for Synergy: How Regional Endowments Drive Digital and Innovation Network Resilience		
投稿期刊	Technology Analysis & Strategic Management		
作者位次	第一作者	当前状态	返修中

证明材料

163网
mail.163.com

xvxueguo@163.com

设置 手机App 下载桌面端 参与调研 自助查询 会员中心 开通邮箱会员

支持邮件全文搜索 AI 搜

首页 通讯录 AI工具箱 收件箱 企业邮箱 26612352...

收信 写信

收件箱

红旗邮件

待办邮件

智能标签

星标联系人邮件

草稿箱

已发送

其他2个文件夹

邮件标签

邮箱中心

超大附件

邮箱附件

其他工具

邮件追踪

简历优化

智能面试顾问

求职信息订阅

PDF转换工具

发票助手

企业邮箱

<< 返回 回复 回复全部 转发 删除 举报 拒收 标记为 移动到 更多

266123520 (Technology Analysis & Strategic Management) A revise decision has been made on your submission

发件人: Technology Analysis & Strategic Management<onbehalfof@manuscriptcentral.com>

(由 0101019fbc323544-e2522cd5-76d5-4dea-849b-9a098c1eba74-000000@outbound.manuscriptcentral.com 代发, 帮助)

收件人: 我<xvxueguo@163.com>

抄送人: luca.ferri<luca.ferri@unina.it>

时 间: 2026年08月01日 15:20 (星期六)

翻译成中文

01-Aug-2026

Dear Xueguo Xu,

Your manuscript entitled "Recipes for Synergy: How Regional Endowments Drive Digital and Innovation Network Resilience" which you submitted to Technology Analysis & Strategic Management, has been reviewed. Further comments are included at the bottom of this letter.

You will see that although the referees find some merit in the paper it is required that substantial revisions be done before we can consider it further. Nevertheless, we do hope that you will be able to undertake the additional work on the paper and look forward to receiving a revised manuscript in due course.

PLEASE ENSURE THAT THE TOTAL NUMBER OF WORDS IN YOUR REVISED MANUSCRIPT REMAINS BELOW OUR MAXIMUM PERMITTED LENGTH OF 7,000 WORDS. NOTE THAT THIS TOTAL INCLUDES REFERENCES, TABLES, APPENDICES, ACKNOWLEDGMENTS, AND BIOGRAPHIES.

When you revise your manuscript please highlight the changes you make in the manuscript by using the track changes mode in MS Word or by using bold or coloured text.

同行评审证明

论文标题	Beyond symbolic compliance: A meta-analysis of corporate AI tasks and non-interchangeable sustainability outcomes		
投稿期刊	Business Ethics, the Environment & Responsibility		
作者位次	第一作者	当前状态	同行评审中

证明材料



Beyond symbolic compliance: A meta-analysis of corporate AI tasks and non-interchangeable sustainability outcomes

Under Review

Manuscript ID: 7669286 Article type: Original Article

Publication: Business Ethics, the Environment & Responsibility

This submission is under consideration and cannot be edited. Further information will be emailed to you by the journal editorial office. [Learn about what happens](#) when your submission is under review.

Submission Overview

Submitted 20 July 2026 by Jun Xu

Started 8 July 2026 by System

Phase-Specific Effects of AI in Self-Regulated Learning: A Meta-Analysis of a Decade of Interventions

Abstract: This meta-analysis addresses inconsistent findings on AI's role in self-regulated learning (SRL) by systematically examining 35 empirical studies (2013-2025) through Zimmerman's cyclical model. Results reveal a moderate overall effect of AI interventions ($g=0.507$), with significantly stronger impacts during the task performance phase ($g=0.574$). A key finding is the superior efficacy of generative AI ($g=0.799$) over rule-based and data-driven paradigms across all SRL phases. Critically, this synthesis highlights a "performance-competence divide," where AI excels at scaffolding immediate tasks but its ability to cultivate transferable competence remains unproven. This research offers a nuanced, evidence-based map for designing future AI systems that foster genuine learner autonomy.

Keywords: Self-regulated learning, Artificial intelligence, large language models, Educational technology, Meta-analysis

Research Objective

While artificial intelligence (AI) shows considerable promise for supporting self-regulated learning (SRL), the empirical evidence remains fragmented, with inconsistent findings across different learning phases and a lack of synthesis that accounts for AI's rapid technological evolution. This meta-analysis addresses this critical gap by systematically synthesizing a decade of intervention research (2013-2025). Guided by Zimmerman's (2002) cyclical model, the study's primary objective is to quantify the overall and phase-specific effects of AI support on SRL. It further aims to disaggregate these effects by examining how they vary across three distinct technological paradigms—rule-based, data-driven, and generative AI—and to identify key moderators that shape their effectiveness. The overarching goal is to provide a comprehensive, evidence-based map of AI's evolving role in SRL, moving beyond a monolithic view to offer nuanced insights for designing technologies that foster genuine learner autonomy.

Theoretical Framework

A Cyclical Model of Self-Regulated Learning

This study is grounded in the social cognitive view of self-regulated learning (SRL), which posits that learners proactively manage their cognition, motivation, and behavior to achieve academic goals (Zimmerman, 1989). While SRL theory has evolved to encompass various dynamic perspectives (e.g., Bandura, 1991; Winne & Hadwin, 1998), this meta-analysis adopts Zimmerman's (2002) three-phase cyclical model as its primary analytical lens. This model delineates SRL into three recursive phases: a **forethought** phase involving pre-task analysis and goal setting; a **performance** phase characterized by in-task strategy use and self-monitoring; and a **self-reflection** phase for post-task evaluation and attribution. Its widespread use as an operational framework in technology-enhanced learning research (Wong et al., 2019) provides a robust, process-oriented structure to systematically map and evaluate the effects of diverse AI interventions on the core components of individual learning.

Synthesizing AI's Evolving Role in the SRL Cycle

While prior research affirms the general benefits of technology for supporting learner autonomy (Dent & Koenka, 2016; Dignath & Büttner, 2018), the rapid growth of the SRL field itself (see Figure 1) has outpaced systematic syntheses of AI's specific role. Existing meta-analyses have often concentrated on particular technologies or treated SRL as a unified construct, thereby obscuring how different AI paradigms support the distinct cognitive and metacognitive demands of each regulatory phase (Chen et al., 2024; Lu & Lin, 2024). This study addresses this critical

gap by moving beyond a monolithic view of AI. It synthesizes evidence through the dual lens of Zimmerman’s (2002) model and the technological evolution of AI (see Figure 2)—from rule-based systems to data-driven platforms and generative AI. This approach allows for a nuanced examination of how the alignment between specific AI affordances and the psychological processes within each SRL phase contributes to intervention effectiveness, providing a more detailed map for future research and evidence-based practice.

Method

This study followed the meta-analytic procedures outlined by Pigott and Polanin (2020) and adhered to PRISMA reporting guidelines (Chen et al., 2024).

Data Sources and Inclusion Criteria

A systematic search for empirical studies published between 2013 and 2025 was conducted across four major databases: ERIC, Ei Compendex, Scopus, and Web of Science. The search and screening process, detailed in the flow diagram (see Figure 3), yielded a final sample of 35 studies. To be included, studies had to: (1) examine an AI-supported SRL intervention; (2) use an experimental or quasi-experimental design comparing an intervention to a control group; (3) provide measures of SRL processes or direct outcomes; (4) report sufficient data for effect size calculation; and (5) achieve a Medical Education Research Study Quality Instrument (MERSQI) score of 10 or higher (Sawatsky et al., 2015). The final robust sample (mean MERSQI = 13.07), comprising 31 journal articles and 4 conference proceedings, yielded 133 independent effect sizes for analysis.

Coding and Data Analysis

A systematic coding scheme was developed to extract study characteristics and classify AI paradigms (see Table 1). To enable a phase-specific analysis, SRL outcome variables from each study were mapped onto Zimmerman’s (2002) forethought, performance, or self-reflection phases (see Table 2). The coding process demonstrated strong interrater reliability (Cohen’s $\kappa = .85-.88$) (Landis & Koch, 1977). Due to significant heterogeneity across all subgroups ($I^2 \geq 75\%$), a random-effects model was employed for all analyses (Higgins & Thompson, 2002). Effect sizes were calculated as Hedges’ g to account for small sample sizes, and all analyses were conducted using Stata 18 with robust variance estimation (Hedges & Tipton, 2010).

Results

The analysis of 35 primary studies yielded 133 effect sizes. Initial heterogeneity tests confirmed substantial variance across studies (overall $I^2 = 89.81\%$, $p < .001$), justifying the use of a random-effects model for all subsequent analyses.

The Evolution of AI Paradigms for SRL Support

In response to RQ1, our chronological analysis of the included studies identified three co-existing technological paradigms for AI-supported SRL, the distribution of which is illustrated in Figure 4. The initial paradigm consists of rule-based systems, which rely on predefined expert knowledge to provide structured but relatively rigid scaffolding (e.g., Duffy & Azevedo, 2015). This was followed by the emergence of data-driven personalization systems, characterized by their capacity to analyze real-time learner data to dynamically tailor feedback and learning paths (e.g., Harati et al., 2021). The most recent paradigm features generative AI ecosystems, which leverage large language models (LLMs) to act as proactive, dialogic partners in the learning process (e.g., Ng et al., 2024).

Overall and Phase-Specific Effects of AI Interventions

As detailed in Table 3, AI-supported interventions demonstrated a moderate and statistically significant overall effect on SRL ($g = 0.507$, 95% CI [0.384, 0.630]). The impact of AI varied across the SRL cycle, with the strongest effect observed during the performance phase ($g = 0.574$). The effects were smaller for the self-reflection ($g = 0.464$)

and forethought phases ($g = 0.401$). Sensitivity analyses confirmed the stability of these estimates. However, visual inspection of the funnel plots (see Figure 5 and Figure 6) and Egger's regression tests indicated the potential for publication bias, particularly for the overall sample and the performance phase (see Table 4), suggesting results should be interpreted as potential upper boundaries.

Subgroup and Moderator Analyses

Subgroup analyses by AI paradigm, which reflect a clear technological progression over the past decade, revealed significant differences in effectiveness (see Table 5). Generative AI interventions yielded a large overall effect ($g = 0.799$), which was substantially greater than those of rule-based ($g = 0.534$) and data-driven systems ($g = 0.318$). This advantage for generative AI was consistent across all SRL phases, with a particularly strong effect on performance ($g = 0.938$). Further moderator analyses (see Table 6) identified several contextual factors influencing intervention effectiveness. Significantly larger effects were found for interventions conducted with secondary school students ($g = 0.636$), in natural science disciplines ($g = 0.753$), within fully online environments ($g = 0.732$), and for durations of less than two weeks ($g = 0.776$).

Discussion and Significance

This meta-analysis moves the field beyond the question of whether artificial intelligence supports SRL to provide a nuanced, evidence-based account of how, when, and which types of AI are most effective. By synthesizing a decade of research through the dual lenses of Zimmerman's (2002) SRL model and AI's technological evolution, this study offers significant theoretical and practical contributions.

Bridging the Performance-Competence Divide: A Core Challenge

The central finding of this study is the identification of a critical "performance-competence divide." While our results confirm that AI interventions yield a moderate overall benefit ($g = 0.507$), their impact is strongest during the in-task performance phase. This suggests that AI excels at providing immediate, in-the-moment scaffolding. However, our synthesis of behavioral evidence indicates that these supported regulatory behaviors often decay once the technology is withdrawn (e.g., Bouchet et al., 2016; Long & Aleven, 2013). This pattern echoes Salomon et al.'s (1991) seminal distinction between the effects with technology versus the lasting cognitive residue, or effects of technology. Our findings empirically substantiate this divide in the context of modern AI, suggesting current systems are more adept at acting as cognitive partners than as tools for internalizing durable skills.

Furthermore, this study demonstrates that AI is not a monolithic entity. The superior efficacy of generative AI, especially in the historically challenging forethought ($g = 0.709$) and self-reflection ($g = 0.826$) phases, signals a paradigm shift. Its effectiveness stems from its capacity for dialogic interaction, which can scaffold the nuanced processes of goal-setting and causal attribution in ways that passive data dashboards cannot (Lee et al., 2024; Pan et al., 2025). This underscores that AI's value lies not in its raw power, but in the precise alignment between its affordances and the cognitive demands of each SRL phase (see Table 5).

Implications for Design: Engineering AI to Cultivate Autonomy

These insights carry profound implications for designing the next generation of educational technologies. To bridge the performance-competence divide, the field must move beyond creating systems that risk fostering dependency and instead engineer AI that explicitly cultivates learner autonomy. Our findings advocate for intelligent integration strategies, such as combining the stable, structural scaffolding of rule-based systems with the adaptive, dialogic support of generative AI to address the full SRL cycle. Critically, AI systems must be designed with mechanisms that promote skill internalization. This includes implementing adaptive scaffolds that strategically fade over time, utilizing guidance-based approaches that provide hints rather than direct answers (Lee et al., 2025), and creating opportunities for "productive failure" where support is temporarily withdrawn to encourage autonomous problem-solving.

A Research Agenda: Moving Beyond Immediate Performance

Finally, our work calls for a fundamental shift in how the field researches and assesses AI-supported SRL. The moderator analysis revealing the outsized impact of short-term interventions (<2 weeks, $g = 0.776$) points to a prevalence of novelty effects that may mask the development of lasting skills. To generate more valid insights, future research must adopt longitudinal designs to track whether scaffolded behaviors persist after the intervention ends. This necessitates new assessment frameworks that prioritize delayed, unsupported performance as the ultimate indicator of success. The critical question for the field must evolve from “Does AI improve learning outcomes?” to “Does AI cultivate learners who can regulate their own learning independently?”. Only by addressing this question can we harness AI’s potential to develop the authentic, transferable autonomy that underpins lifelong learning.

Figures and Tables

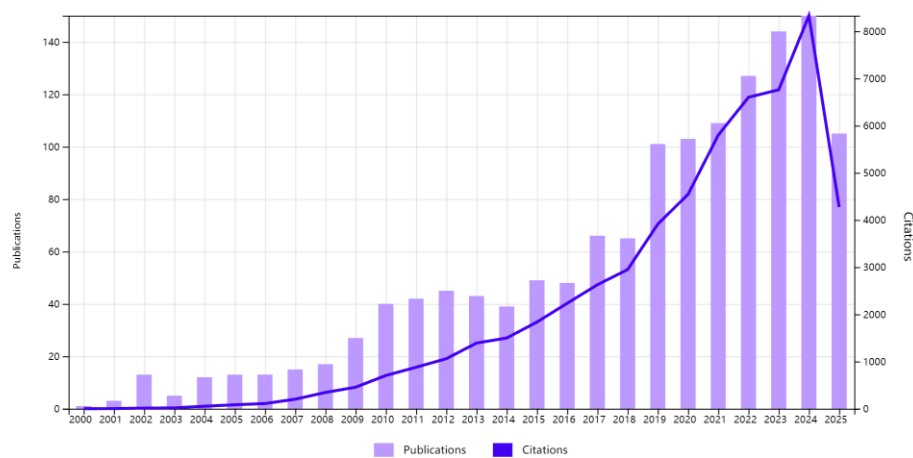


Figure 1. Annual Publications and Citation Trends in Self-Regulated Learning Since the 21st Century

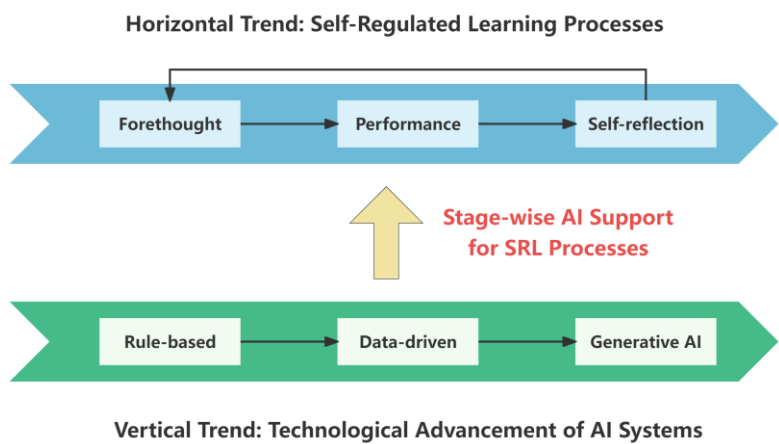


Figure 2. The Two Key Dimensions of This Study

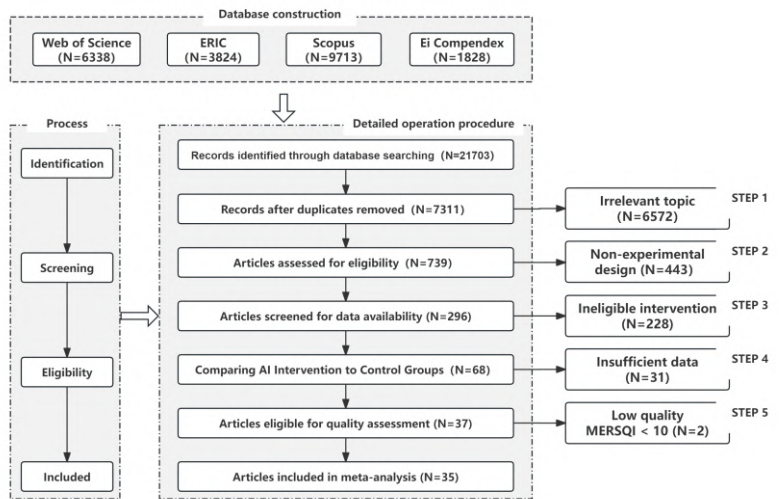


Figure 3. Search flow diagram

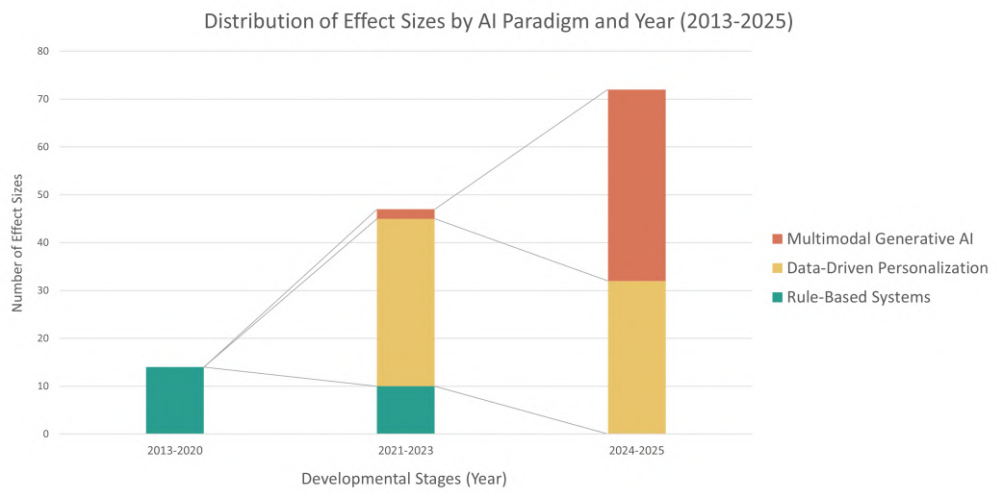


Figure 4. Distribution of Effect Sizes by AI Paradigm and Year (k=133 effect sizes)

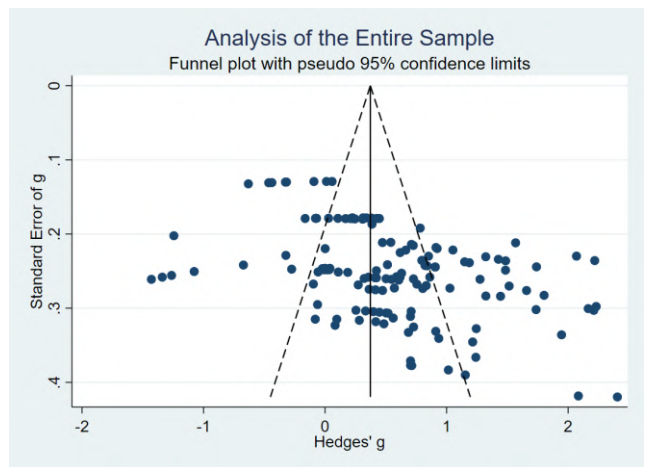


Figure 5. Funnel Plot of Overall Sample

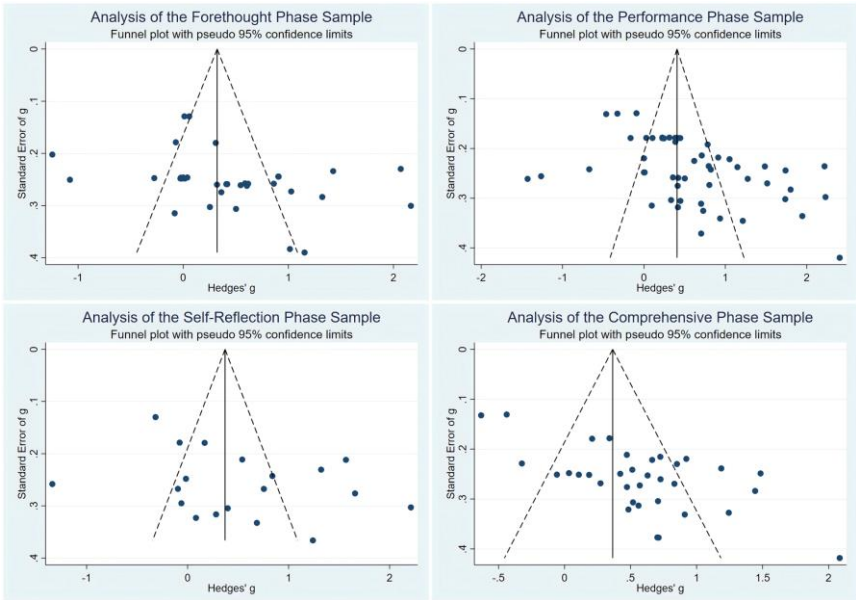


Figure 6. Funnel Plot of Combined Phases

Table 1. Lists of codes for the analysis of selected articles

Dimension	Item	Description
Participant Characteristics	Region	Text records, e.g., the United States, China, Japan, etc.
	Sample Size	(1) 1–50; (2) 51–100; (3) 101–300.
	Educational Level	(1) Primary school; (2) Junior and senior high school; (3) Institutions of higher education.
	Discipline	(1) Natural Science (including science, mathematics, physics, biology, geography); (2) Social Science (including politics, education, psychology, linguistics).
	Learning Environment	(1) Online learning settings; (2) Blended learning settings.
Content Characteristics	AI Intervention Method	A system that uses computational models (e.g., rule-based logic, machine learning, Generative Modeling) to provide adaptive or personalized feedback and scaffolding for SRL.
	SRL Variables	Variables related to SRL recorded in the literature, such as self-efficacy, metacognition, attention focus, and time management.

Table 2. Mapping of SRL Indicators to Phases

Phase	Theoretical Definition	Indicator	Definition
Forethought (Pre-task)	The preparatory phase involving task analysis and mobilizing motivational beliefs.	Goal setting and strategy planning Self-efficacy Task value and interest Outcome expectation	Establishing clear learning goals and implementation paths. Confidence in one’s ability to complete tasks. Awareness of the significance and enjoyment of learning tasks. Predictions of the effort and potential benefits of learning.
Performance (During task)	The execution phase involving self-control (e.g., strategy use) and self-observation.	Task strategies Attention focus and time management Self-observation and monitoring Seeking help	Applying cognitive and metacognitive skills (e.g., summarizing, monitoring). Focusing on tasks and allocating study time effectively. Continuously tracking one’s performance and strategy usage. Identifying difficulties and actively seeking external assistance.
Self-Reflection (Post-task)	The evaluative phase involving self-judgment of performance and adaptive self-reactions.	Self-evaluation Causal attribution Adaptive inference and	Comparing outcomes with standards or initial expectations. Interpreting successes or failures as a basis for strategy modification or effort adjustment. Adjusting future study behavior based on

Comprehensive Phase	An assessment of the overall effectiveness of an SRL-driven task, using measures not attributable to a single phase.	response	reflective results.
		Academic achievement	Test results or indicators of academic performance.
		Skill acquisition and transfer	Application of skills and knowledge in new contexts.
		Learner satisfaction and engagement	Emotional responses and investment in the learning process.

Table 3. Random-Effects Model for AI Interventions in SRL Phases

Variable	Effect size and 95% CI					Heterogeneity test				Tau-squared		
	k	Hedges' g	SE	Lower	Upper	Q	df(Q)	p	I ²	Tau-squared	SE	Tau
Forethought Phase	33	0.401	0.123	0.160	0.643	271.43	32	0.0000	88.95	0.415	0.133	0.644
Performance Phase	51	0.574	0.110	0.358	0.789	533.81	50	0.0000	92.05	0.474	0.119	0.688
Self-Reflection Phase	18	0.464	0.196	0.079	0.848	194.45	17	0.0000	91.81	0.594	0.256	0.771
Comprehensive Phase	31	0.529	0.103	0.328	0.730	204.14	30	0.0000	81.93	0.337	0.117	0.581
Overall	133	0.507	0.063	0.384	0.630	1208.24	132	0.0000	89.81	0.432	0.069	0.657

Table 4. Measures of publication bias

Self-Regulated Learning Process	k	Egger's				Fail-Safe N	
		Score	SD	z	p	Classic	Orwin
Forethought Phase	33	5.15	2.270	2.27	0.0232	594	930
Performance Phase	51	5.94	1.622	3.66	0.0003	3167	1945
Self-Reflection Phase	18	4.08	3.333	1.22	0.2213	218	555
Comprehensive Phase	31	5.86	3.384	1.73	0.0834	843	930
Overall	133	4.66	1.186	3.93	0.0001	5908	4451

^a k, number of studies

^b **Classic Fail-Safe N:** Represents the number of null-effect studies needed to raise the p-value above the significance threshold ($\alpha = 0.05$).

^c **Orwin's Fail-Safe N:** Represents the number of missing studies required to reduce the effect size to a predefined trivial level (ES = 0.01).

Table 5. Combined Effect Sizes for AI Developmental Stages and SRL Phases (Hedges' g)

SRL Process	AI Phase	k	Hedges' g	95% CI	I ²	p	Z
Forethought	Rule-based	2	-	-	-	-	-
	Data-driven	15	0.083	[-0.301, 0.466]	92.53 %	0.6708	0.43
	Generative	16	0.709	[0.437, 0.981]	76.84 %	0.0000	5.11
Performance	Rule-based	6	-	-	-	-	-
	Data-driven	32	0.352	[0.080, 0.624]	93.45 %	0.0112	2.54
	Generative	13	0.938	[0.549, 1.328]	85.64 %	0.0000	4.71
Self-Reflection	Rule-based	2	-	-	-	-	-
	Data-driven	8	0.196	[-0.387, 0.779]	94.24 %	0.5105	0.66
	Generative	8	0.826	[0.247, 1.405]	88.08 %	0.0051	2.80
Comprehensive	Rule-based	13	0.429	[0.197, 0.662]	59.66 %	0.0003	3.61
	Data-driven	13	0.580	[0.190, 0.971]	90.94 %	0.0036	2.91
	Generative	8	0.682	[0.416, 0.938]	60.22%	0.0000	5.02
Overall	Rule-based	23	0.534	[0.339, 0.730]	66.77 %	0.0000	5.34
	Data-driven	68	0.318	[0.134, 0.501]	93.07 %	0.0006	3.41
	Generative	42	0.799	[0.604, 0.994]	82.34 %	0.0000	8.03

^a Caution is advised for subgroups with k < 10, while those with k < 7 may have limited reliability.

Table 6. Subgroup Analysis Results of Meta-Analysis(Hedges' g)

Subgroup	k	g	95 % CI	Z (p)	Between p*
Educational level					0.221 (3.02)
Primary school	14	0.304	[0.060 , 0.548]	2.44 (0.015)	
Junior and senior high school	11	0.636	[0.284 , 0.989]	3.54 (0.001)	
Higher education	108	0.517	[0.374 , 0.660]	7.09 (< 0.001)	
Subject					< 0.001 (21.09)
Natural Science	73	0.753	[0.576 , 0.929]	8.36 (< 0.001)	
Social Science	56	0.211	[0.061 , 0.360]	2.77 (0.006)	
Combined subject areas	129	0.516	[0.390 , 0.643]	8.00 (< 0.001)	
Learning space					0.087 (2.93)
Online learning	37	0.732	[0.376 , 1.087]	4.04 (< 0.001)	
Blended learning	96	0.409	[0.309 , 0.509]	8.02 (< 0.001)	
Experimental duration					0.002 (12.44)
Less than 2 weeks	45	0.776	[0.551 , 1.002]	6.88 (< 0.001)	
2 – 10 weeks	43	0.361	[0.222 , 0.500]	7.64 (< 0.001)	
> 10 weeks	45	0.374	[0.139 , 0.609]	0.32 (< 0.001)	
Overall	133	0.507	[0.384 , 0.630]	8.08 (< 0.001)	—

Funding

This study is supported by a 2021 major project from the National Social Science Foundation of China (grant number 21 and ZD328).

Acknowledgments

This research has been funded by the Research Council of Finland (formerly Academy of Finland) grant numbers 324381, 350249, the University of Oulu profiling project Profi7 Hybrid Intelligence - 352788. This work/Part of the work was carried out with the support of Leaf Research Infrastructure, University of Oulu, Finland.

References

- Bandura, A. (1991). Social cognitive theory of self-regulation. *Organizational Behavior and Human Decision Processes*, 50(2), 248–287. [https://doi.org/10.1016/0749-5978\(91\)90022-L](https://doi.org/10.1016/0749-5978(91)90022-L)
- Bouchet, F., Harley, J. M., & Azevedo, R. (2016). Can Adaptive Pedagogical Agents' Prompting Strategies Improve Students' Learning and Self-Regulation? In A. Micarelli, J. Stamper, & K. Panourgia (Eds.), *Intelligent Tutoring Systems* (pp. 368–374). Springer International Publishing. https://doi.org/10.1007/978-3-319-39583-8_43
- Chen, S., Cheng, H., & Huang, Y. (2024). Emotion Recognition in Self-Regulated Learning: Advancing Metacognition Through AI-Assisted Reflections. In D. Kourkoulou, A.-O. (Olnancy) Tzirides, B. Cope, & M. Kalantzis (Eds.), *Trust and Inclusion in AI-Mediated Education: Where Human Learning Meets Learning Machines* (pp. 185–212). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-64487-0_9
- Dent, A. L., & Koenka, A. C. (2016). The Relation Between Self-Regulated Learning and Academic Achievement Across Childhood and Adolescence: A Meta-Analysis. *Educational Psychology Review*, 28(3), 425–474. Scopus. <https://doi.org/10.1007/s10648-015-9320-8>
- Dignath, C., & Büttner, G. (2018). Teachers' direct and indirect promotion of self-regulated learning in primary and secondary school mathematics classes – insights from video-based classroom observations and teacher interviews. *Metacognition and Learning*, 13(2), 127–157. Scopus. <https://doi.org/10.1007/s11409-018-9181-x>
- Duffy, M. C., & Azevedo, R. (2015). Motivation matters: Interactions between achievement goals and agent scaffolding for self-regulated learning within an intelligent tutoring system. *Computers in Human Behavior*, 52, 338–348. <https://doi.org/10.1016/j.chb.2015.05.041>

-
- Harati, H., Sujo-Montes, L., Tu, C.-H., Armfield, S. J. W., & Yen, C.-J. (2021). Assessment and Learning in Knowledge Spaces (ALEKS) Adaptive System Impact on Students' Perception and Self-Regulated Learning Skills. *Education Sciences*, 11(10), Article 10. <https://doi.org/10.3390/educsci11100603>
- Hedges, L. V., & Tipton, E. (2010). Meta-analysis. In A. Steptoe (Ed.), *Handbook of Behavioral Medicine* (pp. 909–921). Springer New York. https://doi.org/10.1007/978-0-387-09488-5_58
- Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11), 1539–1558. <https://doi.org/10.1002/sim.1186>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 159–174.
- Lee, H.-Y., Chen, P.-H., Wang, W.-S., Huang, Y.-M., & Wu, T.-T. (2024). Empowering ChatGPT with guidance mechanism in blended learning: Effect of self-regulated learning, higher-order thinking skills, and knowledge construction. *International Journal of Educational Technology in Higher Education*, 21(1), 16. <https://doi.org/10.1186/s41239-024-00447-4>
- Lee, Y.-F., Hwang, G.-J., & Chen, P.-Y. (2025). Technology-based interactive guidance to promote learning performance and self-regulation: A chatbot-assisted self-regulated learning approach. *Educational Technology Research and Development*. <https://doi.org/10.1007/s11423-025-10478-x>
- Long, Y., & Aleven, V. (2013). Supporting Students' Self-Regulated Learning with an Open Learner Model in a Linear Equation Tutor. In H. C. Lane, K. Yacef, J. Mostow, & P. Pavlik (Eds.), *Artificial Intelligence in Education* (pp. 219–228). Springer. https://doi.org/10.1007/978-3-642-39112-5_23
- Lu, W., & Lin, C. (2024). Meta-Analysis of Influencing Factors on the Use of Artificial Intelligence in Education. *The Asia-Pacific Education Researcher*. <https://doi.org/10.1007/s40299-024-00883-w>

-
- Ng, D. T. K., Tan, C. W., & Leung, J. K. L. (2024). Empowering student self-regulated learning and science education through CHATGPT: A pioneering pilot study. *British Journal of Educational Technology*, 55(4), 1328–1353. <https://doi.org/10.1111/bjet.13454>
- Pan, M., Lai, C., & Guo, K. (2025). Effects of GenAI-empowered interactive support on university EFL students' self-regulated strategy use and engagement in reading. *The Internet and Higher Education*, 65, 100991.
- Pigott, T. D., & Polanin, J. R. (2020). Methodological Guidance Paper: High-Quality Meta-Analysis in a Systematic Review. *Review of Educational Research*, 90(1), 24–46.
<https://doi.org/10.3102/0034654319877153>
- SALOMON, G., PERKINS, D. N., & GLOBERSON, T. (1991). Partners in Cognition: Extending Human Intelligence with Intelligent Technologies. *Educational Researcher*, 20(3), 2–9.
<https://doi.org/10.3102/0013189X020003002>
- Sawatsky, A. P., Beckman, T. J., Edakkanambeth Varayil, J., Mandrekar, J. N., Reed, D. A., & Wang, A. T. (2015). Association Between Study Quality and Publication Rates of Medical Education Abstracts Presented at the Society of General Internal Medicine Annual Meeting. *Journal of General Internal Medicine*, 30(8), 1172–1177. <https://doi.org/10.1007/s11606-015-3269-7>
- Winne, P., & Hadwin, A. (1998). Studying as Self-Regulated Learning. In *Metacognition in Educational Theory and Practice* (Vol. 93, pp. 277–304).
- Wong, J., Baars, M., Davis, D., Zee, T. V. D., Houben, G.-J., & Paas, F. (2019). Supporting Self-Regulated Learning in Online Learning Environments and MOOCs: A Systematic Review. *International Journal of Human–Computer Interaction*. <https://www.tandfonline.com/doi/abs/10.1080/10447318.2018.1543084>
- Zimmerman, B. J. (1989). A social cognitive view of self-regulated academic learning. *Journal of Educational Psychology*, 81(3), 329–339. <https://doi.org/10.1037/0022-0663.81.3.329>

Zimmerman, B. J. (2002). Becoming a Self-Regulated Learner: An Overview. *Theory Into Practice*, 41(2), 64–70.

https://doi.org/10.1207/s15430421tip4102_2

2022 10th International Conference on Information and Education Technology (ICIET 2022)

Notification of Acceptance

ICIET 2022

April 9-11, 2022 | Matsue, Japan

www.iciet.org

Dear Mr./Ms. Chengliang Wang, Jian Dai, Yuying Luo and Lipeng Wen,

Paper ID: T134

Paper Title: Systematic Analysis of Research in Educational Management from the Perspective of Bibliometrics and Scientometrics

Congratulations! The review process for 2022 10th International Conference on Information and Education Technology (ICIET 2022) has been completed. Based on the recommendations of the reviewers and the Technical Program Committees, we are pleased to inform you that your paper identified above has been accepted for publication and oral presentation. You are cordially invited to present the paper orally at ICIET 2022 to be held in Matsue, Japan (*hybrid conference*) from April 9 to 11, 2022.

Note: The organizing committee of ICIET has decided to host the conference partially virtual due to the spread of coronavirus. Special discounts are available for online attendance. Please download the registration form below for more details.

***PUBLICATION:** Peer reviewed and presented papers of ICIET 2022 will be published in the **Conference Proceedings by IEEE**, which will be submitted for Ei Compendex and Scopus.

***REGISTRATION:** Please see page 2.



Registration Instruction

***In order to register the conference and have your paper included in the proceeding successfully, you must finish following 5 steps.**

- 1. Revise your final paper according to the Review Comments carefully.
- 2. Format the paper according to the Template.
<http://www.iciet.org/pubform.docx>
- 3. Fill in the Registration Form and complete the payment of your registration fee.
<http://www.iciet.org/author-reg.docx>
- 4. Sign the IEEE Copyright form.
<http://www.iciet.org/IEEE-copyright.pdf>
- 5. Send your **final papers** (both Word and PDF format), **copyright form** (PDF format), **filled registration form** (Word format) **and the proof of payment** (.jpg format) to icietconf@vip.163.com **(Before March 25, 2022).**

Please strictly adhere to the format specified in the conference template while preparing your final paper. If you have any problem in preparing registration documents, please feel free to contact us via icietconf@vip.163.com. For the most updated information on the conference, please check it at <http://www.iciet.org/news.html>. The Conference Program may be available at the end of March.

Finally, we would like to further extend our congratulations to you and we are looking forward to meeting you in ICIET 2022.